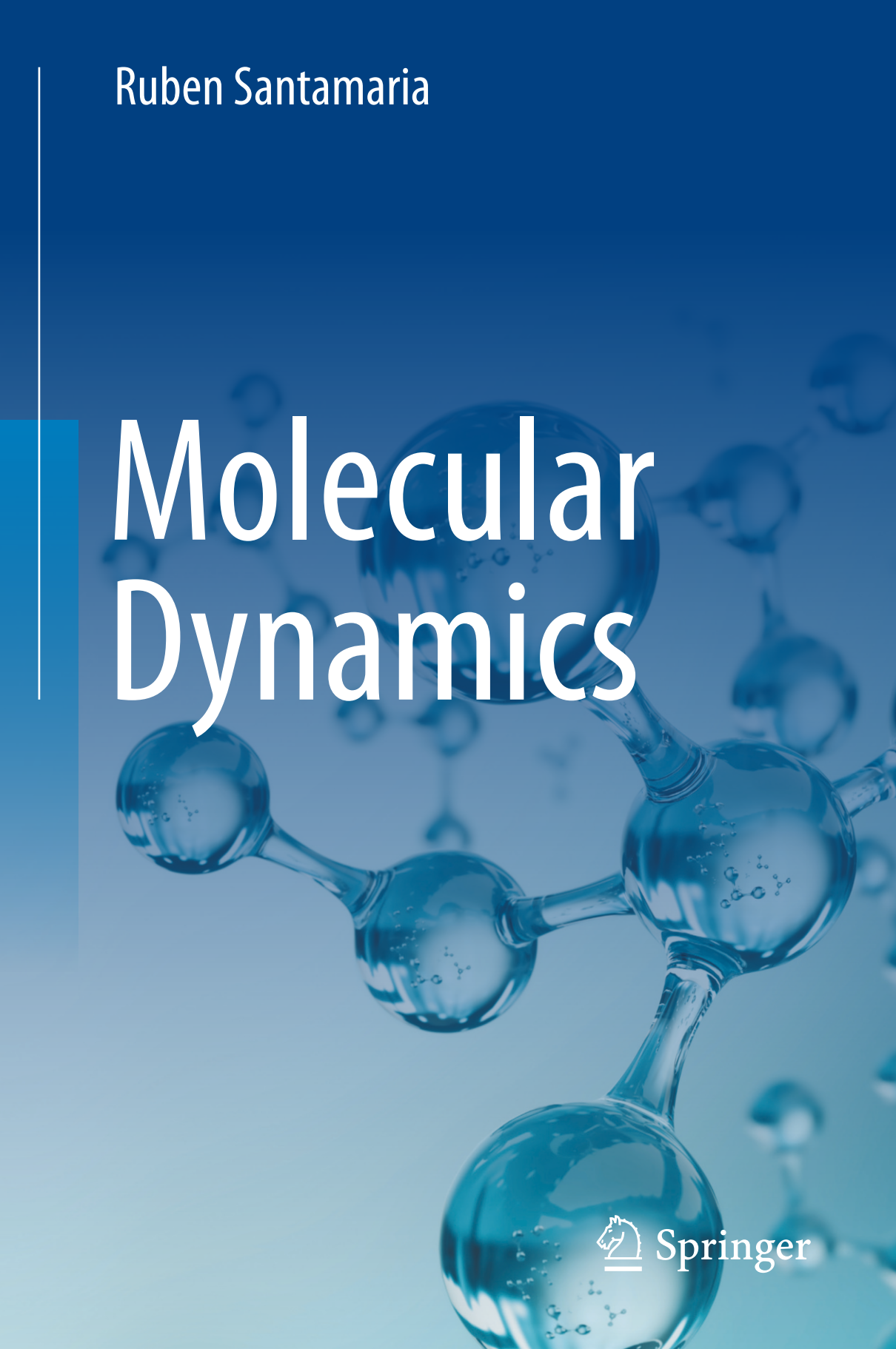
Ruben Santamaria

Molecular Dynamics



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With Love To

Abril
Andrea
Mariana

I would like to express my gratitude for the motivation given by my parents, brother, and sisters, as well as the guidance and knowledge provided by my thesis supervisors, teachers, and colleagues. Additionally, I appreciate the experiences gained from working with my thesis students and friends. I sincerely thank each and every one of them from the bottom of my heart.

Preface

Molecular dynamics is a broad scientific discipline that covers multiple fields, from physics to biology. This is because molecular dynamics can effectively describe the workings of nature from the atomic perspective. The book's organized structure provides consistent and intelligible explanations, making it easy for readers with diverse backgrounds to understand and make sense of molecular dynamics. The book is self-contained, covering the fundamental concepts of mechanics, while also catering to experts by including the required mathematical formalism. Hence, the book is suitable for both non-experts and experts alike.

The book covers a range of topics, from classical mechanics to quantum mechanics, and from few-body systems to many-body systems. It is particularly useful for scientists studying the connection between small and large particle systems, whether in a state of equilibrium or not. The book offers discussions at various levels of theory, allowing for the exploration of complex chemical reactions through first-principles molecular dynamics, while relaxed levels of theory permit the simulation of atomic interactions with model potentials. The book is valuable for academics conducting research at the intersection of molecular dynamics, thermodynamics, and fluid dynamics.

The book is designed to be educational for both teachers and students, with figures, illustrations, problems, solutions, and suggested readings included. It covers the latest developments in molecular dynamics and is a valuable reference for researchers and experts in various scientific fields who share an interest in exploring and characterizing phenomena at the atomic level. The book's discussions make it suitable for researchers and students in physics, chemistry, biology, and interdisciplinary subjects like nanoscience, materials engineering, molecular genetics, and more. Additionally, the book provides the formal and necessary foundations to encourage readers to expand their knowledge of the microscopic world. By learning and utilizing the theory of molecular dynamics, dedicated researchers can make exciting new discoveries in their fields.

Mexico City, Mexico — Ruben Santamaria

Prologue

Molecular dynamics embodies the laws of physics, and thus, allows the study of atomic-level processes. By understanding how microscopic attributes of nature lead to macroscopic properties of matter, we can design new compounds and improve processes. Molecular dynamics enable us to achieve this understanding. The ultimate goal of molecular dynamics is to use natural laws to generate new knowledge, create innovative technologies, and enhance the quality of life.

This book covers the fundamentals of molecular dynamics, integrated by formal concepts, diverse mechanical variables, and dynamic expressions. Due to the intricate nature of atomic-level processes, various levels of theory are unfolded to investigate and characterize the complex world of atoms. Molecular dynamics is a rapidly expanding field, thanks to both the scientific advancements in atomic-level research and the swift progress of computer technology.

The book is constituted by five main parts, which are integrated by one or more chapters:

- Basics of Classical Mechanics
- Basics of Quantum Mechanics
- First-Principles Molecular Dynamics
- Classical Molecular Dynamics
- Time Evolution Operators

Basics of Classical Mechanics

This part of the book begins by discussing the basic principles of classical mechanics, as outlined in Newton's theory. This is because Newtonian mechanics provides a foundation for understanding particle motion and the mechanical variables that are used to describe particle systems. When combined with model interaction potentials, this approach to molecular dynamics becomes very versatile. It can be applied to systems with few or many particles, regardless of whether they are in thermodynamic equilibrium or not. However, when dealing with large numbers of particles, it is important to pay attention to the collective behavior of the system and to use statistical methods to analyze it. By doing so, it is possible to establish a connection between molecular dynamics and statistical mechanics, and to compare

the ensemble averages of the latter with the time averages of the former. In this context, it is important to understand how microscopic and macroscopic properties of matter are related. One way to do this is by using different types of mechanical descriptors, which help to classify and distinguish the state of a system from a dynamic point of view. The tools to describe macroscopic bodies are conveyed in the rigid body approximation. The last chapter introduces the Lagrangian, Hamiltonian, and Hamilton-Jacobi theories to not only provide the elegant formalisms and the different paths in the formulation of the classical equations of motion, but also reach a point where it is possible to create novel approaches in molecular dynamics. The connections among all those classical theories are briefly analyzed.

Basics of Quantum Mechanics

This part of the book is an introduction to quantum theory. It explains the fundamental concepts and equations of quantum mechanics that are required to understand how atoms behave and are described at the quantum level. It covers topics such as the dual nature of matter, the way energy is quantized, and the behavior of microscopic particles with spin. The main equation used in quantum molecular dynamics is the time-dependent wave equation, which is eventually simplified into the time-independent wave equation.

First-Principles Molecular Dynamics

In this part of the book, we explore the time-dependent wave equations of electrons and nuclei, which are essential for first-principles molecular dynamics simulations. We use a self-consistent field approach and later achieve a stationary wave equation for the electrons. To bridge the gap between quantum and classical molecular dynamics, we discuss the mixed quantum-classical molecular dynamics scheme. In this respect, we investigate the classical limit of nuclear motion and derive Newtonian equations of motion for the nuclei. Through these chapters, we connect quantum molecular dynamics to classical molecular dynamics, thus exploring flexible approaches from rigorous approaches.

Classical Molecular Dynamics

In this part of the book, we focus on classical equations of motion for atoms and use model functions for atomic interactions, avoiding the wave equation of electrons. We discuss a variety of model interaction potentials, like those for biological and non-biological molecules. These potentials simplify simulations, making it possible to work with many-body systems since electron effects are parameterized. We also dedicate a chapter to approximations that simulate the behavior of extended systems like gases, fluids, and solids. These approximations are important for

determining the dynamical and thermodynamic properties of compounds immersed in an extended sample.

Time Evolution Operators

In this part of the book, we focus on how particle systems evolve over time. We use the Poisson-bracket formalism to obtain classical time evolution operators, which can be applied in the mixed quantum-classical molecular dynamics approach. By calculating atomic forces on the fly, time integrators can be used to evolve the positions and momenta of particles over time. These integrators have many benefits: they are general, time-reversible, computationally efficient, and simplify simulations by avoiding mathematical complications.

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Part I

Basics of Classical Mechanics



Principles of Classical Dynamics

1

Classical Mechanics is the theoretical framework of physics which describes the mechanical and dynamical aspects of the classical particles and objects. This chapter and the subsequent chapters present the fundamental concepts of classical mechanics such as space, time, mass, and more and introduce the basic variables of linear momentum, angular momentum, and torque from the Newtonian perspective. The notion of time and time reversibility are established. The importance of a reference system of coordinates and the transformation of the physical variables between different inertial reference systems are shown.[1]

1.1 Newtonian Dynamics

Classical mechanics establishes the underlying concepts, mechanical elements, and basic equations on the dynamics of particles. It is also given the name of *Newtonian dynamics* in honor to Sir Isaac Newton, who formulated the laws of classical mechanics. There are other formulations in classical mechanics, like those proposed by Lagrange, Hamilton, Jacobi, Poisson, and others. Those treatments were proposed after the conception of Newtonian dynamics and provide universal concepts on the mechanical aspects of the particles, in addition to their own particular equations of motion. The different formulations of classical mechanics are in one way or another important tools in the analysis of particle systems, and therefore, they all are of interest in molecular simulations.[2]

One of the main purposes of molecular dynamics is to compute the positions, speeds, and forces at every instant of time of the particles. The Newtonian theory applies to particles traveling with velocities sufficiently smaller than the

[1] The discussion follows similar topics to those of: Alonso and Finn [1] and Halliday et al. [2].

[2] A law in physics is a scientific proposal which tells the regular patterns in the behavior of physical bodies and matter under precise conditions, by establishing the relations between different physical variables. Once the law is formulated, it helps us predict the behavior of matter.

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speed of light because, according to the theory of relativity, the bodies suffer length contraction and time dilation at high speeds, which are features beyond the consideration of Newtonian mechanics. The "exotic states" of matter and the particles with very low or very high energies are excluded in Newtonian theory. In similar form, the gravitational effects are out of the description because they are negligible at the atomic level: the electric force, f_e, and the gravitational force, f_g, of a proton interacting with an electron separated by a distance of 1 $Bohr$ give the force ratio $f_e \sim 10^{39} f_g$. Another example illustrating the large differences in the forces is the magnetic force. The magnetic force of a magnet is capable of attracting a small metallic sphere, overcoming the gravitational force exerted by the entire planet. In our classical approach, it is assumed that the particles and physical events have character of "ordinary particles" and "ordinary events," which are properly described by Newtonian mechanics.

Finally, there are entities that are not defined in terms of more basic concepts, variables, or objects in Newtonian theory. They are considered as *fundamental entities*. The ones of interest in molecular dynamics are space, time, mass, energy, and electric charge. It has been postulated that the fundamental entities emerged at the first instances on the creation of the Universe, with relation to each other. Such fundamental entities play important roles in molecular dynamics, and thus, they are considered in our discussion.

1.2 Space and Time

Space and time are formally intertwined together into a single framework referred to as *spacetime*. The spacetime concept is mathematically well accepted since it leads to invariant metrics [3]. The spacetime is conceived as a set of four independent dimensions. Three of them form the ordinary $3D$ *space*, where we are immersed and can freely move, and the other dimension is *time*. We cannot freely move in this last dimension. Time permeates the $3D$ space and never stands still. Time is not necessarily regular and may be measured with time ticks at different rates. According to the theory of relativity, time is not absolute and depends on the relative motions of the observers. In consequence, simultaneous events are no longer so for reference frames mainly moving with high relative speeds.

The four dimensions are neither perceptible nor distinguishable. The spacetime may be considered either an intellectual artifact, with the purpose to construct a conceptual structure of the Universe, or a real object, like a $4D$ limitless fabric with indistinguishable regions, no discernible geometrical shape, and with existence with no reference to any other object. The lack of perception of spacetime demands measurements of relative type among the bodies. The above two points (on conceiving the spacetime as a conceptual structure or a real structure) are in contradiction. In spite of that, they have been used to establish the concepts and formulations that explain natural phenomena, simultaneously allowing to reach a deep understanding of the Universe.

1.3 Mass

In principle, the entities of the Universe may be classified into two classes: *particles of matter* and *particles of light*, or *photons*. The particles of matter have *mass*. Mass is not a fundamental property but an emergent feature of a subatomic (Higgs quantum field) process due to the change in the energy landscape of vacuum. Yet mass is of central importance. From a classical perspective, it is understood as the property of a body to resist changes in its state of motion. Mass is a physically measurable quantity, and therefore, is a *physical observable*.

According to the theory of general relativity, mass and light warp the $4D$ spacetime, Fig. 1.1. Mass and energy are equivalent, with the possibility of transforming one onto another due to the high-energy physics expression:

$$E = mc^2 \tag{1.1}$$

The sub-atomic experiments on particle production exemplify the transformation of mass into energy and vice versa. A matter particle is considered an energy-condensed structure. The mass is subjected to the passage of time, but the light is not. The matter with mass travels slower than light, which has no rest mass. In a closed system, the total energy of the particles (which also includes the energy equivalent of the particle masses) is preserved with time. The mass immersed in a gravitational field acquires *weight*, which is a type of force acting on the mass.

$$\mathbf{w} = m\mathbf{g} \tag{1.2}$$

The factor $\mathbf{g}$ is the gravitational acceleration. In the absence of gravity, the object has mass but not weight. A body isolated from any interaction maintains the initial state of motion, but the state of motion changes under the action of a force, with an acceleration proportional to the mass. A heavy mass is accelerated less rapidly than a light mass due to the different *inertias* that the bodies with distinct masses present. An *inertial mass* represents the resistance of the body to change its state of motion, while the *gravitational mass* is the capacity of the body to curve the spacetime due

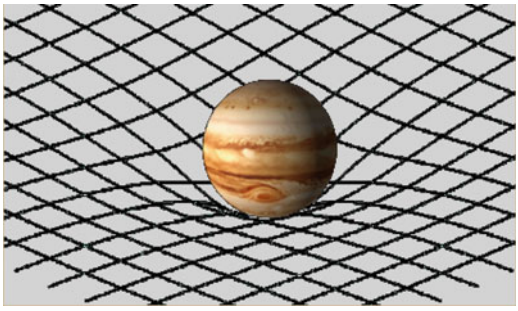


Fig. 1.1 According to the theory of general relativity, any mass bends the $4D$ space

to its gravitational field. In spite of the different conceptions, the inertial mass and gravitational mass are equivalent.

The inertial mass is determined in relative terms, by making a couple of bodies interact, assigning to one of them the unit mass, measuring the appropriate dynamic variables of the experiment, and determining from the mathematical expressions the unknown mass of the second body. On the other hand, the gravitational mass is measured with a balance in the presence of a gravitational field. Let us consider a falling body. If the body weight is $m_g\mathbf{g}$ with m_g the gravitational mass and $\mathbf{a}$ the acceleration of the inertial mass m_i, then we have:

$$\mathbf{a} = (m_g/m_i)\,\mathbf{g} \tag{1.3}$$

When dealing with a different body whose gravitational mass is m'_g and its inertial mass is m'_i, we also have $\mathbf{a} = (m'_g/m'_i)\,\mathbf{g}$. Under equal experimental conditions of falling objects, the accelerations $\mathbf{a}$ imprinted to the primed and unprimed objects are observed to be the same. This leads to the *free fall principle*.

Statement 1.1 Free fall principle: all the bodies fall at the same rate.

Problem 1.1 If the force of gravity depends on the object mass, m, in the form $f = GM_{planet}m/r^2$, show that two bodies with different masses fall by the same rate.

Solution The gravitational force of M_{planet} exerted on body 1 is $f_1 = GM_{planet}m_1/r^2$, and on body 2 is $f_2 = GM_{planet}m_2/r^2$. The acceleration of m_1 and m_2 are $a_1 = f_1/m_1 = GM_{planet}/r^2$ and $a_2 = f_2/m_2 = GM_{planet}/r^2$. Thus, $f1$ and $f2$ are different, but $a_1 = a_2$.

The ratio m_g/m_i in Eq. 1.3 is a constant and may be given the value of 1 to assert that the inertial mass and the gravitational mass own the same value. This in turn reveals the *equivalence principle* (in the weak form) referent to the inertial and gravitational masses under the presence of a gravitational field of local character. The equivalence principle also points out the equivalence between a gravitational field and an accelerated frame of reference, Fig. 1.2 [4].

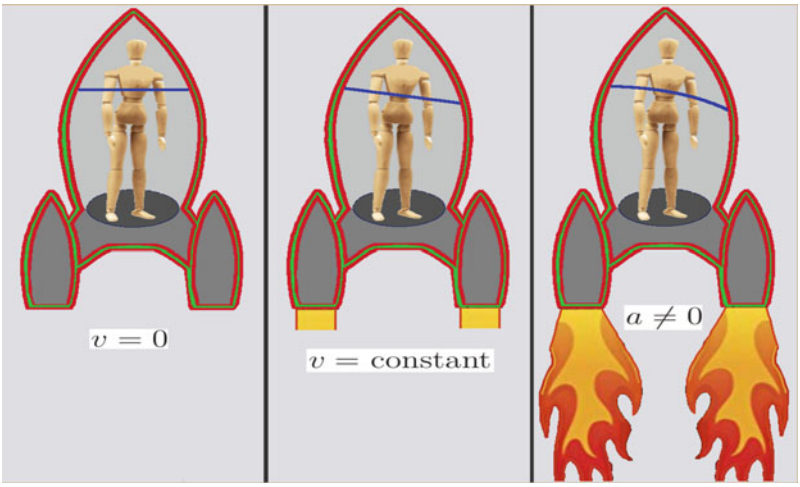


Fig. 1.2 A compartment with an observer in the interior is accelerated in the upward direction. The observer registers a laser light to curve downward. A similar phenomenon is watched when a planet with strong gravity bends a light beam. Thus, the gravitational force exerted on the light beam in the downward direction may be interpreted in terms of the force exerted on the compartment in the upward direction. However, in contraposition with the particles with masses, the photons describing the light beam have no rest mass, and the light bending is associated with the curvature of spacetime created by the planet

Statement 1.2 Equivalence principle (in the weak form): the acceleration of gravity acting on a body and registered in a small spacetime region is equivalent to a uniform acceleration of the body produced by an external force.

The mass is a property that makes the dynamics of atoms different: atoms with small and large masses have different inertias in the dynamics of molecular systems. The atomic mass, abbreviated amu, with value:

$$amu = 1.660,538,92 \times 10^{-27} Kg$$

is the reference unit mass at the atomic level. The mass may be also given in terms of energy units due to the mass-energy equivalence. One gram of mass is close to 89.9 TJ of energy ($1T = 10^{12}$). The large energy value of the mass is due to the light-speed factor taking part in the expression $E = mc^2$.[3]

[3] A quantity of 46.6 mg of mass is required to approximately produce a kiloton of TNT. The energy released by the first atom bomb of World War II was close to a dozen TNT kiloton, coming from $\sim 64\, Kg$ of enriched uranium.

1.4 Energy

From the mathematical perspective, the energy is a scalar quantity whose conjugate variable is the time. From the physical perspective, the energy is the capacity of a system to do work. The energy satisfies some important principles.

Statement 1.3 The energy is always preserved in isolated systems, and it is neither created nor destroyed but only transformed onto other energy forms.

The conservation of a physical observable with time is associated with a symmetrical aspect of nature. The symmetry corresponding to the conservation of the energy is the translational property of time. An event that occurred yesterday remains unchanged if we consider that it occurred any other day due to the translational symmetry of time. Under a general appraisal, the events occurring in nature are independent of the time coordinate, as the intervals of time are indistinguishable from each other. The laws of physics do not change under space and time translation (this is considered a symmetrical aspect of nature), leading to the conservation of energy (this is the conservation of a physical observable). In a closed system, the energy is preserved; nevertheless, when the system is open (this is with external interactions), the energy changes.[4] The energy-time relation is entangled and is present in every aspect of nature. The Heisenberg principle of quantum theory represents an example of the energy-time entanglement, where the limits imposed by nature to the simultaneous measurements of the energy and time are faced: it is not possible to measure the time without affecting the energy and vice versa.

The energy is conveniently classified according to the state of the object. The object may be static or non-static, and the type of energy may be in that order associated with *potential energy* or *kinetic energy*. The potential energy depends on the position coordinates and may also depend on the velocities, while the kinetic energy depends on the speeds. An example of the two basic energy forms is provided by a non-static pendulum. At the turning points of the oscillation, the energy of the body is potential energy, and at the middle point of the turning points, it is kinetic energy. The potential energy is determined by the location of the object in space under the presence of a forcefield, and the kinetic energy is defined by the motion of the object. The energy is not absolute and is measured with respect to a given state of the object. For the pendulum, the point where the body reaches the maximum

[4] Interestingly, in going from a closed system to an open one, we may insist on considering the system as a closed one; however, the external interactions modify the physical behavior of the system, and the modified behavior of the "closed system" may be attributed to a change of the physical forces with time, leading to an energy change. The symmetrical aspects on the nature of the system were modified, with the consequent change in the energy.

speed is frequently considered as the reference point of *zero-potential energy*. There are no rules to establish a reference point of zero-potential energy, but an adequate reference point facilitates the computation of the energy when the system evolves with time.

In the dynamics of classical particles, we have continuous energy conversions of the type potential-kinetic energy and vice versa and energy transfer from one body to another as well. The conversions of energy commonly observed at the macroscopic level are interpreted as *friction* and *thermal energy*. A system with energy may be used as a source of work or heat energy for other systems to overcome barriers imposed by the potential energy and induce, for example, the chemical transformation of compounds. The first law of thermodynamics establishes the energy balance of the macroscopic systems, including the possibility of exchanging energy through matter transfer. The first law is formulated in terms of the thermodynamic variables; still, it is the result of the interactions taking place at the microscopic level. In this respect, the theory of thermodynamics has molecular origins.

Many questions of philosophical character remain. It is a mystery the type of energy that created the Universe in the initial Big Bang event. Still, we know that the energy is preserved in the Universe. For confined systems of particles, the energy is quantized; nonetheless, it acquires continuous values at macroscopic scales, where Newtonian theory is acceptable. The form of energy available for work is getting reduced, since most thermodynamic processes are *irreversible* and dissipate energy in ways that cannot be recovered and reused.[5] In spite of the knowledge gathered on the energy, the question on the type of dominant energy that shall persist in the last episode of the Universe remains unclear.

1.5 Electric Charge

In the standard model of subatomic particles, the electric charge represents the coupling strength of the particle to the electromagnetic field and, in this way, to space. Fundamental particles like quarks interact among each other to give integer values to the charges of the nucleons. In the formulation of molecular dynamics, the electric charge shall be considered a fundamental quantity, and for such a reason, it is not defined in terms of other physical variables. The magnitude and sign of the electric charge associated with a particle are determined experimentally. It is not a general property because there are particles with no charge. The electric charge is independent of the motion state of the particle. The conservation of the electric charge of a molecular system is formulated by an equation of continuity in terms of the charge density and the electric-current flux. At the quantum mechanical level, the charge conservation is due to a phase invariance of the wave representing the particle. There are particles with *positive charge* and *negative charge*, which may have equal or different masses. The charge comes in multiples of the elementary

[5] A *reversible process* is a phenomenon where energy dissipation takes no place in the system.

electric charge, $e = 1.602,176,634 \times 10^{-19}\, C$ in SI units, except for some sub-atomic particles. The proton and electron have positive and negative charges, respectively. The particles with no charge are the *neutral particles*, like the neutron. A charged particle produces an *electromagnetic field* that exerts electromagnetic forces on charged particles, but has no interaction with neutral particles. The electrostatic interaction is ruled by the *Coulomb's law*, which is similar to the gravitational force $\mathbf{f}_g = -G\, m_1 m_2\, \hat{\mathbf{n}}/|\mathbf{r}_1 - \mathbf{r}_2|^2$. The Coulomb's law states that the electric interaction decreases with the square of the interparticle distance.

$$\mathbf{f} = \frac{q_1 q_2}{4\pi\varepsilon}\, \frac{\hat{\mathbf{n}}}{|\mathbf{r}_1 - \mathbf{r}_2|^2} \quad ; \quad \hat{\mathbf{n}} = \frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|} \tag{1.4}$$

$\mathbf{f}$ is the mutal force between a pair of particles with charges q_1 and q_2 located at $\mathbf{r}_1$ and $\mathbf{r}_2$, with interparticle distance $|\mathbf{r}_1 - \mathbf{r}_2|$. The *permittivity of space* is ε, while the *Coulomb constant* $k_c = 1/(4\pi\varepsilon)$ has value $k_c e^2 = 2.30708 \times 10^{-28}\, J \cdot m$. The electromagnetic forces are capable of accelerating charged objects resembling the acceleration of the gravitational field. At sufficiently small dimensions, we are not able to tell if the acceleration of a particle is due to a gravitational force or an electromagnetic force.

The nucleus of an atom is made of protons and neutrons. The protons and neutrons are more spatially localized than the electrons. Such a feature imparts to some extent a classical character to the nucleus, and under certain conditions, it may be visualized as a point particle with a net mass and a net charge. On the other side, the electron is depicted as a negatively charged cloud surrounding the nucleus. In the dynamics of particles, the nuclei of the molecular system may move from one place to another carrying their own positive charges and forcing the redistribution of the electron cloud. It is common to find the electric charge irregularly distributed in molecular systems. In spite of that, all the charges of the sub-systems add up to preserve the original charge of the particle system. For example, the amount of charge remains the same in chemical reactions. When the charges of different signs are spatially separated in the molecule, the system is in a *polarized state*. The macroscopic objects are usually neutral, but exhibit internal polarization. The polarization may have positive or negative directions, where the sign of the polarization is defined in arbitrary manner for a given direction. In some materials, like the metals, the charge mobility is observed under certain circumstances, leading to electrical conductivity. The exotic states in quantum mechanics that involve electrical conductivity are, for example, the superconductivity and the Meissner levitation of magnets, which are not explained under the classical version of mechanics.

The fundamental variables of space, time, mass, energy, and electric charge represent universal concepts in molecular dynamics. In principle, they are intertwined together, with physical meaning when the particles interact. The details of the fundamental variables proliferate, yet the basic aspects deployed here are the ones of main concern in molecular dynamics.

1.6 Reference System of Coordinates

The description of a set of particles requires a *system of coordinates*. The purpose of the system of coordinates is to assign a position to every particle in $3D$ space. In classical mechanics, the atoms are considered tiny entities, represented by points with no internal structure. The position of a particle is unambiguously defined by a set of three real numbers in $3D$ space at a given instant of time. The position is a vector with units of distance. The most common reference frame of coordinates is the Euclidean one, using the orthogonal coordinates x, y, and z to specify the position vector $\mathbf{r}$ of the particle.

$$\mathbf{r} = (x, y, z) \tag{1.5}$$

The description may be always migrated to the moving reference frame of the particle to simplify the equations of motion. Different reference frames are expected to move independently and in rectilinear form with respect to each other; otherwise, external forces are present and should be considered in the systems. The rules to change reference frames that move with a relative velocity, v, are defined by the *Galilean transformations*. In the theory of special relativity, the proper transformations are the *Lorentz transformations* as they consider the dilation and contraction of spacetime. The Galilean and Lorentz transformations are equivalent when the velocity v is smaller than the light speed, c, that is, $v \ll c$.

The particle motion can be described by a sequence of positions of the particle. To do this, we require the introduction of a parameter, t, to impose an order in the positions of the particle. The parameter t increases monotonically, and it is the time. It is with the time that we are able to define the *time evolution* of the particle from its *motion* . By using the time, we keep different events apart, even though they may occur at the same point in space. In this context, we have the opportunity to unequivocally speak of the past, the present, and the future events. Once the parameter t is introduced, the different positions acquire a sequential order like the frames of a movie, defining the *particle trajectory*.

The Newtonian reference frame of coordinates assumes that the space and time are dimensions independent of each other, treating them in homogeneous and isotropic manner.[6] A reference system of coordinates that satisfies these properties ensures identical functional forms of the laws of physics, receiving the name of *inertial reference frame*. The difference between an inertial reference frame and a non-inertial one implies the presence of a force acting on the non-inertial reference frame rather than on the particle. In consequence, the particle motions in the inertial

[6] An *isotropic medium* has the same properties in all directions, and a *homogeneous medium* has the same properties in all locations. In the uniform translational motion of objects, we require an isotropic medium. A space with a symmetrical protuberance (where we may be positioned) is homogeneous, but the space is not isotropic due to the protuberance. In the uniform rotation of objects around the protuberance, we require a homogeneous medium. None of such concepts imply the other.

and the non-inertial reference systems are different. As an example, and in the case of a non-inertial reference frame, consider a particle moving on the planar surface of a rotating disc, and in the case of an inertial reference frame, consider the same particle moving on the planar surface of a static disc. From the perspective of the observer in the inertial frame, the force that makes the particle trajectory different is sensed as a *fictitious force*. In the absence of forces, a particle travels with constant speed in straight line. Such an observation allows to alternatively define an inertial reference frame as one that moves with the particle subjected to no forces. In this situation, the particle is observed to be at rest, remaining at rest in the absence of external forces. In the case of a particle moving with uniform motion with respect to the inertial reference frame, it will continue doing so unless a force acts upon the particle. The laws of physics are identical in the inertial reference frames, acquiring their simplest functional forms. The conceptions may be equally inverted to indicate that the laws of physics define the inertial reference frames.

Under particular conditions, the concept of an inertial reference frame may be generalized. Consider two bodies equally accelerated in linear form. We situate our reference frame on one of the bodies, observing that the second body stays at rest with respect to our reference frame. In this context, the chosen reference frame may be taken as an inertial one. An exception occurs when we deal with a beam of light, which exhibits a fixed speed independently of the state of motion of the emitting body. A case like this is considered in the theory of special relativity.

Our planet rotates, and thus, the reference frame of coordinates in the planet is not an inertial reference frame of coordinates. However, we may in the laboratory consider the distant starts as "fixed objects" to establish an inertial reference frame of coordinates (strictly speaking, the distant starts are not fixed due to an expanding Universe with the possible intervention of forces from dark matter and dark energy). This reference frame of coordinates is of terrestrial use and is relatively adequate in most experiments.

1.7 Newtonian Time

The time is a complex concept that remains under debate. It may be considered as an essential component of the Universe which introduces a new dimension and is independent of any event. Nonetheless, it may be conceived in a different way, this is, as a human contrivance to describe the particle dynamics. No time quantization that reveals the time as a physical variable in quantum mechanics has been firmly established at this stage, although there are investigations in this respect [5]. The laws of physics are time reversible, and there is no time direction favored over another. Contrary to the mechanical formulation, there is apparently a time arrow in thermodynamic theory. The second law of thermodynamics postulates the increase of entropy in one time direction. Yet the time direction does not necessarily correspond to the direction of entropy.

In Newtonian mechanics, the time t is conceived as the parameter that labels the positions of the particle in an event. In general, any physical variable of the

particle (that changes values due to the interaction of the particle with other entities) may carry the label t. We consider t as the ordinary time, measuring it with a clock that gives regular ticks. The quantity t considered in this way is referred to as the *Newtonian time*. Both the time and the reference system of coordinates are indispensable components to characterize the particle dynamics.

Before introducing the general principles of classical mechanics, it is convenient to define the basic variables that describe the different types of motion of particles.

1.8 Linear Motion

The combination of the position and time defines the *velocity vector*, $\mathbf{v}$. When a body moves, the velocity of the body corresponds to the displacement $\Delta\mathbf{r}$ divided by the interval of time Δt: $\Delta\mathbf{r}/\Delta t = [\mathbf{r}(t + \Delta t) - \mathbf{r}(t)]/\Delta t$. It corresponds to the expression of a line slope. In the limit of small time intervals, $\Delta t \to 0$, the time derivative denotes the instantaneous velocity.

$$\mathbf{v}(t) = \lim_{\Delta t \to 0} [\mathbf{r}(t + \Delta t) - \mathbf{r}(t)]/\Delta t = d\mathbf{r}(t)/dt = \dot{\mathbf{r}}(t) \tag{1.6}$$

The velocity owns units of $distance/time$. The magnitude of the velocity, $|\mathbf{v}|$, is the *particle speed*. We are assuming no warped spatial or temporal dimensions in the kinematic expressions when the particle moves with high speed; otherwise, the theory of general relativity should be employed. *Acceleration* is the rate of change of the velocity with respect to time.

$$\mathbf{a}(t) = d\mathbf{v}(t)/dt = \dot{\mathbf{v}}(t) = \ddot{\mathbf{r}}(t) \tag{1.7}$$

Acceleration owns units of $distance/time^2$, or $velocity/time$. A change in the magnitude or the direction of the velocity with time involves an acceleration, $\mathbf{a}$, which is a vector proportional to the *force vector*, $\mathbf{f}$, acting on the particle. The mass m is the factor of proportionality.

$$\mathbf{f}(t) = m\, d\mathbf{v}(t)/dt \tag{1.8}$$

Consider a set of small displacements $x_i = v_i \delta t_i$ along the x axis in the time interval δt_i. The particle is initially located at x_0. The total displacement at the instant of time t is given by the sum of the individual displacements.

$$x = x_0 + \textstyle\sum_i v_i \delta t_i \tag{1.9}$$

A similar expression is displayed for the continuous (or infinitesimally small) displacements, but the sum symbol $\sum$ is replaced by the integral symbol $\int$ in this

case.[7]

$$x = x_0 + \int_{t_0}^{t} v(t)\, dt \tag{1.10}$$

If the speed is constant in the time interval $t - t_0$, the total displacements 1.9 and 1.10 are $x(t) = x_0 + v(t - t_0)$. When the position is replaced by the velocity, and the velocity by the acceleration in the last two expressions, we achieve the velocity in terms of the acceleration.

$$v = v_0 + \sum_i a_i \delta t_i \quad ; \quad v = v_0 + \int_{t_0}^{t} a(t)\, dt \tag{1.11}$$

An alternative expression relating the variables x, v, and a is obtained from the identities $dv = adt$ and $v = dx/dt$. By multiplying both expressions, we have $vdv = adx$. The integration of this equation gives the velocity v in terms of the acceleration a, which in turn depends on the position x.

$$v^2/2 = v_0^2/2 + \int_{x_0}^{x} a(x)\, dx \tag{1.12}$$

There is *accelerated motion* when the velocity and acceleration exhibit the same directions; otherwise, it is a *retarded motion*. There is *uniform acceleration* when the motion is rectilinear with constant acceleration. This is the case where the velocity is:

$$v = v_0 + a(t - t_0)$$

By inserting this identity in Eq. 1.10 the position of the particle is:

$$x(t) = x_0 + v_0(t - t_0) + a(t - t_0)^2/2$$

When the time interval $t - t_0$ is small, we may consider two consecutive times in the trajectory of the particle and obtain the discrete version of the positions.

$$x(t_i) = x(t_{i-1}) + v(t_{i-1})[t_i - t_{i-1}] + \tfrac{1}{2}\, a[t_i - t_{i-1}]^2 \tag{1.13}$$

The acceleration produced by an external force acting on the particle is constant in the above time interval $[t_{i-1}, t_i]$. Thus, the classical particle responds to the external (or environmental) force, and there is a *cause-effect action* on the particle. In general, the force exerted on the particle produces translational motion. A force may also produce the rotation of the particle. Before studying the particle in rotation,

[7] The letter S is employed in different forms to denote a summation. We use the (broken) Greek letter $\sum_i$ in the case of a discrete variable and the (soft) Latin letter $\int$ in the case of a continuous variable. There is a summation over the corresponding type of variable in every case.

it is convenient to introduce the *torque*, defined as the cross product of the position vector $\mathbf{r} = (x, y, z)$ with the force vector $\mathbf{f} = (f_x, f_y, f_z)$.

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{f} = \begin{vmatrix} u_x & u_y & u_z \\ r_x & r_y & r_z \\ f_x & f_y & f_z \end{vmatrix} \tag{1.14}$$

The unit vector $\hat{\mathbf{u}} = (u_x, u_y, u_z)$ is perpendicular to the plane formed by the vectors $\mathbf{r}$ and $\mathbf{f}$, with direction defined by the right-hand rule. Expression 1.14 is a symbolic form to describe the torque given by:

$$\boldsymbol{\tau} = (r_y f_z - r_z f_y)u_x + (r_z f_x - r_x f_z)u_y + (r_x f_y - r_y f_x)u_z \tag{1.15}$$

The torque is important in investigating the rotation of a particle around an axis when a force is applied on the particle. The lever-arm is computed from the rotation axis. When the forces are *concurrent* (i.e., they all have an intersection in a single point) [6], the total torque is the net force $\mathbf{f}_r$ in cross product with the particle position. In this special case, the torque is equally obtained by adding the torques produced by the individual forces.

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{f}_r = \mathbf{r} \times \textstyle\sum_i \mathbf{f}_i = \sum_i \mathbf{r} \times \mathbf{f}_i = \sum_i \boldsymbol{\tau}_i \tag{1.16}$$

The net force produces the rotation of the particle and is equal to the rotation produced by the concurrent forces. This is not necessarily the case of *non-concurrent forces* acting, for example, on a rigid body, which is considered as a set of particles that keep their distances fixed, with no possible deformation, and showing spatial extension. In general, the net effect produced by the individual forces $\mathbf{f}_i$ acting on different points of the rigid body cannot be reproduced by the single force $\mathbf{f}_r$ obtained from the summation of the individual forces. In the situation of the rigid body, the net force $\mathbf{f}_r = \mathbf{f}_1 + \mathbf{f}_2 + \cdots$ and the net torque $\boldsymbol{\tau}_r = \boldsymbol{\tau}_1 + \boldsymbol{\tau}_2 + \cdots$ are not necessarily perpendicular to each other, and expressions 1.16 are not well defined.

Problem 1.2 Consider a number of individual forces $\mathbf{f}_i$ exerted on different points of a rigid body. Show that the torque is preserved when the forces are applied along the *line of action*, namely, the parallel line to the net force $\mathbf{f}_r$. Find the line of action when the position $\mathbf{r}$ of a particle and the force $\mathbf{f}$ over that particle are coplanar.

Solution The net force and torque on the rigid body are $\mathbf{f}_r = \sum_i \mathbf{f}_i$, and $\boldsymbol{\tau} = \sum_i (\mathbf{r}_i - \mathbf{r}) \times \mathbf{f}_i$, with reference point $\mathbf{r}$. When the reference point is changed by an amount $\lambda \mathbf{f}_r$, which is a parallel vector to force $\mathbf{f}_r$, we have:

(continued)

$$\sum_i (\mathbf{r}_i - \mathbf{r} + \lambda \mathbf{f}_r) \times \mathbf{f}_i = \sum_i (\mathbf{r}_i - \mathbf{r}) \times \mathbf{f}_i + \lambda \mathbf{f}_r \times \sum_i \mathbf{f}_i$$

From the properties of the cross product, $\mathbf{A} \times \mathbf{B} = AB \sin(\theta)\, \hat{\mathbf{n}}$, where θ is the angle between the vectors $\mathbf{A}$ and $\mathbf{B}$ and $\hat{\mathbf{n}}$ is the unit vector perpendicular to vectors $\mathbf{A}$ and $\mathbf{B}$ in the direction given by the right-hand rule. The last term is $\lambda \mathbf{f}_r \times \sum_i \mathbf{f}_i = \lambda \mathbf{f}_r \times \mathbf{f}_r = 0$, and the original expression $\boldsymbol{\tau} = \sum_i (\mathbf{r}_i - \mathbf{r}) \times \mathbf{f}_i$ is recovered. Thus, the torque exhibits no change along the line of action $\lambda \mathbf{f}_r$. On the other hand, when $\mathbf{r}$ and $\mathbf{F}$ are coplanar ($z = 0$ and $f_z = 0$ by supposition), the torque is $\tau = x f_y - y f_x$, where f_x and f_y are the rectangular components of the force. This expression is the equation of the action line, with x and y the variables describing it, $y = (f_y/f_x)x - \tau/f_x$.

The equilibrium of a particle demands a zero net force, and the equilibrium of a rigid body additionally requires a zero torque. When the rigid body is static, no translational neither rotational motions are observed.

$$\sum_i \mathbf{f}_i(t) = 0 \quad ; \quad \sum_i \boldsymbol{\tau}_i(t) = 0$$

1.9 Angular Motion

When a particle travels along a curvilinear trajectory, the kinematic equations of the particle change with respect to those of the linear motion. Consider a particle moving along a small arc Δs. The initial and final points of the trajectory are $\mathbf{r}_i$ and $\mathbf{r}_f$. The vector joining the two points along a straight line is $\Delta \mathbf{r}$, Fig. 1.3. The velocity of the particle is given in terms of the position difference $\Delta \mathbf{r} = \mathbf{r}_f - \mathbf{r}_i$.

$$\mathbf{v} = \frac{\Delta \mathbf{r}}{\Delta t} = \frac{\Delta \mathbf{r}}{\Delta s} \frac{\Delta s}{\Delta t} \tag{1.17}$$

When the arc segment is sufficiently small, the vector $\Delta \mathbf{r}$ and the arc segment Δs define the tangent unit vector $\mathbf{u}_t$ to the trajectory, while the fraction $\Delta s/\Delta t$ is the particle speed along the arc segment.

$$\mathbf{v} = \mathbf{u}_t v \quad ; \quad \mathbf{u}_t = \lim_{\Delta s \to 0} \frac{\Delta \mathbf{r}}{\Delta s} \quad ; \quad v = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} \tag{1.18}$$

The vector $\mathbf{v}$ is the particle velocity; it changes with time since both v and $\mathbf{u}_t$ change with time. The acceleration demands differentiation of the velocity with respect to time.

$$\mathbf{a} = \frac{d(v\mathbf{u}_t)}{dt} = \frac{dv}{dt}\mathbf{u}_t + v \frac{d\mathbf{u}_t}{dt} \tag{1.19}$$

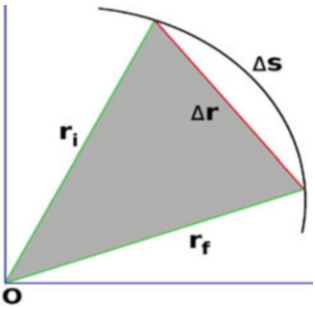


Fig. 1.3 A particle travels along a circular path. The initial and final vectors $\mathbf{r}_i$ and $\mathbf{r}_f$ describe the position of the particle at different instants of time. When the arc segment Δs is sufficiently small, the vector $\Delta\mathbf{r}$ and arc segment Δs define a tangent vector, $\Delta\mathbf{r}/\Delta s$, to the circular path

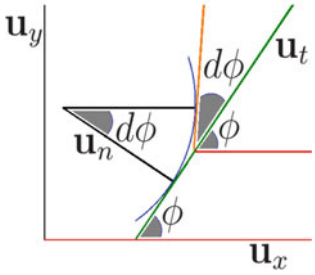


Fig. 1.4 A particle travels along a circular path. The quantities $\mathbf{u}_t$ and $\mathbf{u}_n$ are the tangent and normal unit vectors to the particle trajectory. They are expressed as $\mathbf{u}_t = \mathbf{u}_x \cos\phi + \mathbf{u}_y \sin\phi$ and $\mathbf{u}_n = -\mathbf{u}_x \sin\phi + \mathbf{u}_y \cos\phi$. The angle ϕ is formed by the vectors $\mathbf{u}_t$ and $\mathbf{u}_x$. The tangent and normal vectors are orthogonal to each other

The acceleration is a vector usually described in terms of the Cartesian components $\mathbf{u}_x$ and $\mathbf{u}_y$. Nevertheless, it is convenient to use the tangent vector $\mathbf{u}_t$ and the normal vector $\mathbf{u}_n$ in the case of the angular motion. The tangent and normal vectors are perpendicular to each other and form a basis set for the curvilinear coordinates, Fig. 1.4. By imposing the perpendicular condition $\mathbf{u}_t \cdot \mathbf{u}_n = 0$, the form of the normal vector $\mathbf{u}_n$ is deduced.

$$\mathbf{u}_t = \mathbf{u}_x \cos\phi + \mathbf{u}_y \sin\phi \quad ; \quad \mathbf{u}_t \cdot \mathbf{u}_n = (\mathbf{u}_x \cos\phi + \mathbf{u}_y \sin\phi) \cdot (\mathbf{u}_x a + \mathbf{u}_y b) = 0 \tag{1.20}$$

The components of $\mathbf{u}_n$ that satisfy the above expression are $a = -\sin\phi$ and $b = \cos\phi$. The possible signs $(-,+)$ or $(+,-)$ of (a,b) define the inward or outward directions of vector $\mathbf{u}_n$ with respect to the arc segment. $\mathbf{u}_n$ is chosen to point toward the radial center of the arc Δs, and therefore:

$$\mathbf{u}_n = -\mathbf{u}_x \sin\phi + \mathbf{u}_y \cos\phi$$

The following step is to compute $d\mathbf{u}_t/dt$ to determine the acceleration 1.19. By using the unit vectors $\mathbf{u}_t$ and $\mathbf{u}_n$, and taking consideration of the angle $d\phi$, whose definition in radian units is $d\phi = ds/\rho$, with the angle $d\phi$ displaying the arc portion ds and ρ the curvature radius of such an arc, the time derivative $d\mathbf{u}_t/dt$ is:

$$\frac{d\phi}{dt} = \frac{d\phi}{ds}\,\frac{ds}{dt} = \frac{1}{\rho}\,v \quad ; \quad \frac{d\mathbf{u}_t}{dt} = \left(-\mathbf{u}_x \sin\phi + \mathbf{u}_y \cos\phi\right)\frac{d\phi}{dt} = \mathbf{u}_n\,\frac{d\phi}{dt} = \mathbf{u}_n\,\frac{v}{\rho} \tag{1.21}$$

The incorporation of the last result in Eq. 1.19 gives the acceleration in terms of the unit vectors.

$$\mathbf{a} = a_t\,\mathbf{u}_t + a_n\,\mathbf{u}_n \quad ; \quad a_t = dv/dt \quad ; \quad a_n = v^2/\rho \tag{1.22}$$

The tangent component of the acceleration is given by the change of the speed over time, and the normal component is associated with the change in directions of the particle velocity, as $\mathbf{u}_n$ is related to $d\mathbf{u}_t/dt$ in Eq. 1.21. The normal component is called the *centripetal acceleration*. A similar procedure may be followed using the unit radial vector $\mathbf{u}_r$ and the unit transversal vector $\mathbf{u}_\theta$.

Problem 1.3 Explore the situations when, on the one hand, the tangent acceleration is zero ($a_t = 0$) and, on the other, the normal acceleration is zero ($a_n = 0$), in the angular motion of a particle.

Solution When $a_t = 0$, we have $dv/dt = 0$, and v is constant. In this case, the particle is in uniform circular motion because the speed is constant. In contrast, when the normal acceleration is zero, namely, $a_n = 0$, we may either have $v = 0$ or $\rho = \infty$. By avoiding the trivial case ($v = 0$), we find that the particle travels along a linear path when $\rho = \infty$.

The curvature radius ρ is constant in a circular motion. The expression $d\phi = ds/\rho$ is used to obtain the angular velocity and the angular acceleration. By using expression 1.18 of v, it is possible to write it in terms of the *angular velocity*, ω.

$$v = ds/dt = \rho\omega \quad ; \quad \omega = d\phi/dt \tag{1.23}$$

The *angular acceleration* is calculated from the angular velocity by differentiation with respect to the time.

$$\alpha = d\omega/dt = d^2\phi/dt^2 \tag{1.24}$$

On the other hand, the components of $\mathbf{a}$ acquire different forms when ρ is constant. By using the expressions of a_t and a_n of Eq. 1.22 in the form $a_t = dv/dt = \rho\, d\omega/dt = \rho\, d^2\phi/dt^2 = \rho\,\alpha$ and $a_n = v^2/\rho = \rho\omega^2$, the expression of the acceleration is written in terms of ρ and ϕ.

$$\mathbf{a} = \rho\,\alpha\,\mathbf{u}_t + \rho\,\omega^2\,\mathbf{u}_n = \rho\left(\frac{d^2\phi}{dt^2}\right)\mathbf{u}_t + \rho\left(\frac{d\phi}{dt}\right)^2\mathbf{u}_n \tag{1.25}$$

When ω is constant, the particle describes *uniform circular motion*. In this case, ω of Eq. 1.23 is integrated to show that ϕ depends linearly on the time t.

$$\phi = \phi_0 + \omega(t - t_0) \tag{1.26}$$

A system like this exhibits *periodic motion* and may be used as a clock to give clock ticks and measure the Newtonian time. The *period* of the particle motion is defined as the required time to achieve a complete cycle, and the *frequency* is the number of cycles registered in a time unit. If the particle performs n cycles in the time interval Δt, the oscillation period is $P = \Delta t/n$, and the frequency is $\nu = n/\Delta t$. Both quantities are related to each other.

$$P = 1/\nu \tag{1.27}$$

If we impose the initial conditions $\phi_0 = 0$ and $t_0 = 0$, then Eq. 1.26 is reduced to $\phi = \omega t$. A complete cycle of the circular motion occurs when $\phi = 2\pi$ and $t = P$.

$$\omega = 2\pi/P = 2\pi\,\nu \tag{1.28}$$

The *uniform accelerated circular motion* corresponds to the case where α is constant. In this case, α of Eq. 1.24 is integrated to show that ω depends linearly on the time t.

$$\omega = \omega_0 + \alpha(t - t_0) \tag{1.29}$$

If we consider that $\omega = d\phi/dt$, and use the above equation of ω, then it is possible to integrate and obtain an expression referent to the evolution of the angle ϕ.

$$\phi = \phi_0 + \omega_0(t - t_0) + \tfrac{\alpha}{2}(t - t_0)^2 \tag{1.30}$$

A discrete version of this equation is achieved by following similar steps to those in the deduction of Eq. 1.13.

1.10 Descriptions Between Inertial Reference Frames

In the following, we describe the motion of a particle according to two observers positioned on different inertial reference frames. The inertial reference frames have motion relative to each other with constant speed V_x along the x axis. The positions of the particle in the non-primed and primed reference frames are (x, y, z), and

$(x', y', z',)$. Such positions differ in the x components and are identical in the other dimensions.

$$x' = x - V_x t \quad ; \quad y' = y \quad ; \quad z' = z \quad ; \quad t' = t \tag{1.31}$$

The Galilean transformations rule this change of reference frames. The events in both frames are simultaneous since the observers register the same time (a low speed V_x with respect to the speed of light c is assumed). The relative velocities and accelerations are deduced from the above expressions by differentiation in time.

$$\begin{aligned} v'_x &= v_x - V_x \; ; \; v'_y = v_y \; ; \; v'_z = v_z \\ a'_x &= a_x \qquad\quad ; \; a'_y = a_y \; ; \; a'_z = a_z \end{aligned} \tag{1.32}$$

If we consider that the acceleration multiplied by the particle mass is the force exerted on the particle, then the last line of expressions indicates that $\mathbf{f}' = \mathbf{f}$. Thereby, the force is an invariant quantity under transformations involving a *uniform translational motion*.

The second situation of interest considers two reference frames sharing a common origin, but one of them is in *uniform angular motion* ($\omega =$ constant) with respect to the other. The rotating frame is a *non-inertial reference frame*. The particle velocities and accelerations should be related in the two reference frames. Before, it is convenient to write the central expression $v = \rho\omega$, Eq. 1.23, of the angular motion in vector form to facilitate the algebra. This is done by considering ρ as the projection of vector $\mathbf{r}$ onto the circular plane described by the particle, Fig. 1.5. According to such a geometry $\rho = r \sin\gamma$.

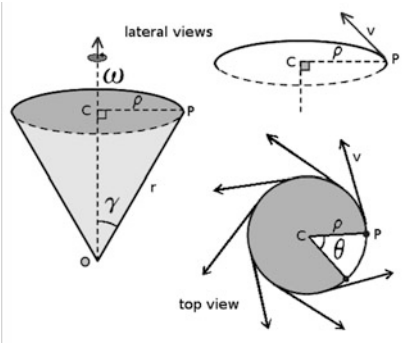


Fig. 1.5 The tangent velocity $\mathbf{v}$, and the angular velocity $\boldsymbol{\omega}$ of a particle with angular motion are shown in the lateral views. The variable $\mathbf{r}$ is the position vector of the particle, and ρ is the projection of $\mathbf{r}$ onto the plane described by the particle's motion. The conical angle is γ, and the angle on the plane defined by the particle with respect to an initial position is θ. Several velocity vectors of the particle at different instants of time are also illustrated in the top view

$$\mathbf{v} = \rho\omega\,\hat{\mathbf{n}} = r\omega\sin(\gamma)\,\hat{\mathbf{n}} = \boldsymbol{\omega}\times\mathbf{r} \tag{1.33}$$

$\boldsymbol{\omega}$ is the vector perpendicular to the circular plane passing through the center, γ is the angle between $\mathbf{r}$ and $\boldsymbol{\omega}$, and the unit vector $\hat{\mathbf{n}}$ is perpendicular to the vectors $\boldsymbol{\omega}$ and $\mathbf{r}$, with direction given by the right-hand rule. The acceleration is obtained by differentiating $\mathbf{v}$ with respect to time, keeping in mind that $\boldsymbol{\omega}$ is constant.

$$\mathbf{a} = \boldsymbol{\omega}\times\mathbf{v} = \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r}) \tag{1.34}$$

The following step is to find the positions and velocities registered by the observers in their own reference frames.

$$\begin{aligned} \mathbf{r} &= \mathbf{u}_x x + \mathbf{u}_y y + \mathbf{u}_z z = \mathbf{u}'_x x' + \mathbf{u}'_y y' + \mathbf{u}'_z z' \\ \mathbf{v} &= \mathbf{u}_x v_x + \mathbf{u}_y v_y + \mathbf{u}_z v_z \quad ; \quad \mathbf{v}' = \mathbf{u}'_x v'_x + \mathbf{u}'_y v'_y + \mathbf{u}'_z v'_z \end{aligned} \tag{1.35}$$

Both reference frames share a common origin of coordinates, but the particle position $\mathbf{r}$ is described by the different unit vectors of the reference frames. The relative circular motion implies a relation between the particle velocities $\mathbf{v}$ and $\mathbf{v}'$ and between the particle accelerations $\mathbf{a}$ and $\mathbf{a}'$. An observer in one of the reference frame considers the other observer in a rotating reference frame with angular velocity $\boldsymbol{\omega}$. The particle velocity is $\mathbf{v}$:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \sum_i \left(\frac{d\mathbf{u}'_i}{dt}\,r'_i + \mathbf{u}'_i\,\frac{dr'_i}{dt}\right) \quad ; \quad r'_i = x', y', z' \tag{1.36}$$

Each unit vector $\mathbf{u}'_i$ rotates with velocity $d\mathbf{u}'_i/dt$. By considering Eq. 1.33, the first term is described by $(\boldsymbol{\omega}\times\mathbf{u}'_i)\,r'_i$, where the basis vector $\mathbf{u}'_i$ plays the role of the rotating vector $\mathbf{r}$ in Eq. 1.33. The second term of the sum is the velocity $\mathbf{v}'$, Eq. 1.35, of the particle in the rotating reference frame.

$$\begin{aligned} \mathbf{v} &= \boldsymbol{\omega}\times(\mathbf{u}'_x\,x' + \mathbf{u}'_y\,y' + \mathbf{u}'_z\,z') + (\mathbf{u}'_x\,\frac{dx'}{dt} + \mathbf{u}'_y\,\frac{dy'}{dt} + \mathbf{u}'_z\,\frac{dz'}{dt}) \\ &= \mathbf{v}' + \boldsymbol{\omega}\times\mathbf{r}' = \mathbf{v}' + \boldsymbol{\omega}\times\mathbf{r} \end{aligned} \tag{1.37}$$

Expression 1.35 was used to replace $\sum \mathbf{u}'_i\,r'_i$ with $\mathbf{r}$. Thus, the velocity of a particle described by an observer in the non-primed reference frame of coordinates is the same as the velocity measured in the primed reference frame of coordinates plus a contribution coming from the relative angular motion of the reference frames. The procedure to obtain Eq. 1.37 is a general one and implicitly defines a mnemonic rule on the derivation of the vectors.

$$\frac{d}{dt}\mathbf{b} = \frac{d}{dt}\mathbf{b}' + \boldsymbol{\omega}\times\mathbf{b} \tag{1.38}$$

where $\mathbf{b}$ and $\mathbf{b}'$ are vectors in the non-primed and primed reference frames. When $\mathbf{b} = \mathbf{r}$, expression 1.37 is recovered. The following step is to compute the acceleration of the particle by either differentiating the velocity 1.37 or using the rule 1.38, under the consideration of constant $\boldsymbol{\omega}$.

$$\frac{d\mathbf{v}}{dt} = \frac{d\mathbf{v}'}{dt} + \boldsymbol{\omega} \times \frac{d\mathbf{r}}{dt} \tag{1.39}$$

The last term $d\mathbf{r}/dt$ is the velocity $\mathbf{v}$ and is replaced by the expression $\mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}$. On the other hand, the first term $d\mathbf{v}'/dt$ may be calculated from Eq. 1.35, where the velocity $\mathbf{v}'$ is shown in terms of the basis vectors $(\mathbf{u}'_x, \mathbf{u}'_y, \mathbf{u}'_z)$, which change with time. By following similar steps referent to the derivation of Eq. 1.37, and gathering all the contributions together, we have:

$$\begin{aligned} \mathbf{a} &= \mathbf{a}' + \boldsymbol{\omega} \times \mathbf{v}' + \boldsymbol{\omega} \times (\mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}) \\ &= \mathbf{a}' + 2\boldsymbol{\omega} \times \mathbf{v}' + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \end{aligned} \tag{1.40}$$

The observers register different accelerations in their reference frames. The discrepancies are due to the *Coriolis acceleration*, $2\boldsymbol{\omega} \times \mathbf{v}'$, and the centripetal acceleration $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$. The force exerted on the particle is not an invariant quantity in this case since the accelerations are different in the reference frames. Therefore, it is not possible to formulate a conservation principle of the force.

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Foundations of Newtonian Dynamics

2

The general purpose of molecular dynamics is to characterize the motion of each particle of a particle system in response to the interaction with the other particles and fields. This chapter presents the fundamental laws on the dynamics of particles according to *Newtonian theory*. In the Newtonian approach, the atoms are represented by points with no internal structure, and the mass and charge are the intrinsic properties that distinguish them. Such properties are assumed to be independent of the motion state of the atoms here. In the Newtonian approach, the atoms are also referred to as the particles of the molecular system.

This chapter introduces additional concepts and basic variables to those already defined in the previous chapter. The new concepts and the basic variables are important to establish Newton's laws and derive the equations of motion of the particles. The laws are formulated after observations and experimentation and by considering some idealizations in the proposal of the theory. The Newtonian laws are used to explain the behavior of the classical particles. The linear and angular motions of a particle are discussed from the concept of force. The time reversibility of the motion equations of small and large particle systems is analyzed. We make reference to the concepts of impulse, work, and power and classify important types of mechanical energies. The Newtonian theory constitutes one of the fundamental approaches of molecular dynamics to derive the functional forms of the forces, establish the equations of motion, and solve them to determine the dynamics of the system.

2.1 First Newton's Law

Our everyday experience indicates that a particle changes its state of motion when there is an interaction affecting it. In this regard, it is important to establish the cause-effect relationship of the particle motion in a formal way. To do this, we introduce the concept of *force*, understood as the entity responsible for producing the change in the state of motion of the particle. Force has a magnitude and direction,

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and therefore, it is a vector. Force is due to the correlation of the particle with the other particles and fields, like the gravitational field, electromagnetic field, etc. The macroscopic systems are made of many particles, and the force on a single particle that builds the body is, in fact, the result of the many forces exerted on that particle by the other particles of the system.

A reference system of coordinates is required to evaluate an external action on the particle state of motion. The simplest reference system of coordinates is positioned on an ideally isolated body, that is, a body free of interaction with any other entity in the Universe, and moving along a straight line with constant speed. According to the Galilean way of thinking, we have the opportunity to choose an inertial reference frame where the particle is observed at rest, for simplicity. The equivalence among the different inertial reference frames makes the particle at rest physically indistinguishable from any other state of motion registered on a different inertial reference system of coordinates. The particle owns the same inertia in the different inertial reference frames. In our particular inertial reference frame, the particle should remain static as long as it is free of forces from other physical entities. Such an assessment leads to the first law of Newtonian dynamics.

Statement 2.1 First Newton's law: a body stays at rest or continues moving uniformly in a straight line unless acted upon by an external force that changes the body state of motion.

In order to provide a mathematical formulation of the first law and be able to evaluate the motion changes of the particle, it is convenient to introduce the concept of *linear momentum*. The linear momentum, $\mathbf{p}$, is a vector proportional to both the scalar mass m and the vector velocity $\mathbf{v}$ of the particle.

$$\mathbf{p} = m\mathbf{v} \tag{2.1}$$

The mass distinguishes the particle, and in our case, it is independent of the motion state. In terms of the linear momentum, the first principle of dynamics reads as follows: a *free particle* moves with *constant momentum*. This principle implicitly assumes the homogeneity of space and time to ensure the *dynamic equilibrium* of the particle, provided it has no interaction with any other physical entity. When the particle exhibits a momentum change, we infer that an external force acted upon the particle. The first law can be generalized to systems constituted by two or more particles, in spite of the possibility that the particles integrating the system may interact with themselves, because the total momentum of the system corresponds to that of an isolated body, but written in terms of the individual particle contributions.

$$\mathbf{P} = \sum_i \mathbf{p}_i = \sum_i m_i \mathbf{v}_i \tag{2.2}$$

The first law is translated to a version which involves the *conservation of linear momentum*.

Statement 2.2 Conservation principle of the linear momentum: the total momentum of an isolated system of particles is preserved in time.

The linear momentum is the central variable in the conservation principle. The conservation principles are important to identify constants of motion and point out the symmetrical aspects of nature. In molecular dynamics, we are interested in such constants, as they help to both monitor the way in which the molecular simulation unfolds and characterize the particle system. For instance, consider an isolated system of particles where the momentum of a subset of particles increases. The momentum of the complementary subset of particles is expected to decrease to keep the total momentum of the isolated system with no change. New particles have been discovered in experiments of sub-atomic physics by applying the conservation principle of linear momentum. Concurrently, such experiments represent an indirect proof of the conservation principle of linear momentum. The conservation principle of the linear momentum also leads to the redefinition of mass from a dynamic perspective: if we have two isolated particles with masses m and m_{ref}, this last one is the reference mass, and the particles interact in the x axis; then, after measuring their speeds, the mass m is determined from the expression $m = m_{ref}\, v_{ref}/v$.

Newton's first law allows to identify the isolation of a particle, and thereby, it is useful to determine inertial reference frames: if a particle has no change of its momentum, then from the first law, we conclude that it is an isolated object, and it may be used to define an inertial reference frame of coordinates. Based on the close link between an isolated body and an inertial reference frame, the first Newton's law is also called the *law of inertia*. From a Galilean perspective, our observations and measurements performed in different inertial frames are equivalent, and the laws of physics formulated in the inertial reference frames have no change of their functional forms; otherwise, a particular inertial reference frame is favored or distinguished over the others. Such an observation leads to the *invariance of the physical laws*.

Statement 2.3 Invariance of the laws of physics: the laws of physics are mathematically invariant in inertial reference frames.

2.2 Second Newton's Law

The purpose of introducing the concept of force is to evaluate the environmental influence and the action of an external factor on the particle state of motion, without paying too much attention to the details that provoke the motion disruption of the particle. The force exerted on a particle does not necessarily imply a physical contact with the particle. The force exerted by the planet makes the bodies fall to the planet surface due to the gravitational force without ever exercising a physical contact. The force is a thrust or drag capable of changing the particle velocity in magnitude and direction. The acceleration of the particle is due to a force, and the force and acceleration have the same direction.

In a system of interacting particles, the trajectory of a particle is determined by all the acting forces on such a particle. In the special case where the addition of forces exerted on the particle is null, there is a dynamic balance, and the particle is in *static equilibrium*. In order to determine the momentum change of the particle under the action of a non-zero net force, we consider the linear momentum of the ith particle in separate form from the momenta of the other particles.

$$\mathbf{p}_i + \sum_{j \neq i} \mathbf{p}_j = \text{ constant} \tag{2.3}$$

The total momentum of the isolated system of particles is preserved at a different instant of time.

$$\mathbf{p}'_i + \sum_{j \neq i} \mathbf{p}'_j = \text{ constant} \tag{2.4}$$

The primed and non-primed variables of Eqs. 2.3 and 2.4 indicate the momenta at different instants of time. We subtract the quantity $\mathbf{p}'_i$ from $\mathbf{p}_i$ of the ith particle.

$$\mathbf{p}_i - \mathbf{p}'_i = -\left[\sum_{j \neq i} \left(\mathbf{p}_j - \mathbf{p}'_j\right)\right] \quad \Rightarrow \quad \Delta\mathbf{p}_i = -\sum_{j \neq i} \Delta\mathbf{p}_j \tag{2.5}$$

The expression indicates that an interaction of the ith particle with the other particles, which constitute the environment for the ith particle, is a process where the exchange of momenta may occur. A time interval has been considered in Eq. 2.5, and we may deal with the rate of change of the linear momentum with time.

$$\frac{\Delta\mathbf{p}_i}{\Delta t} = -\sum_{j \neq i} \frac{\Delta\mathbf{p}_j}{\Delta t} \tag{2.6}$$

In addition to the average value of the momentum, the instantaneous variation of the momentum is also of interest. The instantaneous variation is obtained by taking the limit $\Delta t \to 0$.

$$\frac{d\mathbf{p}_i}{dt} = -\sum_{j \neq i} \frac{d\mathbf{p}_j}{dt} \tag{2.7}$$

The mathematical definition of the force is introduced to describe the changes of momenta of the interacting particles at a given instant of time.

$$\frac{d\mathbf{p}_i}{dt} = \mathbf{f}_i \quad ; \quad \mathbf{f}_i = -\sum_{j \neq i} \frac{d\mathbf{p}_j}{dt} \tag{2.8}$$

The change of the momentum of the ith particle is attributed to the force $\mathbf{f}_i$ exerted by its environment (in this case, integrated by the other particles). In a similar form, the momentum change of the sub-system composed by all the particles, except the ith one which in this last case plays the role of the environment, is attributed to the force exerted by the ith particle on the sub-system. In either case, the force exerted upon the particle or the sub-system of particles corresponds to the time derivative of the momentum. The generalization of the results to include different types of forces on classical objects establishes the second Newton's law.[1]

Statement 2.4 Second Newton's law: the force exerted upon a particle results in the change of the particle's momentum.

If a body composed by many particles changes the state of motion as a single object, it is due to the influence of an external force. The second law is complementary to the first law by indicating that the preservation of the state of motion (established by the first law) is altered by the implication of an external force (established by the second law). The second Newton's law may be shown in terms of the particle position when we take the relation $\mathbf{v}_i = d\mathbf{r}_i/dt$ between the velocity $\mathbf{v}_i$ and position $\mathbf{r}_i$ into consideration.

$$\mathbf{f}_i = d^2(m_i \mathbf{r}_i)/dt^2 \tag{2.9}$$

For interactions where the mass is constant, the equation of the force transforms onto:

$$\mathbf{f}_i = m_i \frac{d^2\mathbf{r}_i}{dt^2} = m_i \frac{d\mathbf{v}_i}{dt} = m\mathbf{a}_i \tag{2.10}$$

The acceleration is proportional to the force and inversely proportional to the mass. By applying a force, we can characterize the amount of mass of two particles: if two particles are embedded in the same environment and are acted upon by equal forces

[1] The quantum origins of the basic forces are not fully clear. The basic forces in nature are the strong force, weak force, electromagnetic force, and gravitational force. The strong force keeps the nucleons together, and the weak force is responsible for the radioactive decay. The two forces act at short distances, typical of nuclear sizes; instead, the electromagnetic and the gravitational forces are long-range forces acting, respectively, on charged particles and masses. In principle, the classical forces are explained in terms of the fundamental interactions.

but their accelerations are different, we conclude that their masses are different. On the other side, if $m_1 + m_2 = m_3$, then the acceleration of the system $m_1 + m_2$ is the same as the acceleration of the particle with mass m_3 when identical forces are acted upon the two systems. The force satisfies the *superposition principle*, which indicates that the acceleration of a particle produced by several forces $\mathbf{f}_i$ is equivalent to the acceleration obtained by a unique force, which corresponds to the net force $\mathbf{f}_{net}$.

$$\sum_i \mathbf{f}_i = m\mathbf{a} = \mathbf{f}_{net} \quad ; \quad \mathbf{f}_{net} = \sum_i \mathbf{f}_i \tag{2.11}$$

The unit of force is the *Newton*, corresponding to $Kg\,m/s^2$ in the MKS system of units. The second law is compatible with the first law when the particle is free to move or it is not interacting with anything else. In this case, the net force is zero, $\mathbf{f}_{net} = 0$; the acceleration is also zero, $\mathbf{a} = 0$; and the momentum $\mathbf{p}$ is constant. While the mass quantifies the inertia of a particle, the force quantifies the environmental influence on the particle. Both entities are related in the second Newton's law, which, additionally, is mathematically invariant in any other inertial reference frame.

The particles are correlated by their interactions, and such interactions are considered by the *interatomic potentials*. There is a type of forces that are deduced from interatomic potentials and another type of forces with no association to an interatomic potential. In the case of a force derived from an interatomic potential, we have the following prescription to get the force:

$$\mathbf{f}_i = -\nabla_i V \tag{2.12}$$

The force $\mathbf{f}_i$ acts on particle i and is due to a change of the potential V with the particle position. In order to illustrate the effects of a potential on the particle, consider a ball located somewhere in a mountain. The ball displacement depends on the local steepness where the ball is positioned. From the physical perspective, a force is produced when the particle, which is immersed in the potential, changes positions. In this regard, the motion of the ith particle may be explained in terms of the interaction potential.

$$-\nabla_i V = m_i\, \mathbf{a}_i$$

Whether the Newton's law is written in terms of a force or in terms of an inter-particle potential, it is a central equation in molecular dynamics. When the functional form of the force or the inter-particle potential is known, we refer to such a central equation as the *equation of motion* of the particle.

2.3 Third Newton's Law

When two particles interact, the force between them acts equally on both particles. This is due to the fact that the force is not defined on a single particle, but it is exerted on the two interacting particles with no directional preference along the line connecting them. The directional symmetry of the force in the dynamics of the particle is formulated in the third Newton's law, known as the *action-reaction law*.

Statement 2.5 Third Newton's law: when two particles interact, the force exerted on one particle is equal in magnitude but opposite in direction to the force exerted on the other particle.

By introducing the force of Eq. 2.8 exerted on the ith particle, $\mathbf{f}_i = d\mathbf{p}_i/dt$, and the force exerted on the object, which is constituted by all the other particles, $\mathbf{f} = \sum_{j\neq i} d\mathbf{p}_j/dt$, the symmetry in the forces is observed.

$$\frac{d\mathbf{p}_i}{dt} = -\sum_{j\neq i} \frac{d\mathbf{p}_j}{dt} \quad \Rightarrow \quad \mathbf{f}_i = -\mathbf{f} \tag{2.13}$$

In the particular case of dealing with two particles, we have the interaction of the ith particle with the jth one. By taking $\mathbf{f}_j = d\mathbf{p}_j/dt$, we obtain the symmetric condition on the forces.

$$\frac{d\mathbf{p}_i}{dt} = -\frac{d\mathbf{p}_j}{dt} \quad \Rightarrow \quad \mathbf{f}_i = -\mathbf{f}_j \tag{2.14}$$

The summation $\mathbf{f}_i + \mathbf{f}_j$ is zero as the internal forces are balanced in an isolated system of particles; otherwise, a self-acceleration of the particle system occurs, contradicting the first Newton's law. The particles of the system interact by exchanging momentum while simultaneously preserving the total momentum of the system. The conservation of the total momentum is verified by integrating the equations of the internal forces.

The electrostatic interactions among the particles are not shown instantaneously, because the interactions propagate with a velocity equal to the speed of light. The finite speed propagation of the interactions results on a time delay, invalidating the third Newton's law. Nevertheless, when the particles are not far away, the third law may be considered as an acceptable law.

2.4 Reduced Mass of a Two-Particle System

When two particles interact, it is possible to have a simplified description of the two-particle system by introducing the *reduced mass*. In order to illustrate it, consider the equations of motion of the two interacting particles.

$$d\mathbf{v}_1/dt = \mathbf{f}_{12}/m_1 \quad ; \quad d\mathbf{v}_2/dt = \mathbf{f}_{21}/m_2 \tag{2.15}$$

$\mathbf{f}_{ij}$ is the force exerted by the jth particle on the ith particle. By subtracting the equation on the right from the equation on the left, and using the third Newton law $\mathbf{f}_{12} = -\mathbf{f}_{21}$, the relative acceleration between the two bodies is obtained.

$$\mathbf{a}_{12} = \left(\tfrac{1}{m_1} + \tfrac{1}{m_2}\right)\mathbf{f}_{12} \quad ; \quad \mathbf{a}_{12} = d(\mathbf{v}_1 - \mathbf{v}_2)/dt \tag{2.16}$$

The two-particle system is reduced to the description of a single particle by introducing the reduced mass, μ, in terms of the individual masses.

$$1/\mu = 1/m_1 + 1/m_2 \tag{2.17}$$

The reduced mass is an effective mass. By making use of μ, the equation of the effective (single) particle is derived from expression 2.16.

$$\mathbf{f}_{12} = \mu\mathbf{a}_{12} \tag{2.18}$$

The reduced mass is important to simplify the description of a particle pair. When the mass of one of the particles, say m_1, is sufficiently heavier than the mass of the other particle, m_2, the reduced mass may be approximated by the light mass m_2.

$$\tfrac{1}{\mu} = \tfrac{1}{m_1} + \tfrac{1}{m_2} \simeq \tfrac{1}{m_2} \quad ; \quad m_2 \ll m_1 \tag{2.19}$$

Thereby, in the interaction of a heavy mass with a light mass, the light mass is the one essentially accelerated when the two particles are observed from a reference system of coordinates fixed at some point on the line joining the two particles.

2.5 Time Reversibility

The *time reversibility* of the laws of physics is intimately linked to the conservation of the energy. The time reversible property indicates that the laws of physics are immutable to the passage of time, favoring no time arrow and ensuring the preservation of the energy when the system is isolated. In this connection, the equations of motion are capable of describing the reverse course of events which were unfolded by the particle dynamics [1]. In order to determine the required conditions that maintain the equations of motion time symmetric, let us consider the instantaneous position of a particle, $x = x_0 + vt + at^2/2$. The velocity and

acceleration satisfy the expressions $dx/dt = v + at$ and $dv/dt = a$ (the mass is assumed to be constant and is avoided). When the change of t by $-t$ is produced, and the velocity is reversed, $v \to -v$, the particle trajectory is recovered in any of the previous equations, for instance, $x = x_0 + (-v)(-t) + a(-t)^2/2$.

This property is confirmed by considering that in the forward-time direction, the trajectory of a particle is described by the variable $\mathbf{r} = \mathbf{r}(t)$, and in the reverse-time direction, the trajectory is described by the primed variable $\mathbf{r}' = \mathbf{r}'(t_f - t)$. The time variable t is used with opposite signs in the previous cases. In the forward direction, the particle travels from time $t_i = 0$ to $t_f = 1$ and in the reverse direction from time $t_f = 1$ to $t_i = 0$. The initial and final conditions on the trajectories are $\mathbf{r}(t_i) = \mathbf{r}'(t_f - t_i = 1) = \mathbf{r}'(t_f)$ and $\mathbf{r}(t_f) = \mathbf{r}'(t_f - t_f = 0) = \mathbf{r}'(t_i)$. By employing the chain rule, the forward and reverse velocities are the same, except for the change of signs (or directions).

$$\frac{d\mathbf{r}(t)}{dt} = \frac{d\mathbf{r}'(t_f - t)}{d(t_f - t)} \frac{d(t_f - t)}{dt} = -\frac{d\mathbf{r}'(t')}{dt'}$$

A similar situation is observed by directly dealing with the time $-t$ instead of $t_f - t$: again, if $\mathbf{r}'$ is the reverse time trajectory and t' is the reverse time, that is, $t' = -t$, then $d\mathbf{r}(t)/dt = [d\mathbf{r}'(-t)/d(-t)]\,[d(-t)/dt] = -d\mathbf{r}'(t')/dt'$. The acceleration that participates in the equation of motion remains invariant due to repeated derivations, $d^2\mathbf{r}(t)/dt^2 = d^2\mathbf{r}'(t')/dt'^2$. Thus, for a particle describing the trajectory $\mathbf{r}(t)$ with velocity $\mathbf{v}(t)$, the time reversibility is produced by reversing both the time and the velocity.

$$t \to -t \quad ; \quad \mathbf{v} \to -\mathbf{v}$$

In principle, it is possible to stop a particle traveling in the forward direction and reverse its velocity with the same magnitude that it owned in the forward direction. To include the necessary parts of this experiment, we also produce a specular reflection of the Newtonian clock with the purpose to reverse the ticks of that clock, Fig. 2.1. Thus, the reverse trajectory may be visualized as a movie of the particle running in the backward direction.

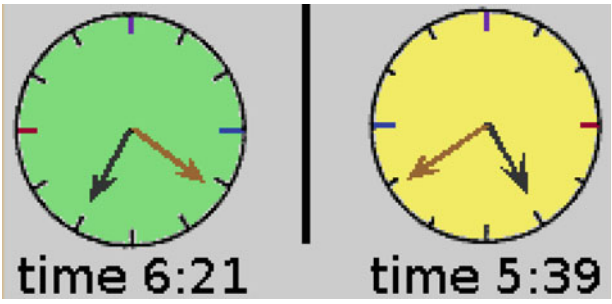


Fig. 2.1 Two clocks are shown. Each one is the specular image of the other. The hands of the clock on the right run clockwise, with time 5:39 hours, and the hands of the clock on the left run counter-clockwise, with time 6:21 hours

A similar situation may be done for two or more interacting particles, and again, the behavior of the particles is shown to be time reversed for an observer. Formally, if the variables $\mathbf{r}(t)$ and $\mathbf{v}(t)$ are solutions of the equations of motion, then the variables $\mathbf{r}(t')$ and $\mathbf{v}'(t')$ are also solutions of such equations.

$$\left.\begin{matrix}[\mathbf{r}(t), \mathbf{v}(t)] \\ [\mathbf{r}(t'), \mathbf{v}'(t')]\end{matrix}\right\} \mathbf{f}(t) = m\,\frac{d\mathbf{v}(t)}{dt} = m\,\frac{d^2\mathbf{r}(t)}{dt^2} = m\,\frac{d^2\mathbf{r}'(t')}{dt'^2} = m\,\frac{d\mathbf{v}'(t')}{dt'} = \mathbf{f}(t') \tag{2.20}$$

The existence of the solution $[\mathbf{r}(t'), \mathbf{v}'(t')]$ gives the possibility to observe and produce the reverse order of events. The invariance of the Newtonian equations of motion under the above transformations of the time and velocity allows to establish an equivalence between the forces:

$$\mathbf{f}(\mathbf{r}, \mathbf{v}, t) = \mathbf{f}(\mathbf{r}, \mathbf{v}', t') \tag{2.21}$$

In order to observe the reverse trajectory of a macroscopic object formed by many microscopic particles, we have to apply the transformations on the time and the velocity of every microscopic particle. In this respect, the macroscopic objects favor no time arrow.

Problem 2.1 The position and velocity of a particle are $\mathbf{r}$ and $\mathbf{v}$. Suppose that the force on the particle is $\mathbf{f} = \alpha\mathbf{r} \times \mathbf{v}$, where α is a constant. Show that the system is not time reversible.

Solution From the expression of the force, we have that $\mathbf{f}(\mathbf{r}, \mathbf{v}, t) = \alpha\mathbf{r} \times \mathbf{v}$ is different from $\mathbf{f}(\mathbf{r}, -\mathbf{v}, -t) = -\alpha\mathbf{r} \times \mathbf{v}$. Thus, when the time and velocity are reversed, the forces are different, and the system is not time reversible.

Frequently, the characterization of a system integrated by a large number of particles is done with *statistical mechanics* or *thermodynamics* instead of Newtonian theory. We introduce simplifications in statistical mechanics and thermodynamics to reach more amenable expressions and procedures to deal with. However, by doing so, there is a loss of information since many degrees of freedom (usually those associated with the environment) are simplified, acquiring a statistical nature. In such a reductionism, a complete description of the system is abandoned. In particular, we miss the capability to analyze the particle trajectories in the forward and backward time directions, because there are not enough details of the forces acting on the

particles. In this process, the evaluation of the energy is strictly affected, resulting in conflict with the energy conservation and the time symmetry of the Newtonian approach.[2] Yet in order to characterize the system at the macroscopic scale, new concepts with macroscopic meaning like the temperature, friction, heat, entropy, and more are introduced. They have statistical character, resulting from averaging the behavior of the microscopic particles when the system is energy equilibrated, because in the *equilibrium state*, the macroscopic variables are observed to be *stationary parameters*, that is, time-indifferent parameters. The energy relaxed states of the many-particle system classify as *thermodynamically stationary states*. In short, we face practical problems in solving a myriad of coupled equations of motion for the accurate description of a many-particle system in Newtonian theory; nevertheless, statistical mechanics and thermodynamics allow the characterization of the many-particle system from a statistical perspective, with the cost of missing some information, like the time reversibility of the macroscopic particle system.

2.6 Angular Momentum and Torque

In this section, we discuss the conservation law of the angular momentum since it is frequently applied in molecular dynamics. Before, it is convenient to introduce the following concepts to classify a particle system.

- When the system is free of interactions with the rest of the Universe, it is an *isolated system*.
- When the system is in thermal or mechanical contact with the environment and there is no exchange of matter, we have a *closed system*, like a closed jar of coffee in contact with the surroundings.
- When the system exchanges matter and energy, it is an *open system*, like a jar of coffee without being closed.

The angular momentum $\boldsymbol{\ell}$ and torque $\boldsymbol{\tau}$ are physical observables of interest to characterize a particle system. The angular momentum and torque are both proportional to the mass m of the particle and involve the cross product of the position vector $\mathbf{r}$ and the velocity $\mathbf{v}$ for the angular momentum and the position vector $\mathbf{r}$ and the acceleration $\mathbf{a}$ for the torque.

$$\boldsymbol{\ell} = \mathbf{r} \times m\mathbf{v} = \mathbf{r} \times \mathbf{p} \quad ; \quad \boldsymbol{\tau} = \mathbf{r} \times m\mathbf{a} = \mathbf{r} \times \mathbf{f} \tag{2.22}$$

[2] The lack of a strict energy conservation of a particle system is commonly due to the oversimplification of the degrees of freedom associated with the external factors that affect the particle system. Still, it is possible to introduce modified forces among the particles to compensate for the environmental effects and formulate the preservation of a modified energy: Santamaria and Soullard [2].

The angular momentum and torque are vectors that depend on the reference system of coordinates. The angular momentum $\boldsymbol{\ell}$ is related to the torque $\boldsymbol{\tau}$ by a time derivative due to the use of $\mathbf{v}$ in the angular momentum and $\mathbf{a}$ $(= d\mathbf{v}/dt)$ in the torque.

$$\boldsymbol{\tau} = d\boldsymbol{\ell}/dt \tag{2.23}$$

For a many-particle system, the individual terms $\boldsymbol{\ell}_i = \mathbf{r}_i \times m_i \mathbf{v}_i$ and $\boldsymbol{\tau}_i = \mathbf{r}_i \times m_i \mathbf{a}_i$ are added to obtain the total angular momentum, $\mathbf{L}$, and the total torque, $\boldsymbol{\tau}_{tot}$.

$$\mathbf{L} = \textstyle\sum_i \mathbf{r}_i \times m_i \mathbf{v}_i \quad ; \quad \boldsymbol{\tau}_{tot} = \sum_i \mathbf{r}_i \times m_i \mathbf{a}_i \tag{2.24}$$

Some constraints on the dynamics of particles may be introduced by establishing initial values of $\mathbf{L}$ and $\boldsymbol{\tau}_{tot}$. For a system of n particles, the angular momentum components are:

$$\mathbf{L}_x(t) = \textstyle\sum_{i=1}^{n} m_i \, [y_i(t) \, v_i^z(t) - z_i(t) \, v_i^y(t)]$$

$$\mathbf{L}_y(t) = \textstyle\sum_{i=1}^{n} m_i \, [z_i(t) \, v_i^x(t) - x_i(t) \, v_i^z(t)]$$

$$\mathbf{L}_z(t) = \textstyle\sum_{i=1}^{n} m_i \, [x_i(t) \, v_i^y(t) - y_i(t) \, v_i^x(t)]$$

where m_i is the mass of the ith particle and (x_i, y_i, z_i) and (v_i^x, v_i^y, v_i^z) are the position and velocity vectors. A similar expression is obtained for the torque $\boldsymbol{\tau}_{tot}$. The total angular momentum of an isolated system is preserved with time because in this case, there are no external torques on the system, $\boldsymbol{\tau}_{tot} = \partial \mathbf{L}/\partial t = 0$, and $\mathbf{L} =$ constant.

Statement 2.6 Conservation principle of the angular momentum: the total angular momentum of an isolated system of particles is preserved with time.

In the closed and open systems, the angular momentum of the particle system may change with time due to the influence of external factors.

2.7 Impulse, Work, and Power

The interaction of a particle with the rest of the Universe is described by the force $\mathbf{f}$, satisfying the equation $\mathbf{f} = d\mathbf{p}/dt$. The integration of the force with respect to the time defines the *impulse*, $\mathbf{I}$.

$$\mathbf{I} = \int_{p_0}^{p} dp = \int_{t_0}^{t} \mathbf{f}\, dt \qquad \Rightarrow \qquad \mathbf{I} = \mathbf{p} - \mathbf{p}_0 = \int_{t_0}^{t} \mathbf{f}\, dt \tag{2.25}$$

The impulse is a vector (with direction and magnitude) and corresponds to the change of momentum due to a force acting on the particle. A large impulse is produced on the particle when either a strong force is applied during a short time interval or a weak force is applied during a long time interval. By integrating once more, the position of the particle is determined.

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \tfrac{1}{m} \int_{t_0}^{t} I\, dt \tag{2.26}$$

The time evolution of the particle is obtained when the impulse in terms of the time is known. A force exerted on the particle produces a *work* along the particle displacement.

$$dW = \mathbf{f} \cdot d\mathbf{r} = f\, dr \cos(\theta) \tag{2.27}$$

θ is the angle between the force $\mathbf{f}$ and the displacement element $d\mathbf{r}$. When $\theta = \pi/2$, there is no work done by the force, and when $\theta = 0$, the maximum work on the particle is performed. The total work in moving the particle from point A to point B is given by the sum of infinitesimal displacements.

$$W = \textstyle\sum_i \mathbf{f}_i \cdot d\mathbf{r}_i \qquad \Rightarrow \qquad W = \int_A^B \mathbf{f} \cdot d\mathbf{r} \tag{2.28}$$

The integral that defines the work is a *line integral* along the trajectory of the particle. Contrary to the expression of the impulse, where we require to know the force in terms of the time, the expression of the work requires to know the force in terms of the position coordinates (x, y, z). Under the situation of a constant force, we have:

$$W = \mathbf{f} \cdot (\mathbf{r}_B - \mathbf{r}_A) \tag{2.29}$$

The work depends on the spatial separation of points A and B. When the force is the addition of several forces, $\mathbf{f} = \mathbf{f}_1 + \mathbf{f}_2 + \cdots$, the work on the particle is the addition of the works produced by the individual forces.

$$dW = (\mathbf{f}_1 + \mathbf{f}_2 + \cdots) \cdot d\mathbf{r} = dW_1 + dW_2 \cdots \qquad ; \qquad dW_i = \mathbf{f}_i \cdot d\mathbf{r} \tag{2.30}$$

The work on a particle may be done slowly or fast. The work per unit of time defines the *instantaneous power*.

$$P = \tfrac{dW}{dt} = \mathbf{f} \cdot \tfrac{d\mathbf{r}}{dt} = \mathbf{f} \cdot \mathbf{v} \tag{2.31}$$

In many physical processes, the power plays a more important role than the work itself.

2.8 Kinetic and Potential Energies

The work performed on a particle is written in terms of the speed $\mathbf{v}$.

$$dW = \mathbf{f} \cdot d\mathbf{r} = m\,\frac{d\mathbf{v}}{dt} \cdot d\mathbf{r} = m\mathbf{v} \cdot d\mathbf{v} \tag{2.32}$$

The integration of the above expression gives the total work on moving the particle from point A to point B in terms of the speed.

$$W = \int_A^B m\mathbf{v} \cdot d\mathbf{v} = \frac{m\mathbf{v}^2(B)}{2} - \frac{m\mathbf{v}^2(A)}{2} \tag{2.33}$$

The work is given by the quantity $m\mathbf{v}^2/2$ evaluated at the points A and B, and in this expression, the work is independent of the nature of the force. The term $m\mathbf{v}^2/2$ is proportional to the square of the momentum, $m\mathbf{v}^2/2 = \mathbf{p}^2/(2m)$, and receives the name of *kinetic energy*. Expression 2.33 written in the form $W = [\mathbf{p}^2(B) - \mathbf{p}^2(A)]/(2m)$ shows a quadratic dependence on the momentum. In contrast, expression 2.25 of the impulse, $\mathbf{I} = \mathbf{p}(B) - \mathbf{p}(A)$, shows a linear dependence on the momentum. The kinetic energy is an energy of motion represented by $E_k = \mathbf{p}^2/(2m)$. The work is related to the change of kinetic energy.

$$W = E_k(B) - E_k(A) \quad ; \quad E_k = \mathbf{p}^2/(2m) \tag{2.34}$$

This expression is general with many applications in mechanics. In the following lines, we classify the force in terms of the type of work performed on the particle. When the work done on a particle is independent of the particle trajectory but it depends on the initial and final positions of the particle, the force responsible for that work is recognized as a *conservative force*. An example of a conservative force is the gravitational force. In the case of a conservative force, the work may be expressed in terms of a function of the particle coordinates, which receives the name of *potential energy* and is designated by E_p. Thus, E_p also represents energy which is gathered with the possibility to perform work. The existence of conservative forces allows to write the work on moving the particle from point A to point B as the change of potential energies.

$$W = E_p(A) - E_p(B) \tag{2.35}$$

When the initial and final points coincide, $A = B$, the work is zero as a result of the conservative force acting on the particle. The point of *zero potential energy* is arbitrarily decided, as the energy difference $E_p(A) - E_p(B)$ is the relevant quantity and constitutes a physical observable. According to Eqs. 2.28 and 2.35 and the fundamental theorem of calculus, the function E_p is a scalar function, continuous in the interval $[A, B]$, and is the anti-derivative of $\mathbf{f}$.

$$W = \int_A^B \mathbf{f} \cdot d\mathbf{r} = E_p(A) - E_p(B) \tag{2.36}$$

Any other anti-derivative of $\mathbf{f}$, say E'_p, is equal to E_p except for a constant c, $E_p = E'_p + c$. The constant c allows to arbitrarily define the reference point of potential energy as previously mentioned. The function $\mathbf{f}$ is written in the anti-derivative form:

$$\mathbf{f} = -\nabla E_p(\mathbf{r}) \tag{2.37}$$

This expression is an alternative definition of the potential energy E_p by emphasizing the fact that a conservative force $\mathbf{f}$ depends on the positions solely. The potential energy constitutes a force field where the interactions among the particles are of non-contact type and receive the name of *forcefield*. The forcefield tends to take the particle to the position of minimum potential energy. Any work performed on the particle has to counteract the forcefield at every point in space, since the work attempts to increase the potential energy. This is a reason to have a negative sign in the relation of the force to potential energy. Such a fact also explains the reverse order of the potential energies in Eq. 2.35 with respect to that of the kinetic energies in Eq. 2.34. In addition to expressions 2.36–2.37, the curl operator may be used to define a conservative force:

$$\nabla \times \mathbf{f}(\mathbf{r}) = 0 \tag{2.38}$$

where it is understood that the curl of the gradient of a scalar function, $\nabla \times (\nabla E_p)$, is zero. There are several examples of forcefields such as the gravitational field, electric field (the magnetic field acting on a charged particle is a non-defined case), the force exerted on a body by a spring, etc. In all those cases, there is a potential energy associated with the particle position. The potential energy acquired by the particle in the forcefield is independent of the trajectory that we chose to move the particle from point A to point B. More precisely, the amount of energy $E_p(A) - E_p(B)$ is constant regardless of the trajectory $A \leftrightarrow B$. It is possible to recover the energy that was inverted in moving the particle from point A to point B because the conservative force preserves the potential energy of the particle in the forcefield.

There are forces that satisfy none of expressions 2.37–2.38. They are classified as *non-conservative forces*. Examples of such forces are those with dependence on the particle velocity, the forces changing in time, the friction forces, and the viscous forces. The non-conservative forces are usually determined from statistical considerations on the particle interactions, and do not constitute a forcefield. Consider a solid sphere immersed in a fluid. The viscosity of the fluid is defined by the statistical average of the interactions on the sphere. The viscosity of the fluid only acts on the external particles of the sphere because it is not a forcefield [3]. In contrast, the gravitational force is a forcefield acting on all the atoms of the solid sphere, including the internal atoms of the sphere.

2.9 Energy Conservation

The energy is a scalar quantity which establishes a relation of fundamental variables. It may be transformed onto different types of energy and even disassembled onto different energy terms.

We have the possibility of writing the work in terms of changes of either the kinetic energy or the potential energy. Still, we may combine both expressions using Eqs. 2.34 and 2.35.

$$E_k(B) - E_k(A) = E_p(A) - E_p(B)$$

By reorganizing terms, we obtain the energies of the particle at point A separated from the energies of the particle at point B, reaching the expression $E_k(B) + E_p(B) = E_k(A) + E_p(A)$. Since the points A and B are arbitrary, we conclude that the kinetic energy plus the potential energy is preserved at any instant of time and at any point in space under the consideration of conservative forces. Still, the kinetic energy may be converted onto potential energy, and vice versa, as a form of reversibility.[3]

$$E_k + E_p = \text{ constant} \tag{2.39}$$

Problem 2.2 A particle in a forcefield moves in the x direction. The total energy is $E = m(dx/dt)^2/2 + E_p(x)$. Find the equation of dx in terms of dt, and obtain the time evolution $x(t)$ of the particle by assuming $E_p = -fx$, where f is a force.

Solution The expression of the energy contains dx/dt. In this regard, we solve for the differential dx in terms of dt.

$$\left(\frac{m}{2}\right)^{1/2} \int_0^x \frac{dx}{[E - E_p(x)]^{1/2}} = \int_0^t dt = t$$

This is a general equation in terms of the total energy and potential energy of a particle moving in the x axis. By taking the potential energy as $E_p = -fx$, the previous expression is integrated using the fact $\int dx/\sqrt{ax+b} = (2/a)\sqrt{ax+b}$.

(continued)

[3] The conservation of the mechanical energy, $E_k + E_p$, is a strict expression. In contrast, the conservation of the thermodynamic energy proposed by the first law of thermodynamics is not a strict expression due to the use of average values. This makes the first law of thermodynamics a law on the statistical balance of thermodynamic energies (exclusive for states in thermodynamic equilibrium), rather than a law on energy conservation.

$$\left(\frac{m}{2}\right)^{1/2} \frac{2}{f} \left[(E + fx)^{1/2} - E^{1/2}\right] = t$$

By solving for the position x, the result is $x(t) = (2E/m)^{1/2}\, t + (f/m)\, t^2/2$. We impose the initial conditions on the particle motion. At the initial time $t = 0$, the position is $x = 0$, and the total energy is $E = mv_0^2/2$. We also have $f/m = a$. In this connection, the final result is $x(t) = v_0 t + at^2/2$.

The energy conservation is employed to characterize physical systems.[4] In addition to the classification of energy terms, we have the classifications of forces. When the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{f}$ is zero, with $\mathbf{r} \neq 0$ and $\mathbf{f} \neq 0$, then the vectors $\mathbf{r}$ and $\mathbf{f}$ are parallel. Under such conditions, the force is known as a *central force*. It may be also stated that for central forces, the torque is null. Many interaction potentials depend on the distance r solely, making the force to also depend on the distance. This is the case of bodies in orbital motion. In this situation, the total energy of the particle may be unfolded, for example, in polar coordinates. By using the expression of the speed in polar coordinates $v^2 = v_r^2 + v_\theta^2$, with $v_\theta = r d\theta/dt$, and the angular momentum L given by:

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v}_\theta = mr v_\theta\, (\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}}) \qquad \Rightarrow \qquad m v_\theta^2/2 = L^2/(2mr^2)$$

the total energy is:

$$\begin{aligned} E &= \frac{m}{2}\left(\frac{dr}{dt}\right)^2 + \frac{L^2}{2mr^2} + E_p(r) = \frac{m}{2}\left(\frac{dr}{dt}\right)^2 + E_{eff}(r) \;\; ; \\ E_{eff}(r) &= \frac{L^2}{2mr^2} + E_p(r) \end{aligned} \tag{2.40}$$

where the effective potential energy E_{eff} in terms of the radius r was defined. The angular term $L^2/(2mr^2)$ is positive and increases the magnitude of the potential energy E_p at every point r. By following similar steps to those performed in problem 2.2, and replacing E_p by E_{eff}, the equation of dr in terms of t is achieved.

$$\left(\frac{m}{2}\right)^{1/2} \int_{r_0}^{r} \frac{dr}{[E - E_{eff}(r)]^{1/2}} = t \tag{2.41}$$

After integration, it provides the variable r in terms of t. On the other hand, if we consider the relation between θ and the angular momentum, $d\theta/dt = L/(mr^2)$, then the angle θ in terms of t is:

[4] The energy is strictly not preserved due to the expansion of spacetime.

$$\theta = \theta_0 + \int_0^t \frac{L}{mr^2}\, dt \tag{2.42}$$

In order to have an equation of r in terms of θ, we employ $(dr/dt)(d\theta/dt)^{-1}$. By using the first expression of Eqs. 2.40 for dr/dt and $d\theta/dt = L/(mr^2)$, we reach:

$$\frac{dr}{d\theta} = \left[\frac{2}{m}\left(E - E_{eff}(r)\right)\right]^{1/2}\left(\frac{mr^2}{L}\right) \quad \Rightarrow \quad \int_{r_0}^{r} \frac{dr}{\left[2(E-E_{eff}(r))/m\right]^{1/2}(mr^2/L)} = \theta - \theta_0 \tag{2.43}$$

The last expression is the equation of the orbit, $r = r(\theta)$. The total work exerted on a body is different from zero when non-conservative forces act on the body in a closed-loop trajectory due to an irreversible transfer of energy, which results on a net energy loss. The non-conservative forces are usually of statistical nature, where we pay no attention to the individual particle interactions but to the effective force of the surrounding particles on the particle of interest. In spite of the possibility of associating a non-conservative force to a statistical average force of the surroundings (acting on the particle), a non-conservative force is also characterized as a force which is not derivable from a potential energy function.

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Many-Particle Systems

3

Most of the systems of our real world are constituted by many particles. The purpose of the present chapter is to introduce the relevant variables that characterize a *many-body system*. Many-body systems are important to link Newtonian mechanics to statistical mechanics and thermodynamics.

In a first stage, the reference frame of a many-particle system is established, to later define the angular momentum and torque of the system. The different types of energies of a many-particle system are considered. Heat is introduced as an external factor to formulate an energy balance expression of the system in interaction with its environment. In a second stage, with the purpose to close gaps between Newtonian mechanics and statistical mechanics, the equivalence between the time averages and the ensemble averages of physical observables is discussed. The mechanical variables of pressure, volume, temperature, and work, which are common in molecular dynamics and statistical mechanics, are derived in the last sections. The combination of the pressure, volume, and temperature into a single expression leads to an equation of state of the particle system.[1]

3.1 Reference Frame of a Many-Particle System

A reference frame of coordinates is required to describe a system of many particles. A general approach to define the reference frame is to consider all the particles without granting particular privileges to any specific particle. In this respect, the position of the *center of mass* (cm) is chosen as the position of the many-particle system.

$$\mathbf{r}_{cm} = \sum_i m_i \mathbf{r}_i / m_{tot} \quad ; \quad m_{tot} = \sum_i m_i \tag{3.1}$$

[1] The discussion follows similar topics to those of: Alonso and Finn [1].

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In this expression m_i and $\mathbf{r}_i$ are the mass and position of the ith particle. The velocity of the *cm* is obtained by derivation of the position variables with respect to time.

$$\mathbf{v}_{cm} = \sum_i m_i \mathbf{v}_i / m_{tot} \tag{3.2}$$

The momentum of the particle system is defined by the product of the body mass with the body velocity.

$$\mathbf{p}_{tot} = m_{tot} \mathbf{v}_{cm} = \sum_i m_i \mathbf{v}_i . \tag{3.3}$$

The entire system is visualized as a single particle with mass m_{tot} and speed $\mathbf{v}_{cm}$. There is a relative motion of the individual particles with respect to the *cm* position. The internal forces produce no change of $\mathbf{p}_{tot}$ because the third Newton law makes them zero by pairs, $\mathbf{f}_{ij} + \mathbf{f}_{ji} = 0$, Eq. 2.14. Thus, the total momentum $\mathbf{p}_{tot}$ is maintained constant when the system is isolated. Under such circumstances, it is convenient to locate the origin of coordinates at the position of the *cm* to have an inertial reference frame where the total linear momentum of the isolated system of particles is zero. However, if an atom or set of atoms changes its momentum by virtue of an external force, the total momentum of the system given by Eq. 3.3 is not conserved. This occurs, for instance, when a protein interacts with external agents, like the atoms of the solvent. There is an exchange of momenta between the different systems of particles (namely, the protein and the particles of the solvent). Still, it is possible to consider a single system of particles formed by the two sub-systems. Let us denote with the letter S the system of particles and with the letter S' the particles of the external system. The composed system is $S + S'$, and it is isolated and preserves the total momentum.

$$\mathbf{p}_{tot} = \mathbf{p}^S_{tot} + \mathbf{p}^{S'}_{tot} = \text{ constant} \tag{3.4}$$

A change of the momentum $\mathbf{p}^S_{tot}$ implies a change of the momentum $\mathbf{p}^{S'}_{tot}$.

$$\sum_{i\,\varepsilon\,S} \Delta \mathbf{p}^S_i = - \sum_{i\,\varepsilon\,S'} \Delta \mathbf{p}^{S'}_i \tag{3.5}$$

By differentiating every term of Eq. 3.5 with respect to time, the mutual external forces acting on each system are obtained.

$$\begin{aligned} \tfrac{d}{dt}\, \mathbf{p}^S_{tot} &= \mathbf{f}_{ext} \quad ; \quad \mathbf{f}_{ext} = -\tfrac{d}{dt}\, \mathbf{p}^{S'}_{tot} \\ \tfrac{d}{dt}\, \mathbf{p}^{S'}_{tot} &= \mathbf{f}'_{ext} \quad ; \quad \mathbf{f}'_{ext} = -\tfrac{d}{dt}\, \mathbf{p}^{S}_{tot} \end{aligned} \tag{3.6}$$

The composed system is isolated, the total force is zero, $\mathbf{f}_{ext} + \mathbf{f}'_{ext} = 0$, and the third Newton law in the subsystems S and S' is satisfied, $\mathbf{f}_{ext} = -\mathbf{f}'_{ext}$. The external force $\mathbf{f}_{ext}$ acts on the *cm* of system S producing the acceleration $\mathbf{a}^S_{cm}$.

$$\mathbf{f}_{ext} = \frac{d}{dt}\,\mathbf{p}^S_{tot} = \frac{d}{dt}\,\sum_{i\,\varepsilon\,S} m_i \mathbf{v}_i = m^S_{tot}\,\frac{d\mathbf{v}^S_{cm}}{dt} = m^S_{tot}\,\mathbf{a}^S_{cm} \tag{3.7}$$

The external force $\mathbf{f}_{ext}$ acting on the whole system S is equivalently considered as the outcome of all the individual external forces $\mathbf{f}_i$ acting on the single particles.

$$\mathbf{f}_{ext} = \frac{d}{dt}\,\sum_{i\,\varepsilon\,S} m_i \mathbf{v}_i = \sum_{i\,\varepsilon\,S} \mathbf{f}_i \tag{3.8}$$

The internal forces among the particles of the S system have no effect on the total momentum of such a system. However, the external forces to the S system are the forces capable of producing a non-zero contribution to the total momentum of the S system.

3.2 Angular Momentum and Torque of a Many-Particle System

The angular momentum $\boldsymbol{\ell}$ of a particle is $\boldsymbol{\ell} = \mathbf{r} \times \mathbf{p}$, where $\mathbf{r}$ and $\mathbf{p}$ are the position and momentum of the particle with respect to a given reference frame. In a system of particles, the total angular momentum includes the contribution $\boldsymbol{\ell}_i$ of every particle.

$$\mathbf{L} = \sum_i \boldsymbol{\ell}_i = \sum_i \mathbf{r}_i \times \mathbf{p}_i \tag{3.9}$$

The change of angular momentum $\mathbf{L}$ of a many-particle system with time is related to the external torque $\boldsymbol{\tau}_{ext}$.

$$\frac{d}{dt}\,\mathbf{L} = \sum_i \frac{d}{dt}\,\boldsymbol{\ell}_i = \sum_i \mathbf{r}_i \times \mathbf{f}_i = \sum_i \boldsymbol{\tau}_{i,ext} = \boldsymbol{\tau}_{ext} \tag{3.10}$$

The many-particle system is described using a system of coordinates, and the angular momentum $\boldsymbol{\ell}_i$ and torque $\boldsymbol{\tau}_{i,ext}$ are described with respect to such a system of coordinates. The internal forces $\mathbf{f}_{ji}$ between the arbitrary ith and jth particles generate no change of $\mathbf{L}$ because the internal forces lie along the separation vector $\mathbf{r}_{ji} = \mathbf{r}_j - \mathbf{r}_i$, resulting on zero contribution, $\mathbf{r}_{ji} \times \mathbf{f}_{ji} = 0$. Thus, Eq. 3.10 implies zero contribution of the internal forces. In addition, the angular momentum of an isolated system is preserved over time because the external forces are zero in this case.

$$\boldsymbol{\tau}_{ext} = 0 \quad \Rightarrow \quad \mathbf{L} = \text{constant} \tag{3.11}$$

A link between the angular momentum in the reference frame of the laboratory, $\mathbf{L}$, and the angular momentum in the *cm* reference frame, $\mathbf{L}_{cm}$, is obtained when the change of coordinates $\mathbf{r}_i = \mathbf{r}'_i + \mathbf{r}_{cm}$ and velocities $\mathbf{v}_i = \mathbf{v}'_i + \mathbf{v}_{cm}$ is employed in Eq. 3.9. If we consider that the non-primed variables describe the particles in the reference frame of the lab, and the primed variables describe the particles in the *cm* reference frame, then we have:

$$\mathbf{L} = \mathbf{L}_{cm} + m_{tot}(\mathbf{r}_{cm} \times \mathbf{v}_{cm}) \quad ; \quad m_{tot} = \sum_i m_i \quad ; \quad \mathbf{L}_{cm} = \sum_i [\mathbf{r}'_i \times \mathbf{p}'_i] \tag{3.12}$$

We employed the facts $\mathbf{r}'_i \times \mathbf{p}_{cm} = 0$ as $\mathbf{p}_{cm}$ is static for $\mathbf{r}'_i$, and $\mathbf{r}_{cm} \times \mathbf{p}'_i = 0$ as $\mathbf{r}_{cm}$ is the origin for $\mathbf{p}'_i$, where the position and velocity of the center of mass are $\mathbf{r}_{cm}$ and $\mathbf{v}_{cm}$. In an isolated system, $\mathbf{L}$ and $\mathbf{L}_{cm}$ are time indifferent. Thus, the angular momenta in the two inertial reference frames are conserved for the isolated system of particles. By differentiating the above expression, the relation between torques is found.

$$\boldsymbol{\tau}_{ext} = \frac{d}{dt}\mathbf{L}_{cm} + \mathbf{r}_{cm} \times \mathbf{f}_{ext} \tag{3.13}$$

Here, we also employed the facts $(d\mathbf{r}_{cm}/dt) \times \mathbf{v}_{cm} = \mathbf{v}_{cm} \times \mathbf{v}_{cm} = 0$ and $d(m_{tot}\mathbf{v}_{cm})/dt = \sum_i m_i \, d\mathbf{v}_i/dt = \sum_i \mathbf{f}_i = \mathbf{f}_{ext}$. We require to associate a torque to the term $d\mathbf{L}_{cm}/dt$. To do this, we use Eq. 3.10 to establish the relation between the primed and non-primed variables.

$$\boldsymbol{\tau}_{ext} = \sum_i (\mathbf{r}'_i + \mathbf{r}_{cm}) \times \mathbf{f}_i = \boldsymbol{\tau}_{cm} + \mathbf{r}_{cm} \times \mathbf{f}_{ext} \quad ; \quad \boldsymbol{\tau}_{cm} = \sum_i (\mathbf{r}'_i \times \mathbf{f}_i) \tag{3.14}$$

The net torque exerted on the particles as observed from the reference frame of the center of mass is $\boldsymbol{\tau}_{cm}$. From Eqs. 3.13 and 3.14, it is concluded that:

$$d\mathbf{L}_{cm}/dt = \boldsymbol{\tau}_{cm} \tag{3.15}$$

The torque exerted on the cm is $\mathbf{r}_{cm} \times \mathbf{f}_{ext}$.

3.3 Mechanical Energies of a Many-Particle System

The interest now comes to obtain an expression of the energy of a many-particle system. The process starts from the Newtonian equations of motion which should be transformed onto an expression of the energy. The equation of motion of the ith particle is given by:

$$m_i \mathbf{a}_i = \mathbf{f}_i + \sum_{j=1, i \neq j}^{N} \mathbf{f}_{ij} \tag{3.16}$$

The force $\mathbf{f}_i$ represents the external force acting on the ith particle, and the summation on the jth index of $\mathbf{f}_{ij}$ gives the total internal force, $\mathbf{f}_i^{int}$, exerted on such a particle. The internal force may be obtained from a potential energy function E_{pot} by differentiation, $\mathbf{f}_i^{int} = -\nabla_i E_{pot}(\{\mathbf{r}_j\})$. The function E_{pot} depends on all the coordinates of the particles, $\{\mathbf{r}_j\}$. In the process of transforming the equation of motion onto and energy equation, it is necessary to multiply each term of Eq. 3.16 by the displacement vector $d\mathbf{r}_i$.

$$m_i \mathbf{a}_i \cdot d\mathbf{r}_i = \mathbf{f}_i \cdot d\mathbf{r}_i + \sum_{j=1}^{N} \mathbf{f}_{ij} \cdot d\mathbf{r}_i \quad ; \quad i \neq j$$

All the equations of motion of the particles should be added up. Before, we illustrate the way in which the third Newton law is considered in the addition of equations. Let us deal with the internal and external forces on the ith and jth particles as given by Eq. 3.16.

$$m_i \mathbf{a}_i \cdot d\mathbf{r}_i + m_j \mathbf{a}_j \cdot d\mathbf{r}_j = \mathbf{f}_i \cdot d\mathbf{r}_i + \mathbf{f}_j \cdot d\mathbf{r}_j + \mathbf{f}_{ij} \cdot (d\mathbf{r}_i - d\mathbf{r}_j) + \cdots$$

The third Newton law was employed in the last term of the above expression. It is convenient to use the identities $\mathbf{a}_i \cdot d\mathbf{r}_i = (d\mathbf{v}_i/dt) \cdot d\mathbf{r}_i = d\mathbf{v}_i \cdot \mathbf{v}_i$ to introduce the velocities of the particles and integrate to apply the concepts of work and kinetic energy.

$$\int_{v_{a_i}}^{v_{b_i}} m_i \mathbf{v}_i \cdot d\mathbf{v}_i + \int_{v_{a_j}}^{v_{b_j}} m_j \mathbf{v}_j \cdot d\mathbf{v}_j = \int_a^b \mathbf{f}_i \cdot d\mathbf{r}_i + \int_a^b \mathbf{f}_j \cdot d\mathbf{r}_j + \int_a^b \mathbf{f}_{ij} \cdot d\mathbf{r}_{ij} + \cdots$$

The difference $d\mathbf{r}_i - d\mathbf{r}_j$ is $d\mathbf{r}_{ij}$. By similarly dealing with all the particles, we reach an expression referent to the balance of the kinetic energy E_{kin}:

$$\begin{gathered} E_{kin} - E_{kin}^{(0)} = W_{ext} + W_{int} \\ E_{kin} = \textstyle\sum_i \frac{1}{2} m_i v_i^2 \quad ; \quad W_{ext} = \sum_i \int_a^b \mathbf{f}_i \cdot d\mathbf{r}_i \quad ; \quad W_{int} = \sum_{i<j}^N \int_a^b \mathbf{f}_{ij} \cdot d\mathbf{r}_{ij} \end{gathered} \tag{3.17}$$

The kinetic energy E_{kin} of the first line is given with reference to the initial kinetic energy, $E_{kin}^{(0)}$. This last energy is due to integrating and evaluating the result at the lower integration limit. The external forces exerted on the particles produce the external work W_{ext}. The last expression is the internal work W_{int}, due to internal forces. According to the expression in the first line, any change of the total kinetic energy comes from the total work $W_{ext} + W_{int}$ exerted on the particles. When the internal forces acting on the particles are conservative, the internal work may be written as the difference of internal potential energies, $W_{int} = E_{pot}^{(0)} - E_{pot}$, Eq. 2.35. The potential energy depends on the interparticle distances and is independent of the reference frame of coordinates. The *proper energy* of the system is defined by:

$$U = E_{kin} + E_{pot} \qquad \text{proper energy}$$

The work exerted by the external forces on the system results on the change of the proper energy.

$$W_{ext} = U - U^{(0)} \quad ; \quad U = E_{kin} + E_{pot} \tag{3.18}$$

The energy increases, $U > U^{(0)}$, when the work is exerted on the particle system, $W_{ext} > 0$. If the particles perform work, $0 > W_{ext}$, the energy diminishes, $U^{(0)} > U$. When there are no external forces, $W_{ext} = 0$, the system is isolated, and the proper energy U of the system is preserved in time. In an isolated particle

system, the energy preservation demands that any change of the kinetic energy concurrently occurs with a change of the potential energy of identical magnitude and opposite sign. By choosing a description of the kinetic energy from the position of the center of mass, and recalling that the internal potential energy is independent of the reference frame, it is possible to define the *internal energy* of the system.

$$U_{int} = E_{kin,cm} + E_{pot} \qquad \text{internal energy} \tag{3.19}$$

Problem 3.1 Find the expression relating the proper energy $U = E_{kin} + E_{pot}$ to the internal energy $U_{int} = E_{kin,cm} + E_{pot}$ in a reference frame of coordinates located at the center of mass.

Solution The kinetic energy given in the reference frame of the laboratory is $E_{kin} = \sum_i m_i v_i^2/2$. The velocity $\mathbf{v}_i$ can be written in the form $\mathbf{v}_i = \mathbf{v}'_i + \mathbf{v}_{cm}$, where $\mathbf{v}_{cm}$ is the velocity of the *cm* and $\mathbf{v}'_i$ is the velocity of the ith particle with respect to *cm*. The insertion of the expression of $\mathbf{v}_i$ in the kinetic energy leads to:

$$\begin{aligned} E_{kin} &= \textstyle\sum_i \frac{1}{2} m_i \left(v'_i + v_{cm}\right)^2 \quad ; \quad m_{tot} = \textstyle\sum_i m_i \\ &= E_{kin,cm} + \tfrac{1}{2} m_{tot} v_{cm}^2 \quad ; \quad E_{kin,cm} = \textstyle\sum_i \frac{1}{2} m_i \left(v'_i\right)^2 \end{aligned}$$

The total primed momentum, $\sum_i m_i \mathbf{v}'_i$, is zero in the reference frame of the *cm*. In this regard, the kinetic energy E_{kin} in the reference frame of the laboratory is written in terms of the internal kinetic energy $E_{kin,cm}$ in the reference frame of the *cm*, plus a kinetic energy of translation of the *cm*, $m_{tot} v_{cm}^2/2$. On the other side, the potential energy, E_p, depends on the particle distances and exhibits no change with respect to the two different reference frames. By adding the potential energy to the above expression, the proper energy U is written in terms of the internal energy U_{int}.

$$U = U_{int} + \tfrac{1}{2} m_{tot} v_{cm}^2 \quad ; \quad U_{int} = E_{kin,cm} + E_{pot}$$

In an isolated system, U is constant, the translation of the system is constant, $m_{tot} v_{cm}^2/2 =$ constant, and the internal energy U_{int} is equally constant. Thereby, the energy is conserved in any inertial reference frame used to compute the energy.

In a diluted gas at high temperature, the particles have large speeds, and the potential energy of the particles is small with respect to the kinetic energy. In this case, the conservation of internal energy is essentially a conservation of kinetic energy. On the other hand, when the external forces in Eq. 3.18 are conservative, such as a

molecule in a gravitational field, the external forces are obtained from a potential energy, $W_{ext} = E^{(0)}_{pot,ext} - E_{pot,ext}$, and it is possible to introduce the *total energy* E of the system.

$$E = U + E_{pot,ext} \qquad \text{total energy} \tag{3.20}$$

The preservation of the total energy is formulated from Eqs. 3.18 and 3.20 by $E = E^{(0)}$, or $E =$ constant. The presence of external forces makes the energy E different from U.

$$E = U + E_{pot,ext} = E_{kin} + E_{pot} + E_{pot,ext} \tag{3.21}$$

In the absence of external forces or in an isolated system of particles, the proper energy U is preserved, namely, $U = U^{(0)}$ from Eq. 3.18, and in the presence of conservative external forces, the total energy E is conserved, namely, $E = E^{(0)}$.

$$\text{particle system} \begin{cases} \text{proper energy:} & U = \text{constant energy} \quad \text{no external forces} \\ \text{internal energy:} & U_{int} = \text{constant energy} \quad \text{no external forces} \\ \text{total energy:} & E = \text{constant energy} \quad \text{conservative external forces} \end{cases}$$

3.4 Transformation of the Energy Components

When an interaction between two particle systems occurs, a transformation of the kinetic energy onto potential energy, and vice versa, may be observed. A way to evaluate the energy transformation is by introducing the q *balance factor*. If the energy is constant, $E^a_{kin} + E^a_{pot} = E^b_{kin} + E^b_{pot}$, and rewritten in the form $E^a_{kin} - E^b_{kin} = E^b_{pot} - E^a_{pot}$, then the q balance factor is defined as:

$$q = E^a_{kin} - E^b_{kin} = E^b_{pot} - E^a_{pot} \tag{3.22}$$

The variables with index b represent the system before the reaction, and the variables with index a represent the system after the reaction. When $q < 0$, there is a lowering of kinetic energy and an increase of potential energy.

$$q < 0 \quad \Rightarrow \quad E^a_{kin} < E^b_{kin} \quad ; \quad E^b_{pot} < E^a_{pot}$$

The opposite situation, $0 < q$, corresponds to a lowering of potential energy and an increase of kinetic energy.

$$0 < q \quad \Rightarrow \quad E^a_{kin} > E^b_{kin} \quad ; \quad E^b_{pot} > E^a_{pot}$$

In both cases, the interaction is an *inelastic collision*. When $q = 0$, there is no transformation of the energy components, and the interaction between the two systems is an *elastic collision*.

The conservation of physical variables such as the linear momentum, angular momentum, and energy are formulated in classical mechanics in terms of positions and velocities of the particles. In this regard, a systems which is suspected to satisfy a conservation law may be examined from the perspective of positions and velocities of the particles, since these are basic variables to describe particle systems.

3.5 Energy Balance Equation

There is an additional energy ingredient of macroscopic nature capable of changing the energy of a particle system. It is called *heat* and plays an important role for objects in contact with different temperatures. Heat is a primitive concept in thermodynamics only determined through observation, arising from the energy diffusion between bodies with a temperature gradient at their contact points. In Newtonian mechanics, heat is the average exchange energy between the system and the surrounding environment due to the individual exchange of energy of the particles of the system with those of the environment [2]. In statistical mechanics, heat is the energy flowing between the heat source and the system of interest due to their different temperatures.

Heat is characterized as an energy in transition, flowing from a body with high temperature, known as the hot body, to another body with lower temperature, known as the cold body. Such a type of flow is stated by the *zero law of thermodynamics*. The energy transfer occurs via the exchange of kinetic and potential energies, collisions, radiation, and other energy-transfer channels between two or more connected systems with a given temperature difference. It is not possible to quantify the heat of a body because the heat is an energy in transition.

Frequently, the macroscopic environment plays the role of a heat source and is referred to as the *heat reservoir*. Heat is associated with forces of non-conservative nature, namely, those forces which are not deduced from a potential energy function and act on the particle system. On the other hand, an external work W_{ext} exerted on the system is supposed to be generated by conservative forces. Heat, represented by the symbol Q (of quantity), must be considered in expression 3.18 to properly establish an *energy balance* of the closed system.

$$W_{ext} + Q = U - U^{(0)} \tag{3.23}$$

The above expression indicates that a change of energy U is produced by: (i) the heat flux between the heat source and the system, and (ii) the work performed on the system or by the system (the system energy may be also changed by adding or subtracting matter, but it is supposed to be constant in this discussion). Heat is created to the detriment of other energy forms: it is positive when the particle system increases its internal energy (heat is absorbed by the system) and negative

when the particle system reduces its internal energy (heat is given by the system). The experiment where the mechanical equivalent of heat is established puts in evidence the transformation of mechanical energy onto heat and vice versa. The energy cannot be created neither destroyed but transformed onto other energy forms. The many phenomenological evidences on the transformation of the energy allow to postulate the energy balance principle, Eq. 3.23, establishing the *first law of thermodynamics*. Yet it is possible to derive it from first principles [3].

In addition to the heat, the energy balance demands the consideration of the *external work*, W_{ext}. A differential work, dW, is related to an external force $\mathbf{f}$ by $dW = \mathbf{f} \cdot d\mathbf{r}$. If the external forces increase the system internal energy, then the work W_{ext} is considered to be positive, and it is negative when the system performs work. The previous proposals on the signs of heat and work follow the $IUPAC$ conventions.[2] The work may be re-defined by introducing W_{sys} in the form $W_{sys} = -W_{ext}$, with the purpose to have a positive sign when the particle system performs a work on the environment.

$$Q - W_{sys} = U - U^{(0)} \tag{3.24}$$

Problem 3.2 The motion of a Brownian particle in a fluid is given by the Langevin expression $m\, d^2\mathbf{r}/dt^2 = \mathbf{f}(t) - \xi m\mathbf{v}(t) + \mathbf{g}(t)$, where $\mathbf{f}(t)$ is a systematic (external) force exerted on the particle, ξ is the medium viscous coefficient, and $\mathbf{g}(t)$ is the stochastic punch of the environment on the particle at the instant of time t. Obtain the energy equation of the particle.

Solution The net force is multiplied by the particle displacement $d\mathbf{r}$, and integrated from point A to point B, to evaluate the particle energy in such a process:

$$\int_A^B m\mathbf{v} \cdot \mathbf{dv} = \int_A^B \mathbf{f}(t) \cdot d\mathbf{r} + \int_A^B [-\xi m\mathbf{v}(t) + \mathbf{g}(t)] \cdot d\mathbf{r}$$

where the identity $\mathbf{a} \cdot d\mathbf{r} = \mathbf{v} \cdot d\mathbf{v}$ was used. The term on the left is the kinetic energy difference between points B and A. The first term on the right is the external work, W_{ext}, and the second is the energy Q due to the interaction with the environment, like the friction forces and the medium strikes.

$$E_{kin}^{(B)} - E_{kin}^{(A)} = W_{ext} + Q$$

[2] The International Union of Pure and Applied Chemistry ($IUPAC$) establishes the standards of names, symbols, units, and more.

There are different pathways to take a system from an initial state to a predetermined final state, for instance, by changing either the pressure, temperature, or volume. The thermodynamic pathways convey energy and define the amount of heat and work given to or received from the particle system. In the process of changing states, the system may visit intermediate states. The work W_{ext} and heat Q have no dependence on the coordinates of the system because they are external factors to the system, and therefore, they are not considered as state functions of the particle system (under a mathematical conception, the infinitesimal changes of W_{ext} and Q are inexact differentials denoted by $đW_{ext}$ and $đQ$) [4]. Thus, the particle system in equilibrium is not characterized by the heat and work;[3] however, the difference between the amount of heat and the amount of work $Q - W_{sys}$ defines the internal energy change $U - U^{(0)}$ of the system. The internal energy U depends on the particle coordinates, and it classifies as a state function of the system (under a mathematical conception, the infinitesimal changes of U are exact differentials denoted by dU).

The first law of thermodynamics relates the internal energy changes of the system to the statistical forces produced by the macroscopic environment. In this respect, the work and heat represent two forms of exchanging energy between the system and the environment. Once the system changes its energy and reaches a state of equilibrium, it is not possible to tell whether such a state was achieved after doing work, transferring heat, or using both because, in addition to the time irreversibility of thermodynamic theory, the work and heat are not state functions.

3.6 Statistical and Time Averages of Physical Observables

When a system is constituted by a large number of particles, the physical variables of the individual particles are not considered observables anymore due to the complexity in measuring the position and velocity of every particle, except under particular situations as in the case of a *Brownian particle*, which is notably different in geometrical size or inertial mass from the other particles. In a many-particle system, the *averages of physical variables* acquire significance, rather than the individual physical variables of the particles. The averages of the mechanical variables may be computed in different ways after using either:

- a large number of similar particle systems, called *replicas*, and performing a *statistical average*, also known as an *ensemble average*, of the mechanical variables over the replicas
- a single molecular system and performing a *time average*, also known as a *trajectory average*, of the mechanical variables over the trajectory.

The field of physics that deals with many particles, their replicas, and the ensemble averages is *statistical mechanics*. According to statistical mechanics, the macro-

[3] In statistical mechanics, a system subjected to heat transfer achieves a new level of *disorder*.

scopic observables of a many-body system are defined by the ensemble averages on the basic mechanical variables using a large number of replicas of the particle system. In contrast, the macroscopic observables in molecular dynamics are defined by time averages of the basic mechanical variables using a single molecular system evolved during a long period of time. The statistical and time averages are important because they represent two forms to establish contact between the microscopic and macroscopic worlds.

Based on the principles of statistical mechanics, the many-particle system is required to be in the state of *thermodynamic equilibrium* to apply statistical mechanics. The thermodynamic equilibrium demands that the rate of energy dissipation and the rate of energy input are balanced and the system-intensive variables, like the pressure P, temperature T, and chemical potential μ, are constants over time. The equilibrium of P, T, and μ is a way to demand mechanical, thermal, chemical, and diffusive equilibrium. The system in contact with a reservoir may use a permeable wall to any of the previous intensive variables to force its homogeneity in space and time and keep constant values of P, T, and μ. The reservoir is supposed to have large dimensions to be little affected by energy changes associated with the contact with the particle system, or in any case, it should be capable of rapidly fading away any noticeable energy gradients (such processes have been formulated in fluctuation-dissipation theorems). The particle system flows to the state of thermodynamic equilibrium when it is in contact with the reservoir, unless it is prevented from doing so by external forces. The equilibrium state may be reached in different stages, with some macroscopic variables possibly achieving the equilibrium before others, or they all reach the equilibrium almost simultaneously and in their own ways.

Once the thermodynamic equilibrium is achieved by the particle system, its thermal fluctuations are small, and then, the mean value of a physical variable may be identified with the value obtained at the macroscopic scale, in spite of the fact that at the microscopic level there are ceaseless local energy transformations and local flows of matter. The induced forces produced in this way give rise to the small fluctuations observed at the macroscopic scale, yet they are continuously balanced and dissipated.

The concept of *thermodynamic limit* plays an important role in systems with a large number of particles because such a concept defines the required features to deal with a macroscopic system and eventually allows the application of thermodynamic theory on the many-particle system. More precisely, in the thermodynamic limit, the system is considered to be of macroscopic dimensions, $N \to \infty$, with natural fluctuations that are small at the macroscopic scale. Still, the size of the fluctuations is important to determine the stability of the system. In this connection, the thermodynamic equilibrium corresponds to a *macroscopic stationary state*, where the average values of the macroscopic variables show no change over time. When the average values, for example of the pressure, temperature, and chemical potential are the same to those of the surroundings, the system stays in the state of thermodynamic equilibrium insofar as there are no external forces or fields driving it away from the stationary state.

Summing up, the averages of physical variables are calculated when the system is in the state of thermodynamic equilibrium, as the macroscopic variables are stationary and the experiments can be reproduced any number of times. The ensemble averages obtained in the state of thermodynamic equilibrium correspond to the measurements performed in the laboratory. Such experimental measurements are also reproduced by computing the trajectory averages in molecular dynamics. The equivalence between ensemble averages and trajectory averages is an important hypothesis, which is briefly analyzed next.

3.7 Ergodic Hypothesis

The equivalence between a time average and a statistical average is postulated and is known as the *ergodic hypothesis*. In order to illustrate it, consider a system with constant NVE values. The discussion follows similar lines to those of Frenkel and Smit [5]. We depart from a trajectory average to reach an ensemble average. Let us denote with $\bar{A}$ the trajectory average of observable A. It is calculated from the sequence of instantaneous values $A(t_1), \cdots, A(t_n)$ registered along the trajectory of the system and when the thermodynamic equilibrium has been achieved in the molecular dynamics simulation.

$$\bar{A}(t) = \frac{1}{n} \sum_{i=1}^{n} A(t_i) \tag{3.25}$$

When the time of the simulation is sufficiently long and the time step is small enough, the time variable may be transformed from a discrete to a continuous variable, and the summation becomes a time integral.

$$\bar{A}(t) = \lim_{\tau \to \infty} \frac{1}{\tau} \int_0^{\tau} A(t)\, dt \tag{3.26}$$

Problem 3.3 The time step in a simulation is $\Delta t = (t_2 - t_1)$, and the particles evolve according to the time grid $t_1, t_2, \cdots, t_{n+1}$. After assuming a uniform grid, show that the numerical integration of f, which is a function of t, is:

$$\int_{t_1}^{t_{n+1}} f(t)dt = \Delta t \left[\frac{f(t_1) + f(t_{n+1})}{2} + \sum_{\ell=1}^{n} f(t_1 + \ell \Delta t) \right]$$

Find the result of integrating $f(t)$ when the grid is non-uniform.

Solution It is possible to compute the integral by considering the area of small rectangles, where a rectangle area is $\Delta t[f(t_i) + f(t_{i+1})]/2$. It is a linear approximation since the function f between $f(t_i)$ and $f(t_{i+1})$ is considered to be a straight line. The sum over all the small areas approximates the total

(continued)

area under the function $f(t)$.

$$\begin{aligned}\int_{t_1}^{t_{n+1}} f(t)dt &= \tfrac{\Delta t}{2}[f(t_1)+f(t_2)]+\tfrac{\Delta t}{2}[f(t_2)+f(t_3)]+\cdots \\ &\quad +\tfrac{\Delta t}{2}[f(t_n)+f(t_{n+1})] \\ &= \tfrac{\Delta t}{2}f(t_1)+\Delta t\,[f(t_2)+f(t_3)+\cdots+f(t_n)]+\tfrac{\Delta t}{2}f(t_{n+1})) \\ &= \Delta t\left[\tfrac{f(t_1)+f(t_{n+1})}{2}+\textstyle\sum_{\ell=1}^{n} f(t_1+\ell\Delta t)\right]\end{aligned}$$

When the grid is non-uniform, it is not possible to factorize the time step, and we have:

$$\int_{t_1}^{t_{n+1}} f(t)dt = \tfrac{1}{2}\textstyle\sum_{i=1}^{n}(t_{i+1}-t_i)\,[f(t_{i+1})+f(t_i)]$$

This numerical integration scheme is known as the *trapezoidal integration method*. It is a linear method with deviations proportional to the second derivative of f. The accuracy is improved when the time step is sufficiently small [6].

A simulation may be performed in a single large run or in several small runs to visit all the permissible configurations of the system that are compatible with the imposed macroscopic conditions NVE. Let us consider several simulation runs. They are started from different initial conditions $\{\mathbf{r}(t_i), \mathbf{p}(t_i)\}$; nevertheless, they all are capable of reaching the NVE state of thermodynamic equilibrium of the system. It is possible to add all the results of the many molecular dynamics simulations in the thermodynamic equilibrium and obtain the average value $\bar{A}$. The simulations and the summation process are represented by:

$$\begin{aligned}\bar{A} = \tfrac{1}{M}\Bigg\{&\text{initial condition (1)} \quad : \quad \overbrace{A^{(1)}(t=1)+\cdots+A^{(1)}(t=N_1)}^{1^{st}\text{ simulation, }N_1\text{ steps}} + \\ &+\text{initial condition (2)} \quad : \quad \overbrace{A^{(2)}(t=1)+\cdots+A^{(t=2)}(t=N_2)}^{2^{nd}\text{ simulation, }N_2\text{ steps}} +\cdots \\ &+\text{initial condition (n)} \quad : \quad \overbrace{A^{(n)}(t=1)+\cdots+A^{(n)}(t=N_n)}^{n^{th}\text{ simulation, }N_n\text{ steps}}\Bigg\}\end{aligned} \tag{3.27}$$

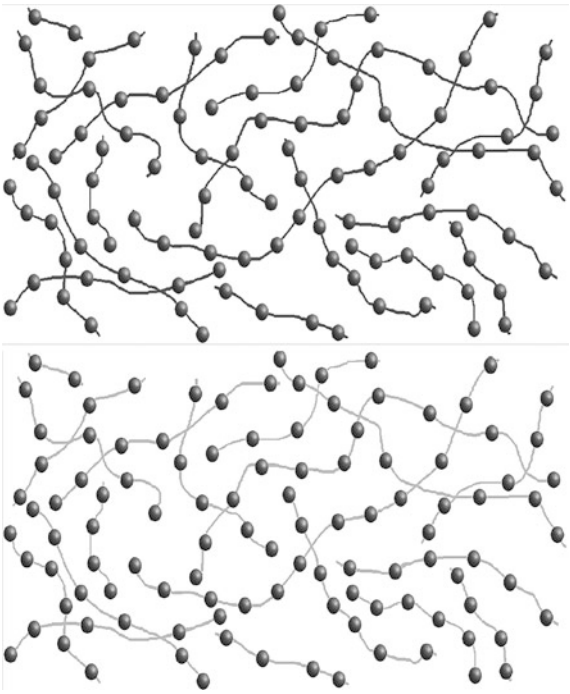


Fig. 3.1 The sketch on the top illustrates different trajectories in the multidimensional space of positions and momenta of a system that contains many particles. The trajectories are depicted with lines, and the entire system is depicted with a ball. The different balls along the trajectories are snapshots of the system, supposed to be in thermodynamic equilibrium. The trajectories are generated using molecular dynamics. A single large trajectory or many small trajectories (created from different initial conditions) cover the multidimensional space of positions and momenta. The white islands are incompatible regions with the macroscopic observables of the system.
The sketch on the bottom illustrates that, once the multidimensional space is covered, the time has no meaning, the trajectories may be dismissed (faded away in the sketch), and the time averages may be replaced by statistical averages, because we have knowledge of the different microscopic states or replicas which are compatible with the macroscopic state of the system.
In simulations like these, molecular dynamics meets statistical mechanics, and the average process of physical variables may follow the rules of statistical mechanics

The terms $A^{(i)}(t = 1),\ A^{(i)}(t = 2),\ \cdots,\ A^{(i)}(t = N_i)$, with $i = 1, \cdots, n$, represent the values of A obtained in the ith simulation at the instants of time $t = 1, \cdots, N_i$. In principle, we obtain all the M configurations, $M = N_1 + N_2 + \cdots + N_n$, that produce the NVE macroscopic state. When such configurations are registered and numbered, the time has no more significance, and expression 3.27 may be considered to represent a statistical average over the replicas of the system, Fig. 3.1. In this respect, the time is dismissed, and we only maintain the indexes of A to identify the values obtained from the different replicas of the statistical ensemble.

$$\bar{A} = [A^{(1)} + \cdots + A^{(N_1)} + A^{(N_1+1)} + \cdots + A^{(N_1+N_2)} + \cdots + A^{(M)}] / M \tag{3.28}$$

Now, the nomenclature $A^{(i)}$ indicates the value of observable A in the ith configuration determined from the set of variables $\{\mathbf{r}, \mathbf{p}\}$ of all the particles. Procedures 3.27 and 3.28 establish a point of contact between molecular dynamics and statistical mechanics. The summation 3.28 over replicas may be avoided by introducing a weight function that contains the relevant information of the replicas to compute the statistical average. In statistical mechanics, the statistical average is more precisely referred to as the ensemble average, denoted in the form $\langle A \rangle$ to distinguish it from the geometric mean $\bar{A}$ in molecular dynamics, and the weight function is known as the *probability distribution function*. This function should be normalized and is used to compute the average of the observable. When $M \to \infty$, expression 3.28 is written as an integral over both the coordinates and momenta of the particles.

$$\begin{gathered} \langle A(\{\mathbf{r}, \mathbf{p}\}) \rangle = \frac{1}{Z} \int \rho_{NVE}\ A(\{\mathbf{r}, \mathbf{p}\})\, d\Gamma \\ Z = \int \rho_{NVE}\, d\Gamma \quad ; \quad \Gamma = d^3 r_1 \cdots d^3 r_N\, d^3 p_1 \cdots d^3 p_N \end{gathered} \tag{3.29}$$

The function ρ_{NVE} is the probability distribution function with dependence on the set of variables $\{\mathbf{r}, \mathbf{p}\}$ of all the particles and is characteristic of the particle system under the given macroscopic conditions, NVE in our case. The factor Z is the normalization factor and contains information of the microscopic states, or *microstates*, through ρ_{NVE}. By recalling that the average value of the observable A is expressed in the form $\bar{A}$ in molecular dynamics and $\langle A \rangle$ in statistical mechanics, the ergodic hypothesis of the particle system is abbreviated by:

$$\bar{A}(\{\mathbf{r}(t), \mathbf{p}(t)\}) = \langle A(\{\mathbf{r}, \mathbf{p}\}) \rangle \tag{3.30}$$

The averages of an observable A computed in the form $\bar{A}$ or $\langle A \rangle$ are equivalent. When the system is in thermodynamic equilibrium, it remains indefinitely in that stable state, and the ergodic hypothesis is satisfied. Under the action of small and transient external forces, the system in thermodynamic equilibrium recovers the stability. However, when the external forces are large, they produce large fluctuations capable of driving the particle system out equilibrium, changing the macroscopic parameters that characterize the system. Contrary to the system in the thermodynamic-equilibrium state, a system in a *metastable state* may be driven out of equilibrium by small forces. When the system is in a *non-equilibrium state*, the concepts of thermodynamic equilibrium and ergodicity have no application, making the thermodynamics of the system meaningless. Yet the system in the out-of-equilibrium state may be investigated with molecular dynamics since it only has dependence on the fundamental interactions of the particles.

3.8 Breaking the Ergodic Hypothesis

The ergodic hypothesis is characteristic of systems in thermodynamic equilibrium. In this regard, it is not fully satisfied when the particle system is out of thermodynamic equilibrium, since the mean values of physical variables have large fluctuations and, in consequence, an average value is not properly defined. The thermodynamic equilibrium of a system is a requirement that in most instances is difficult to obtain in the laboratory. Nevertheless, the equilibrium condition may be relaxed by demanding relative temporal equilibrium in such a way that, after assessing the temporal stability of the macroscopic parameters, any required measurement of a microscopic variable may be performed without time restrictions on the physical system (in principle, the time interval of stability implied by thermodynamics is a relative interval of time). The system may not be required to be isolated, and the measurements may be performed on an open system when it is in thermodynamic equilibrium with the surroundings.

The systems with a large number of particles and in thermodynamic equilibrium show statistical deviations of order $N^{-1/2}$, with N the number of particles. The systems with a small number of particles like the atomic clusters exhibit large fluctuations of their properties with respect to the average values of those properties. Thus, care should be exercised in the study of small systems.[4] Some examples of systems out of thermodynamic equilibrium are those in the instances of a chemical reaction, a phase transition, and more. Molecular dynamics may be applied to investigate the behavior of small and large particle systems regardless of those circumstances.

3.9 Velocity Distribution Function

In a molecular dynamics simulation, the atoms move and each of them has a velocity. When the system is in thermal equilibrium, the speeds of the atoms satisfy the *Maxwell-Boltzmann speed distribution function*:

$$f(v) = \frac{4}{\sqrt{\pi}} \left[\frac{m}{2k_B T}\right]^{\frac{3}{2}} v^2 \exp\left[\frac{-mv^2}{2k_B T}\right] \tag{3.31}$$

where m is the particle mass, k_B is the Boltzmann constant, v is the velocity, and T is the temperature. The most probable speed, v_p; mean speed, $\overline{v}$; and root mean square speed, v_{rms}, deduced from the Maxwell-Boltzmann distribution are, respectively:

$$v_p = \sqrt{2k_B T/m} \quad ; \quad \overline{v} = \sqrt{8k_B T/(\pi m)} \quad ; \quad \overline{v}_{rms} = \sqrt{3k_B T/m}$$

[4] The clusters are of particular interest because they introduce the *molecular size* as a new physical observable. The clusters change their properties with size. Such a feature fades away toward the macroscopic scale, and therefore, has no counterpart in macroscopic systems.

These speeds are temperature dependent. In a simulation, the initial speeds, usually referred to as the *offset speeds*, are frequently assigned from a Maxwell–Boltzmann distribution function. This is done by taking in random manner the speeds from a histogram that resembles a Maxwell-Boltzmann speed distribution function, where the abscissa of the distribution function corresponds to the speed intervals and the ordinate to the number of speeds that classify in the intervals.

The temperature, pressure, and volume are important features employed to characterize the states of thermodynamic equilibrium at the macroscopic scale. The goal of the following sections is to discuss those variables from a mechanical perspective, namely, in terms of the basic variables of position and velocity of the particles. The temperature, pressure, and volume may be also evaluated in statistical mechanics after using the appropriate distribution functions, giving results consistent with their mechanical counterparts.

3.10 Temperature of a System of Particles

In this section, we deal with the temperature since it is one of the properties that characterizes the macroscopic state of a many-particle system. *Temperature*, T, is a scalar parameter which indicates the ability of a system to give and receive heat from the surroundings: the higher the temperature of the system, the better the ability to receive and give energy. Temperature is one of the factors that we use to determine the thermal equilibrium of the bodies. The concept of temperature is different from that of heat. The difference may be illustrated with an example: two metallic rods with equal geometry and different composition may have equal temperatures but transmitting different amounts of heat to other bodies.

Suppose that a closed system has a higher temperature than that of its environment. The energy which the system gives to the environment is transferred in the form of heat. The energy transfer decreases the kinetic energy of the system, cooling the system down. The opposite situation occurs when the system temperature is lower than that of the environment, warming the system up. According to our observations, when body C is in thermal equilibrium with bodies A and B in separate form, then body A and body B are found in thermal equilibrium when they are in contact with each other. The *zero law* of thermodynamics defines the temperature as a system property by introducing the thermal equilibrium between bodies. The zero law may be explained from a statistical perspective by using the *equipartition of the energy*, which indicates that for a system in thermal equilibrium, the average kinetic energy of each independent degree of freedom is $kT/2$, and an additional amount of $kT/2$ energy for the average potential energy of each independent degree of freedom, inasmuch as the kinetic and potential energies are quadratic functions of the independent degrees of freedom. Thus, the temperature of a system of particles may be obtained from either the average kinetic energy or the average potential energy of the independent degrees of freedom. Yet in many particle systems, the potential energy does not necessarily contain quadratic terms of the independent degrees of freedom, and then, it is convenient to determine the

temperature from the average kinetic energy, which usually involves quadratic terms of the independent degrees of freedom of motion.

From the phenomenological perspective, we observe that when the average velocity of the particles increases (decreases), the temperature linearly increases (decreases). In other words, the temperature and the average kinetic energy exhibit a linear dependence on each other. Still, it takes different amounts of energy to reach the same average kinetic energy in different particle systems: it takes some amount of energy to increase the average kinetic energy of the particles of an ideal gas and a different amount of energy to reach the same average kinetic energy of a system of bonded particles, where the potential energy among the particles modifies in an effective way the inertia of the particles. Thus, according to our observations and the equipartition theorem, the *kinetic expression of the temperature* may be obtained from the microscopic motion of the particles without regard to the potential energy.

$$\langle E_{kin}(t)\rangle = \frac{(3N-N_c)}{2}\, k_B T \quad \Rightarrow \quad T = \frac{2}{(3N-N_c)k_B} \left\langle \sum_{i=1}^{N} \frac{1}{2}\, m_i \mathbf{v}_i^2(t) \right\rangle \tag{3.32}$$

There is an equivalence between a trajectory average and an ensemble average in the above expression (hence, we also use brackets to represent time averages). The factor $3N$ is the number of degrees of freedom of the N particles, and N_c is the number of constrained degrees of freedom. The kinetic energy is not only averaged over a sufficiently long period of time but also averaged with respect to the non-constrained degrees of freedom, $3N - N_c$, in Eq. 3.32. Except for the units introduced by the constant k_B, the temperature may be considered as an energy per degree of freedom. Such an energy represents the equipartition of the kinetic energy due to the homogeneous environment in which the particles are found in space and time for the system in thermodynamic equilibrium. This fact establishes the zero law of thermodynamics with reference to the kinetic energy per degree of freedom. The factor k_B is the Boltzmann constant.

$$k_B = 1.380,649 \times 10^{-23}\ J/K.$$

k_B introduces the proper units to link the energy to the temperature. Apart from units, if the k_B factor is omitted, then the temperature is of similar order to that of the kinetic energy per degree of freedom. In this respect, the k_B factor is sufficiently small ($\sim 10^{-23}$) to connect the macroscopic temperature with the microscopic kinetic energy per degree of freedom. On the other hand, the symbols m_i and $\mathbf{v}_i$ of Eq. 3.32 represent the mass and velocity of the ith particle. Temperature is measured using the reference frame of the center of mass to avoid contributions of the translational kinetic energy of the whole system. The temperature difference between a system and its environment vanishes in time, reaching a *thermal equilibrium* and making the average kinetic energy of the system stationary. The thermal equilibrium should not be confused with the thermodynamic equilibrium

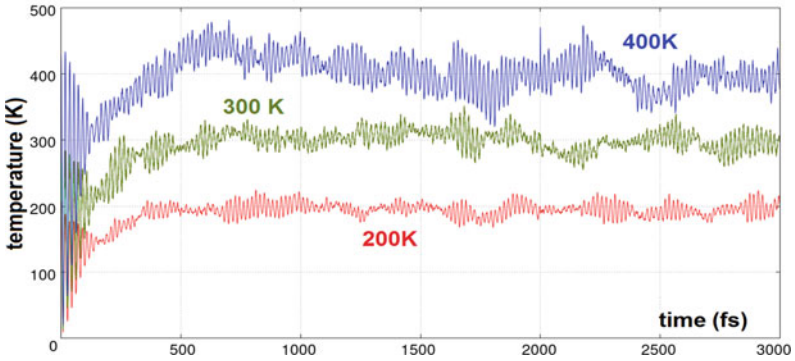


Fig. 3.2 The figure illustrates several temperature plots of a many-particle system with time. The equilibrium temperatures are 200, 300, and 400 K. In a first stage, the system is cold and eventually starts heating up. After a period of time, which depends on the system under investigation, the thermal equilibrium is reached. The thermal fluctuations are relatively small with respect to the equilibrium temperature but increase with the reference temperature

as in this last case, all the mechanical observables are stationary, while in the thermal equilibrium, the temperature is stationary. The state of thermal equilibrium is characterized by a Maxwell-Boltzmann distribution function of particle speeds. In short, expression 3.32 is compatible with our observations and consistent with the zero law of thermodynamics.

In the theory of molecular dynamics, the equation of motion of each particle is integrated at every time step to generate a particle trajectory. The time average of the kinetic energy is calculated from the sequence of values $E_{kin}(t_1), \cdots, E_{kin}(t_n)$ for the system in thermodynamic equilibrium.

$$\langle E_{kin}(t) \rangle = \frac{1}{n} \sum_{k=1}^{n} E_{kin}(t_k) \tag{3.33}$$

From the instantaneous values of the kinetic energy, the temperature may be computed instantaneously, Fig. 3.2.

$$T = \frac{1}{n} \sum_{k=0}^{n} T(t_k) \quad ; \quad T(t_k) = \alpha \sum_{i=1}^{N} m_i \mathbf{v}_i^2(t_k)/2 \tag{3.34}$$

Where $\alpha = 2/(3Nk_B)$. The average kinetic energy of an isolated system is stationary; however, the temperature of a system out of equilibrium is undefined.

Problem 3.4 Consider a gas of N particles with equal masses. Show that $\langle v_{rms}^2(t) \rangle$ is proportional to the temperature, where v_{rms} is the root mean square (rms) speed of the particles.

(continued)

Solution The kinetic energy is $E_{kin} = \sum_i m_i \mathbf{v}_i^2/2$. If the gas contains particles with equal masses, then it is possible to take the mass out of the summation, $E_{kin} = (m/2)\sum_i \mathbf{v}_i^2$. The summation corresponds to the *rms* speed of the particles, $v_{rms}^2 = (\mathbf{v}_1^2 + \mathbf{v}_2^2 + \cdots)/N$. Thus, $E_{kin} = Nmv_{rms}^2/2$. On the other hand, the temperature is proportional to the kinetic energy, Eq. 3.32. Therefore, T is proportional to v_{rms}^2, $T = \alpha\langle E_{kin}(t)\rangle = (N\alpha m/2)\langle v_{rms}^2(t)\rangle$. By solving for v_{rms}^2 in terms of T, we have:

$$\frac{m}{2}\left\langle v_{rms}^2(t)\right\rangle = \frac{3}{2}k_B T$$

In this particular case, we may think on the system temperature as the temperature of a single particle with internal structure, mass m, and speed v_{rms}.

3.11 Temperature Scaling as a Thermostat

In many simulations, it is necessary to stabilize the temperature or change it to a different value. For instance, it is necessary to preset the reference temperature T in simulations with fixed NVT values, where N is number of particles and V is the system volume. In the laboratory, the temperature is regulated using a *thermostat*, which is a device that automatically tunes the temperature by opening and closing switches to permit contact of the system with a heat reservoir. In molecular dynamics, there is no such device, but it may be simulated to control the heat flow and set the required temperature. The simplest implementation of a thermostat in molecular dynamics is based on an approach where the particle velocities are scaled [7]. The scaling of the velocities changes the kinetic energy, which in turn determines the system temperature. In order to implement this thermostat, the scale factor λ is introduced using the fact that the kinetic energy is a homogeneous function of second order in the velocities.[5] It is also convenient to define the old and the new temperatures.

$$T_{new} = \frac{2}{3Nk_B}\left\langle \sum_{i=1}^{N} \frac{1}{2} m_i (\lambda \mathbf{v}_i)^2 \right\rangle \quad ; \quad T_{old} = \frac{2}{3Nk_B}\left\langle \sum_{i=1}^{N} \frac{1}{2} m_i \mathbf{v}_i^2 \right\rangle \tag{3.35}$$

The value of λ is determined from the new and old temperatures of the system in the form $T_{new} = \lambda^2 T_{old}$. The heat flow between the heat reservoir and the particles is regulated by λ. The quantity $\Delta T = T_{new} - T_{old}$ is the temperature change produced by the thermostat.

[5] f is a homogeneous function of order n when it satisfies the expression $f(\alpha\mathbf{v}) = \alpha^n f(\mathbf{v})$, where n is an integral number and $\mathbf{v}$ is the independent variable.

$$\Delta T = \frac{2}{3Nk_B}\left[\left\langle \sum_{i=1}^{N} \frac{1}{2} m_i (\lambda \mathbf{v}_i)^2 \right\rangle - \left\langle \sum_{i=1}^{N} \frac{1}{2} m_i \mathbf{v}_i^2 \right\rangle\right] = (\lambda^2 - 1)\, T_{old}(t)$$

The new temperature T_{new} may be achieved in different ways during the simulation. For example, T_{old} may be changed suddenly by scaling the velocities in a single time step, or it may be gently changed in several time steps of the dynamics by fractions of the scale factor until reaching T_{new}. In the scaling process, it may be also convenient to distinguish particles with high and low speeds, with the purpose of selectively applying the scale factor (or several scale factors) to the fastest and slowest particles of the system, and homogenize the particle speeds. The scale factor may also increase or reduce the magnitudes of the velocities in a predetermined direction solely. Yet the way of introducing the thermostat effects is not genuine enough because there is no consideration of the temperature fluctuations affecting the particle system. In this respect, the scaling of velocities is only used to relatively rapidly achieve the new temperature of reference from an old temperature of the particle system, without employing the Maxwell-Boltzmann distribution function in the assignation of the new particle speeds.

3.12 Temperature Fluctuations

The thermal fluctuations are important to determine the stability of the system, Fig. 3.3. They are evaluated by computing the root mean square deviation ($rmsd$) of the temperature with respect to the reference temperature T_{ref} during a period of m time steps of the simulation.

$$rmsd(m) = \sqrt{\sum_{i=1}^{m} (T_i - T_{ref})^2 / m} \tag{3.36}$$

The temperature at the ith time step is T_i. The $rmsd$ factor, which has units of temperature and depends on the number m of time steps, shows the temperature variations of the system. In a simulation with say $n = 16{,}000$ steps, the $rmsd$ factor

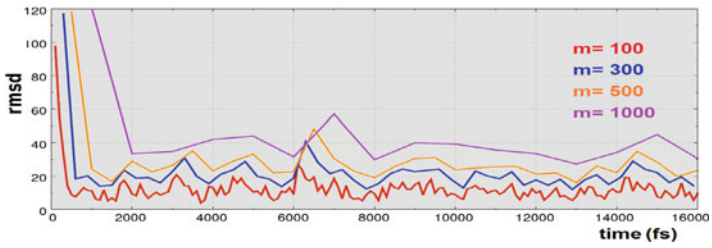


Fig. 3.3 The $rmsd$ factor, Eq. 3.36, is shown over time. The simulation consists of 16,000 time steps with a reference temperature of 200 K. The $rmsd$ factor is calculated with values of m equal to 100, 300, 500, and 1000. The $rmsd$ plots with small values of m show more structure than the $rmsd$ plots with large values of m

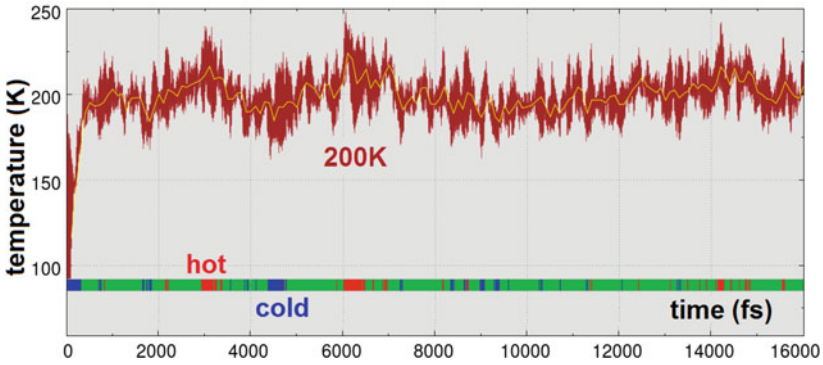


Fig. 3.4 The time evolution of the temperature is illustrated. The reference temperature of the simulation is 200 K. The thin line running along the temperature curve represents the average temperature every 100 steps of the simulation. The color band in the lower part of the plot is the temperature spectrum. It shows the intervals of time where the particle system is hot ($T > T_{ref} + \Delta T$, red color), cold ($T < T_{ref} - \Delta T$, blue color), or in the acceptable temperature range ($T \epsilon [T_{ref} - \Delta T, T_{ref} + \Delta T]$, green color) for a user-defined temperature range Δt

may be computed every $m = 100$ steps ($n \gg m$) and plotted to observe the thermal variations in time. When m is small, the $rmsd$ factor shows structure referent to the partial temperature deviations. By using a large number m, the $rmsd$ factor averages the partial temperature deviations and shows less structure, Fig. 3.3. In the region of thermodynamic equilibrium, the $rmsd$ factor shows relative stability. A complementary way to visualize the temperature deviations consists on plotting a *temperature spectrum* for better analysis of the simulation, Fig. 3.4, or creating a histogram of the particle speeds, which is expected to resemble a Maxwell-Boltzmann distribution function when the system reaches the thermodynamic equilibrium.

Once the system of particles is thermally equilibrated, it stays in an isothermal state, and the sampling of important physical variables may be performed. The sampling may follow the *Nyquist-Shannon theorem*, which indicates that a sampling frequency equal to twice the vibrating frequency of the fastest-moving particles should be considered to obtain accurate information of the physical variables with time. For instance, if we have the behavior of a variable like $y(t) = \cos(2\pi t) + \cos(4\pi t) + \cos(5\pi t)$, then the corresponding angular frequencies are $\omega_1 = 2\pi$, $\omega_2 = 4\pi$, $\omega_3 = 5\pi$. By considering that $\omega = 2\pi f$, the maximum frequency is $f_{max} = f_3 = 5/2$. No further information is acquired by increasing the sampling rate beyond twice the system bandlimit. Based on the theorem, it is possible to acquire important information from the simulations with minimal frequency sampling, $f_{samp} = 2 f_{max}$.[6]

[6] Harry Nyquist in 1928 and Claude Shannon in 1949 proposed the theorem after investigating telegraph communications in a transmission channel. According to the theorem, a bandlimit

3.13 Pressure and Volume

When a system with a large number of particles is confined in a sealed container, the volume of the container determines the confining volume V of the particle system. The volume is linked to the pressure. Inside the container, the particles are constantly interacting and colliding with the walls of the container. The time interval between successive collisions of the particles with the confining walls is sufficiently small from the macroscopic perspective that every wall is essentially subjected to a constant force. The constant force originates a *pressure* P on the wall and is calculated by:

$$P = \mathbf{f} \cdot \hat{n}/A \tag{3.37}$$

The surface area where the force is acting on is A, and the direction perpendicular to the surface area is indicated by the unit vector $\hat{n}$. The definition of pressure 3.37 has statistical character as it depends on the net force and the total surface area. In the description of a movable piston, the pressure is written in terms of the external work by using the identity $dW_{ext} = -\mathbf{f} \cdot \hat{n}\, dl$.

$$P = -\frac{dW_{ext}}{dV} \quad ; \quad dV = Adl \tag{3.38}$$

The quantity dl is the infinitesimal displacement of the wall produced by the force $\mathbf{f}$. By writing $W_{ext} = U - U^{(0)}$, Eq. 3.18, the pressure is expressed in terms of the energy U of the system.

$$P = -\frac{\partial U}{\partial V} \tag{3.39}$$

When the energy U in terms of the volume V is known, it is possible to compute the pressure. In a compression process, the volume changes from an initial volume V_{ini} to a final volume V_{fin}. With these limits, the external work is calculated by integrating the pressure, Eq. 3.38,

$$W_{ext} = -\int_{V_{ini}}^{V_{fin}} P(V)\, dV \tag{3.40}$$

One of the simplest situations occurs when the pressure is constant and the change in volume is ascribed to the temperature. In this case, the external work is $W_{ext} = P(V_{ini} - V_{fin})$. In other occasions, under conditions of constant temperature and constant amount of matter, the pressure is found to be inversely proportional to the volume, and thus, the pressure increases when the volume decreases and vice versa. In addition, if it happens that for any two points the identity $P_1 V_1 = P_2 V_2$

function can be reconstructed when the sampling frequency f_{samp} is equal to or greater than twice the bandlimit f_{max} of the signal, $f_{samp} \geq 2 f_{max}$. The theorem is used to sample continuous (analog) signals and produce reliable discrete (digital) signals.

is satisfied, which is one of the features of an *ideal gas*, then we have $PV = c$ or equivalently $P = c/V$, where c is a constant. This expression is acceptable for systems under moderate pressures and temperatures. The *Boyle-Mariotte law* is obtained by integrating the pressure, Eq. 3.40, with $P = c/V$.

$$W_{ext} = c \ln(V_{ini}/V_{fin}) \tag{3.41}$$

The deviations from the Boyle-Mariotte law occur under extreme conditions. For instance, at high temperature, the relativistic effects are important, and at high pressure and low temperature, the quantum mechanical effects are important. At moderate pressures and temperatures, the molecules behave in classical ways because the interparticle distances are greater than the *thermal de Broglie wavelength*, which is considered a reference parameter to distinguish quantum particles from classical ones.

$$\lambda_T = h/(2\pi m k_B T)^{1/2} \tag{3.42}$$

Problem 3.5 Consider a system of confined diatomic molecules. The following two expressions are equations of state for the system of particles. The first expression deals with molecules and the second one with atoms.

$$E_{mol} = a + b \exp\left(\alpha V_{mol}^{1/3}\right) + c\, V_{mol}^{1/3} \exp\left(\alpha V_{mol}^{1/3}\right) \quad \text{molecule}$$
$$E_{atm} = a' + b' \exp\left(\alpha' V_{atm}^{1/3}\right) + c'\, V_{atm}^{1/3} \exp\left(\alpha' V_{atm}^{1/3}\right) \quad \text{atom}$$

Show their relation through the energy parameters, and demonstrate that the pressures derived from them are the same.

Solution If the expressions are equivalent, then we should find a transformation between the set of parameters a, b, c, α of the molecular case and the set of parameters a', b', c', α' of the atomic case. We depart from the fact that the number of particles and volumes is related by:

$$n_{atm} = 2\, n_{mol} \quad ; \quad V_{atm} = \frac{V}{n_{atm}} \quad ; \quad V_{mol} = \frac{V}{n_{mol}} = \frac{2V}{n_{atm}} = 2\, V_{atm}$$

When E_{conf} is the energy of confinement, a relation between the energies of the system is established.

$$E_{atm} = \frac{E_{conf}}{n_{atm}} \quad ; \quad E_{mol} = \frac{E_{conf}}{n_{mol}} \quad ; \quad E_{mol} = 2\, E_{atm}$$

(continued)

The identities $n_{atm} = 2\,n_{mol}$, $V_{mol} = 2\,V_{atm}$, and $E_{mol} = 2\,E_{atm}$ are used to determine a connection between the equations of state. We start by writing the energy of the molecules E_{mol} in terms of the volume V_{atm}.

$$E_{mol} = a + b\,\exp\left(2^{1/3}\alpha V_{atm}^{1/3}\right) + 2^{1/3}\,c\,V_{atm}^{1/3}\,\exp\left(2^{1/3}\alpha V_{atm}^{1/3}\right)$$

If we consider that $2\,E_{atm} = E_{mol}$, then the coefficients $a' = a/2$, $b' = b/2$, $c' = 2^{1/3}(c/2)$ and $\alpha' = 2^{1/3}\alpha$ may be defined. In this way, we recover E_{atm} from E_{mol}.

$$E_{atm} = a' + b'\exp\left(\alpha' V_{atm}^{1/3}\right) + c' V_{atm}^{1/3}\,\exp\left(\alpha' V_{atm}^{1/3}\right)$$

The pressure is calculated as:

$$P_{mol} = -\frac{\partial E_{mol}}{\partial V_{mol}} = -\frac{\partial(2E_{atm})}{\partial(2V_{atm})} = P_{atm}$$

The pressures to which the molecules and atoms are subjected to are the same.

3.14 The Virial and the Equation of State

The purpose of this section is to deduce a relation between the kinetic energy of a system of particles and the forces acting on the particles. Such a relation is the *virial theorem*. The virial theorem allows to formulate an equation of state of the system in terms of the pressure, volume, and temperature. The departure point is from the scalar quantity A.

$$A = \textstyle\sum_i m_i \mathbf{v}_i \cdot \mathbf{r}_i \tag{3.43}$$

The virial applies to systems in equilibrium and out of thermodynamic equilibrium since A depends on the basic variables of position and velocity. It is necessary to consider the time derivative of A to obtain the forces and kinetic energy of the particles.

$$\frac{d}{dt}A = \textstyle\sum_i m_i \mathbf{a}_i \cdot \mathbf{r}_i + \sum_i m_i v_i^2$$

It is convenient to consider the forces over the ith particle. This particle is subjected to the internal force $\mathbf{f}_{ij}$ from the jth particle and the external force $\mathbf{f}_i$ as well. If we consider that the j particle is in turn subjected to the force $\mathbf{f}_{ij}$ of particle i, then we have terms like $\mathbf{f}_{ij} \cdot \mathbf{r}_i + \mathbf{f}_{ji} \cdot \mathbf{r}_j$, which can be written as $\mathbf{f}_{ij} \cdot (\mathbf{r}_i - \mathbf{r}_j)$ after using the third Newton's law. The second contribution, $\sum m_i v_i^2$, is proportional to the kinetic

energy E_{kin}. The expression becomes:

$$\frac{d}{dt}A = 2E_{kin} + \sum_i \mathbf{f}_i \cdot \mathbf{r}_i + \sum_{i,j} \mathbf{f}_{ij} \cdot \mathbf{r}_{ij} \quad ; \quad \mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$$

We are interested in the time averages of every term. The average term of the left is zero.

$$\langle E_{kin} \rangle = -\tfrac{1}{2}\langle \sum_i \mathbf{f}_i \cdot \mathbf{r}_i \rangle - \tfrac{1}{2}\langle \sum_{i,j} \mathbf{f}_{ij} \cdot \mathbf{r}_{ij} \rangle \tag{3.44}$$

Problem 3.6 The physical observable A is a bounded function with time. Show that the time average of dA/dt is zero.

Solution The observable A does not grow unlimited with time because it is a bounded function. The time average of the derivative dA/dt demands integration over time.

$$\langle \frac{d}{dt}A \rangle = \lim_{\tau\to\infty} \frac{1}{\tau} \int_0^\tau \frac{d}{dt} A \, dt = \lim_{\tau\to\infty} (A - A_0)/\tau = 0$$

The result is zero because τ grows without limits, $\tau \to \infty$.

Expression 3.44 is one of the several versions of the virial theorem. It relates the kinetic energy to the external and internal forces. The following goal is to use the virial theorem to establish the equation of state of the particles. The *equation of state* is important as it relates the pressure P, volume V, and temperature T of the particle system. The parameters P, V, and T are described using the positions and velocities of the particles, and therefore, are related to each other. In order to formulate the equation of state, the particles are assumed to be confined in a cubic box of length ℓ, with one of the vertex located at the origin and three edges resting on the x, y, and z axes. The six faces forming the container are identified with the planes, Fig. 3.5.

$$\begin{array}{lll} ABGH : (x, y, 0) & ; & BCFG : (\ell, y, z) \\ ABCD : (x, 0, z) & ; & EFGH : (x, \ell, z) \\ ADEH : (0, y, z) & ; & CDEF : (x, y, \ell) \end{array} \tag{3.45}$$

When a particle collides with a wall, the wall exerts a force on the particle in the perpendicular direction to the wall surface. The contribution to the virial of a particle colliding with surface $ABGH$ is null because $\mathbf{f} \cdot \mathbf{r} = -(0, 0, f) \cdot (x, y, 0) = 0$. The negative sign comes from a force pointing in opposite direction to the particle motion. Similar results are obtained for the surfaces $ABCD$ and $ADEH$. The

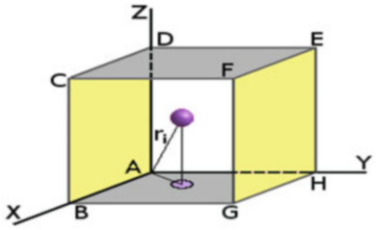


Fig. 3.5 The figure shows a particle confined in a cubic box. The position vector of the particle is $\mathbf{r}_i$. When the particle reaches a wall, say the $ABGH$ surface, the position of the particle is $\mathbf{r}_i = (x, y, 0)$, and the force exerted by the wall on the particle is $\mathbf{f}_i = (0, 0, -f)$

contributions to the virial of the particles colliding on the other surfaces are not zero. For instance, $\mathbf{f} \cdot \mathbf{r} = -(f, 0, 0) \cdot (\ell, y, z) = -f\ell$ for the $BCFG$ surface.

The summation of the forces exerted by the walls $BCFG$, $EFGH$ and $CDEF$ is $-3f\ell = -3PV$, where the identities $P = f/A$, and $V = \ell^3 = A\ell$ were used. The result is inserted in the term of the external forces in the viral expression. By solving for PV, we have:

$$PV = \tfrac{2}{3}\langle E_{kin}\rangle + \tfrac{1}{3}\langle \textstyle\sum_{i,j} \mathbf{f}_{ij} \cdot \mathbf{r}_{ij}\rangle \tag{3.46}$$

When the system is in thermodynamic equilibrium, the average kinetic energy is proportional to the temperature T of the system, Eq. 3.32.

$$PV = Nk_BT + \tfrac{1}{3}\langle \textstyle\sum_{i,j} \mathbf{f}_{ij} \cdot \mathbf{r}_{ij}\rangle \tag{3.47}$$

This is the equation of state of the system. It establishes a relation between P, V, and T, where the internal forces also play an important role. When the contribution of the forces is negative (positive), the pressure is reduced (increased). When the contribution of the forces is sufficiently small, the equation of state of an ideal gas is obtained.

$$PV = Nk_BT \tag{3.48}$$

Several important results are deduced from the *ideal gas law* when either the temperature, pressure, or volume is constant. For two points, (P_1, V_1, T_1) and (P_2, V_2, T_2), of the particle system, under the consideration that Nk_B is constant, we establish different laws:

$$\begin{array}{lll} T_1 = T_2 \Rightarrow & P_1V_1 = P_2V_2 & \text{Boyle's law} \\ V_1 = V_2 \Rightarrow & P_1/T_1 = P_2/T_2 & \text{Gay-Lussac's law} \\ P_1 = P_2 \Rightarrow & V_1/T_1 = V_2/T_2 & \text{Charles' law} \end{array} \tag{3.49}$$

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Mechanical Descriptors

4

Many particle properties change their values in space and time in molecular dynamics. In this regard, it is of interest to introduce additional descriptors for the analysis and characterization of the particle system in the simulations. We discuss the caloric curve from an energetic perspective. A way to evaluate the fluctuations and deviations of the particle positions with respect to reference points is presented with the purpose to determine the changes in the orientations of the molecules, assess the structural changes, and help disclose phase transitions. The correlation functions are discussed since they are tools to reveal the interconnection of physical variables in space and time.

4.1 Caloric Curve

When the total energy of the particle system in terms of the temperature is determined, we obtain the *caloric curve*. The total energy E is given in terms of the kinetic energy and potential energy.

$$\langle E(T)\rangle = \langle E_k\rangle + \langle E_p\rangle$$

The brackets represent time averages of E_k and E_p over the simulation with temperature T. The phase transitions may be monitored by means of the caloric curve. A direct change of $\langle E(T)\rangle$ implies a *first-order transition*, and a change in the derivative of $\langle E(T)\rangle$ corresponds to a *second-order transition*. At low temperatures, the system is in the solid-like region, with a harmonic behavior of the particles. When the temperature increases, the total energy increases linearly, and as the temperature is further increased, the particle system reaches a transition region where the slope of $\langle E(T)\rangle$ changes. The plot of $\langle E(T)\rangle$ vs T shows a leap because the liquid phase grows at the expense of the solid one plus the actions of the external factors. The leap corresponds to the melting point (a first-order transition), where the solid transforms onto a liquid due to the fusion heat. There may be

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an overestimation of the *melting temperature* due to a response lag (or hysteresis effect) of the interacting particles. In the liquid-like region, the caloric curve keeps increasing by increasing the temperature.

4.2 Interatomic Distance Fluctuations

An additional descriptor of interest is δL, which defines the *interatomic distance fluctuations*.

$$\delta L = \frac{2}{N(N-1)} \sum_{i>j}^{N} \langle r_{ij} \rangle^{-1} \sqrt{\langle r_{ij}^2 \rangle - \langle r_{ij} \rangle^2} \tag{4.1}$$

The quantity r_{ij} is the distance between the ith and jth particles, the number of atoms in the molecule is N, and the brackets indicate time averages. By assuming n steps in the molecular dynamics simulation, the averages are numerically computed in the form:

$$\left\langle r_{ij}^k \right\rangle = \left[r_{ij}^k(t_1) + r_{ij}^k(t_2) + \cdots + r_{ij}^k(t_n) \right] / n \quad ; \quad k = 1, 2 \tag{4.2}$$

The quantity δL of Eq. 4.1 gives the fluctuations of the interatomic distances. It is also recognized as the *Lindemann index* [1]. It is important to: (i) elucidate the disorder provoked by the temperature effects in molecules, like the proteins, (ii) determine the moments of melting of a cluster or solid, and (iii) establish the phase transition of a system of particles. The parameter δL grows quasi-linearly with an increase in temperature in the solid-like region. However, in the moments of the solid-liquid transition, the particles acquire high kinetic energy to overcome the potential energy barriers, and the interatomic distances increase rapidly, producing a sharp growth of δL. The parameter δL is stabilized after some time in the liquid region.

Some specific values of the Lindemann index have been proposed as threshold values to determine the solid-liquid phase transitions. The value $\delta L \simeq 0.1$ puts in evidence the fusion of some macroscopic solids; nevertheless, that value does not necessarily apply to finite size systems like the atomic clusters. A practical way to establish the melting of a system of atoms is to consider values of δL in the interval $[0.20, 0.25]$; this interval was determined experimentally in crystals. The melting is supposed to occur at a temperature T where the average amplitude $\sqrt{\langle A^2 \rangle}$ of the particle vibrations is greater than the product $\langle r_{ij} \rangle \delta L/2$.

$$\text{if} \quad \sqrt{\langle A^2 \rangle} > (\langle r_{ij} \rangle \, \delta L)/2 \quad \Rightarrow \quad \text{melting of the particle system} \tag{4.3}$$

The Lindemann criterion is equally applied to a double DNA helix when, for instance, it is unfolded into separate strands at high temperature, resembling a type of melting process.

4.3 Root Mean Square Deviation of Positions

Another important parameter to describe the structural changes of the molecules is the *root mean square deviation* ($rmsd$) of positions. The $rmsd$ evaluates the statistical deviation of a molecular conformation at the instant of time t from a reference molecular conformation taken at the instant of time t_0.

$$rmsd(t) = \left[\frac{1}{N}\sum_{i=1}^{N}[\mathbf{r}_i(t) - \mathbf{r}_i(t_0)]^2\right]^{1/2} \tag{4.4}$$

The vector $\mathbf{r}_i(t)$ is the position of the ith particle at time t, and the reference position of this particle is $\mathbf{r}_i(t_0)$. The summation runs over the number of particles in the system, N. The $rmsd$ is equally applied to other quantities, such as the atomic velocities, forces, etc. The $rmsd$ is used to estimate the *structural similarity* of different molecules, which should be translated and rotated to maximize the position coincidence. In the case of proteins, the α carbons may be considered in the evaluation of the structural similarity, because they form the protein backbone. The $rmsd$ is small for atoms in the solid phase; it rises linearly for atoms in the liquid phase with time and is stabilized after a long period of time.

4.4 Orientational Order Parameter

In molecular dynamics, the molecular structures continuously change. The structural changes may be evaluated using the *orientational order parameter*. It has the form of a quadrupole expression and employs the vector direction between atom pairs to define the orientational order. The unit direction vector $\hat{u}_{ij}$ between the ith and jth particles is:

$$\hat{u}_{ij}(t) = \frac{\mathbf{r}_i(t)-\mathbf{r}_j(t)}{|\mathbf{r}_i(t)-\mathbf{r}_j(t)|}$$

The vector $\hat{u}_{ij}(t)$ defines the axis of the atom pair at the instant of time t. The reference vector $\hat{n}_{ij}(t)$ is required to gauge the orientation of $\hat{u}_{ij}$. The reference vector $\hat{n}_{ij}$ is defined in terms of the time average of $\hat{u}_{ij}$.

$$\hat{n}_{ij} = \langle\hat{u}_{ij}(t)\rangle \,/\, |\langle\hat{u}_{ij}(t)\rangle|$$

The orientational order parameter for the pair ij of atoms with the form of a quadrupole expression is [2]:

$$O_{ij} = \frac{3}{2}\left\langle[\hat{u}_{ij}(t)\cdot\hat{n}_{ij}]^2 - \frac{1}{3}\right\rangle \tag{4.5}$$

The parameter O_{ij} involves the scalar product of the vector of interest $\hat{u}_{ij}$ with the reference vector $\hat{n}_{ij}$. The coefficients in the definition of O_{ij} introduce the

normalization factor. In a molecule, there may be many atom pairs, and it is convenient to consider them by summing over all atom pairs, N_p.

$$O = \langle O_{ij} \rangle = \frac{1}{N_p} \sum_{i \neq j} O_{ij} \tag{4.6}$$

When the structural changes of the molecule are small, the vectors $\hat{u}_{ij}$ remain essentially the same, and $\langle (\hat{u}_{ij} \cdot \hat{n}_{ij})^2 \rangle = 1$, resulting on a maximal orientational order, $O = 1$. When the pairs of atoms point with equal probability in every direction, the result $\langle (\hat{u}_{ij} \cdot \hat{n}_{ij})^2 \rangle = 1/3$ is obtained, producing a minimal value, $O = 0$, characteristic of a low orientational order of the particles. When all the vectors u_{ij} point in the same direction ($\hat{n}_{ij} = \hat{n}$), the *nematic order* parameter is reproduced by Eq. 4.6 [3].

4.5 Probability Distribution Functions

Suppose a system of N particles in the fluid state and confined in a volume V. The *distribution function* of the N particles is ρ_N, which depends on the coordinates $(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$. The function ρ_N is normalized to unity.

$$\int\int \rho_N(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)\, d\mathbf{r}_1 d\mathbf{r}_2 \cdots d\mathbf{r}_N = 1 \tag{4.7}$$

The *reduced distribution function* of order k is obtained from ρ_N by integration.

$$\rho_k(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_k) = \frac{N!}{(N-k)!} \int\int \rho_N(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)\, d\mathbf{r}_{k+1} d\mathbf{r}_{k+2} \cdots d\mathbf{r}_N \tag{4.8}$$

In particular, the *one-body distribution function* is obtained after integration over all variables, except one of them.

$$\rho_1(\mathbf{r}) = N \int\int \rho_N(\mathbf{r}, \mathbf{r}_2, \cdots, \mathbf{r}_N)\, d\mathbf{r}_2 \cdots d\mathbf{r}_N \tag{4.9}$$

The one-body distribution function is the fluid density, $\rho_1(\mathbf{r}) = \rho$, which is constant. The *two-body distribution function* is similarly computed.

$$\rho_2(\mathbf{r}_1, \mathbf{r}_2) = N(N-1) \int\int \rho_N(\mathbf{r}_1, \cdots, \mathbf{r}_N)\, d\mathbf{r}_3 \cdots d\mathbf{r}_N \tag{4.10}$$

The two-body distribution function in terms of the one-body distribution function is written as:

$$\rho_2(\mathbf{r}_1, \mathbf{r}_2) = \rho_1^2\, g(\mathbf{r}) \tag{4.11}$$

We have implicitly defined the *correlation function*, $g(\mathbf{r})$, in terms of the distance vector $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$. The correlation function gives the probability to find a number of particles in volume $d\mathbf{r}$ around position $\mathbf{r}$ when a different particle, say

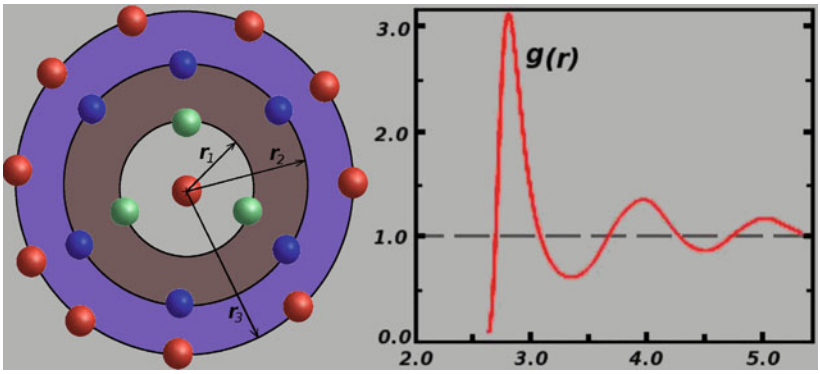


Fig. 4.1 An arbitrary particle of the system is taken as a reference point to make interparticle-distance measurements. The process is repeated on all the particles to have many reference points. By plotting a histogram of the whole set of distances, the average radial packing of particles around any particle is determined. The function describing the packing is the correlation distribution function, $g(r)$. It is interpreted as the probability to find a number of particles in volume $d\mathbf{r}$ around position $\mathbf{r}$ when a different particle is located at $\mathbf{r} = 0$. The correlation distribution function frequently exhibits peaks, indicating the local particle density and neighbor abundance

a central reference particle, is located at $\mathbf{r} = 0$, Fig. 4.1. The correlation function characterizes the structure of the system as either a solid, amorphous solid, liquid, or gas, because g provides the spatial distribution of the particles, which in turn depends through the internuclear distances r_{ij} on the temperature. For a crystal, the correlation function g exhibits separate peaks not connected to each other (like a half-wave rectified electric signal) and is related to the structure factor of the crystal. For a liquid, g reveals a curve with structure like that in Fig. 4.1. For a gas, g shows a continuous curve with almost no structure (like a DC electric signal with small ripples).

When the correlation function is known, it helps describe the interactions of *two-body potentials*. Suppose that $U = \sum_{i,j} u(\mathbf{r}_i, \mathbf{r}_j)$, $i < j$, describes the interactions between particle pairs. The average interaction energy is computed by:

$$\langle U \rangle = \sum_{\substack{i,j \\ i<j}} \iint u(\mathbf{r}_i, \mathbf{r}_i)\, \rho_N(\mathbf{r}_1 \cdots \mathbf{r}_N)\, d\mathbf{r}_1 \cdots d\mathbf{r}_N \tag{4.12}$$

The integration variables are dummy variables (they play the role of placeholders solely), and the summations can be reduced to a single term multiplied by the binomial coefficient $N(N-1)/2$, which accounts for the total number of pairs without repeating any of them.

$$\begin{aligned} \langle U \rangle &= \tfrac{N(N-1)}{2} \iint d\mathbf{r}_1 d\mathbf{r}_2\, u(\mathbf{r}_1, \mathbf{r}_2) \iint \rho_N(\mathbf{r}_1 \cdots \mathbf{r}_N)\, d\mathbf{r}_3 \cdots d\mathbf{r}_N \\ &= \tfrac{1}{2} \iint \rho_2(\mathbf{r}_1, \mathbf{r}_2)\, u(\mathbf{r}_1, \mathbf{r}_2)\, d\mathbf{r}_1 d\mathbf{r}_2 \end{aligned}$$

The two-body distribution function may be written in terms of the correlation function as $\rho^2 g$. If the interaction potential u depends on the particle distance $r = |\mathbf{r}_2 - \mathbf{r}_1|$, then it is possible to perform an additional integration to finally determine the average potential energy.

$$\begin{aligned}\langle U\rangle &= \tfrac{1}{2}\iint d\mathbf{r}_1 d\mathbf{r}_2\, \rho^2 g(|\mathbf{r}_2-\mathbf{r}_1|)\, u(|\mathbf{r}_2-\mathbf{r}_1|) \\ &= \tfrac{N^2}{2V}\int_0^\infty g(\mathbf{r})\, u(\mathbf{r})\, d\mathbf{r} = \tfrac{2\pi N^2}{V}\int_0^\infty g(r)\, u(r)\, r^2 dr\end{aligned} \tag{4.13}$$

In order to investigate the asymptotic limit of the correlation function, let us again consider N particles confined in volume V. The fraction of particles $\langle n_{r,r+dr}\rangle$ in the volume element dV, with respect to a central reference particle positioned at a distance r, is:

$$\langle n_{r,r+dr}\rangle = \tfrac{N}{V}\, g(r) dV$$

The volume element dV, keeping linear terms in dr, is:

$$dV = \tfrac{4\pi}{3}[(r+dr)^3 - r^3] = \tfrac{4\pi}{3}(3r^2 dr + 3r dr^2 + dr^3) \approx 4\pi r^2 dr$$

By using the result $dV = 4\pi r^2 dr$, we observe $g(r) = (V/N)\langle n_{r,r+dr}\rangle / dV$. When $\langle n_{r,r+dr}\rangle \to N$, we have $dV \to V$, and therefore $g(r) = 1$. In extended systems $g(r)$ tends to 1 as r increases [4]. In atomic clusters, $g(r)$ tends zero as r increases because clusters have finite sizes.

Problem 4.1 Write the *virial equation of state* $PV = Nk_BT + \langle \sum_{i,j=1}^{N} \mathbf{r}_{ij} \cdot \mathbf{f}_{ij}\rangle/3$, Eq. 3.47, in terms of the correlation distribution function $g(r)$, and $U = \sum_{i,j} u(\mathbf{r}_i, \mathbf{r}_j)$, $i < j$.

Solution By considering conservative forces $\sum_i \mathbf{r}_i \cdot \nabla_i U = \sum_{i<j} r_{ij}\, \partial u(r_{ij})/\partial r_{ij}$, and following similar steps in the deduction of $\langle U\rangle$, we have:

$$\begin{aligned}PV &= Nk_BT - \tfrac{1}{3}\sum_{\substack{i,j\\ i<j}} \iint r_{ij}\, \tfrac{\partial u(r_{ij})}{\partial r_{ij}}\, \rho_N(\mathbf{r}_1 \cdots \mathbf{r}_N)\, d\mathbf{r}_1 \cdots\cdots d\mathbf{r}_N \\ &= Nk_BT - \tfrac{1}{3}\left(\tfrac{N^2}{2V}\right)\int g(r)\, r\, \tfrac{\partial u(r)}{\partial r}\, (4\pi r^2)\, dr \\ &= Nk_BT - \tfrac{2\pi N^2}{3V}\int_0^\infty g(r)\, r^3\, \tfrac{\partial u(r)}{\partial r}\, dr\end{aligned}$$

4.6 Correlation Functions

The particles interact and move in a dynamic system, and the physical variables continuously change their magnitudes in space and time. In many instances, two or more physical variables may be correlated, and it is difficult to observe their relation due to the large number of particles in an apparent chaotic motion. In this respect, it is important to discuss the way to evaluate the correlation of the physical variables and classify them according to their dependencies.

The correlation of the physical variables may be of spatial or temporal nature. The *time correlation function* C_{AB} gives the interconnection of the physical variables **A** and **B** by evaluating **A** at the instant of time t_1 and **B** at the instant of time t_2 and computing the average of their scalar product.

$$C_{AB}(t_1, t_2) = \frac{1}{3} \langle \mathbf{A}(t_1) \cdot \mathbf{B}(t_2) \rangle \tag{4.14}$$

The time may be replaced by a spatial coordinate to define a *spatial correlation function*. The scalar product in the correlation function 4.14 is written as:

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}(t_1)| \, |\mathbf{B}(t_2)| \, \cos\theta$$

The angle between vectors is θ. The scalar product is positive when both vectors point in the same direction and negative when they point in opposite directions. Thus, the correlation function considers the positive and negative values of $\mathbf{A}(t_1) \cdot \mathbf{B}(t_2)$ at different instants of time in the simulation. To some extent, the correlation function establishes the degree of similarity of the physical variable **A** with the physical variable **B**, with some time delay between such variables included. The variables **A** and **B** are dissimilar when there is no privileged orientation or all the orientations between them are equally possible in an isotropic environment. In this case, the value of variable **A** implies no value of variable **B** and vice versa. In general, the correlations fade away with many collisions of the particles, and in this connection, the correlation function gives the time persistence of the $A \leftrightarrow B$ relation before it vanishes.

The brackets of Eq. 4.14 indicate either an ensemble average or time average. When we deal with an ensemble average, Eq. 4.14 is [5]:

$$C_{AB}(t_1, t_2) = \langle A(t_1) B(t_2) \rangle = \int_\Gamma f(\Gamma) \, A(\Gamma; t_1) \, B^*(\Gamma; t_2) \, d\Gamma$$
$$\Gamma = q_1, \cdots, q_{3N}, p_1, \cdots, p_{3N} \quad ; \quad d\Gamma = dq_1 \cdots dq_{3N} \, dp_1 \cdots dp_{3N} \tag{4.15}$$

The joint probability distribution function is f, and the physical observables are A and B. The asterisk on B signals the complex conjugate of that variable. The distribution function defines the probability of finding the observables A and B with values in $[A, A + dA]$ and $[B, B + dB]$. The integral is performed over all the phase space variables, when the system is in thermodynamic equilibrium. Equation 4.15 may be compared with an ensemble average over a single variable.

$$\langle A(t)\rangle = \int_\Gamma A(P)\, f(P)\, d\Gamma \tag{4.16}$$

The phase-space point $P = (q_1, \cdots, q_n, p_1, \cdots, p_n)$, where q_i and p_i are *generalized coordinates* of particle i, determines the microscopic state of the classical system. Function f is the canonical distribution function for the system with fixed number of particles, volume, and subjected to a given temperature. The product $f(P)\, d\Gamma = f(q_1, \cdots, q_n, p_1, \cdots, p_n)\, dq_1 \cdots dq_n dp_1 \cdots dp_n$ gives the probability to find particle i with coordinates between q_i and $q_i + dq_i$ and momentum between $p_i + dp_i$, where $i = 1, \cdots, n$ [6]. The correlation function 4.15 may be considered a generalization of the ensemble average 4.16 as Eq. 4.15 contains a pair of observables with their respective times. When the system is ergodic, the ensemble averages $\langle A\rangle$ and $C_{AB}(t_1, t_2)$ may be replaced by time averages obtained from a molecular dynamics simulation.

$$\begin{aligned} \bar{A} &= \lim_{T\to\infty} \tfrac{1}{T} \int_0^T A(t)\, dt \\ C_{AB}(t_1, t_2) &= \overline{A(t_1)B(t_2)} = \lim_{T\to\infty} \tfrac{1}{T} \int_0^T A(t + t_1)B^*(t + t_2)\, dt \end{aligned} \tag{4.17}$$

The function C_{AB} is the outcome of averaging the product AB over the time interval of thermodynamic equilibrium. The averaging process considers possible retarded effects between A and B by dealing with times t_1 and t_2. The difference $t_2 - t_1$ is a time delay in the response of every particle to the forces produced by the other particles, which is typical of crowded environments. The correlation function may exhibit a *cause-effect relationship* between the variables A and B. Yet care should be exercised because the correlation of variables does not necessarily imply that one variable causes an effect on the other.

By using the correlation functions, it is possible to tell when the motion of a particle has a random aspect. Thus, some of the important applications of correlation functions are given on the characterization of particle dynamics in environments with fluctuations in the density, temperature, pressure, etc. For instance, consider a body in a fluid. By supposition, it has bigger size than the sizes exhibited by the fluid particles. We barely have predictive capacity of the body's trajectory because the body is constantly kicked by a myriad of surrounding fluid particles, making the body describe a random trajectory. Such a trajectory has little meaning to us. In this regard, a statistical characterization of both the environmental forces acting on the body and the body's trajectories is convenient.

The correlation function C_{AB} is also distinguished with the name *cross-correlation function*. When the variables A and B in the cross-correlation function are the same, that function is called *autocorrelation function* and is denoted with a single index, C_A. The autocorrelation function is the self-correlation of A at different instants of time or different spatial positions. In general, transport quantities such as the diffusion, viscosity, and thermal conductivity of a fluid have

their own representations in autocorrelation functions. For instance, the diffusion of a particle is computed after evaluating the autocorrelation function of velocities. Such autocorrelation functions are able to characterize stochastic processes in stationary states. The general scheme that establishes the relations between the autocorrelation functions and the transport quantities is the *Green-Kubo approach*.

The autocorrelation function of variable A in the quantum mechanical case is:

$$C_{AB}(t_1, t_2) = \sum_n p_n \langle n|A(t_1)A(t_2)|n\rangle \quad ; \quad p_n = Z^{-1} \exp[-E_n/(k_B T)] \tag{4.18}$$

E_n is the system energy in the nth state, T the equilibrium temperature, k_B the Boltzmann constant, p_n the canonical probability function, and $Z = \sum_n \exp[-E_n/(k_B T)]$ the normalization factor.

Problem 4.2 The equation of motion of the harmonic oscillator is $m\ddot{q} = -kq$, with solution $q(t) = q(0)\cos(\omega t)$, and $\omega^2 = k/m$. The distribution function of the harmonic oscillator is $f = \exp[-E_{tot}/(k_B T)]/Z$, where E_{tot} is the total energy of the oscillator and Z is the normalization constant. Obtain the autocorrelation function $\langle q(t)q(0)\rangle$.

Solution The insertion of the energy in f gives $f = Z^{-1} \exp[-(p^2/m + m\omega^2 q^2)/(2k_B T)]$. From the solution of the equation of motion, the average is $\langle q(t)q(0)\rangle = \langle q^2(0)\rangle \cos(\omega t)$. We compute the value of $\langle q^2\rangle$ using the probability function f.

$$\langle q^2\rangle = \tfrac{1}{Z} \int_{-\infty}^{\infty} \exp(-\tfrac{p^2}{2mk_B T}) dp \int_{-\infty}^{\infty} q^2 \exp(-\alpha q^2) dq \; ; \; \alpha = m\omega^2/(2k_B T)$$

$$= \int_{-\infty}^{\infty} q^2 \exp(-\alpha q^2) dq \; / \int_{-\infty}^{\infty} \exp(-\alpha q^2) dq$$

We use the results $\int_{-\infty}^{\infty} \exp(-\alpha q^2) dq = \sqrt{\pi/\alpha}$, and $\int_{-\infty}^{\infty} q^2 \exp(-\alpha q^2)\, dq = 1/(2\alpha)\sqrt{\pi/\alpha}$ to simplify $\langle q^2\rangle$.

$$\langle q^2\rangle = k_B T/(m\omega^2)$$

In this connection, the autocorrelation function of the position in time is:

$$\langle q(t)q(0)\rangle = \langle q^2(0)\rangle \cos(\omega t) = \tfrac{k_B T}{m\omega^2} \cos(\omega t)$$

4.7 Properties of Correlation Functions

The correlation functions satisfy several properties that may be used for the better understanding of particle systems. The properties are based on the fact that the particle system is in an stationary state, and thus, we are able to apply probability distribution functions in the evaluation of ensemble averages. Yet according to the ergodic hypothesis, the correlation functions may be equivalently computed from trajectory averages in molecular dynamics since the ensemble averages and the trajectory averages are equivalent under conditions of thermodynamic equilibrium [7].

One of the properties of the correlation functions involves *conjugate symmetry*. If we consider the change of variables $x = t - \tau$ in:

$$C_{AB}(\tau) = \int_{-\infty}^{\infty} A(t)B^*(t-\tau)dt \tag{4.19}$$

we obtain $C^*_{AB}(\tau) = \int_{-\infty}^{\infty} B(x)A^*(x+\tau)dx$. By replacing τ by $-\tau$, we reach $C^*_{AB}(-\tau) = \int_{-\infty}^{\infty} B(x)\ A^*(x-\tau)dx$. The integral corresponds to the definition $C_{BA}(\tau)$. Hence, the conjugate symmetry indicates that:

$$C^*_{AB}(-\tau) = C_{BA}(\tau) \tag{4.20}$$

When $A = B$, the autocorrelation function is even in time, $C_A(\tau) = C_A(-\tau)$. Another property is the transposition of the independent variables in the expression of the correlation function.

$$C_{AB}(t_1, t_2) = C_{AB}(\Delta t, 0) \quad ; \quad \Delta t = t_1 - t_2 \tag{4.21}$$

The correlation function is also an even function in time, $C_{AB}(t, 0) = C_{AB}(0, -t)$. The functions A and B are orthogonal when $C_{AB}(\tau = 0) = \int_{-\infty}^{\infty} A(t)B^*(t)dt = 0$. The correlation function should not be confused with the *convolution function*. The convolution equally represents a mathematical operation on a pair of functions, specifically, on the product AB. It is defined by (the discrete version is also shown):

$$A(t) * B(t) = \int_{-\infty}^{\infty} A(\tau - t)B(t)\,dt \quad ; \quad F * I(x) = \sum_{i=-N}^{N} F(i)I(x-i) \tag{4.22}$$

There is a sign flip in an integration variable. In addition to the properties of the correlation function, the autocorrelation function also satisfies some properties. It is always equal or greater than zero when it is evaluated at equal times, $t_1 = t_2 = t$, commutative in the times t_1 and t_2, constant for absolute times, and invariant with respect to time shifts.

$$\begin{aligned} &\langle A(t)A(t)\rangle = \langle A^2(t)\rangle \geq 0\ ; \qquad \langle A(t_1)A(t_2)\rangle = \langle A(t_2)A(t_1)\rangle \\ &\langle A(0)A(0)\rangle = \text{constant} \quad ; \langle A(t_1)A(t_2)\rangle = \langle A(t_1+\Delta t)A(t_2+\Delta t)\rangle \end{aligned} \tag{4.23}$$

A time shift, Δt, may be moved from an argument to the other with a change of signs.

$$\begin{aligned} C_A(t, t+\Delta t) &= \langle A(t)A(t+\Delta t)\rangle = \langle A(t-\Delta t)A(t)\rangle = C_A(t-\Delta t, t) \\ C_A(t_1, t_2) &= C_A(\Delta t, 0) \quad ; \quad \Delta t = t_1 - t_2 \end{aligned} \tag{4.24}$$

The autocorrelation function is stationary due to the dependence on relative times. After a long time, $\Delta t \to \infty$, the particles become uncorrelated due to the continuous collisions with many other particles. The time to reach that asymptotic limit is characteristic of the particle system and its environment. The asymptotic form of the autocorrelation function is written as the product of the individual averages of the physical variable.

$$C_A(t_1, t_2) = \langle A(t_1)A(t_2)\rangle \Big|_{\Delta t \to \infty} = \langle A(t_1)\rangle \langle A(t_2)\rangle = \langle A\rangle^2 \tag{4.25}$$

When the fluctuations of observable A with respect to its average value $\langle A\rangle$ are required, the correlation function of the deviation $\delta A = A - \langle A\rangle$ may be computed.

$$\begin{aligned} \langle \delta A(t_1)\delta A(t_2)\rangle &= \lim_{T\to\infty} \tfrac{1}{T} \int_0^T [A(t_1+t) - \langle A\rangle]\,[A(t_2+t) - \langle A\rangle]\, dt \\ &= \langle A(t_1)A(t_2)\rangle - \langle A\rangle^2 \end{aligned} \tag{4.26}$$

From Eqs. 4.25 and 4.26, the fluctuations of A at different instants of time are uncorrelated in the asymptotic time limit, $C_{\delta A}(\Delta t \to \infty) = 0$. When the times t_1 and t_2 are zero, the *variance* σ is obtained as a special case of the autocorrelation function of δA.

$$\sigma^2 = \langle \delta A(0)\, \delta A(0)\rangle = \langle A^2(0)\rangle - \langle A\rangle^2 \tag{4.27}$$

In many instances, we may be interested in analyzing a specific direction of a vector variable, for example, a particular direction of the dipole moment. Under such a situation, the projection of variable $\mathbf{A}$ onto the direction of interest, say $\hat{\mathbf{n}}$, is:

$$C_A(t) = \langle (\mathbf{A}(0)\cdot\hat{\mathbf{n}})\,(\mathbf{A}(t)\cdot\hat{\mathbf{n}})\rangle \tag{4.28}$$

In particular, the correlations of the velocities usually fade away in time due to the particle collisions. Such a decay permits the integration of the velocity autocorrelation function in time because it is a bounded function from above by a finite value. When it is integrated using a large time interval, $[t = 0, t \to \infty]$, it results in the diffusion constant.

For a vector with components $x_1, x_2, \cdots, x_n$, the corresponding *correlation matrix* with elements ij and dimension $n \times n$ is:

$$C_{ij}(r,s) = \langle x_i(r)\, x_j(s)\rangle \tag{4.29}$$

The variables r and s belong to a mathematically well-behaved space. The correlation matrix may inherit the symmetries of the distribution function employed in the evaluation of ensemble averages. For the spatial coordinates r and s, the translational and rotational symmetries are shown, in that order, in the correlation function as:

$$C_{ij}(r,s) = C_{ij}(r-s) \quad ; \quad C_{ij}(r,s) = C_{ij}(|r-s|) \tag{4.30}$$

In the situation of $2D$ discrete data, the correlation function is written in the form:

$$F \circ I(x,y) = \sum_{i=-N}^{N} \sum_{j=-N}^{N} F(i,j)\, I(x+i, y+j) \tag{4.31}$$

The function F plays the role of a filter, and function I represents the pixels of an image. For a given central pixel, the correlation function and the convolution function combine the neighboring pixels to replace the central pixel.[1] The correlation and convolution functions are ubiquitous in molecular dynamics and image processing and are employed in many codes of machine learning.

The autocorrelation function is related to the *Fourier transform*. In order to have this relation, we recall the Fourier transform of the variable A and insert it into the autocorrelation function.

$$C_A(t) = \int_{-\infty}^{\infty} A(t+\tau) A^*(\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} A(\omega_2)\, e^{-i\omega_2(t+\tau)}\, d\omega_2 \right) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} A(\omega_1)\, e^{-i\omega_1 \tau}\, d\omega_1 \right)^* d\tau$$

By using the expression of the Dirac function $\int \exp[-i(\omega_2 - \omega_1)\tau]\, d\tau/(2\pi) = \delta(\omega_2 - \omega_1)$:

$$C_A(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |A(\omega)|^2\, e^{-i\omega t}\, d\omega = \mathcal{F}[|A(\omega)|^2](t) \tag{4.32}$$

The Fourier transform is $\mathcal{F}$. Therefore, the autocorrelation function has a spectral decomposition and may be computed from $|A|^2$ in the frequency domain. By dealing with:

$$C_{AB}(t) = \int A(t+\tau) B^*(\tau)\, d\tau$$

and following the procedure presented above, the *cross-correlation theorem* is obtained.

[1] D. Jacobs [8]. Many freely distributed programs for image manipulation allow the use of convolution to modify the original images.

$$\int A(t+\tau)B^*(\tau)\,d\tau = \frac{1}{2\pi}\int_{-\infty}^{\infty} A(\omega)B^*(\omega)\,e^{-i\omega t}\,d\omega = \mathcal{F}[A(\omega)B^*(\omega)](t) \tag{4.33}$$

The Wiener-Khinchin theorem is a particular result of the cross-correlation theorem.

4.8 Vibrational Spectra from Autocorrelation Functions

In many dynamic systems, the particles self-correlate their motions, and it is important to determine the main vibrational frequencies to characterize the system (the synchronization of many metronomes represents an example on the self-correlation of dynamic bodies). The velocity autocorrelation function, $C_{\mathbf{v}}(t) = \langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle$, is appropriate to reveal the collective vibrations of the particles, Fig. 4.2. In particular, the *power spectrum* is defined in terms of the velocity autocorrelation function. By starting from $C_{\mathbf{v}}(t)$, applying the Fourier transform to the time function, we obtain:

$$F(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle\, e^{i\omega t}\,dt \tag{4.34}$$

The power spectrum provided by the Fourier transform, $F(\omega)$, includes all the frequencies of the particle system. The peaks of the spectrum are associated with intense vibrational modes, which are frequently the ones of interest. Thus, the velocity autocorrelation function embodies the vibrational characteristics of the system, and in this way, it shows the spreading of the energy.

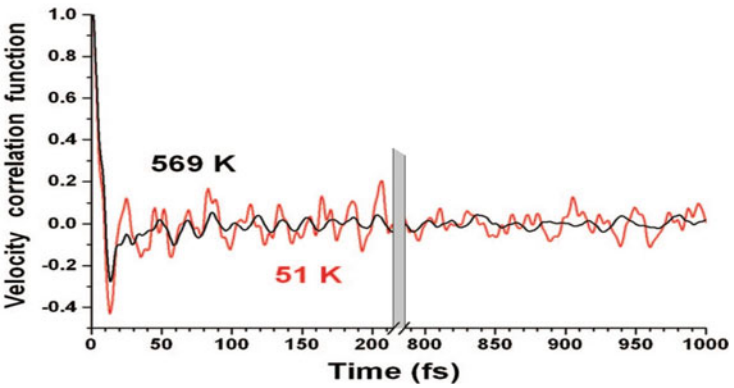


Fig. 4.2 The velocity autocorrelation functions of 13 hydrogen molecules in confinement and under temperatures of $51K$ and $569K$ are depicted. At $51K$, the molecules vibrate with large amplitudes around their equilibrium positions, giving evidence of the atomic localization, which is a characteristic behavior of a solid with strong interactions. At 569 K, the atoms collide, producing frequent changes of the velocity directions and leading to atomic delocalization. The velocity autocorrelation function is an indicator on the melting of a molecular system [9]

The absorption of infrared radiation produces atomic vibrations. Similarly, the inelastic scattering of visible light produces atomic vibrations [10]. The autocorrelation function of the molecular dipole moment gives information of the infrared spectrum, and the autocorrelation function of the molecular polarizability gives information of the Raman spectrum. In this last case, the molecular simulation demands an electric field to determine the polarizabilities. The anharmonic vibrations are inherently considered in the vibrational spectra determined from the correlation functions because the atoms are not constrained to move in a harmonic potential [11].

Problem 4.3 Show that the velocity autocorrelation function is related to the position autocorrelation function by using the Fourier and inverse-Fourier transforms:

$$X(\omega) = \mathcal{F}[x(t)] = \int_0^\infty x(t)\, e^{-i\omega t}\, dt;$$

$$x(t) = \mathcal{F}^{-1}[X(i\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(i\omega)\, e^{i\omega t} d\omega$$

Solution We know that $\dot{x}(t) = dx(t)/dt$. By applying the Fourier transform to the positions, and renaming the temporal variables t and $t+\tau$, we obtain:

$$\begin{aligned}\int_{-\infty}^{\infty} \langle \dot{x}(\tau)\, \dot{x}(t+\tau)\rangle\, dt &= \int_{-\infty}^{\infty} \langle \frac{d}{dt_1}\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} X(i\omega)\, e^{i\omega t_1} d\omega\right) \\ &\quad \times \frac{d}{dt_2}\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} X(i\omega)\, e^{i\omega t_2} d\omega\right)\rangle_{t_1=t_2=t}\, dt \\ &= (i\omega)^2 \int_{-\infty}^{\infty} \langle x(\tau)\, x(t+\tau)\rangle\, dt\end{aligned}$$

The autocorrelation function of the velocities is related to the autocorrelation function of positions by integration in time.

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Rigid Body

5

The purpose of this chapter is to determine the main equations of a body integrated by a number of particles which maintain their distances fixed from each other and give shape and volume to the body. A particle system like this is known as a *rigid body*, formed by many particles with inter-particle distances fixed. The motion equation and energy of the rigid body depend on the spatial arrangement of the particles, and henceforth, a set of geometric parameters are introduced to characterize the rigid body's degrees of freedom. Two main degrees of freedom are identified: the translational motion, where the particles of the rigid body describe parallel trajectories, and the rotational motion, where the particles rotate all together as a single entity. The translational motion is frequently omitted as it is possible to work in an inertial reference frame undergoing translational motion with the rigid body. In this respect, we analyze the rotating motion of the rigid body in terms of the angular momentum and the torque, which is produced by external forces acting on the body.

5.1 Angular Momentum of a Rotating System of Particles

The description of the circular motion requires an axis of reference, and therefore, the z axis is taken as the rotation axis of the particles that form the body. The reference frame of coordinates is depicted in Fig. 5.1 (similar to the reference frame of Fig. 1.5). The ith particle has position $\mathbf{r}_i = (x_i, y_i, z_i)$ and velocity $\mathbf{v}_i = \boldsymbol{\omega} \times \mathbf{r}_i$. The angular velocity $\boldsymbol{\omega}$ is constant in an isolated system and is the same for all the particles of the body. The velocity magnitude is $v_i = \omega r_i \sin(\gamma_i)$, where γ_i is the angle between vectors $\boldsymbol{\omega}$ and $\mathbf{r}_i$. On the other hand, the angular momentum $\boldsymbol{\ell}_i$ is perpendicular to both $\mathbf{r}_i$ and $\mathbf{v}_i$, because it is the result of the cross product $\boldsymbol{\ell}_i = \mathbf{r}_i \times m_i \mathbf{v}_i$, with magnitude $|\boldsymbol{\ell}_i| = m_i r_i v_i \sin(\pi/2) = m_i r_i v_i$. The component of $\boldsymbol{\ell}_i$ in the z axis is:

$$\ell_{iz} = |\boldsymbol{\ell}_i| \cos(\pi/2 - \gamma_i) = m_i r_i v_i \sin(\gamma_i)$$

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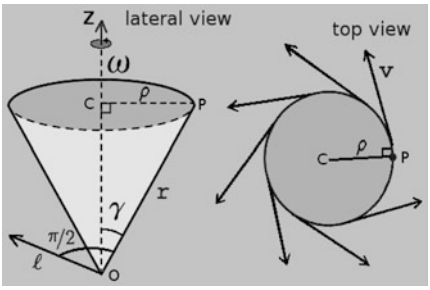


Fig. 5.1 A particle P of a rigid body rotates around the z axis. It has an instantaneous position $\mathbf{r}$, tangent velocity $\mathbf{v}$, angular velocity $\boldsymbol{\omega}$, and angular momentum $\boldsymbol{\ell}$. The quantity γ is the conical angle, and $\pi/2$ is the angle between $\boldsymbol{\ell}$ and the plane formed by $\mathbf{r}$ and $\mathbf{v}$

The trigonometric addition $\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$ was used in the last identity. It is convenient to simplify the expression of the component ℓ_{iz} in terms of ρ_i, the distance from the z axis to the particle position. To do this, we consider the projection of $\mathbf{r}_i$ onto the plane described by the particle motion, $r_i \sin(\gamma_i) = \rho_i$. Using this identity, the component of the angular momentum in the z axis transforms onto $\ell_{iz} = m_i r_i v_i \sin(\gamma_i) = m_i r_i [\omega r_i \sin(\gamma_i)] \sin(\gamma_i) = m_i \omega \rho_i^2$. The same procedure applies to all the particles of the body, and a summation over the number of particles gives the total z component of the angular momentum.

$$L_z = \textstyle\sum_i m_i \omega \rho_i^2 \tag{5.1}$$

$\sum_i m_i \rho_i^2$ defines the *moment of inertia* of the body with respect to the axis of rotation and is designated with the letter I (not to be confused with the symbol of impulse on a particle, which is a vector). The component L_z is now written in terms of I and ω.

$$L_z = I\omega \quad ; \quad I = \textstyle\sum_i m_i \rho_i^2 \tag{5.2}$$

The inertia moment I is a geometric factor of the system defined by the masses and positions of the particles with respect to the rotation axis. Therefore, I depends on the *mass distribution*. We may deal with the rotation of the body under more general terms to introduce the additional components of the angular momentum. Consider a particle with mass m rotating around an axis with angular velocity $\boldsymbol{\omega}$. The angular momentum of the particle is $\mathbf{L} = \mathbf{r} \times \mathbf{p} = m\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r})$. Contrary to the linear momentum $\mathbf{p}$, the angular momentum $\mathbf{L}$ is not an intrinsic property of the particle, and for such a reason, the reference frame of coordinates should be specified when the angular momentum and torque are discussed. The components of the angular momentum (L_x, L_y, L_z) are determined from the *vector triple product* $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$. In this regard, by identifying vectors $\mathbf{a} = \mathbf{r}$, $\mathbf{b} = \boldsymbol{\omega}$, $\mathbf{c} = \mathbf{r}$ and applying the vector triple product, we have $\mathbf{L} = m\boldsymbol{\omega}(\mathbf{r} \cdot \mathbf{r}) - m\mathbf{r}(\mathbf{r} \cdot \boldsymbol{\omega})$. The components of $\mathbf{L}$ are (L_x, L_y, L_z). For instance, the x component is:

$$L_x = m\omega_x(\mathbf{r}\cdot\mathbf{r}) - mx(\mathbf{r}\cdot\boldsymbol{\omega}) = m\omega_x r^2 - mx(x\omega_x) - mx(y\omega_y) - mx(z\omega_z)$$
$$= m(r^2 - x^2)\omega_x - m(xy)\omega_y - m(xz)\omega_z = I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z$$
$$= \sum_{i=1}^{3} I_{xi}\omega_i$$

It was defined above $I_{xx} = m(r^2 - x^2)$, $I_{xy} = m(xy)$, $I_{xz} = m(xz)$. A similar situation is obtained for the other components, in such a way that the angular momentum components associated with a general rotation of the body may be written in the form:

$$\mathbf{L} = \left(\sum_{i=1}^{3} I_{xi}\omega_i \ , \sum_{i=1}^{3} I_{yi}\omega_i \ , \sum_{i=1}^{3} I_{zi}\omega_i\right)$$

This expression may be written in matrix form.

$$\begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = \begin{pmatrix} I_x & I_{xy} & I_{xz} \\ I_{yx} & I_y & I_{yz} \\ I_{zx} & I_{zy} & I_z \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} \tag{5.3}$$

Equation 5.3 indicates that the inertia moment is a $3D$ tensor, $\overline{I} = I_{ij}$, with components:

$$I_x = \sum_i m_i(y_i^2 + z_i^2) \ ; \ I_y = \sum_i m_i(x_i^2 + z_i^2) \ ; \ I_z = \sum_i m_i(x_i^2 + y_i^2)$$
$$I_{xy} = -\sum_i m_i x_i y_i \quad ; \quad I_{yz} = -\sum_i m_i y_i z_i \quad ; \quad I_{xz} = -\sum_i m_i x_i z_i \tag{5.4}$$

The term $(y_i^2 + z_i^2)$ defining I_x is the distance of the ith particle to the x axis. The diagonal element I_x is the inertia moment in the x axis, with similar results for I_y and I_z. The off diagonal elements, I_{xy}, I_{yz}, ..., are the *inertia moment products*. Six quantities define the body rotation since the tensor I_{ij} is a real-symmetric matrix. Equation 5.3 is written in matrix notation, $\mathbf{L} = \overline{I}\boldsymbol{\omega}$. According to matrix theory, by carrying a transformation, it is possible to deal with the diagonal form of matrix I_{ij}. The diagonalization of I_{ij} is determined by solving the eigenvalue problem $\overline{I}\boldsymbol{\omega}_i = \lambda_i\boldsymbol{\omega}_i$, where $i = x, y, z$, and the eigenvectors $\boldsymbol{\omega}_i$ and eigenvalues λ_i are real. A diagonalization process is illustrated in Sect. A.1. The eigenvalue solutions may be reached for bodies with simple and symmetric shapes or complicated geometries.

$$I_{ij} = \begin{pmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{pmatrix} \tag{5.5}$$

In short, for any shape and rotation of the body, it is possible to find three *rotation axes* where the inertia moment products are zero. The angular momentum in terms of the inertia moments associated with the three preferent axes is:

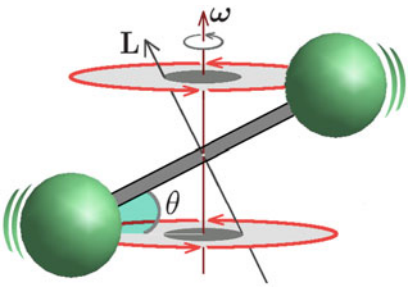


Fig. 5.2 Two symmetric bodies rotate around a vertical axis. The line joining the bodies makes an angle θ with the horizontal plane. The angular momentum $\mathbf{L}$ is not parallel to the angular velocity $\boldsymbol{\omega}$, as this last vector lies along the vertical axis. However, when the line joining the particles is perpendicular to the vertical axis of rotation, the angular momentum $\mathbf{L}$ is parallel to vector $\boldsymbol{\omega}$

$$\mathbf{L} = \mathbf{u}'_x I_x \omega_x + \mathbf{u}'_y I_y \omega_y + \mathbf{u}'_z I_z \omega_z \tag{5.6}$$

The unit vectors $\mathbf{u}'_i$, where $i = x, y, z$, are in motion with respect to a fixed reference frame of coordinates in the lab. The unit vectors are mutually orthogonal and, in turn, fixed in the body, which is rotating, to describe the three directions of independent rotations in $3D$ space. Any rotation of the body can be described in terms of the three main axes, which receive the name of *principal axes of rotation*. The inertia moments I_x, I_y, and I_z with respect to such axes are the *principal moments of inertia*. The components of the angular velocity $\boldsymbol{\omega}$ in the principal axes of rotation are ω_x, ω_y, and ω_z. In this regard, the angular momentum of Eq. 5.6 has general character for any rotating body and indicates that $\mathbf{L} = (L_x, L_y, L_z)$ and $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)$ are not in general parallel, Fig. 5.2. Under special circumstances, the body may rotate around a principal axis. In this case, the vectors $\mathbf{L}$ and $\boldsymbol{\omega}$ are parallel, and expression 5.6 of the angular momentum becomes $\mathbf{L} = I\boldsymbol{\omega}$, where the factor of proportionality is I, namely, the principal moment of inertia for the respective principal axis of rotation. This last expression shows that $\mathbf{L}$ points in the direction of the rotation axis. For symmetric bodies, the axes of symmetry correspond to the principal axes of rotation since a rotation of the body around an axis of symmetry keeps the angular momentum parallel to the rotating axis, and $\mathbf{L} = I\boldsymbol{\omega}$.

When the body is dense, we introduce the density ρ to relate the infinitesimal mass dm to the infinitesimal volume dV by $dm = \rho dV$ and transform the summation over the number of particles onto an integral over the body volume for the inertia moment of expression 5.2.

$$I = \int_{\text{vol}} \rho(\mathbf{r})\, r^2 dV \tag{5.7}$$

When the mass density ρ is constant, the inertia moment is $I = \rho \int r^2 dV$. The body geometric features and spatial extension are considered by the integral over the body's volume.

5.2 External Torques Acting on a Rotating Body

The purpose of this section is to use the inertia moment I in the equation of motion of a rotating rigid body to get it simplified as, otherwise, the equation of motion of every particle that constitutes the body has to be specified. The equation of motion of a rotating body is proposed from the expression $d\mathbf{L}/dt = \boldsymbol{\tau}$, relating the angular momentum $\mathbf{L}$ to the torque $\boldsymbol{\tau}$. The angular momentum and torque are described from a common inertial reference frame. When the body is formed by many particles, the total angular momentum and total torque are:

$$\mathbf{L} = \sum_i \boldsymbol{\ell}_i \quad ; \quad \boldsymbol{\tau} = \sum_i \boldsymbol{\tau}_i \tag{5.8}$$

$\boldsymbol{\ell}_i$ and $\boldsymbol{\tau}_i$ are the angular momentum and torque on the ith particle. The individual torque $\boldsymbol{\tau}_i$ is due to the external forces acting on the ith particle. In a general situation, the body rotates around an arbitrary axis with simultaneous translatory motion. An example of such a situation is a rotating yoyo, where the yoyo falls due to gravity. It is convenient to consider the body rotation axis as the z axis and work with the expression $dL_z/dt = \tau_z$ to describe the body rotation. The component L_z may be replaced by the product $I\omega$, Eq. 5.2. Further simplification is achieved when the inertia moment I is time indifferent.

$$I\, d\omega/dt = \tau_z \tag{5.9}$$

The torque component τ_z is zero for constant angular velocity. In order to keep the direction of the total angular momentum in the z axis, the torque components τ_x and τ_y need to be different from zero to counteract the centrifugal forces exerted on the particles moving in a circular path. Therefore, in addition to τ_z, the torque components τ_x and τ_y participate in the stability of the body rotation.

The description of the particle system is substantially reduced when the rotation axis corresponds to a principal axis of rotation as, in this case, the total angular momentum is written in the form $\mathbf{L} = I\boldsymbol{\omega}$, being the vectors $\mathbf{L}$ and $\boldsymbol{\omega}$ parallel. The equation of rotation is:

$$d(I\boldsymbol{\omega})/dt = \boldsymbol{\tau} \tag{5.10}$$

For zero torque, the product $I\boldsymbol{\omega}$ is preserved. The expression $I\boldsymbol{\omega} =$ constant discriminates between a rigid body and a non-rigid one. A non-rigid body demands an increase of the inertia moment I for a decreasing value of $\boldsymbol{\omega}$, and vice versa, to keep the product $I\boldsymbol{\omega}$ constant. On the other side, the inertia moment I of a rigid body

is fixed, and the condition $\boldsymbol{\tau} = 0$ implies a constant angular velocity, $\boldsymbol{\omega} =$ constant. Examples of these two cases correspond to the different spins that a cooked egg and a raw egg exhibit. A cooked egg is solid and spins relatively easy, while the liquid of the raw egg takes time to acquire the spin and also slows down the rotation by friction.

In many situations it is convenient to take the body center of mass as the origin of coordinates to describe the angular momentum. Once the reference frame is located at that point, any external torque should be also described from the center of mass to have consistent descriptions of the angular momentum and the external torques.

$$d\mathbf{L}_{cm}/dt = \boldsymbol{\tau}_{cm} \tag{5.11}$$

Again, the equation is simplified when the rotation axis of the body is a principal axis of rotation because $\mathbf{L}_{cm} = I_{cm}\boldsymbol{\omega}$, and the equation of motion is transformed onto $I_{cm}d\boldsymbol{\omega}/dt = \boldsymbol{\tau}_{cm}$. In the absence of external torques, the angular momentum may be considered an intrinsic property of the body because $\mathbf{L}_{cm}$ shows no change. In such a case, the angular momentum is referred to as the *body spin*. A spinning body may additionally have an orbital angular momentum, like that of a spinning planet in orbital motion around the Sun.

Problem 5.1 Suppose that a body is translating and rotating. The axis of rotation passes through the body's center of mass (cm). Show that the angular momentum, $\mathbf{L}$, may be written in terms of an orbital component plus a spin component.

Solution The body's angular momentum is $\mathbf{L} = \int \mathbf{r} \times \mathbf{v}\, dm$. The factor dm is a mass element of the body. The total body's velocity is decomposed in $\mathbf{v} = \mathbf{v}_{cm} + \mathbf{v}_{rot}$, with $\mathbf{v}_{cm}$ the *cm* translational velocity and $\mathbf{v}_{rot}$ the body's rotational velocity with reference to the axis passing through the *cm*. We have $\mathbf{v}_{rot} = \boldsymbol{\omega} \times \mathbf{r}'$, where $\mathbf{r}'$ is the position of the mass element dm with respect to the *cm* reference frame of coordinates. The total velocity reads $\mathbf{v} = \mathbf{v}_{cm} + \boldsymbol{\omega} \times \mathbf{r}'$, and the angular momentum becomes:

$$\begin{aligned}\mathbf{L} &= \int (\mathbf{r}_{cm} + \mathbf{r}') \times (\mathbf{v}_{cm} + \boldsymbol{\omega} \times \mathbf{r}')\, dm \\ &= \mathbf{L}_{cm} + \int \mathbf{r}' \times (\boldsymbol{\omega} \times \mathbf{r}')\, dm\end{aligned}$$

$\mathbf{L}_{cm} = m_{tot}(\mathbf{r}_{cm} \times \mathbf{v}_{cm})$ and the body's total mass is m_{tot}. In this regard, the angular momentum has components $\mathbf{L}_{orb} = \mathbf{L}_{cm}$ and $\mathbf{L}_{spin} = \int \mathbf{r}' \times (\boldsymbol{\omega} \times \mathbf{r}')\, dm$.

In the following lines, we are interested in describing the torque from different reference frames. To do this, consider a pair of reference frames with common origin. One of them has angular motion with respect to the other. The torque in the non-primed reference frame of coordinates, $\boldsymbol{\tau} = d\mathbf{L}/dt$, is related to the torque in the rotating (or primed) reference frame of coordinates, $\boldsymbol{\tau}' = d\mathbf{L}'/dt$. The relation is established by using the mnemonic rule 1.38, $d\mathbf{b}/dt = d\mathbf{b}'/dt + \boldsymbol{\omega} \times \mathbf{b}$. The angular momentum $\mathbf{L}$ replaces vector $\mathbf{b}$ in Eq. 1.38. Since the reference frame co-rotates with the body, and the axes of the rotating reference frame are assumed to be the body principal axes of rotation, the quantity $\boldsymbol{\omega}$ is the angular velocity of the primed reference frame. The relation between the two torques is:

$$\boldsymbol{\tau} = \boldsymbol{\tau}' + \boldsymbol{\omega} \times \mathbf{L} \tag{5.12}$$

This identity is known as the *Euler equation* for the torques computed in the different reference frames. The Cartesian components of the equation are:

$$\begin{gathered} \tau_x = I'_x \frac{d\omega_x}{dt} + (I'_z - I'_y)\omega_y\omega_z \quad ; \quad \tau_y = I'_y \frac{d\omega_y}{dt} + (I'_x - I'_z)\omega_x\omega_z \\ \tau_z = I'_z \frac{d\omega_z}{dt} + (I'_y - I'_x)\omega_x\omega_y \end{gathered} \tag{5.13}$$

The Euler equations are simplified when there is no angular acceleration because in this case $d\boldsymbol{\omega}/dt = 0$. Still, there are other terms of the torque components (τ_x, τ_y, τ_z) that are not zero and act on the body. If the body's rotating axis is the z' principle axis of rotation, $\omega_z \neq 0$, then all the torque components τ_i in Eqs. 5.13 are zero because the angular velocity components ω_x and ω_y are zero, and the body rotates steadily due to the absence of torque components.

5.3 Total Energy of a Rotating Rigid Body

The following quantity of interest is the total energy of the rigid body. It involves the kinetic energy and potential energy. The purpose is to separate the translational kinetic energy from the rotational kinetic energy in the expression of the total kinetic energy and separate the external potential energy from the internal potential energy in the expression of the total potential energy. To do this, it is convenient to establish a reference frame of coordinates in the body's center of mass (cm). The rotational kinetic energy is given in terms of the translational velocity, $\mathbf{v}_{cm}$, and rotational velocity, $\boldsymbol{\omega} \times \mathbf{r}'$.

$$E_{kin} = \tfrac{1}{2} \int dm \, (\mathbf{v}_{cm} + \boldsymbol{\omega} \times \mathbf{r}')^2 \tag{5.14}$$

The translational kinetic energy is that of the center of mass, $m_{tot}\mathbf{v}_{cm}^2/2$, with m_{tot} the mass of the rigid body. The rotational velocity is $\boldsymbol{\omega} \times \mathbf{r}'$, where $\mathbf{r}'$ is the position of the mass element dm with respect to the cm reference frame of coordinates and

$\boldsymbol{\omega}$ is the angular velocity. We use the angular momentum $\mathbf{L}$ and the angular velocity $\boldsymbol{\omega}$ to rewrite the kinetic energy.

$$E_{kin} = \tfrac{1}{2}\, m_{tot}\mathbf{v}_{cm}^2 + \tfrac{1}{2}\boldsymbol{\omega} \cdot \int dm\ \mathbf{r}' \times (\boldsymbol{\omega} \times \mathbf{r}') = \tfrac{1}{2}\, m_{tot}\mathbf{v}_{cm}^2 + \tfrac{1}{2}\,\boldsymbol{\omega} \cdot \mathbf{L} \tag{5.15}$$

We have translational and rotational components of the kinetic energy.

Problem 5.2 Obtain the kinetic energy 5.15 from expression 5.14 in the case of many particles forming the body, namely, the particles discrete case.

Solution The discrete version of Eq. 5.14 (omitting primes) is $E_{kin} = m_{tot}\mathbf{v}_{cm}^2/2 + \sum_i m_i(\boldsymbol{\omega} \times \mathbf{r}_i)^2/2$. The product of vectors in this expression has the general form $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d})$. We identify vectors: $\mathbf{a} = \mathbf{c} = \boldsymbol{\omega}$ and $\mathbf{b} = \mathbf{d} = \mathbf{r}_i$. The purpose is to demonstrate that $\mathbf{a} \cdot [\mathbf{b} \times (\mathbf{c} \times \mathbf{d})]$ is equivalent to $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d})$. We start from the expression:

$$\mathbf{b} \times (\mathbf{c} \times \mathbf{d}) = (\mathbf{b} \cdot \mathbf{d})\mathbf{c} - (\mathbf{b} \cdot \mathbf{c})\mathbf{d}$$

By carrying the dot product of $\mathbf{a}$ with every term of the equation, we have:

$$\begin{aligned} \mathbf{a} \cdot [\mathbf{b} \times (\mathbf{c} \times \mathbf{d})] &= (\mathbf{b} \cdot \mathbf{d})(\mathbf{a} \cdot \mathbf{c}) - (\mathbf{b} \cdot \mathbf{c})(\mathbf{a} \cdot \mathbf{d}) \\ &= (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c}) \\ &= (\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) \end{aligned}$$

By using this result, the kinetic energy is $E_{kin} = m_{tot}\mathbf{v}_{cm}^2/2 + (\boldsymbol{\omega}/2) \cdot \sum_i m_i \mathbf{r}_i \times (\boldsymbol{\omega} \times \mathbf{r}_i)$. The factor $\boldsymbol{\omega} \times \mathbf{r}_i$ represents the particle rotational velocity. Thus, the summation gives the total angular momentum of the body, $E_{kin} = m_{tot}\mathbf{v}_{cm}^2/2 + \boldsymbol{\omega} \cdot \mathbf{L}/2$.

We know that when the rotating axis is a principal axis of rotation passing through the center of mass, we have $\mathbf{L} = I\boldsymbol{\omega}$. The equation of the kinetic energy is simplified to $E_{kin} = m_{tot}\mathbf{v}_{cm}^2/2 + L^2/(2I)$. The total kinetic energy may be alternatively written as $E_{kin} = m_{tot}\mathbf{v}_{cm}^2/2 + I\omega^2/2$. If we use the angular momentum $\mathbf{L} = \overline{I}\boldsymbol{\omega}$ in terms of the inertia moment tensor $\overline{I}$, Eq. 5.3, the kinetic energy in matrix form is:

$$E_{kin} = \boldsymbol{\omega}^t\, \overline{I}\, \boldsymbol{\omega}/2 \tag{5.16}$$

where the translational kinetic energy is neglected for the isolated rigid body and $\boldsymbol{\omega}^t$ is the transpose of the angular velocity. Equation 5.16 in terms of the components of $\boldsymbol{\omega}$ and $\overline{I}$ is:

$$E_{kin} = \frac{1}{2}(\omega_x, \omega_y, \omega_z) \begin{pmatrix} I_x & I_{xy} & I_{xz} \\ I_{yx} & I_y & I_{yz} \\ I_{zx} & I_{zy} & I_z \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} \\ = \frac{1}{2}(I_x\omega_x^2 + 2I_{xy}\omega_x\omega_y + 2I_{xz}\omega_x\omega_z + I_y\omega_y^2 + 2I_{yz}\omega_y\omega_z + I_z\omega_z^2) \tag{5.17}$$

By assuming a system where the inertia moment products have null contribution, and using expression 5.6 of the angular momentum, the kinetic energy is written in terms of the principal moments of inertia solely.

$$E_{kin} = \left(L_x^2/I_x + L_y^2/I_y + L_z^2/I_z\right)/2 \tag{5.18}$$

The internal potential energy of the rigid body is constant. However, in order to formulate the total energy of the system, we are required to also deal with any external action on the rigid body capable of producing a change of its kinetic energy. In this regard, we recall the energy balance $W_{ext} + Q = E_{kin} - E_{kin}^{(0)}$, Eq. 3.23, where the external work W_{ext} is due to external conservative forces and Q is due to external non-conservative forces. The external work is expressed as the difference of external potential energies, $W_{ext} = E_{pot,ext}^{(0)} - E_{pot,ext}$.

$$E_{kin} + E_{pot,ext} = E_{kin}^{(0)} + E_{pot,ext}^{(0)} + Q \tag{5.19}$$

When $Q = 0$, the total energy E_{tot} at different instants of time is preserved.

$$E_{tot} = \frac{1}{2}\, m_{tot}\mathbf{v}_{cm}^2 + \frac{1}{2}\, I_{cm}\omega^2 + E_{pot,ext} = \text{ constant} \tag{5.20}$$

When the force of gravity is the external potential energy, we have $E_{pot,ext} = m_{tot}gh_{cm}$, where h_{cm} is the position of the mass center with respect to ground level.

$$E_{tot} = \frac{1}{2}\, m_{tot}\mathbf{v}_{cm}^2 + \frac{1}{2}\, I_{cm}\omega^2 + m_{tot}gh_{cm} = \text{constant}$$

A.1 Appendix: Matrix Diagonalization

Suppose that A is a matrix with dimension $n \times n$, and it can be diagonalized. The *diagonalization of matrix A* demands the writing of A in terms of matrices A_d and T.

$$A = T\,A_d\,T^{-1} \tag{5.21}$$

A_d is the diagonal matrix whose components are the eigenvalues of A, while T is the transformation matrix whose components are the *eigenvectors* of A. The transformation matrix T acting on both sides of A gives A_d.

$$T^{-1}AT = A_d \tag{5.22}$$

In the first stage, the *eigenvalues* of A are obtained from the determinant:

$$|A - \lambda I| = 0 \tag{5.23}$$

I is the unit matrix. The expansion and evaluation of determinants are given in Sect. B.1 of Chap. 10. The determinant unfolds onto its *characteristic polynomial* of order n in λ. The polynomial equation may be solved for the roots or eigenvalues $\lambda_1, \cdots, \lambda_n$. In the second stage, we observe that $A\mathbf{V} = \lambda\mathbf{V}$, and therefore, $(A - \lambda I)\mathbf{V} = 0$. The matrix of coefficients is $(A - \lambda I)$, and the vector of variables is $\mathbf{V}$, supposed to be a non-trivial vector. We must work with each eigenvalue separately to obtain the corresponding eigenvector. The first matrix equation $(A-\lambda_1 I)\mathbf{V}_1 = 0$, where $\mathbf{V}_1$ is not $\mathbf{0}$, is proposed.

$$\begin{pmatrix} A_{11} - \lambda_1 & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} - \lambda_1 & \cdots & A_{2n} \\ \cdots & & & \cdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} - \lambda_1 \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{12} \\ \cdots \\ v_{1n} \end{pmatrix} = 0 \tag{5.24}$$

The components of the eigenvector $\mathbf{V}_1 = (v_{11}, \cdots, v_{1n})$ are obtained from the system of n equations with n unknown variables implied by the first matrix equation. Similarly, the components of the linear independent eigenvector $\mathbf{V}_2 = (v_{21}, \cdots, v_{2n})$ are obtained from the second matrix equation $(A - \lambda_2 I)\mathbf{V}_2 = 0$ and so on. In principle, the number of eigenvectors is n, and they are linearly independent from each other (when there are repeated eigenvalues, the linear independence property may be additionally used to find the eigenvectors). After finishing the whole process, we have the n eigenvalues and the corresponding eigenvectors of A.

$$\begin{aligned} \lambda_1 \ ; \ \mathbf{V}_1 &= (v_{11}, \cdots, v_{1n}) \\ \lambda_2 \ ; \ \mathbf{V}_2 &= (v_{21}, \cdots, v_{2n}) \\ \cdots & \qquad \cdots \\ \lambda_n \ ; \ \mathbf{V}_n &= (v_{n1}, \cdots, v_{nn}) \end{aligned}$$

The matrices A_d and T are consequently determined from the eigenvalues and eigenvectors.

$$A_d = \begin{pmatrix} \lambda_1 & 0 & 0 & \cdots \\ 0 & \lambda_2 & 0 & \cdots \\ \cdots & & & \cdots \\ 0 & 0 & 0 & \lambda_n \end{pmatrix} \ ; \ T = \begin{pmatrix} v_{11} & v_{21} & \cdots & v_{n1} \\ v_{12} & v_{22} & \cdots & v_{n2} \\ \cdots & & & \cdots \\ v_{1n} & v_{2n} & \cdots & v_{nn} \end{pmatrix} \tag{5.25}$$

The order in writing the eigenvalues follows the order in writing the eigenvectors and vice versa. The inverse matrix T^{-1} is determined from the condition on the matrix product $TT^{-1} = I$. Once we determined A_d, T, and T^{-1}, the identity $A = TA_dT^{-1}$ may be confirmed. Finally, the diagonalization of matrices is a process among many processes in linear algebra, but it acquires especial importance in quantum physics, for instance, for the determination of state energies, the finding of vibrational modes and frequencies, etc.



Analytical Mechanics

6

The formulations presented in this chapter establish the principles of analytical mechanics. The chapter starts by introducing the action as the fundamental function that contains the physical information of the particle system. By considering the stationary principle on the action, the Lagrangian theory is derived, paying special attention to the equations of motion. Based on the Lagrangian formalism, we show that a symmetry in nature implies a conserved observable via Noether's theorem. Later, by using a Legendre transformation, the relation between the Lagrangian function and the Hamiltonian function is achieved. Both functions codify the physical information of the system but employ different conjugate variables. A formulation of mechanics in terms of the Hamiltonian, which is equivalent to Lagrangian mechanics, is now established with the respective equations of motion. The Hamiltonian is demonstrated to be a constant of motion, corresponding to the energy of the system when the Lagrangian is time independent. The possibility of dealing with different Hamiltonians of a given system takes us to the discussion of the canonical transformations with the purpose to link the different Hamiltonian functions. The specific canonical transformation where the basic conjugate variables of the system are constants introduces the Hamilton-Jacobi scheme as a branch of Hamiltonian theory where the solutions of the equations of motion may be found in a distinct manner. The Hamilton-Jacobi theory not only brings the connection back to the action function but also provides the first relative contact of classical mechanics with quantum mechanics.

The particles dynamics is analyzed according to the Poisson formulation, because this offers the opportunity to derive a classical time propagator from the Liouville operator. The functional form of the propagator resembles that of a quantum mechanical propagator and allows to have the time evolution of physical observables in terms of the positions and momenta. All the previous mechanical formulations are linked together and are equivalent to Newtonian theory. Yet, every formulation provides from different perspectives the fundamentals of analytical mechanics. The analytical theories of mechanics shall be used in later chapters to establish contact

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with quantum theory. The chapter is finished by discussing the general and modern concepts of mechanics.[1]

6.1 Action Function

In daily life, an action is the accomplishment of a task using a force in an interval of time. In mechanics, it has a more precise meaning. The *action* is a mathematical functional that, by using the function L, contains the physical characteristics of the particle system, such as the number of particles, their types, forms of kinetic energy, and kinds of interactions, and from which the information can be extracted in systematic manner to establish the equations of motion of the particles. The method to obtain information from the action is general and plays a central role not only in mechanics but also in other fields of physics like in electromagnetism, general relativity, high-energy physics, etc. The action is represented by the letter S; it is proposed as an integral of the function L over time and depends on the time limits t_i and t_f.

$$S = \int_{t_i}^{t_f} L(\{q_i\}, \{\dot{q}_i\}, t)\, dt \tag{6.1}$$

The function L is the *Lagrangian*. The Lagrangian provides key information on the motion and interaction aspects of the particles. It is given as the balance between the kinetic energy, T, and potential energy, V. The units of S are those of angular momentum ($energy \times time$).

$$L = T(\{\dot{q}_i\}) - V(\{q_i\}) \tag{6.2}$$

The formulation in terms of the action demands a bounded trajectory in time and space and is invariant to the coordinates representation. In this respect, the variables q_i and $\dot{q}_i$ (the dot indicates differentiation in time) are the *generalized coordinates* and *generalized velocities* in the action. They constitute the independent degrees of freedom of the theory and may have units that are not necessarily those of position and velocity. By dealing with the action, it is possible to determine the equations of motion of the particles. When such equations are solved, we are able to describe the variables q_i and $\dot{q}_i$ with time. The time evolution of the system corresponds to the time evolution of the particles, supposed to evolve along the path that goes from $\{q_j(t_i), \dot{q}_j(t_i)\}$ to $\{q_j(t_f), \dot{q}_j(t_f)\}$, with t_i and t_f the initial and final times. The sets of generalized coordinates and generalized velocities define the state of the system.

[1] The reader may also follow the discussions of: Goldstein [1] and Arnold [2].

6.2 Principle of Stationary Action

The path followed by the particles is observed to make the action an extremum. In this regard, the action is postulated to be *stationary* for the real trajectory of the particles. The origin of the postulate comes from experimentation and calculations on both the paths of diffracted light at the interface of two mediums and the paths described by the particles in time and space. In reference to the paths described in time and space, consider a bead that slides without friction along a thin wire due to the action of gravity. The bead departs with zero speed from point A at a fixed height and reaches the lowest point B describing a particular trajectory, Fig. 6.1.

The experiments show that the sliding bead moves along the trajectory with the shortest traveling time. In mathematical terms, that path is an *extremum trajectory*, and in the case of the sliding bead, the trajectory corresponds to a *brachistochrone*.[2] The brachistochrone trajectory of the sliding bead is intimately related to the trajectory of a light beam diffracted at the interface of two mediums.[3] These observations are generalized by assuming similar behavior of the particles at the microscopic level, leading to a more general scenario. Thereby, the *principle of least action*, also known as the *Hamilton's principle*, is a fundamental piece of mechanics, allowing in a formal and systematic way to mathematically reach the equations of motion of the particles in general.

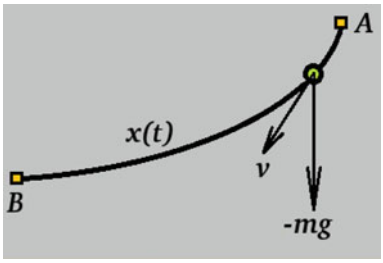


Fig. 6.1 A bead with mass m slides along a wire without friction from point A to point B under the action of gravity. The principle of stationary action indicates that among many possible trajectories, the path with minimum time of descent is the favored trajectory by nature

[2] In common jargon, nature is efficient as it considers the least-time path in the time evolution of a particle.

[3] The name brachistochrone is due to the Greek language (brachisto ∼ shortest, and chrone ∼ time). The trajectory that makes the action an extremum was found by Johann Bernoulli and corresponds to a brachistochrone curve. On the other hand, when light crossing two mediums with different indexes of refraction travels from point A to point B, it does so taking the fastest path. The light path is determined using the Fermat's principle and the Snell's law. The brachistochrone trajectory and the light path are closely related to each other: Levi [3].

Statement 6.1 Principle of stationary action: the motion of a particle from time t_1 to time t_2 is described by the stationary solution of the action. The action is S and is given as an integral of the Lagrangian over time.

$$S = \int_{t_1}^{t_2} L\, dt = \int_{t_1}^{t_2} (T - V)\, dt \tag{6.3}$$

The Lagrangian L is $T - V$, where T is the motion energy and V is the interaction energy. The stationary condition is $\delta S = 0$.

In contraposition to Newtonian mechanics, which is based on a cause-effect paradigm for describing the motion of the particles under the action of vector forces, analytical mechanics considers a common pattern in the motion of the particles, because it deals with the whole path of a particle in an interval of time in the definition of the action, postulating that the trajectory followed by the particle is a path of least time and formally establishing a variational principle on the action. The action has dependence on a Lagrangian function, which introduces the information of the system through the energy difference $T - V$, a scalar quantity which must have small variations over time, thus avoiding to directly deal with vector forces. The Lagrangian equations of motion are deduced from the condition:

$$\delta S = 0 \tag{6.4}$$

The principle of least action applies to the particle system in any time interval, and therefore, the time evolution of the system satisfies the Lagrangian equations of motion at every instant of time. The stationary condition indicates that the action may represent a minimal, maximal, or saddle point of a multidimensional surface generated by the generalized coordinates and generalized velocities. However, we consider a minimal value of S for the trajectory with the shortest traveling time. From the physical point of view, the stationary principle on the action replaces the Newtonian principles and is of special interest when the forces have little sense, such as in quantum mechanics. From a mathematical perspective, the stationary principle constitutes a variational principle that introduces an alternative mechanical formulation of Newtonian theory. The principle of stationary action demands the functional derivative of S to be zero, $\delta S = 0$, for small perturbations of $L = T - V$ in terms of the generalized coordinates and generalized velocities. The proper mathematical field to deal with the stationary condition on a function of several variables is *calculus of variations*. In the process of connecting calculus of variations to differential calculus, and imposing the stationary condition on the action function, we shall reach differential equations of the generalized variables q_i and $\dot{q}_i$. Such differential expressions are the equations of motion of the particles. This formulation makes use of the Lagrangian function and is known as *Lagrangian*

theory. After applying the initial conditions on the dynamics of the system, the solutions of the differential equations are unique and provide the time evolution of the particles, namely, the particle trajectories with time.

The Lagrangian theory has applications in different domains of physics, as in electromagnetic theory, quantum mechanics, relativistic mechanics, quantum field theory, etc. The quantum path integral method has been found to explain to some extent the principle of stationary action. This method takes every path of a quantum particle, like that of an electron or photon, into consideration to estimate the different probabilities to reach a final quantum state from an initial one. Here, the Schrödinger quantum equation plays the role of the diffusion equation. The wave phases far from a minimum-action path are found to cancel each other, only surviving the wave phases close to the minimum-action path. In the classical limit, when $S(x) \gg \hbar$ and $\hbar$ is the Planck constant, the path integral method leads to the principle of stationary action and explains the Fermat's principle of optics, which specifically states that a light beam travelling from point A to point B takes the fastest trajectory.

6.3 Classifying Molecular Systems

There are general definitions and important classifications which are required for the characterization of the particle systems. The definitions and the classifications shall be used later in the formulations of analytical mechanics. The system of particles may be able to preserve the energy when the forces exerted upon the particles are conservative, because they are derived from a potential function V in the form $\mathbf{f}_i = -\nabla_i V(\{\mathbf{r}_k\})$. When the forces are not associated with a potential function, they are non-conservative forces, and the system may absorb or dissipate energy.[4] On the other side, a system may be subjected to forces that impose constraints, for example, those forces that keep selected atomic bonds rigid. There are different types of constraints, and they have to be classified as well. The constraints are *holonomic* when they are written in terms of the particle coordinates, including the time, or when the constraints are written as differential equations. An example of constraints relating the coordinates is:

$$f(q_i, t) = 0 \tag{6.5}$$

An holonomic constraint satisfies the identity $df/dt = 0$. The constraints may be expressed in differential form.

$$df = \sum_j \frac{\partial f}{\partial q_j} dq_j + \frac{\partial f}{\partial t} dt \tag{6.6}$$

[4] Suppose a great number of particles interacting with a relatively large body. The individual particle forces are conservative and decelerate the body's velocity. If the interactions with the body are statistically averaged, we may write the average force as a frictional force, $-\eta v$, where η is the friction factor and v is the body's velocity. The negative sign indicates the opposition of the statistical force to the body's motion. The frictional force is non-conservative, in spite of representing a statistical average of the conservative forces. In summary, a simplified description of many conservative forces may result on a single non-conservative force acting on the body.

When the constraints are written in terms of mathematical inequalities, the constraints are *non-holonomic*. An example of a non-holonomic constraint involves a moving particle on the external surface of a sphere, in such a way that the particle is in contact with the sphere surface ($x^2 + y^2 + z^2 \geq R$, where the coordinates of the particle are (x, y, z) and the radius of the sphere is R). The particle moves and eventually falls from the sphere at some stage due to the action of gravity. The non-holonomic constraints may additionally include the velocities, $f(q_i, \dot{q}_i, t) = 0$. In general, it is difficult to transform non-holonomic expressions onto holonomic constraints, as the non-holonomic expressions are usually not integrable. There is no general method to deal with non-holonomic constrains. Yet, in many instances, it is possible to apply several methods at different stages of the problem for describing the particle motion. The Lagrangian equations of motion consider holonomic and non-holonomic constraints that are integrable.

The constraints may or may not show time dependence. In the case of an explicit time dependence, the system classifies as *rheonomous*; otherwise, it is *scleronomous*. The minimum number of independent variables required to specify the state of a particle system is referred to as the *degrees of freedom* of the system. In the general case, there are $3N$ spatial degrees of freedom of a system composed by N particles. If there are m spatial constraints relating the variables, then the number of independent degrees of freedom is reduced to $3N - m$. The spatial degrees of freedom are given by the independent variables $q_i(t)$, which all together built a mathematical space identified as the *configuration space*. When the momenta p_i are included in the description, the space formed by all the sets of variables $\{q_i(t), p_i(t)\}$ is the *phase space*. Without constraints on the coordinates and momenta, the number of degrees of freedom is $6N$. The set of $6N$ variables is visualized all together as forming a single point in the multidimensional phase space. In this case, the time evolution of the system is described by the trajectory of that single point in phase space, Fig. 6.2.

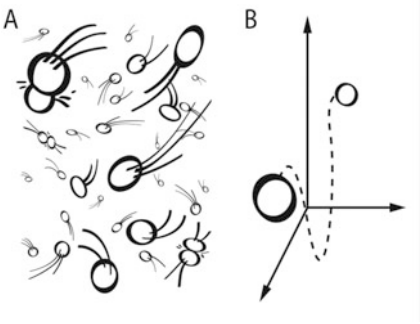


Fig. 6.2 Inset A: the phase space is a mathematical space with dimension determined by the total number of degrees of freedom of the particles. A state of the particle system is visualized as a single point in the multidimensional phase space. Inset B: the time evolution of the particle system defines the trajectory of the single point in phase space. In this regard, the phase space point is associated with the dynamical state of the mechanical system

6.4 Lagrange's Equations of Motion

Consider a set of interacting particles. The kinetic energy of the system is T, and the interaction among the particles is described by the potential energy V, a scalar function that has dependence on the coordinates q_i, velocities $\dot{q}_i$, and time t. The purpose of this section is to deduce the Lagrange equations from the energy terms T and V. The Lagrangian formulation is an alternative approach to the Newtonian equations to describe a system of particles.

The Lagrange equations may be derived either from the differential principle of d'Alembert or the integral principle of Hamilton. In the d'Alembert's approach, the virtual displacements $\delta\mathbf{r}_i$ occur in an instant of time t, not in the time interval dt when the real forces change. Yet, the virtual displacements are supposed to be compatible with the constraints and the real forces acting on the system. An important advantage of the d'Alembert's approach is the deduction of the Lagrangian equations of motion from the Newtonian equations of motion, giving evidence of the connection between Newtonian mechanics and Lagrangian mechanics. However, in the present section, we depart from the Hamilton's integral principle because it is a common way to deduce the equations of motion in many fields of physics.

According to the Hamilton's principle, the stationary requirement on the action, $\delta S = 0$, represents a formal equation of motion. There are no perceptible forces pushing the particles in the principle of least action, and in spite of that, the particles follow an optimal path.[5] For practical reasons, we favor differential equations of motion over an integral equation of motion. However, in order to reach a differential form of the stationary condition, where the forces become visible, the calculus of variations should be used. It is an important mathematical tool that provides differential equations in the search of extremal points (we use it to derive, e.g., the Laplace equation, the Helmholtz equation, and the wave equation, among others, and also solve problems like the Dido's isoperimetric problem and more).

The fact that the action function S is stationary indicates that small variations from the central trajectory precisely end up in the central trajectory, Fig. 6.3.

The correct nature of the perturbed paths is ensured by requiring mathematically well-behaved functions α_i, together with the introduction of the small parameter ε. Suppose that $q_i(t, 0)$ is the central trajectory and $q_i(t, \varepsilon)$ is the perturbed trajectory. Their relation is $q_i(t, \varepsilon) = q_i(t, 0) + \varepsilon\alpha_i(t)$. The α_i functions are continuous and non-singular, with continuous first and second derivatives, vanishing at the fixed boundary points t_1 and t_2, $\alpha_i(t_1) = 0$ and $\alpha_i(t_2) = 0$. We deal with the infinitesimal character of the variations by transforming the condition $\delta S = 0$ onto the condition $(\partial S/\partial\varepsilon)\, d\varepsilon = 0$ up to first order, where differential calculus is employed for finding extremal points. In this connection, the δS variation is replaced by partial derivatives.

[5] The unknown properties of spacetime in some sense impose the particle dynamics.

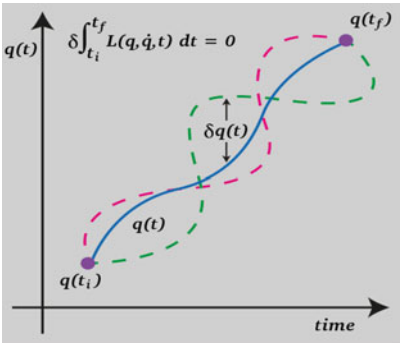


Fig. 6.3 Nature favors processes where the mechanical systems evolve along stationary states of the action. This is, the evolution of the particles between two states follows an extremum path, $\delta S = \delta \int L(q, \dot{q}, t)\, dt = 0$, in phase space. This constitutes the principle of least action, and is based on the balance of the kinetic energy and the potential energy present in the expression of the Lagrangian, L

$$\frac{\partial S}{\partial \varepsilon}\, d\varepsilon = \int_{t_1}^{t_2} \sum_i \left[\frac{\partial (T-V)}{\partial q_i} \frac{\partial q_i}{\partial \varepsilon}\, d\varepsilon + \frac{\partial (T-V)}{\partial \dot{q}_i} \frac{\partial \dot{q}_i}{\partial \varepsilon}\, d\varepsilon \right] dt$$

The integration-by-parts technique, $\int u\, dv = uv - \int v\, du$, is applied to the second term.

$$\int_{t_1}^{t_2} \underbrace{\frac{\partial}{\partial \dot{q}_i}(T-V)}_{u} \underbrace{\frac{\partial}{\partial t} \frac{\partial q_i}{\partial \varepsilon}\, dt}_{dv} = \frac{\partial (T-V)}{\partial \dot{q}_i} \frac{\partial q_i}{\partial \varepsilon} \Bigg|_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\partial q_i}{\partial \varepsilon} \frac{d}{dt} \left(\frac{\partial (T-V)}{\partial \dot{q}_i} \right) dt$$

The factors $\partial q_i / \partial \varepsilon$ are the α_i functions in the expansion of $q_i(t, \varepsilon)$. The α_i functions vanish at the instants of time t_1 and t_2 because $\alpha_i(t_1)$ and $\alpha_i(t_2)$ are fixed points. The surviving terms are collected.

$$\frac{\partial S}{\partial \varepsilon}\, d\varepsilon = \int_{t_1}^{t_2} \sum_i \left[\frac{\partial (T-V)}{\partial q_i} - \frac{d}{dt} \frac{\partial (T-V)}{\partial \dot{q}_i} \right] \frac{\partial q_i}{\partial \varepsilon}\, d\varepsilon\, dt$$

In order to impose the stationary condition of the action, the term $(\partial S / \partial \varepsilon) d\varepsilon$ is replaced by δS and similarly $(\partial q_i / \partial \varepsilon) d\varepsilon$ by δq_i, making the expression zero.

$$\delta S = \int_{t_1}^{t_2} \sum_i \left[\frac{\partial (T-V)}{\partial q_i} - \frac{d}{dt} \frac{\partial (T-V)}{\partial \dot{q}_i} \right] dt\, \delta q_i = 0$$

Since the n degrees of freedom q_i are independent of each other, the variations δq_i are also independent. The way to satisfy the condition $\delta S = 0$ under general terms is by making each coefficient of δq_i zero.

$$\frac{d}{dt} \left[\frac{\partial (T-V)}{\partial \dot{q}_i} \right] - \frac{\partial (T-V)}{\partial q_i} = 0 \quad ; \quad i = 1, \cdots, n$$

It is time to introduce the scalar-valued *Lagrangian function* $L = T - V$. The equations are abbreviated using this definition, and $i = 1, \cdots, n$

$$\frac{d}{dt}\left[\frac{\partial L(\{q_i\},\{\dot{q}_i\},t)}{\partial \dot{q}_i}\right] - \frac{\partial L(\{q_i\},\{\dot{q}_i\},t)}{\partial q_i} = 0 \tag{6.7}$$

These are the *Lagrange equations of motion* which extremize the action. They represent a set of coupled second-order differential equations for the independent degrees of freedom. The solutions describe the behavior of the mechanical system. The term in square parenthesis is known as the *generalized momentum*, p_i and is the conjugate variable of q_i (both form a duality under a Fourier transformation).[6]

$$p_i = \partial L / \partial \dot{q}_i \tag{6.8}$$

The generalized momentum p_i corresponds to the linear momentum when the Cartesian coordinates are used, but it may not necessarily represent the linear momentum when different coordinates are employed. If it happens in Eq. 6.7 that $\partial L / \partial q_i = 0$ for some q_i, the variable q_i is a *cyclic variable* or an *ignorable variable*, since q_i is not present in L, and p_i is conserved along the physical path.

$$\frac{dp_i}{dt} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = 0 \qquad \Rightarrow \qquad p_i = \text{ constant} \tag{6.9}$$

The Lagrangian equations of motion differ from the Newtonian equations of motion in the use of both the scalar function L and the generalized variables. The Lagrangian equations of motion have general character but become specific when T and V are proposed for the system under study. For instance, the Lagrangian equations of a planet moving in the Solar System are different to the Lagrangian equations of a charged body moving in an electric field because such systems differ in the types of interaction. The quantum states, like the excited molecular states, are excluded by the classical Lagrange equations of motion.

[6] In order to explain the conjugation of the variables q and p and simultaneously the independence of each other, suppose that a first observer in an inertial reference frame knows the position q and momentum p of a particle: q is an absolute quantity and p is a relative one, as it depends on relative positions and relative clock times (the Newtonian time intervals of different clocks may not necessarily be the same, yet we suppose equal clock times for the observers). If the first observer measures p at $q(t)$, a second observer may not ensure that p corresponds to the particle at $q(t)$, because the second observer is supposed to only know the position $q(t)$ and requires to know the change of positions and the relative time clocks registered at the location $q(t)$ by the first observer to deduce p at $q(t)$. In fact, the two observers require an equation (of motion) to link not only the variables $q(t) \Leftrightarrow p(t)$ but also their observations. The initial conditions uniquely determine such a relation and help synchronize the variables of the two observers, $q(t_0) \Leftrightarrow p(t_0)$ at a given instant of time, say $t_0 = 0$. Thus, q and p are not only independent variables but also conjugate variables, as both are required to define the particle motion.

Problem 6.1 A particle with mass m and instantaneous position $\mathbf{r} = (x, y, z)$ moves freely. The Lagrangian is $L = m\dot{\mathbf{r}}^2/2$. An observer rotates around the z axis with frequency $\boldsymbol{\omega} = (0, 0, \omega)$. The coordinates of the particle according to the observer are:

$$x' = x\cos(\omega t) + y\sin(\omega t) \quad ; \quad y' = y\cos(\omega t) - x\sin(\omega t) \quad ; \quad z' = z$$

Deduce the equations of motion in the rotating (primed) reference system.

Solution The Lagrangian in the primed reference system should be proposed to eventually deduce the equations of motion. We start by differentiating the primed position coordinates with respect to time.

$$\dot{x}' = \dot{x}\cos(\omega t) - \omega x\sin(\omega t) + \dot{y}\sin(\omega t) + \omega y\cos(\omega t)$$
$$\dot{y}' = \dot{y}\cos(\omega t) - \omega y\sin(\omega t) - \dot{x}\sin(\omega t) - \omega x\cos(\omega t) \quad ; \quad \dot{z}' = \dot{z}$$

The quantity $\dot{\mathbf{r}}^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2$ is written in terms of the primed variables in the form:

$$\dot{x}' - \omega y' = \dot{x}\cos(\omega t) + \dot{y}\sin(\omega t) \quad ; \quad \dot{y}' + \omega x' = \dot{y}\cos(\omega t) - \dot{x}\sin(\omega t)$$
$$\dot{\mathbf{r}}^2 = (\dot{x}' - \omega y')^2 + (\dot{y}' + \omega x')^2 + (\dot{z}')^2$$

The result is used to build the transformed Lagrangian.

$$L = \tfrac{m}{2}[(\dot{x}' - \omega y')^2 + (\dot{y}' + \omega x')^2 + (\dot{z}')^2] = \tfrac{m}{2}(\dot{\mathbf{r}}' + \boldsymbol{\omega} \times \mathbf{r}')^2$$

The equation of motion is determined from $d(\partial L/\partial \dot{q}')/dt - \partial L/\partial q' = 0$.

$$\tfrac{d}{dt}\left(\tfrac{\partial L}{\partial \dot{x}'}\right) - \tfrac{\partial L}{\partial x'} = m(\ddot{x}' - 2\omega\dot{y}' - \omega^2 x') = 0$$
$$\tfrac{d}{dt}\left(\tfrac{\partial L}{\partial \dot{y}'}\right) - \tfrac{\partial L}{\partial y'} = m(\ddot{y}' + 2\omega\dot{x}' - \omega^2 y') = 0 \; ; \; \tfrac{d}{dt}\left(\tfrac{\partial L}{\partial \dot{z}'}\right) - \tfrac{\partial L}{\partial z'} = m\ddot{z}' = 0$$

We add the previous expressions to have a single equation of motion in vector form.

$$m[\ddot{\mathbf{r}}' + 2\omega(-\dot{y}', \dot{x}', 0) + \omega^2(-x', -y', 0)] = m[\ddot{\mathbf{r}}' + 2\boldsymbol{\omega} \times \dot{\mathbf{r}}' + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}')$$
$$= 0$$

This is the equation of a particle with angular motion.

6.5 Newtonian Equations of Motion from Lagrange Theory

In order to illustrate the application of Lagrangian theory, consider a system of particles with kinetic energy $T = \sum_i m_i v_i^2/2$, interaction potential V with dependence on the spatial coordinates of the particles, such as $V = \sum_i k x_i^2/2$. The Cartesian coordinates are conveniently considered the generalized variables in this case. The Lagrangian equation of motion of the ith particle, with position x_i and speed v_{ix} in the x axis, acquires the form:

$$\begin{gathered} \frac{d}{dt}\frac{\partial L}{\partial v_{ix}} - \frac{\partial L}{\partial x_i} = \frac{d}{dt}\frac{\partial}{\partial v_{ix}} \sum_j \frac{m_k}{2}\left(v_{jx}^2 + v_{jy}^2 + v_{jz}^2\right) + \frac{\partial V}{\partial x_i} = 0 \\ m_i \frac{dv_{ix}}{dt} = -\frac{\partial V}{\partial x_i} \quad ; \quad \frac{\partial V}{\partial x_i} = k x_i \end{gathered} \tag{6.10}$$

The equation of motion of particle i is that of the harmonic oscillator, $m_i d^2 x_i/dt^2 = -k x_i$. In terms of the energy functions T and V, and the conjugate variables q and $\dot{q}$ of the Lagrangian, it is observed that:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}}\right) = -\frac{\partial V}{\partial q} \quad \Rightarrow \quad \frac{dp}{dt} = -\frac{\partial V}{\partial q} \tag{6.11}$$

The Lagrange's equations of motion are originated from a universal principle, and they are expected to be invariant with respect to different types of coordinates and applicable to all the mechanical systems. The Newtonian laws appear to be implicitly included in the principle of stationary action. In this regard, the Lagrangian equations of motion are presented as an alternative to the Newtonian equations of motion. The change of formulations, from Newtonian theory to Lagrangian theory and vice versa, is particularly useful because some problems are more amenable under certain formulations than others.

Problem 6.2 The Lagrangian of a particle is given in spherical coordinates $L = m\left(\dot{r}^2 + r^2\dot{\theta}^2\right)/2$. Obtain the equations of motion from L. Consider the solution $z = re^{i\theta}$, and show that it corresponds to a free moving particle.

Solution The independent degrees of freedom are the radial distance r and the angular variable θ. The equations of motion of these variables are $d(\partial L/\partial \dot{r})/dt - \partial L/\partial r = 0$, and similarly for θ. After performing the derivations, the equations of motion are:

$$\ddot{r} - r\dot{\theta}^2 = 0 \quad ; \quad \ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta}^2 = 0$$

(continued)

If $z = re^{i\theta}$ then, by considering the general expression $de^{if(t)}/dt = ie^{if(t)}\,df(t)/dt$, the second derivative of z is $\ddot{z} = (\ddot{r} - r\dot{\theta}^2) + i(2\dot{r}\dot{\theta} + r\ddot{\theta})$. According to the equations of motion, each term is zero. Thus, $\dot{z} =$ constant, indicating that the real and imaginary parts of $z = r\cos\theta + ir\sin\theta$ depend linearly on t.

$$r\cos\theta = \alpha t + \beta \quad ; \quad r\sin\theta = \alpha' t + \beta'$$

The variables r and θ are coupled in the solutions. By doing $x = r\cos\theta$ and $y = r\sin\theta$, we obtain $x = \alpha t + \beta$ and $y = \alpha' t + \beta'$. Therefore, L describes a free particle in motion.

6.6 Non-uniqueness of the Lagrangian

In principle, it is possible to propose different Lagrangians of a single mechanical system and obtain the same trajectories of the particles. For instance, we may use L or αL, where α is a real number, and the Lagrangian equations of motion are the same. The Lagrangians may differ by more than a constant factor. Suppose that L is the Lagrangian of a system of particles and L' is a different Lagrangian of the same system of particles. The interest is to demonstrate that the different Lagrangians are related by the time derivative of a function, which depends on the generalized coordinates and time. We start from the stationary action.

$$\delta \int_{t_1}^{t_2} L(\{q_i, \dot{q}_i\}, t)\, dt = 0$$

We assume that L' is related to L in the form:

$$L'(\{q_i, \dot{q}_i\}, t) = L(\{q_i, \dot{q}_i\}, t) + \frac{dF(\{q_i\},t)}{dt} \tag{6.12}$$

F is a mathematically well-behaved function of the generalized coordinates and time. If L' is the Lagrangian of the same particle system, then it also makes the action stationary.

$$\delta \int_{t_1}^{t_2} L'(\{q_i, \dot{q}_i\}, t)\, dt = \delta \int_{t_1}^{t_2} L(\{q_i, \dot{q}_i\}, t)\, dt + \delta F(\{q_i\}, t_2) - \delta F(\{q_i\}, t_1) = 0$$

The F function shows no variation at t_1 and t_2, and $\delta F(\{q_i\}, t_i) = 0$, with $i = 1, 2$.

$$\delta \int_{t_1}^{t_2} L(\{q_i, \dot{q}_i\}, t)dt = \delta \int_{t_1}^{t_2} L'(\{q_i, \dot{q}_i\}, t)dt = 0 \tag{6.13}$$

It is observed that L' is equivalent to the Lagrangian L. In short, a system of particles may be described by different Lagrangians; however, they are related to each other. The Lagrangian equations, Eq. 6.7, are invariant under the different representations of L.

$$\frac{d}{dt}\left[\frac{\partial L(\{q_i\},\{\dot{q}_i\},t)}{\partial \dot{q}_i}\right] - \frac{\partial L(\{q_i\},\{\dot{q}_i\},t)}{\partial q_i} = \frac{d}{dt}\left[\frac{\partial L'(\{q_i\},\{\dot{q}_i\},t)}{\partial \dot{q}_i}\right]$$
$$-\frac{\partial L'(\{q_i\},\{\dot{q}_i\},t)}{\partial q_i} = 0 \quad ; \quad i = 1,\cdots,n$$

Problem 6.3 A particle with mass m and charge e moves in an electromagnetic field. The electric field, $\mathbf{E}$, and magnetic field, $\mathbf{B}$, are given in terms of the magnetic vector potential, $\mathbf{A}$, and the electric scalar potential, ϕ.

$$\mathbf{E} = -\nabla\phi - \tfrac{1}{c}\tfrac{\partial \mathbf{A}}{\partial t} \quad ; \quad \mathbf{B} = \nabla \times \mathbf{A}$$

Both $\mathbf{A}$ and ϕ depend on the coordinate $\mathbf{r}$ and time t. The speed of light is c. The Lagrangian of the particle is:

$$L = m\dot{\mathbf{r}}^2/2 - e\phi + e\,\dot{\mathbf{r}}\cdot\mathbf{A}/c$$

Show that the force exerted on the particle depends on the velocity and corresponds to the Lorentz force.

Solution The potential is a velocity-dependent function; still, the Lagrangian approach is appropriate to establish the equation of motion. The strategy is to obtain the equation of motion and, later, the force acting on the particle. We compute the generalized momentum from the Lagrangian, $p = \partial L/\partial \dot{r}$.

$$\mathbf{p} = m\dot{\mathbf{r}} + e\mathbf{A}/c$$

We are in the position to formulate the Lagrangian equation of motion $d\mathbf{p}/dt - \partial L/\partial r = 0$, taking care of the vectors in the derivation.

$$\frac{d}{dt}(m\dot{\mathbf{r}} + \frac{e}{c}\mathbf{A}) + e\nabla\phi - \frac{e}{c}\nabla(\dot{\mathbf{r}}\cdot\mathbf{A}) = m\ddot{\mathbf{r}} + \frac{e}{c}\sum_j \frac{\partial \mathbf{A}}{\partial \mathbf{r}_j}\cdot\dot{\mathbf{r}}_j + \frac{e}{c}\frac{\partial \mathbf{A}}{\partial t} + e\nabla\phi$$
$$-\frac{e}{c}\nabla(\dot{\mathbf{r}}\cdot\mathbf{A}) = 0$$

(continued)

For two arbitrary vector functions, $\mathbf{F}$ and $\mathbf{G}$, we have the vector identity $\nabla(\mathbf{F}\cdot\mathbf{G}) = \mathbf{F}\times(\nabla\times\mathbf{G}) + (\mathbf{F}\cdot\nabla)\mathbf{G} + \mathbf{G}\times(\nabla\times\mathbf{F}) + (\mathbf{G}\cdot\nabla)\mathbf{F}$. By choosing $\mathbf{F} = \dot{\mathbf{r}}$ and $\mathbf{G} = \mathbf{A}$, we obtain:

$$m\ddot{\mathbf{r}} + e\left(\nabla\phi + \frac{1}{c}\frac{\partial\mathbf{A}}{\partial t}\right) + \frac{e}{c}\left[(\dot{\mathbf{r}}\cdot\nabla)\mathbf{A} - \nabla(\dot{\mathbf{r}}\cdot\mathbf{A})\right]$$
$$= m\ddot{\mathbf{r}} + e\left(\nabla\phi + \frac{1}{c}\frac{\partial\mathbf{A}}{\partial t}\right) - \frac{e}{c}\left[\dot{\mathbf{r}}\times(\nabla\times\mathbf{A})\right] = 0$$

The expressions of the fields $\mathbf{E}$ and $\mathbf{B}$ are employed in the equation.

$$m\ddot{\mathbf{r}} = e\mathbf{E} + (e/c)\,\dot{\mathbf{r}}\times\mathbf{B}$$

This is the equation of motion of the particle; it includes the velocity and is known as the *Lorentz force*, an important expression in electromagnetic theory.

An interesting example of the non-uniqueness property of the Lagrangian is the *gauge invariance* in electrodynamics. According to electromagnetic theory, the vector and scalar potentials, $\mathbf{A}$ and ϕ, are non-unique, and different potentials reproduce the same electric and magnetic fields.

$$\mathbf{A}' = \mathbf{A} + c\,\nabla\Lambda \quad ; \quad \phi' = \phi - \partial\Lambda/\partial t$$

The function Λ may be a function of space and time; it is the gauge. The transformed Lagrangian L' may be expressed in terms of L, but the expression is simplified when the total derivative of Λ with respect to time is used.

$$L' = m\dot{\mathbf{r}}^2/2 - e\left(\phi - \tfrac{\partial\Lambda}{\partial t}\right) + \tfrac{e}{c}\,\dot{\mathbf{r}}\cdot(\mathbf{A} + c\nabla\Lambda) = L + e\left(\tfrac{\partial\Lambda}{\partial t} + \dot{\mathbf{r}}\cdot\nabla\Lambda\right)$$
$$= L + e\,d\Lambda/dt \quad ; \quad L = m\dot{\mathbf{r}}^2/2 - e\phi + e\,\dot{\mathbf{r}}\cdot\mathbf{A}/c$$

This takes us back to Eq. 6.12. In short, the gauge invariance is consistent with the non-uniqueness property of the Lagrangian. Interestingly, the gauge shift in the electromagnetic field is coupled with the phase shift in the quantum wave function of a particle with electric charge. Such a coupling establishes the conservation of electric charge.

6.7 Invariance of the Lagrange Equations of Motion

The next interest is to show that the Lagrange equations maintain the same functional form under a transformation of the conjugate variables, thus making the Lagrangian theory independent of the types of coordinates used to describe the system. We start by assuming two sets of variables $\{q_i\}$ and $\{x_i\}$, which can be transformed onto one another.

$$\begin{aligned} q_i &= q_i(\{x_\alpha\}, t) \quad ; \quad x_\alpha = x_\alpha(\{q_i\}, t) \\ \frac{dq_i}{dt} &= \sum_\alpha \frac{\partial q_i}{\partial x_\alpha}\dot{x}_\alpha + \frac{\partial q_i}{\partial t} \quad ; \quad \frac{dx_\alpha}{dt} = \sum_i \frac{\partial x_\alpha}{\partial q_i}\dot{q}_i + \frac{\partial x_\alpha}{\partial t} \end{aligned} \tag{6.14}$$

We require to show the equivalence between the Lagrange equations in the variables q_i and x_α by making use of the above transformations. To do this, the variables x_α are assumed to satisfy the Lagrange equations. We now have to show that the variables q_i satisfy the Lagrange equations, in other words, that the following expression is zero:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = \sum_\alpha \left\{ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_\alpha}\frac{\partial \dot{x}_\alpha}{\partial \dot{q}_i}\right) - \left[\frac{\partial L}{\partial x_\alpha}\frac{\partial x_\alpha}{\partial q_i} + \frac{\partial L}{\partial \dot{x}_\alpha}\frac{\partial \dot{x}_\alpha}{\partial q_i}\right]\right\}$$

We use the last identity of Eqs. 6.14 in the last contribution where $\dot{x}_\alpha$ is present.

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = \sum_\alpha \Bigg\{ & \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_\alpha}\right)\frac{\partial \dot{x}_\alpha}{\partial \dot{q}_i} + \frac{\partial L}{\partial \dot{x}_\alpha}\frac{d}{dt}\left(\frac{\partial \dot{x}_\alpha}{\partial \dot{q}_i}\right) - \\ & - \left[\frac{\partial L}{\partial x_\alpha}\frac{\partial x_\alpha}{\partial q_i} + \frac{\partial L}{\partial \dot{x}_\alpha}\frac{\partial}{\partial q_i}\sum_j\left(\frac{\partial x_\alpha}{\partial q_j}\dot{q}_j + \frac{\partial x_\alpha}{\partial t}\right)\right]\Bigg\} \end{aligned}$$

By carrying the derivation $d(\partial \dot{x}_\alpha/\partial \dot{q}_i)/dt$ in the last term of the first line and applying the derivative $\partial/\partial q_i$ in the last term of the second line, we have:

$$\begin{aligned} \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \sum_\alpha \Bigg[& \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_\alpha}\right)\frac{\partial \dot{x}_\alpha}{\partial \dot{q}_i} - \frac{\partial L}{\partial x_\alpha}\frac{\partial x_\alpha}{\partial q_i} + \frac{\partial L}{\partial \dot{x}_\alpha}\frac{d}{dt}\left(\frac{\partial x_\alpha}{\partial q_i}\right) \\ & - \frac{\partial L}{\partial \dot{x}_\alpha}\sum_j\left(\frac{\partial^2 x_\alpha}{\partial q_i \partial q_j}\dot{q}_j + \frac{\partial}{\partial q_i}\frac{\partial x_\alpha}{\partial t}\right)\Bigg] \end{aligned}$$

By using the chain rule in the term $d(\partial x_\alpha/\partial q_i)/dt$, it is observed that it reproduces the summation of the last contribution of the above equation but with opposite sign.

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = \sum_\alpha \left[\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_\alpha}\right)\frac{\partial \dot{x}_\alpha}{\partial \dot{q}_i} - \frac{\partial L}{\partial x_\alpha}\frac{\partial x_\alpha}{\partial q_i}\right]$$

Once more, we use the last identity of Eqs. 6.14 to show the equivalence $\partial \dot{x}_\alpha / \partial \dot{q}_i = \partial x_\alpha / \partial q_i$. By using this identity in the above expression, we have:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = \sum_\alpha \left[\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_\alpha}\right) - \frac{\partial L}{\partial x_\alpha}\right] \frac{\partial x_\alpha}{\partial q_i}$$

If the relation between the coordinates q_i and x_α is invertible, then we know that $\partial x_\alpha / \partial q_i \neq 0$, and it is the other coefficient which should be zero. By considering that the variables x_α satisfy the Lagrange equations, we conclude that the variables q_i equally satisfy such equations.

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

6.8 Motion with Constraints

When a system of particles is subjected to restraints, the Lagrangian equations of motion have to consider the imposed conditions to properly describe the particle behaviors. We assume a number r of restraints denoted by functions f_i, involving the variables q_i and time t.

$$f_1(q_1, \cdots, q_n, t) = 0 \quad ; \quad f_2(q_1, \cdots, q_n, t) = 0 \quad \cdots \quad f_r(q_1, \cdots, q_n, t) = 0 \tag{6.15}$$

The constraints f_i may be of holonomic type, and some others may be of non-holonomic type. An example of common holonomic constraints is the rigid bond, $|\mathbf{r}_i(t) - \mathbf{r}_j(t)| = c_{ij}$, where the coordinates of selected particles participate.[7] There is a number r of variables q_j which are not independent of the others. For mathematical manipulations, it is convenient to consider the restraints in their differential forms.

$$\begin{gathered} df_i/dt = 0 \quad \Rightarrow \quad \sum_{j=1}^{n} c_{ij}\, dq_j + c_{it}\, dt = 0 \\ c_{ij} = \partial f_i / \partial q_j \quad ; \quad c_{it} = \partial f_i / \partial t \quad ; \quad i = 1, \cdots, r \end{gathered} \tag{6.16}$$

The r equations on the variables q_j are linear and integrable expressions. It is important that the *differential forms* of the constraints acquire expressions similar to the last identity of the first line. The constraints expressed with inequalities are avoided. The strategy to properly describe the time evolution of the particles consists on incorporating the restraints into the Hamilton's integral to define an

[7] In a restrained bond like $|\mathbf{r}_i(t) - \mathbf{r}_j(t)| = c_{ij}$, the atoms i and j move according to an equation of motion, and the distance between them is concurrently maintained constant by satisfying the constraint that establishes the relation between the variables $\mathbf{r}_i$ and $\mathbf{r}_j$.

unconstrained Lagrangian from which the equations of motion should be deduced. In this regard, the equations of motion derived from the unconstrained Lagrangian should generate the same phase-space points as those deduced from the constrained Lagrangian, where the generalized coordinates also satisfy the r constraints f_i, but in separate form from the ordinary Lagrangian equations of motion.

In this process, the initial independent variables are $n - r$ due to r constraints. By introducing the differential forms of the r constraints into the motion equations, we are able to have n differential equations to solve, with the cost of introducing r Lagrange multipliers. So we now have $n + r$ variables and only n motion equations. However, we also have to satisfy the r constraints in their non-differential forms. By considering these additional equations, we are able to have a system of $n + r$ equations with $n + r$ unknowns.

The first step to incorporate the constraints into the equations of motion is to deal with a number r of additional terms to the n variables q_j and add an equal number of equations with those terms to determine the solutions in unique form. The new terms are denoted by λ_i; they are linked to the coordinates q_j and time t and are referred to as the *Lagrange multipliers*. The number r of multipliers corresponds to the r constraints. The constraints are introduced in the main equations via their differential forms 6.16, because such forms can be accommodated in the integral expression of the action. Since the variations δq_i of the generalized coordinates produce deviations δf_i, we have:

$$\lambda_i\, \delta f_i = \lambda_i \sum_{j=1}^{n} c_{ij}\, \delta q_j = 0 \quad \Rightarrow \quad \int_{t_1}^{t_2} dt \left[\sum_{i=1}^{r} \sum_{j=1}^{n} \lambda_i\, c_{ij}\, \delta q_j \right] = 0 \tag{6.17}$$

The approach to be displayed is known as the *Lagrange method of undetermined multipliers* [4]. The constraints in their differential forms are now included (as a "smart zero") in the kernel of the Hamilton's integral to follow a similar treatment in the deduction of the Lagrangian equations of motion.

$$\int_{t_1}^{t_2} dt \sum_{j=1}^{n} \left[\frac{\partial L}{\partial q_j} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} + \sum_{i=1}^{r} \lambda_i\, c_{ij} \right] \delta q_j = 0 \tag{6.18}$$

We have the freedom to choose the coefficients λ_i. They are proposed in such a way to satisfy the equations:

$$\frac{\partial L}{\partial q_j} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} + \sum_{i=1}^{r} \lambda_i\, c_{ij} = 0 \quad ; \quad j = 1, \cdots, r \tag{6.19}$$

It is with these equations that the multipliers are included in the dynamics. After making zero the first r terms of Eq. 6.18, the Hamilton's integral maintains the functional form for the remaining terms with subindexes $r + 1, r + 2, \cdots, n$.

$$\int_{t_1}^{t_2} dt \sum_{j=r+1}^{n} \left[\frac{\partial L}{\partial q_j} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} + \sum_{i=1}^{r} \lambda_i\, c_{ij} \right] \delta q_j = 0 \tag{6.20}$$

Once the Lagrange multipliers are decided, the coordinates $q_{r+1}, \cdots, q_n$ in the kernel are now independent of each other. Thus, for arbitrary variations of δq_j, the terms in square brackets are equal to zero.

$$\frac{\partial L}{\partial q_j} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j} + \sum_{i=1}^{r} \lambda_i \, c_{ij} = 0 \quad ; \quad j = r+1, \cdots, n \tag{6.21}$$

It is convenient to write Eqs. 6.19 and 6.21 together because they have the same forms.

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = \sum_{i=1}^{r} \lambda_i \, c_{ij} \quad ; \quad j = 1, \cdots, n \tag{6.22}$$

At this stage, the equations of motion include a number r of constraints in their differential forms and using the Lagrange factors λ_i. We have n differential equations of the kind 6.22 but $n + r$ unknowns. In order to have the number of equations equal to the number of variables, we consider the r constraints of type f_i (in their non-differential forms). This procedure establishes a system of $n + r$ variables with $n+r$ equations. Once the solutions of Eqs. 6.15 and 6.22 are obtained consistently, we have the trajectory of the particles simultaneously satisfying the constraints. In this regard, the point $(\mathbf{q}, \boldsymbol{\lambda})$ of the unconstrained Lagrangian contains the point $(\mathbf{q}, 0)$ of the constrained Lagrangian in phase space. The equations are further simplified by using Eq. 6.16 for the coefficients c_{ij}, that is, $c_{ij} = \partial f_i / \partial q_j$, and introducing the *generalized forces*, Q_j, in Eq. 6.22.

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = Q_j \quad ; \quad Q_j = \sum_{i=1}^{r} \lambda_i \, \frac{\partial f_i}{\partial q_j} \quad ; \quad j = 1, \cdots, n \tag{6.23}$$

The Lagrange multipliers λ_i are identified in terms of the basic variables and the generalized forces from the Lagrange multipliers. For conservative systems, we have $Q_j = 0$, because the Lagrange equations already consider the cases with conservative forces via the V potential of the Lagrangian. However, in addition to the constraining forces, the Q_j may equally represent generalized non-conservative forces as they are restrictive forces on the particle dynamics. The relation of the Q_j forces to the standard $\mathbf{f}_j$ forces is established with the transformations of the coordinates $\mathbf{r}_i$ and q_i.

$$\delta \mathbf{r}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \, \delta q_j \quad ; \quad \sum_i \mathbf{f}_i \cdot \delta \mathbf{r}_i = \sum_{i,j} \mathbf{f}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \, \delta q_j = \sum_j Q_j \, \delta q_j$$

The Q_j forces are defined in terms of the standard forces $\mathbf{f}_i$ and positions $\mathbf{r}_i$ in the form:

$$Q_j = \sum_i \mathbf{f}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \tag{6.24}$$

The generalized forces may represent not only forces but also torques. In special cases, the non-conservative forces may be derived from a function, say D, known as a *generalized potential function*. In order to illustrate the Lagrange equations of

motion with a generalized potential function, let us consider *Rayleigh dissipative forces* with dependence on the speeds of the particles and constant coefficients k_x, k_y, k_z.

$$D = \sum_{i=1}^{n} \left(k_x v_{ix}^2 + k_y v_{iy}^2 + k_z v_{iz}^2 \right) /2 \tag{6.25}$$

The generalized potential function may take a linear form on the kinetic energy or a quadratic form on the speeds. It differs from the potential which is quadratic on the position coordinates, $V = \sum_i (k_x x_i^2 + k_y y_i^2 + k_z z_i^2)/2$. The quantity $2D$ is understood as the time rate of energy dissipation due to the friction force $\mathbf{f}^f = -\nabla_v D$.

$$\frac{dE^f}{dt} = -\mathbf{f}^f \cdot \frac{d\mathbf{r}}{dt} = -(\mathbf{f}^f \cdot \mathbf{v}) = (\nabla_v D \cdot \mathbf{v}) = \left(k_x v_x^2 + k_y v_y^2 + k_z v_z^2 \right) = 2D$$

The generalized force associated with the friction force is obtained from Eq. 6.24 and using the fact $\partial \dot{x}_\alpha / \partial \dot{q}_i = \partial x_\alpha / \partial q_i$ from Eqs. 6.14.

$$Q_j = \sum_i \mathbf{f}_i^f \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = -\sum_i \nabla_{v_i} D \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = -\sum_i \nabla_{v_i} D \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j} = -\frac{\partial D}{\partial \dot{q}_j}$$

The equation of motion 6.23 of a particle that includes the Rayleigh dissipative force D is:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = -\frac{\partial D}{\partial \dot{q}_j} \tag{6.26}$$

By considering Cartesian coordinates in the Lagrangian, $L = T - V$, and using the above expression, we identify equations of motion which include *kinematic friction* effects.

$$\begin{gathered} \frac{d}{dt}\frac{\partial L}{\partial v_{ix}} - \frac{\partial L}{\partial x_i} = \frac{d}{dt}\frac{\partial}{\partial v_{ix}} \sum_k \frac{m_k}{2} \left[v_{kx}^2 + v_{ky}^2 + v_{kz}^2 \right] + \frac{\partial V}{\partial x_i} = -\frac{\partial D}{\partial v_{ix}} \\ \frac{dp_{ix}}{dt} = -\frac{\partial V}{\partial x_i} - k_x v_{ix} \end{gathered} \tag{6.27}$$

The *viscous term* is negatively proportional to the particle speed. When the viscous term is neglected, the traditional Newtonian equation of motion is reproduced. A disk rolling down a tilted plane illustrates the importance of the generalized forces, Fig. 6.4. In conclusion, the constraining forces may be incorporated into the equations of motion via the Lagrange method of undetermined multipliers. Such a method is also employed in other fields of science, like in statistical physics to obtain the maximum entropy of a system with a large number of particles.

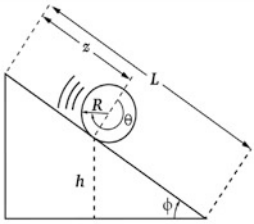


Fig. 6.4 A disk of mass m, radius R, and inertia moment $I = mR^2/2$, rolls down a tilted plane without slipping due to the action of gravity. The plane has length L and makes an angle ϕ with the horizontal axis. The instantaneous position of the disk in the plane is $(L - z)$. The radius R is sufficiently small with respect to the instantaneous height h in the plane ($h \gg R$)

Problem 6.4 A disk with mass m, radius R, and inertia moment $I = mR^2/2$ rolls down a tilted plane without slipping due to the action of gravity. The plane has length L and makes an angle ϕ with the horizontal axis, Fig. 6.4. Obtain the equations of motion.

Solution The kinetic energy of the disk is due to translation and rotation, Sect. 5.3, $T = m\dot{z}^2/2 + I\dot{\theta}^2/2$. The position of the disk on the plane is $(L - z)$, and the height of the disk mass center is $h = (L - z)\sin\phi$. The corresponding potential energy is $V = mg(L - z)\sin\phi$. The Lagrangian is:

$$L = T - V = \tfrac{m}{2}\,\dot{z}^2 + \tfrac{mR^2}{4}\,\dot{\theta}^2 - mg(L - z)\sin\phi$$

The non-sliding constraint on the disk motion relates the translation distance to the rotated distance of the disk by $dz = Rd\theta$, and thus, the constraint is $f = z - R\theta = 0$. The equations of motion in z and θ are determined from Eqs. 6.23.

$$m\ddot{z} - mg\sin\phi - \lambda = 0 \quad ; \quad \tfrac{mR^2}{2}\,\ddot{\theta} + \lambda R = 0 \quad ; \quad f = z - R\theta = 0$$

These are three coupled equations with unknown variables z, θ, and λ. The equations are decoupled by using the last identity in the second equation and solving for λ, which in turn is used in the equation of z. The final expressions are:

$$\ddot{z} = \tfrac{2}{3}\,g\sin\phi \quad ; \quad \ddot{\theta} = \tfrac{2}{3}\,\tfrac{g\sin\phi}{R} \quad ; \quad \lambda = -\tfrac{1}{3}\,mg\sin\phi$$

The restraining forces, Eqs. 6.23, in terms of the basic variables are:

$$Q_z = \lambda\,\tfrac{\partial f}{\partial z} = \lambda = -\tfrac{1}{3}\,mg\sin\phi \quad ; \quad Q_\theta = \lambda\,\tfrac{\partial f}{\partial \theta} = -\lambda R = \tfrac{1}{3}\,mgR\sin\phi$$

6.9 Hamilton's Function

In this section, we define the $\mathcal{H}$ function, known as the *Hamiltonian*. It is a quantity of significance on the same right as the Lagrangian, because it equally encodes the relevant information of the system. The Lagrangian and Hamiltonian are related to each other. In this regard, it is also possible to formulate equations of motion based on the Hamiltonian.[8]

In the following lines, we demonstrate that the Lagrangian and Hamiltonian are linked by a Legendre transformation.[9] To do this, we consider a function $f(x)$ encoding information in terms of the variable x. The Legendre transform is used to obtain an equivalent function, say g, that contains the same information of f but in terms of the variable s, were s is the derivative of f with respect to x, $s = df/dx$. A geometrical interpretation of the Legendre transform is given in Fig. 6.5. In order to link f and g, the functions $x(s)$ and $s(x)$ should be single-valued expressions with a one-to-one mapping relation, showing the possibility of inverting them. The Legendre transformation considers a symmetric relation between the functions f and g and their variables x and s in the form:

$$f(x) + g(s) = sx \tag{6.28}$$

The variables x and s are conjugate to each other, but only one of them is the independent variable. When x plays the role of the independent variable, we have $f(x) + g(s(x)) = s(x)\,x$, and for the case of s, it is $f(x(s)) + g(s) = sx(s)$. By differentiating expression 6.28 with respect to x and s, the symmetric expressions in the variables x and s are observed.

$$\left.\begin{aligned} \tfrac{df(x)}{dx} + \tfrac{dg(s(x))}{ds}\,\tfrac{ds}{dx} &= x\,\tfrac{ds}{dx} + s(x) \\ \tfrac{dg(s)}{ds} + \tfrac{df(x(s))}{dx}\,\tfrac{dx}{ds} &= s\,\tfrac{dx}{ds} + x(s) \end{aligned}\right\} \Rightarrow x(s) = \tfrac{dg(s)}{ds} \quad ; \quad s(x) = \tfrac{df(x)}{dx} \tag{6.29}$$

From the results on the right-hand side, the symmetry between f and g goes to higher derivatives.

$$\left.\begin{aligned} \tfrac{dx}{ds} &= \tfrac{d^2g}{ds^2} \\ \tfrac{ds}{dx} &= \tfrac{d^2f}{dx^2} \end{aligned}\right\} \Rightarrow \left(\tfrac{dx}{ds}\right)\left(\tfrac{ds}{dx}\right) = 1 \quad ; \quad \tfrac{d^2g}{ds^2}\,\tfrac{d^2f}{dx^2} = 1$$

[8] The Lagrangian function stores information of the particle system in the form $L = T - V$. From an initial impression, one may wonder on the possibility of similarly storing that information in a function, say $\mathcal{H}$, of the same particle system in the form $\mathcal{H} = T + V$. Such a possibility motivates the exploration of an equivalent approach in terms of $\mathcal{H}$, as an alternative to L. This is a rational ground for the proposal of a new formulation, which we now recognize as Hamiltonian theory.

[9] In addition to the Fourier and Laplace transforms, the Legendre transform presents an alternative way to characterize a function in terms of its own derivatives: Zia et al. [5].

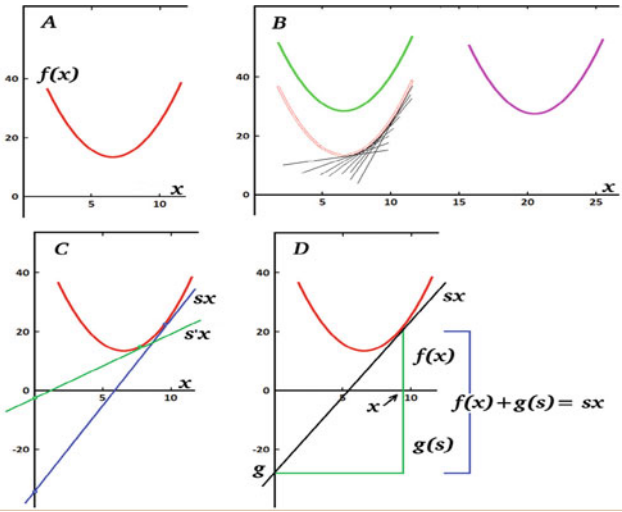


Fig. 6.5 Inset A shows the plot of a function f in terms of the independent variable x. The function is convex and smooth. Inset B shows similar curves, they all originate tangent lines equal to the tangent lines of the curve in A (depicted at the bottom of inset B), but the tangent lines differ in the intersections with the vertical axis. The function $f(x)$ of inset A is implicitly determined when the slopes and the correct intersection points with the vertical axis of the tangent lines are found, inset C. A tangent line at x is shown in inset D. It has slope s, and the intersection point with the vertical axis is $-g$. The quantity g is visualized as a function of the variable s. The identity $y = sx$ is the equation of the line when the origin is at point g. The equation of the tangent line is described by $f(x) = sx - g(s)$, where f depends on x and g depends on s

One of the important properties of the Legendre transform resides on the fact that the transform of g returns the function f and vice versa. The results may be generalized to any number of independent variables. The Legendre transform of the Lagrangian takes us to the formal proposal of the Hamiltonian, understood as an alternative function to code the information of the particle system. In analogy to Eq. 6.28, the link between the Hamiltonian and Lagrangian is given in terms of the generalized variables.

$$\mathcal{H} + L = \sum_i \dot{q}_i \, p_i \quad ; \quad p_i = \frac{\partial L}{\partial \dot{q}_i} \tag{6.30}$$

This is the general definition of $\mathcal{H}$. The derivative of L with respect to the coordinate $\dot{q}_i$ is shown in terms of the generalized momentum p_i, as required by Eq. 6.29. The generalized momentum is an independent degree of freedom that also introduces flexibility to choose the variables to describe the mechanical system. The basic variables of the Lagrangian are q_i, $\dot{q}_i$ and time t, while the basic variables of the Hamiltonian are q_i, p_i, and t.

6.10 Preservation of the Hamiltonian in Time

The next interest is to show the preservation of the Hamiltonian with time. This is especially important since the conserved quantities reduce the number of degrees of freedom and are useful to monitor the proper evolution of the particle dynamics. In this respect, we introduce the definition of a constant of motion: a function f with dependence on the variables q_i, $\dot{q}_i$ and time t is a *constant*, or *integral of motion*, when the total time derivative is zero.

$$\frac{df}{dt} = \sum_i \frac{\partial f}{\partial q_i}\,\dot{q}_i + \frac{\partial f}{\partial \dot{q}_i}\,\ddot{q}_i + \frac{\partial f}{\partial t} = 0 \tag{6.31}$$

This expression represents a way to find the time invariant observables (a different method to find integrals of motion is the Hamilton-Jacobi formulation of Sect. 6.18). If the function $\mathcal{H}$ is invariant with time, its total time derivative should be zero. Let us assume an isolated system of particles. The goal is to verify that $\mathcal{H}$ is a constant of motion. By differentiating the Lagrangian with respect to t, we have:

$$\frac{dL}{dt} = \sum_i \left[\frac{\partial L}{\partial q_i}\frac{dq_i}{dt} + \frac{\partial L}{\partial \dot{q}_i}\frac{d\dot{q}_i}{dt}\right] + \frac{\partial L}{\partial t}$$

According to the Lagrangian equations, the term $d(\partial L/\partial \dot{q}_i)/dt$ replaces $\partial L/\partial q_i$.

$$\frac{dL}{dt} = \sum_i \left[\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\dot{q}_i + \frac{\partial L}{\partial \dot{q}_i}\frac{d\dot{q}_i}{dt}\right] + \frac{\partial L}{\partial t} = \sum_i \frac{d}{dt}\left(\dot{q}_i\frac{\partial L}{\partial \dot{q}_i}\right) + \frac{\partial L}{\partial t}$$

The left- and right-hand-side members are gathered together, and the identity 6.30 is used.

$$\frac{d}{dt}\left(\sum_i \dot{q}_i\,\frac{\partial L}{\partial \dot{q}_i} - L\right) + \frac{\partial L}{\partial t} = \frac{d}{dt}\left(\sum_i \dot{q}_i\,p_i - L\right) + \frac{\partial L}{\partial t} = 0 \quad \Rightarrow \quad \frac{d\mathcal{H}}{dt} = -\frac{\partial L}{\partial t}$$

The requirements to have $d\mathcal{H}/dt$ equal to zero are to: (i) satisfy the Lagrangian equations of motion for every independent degree of freedom, and (ii) have L with no explicit dependence on the time, $\partial L/\partial t = 0$. These requirements make the Hamiltonian a constant of motion in a conservative system.

$$\mathcal{H} = \text{constant} \quad ; \quad \text{conservative system}$$

The form of $\mathcal{H}$ for the system of particles is determined from $L = T - V$, with the kinetic energy given as a homogeneous quadratic function of $\dot{q}_i$.

$$\begin{aligned} &T = \sum_{ij} c_{ij}\dot{q}_i\dot{q}_j \quad ; \quad \frac{\partial T}{\partial \dot{q}_k} = 2\sum_i c_{ik}\dot{q}_i \quad ; \quad 2T = \sum_k \dot{q}_k\,\frac{\partial T}{\partial \dot{q}_k} \\ &\mathcal{H} = \sum_i \dot{q}_i\,p_i - L = 2T - (T - V) = T + V \quad \Rightarrow \quad \mathcal{H} = E \end{aligned} \tag{6.32}$$

The Hamiltonian is identified with the *energy* of the isolated system and is conserved in the particle dynamics. Historically, the energy and momentum were the first

physical observables that were found to be invariant with time after investigating an isolated system of colliding particles. If a frictional force is considered in the potential energy V, then the work done by that force on a particle depends on the trajectory, the system is not conservative, the process is time irreversible, and the Hamiltonian is no more a constant of motion. Furthermore, $\mathcal{H}$ does not represent the energy of the system and is not conserved. This usually occurs when the time is present in the transformation of coordinates. In the following section, we identify the symmetries of nature associated with the preservation of the energy, momentum, and more.

Problem 6.5 A trolley moves, and a particle of mass m is in the interior of the trolley attached to a spring with constant k. An external force keeps the trolley moving with constant speed v. The coordinate describing the particle with respect to a static reference frame is q. The Lagrangian of the particle and the corresponding equation of motion are:

$$L = \frac{m\dot{q}^2}{2} - \frac{k}{2}(q - vt)^2 \quad ; \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) + \frac{\partial L}{\partial q} = m\ddot{q} + k(q - vt) = 0$$

The time-dependent Hamiltonian of the particle is $\mathcal{H} = p^2/(2m) + (k/2)(q - vt)^2$. Define the new variable $q' = q - vt$, which describes the position of the particle in the trolley. Deduce the equation of motion, obtain the transformed Lagrangian and Hamiltonian in terms of q' and the momentum p', and derive a time-independent Hamiltonian.

Solution The equation of motion is obtained by replacing q by $(q' + vt)$. It is $m\ddot{q}' + kq' = 0$. The transformed Lagrangian is similarly computed.

$$L' = (m/2)(\dot{q}' + v)^2 - (k/2)(q')^2$$

The expression of the Hamiltonian requires the calculation of the new momentum.

$$p' = \partial L'/\partial \dot{q}' = m(\dot{q}' + v)$$

The primed Hamiltonian is $\mathcal{H}' = p'\dot{q}' - L' = (m\dot{q}' + mv)\dot{q}' - m(\dot{q}')^2/2 - m\dot{q}'v - mv^2/2 + k(q')^2/2$. This expression is changed by using the momentum p'.

$$\mathcal{H}' = (p')^2 - 2mvp'/(2m) + (k/2)(q')^2$$

(continued)

The Hamiltonian shows no time dependence, and it is not the energy because there is an external force on the trolley that affects the particle. Therefore, the Hamiltonian of a system may not necessarily be the system energy, and the Hamiltonian is not used to define a conservative system.

6.11 Conserved Observables and Symmetries

There are fundamental variables in molecular dynamics, and many of them are conserved over time under the appropriate conditions. The interest in this section is to discuss the conservation of physical variables and their relation to the symmetries observed in nature. To do this, we present the *Noether's theorem*. The theorem is a generalization of the results obtained on cyclic coordinates, where the absence of q_i in the Lagrangian is the cause to preserve the generalized momentum p_i, Eq. 6.9. The Noether's theorem is of central importance to establish the intrinsic links between the conserved observables of particle systems and the associated symmetries of nature in different fields of physics.

Statement 6.2 Noether's theorem: for a system with a continuous symmetry, there is a corresponding time invariant observable determined from the action.

In order to demonstrate the theorem, we obtain the action S after integrating a Lagrangian L, which is perturbed in the generalized coordinates q and time t by small amounts:

$$\begin{aligned} q' = q + \delta q \quad ; \quad t' = t + \delta t \\ \delta q = \varepsilon Q \quad ; \quad \delta t = \varepsilon T \end{aligned}$$

The quantity ε is a small real factor, Q is a real constant playing the role of a *generator of the generalized coordinates*, and T the *generator of the time evolution*.

$$t' = t + \varepsilon T \quad ; \quad q'(t') = f(q(t), \varepsilon) \quad ; \quad \dot{q}'(t') = \tfrac{\partial f(q(t),\varepsilon)}{\partial q}\, \dot{q}(t)$$

$$S(t) = \textstyle\int_{t_1}^{t_2} L(q(t), \dot{q}(t), t)\, dt \quad \rightarrow \quad S(\varepsilon) = \int_{t_1+\varepsilon T}^{t_2+\varepsilon T} L(q'(t'), \dot{q}'(t'), t')\, dt' \tag{6.33}$$

In the process, we shall identify the variables playing the roles of the generators T and Q and the corresponding symmetries. The function f represents the transformation of the coordinate q in going from time t to t'. The strategy is to integrate the perturbed Lagrangian in $S(\varepsilon)$ and achieve a pair of expressions with the same functional form and their difference giving zero. One of them shall be evaluated at time t_1 and the other expression at time t_2. By equating them, we should reach the conclusion that the expression under consideration is unique and time independent. We start from the fact that S is stationary under small perturbations and then we differentiate S with respect to ε.

$$\frac{d}{d\varepsilon}S(\varepsilon) = \int_{t_1+\delta t}^{t_2+\delta t}\left[\frac{\partial L}{\partial q'}\frac{\partial q'}{\partial \varepsilon} + \frac{\partial L}{\partial \dot{q}'}\frac{\partial \dot{q}'}{\partial \varepsilon} + \frac{\partial L(q'(t'),\dot{q}'(t'),t')}{\partial t'}\frac{\partial t'}{\partial \varepsilon}\right]dt' = 0$$

The differentiation of S with respect to ε avoids the direct differentiation of S with respect t. The differentiation is assisted by the chain rule due to the dependencies on the mechanical variables with respect to time:

$$\int_{t_1+\delta t}^{t_2+\delta t}\frac{\partial L}{\partial t'}\frac{\partial t'}{\partial \varepsilon}\,dt' = \int_{t_1+\delta t}^{t_2+\delta t}\frac{\partial L}{\partial t'}T dt' = (\Delta L)T$$

$$\Delta L = L(q(t),\dot{q}(t),t)\Big|_{t=t_1}^{t=t_2}$$

$$\frac{d}{d\varepsilon}S(\varepsilon) = (\Delta L)T + \int_{t_1+\delta t}^{t_2+\delta t}\left[\frac{\partial L}{\partial q'}\left(\frac{\partial f}{\partial q}\frac{\partial q}{\partial t}\frac{\partial t}{\partial \varepsilon} + \frac{\partial f}{\partial \varepsilon}\right) + \frac{\partial L}{\partial \dot{q}'}\frac{\partial}{\partial \varepsilon}\left(\frac{\partial f(q(t),\varepsilon)}{\partial q}\dot{q}(t)\right)\right]dt$$

where the identities of the first line of Eqs. 6.33 were used and are shown in the round parenthesis above. In order to continue evaluating the integral, we display the derivatives of the second term in parenthesis.

$$\frac{\partial L}{\partial \dot{q}'}\frac{\partial}{\partial \varepsilon}\left(\frac{\partial f(q(t),\varepsilon)}{\partial q}\dot{q}(t)\right) = \frac{\partial L}{\partial \dot{q}'}\left[\left(\frac{\partial}{\partial \varepsilon}\frac{\partial f}{\partial q}\right)\dot{q} + \frac{\partial f}{\partial q}\frac{\partial \dot{q}}{\partial \varepsilon}\right]$$
$$= \frac{\partial L}{\partial \dot{q}'}\left[\frac{\partial}{\partial q}\left(\frac{\partial f}{\partial q}\right)\left(\frac{\partial q}{\partial t}\frac{\partial t}{\partial \varepsilon}\right)\dot{q} + \frac{\partial^2 f}{\partial \varepsilon \partial q}\dot{q} + \frac{\partial f}{\partial q}\frac{\partial \dot{q}}{\partial t}\frac{\partial t}{\partial \varepsilon}\right]$$

We also replace derivatives like the following: $\dot{q} = \partial q/\partial t$, $\ddot{q} = \partial \dot{q}/\partial t$, and $\partial t/\partial \varepsilon = -T$.

$$\frac{d}{d\varepsilon}S(\varepsilon)\Big|_{\varepsilon=0} = (\Delta L)T + \int_{t_1}^{t_2}\left[\frac{\partial L}{\partial q'}\left(-\frac{\partial f}{\partial q}\dot{q}T + \frac{\partial f}{\partial \varepsilon}\right)\right.$$
$$\left. + \frac{\partial L}{\partial \dot{q}'}\left(-\frac{\partial^2 f}{\partial q^2}\dot{q}^2 T + \frac{\partial^2 f}{\partial \varepsilon \partial q}\dot{q} - \frac{\partial f}{\partial q}\ddot{q}T\right)\right]_{\varepsilon=0} dt$$

Once the derivatives with respect to ε are performed, we consider $\varepsilon = 0$ (which leads to dropping the primes on the variables). The terms with negative signs are rewritten using the Lagrangian equations of motion.

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial q}\,\dot{q}T\right) = \left(\frac{d}{dt}\,\frac{\partial L}{\partial \dot{q}}\right)\frac{\partial f}{\partial q}\,\dot{q}T + \frac{\partial L}{\partial \dot{q}}\left(\frac{d}{dt}\,\frac{\partial f}{\partial q}\right)\dot{q}T + \frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial q}\,\ddot{q}T$$

$$= \frac{\partial L}{\partial q}\,\frac{\partial f}{\partial q}\,\dot{q}T + \frac{\partial L}{\partial \dot{q}}\left(\frac{\partial^2 f}{\partial q^2}\right)\dot{q}^2 T + \frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial q}\,\ddot{q}T$$

The integral is simplified by inserting the last result in the integral, keeping the old terms of the kernel that carry positive signs inside the integral, and applying the fundamental theorem of calculus on the relation between derivation and integration.[10]

$$\frac{d}{d\varepsilon}S(\varepsilon)\Big|_{\varepsilon=0} = \Delta L\,T - \Delta L'\,T + \int_{t_1}^{t_2}\left[\frac{\partial L}{\partial q}\left(\frac{\partial f}{\partial \varepsilon}\right) + \frac{\partial L}{\partial \dot{q}}\left(\frac{\partial^2 f}{\partial \varepsilon \partial q}\,\dot{q}\right)\right]_{\varepsilon=0} dt \quad ;$$

$$\Delta L' = \frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial q}\,\dot{q}\Big|_{t=t_1}^{t=t_2}$$

Once more, the Lagrangian equations of motion are employed to reduce the integral.

$$\int_{t_1}^{t_2}\left[\frac{\partial L}{\partial q}\left(\frac{\partial f}{\partial \varepsilon}\right) + \frac{\partial L}{\partial \dot{q}}\left(\frac{\partial^2 f}{\partial \varepsilon \partial q}\,\dot{q}\right)\right]_{\varepsilon=0} dt = \int_{t_1}^{t_2}\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial \varepsilon}\right) dt$$

It is now time to collect the final results of the integrals.

$$\frac{d}{d\varepsilon}S(\varepsilon) = \left[L(q(t),\dot{q}(t),t)\,T - \frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial q}\,\dot{q}\,T + \frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial \varepsilon}\right]_{t=t_1}^{t=t_2} = 0$$

In the limit $\varepsilon = 0$, the coordinate q is recovered from f due to the equivalence $f(q,0) = q$, which in turn leads to $\partial f/\partial q = 1$.

$$\left[L(q(t),\dot{q}(t),t)\,T - \frac{\partial L}{\partial \dot{q}}\,\dot{q}\,T + \frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial \varepsilon}\right]_{t=t_2}$$
$$= \left[L(q(t),\dot{q}(t),t)\,T - \frac{\partial L}{\partial \dot{q}}\,\dot{q}\,T + \frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial \varepsilon}\right]_{t=t_1}$$

The expressions on the left- and right-hand sides have equal functional forms, but they are evaluated at different instants of time. In this regard, we have a unique expression preserved with time.

$$L(q(t),\dot{q}(t),t)\,T - \frac{\partial L}{\partial \dot{q}}\,\dot{q}\,T + \frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial \varepsilon} = \text{constant} \tag{6.34}$$

[10] Suppose that f and F are continuous and differentiable functions in the interval $[a, x]$. If the function F is obtained from f by integration, $F(x) = \int_a^x f(y)\,dy$, then f is obtained from F by differentiation, $f(x) = F'(x)$.

This is the main result of the Noether's theorem using a single degree of freedom and for a system of classical particles. It leads to the existence of conserved observables for the symmetries that satisfy the previous result. The integrals of motion are derived from the type of perturbation produced in the generalized coordinates. For example, the translation and rotation of the coordinates are associated with a homogeneous and isotropic space, respectively. The Noether's theorem changes forms in the different fields of physics. For example, it changes its form at the quantum level because the quantized variables are not continuous. The appropriate expression in quantum field theory is known as the Ward-Takahashi equation. The following goal is to illustrate the Noether's theorem by making reference to: (i) the homogeneity of space, (ii) the isotropy of space, and (iii) the invariance of the Lagrangian with the passage of time. These cases show the way in which space and time mold the laws of physics.

6.12 Space Homogeneity

We start by considering the invariance of L with respect to the spatial translation of a particle (if the space shows no change in the direction of the particle translation, then no change of the particle momentum is expected). Consider a Lagrangian, which is symmetric with respect to the translations of the particle coordinates.

$$L(\mathbf{r}_i, \dot{\mathbf{r}}_i, t) = L(\mathbf{r}_i + \delta r\, \hat{\mathbf{e}}, \dot{\mathbf{r}}_i, t) \tag{6.35}$$

The unit vector $\hat{\mathbf{e}}$ indicates the direction of translation. The expression embodies the homogeneity of the physical space since no portion of it is privileged. In order to apply the theorem in the form of Eq. 6.34, we observe that there is no time perturbation, $T = 0$; however, $\varepsilon Q = \delta r\, \hat{\mathbf{e}}$. According to the spatial perturbation, we have $\partial f/\partial \varepsilon = \partial(q + \varepsilon Q)/\partial \varepsilon = Q$, and the preserved variable is:

$$L(q(t), \dot{q}(t), t)\, T - \frac{\partial L}{\partial \dot{q}}\, \dot{q}\, T + \frac{\partial L}{\partial \dot{q}}\, \frac{\partial f}{\partial \varepsilon} = \text{constant} \quad ; \quad T = 0 \quad ; \quad \frac{\partial f}{\partial \varepsilon} = Q$$

$$p \cdot \hat{\mathbf{e}} = \text{ constant} \tag{6.36}$$

The application of the theorem gives the conservation of the linear momentum in the $\hat{\mathbf{e}}$ direction. A similar situation applies to the center of mass of a particle system by summing over the degrees of freedom pointing in the $\hat{\mathbf{e}}$ direction, resulting on the conservation of the total linear momentum of the particle system. The conclusion is the association of the space homogeneity to the preservation of the linear momentum of the particle, a characteristic that was anticipated in Sect. 2.1. The preservation of the linear momentum is broken when, for example, a time-varying external field acts upon the particle as, in some sense, it is equivalent to alter the space homogeneity of the system. Yet the preservation of the linear momentum is not a proof of the emptiness of the physical space (because, in principle, there is a Higgs field permeating the space).

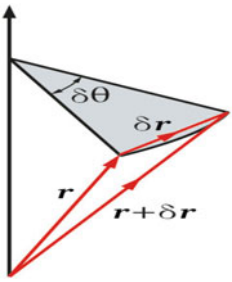


Fig. 6.6 The position of a particle is rotated by an infinitesimal angle $\delta\theta$ around the axis $\hat{\mathbf{e}}$. The change of the vector position $\mathbf{r}$ is given by $\delta\mathbf{r} = \delta\theta\hat{\mathbf{e}} \times \mathbf{r}$. According to Noether's theorem, the conservation of angular momentum demands space isotropy

6.13 Space Isotropy

We analyze the isotropy of space with implications on the rotation of a particle (if the space shows no change in the region of the particle rotation, then no change of the body's angular momentum is expected). Consider the vector $\hat{\mathbf{e}}$ as the rotation axis of the particle. The vector position of the particle is $\mathbf{r}$ (Cartesian coordinates are used in this discussion). The vector perpendicular to the plane described by the two vectors is $\hat{\mathbf{e}} \times \mathbf{r}$. An infinitesimal rotation of the vector position is $\mathbf{r} + \varepsilon\hat{\mathbf{e}} \times \mathbf{r}$, where ε is the real number concerned with the infinitesimal rotation, Fig. 6.6.

It is time to apply the Noether's theorem in the form exposed in Eq. 6.34. There is no time perturbation, $T = 0$. According to the spatial perturbation, we have $\partial f/\partial\varepsilon = \partial(q + \varepsilon Q)/\partial\varepsilon = \hat{\mathbf{e}} \times \mathbf{r}$.

$$\begin{gathered} L(q(t), \dot{q}(t), t)\, T - \frac{\partial L}{\partial \dot{q}}\, \dot{q}\, T + \frac{\partial L}{\partial \dot{q}}\, \frac{\partial f}{\partial \varepsilon} = \text{ constant} \\ \mathbf{p} \cdot (\hat{e} \times \mathbf{r}) = \text{ constant} \end{gathered} \tag{6.37}$$

By using the properties of the cross product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a})$, the term on the left-hand side is written in the form $\hat{\mathbf{e}} \cdot (\mathbf{r} \times \mathbf{p})$, resulting on the conservation of the angular momentum projected onto the $\hat{\mathbf{e}}$ direction of the rotation axis. Therefore, the conservation of the angular momentum is associated with the space isotropy.

6.14 Uniform Passage of Time

The next application of the Noether's theorem involves the uniform passage of time (if the passage of time is uniform, then no change of the particle energy is expected). Suppose that the Lagrangian L remains constant as time passes. We are interested in finding the observable that is conserved for the particle under the above condition. To do this, we follow similar steps to those of the preceding sections. We produce a shift of the time origin, $T = 1$, while the coordinate generator Q is null.

$$L(q(t),\dot{q}(t),t)\,T - \frac{\partial L}{\partial \dot{q}}\,\dot{q}\,T + \frac{\partial L}{\partial \dot{q}}\,\frac{\partial f}{\partial \varepsilon} = \text{constant} \quad ; \quad Q = 0 \quad ; \quad T = 1$$
$$-L + \frac{\partial L}{\partial \dot{q}}\,\dot{q} = \mathcal{H} = \text{ constant} \tag{6.38}$$

where the Hamiltonian $\mathcal{H}$ given according to Eq. 6.30 was used. Since $\mathcal{H}$ is a constant in time, the Hamiltonian is an integral of motion. In other words, the uniform passage of time in an isolated particle results on the conservation of the energy $\mathcal{H} = E$. Such a result is consistent with that of Eq. 6.32. The Noether's theorem indirectly implies the first Newton's law under the supposition that the homogeneity and isotropy of the $3D$ space are altered with the introduction of an external forcefield, thus affecting the motion state of the body. Moreover, the Noether's theorem may be understood as a generalization of the first Newton's law under the supposition that the uniform passage of time is changed by the introduction of an external forcefield capable of changing the spacetime mesh, thus affecting the energy of the body.

The general conclusion is that the linear momentum, angular momentum, and energy are preserved by virtue of the invariance of the Lagrangian with respect to the homogeneity and isotropy of the $3D$ space and the uniform passage of time, respectively. The Noether's theorem is the formal way to indicate that the laws of physics do not change with the location in space and the passage of time. The Noether's theorem is frequently used in the design of new Lagrangians by demanding the proper symmetries and reproducing the physical laws observed under experimentation. For instance, the Lagrangian in the theory of general relativity for a particle in the presence of an electromagnetic field is proposed by:

$$L = -m_0 c\sqrt{-g_{\mu\nu}(x(t))\,\frac{dx^\mu}{dt}\,\frac{dx^\nu}{dt}} + e\,\frac{dx^\mu(t)}{dt}\,A_\mu(x(t))$$

The factor $g_{\mu\nu}$ is the metric tensor of the gravitational potential, the variables x_μ are the spacetime coordinates, and A_μ is the electromagnetic four-potential.

Problem 6.6 The relativistic Lagrangian in the theory of special relativity of a particle subjected to the potential U is given in an inertial reference frame by:

$$L = -m_0 c^2\sqrt{1 - v^2/c^2} - U(x)$$

Obtain the Lagrangian equation of motion, and compare with the Newtonian equation of motion. Consider the motion of the particle in the presence of an electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$, given in terms of the vector potential $\mathbf{A}$ and scalar potential ϕ, namely, $\mathbf{E} = -\nabla\phi - (1/c)\,\partial\mathbf{A}/\partial t$ and $\mathbf{B} = \nabla \times \mathbf{A}$, problem 6.3. Obtain the Lagrangian equation of motion.

(continued)

Solution The generalized coordinates are the Cartesian coordinates.

$$\frac{d}{dt}\left(\frac{\partial L}{\partial v}\right) - \frac{\partial L}{\partial x} = 0 \quad \Rightarrow \quad \frac{d}{dt}(mv) = -\frac{\partial U(x)}{\partial x} \quad ; \quad m = m_0 / \sqrt{1 - \frac{v^2}{c^2}}$$

The equation of motion in the relativistic frame is similar to the Newtonian equation of motion, except for the use of the "relativistic mass," m, defined in terms of the particle speed. The Lagrangian of the Newtonian particle is invariant under Galilean transformations, but the Lagrangian of the relativistic particle is invariant under Lorentz transformations. The Lagrangian of the relativistic particle may be mixed with that of a particle moving in an electromagnetic field, problem 6.3, by considering $U = e\phi - e\,\dot{\mathbf{r}} \cdot \mathbf{A}/c$. The equation of motion corresponds to the Lorentz force law.

$$\frac{d}{dt}(m\mathbf{v}) = e\mathbf{E} + \frac{e}{c}\,\dot{\mathbf{r}} \times \mathbf{B}$$

The preservation of the energy, linear momentum, and angular momentum is used to monitor the proper evolution of the molecular dynamics simulations. The conservation laws (involving observables related to both the space and the time) are formulated in a single conservation law in general relativity based on the stress-energy tensor, because space and time are intertwined together. Some of the main limitations of the Noether's theorem in the present form involve the inability to consider discrete symmetries, like the reflection of a system of coordinates, and the impossibility to deal with discrete variables. Such cases are properly addressed with a similar theorem in quantum mechanics. On the other side, the electric charge conservation of a charged particle is properly addressed by an equivalent theorem in quantum field theory. The extension of the Noether's theorem to other fields of physics results in the proposal of continuity equations. In particular, in dissipative systems, the physical observables fluctuate (i.e., they are not preserved in time and space), because the space homogeneity and the space isotropy are constantly changed by external factors.

6.15 Hamilton's Equations of Motion

The Lagrangian equations of motion are second-order differential equations, and they are frequently difficult to solve. If we carry an analogy with the Newtonian equations of motion, we have to solve equations like $\mathbf{f} = m\,d^2\mathbf{r}/dt^2$ in Lagrangian theory; however, this expression is equivalently written in the form of two first-order differential equations, $\mathbf{p} = md\mathbf{r}/dt$ and $\mathbf{f} = d\mathbf{p}/dt$. In this process, a single equation of second-order has been transformed into two equations of first order, which may be easier to solve in terms of $\mathbf{r}$ and $\mathbf{p}$.

$$\mathbf{f}_i = m_i\, d^2\mathbf{r}_i/dt^2 \quad \Rightarrow \quad \mathbf{p}_i = m_i\, d\mathbf{r}_i/dt \quad ; \quad \mathbf{f} = d\mathbf{p}_i/dt$$

In this section, we present the equations of motion of classical mechanics according to the Hamiltonian theory. In this approximation, the order of the differential equations of motion is reduced with respect to that of Lagrangian mechanics. To do this, the conjugate momentum p_i, instead of the velocity $\dot{q}_i$, and the generalized coordinate q_i constitute the basic variables in the description of the particle system. The generalized momentum p_i is defined in terms of the Lagrangian as before, Eq. 6.8.

$$p_i = \partial L/\partial \dot{q}_i \tag{6.39}$$

The generalized momentum is consistent with the ordinary knowledge on the Newtonian momentum. Suppose that the kinetic energy of a system of particles is $T = \sum_k m_k v_k^2/2$. From Newtonian mechanics, we know that, for example, in the x direction, $p = m\,dx/dt = mv$. On the other side, by considering the Lagrangian as $L = T - V$, where the interaction potential V has no dependence on the velocities, we deduce $p = \partial L/\partial \dot{x} = mv$. Therefore, the expression of the generalized momentum in Lagrangian theory is consistent with our common knowledge on the Newtonian momentum in Cartesian coordinates. By assuming a conjugate momentum p_i for every q_i, the set $\{p_i\}$ is defined. Next, we have to establish the Hamiltonian equations that rule the motion of the particles in terms of the conjugate variables q_i and p_i. Since we are dealing with a change of variables from $\dot{q}_i$ to p_i, the Legendre transformation of the Lagrangian function, Eq. 6.30, is employed.

$$\mathcal{H}(\{q_i\}, \{p_i\}, t) + L(\{q_i\}, \{\dot{q}_i\}, t) = \sum_i p_i \dot{q}_i$$

The transformation is symmetric in L and $\mathcal{H}$ and also in the coordinates q_i and p_i. The Lagrangian and Hamiltonian on the one side, and $\dot{q}_i$ and p_i on the other, classify in the same category according to the Legendre transformation. In order to reach a differential equation, the following step consists on taking the differential of the above expression:

$$\sum_i \left[\frac{\partial \mathcal{H}}{\partial q_i}\, dq_i + \frac{\partial \mathcal{H}}{\partial p_i}\, dp_i\right] + \frac{\partial \mathcal{H}}{\partial t}\, dt = \sum_i (p_i\, d\dot{q}_i + \dot{q}_i\, dp_i) - \sum_i \left[\frac{\partial L}{\partial q_i}\, dq_i + \frac{\partial L}{\partial \dot{q}_i}\, d\dot{q}_i\right] - \frac{\partial L}{\partial t}\, dt$$

The terms $p_i d\dot{q}_i$ and $(\partial L/\partial \dot{q}_i) d\dot{q}_i$ of the right-hand side cancel each other. By using the definition of the conjugate momentum, $p_i = \partial L/\partial \dot{q}_i$, plus the identity deduced from the Lagrangian equation $\partial L/\partial q_i = d(\partial L/\partial \dot{q}_i)/dt - Q_i = dp_i/dt - Q_i = \dot{p}_i - Q_i$, and considering that the variables are independent, the coefficients from

the left-hand side should match those of the right-hand side.

$$\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i} \quad ; \quad Q_i - \dot{p}_i = \frac{\partial \mathcal{H}}{\partial q_i} \quad ; \quad -\frac{\partial L}{\partial t} = \frac{\partial \mathcal{H}}{\partial t} \quad ; \quad i = 1, \cdots, n \tag{6.40}$$

These are the *Hamilton's equations of motion*. The q_i and p_i variables are obtained by integration of these equations with time. The equations form a set of $2n$ first-order differential equations, plus the last expression. If the Hamiltonian is time independent, we require $2n$ initial conditions or n conditions at two different instants of time to determine the solutions in unique form. The equations are in principle easier to solve than the set formed by the Lagrangian equations, which are second-order differential expressions. In this formalism, the role of the Lagrangian is replaced by the Hamiltonian function, $\mathcal{H}$. The new independent variables are q_i and p_i. They are shown in symmetric forms (except for a minus sign and a generalized force, Q_i) in the equations. The Hamilton's formulation constitutes a general approach to study mechanical systems of particles, and in this context, Eqs. 6.40 are the *canonical equations of motion* since they are invariant and valid for any set of conjugate variables q_i and p_i of conservative systems, $Q_i = 0$.[11] The Hamiltonian equations reproduce the Newtonian equations. If we take a system of particles with Hamiltonian $\mathcal{H} = \sum_k (p_{kx}^2 + p_{ky}^2 + p_{kz}^2)/2m_k + V$, then the Hamiltonian equations corresponding to the ith particle in the x direction are:

$$\dot{x}_i = \frac{\partial \mathcal{H}}{\partial p_{ix}} = \frac{p_{ix}}{m_i} \quad ; \quad \dot{p}_{ix} = -\frac{\partial \mathcal{H}}{\partial x_i} = -\frac{\partial V}{\partial x_i} \quad \Rightarrow \quad \mathbf{f}_i = m_i \, d^2\mathbf{r}_i/dt^2$$

Similar expressions are achieved for the other degrees of freedom. The point to remark here refers to the Hamiltonian equations reproducing the Newtonian equations. On the other hand, the procedure to solve the Hamiltonian equations looks in principle simpler, but the building of the Lagrangian is required to create the conjugate momenta from the generalized coordinates and insert the momenta into the expression of the Hamiltonian. This process implies additional work to that required in the Newtonian and Lagrangian approaches. Yet, all the mechanical formulations converge to the same solution when the initial conditions are given.

6.16 Invariance Under Canonical Transformations

The generalized coordinates q_i and generalized momenta p_i are used to describe the system in the Hamiltonian approach. Every pair of conjugate variables q_i and p_i satisfy the Hamilton's equations.

$$\dot{q}_i = \partial \mathcal{H}/\partial p_i \quad ; \quad \dot{p}_i = -\partial \mathcal{H}/\partial q_i$$

[11] The canonical expressions are part of a set of rules that provide the basis of a theoretical framework. The canonical expressions are simple yet sufficiently general to govern the mathematical operations and procedures within the theoretical framework.

The choice of using the variables q_i and p_i is arbitrary, as we could have chosen the variables Q_i and P_i, different to q_i and p_i, with their own Hamiltonian, K, also satisfying the Hamilton's equations.

$$\dot{Q}_i = \partial K/\partial P_i \quad ; \quad \dot{P}_i = -\partial K/\partial Q_i$$

Under the supposition of equal number of degrees of freedom in the transformations of the coordinates, there are different sets of generalized variables to describe the same system. In the case that we require a description in terms of the variables (Q_i, P_i), instead of (q_i, p_i), the variables (q_i, p_i) and the Hamiltonian $\mathcal{H}(\{q_i\}, \{p_i\}, t)$ have to be transformed. In general, the transformed Hamiltonian, K, in terms of the variables (Q_i, P_i), adopts a different functional form to that of $\mathcal{H}$ in terms of the variables (q_i, p_i). The variables (Q_i, P_i) and the transformed Hamiltonian K do not necessarily satisfy the Hamiltonian equations, since the proper mechanical requirements on the transformation of coordinates have not been imposed. The goal of this section is to propose the adequate transformations that maintain the functional forms of the Hamiltonian equations of motion invariant. Such types of transformations are known as *canonical transformations*. Before, it is important to recall the stationary principle of the action.

$$\delta \int_{t_1}^{t_2} L \, dt = 0$$

In place of the Lagrangian, expression 6.30 in terms of the Hamiltonian $\mathcal{H}$ is used.

$$\delta \int_{t_1}^{t_2} L \, dt = \delta \int_{t_1}^{t_2} \left[\sum_i p_i \dot{q}_i - \mathcal{H}(\{q_i\}, \{p_i\}, t)\right] dt = 0$$

If the link between the new variables and old variables is of the kind $Q_i = Q_i(\{q_i\}, \{p_i\}, t)$ and $P_i = P_i(\{q_i\}, \{p_i\}, t)$, then the Hamilton's principle should be satisfied by the variables describing the phase space of the particle system

$$\delta \int_{t_1}^{t_2} \left[\sum_i p_i \dot{q}_i - \mathcal{H}(\{q_i\}, \{p_i\}, t)\right] dt = 0 \quad ;$$

$$\delta \int_{t_1}^{t_2} \left[\sum_i P_i \dot{Q}_i - K(\{Q_i\}, \{P_i\}, t)\right] dt = 0$$

In going from the representation (q_i, p_i) to (Q_i, P_i), we require to build a bridge through a *generating function*. We call such a function G, with mixed dependence on the old and new variables, $G = G(\{q_i\}, \{Q_i\}, t)$. The dependence on the generalized coordinates (q_i, Q_i) may be equally changed to any other possibility, (q_i, P_i), (p_i, Q_i) or (p_i, P_i), leading to the same results. Based on the discussion of Sect. 6.6, where the non-uniqueness of the Lagrangian was attributed to the time derivative of a function, we determine in that context the generating function G. The kernels of the previous integrals are equal, except for the time derivative of a function, which,

in this case, is the generating function.

$$\begin{aligned}
&\textstyle\sum_i p_i \dot{q}_i - \mathcal{H}(\{q_i\}, \{p_i\}, t) \\
&\quad = \textstyle\sum_i P_i \dot{Q}_i - K(\{Q_i\}, \{P_i\}, t) + \frac{d}{dt} G(\{q_i\}, \{Q_i\}, t) \\
&\quad = \textstyle\sum_i P_i \dot{Q}_i - K(\{Q_i\}, \{P_i\}, t) + \sum_i \left(\frac{\partial G}{\partial q_i} \dot{q}_i + \frac{\partial G}{\partial Q_i} \dot{Q}_i \right) + \frac{\partial G}{\partial t}
\end{aligned}$$

By comparing the factors of the independent terms on the left with those on the right, the conditions on the transformation are obtained.

$$\begin{gathered}
p_i = \partial G / \partial q_i \quad ; \quad P_i = -\partial G / \partial Q_i \\
\mathcal{H}(\{q_i\}, \{p_i\}, t) = K(\{Q_i\}, \{P_i\}, t) - \partial G / \partial t
\end{gathered} \tag{6.41}$$

These conditions involve the generating function, the two Hamiltonians, the generalized coordinates of both phase spaces, and time. A transformation that satisfies such requirements is recognized as a canonical transformation since it preserves the functional form of the Hamiltonian equations. By assuming that the generating function G is known, the new generalized variables are deduced. In order to illustrate it, we consider the following points: the first identity ($p_i = \partial G / \partial q_i$) gives n equations for the p_i momenta in terms of q_i, Q_i, and t, namely $p_i = p_i(\{q_i\}, \{Q_i\}, t)$. From these expressions, the generalized coordinates Q_i are obtained by inverting the equations, $Q_i = Q_i(\{q_i\}, \{p_i\}, t)$. The momenta $\{P_i\}$ are obtained from the second set of equations, $P_i = -\partial G / \partial Q_i$, resulting on $P_i = P_i(\{q_i\}, \{Q_i\}, t)$. When one substitutes the $\{Q_i\}$ by the $\{q_i\}$ and $\{p_i\}$ variables, the generalized momenta P_i are shown in terms of the $\{q_i, p_i\}$ variables, that is, $P_i = P_i(\{q_i\}, \{p_i\}, t)$. The Hamiltonian K is written in terms of $\{Q_i\}$ and $\{P_i\}$, and the transformation is accomplished.

$$Q_i = Q_i(\{q_i\}, \{p_i\}, t) \quad ; \quad P_i = P_i(\{q_i\}, \{p_i\}, t) \quad ; \quad K = K(\{Q_i\}, \{P_i\}, t)$$

When the transformation between the variables $\{q_i, p_i\}$ and $\{Q_i, P_i\}$ is known, but the generating function is unknown, the generating function G may be obtained. To do this, it is necessary to have the momenta p_i and P_i in terms of the coordinates q_i, Q_i, and time t, because these are the variables of the G function.

$$p_i = p_i(\{q_i\}, \{Q_i\}, t) \quad ; \quad P_i = P_i(\{q_i\}, \{Q_i\}, t)$$

The final step demands integrating the differential equations 6.41 to determine G. When it is required to work with other combinations of variables, such as (q_i, P_i), (p_i, Q_i) or (p_i, P_i), instead of the combination (q_i, Q_i) discussed above, the canonical transformations change accordingly. In those cases, it is convenient to replace G by one of the generating functions G' for the respective variables of the phase space.

$$
\begin{aligned}
G' &= G(\{q_i\},\{P_i\},t) - \textstyle\sum_i Q_i P_i \qquad ;\ \{q_i, P_i\} \\
G' &= \textstyle\sum_i q_i p_i + G(\{p_i\},\{Q_i\},t) \qquad ;\ \{p_i, Q_i\} \\
G' &= \textstyle\sum_i (q_i p_i - Q_i P_i) + G(\{p_i\},\{P_i\},t) \ ;\ \{p_i, P_i\}
\end{aligned}
\tag{6.42}
$$

The equations of the generating functions can be deduced following similar steps to those illustrated above (where the kernels of the action integrals were related). The transformation of variables may demand the combination of several generating functions. In special cases, by using the proper transformation, it may be also possible to reduce the Hamiltonian to a cyclic form in the coordinates. In general, the Hamiltonian adopts a different functional form after the transformations of the conjugate variables are performed. Still, the Hamiltonian equations of motion remain invariant in the transformed phase spaces.

6.17 Time Reversibility in Hamiltonian Theory

In the motion of particles, the generalized variables change in time. The change of the variables may be interpreted as a transformation of variables due to a time shift, τ. More precisely, the transformation is from the variables $(q(t),\, p(t))$ to the variables $Q(t) = q(t+\tau)$ and $P(t) = p(t+\tau)$. According to Hamilton's principle and the canonical transformations $Q(t) \leftrightarrow q(t')$ and $P(t) \leftrightarrow p(t')$, we have the equivalence:

$$
\delta \int_{t_1}^{t_2} \left[\textstyle\sum_i P_i \dot{Q}_i - K(\{Q_i\},\{P_i\},t)\right] dt = \delta \int_{t_1'}^{t_2'} \left[\textstyle\sum_i p_i \dot{q}_i - \mathcal{H}(\{q_i\},\{p_i\},t')\right] dt'
\tag{6.43}
$$

The equation remains without changes when the variables $(Q(t),\, P(t))$ are replaced by the variables $(q(t+\tau),\, p(t+\tau))$, and the time is $t' = t + \tau$. The primed time, t', on the expression of the right is a dummy variable and is replaced by t.

$$
\begin{aligned}
&\delta \int_{t_1+\tau}^{t_2+\tau} \left[\sum_i p_i \dot{q}_i - \mathcal{H}(\{q_i\},\{p_i\},t+\tau)\right] d(t+\tau) \\
&\qquad = \delta \int_{t_1}^{t_2} \left[\sum_i p_i \dot{q}_i - \mathcal{H}(\{q_i\},\{p_i\},t)\right] dt
\end{aligned}
\tag{6.44}
$$

The expression indicates an invariance with respect to a time shift. Since the time shift produces the same trajectory $(q(t),\, p(t))$ to that with no time shift, the motion of the particles may be understood as a canonical transformation. It is also important to investigate the *time reversibility* of the solutions of the Hamiltonian equations. It is assumed that the Hamiltonian is an even function of the momentum, $\mathcal{H}(q,\, p) = \mathcal{H}(q,\, -p)$, an acceptable hypothesis in many particle systems. In order to invert the dynamics with time, we require two steps: (i) to reverse the momenta

of the particles (e.g., by stopping and pushing the particles backward with the same momenta they had at the instant of stopping them), and (ii) to reverse the ticks of the Newtonian clock (for instance, by producing a specular reflection of the clock) [6]. Thus, we consider a negative time, like $\tau = -t$. The respective equations of motion are:

$$dq_i/d\tau = -\partial\mathcal{H}/\partial p_i \quad ; \quad dp_i/d\tau = \partial\mathcal{H}/\partial q_i \quad ; \quad \tau = -t$$

The variables q_i and p_i are functions of the time and describe the trajectory of the ith particle when the initial conditions are given. It is convenient to define a negative momentum with respect to p_i (such as $\pi_i = -p_i$), because the momentum is a basic variable whose definition involves the time. The Hamiltonian remains unaltered with the new definition of the momentum.

$$dq_i/d\tau = \partial\mathcal{H}/\partial\pi_i \quad ; \quad -d\pi_i/d\tau = \partial\mathcal{H}/\partial q_i \quad ; \quad \pi_i = -p_i$$

These expressions are equivalent to the Hamiltonian equations of motion 6.40, with $Q_i = 0$. The conclusion is that the Hamiltonian equations of motion are recovered in the time-reversed process and the variables $q_i(\tau)$ and $\pi_i(\tau)$ describe the reversed trajectory of the ith particle.

Problem 6.7 Demonstrate that the Lagrangian equations of motion are time reversible.

Solution The Lagrangian equations of motion are $d(\partial L/\partial\dot{q}_i)/dt - \partial L/\partial q_i = 0$. We change signs in the generalized velocity and time.

$$\frac{d}{-dt}\left(\frac{\partial L}{-\partial\dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

After the cancellation of negative signs, the expression reproduces the original Lagrangian equations of motion. Suppose that q_i and q_i' are the forward and reverse trajectories described by particle i. When we are at a given point of the particle trajectory, and a time t elapses, we conclude that $q_i(-t) = q_i'(t)$ and $-\dot{q}_i(-t) = \dot{q}_i'(t)$. In this regard, the Lagrangian equations of motion are time reversible.

6.18 Hamilton-Jacobi Theory

The goal of this section is to describe an alternative approach to the previous theories. It is the *Hamilton-Jacobi formulation* of classical mechanics and corresponds to a branch of Hamiltonian theory. The Hamilton-Jacobi approach is consistent with Newtonian mechanics, represents a tool to find integrals of motion and is considered to be the "closest classical approach" to quantum mechanics. The Hamilton-Jacobi formulation departs from the Hamiltonian. The purpose is to employ a canonical transformation, G, to determine the solutions of the Hamiltonian equations concurrently taking advantage of the system constants of motion. The Hamilton-Jacobi formulation is an abstract scheme, and for such a reason, it shall be illustrated with a specific example such as the harmonic oscillator in Sect. 6.19. The G transformation is supposed to depend on the mixed variables (q_i, P_i, t). It also relates to the Hamiltonians $\mathcal{H}$ and K by:

$$K(Q_i, P_i, t) = \mathcal{H}(q_i, p_i, t) + \partial G(q_i, P_i, t)/\partial t \tag{6.45}$$

The variables (q_i, p_i) and (Q_i, P_i) are obtained from both the equations on the transformation of G (associated with the first identity of Eqs. 6.42, and presented in the first row below) and the equations of motion on the Hamiltonian K (presented in the second row below).

$$\begin{aligned} p_i &= \frac{\partial G}{\partial q_i} \quad ; \; Q_i = \frac{\partial G}{\partial P_i} \\ \dot{P}_i &= -\frac{\partial K}{\partial Q_i} \; ; \; \dot{Q}_i = \frac{\partial K}{\partial P_i} \end{aligned} \tag{6.46}$$

The Hamiltonian of the harmonic oscillator is $\mathcal{H} = p^2/2m + kq^2/2$, where the spring constant k is related to the frequency ω, with $k = m\omega^2$. The Hamiltonian contains the basic variables (q, p) that we shall deal with in Sect. 6.19.

$$\mathcal{H} = p^2/(2m) + (m\omega q)^2/(2m)$$

We are free to choose the transformation G; therefore, it is taken in such a way that the transformed Hamiltonian satisfies the trivial identity $K = 0$. The derivatives $\dot{Q}_i = \partial K/\partial P_i$ and $\dot{P}_i = -\partial K/\partial Q_i$ involve K, and in this case, they are zero. The change of Q_i and P_i in time is null, $\dot{Q}_i = 0$ and $\dot{P}_i = 0$, and they are constants of motion.

$$P_i = \alpha_i \; ; \; Q_i = \beta_i \; ; \; \mathcal{H}(q_i, p_i, t) + \partial G(q_i, P_i, t)/\partial t = 0 \; ; \; i = 1, \cdots, n \tag{6.47}$$

The quantities α_i and β_i are constants. There is a number of n expressions for P_i and an equal number for Q_i (in the case of m constraints, the number of degrees of freedom is reduced to $2n - m$). It is time to introduce the expression of the action S in terms of the function G and the arbitrary constant c. To do this, we

recall from Eqs. (6.12) and (6.30) that $\sum_i p_i \dot{q}_i - \mathcal{H}(\{q_i\}, \{p_i\}, t) = \sum_i P_i \dot{Q}_i - K(\{Q_i\}, \{P_i\}, t) + dG/dt$, $\dot{Q}_i = \dot{P}_i = 0$, and $L = dG/dt$, which under time integration leads to:

$$S(q_i, \alpha_i, t) = G(q_i, P_i, t) + c = G(q_i, \alpha_i, t) + c \tag{6.48}$$

The action function in this approach is understood as the generating function of a canonical transformation where the solutions of the equations associated with the transformed Hamiltonian are constants of motion (in the transformation of variables, we are implicitly assuming $\det(\partial^2 S/\partial q_i \partial P_i) \neq 0$). The S function depends on the canonical coordinates q_i, time t, and constants of motion α_i, with $i = 1, \cdots, n$. The equations of motion in terms of the action function S are deduced from expressions 6.46 and 6.47.

$$\begin{gathered} Q_i = \beta_i = \frac{\partial S(q_i, \alpha_i, t)}{\partial \alpha_i} \quad ; \quad p_i = \frac{\partial S(q_i, \alpha_i, t)}{\partial q_i} \\ \frac{\partial}{\partial t} S(q_1, \cdots, q_n; \alpha_1, \cdots, \alpha_n; t) + \mathcal{H}\left(q_1, \cdots, q_n, \frac{\partial S}{\partial q_1}, \cdots, \frac{\partial S}{\partial q_n}; t\right) = 0 \end{gathered} \tag{6.49}$$

The last expression is a first-order nonlinear equation of partial derivatives on the generating function S. It is called the *Hamilton-Jacobi equation* . The mathematical problem now resides on solving the single equation of partial derivatives to obtain the trajectory of the particles. The way in which the Hamilton-Jacobi equation has been proposed also becomes of help to find conserved quantities in the dynamics of the particles. The identity $p_i = \partial S/\partial q_i$ from Eq. 6.49 is the representation of the canonical momentum in terms of S and is commonly written in the form:

$$\boldsymbol{p}_i = \nabla_i S \tag{6.50}$$

The Hamiltonian of the harmonic oscillator satisfies $\mathcal{H} + \partial G/\partial t = 0$, Eq. 6.47, and the Hamilton-Jacobi equation is:

$$\frac{1}{2m}\left[\left(\frac{\partial S}{\partial q}\right)^2 + (m\omega q)^2\right] + \frac{\partial S}{\partial t} = 0 \tag{6.51}$$

The term $\partial S/\partial q$ replaces the momentum p to have the Hamilton-Jacobi expression written as a partial differential equation for the action function S in terms of q. In more general terms, the Hamilton-Jacobi approach employs the main variables (Q_i, P_i) and the initial conditions (α_i, β_i), to obtain the solutions (q_i, p_i) of the mechanical problem by working out the proper canonical transformation. Under this perspective, the time evolution of the system is considered a canonical transformation from $(\alpha_i, \beta_i) = (q_i(t_0), p_i(t_0))$ to $(q_i(t), p_i(t))$ at any instant of time t.

The following goal is to introduce an abbreviated action. Firstly, we establish a relation between the canonical transformation G and the action function S

as an alternative expression to Eq. 6.48. We differentiate G in time, $dG/dt = \sum_i (\partial G/\partial q_i)(dq_i/dt) + \partial G/\partial t$. By using Eqs. 6.49, where $p_i = \partial S/\partial q_i$ and $\partial S/\partial t = -\mathcal{H}$, we reach:

$$\frac{dG}{dt} = \sum_i \frac{\partial G}{\partial q_i}\frac{dq_i}{dt} + \frac{\partial G}{\partial t} = \sum_i p_i \dot{q}_i - \mathcal{H} \tag{6.52}$$

The expression on the right confirms the Lagrangian function of Eq. 6.30. In this respect, dG/dt is related to the action function S upon integration over time. The action function is sufficiently versatile as it is the departure function of different approaches in obtaining the equations of motion. Now, the *abbreviated action*, W, is introduced by $S = W - \alpha t$.

$$W(q_1, \cdots, q_n; \alpha_1, \cdots, \alpha_n) = S + Et \tag{6.53}$$

Under the supposition of a time-independent Hamiltonian, the equation is written in terms of functions L and $\mathcal{H}$ when the expressions $S = \int L\, dt$ and $\int \mathcal{H}\, dt = Et$ are recalled.

$$W = \int (L + \mathcal{H})\, dt \tag{6.54}$$

The abbreviated action is written in terms of the basic variables with Eq. 6.30.

$$W = \sum_i \int p_i\, dq_i \tag{6.55}$$

For a freely moving particle in the $\mathbf{r}$ direction, with momentum $\mathbf{p}$, the functions S and W that satisfy the Hamilton-Jacobi equation are:

$$W = \mathbf{r}\cdot\mathbf{p} \quad ; \quad S(\mathbf{r}, \mathbf{p}, t) = \mathbf{r}\cdot\mathbf{p} - Et \quad ; \quad p = \sqrt{2mE} \tag{6.56}$$

Problem 6.8 The Hamiltonian principal function S is a power series over time:

$$S(t) = \alpha q - \frac{\alpha^2}{2m}\, t + \frac{k}{2}\, qt^2 - \frac{\alpha k}{6m}\, t^3 - \frac{k^2}{40m}\, t^5$$

The position variable is q, the time is t, and the parameters α, k, and m are constants. Find the Hamiltonian and the solutions (q, p) from the Hamilton-Jacobi equation that employs the function S.

Solution According to the Hamilton-Jacobi expression, we have $\partial S/\partial t + \mathcal{H} = 0$. Thus, by differentiating S with respect to time, we obtain $\mathcal{H}$.

$$-\mathcal{H} = \frac{\partial S}{\partial t} = -\frac{1}{2m}\left(\alpha^2 + \alpha k t^2 + \frac{k^2 t^4}{4}\right) + ktq$$

(continued)

The following step is to find the momentum p to write $\mathcal{H}$ in terms of q and p. To do this, we recall expression 6.50, $p = \partial S/\partial q = \alpha + kt^2/2$.

$$p^2 = \alpha^2 + \alpha k t^2 + k^2 t^4/4$$

The Hamiltonian is now written in terms of the conjugate variables q and p as $\mathcal{H} = p^2/2m - ktq$. It is possible to deduce the Hamiltonian equations of motion.

$$\dot{q} = \frac{\partial \mathcal{H}}{\partial p_i} = p/m \quad ; \quad \dot{p} = -\frac{\partial \mathcal{H}}{\partial q_i} = kt$$

The integration of the last expression gives $p = p_0 + kt^2/2$. This result is inserted in the first expression to integrate the position over time, giving $q = q_0 + p_0 t/m + kt^3/6m$. More results are found in [7].

6.19 Illustrating with the Harmonic Oscillator

The abbreviated action W is of particular help to study the harmonic oscillator. In order to obtain the coordinates q and p of the oscillator in terms of the time, we require to determine the constants of motion Q and P and use W as the canonical transformation to get q and p. Some of the constants of motion are found from the initial conditions, when $t = 0$. These are, for example, the oscillation amplitude, the phase factor, and energy. The conjugate variables q and p are described in terms of such constants of motion. We start from the Hamilton-Jacobi equation 6.51, taking $S = W - Et$ into consideration.

$$\frac{1}{2m}\left[\left(\frac{\partial W}{\partial q}\right)^2 + (m\omega q)^2\right] = \alpha$$

The α term is the constant energy. Among the many types of variables that are used to formulate the equations of the harmonic oscillator, there is a particular set that leads to the solution in a relatively simple manner. We take α as the variable P (it could equally be Q). The function W defines the canonical transformation in going from the variables (Q, P) to the variables (q, p). According to the second identity of Eqs. 6.49 and 6.53, the canonical momentum is $p = \partial W/\partial q$, and this term is present in the above equation of W. By solving for the term $\partial W/\partial q$, we obtain p in terms of q.

$$p = \frac{\partial W}{\partial q} = \left[2m\alpha - (m\omega q)^2\right]^{1/2} \tag{6.57}$$

In order to solve the equation for W, we integrate over the variable q.

$$W = \int \left[2m\alpha - (m\omega q)^2\right]^{1/2} dq \tag{6.58}$$

The following step is to find the variable Q from the expression of S by considering $Q = \beta = \partial S/\partial\alpha$, which corresponds to the first identity of Eqs. 6.49, and $W(q, \alpha) = S(q, \alpha, t) + \alpha t$, Eq. 6.53.

$$\beta = \frac{\partial S}{\partial \alpha} = \frac{\partial (W - \alpha t)}{\partial \alpha} = m \int \left[2m\alpha - (m\omega q)^2\right]^{-1/2} dq - t \tag{6.59}$$

The integral is done according to the result $\int dx/(a^2 - x^2)^{1/2} = \sin^{-1}(x/a)$.

$$\beta = \frac{1}{\omega} \sin^{-1} \sqrt{mw^2 q^2/(2\alpha)} - t \tag{6.60}$$

At this stage, we have the variables $(Q, P) = (\beta, \alpha)$. We now invert the above identity for the variable q and use it in the expression of p, Eq. 6.57.

$$q(t) = \sqrt{2\alpha/(m\omega^2)}\, \sin(wt + \omega\beta) \quad ; \quad p(t) = \sqrt{2m\alpha}\, \cos(\omega t + \omega\beta) \tag{6.61}$$

The solutions describe the oscillatory character of the motion. The terms involving α are the sinusoidal amplitudes, and the terms involving β are the phase factors. In this treatment, the P and Q conjugate variables are the α and β constants of motion, respectively. Thus, by dealing with the equation for W, which relates the variables $P = \alpha$ and q, it was possible to obtain the variables q and p with their explicit time dependence.

6.20 Contact Between Quantum and Classical Mechanics

The Hamilton-Jacobi approach is particularly useful to establish "contact" with the theory of quantum mechanics. In order to illustrate it, we start from quantum mechanics because it is a more general theory. A particle in quantum mechanics is represented by a wave, and the equation ruling the dynamics of the particle is the *time-dependent wave equation*.

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{\mathcal{H}}\psi \tag{6.62}$$

The function $\hat{\mathcal{H}}$ is the *Hamiltonian operator* , and ψ is the *wave function* that describes the particle. For a particle with a given momentum and subjected to potential V, the wave equation is:

$$i\hbar \frac{\partial \psi(q,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(q,t)}{\partial q^2} + V(q)\, \psi(q,t) \tag{6.63}$$

where the Hamiltonian operator was identified by $\hat{\mathcal{H}} = -\hbar^2/(2m)\,\partial^2/\partial q^2 + V(q)$. Due to the time derivative and the spatial derivative, the wave function ψ is expected to be related to an exponential function. Hence, a wave function where the action S plays the role of a phase factor is proposed.

$$\psi = \psi_0 \exp(iS/\hbar) \quad ; \quad \int_{-\infty}^{\infty} |\psi(q,t)|^2\, dq = 1 \tag{6.64}$$

The wave function is normalized and is called the *action wave* when it is given in that form. The factor $\hbar$ is the Planck constant that divides S in $\hbar$ units, making the exponent unitless. Such a factor is characteristic of quantum mechanics (like the Boltzmann constant k_B to statistical mechanics or the speed of light c to relativistic mechanics). However, $\hbar$ is very small in classical systems. Since the particle propagates as a wave, we consider the similarity between the action $S = \mathbf{p}\cdot\mathbf{r} - Et$ of a freely moving particle, Eq. 6.56, and the phase $\mathbf{k}\cdot\mathbf{r} - \omega t$ of a plane wave, where the wave vector is $\mathbf{k}$ and the angular frequency is $\omega = E/\hbar$. In fact, when we divide $(\mathbf{p}\cdot\mathbf{r} - Et)$ by $\hbar$, we obtain $(\mathbf{k}\cdot\mathbf{r} - \omega t)$, with the definitions $\mathbf{k} = \mathbf{p}/\hbar$ and $\omega = E/\hbar$.

The following purpose is to show that S is the action function of classical mechanics. We insert ψ given by Eq. 6.64 in Eq. 6.63, using the expression $\partial\psi/\partial q = (i/\hbar)\psi\,\partial S/\partial q$.

$$-\tfrac{i\hbar}{2m}\tfrac{\partial^2 S}{\partial q^2} + \tfrac{1}{2m}\left(\tfrac{\partial S}{\partial q}\right)^2 + V(q) + \tfrac{\partial S}{\partial t} = 0 \tag{6.65}$$

By assuming that the terms without $\hbar$ are much bigger than those containing $\hbar$, we neglect the first term of the equation.

$$\tfrac{1}{2m}\left(\tfrac{\partial S}{\partial q}\right)^2 + V(q) + \tfrac{\partial S}{\partial t} = 0 \quad \Rightarrow \quad \mathcal{H}(q, \partial S/\partial q) + \tfrac{\partial S}{\partial t} = 0 \tag{6.66}$$

Problem 6.9 The action function is proposed in the form $S = \pm\oint p dr + c$, where c is a constant. The *de Broglie relation* is $\lambda(r) = h/p(r)$. Show that $|\partial\lambda(r)/\partial r| \ll 2\pi$ when $|\hbar(\partial^2 S/\partial r^2)| \ll (\partial S/\partial r)^2$.

Solution Based on the form of the above action function, and assuming the inequality $|\hbar(\partial^2 S/\partial r^2)| \ll (\partial S/\partial r)^2$, we obtain (with $p = \partial S/\partial r$) an expression in terms of the momentum p.

$$|\hbar\,\partial p/\partial r| \ll p^2$$

The inequality is written in the equivalent form $|\partial(\hbar p^{-1})/\partial r| \ll 1$, where $\hbar = h/(2\pi)$. The de Broglie relation is $\lambda(r) = h/p(r)$; therefore,

(continued)

$|\partial\lambda(r)/\partial r| \ll 2\pi$. This inequality is one of the conditions for quantum mechanics to transition into classical mechanics.

The last expression is the Hamilton-Jacobi equation 6.49. Thus, we have evidence of a close contact between classical mechanics and quantum mechanics under the conditions discussed previously.[12] If we proceed in the reverse direction to meet quantum mechanics from the classical formulation, then discrepancies are found. By considering the Hamilton-Jacobi expression 6.66, and using the identity $\psi = \psi_0 \exp(iS/\hbar)$, we obtain:

$$i\hbar \frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{1}{\psi} \left(\frac{\partial\psi}{\partial q}\right)^2 + V\psi \tag{6.67}$$

This expression differs from the wave equation 6.63. Historically, there was no reason to create a new formulation in the years previous to the birth of quantum mechanics because classical mechanics appropriately explained the experiments of that epoch.

6.21 Poisson's Brackets

In this section, we discuss the *brackets approach* of Poisson, understood as an alternative representation of the equations of motion in classical mechanics. From the Poisson brackets, the Liouville operator is introduced to perform the time evolution of the physical observables, which are expressed in terms of the conjugate variables and time.

The Poisson approach departs from an *anti-commutative operator*, denoted with brackets for the arbitrary functions u and v.

$$\{u, v\} = \frac{\partial u}{\partial q} \frac{\partial v}{\partial p} - \frac{\partial v}{\partial q} \frac{\partial u}{\partial p} \tag{6.68}$$

We are interested in applying the Poisson brackets to the Hamiltonian, $\mathcal{H}$, and an arbitrary function, F, both in terms of the generalized variables q and p.

[12] The Hamilton-Jacobi equation is a particular case of the Hamiltonian formulation, which in turn is linked to the Lagrangian formulation by a Legendre transformation. The Lagrange equations are the result of the motion of particles satisfying the principle of stationary action. In this respect, the wave equation of quantum mechanics is believed to implicitly contain the principle of stationary action by reaching the Hamilton-Jacobi equation.

$$\{F, \mathcal{H}\} = \left[\frac{\partial \mathcal{H}}{\partial p}\frac{\partial}{\partial q} - \frac{\partial \mathcal{H}}{\partial q}\frac{\partial}{\partial p}\right] F = \hat{\mathcal{L}} F \quad ; \quad \hat{\mathcal{L}} F = \{F, \mathcal{H}\} = -\{\mathcal{H}, F\} \tag{6.69}$$

This expression defines the *Liouville operator*, $\hat{\mathcal{L}}$, over the function F. The imaginary symbol i may be employed under convention in the definition of the Liouville operator, such as $i\hat{\mathcal{L}}F$ instead of $\hat{\mathcal{L}}F$. We require to differentiate F with respect to time to eventually introduce the equations of motion of the particles into the structure of the Poisson brackets.

$$\frac{d}{dt} F(q, p, t) = \frac{\partial F}{\partial t} + \frac{\partial F}{\partial q}\frac{\partial q}{\partial t} + \frac{\partial F}{\partial p}\frac{\partial p}{\partial t}$$

The time derivatives $\dot{q}$ and $\dot{p}$ are replaced, in that order, by the Hamiltonian derivatives of $\mathcal{H}$ with respect to the variables p and q. Thus, the change of F in time is computed using Hamilton's equations of motion.

$$\frac{d}{dt} F(q, p, t) = \frac{\partial F}{\partial t} + \frac{\partial F}{\partial q}\frac{\partial \mathcal{H}}{\partial p} - \frac{\partial F}{\partial p}\frac{\partial \mathcal{H}}{\partial q} \tag{6.70}$$

The use of the Poisson brackets simplifies the above expression.

$$\frac{d}{dt} F(q, p, t) = \frac{\partial F}{\partial t} + \{F, \mathcal{H}\} \tag{6.71}$$

When $F = \mathcal{H}$, we have $d\mathcal{H}/dt = \partial F/\partial t$. Expression 6.71 is consistent with the Hamiltonian equations of motion for $F = q$ and $F = p$.

$$\frac{d}{dt} q = \{q, \mathcal{H}\} = \frac{\partial \mathcal{H}}{\partial p} \quad ; \quad \frac{d}{dt} p = \{p, \mathcal{H}\} = -\frac{\partial \mathcal{H}}{\partial q}$$

In this regard, Eq. 6.71 is a general equation of motion in the Poisson formulation. Such an equation is the classical version of the quantum mechanical counterpart shown below.

$$\frac{d}{dt} F(t) = \langle \psi(t) | \frac{\partial \hat{F}(t)}{\partial t} | \psi(t) \rangle + \frac{1}{i\hbar} \langle \psi(t) | [\hat{F}, \hat{\mathcal{H}}] | \psi(t) \rangle$$

The quantity $\hat{F}$ is the quantum mechanical operator associated with the physical observable F, and ψ is the wave function of the particle system. The above equation gives the time evolution of the observable at the quantum mechanical level. In principle, there is a correspondence between quantum and classical terms in the classical limit, which is symbolized by $\hbar \to 0$.

$$\frac{1}{i\hbar} \langle \psi(t) | [\hat{F}, \hat{\mathcal{H}}] | \psi(t) \rangle \quad \underset{\hbar \to 0}{\longrightarrow} \quad \{F, \mathcal{H}\}$$

The Newtonian equations of motion are recovered from the quantum mechanical approach when the proper conditions are provided (these are discussed in a later chapter).

Problem 6.10 Deduce the Newtonian equations of motion from Eq. 6.71.

Solution Consider the dynamic variables x and p. We write $F = x$ on the one hand and $F = p$ on the other, in the Poisson brackets of Eq. 6.71.

$$\frac{dx}{dt} = \frac{\partial x}{\partial x}\frac{\partial \mathcal{H}}{\partial p} - \frac{\partial x}{\partial p}\frac{\partial \mathcal{H}}{\partial q} \quad ; \quad \frac{dp}{dt} = \frac{\partial p}{\partial x}\frac{\partial \mathcal{H}}{\partial p} - \frac{\partial p}{\partial p}\frac{\partial \mathcal{H}}{\partial q}$$

Variables x, p are independent of each other, $\partial x/\partial p = 0$, $\partial p/\partial x = 0$. By assuming the Hamiltonian as $\mathcal{H} = \sum_i p_i^2/2m_i + V(\{q_i\})$, the above equations reduce to:

$$\frac{dx_i}{dt} = \frac{p_i}{m_i} \quad ; \quad \frac{dp_i}{dt} = -\frac{\partial V}{\partial q_i}$$

Thereby, the Newtonian equations of motion are reproduced in the Poisson formulation.

In the classical situation, the time evolution of F is obtained from Eq. 6.71, and after taking the Liouville operator into consideration.

$$\frac{d}{dt}F(q, p, t) = \left[\frac{\partial}{\partial t} - \{\mathcal{H}, \cdot\}\right]F = \left[\frac{\partial}{\partial t} + \hat{\mathcal{L}}\right]F \tag{6.72}$$

The total time derivative of a constant of motion, say F, satisfies the expression $dF/dt = 0$ and, consequently, $\partial F/\partial t = \{\mathcal{H}, F\}$, which provides an alternative way to find constants of motion. If the constant of motion shows no explicit time dependence, then $\partial F/\partial t = 0$, and F commutes with the Hamiltonian, $\{\mathcal{H}, F\} = 0$. At the quantum mechanical level, the quantities conserved over time also commute with the Hamiltonian operator, thus maintaining consistency between the quantum and classical theories.

There are additional properties satisfied by the Poisson brackets. According to the definition based on the partial derivatives, we obtain for $u = q_i$ and $v = p_i$ in Eq. 6.68 the binary operation rules.

$$\begin{gathered} \{q_i, q_j\} = 0 \quad ; \quad \{p_i, p_j\} = 0 \quad ; \quad \{q_i, p_j\} = \delta_{ij} \\ \{q_i, u\} = \frac{\partial u}{\partial p_i} \quad ; \quad \{u, p_i\} = \frac{\partial u}{\partial q_i} \end{gathered} \tag{6.73}$$

Other features of the Poisson brackets are the antisymmetric, bilinear, non-commutative product, and permutation properties, which are illustrated with the arbitrary functions u, v, g, and the constants c_1 and c_2.

$$\begin{gathered}\{u,u\}=0 \quad ; \quad \{u,v\}=-\{v,u\} \\ \{c_1u+c_2v,g\}=c_1\{u,g\}+c_2\{v,g\} \quad ; \quad \{uv,g\}=u\{v,g\}+v\{u,g\} \\ \{u,\{v,g\}\}+\{v,\{g,u\}\}+\{g,\{u,v\}\}=0\end{gathered} \tag{6.74}$$

These properties define a *Lie algebra* for the Poisson brackets. The Poisson brackets are invariant under canonical transformations and also include properties of the time derivatives.

$$\tfrac{\partial}{\partial t}\{u,v\}=\{\tfrac{\partial u}{\partial t},v\}+\{u,\tfrac{\partial v}{\partial t}\} \tag{6.75}$$

6.22 Classical Time Propagator

One of the advantages of the Poisson scheme over the other approaches is the introduction of a *classical time propagator* to advance a system of particles forward in time. The time evolution is performed on the individual positions and velocities of the particle system. In contraposition, the time propagator of quantum mechanics evolves the wave function with time. The outcome of evolving a particle system in time is the detailed knowledge of its history. The goal is to propose an arbitrary function, like $F(\{q_i\},\{p_i\},t)$, with dependence on the generalized coordinates and time, and find its evolution with time. Some examples of a function like this are the distribution function in phase space, the mechanical pressure, system volume, temperature, and, in general, any quantity given in terms of the basic variables of position and momentum.

In order to find the classical time propagator, it is supposed that the equation of motion:

$$\partial F/\partial t=0 \quad \Rightarrow \quad dF/dt=\{F,\mathcal{H}\}$$

is satisfied, where the Hamiltonian is time independent. The formal solution of this equation is obtained in iterative form (in the same way as it is performed in the quantum situation for the Dyson time propagator). The result contains nested Poisson brackets and is written as a series expansion.

$$F(t)=F(t_0)+\{F,\mathcal{H}\}_{t=t_0}\,t+\{\{F,\mathcal{H}\},\mathcal{H}\}_{t=t_0}\,\tfrac{t^2}{2!}+\{\{\{F,\mathcal{H}\},\mathcal{H}\},\mathcal{H}\}_{t=t_0}\,\tfrac{t^3}{3!}+\cdots \tag{6.76}$$

Problem 6.11 Obtain the series expansion 6.76 of function F by assuming that $\{F,\mathcal{H}\}$ is time indifferent.

(continued)

Solution We depart from the equation $dF/dt = \{F, \mathcal{H}\}$. It is integrated over time.

$$\int_{t_0}^{t} \frac{d}{dt} F \, dt = \int_{t_0}^{t} \{F, \mathcal{H}\} \, dt \qquad \Rightarrow \qquad F(t) = F(t_0) + \{F(t_0), \mathcal{H}\} \, \Delta t$$

The identity on the left is the expression that we take as a reference point to work out the iteration process. This process starts by inserting $F(t)$, given on the right, into the identity on the left.

$$\begin{aligned} F(t) &= F(t_0) + \int_{t_0}^{t} dt \, \{F(t_0) + \{F, \mathcal{H}\} \, \Delta t, \mathcal{H}\} \\ &= F(t_0) + \{F(t_0), \mathcal{H}\} \, \Delta t + \{\{F(t_0), \mathcal{H}\}, \mathcal{H}\} \, (\Delta t)^2/2 \end{aligned}$$

By inserting this identity into the expression considered as our reference point, we have:

$$\begin{aligned} F(t) &= F(t_0) + \int_{t_0}^{t} dt \, \{F(t_0) + \{F(t_0), \mathcal{H}\} \, \Delta t + \{\{F, \mathcal{H}\}, \mathcal{H}\} \, \tfrac{(\Delta t)^2}{2}, \mathcal{H}\} \\ &= F(t_0) + \{F(t_0), \mathcal{H}\} \, \Delta t + \{\{F(t_0), \mathcal{H}\}, \mathcal{H}\} \, \tfrac{(\Delta t)^2}{2} \\ &\quad + \{\{\{F(t_0), \mathcal{H}\}, \mathcal{H}\}, \mathcal{H}\} \frac{(\Delta t)^3}{3!} \end{aligned}$$

where the properties 6.74 of the Poisson brackets were used. When we continue with the iterations, the series expansion 6.76 is obtained. The same result is achieved by continuous differentiation of $dF/dt = \{F, \mathcal{H}\}$, and using the fact that the nth derivative is the derivative of the $(n-1)$th derivative, namely, $d^n/dt = d(d^{n-1}/dt)/dt$, which implies an iteration process in the Poisson method. For a third derivative, we have:

$$\frac{dF}{dt} = \{F, \mathcal{H}\} \quad ; \quad \frac{d^2F}{dt^2} = \frac{d}{dt} \{F, \mathcal{H}\} = \left\{\frac{dF}{dt}, \mathcal{H}\right\} = \{\{F, \mathcal{H}\}, \mathcal{H}\}$$

$$\frac{d^3F}{dt^3} = \frac{d}{dt} \{\{F, \mathcal{H}\}, \mathcal{H}\} = \left\{\frac{dY}{dt}, \mathcal{H}\right\} = \left\{\{\frac{dF}{dt}, \mathcal{H}\}, \mathcal{H}\right\} = \{\{\{F, \mathcal{H}\}, \mathcal{H}\}, \mathcal{H}\};$$

$$Y = \{F, \mathcal{H}\}$$

Higher derivatives are similarly computed. A Taylor expansion of F is:

$$F(t) = F(t_0) + t \left.\frac{dF}{dt}\right|_{t=t_0} + \frac{t^2}{2!} \left.\frac{d^2F}{dt^2}\right|_{t=t_0} + \cdots$$

(continued)

When the derivatives are replaced by their respective Poisson brackets, the result 6.76 is reached.

The series expansion based on the Poisson brackets plays the role of a generator in the approach of infinitesimal canonical transformations, leading to the time evolution of the dynamical variables [8]. It is convenient to write Eq. 6.76 in a compact form by taking the function F as the common factor of the Poisson brackets from the left side.

$$F(t) = F(t_0)\left[1 + \{\,,\mathcal{H}\}\,t + \{\{\,,\mathcal{H}\},\mathcal{H}\}\,\frac{t^2}{2} + \{\{\{\,,\mathcal{H}\},\mathcal{H}\},\mathcal{H}\}\,\frac{t^3}{3!} + \cdots\right]_{t=t_0} \tag{6.77}$$

The terms in square brackets are symbolically represented altogether by an exponential operator, $\exp(\mathcal{H}\,t)$, acting on the F function from the right-hand side.

$$F(t) = F(t_0)\,\exp(\mathcal{H}\,t)\Big|_{t=t_0} \tag{6.78}$$

The *time propagator*, $\hat{\mathcal{U}}$, is now defined to advance the function F from t_0 to t.

$$F(t) = F(t_0)\,\hat{U}(t_0, t) \quad ; \quad \hat{U}(t_0, t) = \exp(\mathcal{H}\,t) \tag{6.79}$$

This expression is the analog of a quantum mechanical propagator after carrying a *Wick rotation*, where the Newtonian time is replaced by the imaginary time $-it/\hbar$. We now illustrate the use of the Poisson brackets as a time propagator. Consider the Hamiltonian of a particle moving in the z dimension and subjected to gravity effects, $V = -mgz$.

$$\mathcal{H} = p^2/2m - mgz$$

We require to calculate the time evolution of the z coordinate. By making $F = z$, and using Eqs. 6.73 to compute the first Poisson brackets, $\{q_i, F\} = \partial F/\partial p_i$, $\{F, p_i\} = \partial F/\partial q_i$, we implement the time propagator.

$$\{z, \mathcal{H}\} = p/m \quad ; \quad \{\{z, \mathcal{H}\}, \mathcal{H}\} = \{p, \mathcal{H}\}/m = g$$

The following Poisson brackets in the series expansion of the exponential are null because they commute with the constant that results from the application of the previous brackets. Such a procedure represents a way to generate constants of motion using the Poisson approach. We collect the non-zero terms in a single expression according to Eq. 6.77.

$$z(t) = z(t_0) + \{z(t_0), \mathcal{H}\}\, t + \{\{z(t_0), \mathcal{H}\}, \mathcal{H}\}\, t^2/2$$

$$z(t) = z(t_0) + p(t_0)\, t/m + g t^2/2$$

The equation describes a particle falling in a gravitational field with constant acceleration g.

In this chapter, we have discussed the various formulations of analytical mechanics, each of which leads to different equations of motion. However, it is important to note that these different formulations and their corresponding equations of motion are equivalent, resulting in the same time evolution of the particle system and providing a unique solution. While some formulations may be more suitable for solving certain types of problems, it is often necessary to use numerical methods to solve the equations of motion when no analytical solution is available. This will be the focus of a future chapter.

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Part II

Basics of Quantum Mechanics



Wave-Particle Duality of Matter

7

Quantum mechanics is an essential part of our science to understand nature. It emerged in the turn of the 20*th* century and continues growing up to explain phenomena at the atomic and subatomic levels. The purpose of this chapter is to briefly expose the experiments from which the novel ideas emerged and gave birth to quantum mechanics. The first section describes the works that demonstrated the wave character of both the light and the microscopic particles as in the Young's experiment, and contrariwise, the experiments that unveiled the particle character of the electromagnetic waves as in the photo-electron and the Compton experiments, thus establishing the *dual nature of matter*. In the second section, the symmetry between the particles and the waves is confirmed with the Davisson-Germer's experiment and the de Broglie hypothesis. It is rationalized that the particles may be described as waves and the waves may be described as particles. All these experiments point out the need to reformulate the conceptual world to have a wave theory of matter.

7.1 Young's Experiment

In the first centuries of science, the light was conceived as a particle. However, in the Young's experiment, an interference pattern was produced, showing the wave nature of light. The experiment consisted on emulating the undulations and interference observed in water waves and sound pulses. Two coherent light waves were produced from a single source when the original light wave was diffracted by two slits. The pair of waves interfered in constructive and destructive forms, creating white and black fringes on a screen as a result of traveling different optical paths and interfering with relative phases, Fig. 7.1.

The diffraction pattern is not the smooth curve that one expects from the dispersion of small particles and deduced from the addition $I_1 + I_2$ of the individual light intensities, I_i. In fact, the pattern is understood by assuming a *probability density* obtained from a wave, ψ, describing the particles. At the instant of time

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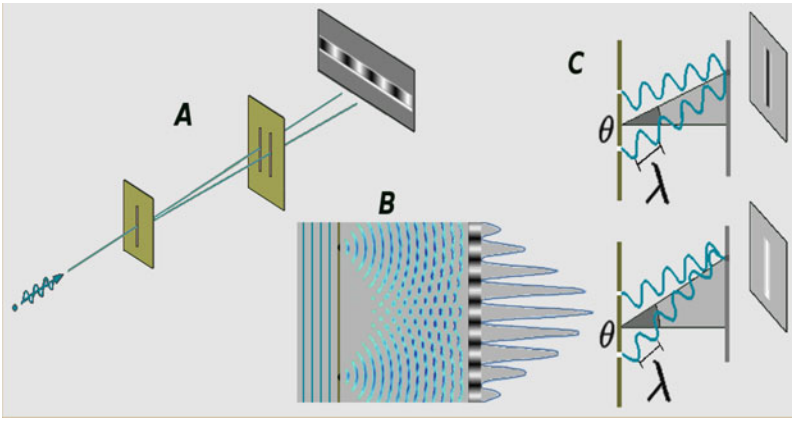


Fig. 7.1 The double-slit experiment demonstrates the wave nature of light, as well as that of the electron, proton, neutron, and sub-atomic particles. Inset A: a single slit makes the beam parallel, and a couple of slits located further ahead produce two sources of coherent beams. Inset B: the beams are described by spherical wavefronts and treated as plane waves at far distances. Inset C: in essence, a wave is made to interfere with a copy of it to produce a diffraction pattern consisting of dark and bright fringes onto a screen. A dark band is due to waves interfering destructively, and a bright band is due to waves interfering constructively

t in the position $\mathbf{r}$ of the screen, the *total wave function* is the superposition of the individual wave functions, $\psi(\mathbf{r}, t) = \psi_1(\mathbf{r}, t) + \psi_2(\mathbf{r}, t)$, where the position $\mathbf{r}$ is a continuous variable and the subindexes of the wave functions distinguish the beams coming from the two slits of the experiment. The probability density is $I(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2$ [1]. The Young's experiment is considered a proof of the wave behavior of light and in general of other microscopic particles capable of producing a diffraction pattern in that type of experiment. For instance, it has been possible to obtain the quantum interference of large objects with masses exceeding 10^4 *amu* in the Young's experiment [2]. In this connection, the Young's experiment is also important to differentiate the behavior of particles acting in the quantum domain from the behavior of particles acting in the classical domain. The Young's experiment demonstrates the need to reformulate our conceptual world at the microscopic level since classical mechanics fails to provide the correct description on the behavior of particles at the atomic and sub-atomic levels.

7.2 Interference of Waves

The double-slit experiment is an experiment that provides information on the wave character of the particles. The beam coherence is required to induce the stationary states of the waves. In principle, the coherence of waves is estimated by evaluating the correlation of functions involving the phase, frequency, amplitude, and intensity.

The key aspect of the Young's experiment is the formation of a diffraction pattern as a consequence of the interference of waves.

The interference represents the addition of waves to create a new wave with an amplitude that depends on the relative phase of the interfering waves. The interference depends on the coherence of the waves, but the coherence is not directly measured in the lab due to the fast oscillations of the electromagnetic waves; nevertheless, it can be indirectly estimated using the correlation between waves. There are spatial and temporal correlations that establish the coherence of waves at different spatial points and instants of time. By making a wave to interfere with a copy of itself using a beam splitter, it is possible to induce a self-interaction at different spatial positions and time instants.

A monochromatic wave is characterized by owing a single frequency, like the light emitted by a laser with a high degree of correlation. A wave composed by waves with different frequencies (and possibly with different amplitudes and phases) as, for example, the white light emitted by an ordinary bulb, has a poor degree of correlation. The interference of waves consisting of different frequencies, but with a fixed phase difference, produces localized pulses. If the waves have different frequencies and no fixed phase difference, then their superposition produces a continuous beam of white light. It is referred to as *white noise* because it is formed by incoherent and poorly correlated waves.

The precise trajectories of the waves before they form the interference pattern are unknown. The interference pattern may be sampled by changing the distance of the screen from the slits.

Finally, other similar phenomena, like the interference of laser light, superconductivity, and superfluidity, represent cases of coherence. They are observed at the macroscopic level. In this connection, the forms in which the waves behave, interfere, and build a diffraction pattern are important for the formulation of quantum mechanics, where probability fields, probability densities, and more have to be considered.

7.3 Photo-Electron Experiment

The metals are characterized by atoms containing a large number of electrons. Many of the electrons orbit the nucleus relatively far away and are weakly bonded. Such electrons can produce an electric current when an electric field is applied to the metal. However, under the action of an intense external force, the electrons may abandon the metal. When light impacts a metal, some electrons are released from the metal surface since the light transports energy, which is transformed onto kinetic energy for the electrons. The released electrons receive the name of *photo-electrons*. In the original photo-electron experiment, the light strikes a metal of sodium atoms and detaches electrons, Fig. 7.2.

More energy is spent on releasing electrons from the internal material layers than from the surface. The released electrons show different values of the kinetic energy E_{kin}, but the surface electrons achieve the maximum kinetic energy. The

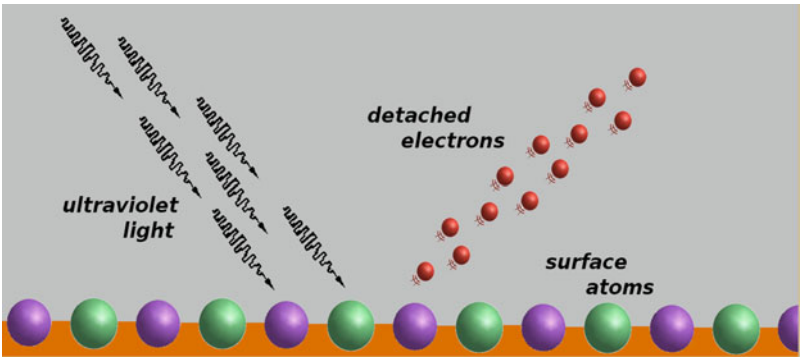


Fig. 7.2 In the photo-electron experiment, a light striking a metal produces a flux of electrons after being detached. The electron detachment depends on the characteristics of the impacting light and the material. For instance, electrons are released from an alkali metal surface when visible light strikes the surface. *X* rays, ultraviolet light, and the infrared beams can detach electrons from other types of surfaces as well

energy distribution of the detached electrons is independent of the light intensity; nevertheless, it depends on the light frequency because a stream of electrons is produced when the light frequency exceeds a threshold value, ν_0, particular of the surface. Moreover, the maximum kinetic energy of the ejected electrons is observed to be proportional to the light frequency ν, where the constant of proportionality is the *Planck's constant*, h, with units of angular momentum.

$$E_{kin}^{max} = h(\nu - \nu_0) \quad ; \quad h = 6.626,07 \times 10^{-34}\ J\,s \tag{7.1}$$

The term $h\nu_0$ is the minimum energy required by an electron to escape from the surface. It is identified as the photo-electron *work function*. If we direct ultraviolet light of 3500 Å wavelength onto a sodium surface, $h\nu_0 = 2.2\,eV$, then the maximum kinetic energy registered is $E_{kin}^{max} = 1.4\,eV$. Interesting phenomena is observed in the photo-electron experiment:

- The number of released photo-electrons increases with the number of photons (or light intensity) for a specific light frequency.
- The maximum kinetic energies of the electrons are the same for high-intensity light and low-intensity light of the same frequency.
- A high-intensity light produces less energetic electrons than a low-intensity light of higher frequency (the high-intensity red light produces the ejection of a large number of electrons with relatively small kinetic energies; in contrast, a low-intensity violet light produces the ejection of a smaller number of electrons with comparatively higher kinetic energies).
- No absorption of a fraction of a photon occurs, and all the photon energy is entirely transferred to the electron.

- The ejection of electrons occurs with no time delay (i.e., there is not a gradual energy accumulation).

The theory of classical electromagnetism is incapable to explain the above observations as it assumes several points:

- A wave is a uniform and continuous distribution of energy with spatial extension. The light spreads uniformly over the surface and is difficult to be absorbed by a spatially localized (tiny) electron.
- For any wave frequency, by increasing the light intensity (the intensity is related to the amplitude E of the electromagnetic wave), the electric force eE is believed to induce large amplitude vibrations of the electron, eventually producing its release from the material.
- The electron emission is presumed to occur after gathering sufficient energy from the illuminating beam. The energy gathering is independent of the light frequency and the threshold frequency of the material.
- It is speculated that a time lag should be registered due to the time it takes an electron to accumulate energy from the striking electromagnetic waves before its ejection. The time lag is supposed to depend on the wave intensity, since low-intensity light provides less energy than a high-intensity light, in a given time interval.

According to Einstein, the phenomena observed in the photo-electron experiment are explained by considering light as a localized-energy entity rather than a continuous distribution of energy (like a wave) in space. The entity transfers energy in the form of discrete energy bundles to the surface. The name photon or *quanta* is given to light that is considered in discrete energy packets and acts as a particle. It is also understood as a moving electromagnetic ripple. The photo-electron experiment indicates that frequency is the main variable in the electron detachment. The emission occurs instantaneously, by overcoming a threshold frequency which is specific to the material.

The photo-electric effect is not the only technique to eject electrons from a metal, because we also have the *thermionic emission*. In this process the metal is subjected to intense heat. When the thermal vibrations exceed the thermionic work function, known to be close in value to the work function of the photo-electric effect, the electrons are ejected (the functioning of incandescent bulbs is based on the thermionic emission process). In this respect, the photo-electric and thermionic emission effects represent two sources for ejecting electrons. Some of the important applications of the photo-electric effect are the photomultiplier tubes and photometers, which work as "electric eyes" to detect ionizing radiation and measure the intensity of light emitted by materials.

Concluding, the usual behavior of light in free space may be that of a wave; however, the photo-electron experiment suggests that a beam of light behaves as a stream of particles, which is distinguished by a frequency (an attribute originally thought to be exclusive of waves). Clearly, we require a new formulation capable

of explaining the interaction of light with matter and, in particular, the energy transfer between light and the microscopic particles, still, without abandoning the classical electromagnetic theory because it provides the appropriate description of light propagation, as well as the interference and diffraction of electromagnetic waves.

7.4 Compton's Experiment

Another work that fundamentally changed our understanding of nature was Compton's experiment. In such an experiment, a beam of X rays is created in a vacuum tube. The electromagnetic radiation impacts a target made of carbon atoms and is scattered by the electrons of the carbon atoms in every direction. The electrons have mass m_0 at rest. The scattered wavelength λ' is larger than the wavelength λ of the incident beam and depends on the scattering angle ϕ, Fig. 7.3.

The shift in the wavelength with the scattering angle is explained by employing the energy and momentum conservation laws in the collision of the incident light with the electron, where the light is assumed as a particle. In this context, the shifted wavelength is given by:

$$\lambda' = \lambda + h/(m_0 c)\,(1 - \cos\phi) \tag{7.2}$$

Compton's experiment additionally confirms the theory of special relativity. In contrast to Compton's experiment, there are experiments that reveal the wave nature of microscopic particles. They are discussed below.

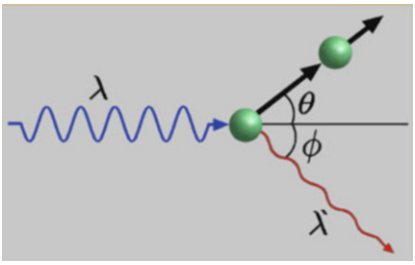


Fig. 7.3 In the Compton's experiment, an electromagnetic wave of high energy, such as an X ray with wavelength λ, impacts an electron of an atom. The electron recoils with angle θ, and the wave is scattered out of the original direction with angle ϕ and wavelength λ'. The interaction is described as a collision between a pair of particles, that is, between a photon and an electron. The experiment illustrates the particle character of the electromagnetic wave in this case

7.5 Davisson-Germer's Experiment

Davisson-Germer's experiment starts with a thermionic emission, where a flux of electrons is emitted from a hot filament and is used as an electron gun. The electrons are accelerated by applying a voltage gradient of a battery and scattered by the atoms of the crystal planes of a nickel-chloride metal. Since the crystal atoms simulate a kind of reflection and regular grating, the electrons are expected to produce a uniform distribution of particles; however, they are observed to produce a distribution of alternating intensity peaks at specific scattering angles, namely, a diffraction pattern. The pattern is explained by *Bragg's law*, which was in principle believed to only apply to the diffraction of waves. The Bragg condition for constructive interference is:

$$2d \sin\theta = n\lambda \quad ; \quad n = 1, 2, \cdots \tag{7.3}$$

The distance between contiguous lattice planes is d, the variable θ is the angle between the incident beam and the crystal planes of the metal, and λ is the associated wavelength of the particles. The angle between the incident beam and the reflected beam is the scattering angle, denoted by ϕ, which is not present in Eq. 7.3. The factor n gives the different intensity peaks of the diffraction pattern. By measuring the relevant parameters in Davisson-Germer's experiment ($d = 0.91\,\text{Å}$, $\theta = 65°$), the maximum intensity ($n = 1$, with direction $\phi = 50°$) is obtained at an accelerating voltage of $V = 54\,eV$. The resulting wavelength is $\lambda \simeq 1.649\,\text{Å}$. There are complications in applying Bragg's law to the electrons diffracted by the inner layers of the crystal as they interfere in complex forms. Similar experiments to Davisson-Germer's experiment have been performed using other types of particles, such as hydrogen and helium, thus determining the wave character of those particles. Summing up, Davisson-Germer's experiment gives testimony of the wave properties of matter particles and also confirms the de Broglie hypothesis, discussed below.

7.6 de Broglie's Hypothesis

The fact that matter particles also act as waves permits to assign wave properties to the microscopic particles. Louis de Broglie hypothesized a wavelength for matter waves by considering both: (i) the relativistic energy $E_p = (p^2c^2 + m_0^2c^4)^{1/2}$ of a particle, where the particle's rest mass is m_0, the momentum is p, and the light speed is c, and (ii) the energy $E_w = h\nu$ of a wave, where ν is the wave frequency and h is the Planck's constant. For small kinetic energies of the particle with respect to the rest mass energy, m_0c^2, the relativistic effects are neglected after doing $m_0 = 0$. By recalling that $\lambda\nu = c$, we have two representations of the wave-particle energy, $E_p = pc$ and $E_w = hc/\lambda$. The equivalence of the energies, $E_p = E_w$, gives the main result $\lambda = h/p$. The particle has momentum $p = mv$, and the wavelength of matter waves is inversely proportional to the momentum.

Table 7.1 de Broglie's wavelength of different bodies

Particle	Mass (Kg)	Speed	λ (Å)
Electron	9.11×10^{-31}	$10^6\, m/s$	7.3
Proton	1.6726×10^{-27}	$10^5\, m/s$	0.04
Tennis ball	0.056	$260\, Km/hr$	1.6×10^{-24}

$$\lambda = h/(mv) \tag{7.4}$$

For macroscopic bodies, the mass is large, and the wavelength is sufficiently small; for sub-atomic particles, the mass is small, and the wavelength is large. Table 7.1 shows the wavelength of different particles. The particles with large wavelengths have wave character. The electron is the particle with largest wave character in the table, while ordinary bodies, such as a tennis ball, have essentially null wave character. Thus, Davisson-Germer's experiment is not the only one to exhibit the wave character of the microscopic particles.

An alternative form of writing the matter wavelength is in terms of the kinetic energy of the particle, $E_{kin} = p^2/(2m)$. By solving for the momentum p in this expression, the de Broglie relation is shown in terms of the kinetic energy.

$$\lambda = h\,(2m\,E_{kin})^{1/2} \tag{7.5}$$

For a macroscopic particle with values $m = 1\,Kg$, and $v = 1\,m/s$, the de Broglie wavelength is $\lambda = 6.63 \times 10^{-24}$ Å. For a microscopic electron, which is accelerated with a voltage $V = 54\,eV$ in Davisson-Germer's experiment, the de Broglie wavelength is $\lambda = 1.669$ Å, and it compares well with $\lambda \simeq 1.649$ Å obtained from the Bragg's law of the previous section. These results have two important consequences: (i) Davisson-Germer's experiment confirms the de Broglie hypothesis, and (ii) bodies with small de Broglie wavelengths like a dust particle, a tennis ball, and other macroscopic objects are properly described by the theory of classical mechanics. However, bodies with large de Broglie wavelengths, like the helium atom, hydrogen atom, electron, proton, and neutron, lead to interference and diffraction patterns, and in this respect, they are properly described by a *wave theory of matter*.

The fact that microscopic particles have a wave character has led to technological applications in unexpected areas, like optics. For instance, by conceiving the electrons as waves, it was possible the invention of the electron microscope. Summing up, the experiments firmly establish the dual aspect of matter: the particles may be characterized using wave variables, and the waves may be characterized using particle variables. In this connection, the de Broglie's hypothesis represents a symmetry of nature since a wave and a particle are rationalized as equivalent entities.

7.7 Bohr's Complementary Principle

The wave-particle duality is an intrinsic property of matter. It is mainly observed in the microscopic particles and light beams, to which we refer in generic form as the entities. The wave and particle aspects have equal validity on an entity; nevertheless, such a type of symmetry is destroyed when there is an interaction with a measuring device because the device registers either the wave character or particle character of the entity. For instance, if we determine the slit where the entity travels through in the Young's experiment, then the entity is revealed with a particle character. In contrast, when no measurement on any of the slits is performed, the entity is permitted to interfere, and its wave aspect is disclosed on the screen. In general, when one of the two behaviors is displayed, the other complementary behavior is inhibited. In this respect, Neils Bohr proposed the *complementary principle* to indicate that it is possible to determine either the wave character or particle character of an entity in a given experiment, but it is impossible to observe both behaviors of the same entity simultaneously. The wave-particle duality of an entity is implicitly considered in the uncertainty principle, yet the uncertainty principle should not be confused with the complementary principle.

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Quantization of the Energy

8

Quantum mechanics describes phenomena which in principle are beyond our common sense. It represents a general subject from which the theory of classical mechanics is derived under special circumstances. The general aspects of quantum mechanics are originated from a scientific frame based on a rigorous mathematical formulation, which allows to realize not only the wave-particle nature of matter but also the *energy quantization* in the microscopic world. Energy quantization was revealed in a few important experiments in the beginning of the 20*th* century. They now represent landmarks in the history of science, as they provide the foundations of quantum mechanics. We introduce in the following sections additional concepts and principles to those already discussed previously. It is with quantum mechanics that we are able to reformulate the old ideas, and establish a new philosophy to understand the world.

8.1 Planck's Energy Equation

The maximum kinetic energy E_{kin}^{max} of the electrons emitted in the photo-electron experiment shows a linear relation with the frequency ν of the striking radiation, Eq. 7.1. The linear relation is independent of the beam intensity, and the slope of the line in this relation is the Planck constant, h. In this regard, the connection of E_{kin}^{max} to ν is ruled by the general *Planck's energy equation*.

$$E = h\nu \tag{8.1}$$

After overcoming a threshold energy, the above expression agrees with the energies reported in the photo-electron experiment. It also represents the basic hypothesis to explain the distribution of frequencies in the blackbody-radiation experiment.

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Planck's expression may be written in terms of the angular frequency, ω, using the expression of $\hbar$.

$$E = \hbar\omega \quad ; \quad \hbar = h/(2\pi) \quad ; \quad \omega = 2\pi\nu \tag{8.2}$$

8.2 Blackbody Radiation Experiment

When a piece of matter is heated, it reaches a temperature above absolute zero, and there is a transformation of the body's thermal energy onto electromagnetic energy, which is dispersed as electromagnetic radiation. This radiation is known as *thermal radiation*. However, a body equally absorbs thermal radiation. When the body or system absorbs and eventually emits all the incident radiation with no dependence on the type of the incident radiation, then the body is known as a *blackbody*. The blackbody is a perfect body made of non-reflective material that absorbs and radiates energy in equal amounts and in all directions in the state of thermal equilibrium, Fig. 8.1. The emission factor of a blackbody is 1. The blackbody becomes a *gray body* when the emission factor is lower than one. The body is a *white body* when it reflects all the incident radiation. In contrast, a perfect *transparent body* transmits all the incident radiation.

When an ordinary body is burned, the energy emitted per unit area and per unit time determines the body's radiation intensity. In the special case of a blackbody at temperature T, the radiation shows a characteristic distribution of continuous frequencies. Moreover, for a given frequency of emitted radiation, the radiation intensity exclusively depends on the body's temperature, regardless of the body's composition and shape. The typical colors of the emitted radiation start from low-frequency invisible radiation (given at room temperature), following red visible light (colloquially referred as heat), passing by bright yellow, dazzling blue-white color,

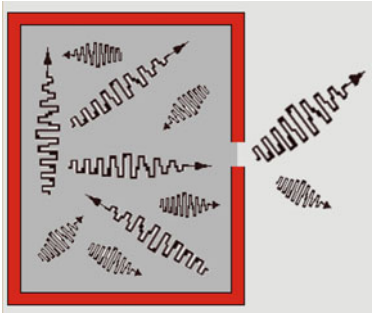


Fig. 8.1 A blackbody is a perfect absorber and radiator of electromagnetic radiation. The figure shows that a tiny hole in the surface of a box can be used to characterize a blackbody. The irradiance of the blackbody illustrates the quantization of the energy, confirms the dual character of light, and satisfies the Planck formula from which the Rayleigh-Jeans law, Stefan-Boltzmann expression, and Wien's displacement law are obtained

high-frequency blue visible light, reaching ultraviolet light, and X rays. The order of the colors corresponds to increasing temperatures and leads to a chromatic diagram in terms of the temperature. A model frequently used to emulate the radiation features of a blackbody is an idealized hole located in one of the walls of an insulated cavity with reflective walls in the interior, Fig. 8.1. The hole is sufficiently small with respect to the cavity size but larger than the wavelength of the incident radiation to avoid reflections. A small hole also ensures negligible perturbations once the thermal equilibrium is achieved inside the cavity. It is through the hole that radiation comes in and goes out of the cavity, simulating the radiation absorption-emission process of a blackbody.

8.3 Rayleigh-Jeans Law

The energy in the cavity takes the form of electromagnetic waves.[1] The waves are unpolarized and take specific forms since the electric and magnetic fields are perpendicular to each other and must satisfy boundary conditions of zero electric field and zero magnetic field at the cavity surfaces. The waves are adjusted to the cavity dimensions. The electromagnetic waves satisfy the wave equation:

$$\frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2} + \frac{\partial^2 E}{\partial z^2} = \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2} \tag{8.3}$$

For a cubic cavity with side length L, in thermal equilibrium, the electromagnetic waves of different frequencies in the cavity are represented by periodic functions forming standing waves, also recognized as the resonant vibrational modes of the cavity.

$$E(x, y, z) = A \sin\left(\frac{n_1 \pi x}{L}\right) \sin\left(\frac{n_2 \pi y}{L}\right) \sin\left(\frac{n_3 \pi z}{L}\right) \sin\left(\frac{2\pi ct}{\lambda}\right) \tag{8.4}$$

By dealing with the escaping energy from the cavity in perpendicular way to the hole, a radiated energy with dependence on the frequency ν is found. From a classical perspective, each vibrational mode represents an independent degree of freedom and has the possibility to exchange energy with the other modes. According to the equipartition theorem of classical statistical mechanics, the vibrational modes have the same energy $k_B T$ in thermal equilibrium. The number of cavity modes per frequency and per volume is proportional to ν^2.

[1] An electromagnetic wave is understood as an electric field and a magnetic field in interaction with mutual perturbations on each other, jointly oscillating and travelling in space with the speed of light. The electric field may been considered as the source of the magnetic field from a relativistic perspective, but both of them are treated as the electromagnetic field, governed by the Maxwell equations.

$$\rho(\nu) = \frac{8\pi}{c^3}\,\nu^2 \tag{8.5}$$

The total radiated energy per volume and per frequency corresponds to the number of modes times the energy per mode.

$$u(\nu, T) = \tfrac{8\pi}{c^3}\,\nu^2 k_B T \tag{8.6}$$

According to this expression, the waves in the cavity with short wavelengths (or high frequencies) contribute more to the energy than the waves with large wavelengths (or low frequencies), as the energy contribution scales quadratically with the frequency, Fig. 8.5. If we base our predictions on classical mechanics, there is no restriction on the radiated energy in acquiring any value. In consequence, for increasing frequencies, the intensity spectrum predicted by classical mechanics increases without limits (and the spectrum rapidly diverges when the frequency approaches the ultraviolet region). The corresponding expression of the radiation intensity or spectral radiance is known as the *Rayleigh-Jeans law*, presently known to be valid at low frequencies solely.

$$I(\nu, T) = (2/c^2)\,\nu^2 k_B T \tag{8.7}$$

In Sect. A.1, expressions 8.5–8.7 are derived.

8.4 Wien's Displacement Law

The experiments on objects that approximately exhibit features of a blackbody show a curve of the radiation intensity with a bell shape that eventually decays in the ultraviolet region of the electromagnetic spectrum. For increasing temperatures, the prominence moves to smaller wavelengths (or higher frequencies). The wavelengths of the peaks are related to each other by the *Wien's displacement law*, which indicates that the product of a wavelength λ_{max} registered at the maximum of the radiation spectrum with the equilibrium temperature T_{eq} is constant.

$$\lambda_{max} T_{eq} = 2.898 \times 10^{-3}\, m K \tag{8.8}$$

There is an analog expression in terms of the maximum frequency, $\nu_{max} T^{-1} = 5.879 \times 10^{10}\, Hz\, K^{-1}$. The Wien's displacement law in its different forms is derived in Sect. A.1.5.

8.5 Ultraviolet Catastrophe

The experimental observations contradict the predictions of classical mechanics referent to the shape of the blackbody spectrum, especially in the region where an infinite number of vibrational modes is conceived from the classical perspective. Such a discrepancy is catastrophic in the sense that it is of fundamental importance in physics, receiving the name of *ultraviolet catastrophe*. However, we find the appropriate explanation of the blackbody radiation in the wave theory of matter and, in particular, using the Planck's law on the energy.

8.6 Planck's Law

In order to explain the maximum radiation intensity for every temperature of the blackbody, Planck postulated an energy quantization based on the vibrational modes in the cavity. Specifically, the electromagnetic waves in the cavity only take discrete packets of energy.

$$E = h\nu,\ 2h\nu,\ 3h\nu,\ \cdots \tag{8.9}$$

The basic energy unit is $h\nu$, and it represents a minimal energy and is called the *quantum* (from the Latin word which means "how much"). According to the energy quantization, the energy states can only acquire integral multiples of the energy $h\nu$, where h is the Planck's constant and ν is the frequency of the vibrational mode. The radiation in the cavity and emitted by the cavity is quantized and comes in energy packets defined by the frequency and Planck's constant. One of the main consequences of assuming the energy quantization is found at high frequencies (or short wavelengths), because high frequency modes have lower probability to occur, imposing a probabilistic upper limit to the radiation intensity. The blackbody's radiative intensity was proposed from experimental results and fitted by Planck in the form:

$$I(\nu, T) = (2h\nu^3/c^2)\,[\exp(h\nu/k_B T) - 1]^{-1} \tag{8.10}$$

The speed of light in the medium (it may be the vacuum) is c, and the Boltzmann constant is k_B. The intensity I is the energy radiated in the normal direction per unit surface area, per unit solid angle, per unit time, and per unit frequency at the given temperature T.[2] The radiated energy is emitted in integral multiples of $h\nu$, becoming more intense at high temperatures for a given frequency, but with upper bounds on the intensity. Planck's law explains why the Rayleigh-Jeans and Stefan-Boltzmann laws are partially valid, Fig. 8.5. The energy quantization is in agreement

[2] Planck's constant is determined by adjusting its value to reproduce the blackbody spectrum in the blackbody expression.

with the experimental facts and eludes the ultraviolet catastrophe (the Planck and Stefan-Boltzmann laws are derived in Sect. A.1).

Problem 8.1 By using Planck's law, Eq. 8.10, referent to the radiation intensity, I, and integrating over the variables (ν, θ, ϕ), deduce the *Stefan-Boltzmann law* expressed by $J = \sigma T^4$, where J is the total radiated power, T is the absolute temperature, and σ is the Stefan-Boltzmann constant. Discuss the possible applications of such a law. The following integrals will be required.

$$J_1 = \int_0^{2\pi} d\phi \int_0^{\pi/2} d\theta \, \sin\theta \, \cos\theta = \pi \quad ; \quad J_2 = \int_0^{\infty} dx \, \frac{x^3}{\exp(x) - 1} = \frac{\pi^4}{15}$$

Solution The blackbody radiates energy with all frequencies in every direction. The total radiated power is obtained by integrating over the variables ν, ϕ, and θ.

$$J = \int_0^{\infty} d\nu \int_0^{2\pi} d\phi \int_0^{\pi/2} d\theta \, \sin\theta \, \cos\theta \, I(\nu, T) \quad ;$$

$$I(\nu, T) = \frac{2h\nu^3}{c^2} \, \frac{1}{\exp(h\nu/k_B T) - 1}$$

The function $\cos\theta$ in the kernel of the integral derives from Lambert's cosine law; it considers the deviations in the radiation from the normal direction to the surface. By making the change of variables $x = (h/k_B T)\nu$, we have:

$$J = \frac{2h}{c^2} \left(\frac{k_B T}{h} \right)^4 J_1 J_2 = \sigma T^4 \quad ; \quad \sigma = \frac{2\pi^5}{15} \, \frac{k_B^4}{c^2 h^3}$$

This is the Stefan-Boltzmann law, where σ is a constant and the radiated power is proportional to T^4. The law is frequently used to estimate the temperature of interstellar objects, such as stars and planets, by measuring their radiated power, Fig. 8.2. The temperature may be similarly assessed by knowing the wavelength at the maximum of the intensity spectrum according to the Wien's displacement law.

The blackbody radiates more energy when the energy supply is increased. The blackbody is an idealized object as the energy emission is totally compensated by the energy absorption, establishing a perfect pattern of the radiation spectrum. Any other body in thermodynamic equilibrium cannot radiate more energy than that of a blackbody. The blackbody spectrum is commonly used to determine the deviations

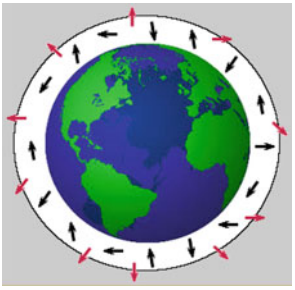


Fig. 8.2 Many interstellar objects like stars, planets, and more radiate light in all directions in the photosphere envelope. Some light is internally reflected (black arrows), and other light escapes to outer space (red arrows). By monitoring the light emitted to outer space and assuming that such interstellar objects behave like a blackbody, their temperatures are determined from the equations of the blackbody

of radiation intensity of real or "non-perfect" objects, generically referred to as gray objects. The deviations from the ideal radiation spectrum are attributed to molecular composition, surface roughness, energy diffusion in the material, and more, which modify the radiation emission of the object.

One of the objects that is close to reproduce the properties of a blackbody is a *black hole*. The black hole acts as a perfect absorber since it absorbs all the electromagnetic waves that come in contact with it, and without any reflection or transmission. The black hole is also a blackbody radiator in a process called *Hawking radiation*. The investigation on the blackbody radiation is old but far from being over. According to the *Fulling-Davies-Unruh effect*, a non-inertial observer registers a non-zero temperature, whereas an inertial observer registers a zero temperature. If a is the acceleration of the non-inertial observer, the Fulling-Davies-Unruh temperature is $T = \hbar a/(2\pi c k_B)$, a quantity similar to Hawking's radiation, where a replaces the gravitational acceleration. In this context, the inertial and non-inertial observers in principle measure different energy states, unexpectedly leading to different numbers of particles in the same vacuum.

8.7 Franck-Hertz Experiment

The quantization of the energy is a fundamental aspect of microscopic matter. The Franck-Hertz experiment gives evidence of the energy quantization in atoms. In such an experiment a flux of electrons is produced to induce the collision of the moving electrons with the electrons bonded to atoms, which are in the gas phase state. The collisions of interest are of inelastic type as they modify the internal atomic structure. When an accelerated electron has the "appropriate energy" and collides with a bonded electron of the atom, the incident electron gives its kinetic energy, and the bonded electron is excited to a higher electronic orbit. The excitation is possible due to the fact that the supplied energy is a quantum of energy that

matches the energy difference of the electronic orbits. The external electrons that come to a halt give no signal of an electric current in the electrodes of the Franck-Hertz experiment, until they are again accelerated by the external voltage. The changes of voltages and the changes of electric currents are related to the excitation energies of the atoms. The excitation energies using the electron bombardment technique reproduce the excitation energies registered in the absorption of photons by atoms in other experiments. In this regard, the change of electronic states is independent of the technique employed, as long as the energy provided to the atom (either by an external particle in collision or by the absorption of an electromagnetic wave) corresponds to an energy quantum of the atom.

The atom returns to the ground state when the electron de-populates the high energy state with energy E_f and decays to the energy state with lower energy E_i. The transition has radiative nature as a photon with frequency ν is emitted in the decay process to preserve the energy.

$$\nu = (E_f - E_i)/h \tag{8.11}$$

The spectral radiation is characteristic of the atom. The emission and absorption spectra of an atom are the same (except, for instance, in the fluorescence phenomenon where a molecule may show a non-radiative transition due to vibrational relaxation, dispersing energy to the environment). A refined Franck-Hertz experiment is required to determine the excitation energies of the inner electronic orbits, because they are masked by the electronic transitions of the first energy levels. Not all the electronic transitions are permitted due to symmetry rules in the angular momentum and the spin of the electrons, Sect. 10.7. Still, among the possible excitations of an atom, the one of general interest is that where the electron makes a transition to the continuum, namely, a transition to the electronic state where the binding energy of the electron is zero, as the electron is released from the atom and the atom is left with a positive charge in this case. The required energy to knock the most weakly bonded electron out of the atom is an energy quantum and corresponds to the *ionization potential* of the atom. Thus, the atoms can be ionized by giving the appropriate kinetic energy to the external electrons in the Franck-Hertz experiment. When the electron density is sufficiently high, the ionized atoms recover their charges by capturing electrons from the dense environment. On the other hand, when the atomic density is low, there is a possibility to produce avalanche effects since not only the external electrons but also the released electrons have large mean free paths, that is, they can travel large distances achieving high speeds and ionizing more atoms by collision. The control of the temperature is equally important to minimize the thermal velocities of the atoms in the experiment. As an example, the formation of an Aurora Borealis is due to the collisions of particles in the atmosphere, resulting in a burst of photons. Concluding, the Franck-Hertz experiment is relevant because it demonstrates the energy quantization of the electron states in atoms.

8.8 Heisenberg's Uncertainty Principle

The theory of classical mechanics is based on what we perceive at macroscopic scales from the physical world. According to classical mechanics, the trajectory of a particle is well-defined when the initial conditions are given, and in consequence, the particle's trajectory is predictable. However, this is not the case of a microscopic particle. The inherent wave aspect of a microscopic particle leads to an indeterminacy in the position, Δx, and momentum, Δp. In this regard, there is a probabilistic factor in the prediction of the particle's trajectory. The indeterminacy in the predictions is not related to the inaccuracy of the measuring devices but to the intrinsic wave behavior of the microscopic particles. This is usually illustrated as follows. Suppose we are interested in determining the particle's position in the laboratory. We point a beam of light to the particle. The radius of the particle is r, a non-zero value, and the wavelength of the light is λ. We choose $\lambda \sim r$. The position of the particle is registered by the detectors when the light is deviated by the particle. However, an electromagnetic wave is a type of energy that can transfer momentum, even though it does not have mass. The energy of a particle, whether it has mass or not and whether it is moving or not, is described using the equation $E = pc + mc^2$. The last term of this equation applies only to particles that have mass and in a steady state. Photons, on the other hand, have no mass but have energy due to their momentum, which is described by the equation $E = pc$. This energy can be transferred to the particle, and we cannot accurately establish the momentum of the microscopic particle. On the other side, if the beam is less energetic and $\lambda \gg r$, the momentum is accurately determined at the expense of an indeterminacy of the particle's position, Fig. 8.3. In general terms, we disturb a particle by measuring its position while the conjugate variable, p, is instantaneously affected and vice versa. Thereby, it is not possible to simultaneously determine the exact position and momentum of the particle.

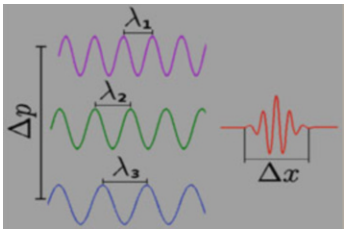


Fig. 8.3 The detection of a particle with radius r is done using a light pulse or wave packet with wavelength $\lambda \sim r$. The pulse is the superposition of many waves. The larger the number of waves forming the pulse, the more localized the pulse is. A small pulse width implies a small uncertainty in detecting the particle's position; nevertheless, the many waves that build the pulse imply a large uncertainty in the momentum according to the theory of Fourier transforms. In contrast, a large pulse width implies a large uncertainty in the particle's position but a small spread of the momentum

If we consider a mass of 1 Kg hanging from a rope with length 10 cm, and oscillating in a gravitational field, $g = 10\ m/s^2$, then the root mean square deviations of the position and momentum computed quantum-mechanically are $10^{-18}\ m$ and $10^{-17}\ kg\,m/s$, respectively. These values are negligible with respect to the dimensions of ordinary macroscopic bodies. This is not the case for microscopic particles [1].

Problem 8.2 A pulse, represented by ψ, is a wave packet formed by the superposition of a large number of waves. The waves have amplitudes A_i and discrete wave numbers k_i. The waves superposition is described by $\psi(x) = \sum_i A_i \cos(k_i x)$. However, by working in the continuum of wave numbers, the wave superposition is represented by a Fourier transform, $\psi(x) = (2\pi)^{-1/2} \int A(k)\, \cos(kx)\, dk$. Propose the wave-amplitude function $A(k)$ to reproduce the shape of a wave pulse.

Solution The Fourier transform may be written in abbreviated form using complex numbers as $\psi' = (2\pi)^{-1/2} \int A(k)\, \exp(ikx)\, dk$, which is a suitable expression to perform the integrations. In this case $\psi(x) = Re[\psi'(x)]$, where Re indicates the real part of the complex wave function ψ'. Next, with the purpose of generating a single and spatially concentrated wave that represents the pulse, the superposition of waves requires modulation. Since we have the freedom to choose the waves amplitude $A(k)$, we choose a Gaussian envelope to only consider those parts of the waves that are inside the Gaussian envelope. A normalized Gaussian distribution, with mean $\langle x \rangle$ and standard deviation σ, is $(2\pi\sigma^2)^{-1/2} \exp\{-[x - \langle x \rangle]^2/(2\sigma^2)\}$. We take the normalized Gaussian distribution to introduce the waves modulation, $A(k) = (2\alpha/\pi)^{1/4} \exp[-\alpha(k - \langle k \rangle)^2]$. By inserting $A(k)$ in the previous Fourier transform and integrating, we obtain $\psi'(x) = (2\pi\alpha)^{-1/4} \exp(i\langle k \rangle x)\ \exp[-x^2/(4\alpha)]$. For constant values of α and $\langle k \rangle$, the function ψ' renders a pulse shape.

The conclusion on the position and momentum of a microscopic particle is summarized in an important principle with name *Heisenberg's uncertainty principle*. It indicates the impossibility of simultaneously determining the particle's position and particle's momentum in an accurate way. In mathematical form, the principle introduces a lower limit in the product of the uncertainties of the particle's position and the particle's momentum.

$$\Delta x\, \Delta p \geq \hbar/2 \tag{8.12}$$

Next, consider a particle represented by the superposition of many waves with a Gaussian modulation to produce a wave packet.

$$\psi = (2\pi)^{-1/2} \int_{-\infty}^{\infty} A(k) \exp(-ikx)\, dk$$
$$A(k) = (2\alpha/\pi)^{1/4} \exp[-\alpha(k - \langle k \rangle)^2]$$

When the integral is performed, we obtain the representation of the particle by a wave packet.

$$\psi = (2\pi\alpha)^{-1/4} \exp(i\langle k \rangle x)\ \exp[-x^2/(4\alpha)] \tag{8.13}$$

where the real part of ψ should be considered. The probability distribution function of the position x is computed from the square of ψ, that is, $|\psi|^2 = \psi^* \psi = (2\pi\alpha)^{-1/2} \exp[-x^2/(2\alpha)]$. It corresponds to a Gaussian function with standard deviation $\sigma_x = \alpha^{1/2}$. On the other hand, the Gaussian envelope of the wave packet is $A(k) = (2\alpha/\pi)^{1/4} \exp[-\alpha(k - \langle k \rangle)^2]$. The probability distribution function of the wave number is $|A|^2 = (2\alpha/\pi)^{1/2} \exp[-2\alpha(k - \langle k \rangle)^2]$. The standard deviation in this case is $\sigma_k = (4\alpha)^{-1/2}$. The standard deviations σ_x and σ_k give the dispersion extents of x and k with respect to the central values $\langle x \rangle$ and $\langle k \rangle$. In other words, the standard deviations give the uncertainties Δx and Δk in the position and wave number. From the expression of the momentum, $p = \hbar k$, the dispersion of the wave number is equivalent to the dispersion of the momentum, $\Delta k = \Delta p/\hbar$. The product $\sigma_x \sigma_k = 1/2$ is now transformed onto $\Delta x \Delta_p = \hbar/2$. This is the Heisenberg uncertainty principle derived from a wave-packet description of the particle. For other types of probability distribution functions, the dispersion increases. Therefore, the general expression which involves the uncertainties is $\Delta x \Delta_p \geq \hbar/2$.

We have assumed that the particle is described by a wave packet, and we know that it satisfies initial conditions on the Heisenberg uncertainty principle, $\Delta x_0 \Delta p_0 = \hbar/2$. The uncertainty in the momentum may be assigned to an uncertainty in the particle's velocity, $\Delta p_0 = m\Delta v_0$. The Heisenberg uncertainty principle in terms of the velocity now reads $\Delta x_0 (m\Delta v_0) = \hbar/2$. By solving for Δv_0, we have $\Delta v_0 = \hbar/(2m\Delta x_0)$. In general, we may write $\Delta x(t) = \Delta v t$ to find that $\Delta x(t) = \hbar t/(2m\Delta x_0)$. We observe from this expression that the uncertainty in the position increases with time, that is, there is a spatial spread of the wave packet. The Heisenberg principle also provides a qualitative picture on the size of subatomic particles, such as the electron and proton. The proton has a large momentum due to its mass, and the electron has a lower momentum due to its small mass. In this respect, the proton is distributed in a smaller spatial region than that of the electron.

Heisenberg's uncertainty relation is a fact that establishes a general principle and applies to any other pair of conjugate variables, like energy and time.

$$\Delta E\, \Delta t \geq \hbar/2 \tag{8.14}$$

The expression indicates the impossibility of exactly determining the energy in a short time interval and vice versa. Expressions 8.12 and 8.14 have applications in high energy physics where, for example, the virtual particles of high energy have

short lifetimes, thus relating the energy, time, and distance ranges of the fundamental forces. Yet the point to be aware of is the impossibility of simultaneously observing both aspects with infinite precision.

In order to use the uncertainty principle and illustrate the way in which it suggests to abandon concepts of classical nature, let us consider an electron in an atom. This system represents a bound state of the electron; however, it could be also the case of a microscopic particle confined in a box. The electron moves in the neighborhood of the nucleus, and it is not localized at the atom center; otherwise, it would have no momentum, and the uncertainties in the position and the momentum would be zero, $\Delta x = 0$, and $\Delta p = 0$, contradicting Heisenberg's uncertainty principle. In this respect, the uncertainty relation implies a non-zero momentum of the electron in the atom. The minimal energy associated to this motion receives the name of *zero-point energy*, and it constitutes a property of confined microscopic particles, even under low-temperature conditions. In classical mechanics, the minimal motion of a particle has zero momentum, and the particle is at rest. In consequence, there is neither a factor of likelihood in determining the particle's position and momentum nor a zero-point energy at the classical level of theory.

Another important result on applying the uncertainty principle in the form $\Delta E\, \Delta t \geq \hbar/2$ involves a spatial region of confinement. Under the classical perspective, the *vacuum* is a region of space empty of matter and energy. This is not the case at the quantum-mechanical level, because the vacuum under confining conditions may have no matter, but there is background energy there. According to the equation $E = mc^2$, matter is a form of energy, and a particle represents an energy-condensed structure occupying a small space region (a photon, in contrast, is understood as a structured energy traveling in space). In this respect, there is a non-zero probability on the transformation of energy onto sub-atomic particles with time. Such sub-atomic particles, known as *virtual particles*, come in pairs of the particle-antiparticle type, and their existence takes just a small fraction of time, making difficult the observations before they annihilate each other. Those vacuum fluctuations are believed to be responsible for the Lamb shift, referent to the spontaneous emission of energy quanta by the nuclei and the atoms when they return to their ground states, and the Casimir effect as well. The contemporary understanding of vacuum is complex, and unsurprisingly, it is related to the spacetime structure of the Universe.

A.1 Appendix: Planck's Radiation Intensity Law

The goal of this section is to provide a formal discussion of the Planck's radiation intensity, and use it to derive other related laws initially proposed in the frame of classical mechanics. The derivation of the classical results from an initial quantum proposal that introduces energy quantization exemplifies the general character of the quantum theory to obtain the classical theory.

A.1.1 Average Energy

Planck hypothesized that the modes and energy spectrum in the blackbody cavity correspond to the modes and energy spectrum of an ensemble of thermalized harmonic oscillators, with a harmonic oscillator for each possible frequency. The photons may be radiated or absorbed in the cavity, achieving an equilibrium at temperature T. The cavity has non-movable walls, is cubic of side length L, and is in thermal equilibrium with the oscillators. In contraposition to a classical energy that changes in continuous form, the energy of the oscillators is supposed to be quantized. We require the probability distribution function, P, of the energy levels of the particle system to obtain the average energy of the photons.

$$\begin{aligned} P &= \exp(-E_n^{(m)}/k_B T) \,/\, \sum_{m=0}^{\infty} \exp(-E_n^{(m)}/k_B T) \\ E_n^{(m)} &= (m + 1/2)(hc) \,/\, (2L)\sqrt{n_1^2 + n_2^2 + n_3^2} \end{aligned} \tag{8.15}$$

The quantum number in the expression of the energy is m. The positive integral numbers n_1, n_2, and n_3 form the vector $\mathbf{n} = (n_1, n_2, n_3)$ with magnitude $n = (n_1^2 + n_2^2 + n_3^2)^{1/2}$. They are related to the wavelengths λ_i in the three different cavity axes using the condition of *stationary waves*, that is, waves that oscillate with time but without traveling in space. When a wave is reflected on a surface, the reflected wave is the wave that would have travelled further ahead with no reflecting surface. The superposition of waves creates a stationary wave when $n_i\, \lambda_i/2 = L$. The 1/2 factor is the minimum length of the wave to show a 180° phase shift.

$$\lambda_i = 2L/n_i \quad ; \quad n_i/(2L) = \nu/c \quad ; \quad i = 1, 2, 3 \tag{8.16}$$

The purpose is to simplify the expression of both the energy and probability distribution function by following the same procedure applied to a particle in a rigid box. We introduce the ε energy and the Z function; this last one is the normalization factor of P in Eq. 8.15.

$$\begin{aligned} E_n^{(m)} &= \left(m + \frac{1}{2}\right)\varepsilon \quad ; \quad \varepsilon = (hc)n \,/\, (2L) \\ Z &= \sum_{m=0}^{\infty} \exp(-E_n^{(m)}/k_B T) = \exp(-\frac{1}{2}\,\frac{\varepsilon}{k_B T}) \sum_{m=0}^{\infty} \left[\exp(\frac{-\varepsilon}{K_B T})\right]^m \\ Z &= \exp[-\varepsilon/(2k_B T)] \,/\, \{1 - \exp[-\varepsilon/(k_B T)]\} \end{aligned} \tag{8.17}$$

The geometric series $\sum_{m=0}^{\infty} r^m = 1/(1-r)$ was used in the last identity to reach the closed expression $\exp(-x) \sum_{m=0}^{\infty} [\exp(-y)]^m = \exp(-x) \,/\, [1 - \exp(-y)]$. The average energy $\langle E \rangle$ is computed from the probability distribution function.

$$\langle E \rangle = \textstyle\sum_{m=0}^{\infty} E_n^{(m)} \exp(-E_n^{(m)}/k_B T) \,/\, Z = -\partial \ln(Z) \,/\, \partial(k_B T) \tag{8.18}$$

By using function Z in terms of ε, Eq. 8.17, in the last identity, the average energy is:

$$\langle E \rangle = -\partial \ln(Z) / \partial(k_B T) = \varepsilon/2 + \varepsilon / \{\exp[\varepsilon/(k_B T)] - 1\} \tag{8.19}$$

A.1.2 Density of States

The average energy of Eq. 8.19 does not introduce the cavity size, neither the energy stored in the blackbody cavity, yet, it is possible to calculate the number of energy states in which the electromagnetic energy of the blackbody is distributed.

The energy states are given in terms of the numbers n_1, n_2, and n_3, which take part in the expression of the energy $E_n^{(m)}$, Eq. 8.15. There are states that are different, like $(n_1 = 2, n_2 = 1, n_3 = 1)$ and $(n_1 = 1, n_2 = 2, n_3 = 1)$, but they give the same energy. The integers n_1, n_2, and n_3 form a $3D$ grid. The blackbody cavity is supposed to contain a large number of energy states, and in order to determine that number, we visualize the number of states as building a big sphere whose center is the origin of the three orthogonal axes. The number of states is proportional to the sphere volume (the particular spherical shape alluded here is not important because an integration over an increasing spherical shell thickness shall be carried out later to cover the whole $3D$ space). However, the numbers n_1, n_2, and n_3 are positive, and this characteristic takes us to concentrate our attention on the positive quadrant of the eight possible quadrants formed by the three orthogonal axes, Fig. 8.4.

The volume of the quadrant corresponds to 1/8 of the spherical volume. If the radius of the sphere is given by $n = (n_1^2 + n_2^2 + n_3^2)^{1/2}$, then the volume of the spherical quadrant is $V = (1/8)(4\pi n^3/3)$, and the differential volume element is $dV = \pi n^2 dn/2$. On the other hand, the direction of the electric field oscillation is twofold polarized (as demonstrated in the Barkla experiment), and it should be considered in the expression of the number of available energy states. Thus, the volume is multiplied by a factor of 2 to take the two polarization states of a photon

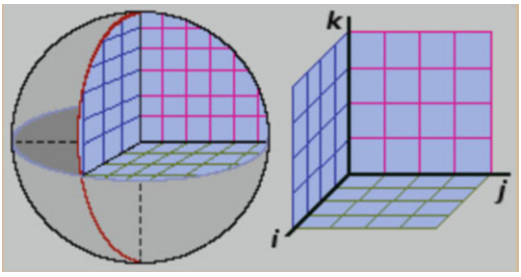


Fig. 8.4 Left sketch: a $3D$ sphere is shown. The quadrant in the sphere and in color is the only one that contains a grid of positive numbers. Right sketch: a magnification of such a grid is shown. An arbitrary vector **n** is expressed as a set of integral numbers (n_1, n_2, n_3). The volume of the quadrant is 1/8 the volume of the sphere

into consideration. The differential volume element is now $dV = \pi n^2 dn$. The next step links dV to the energy ε using the relation between n and ε of Eq. 8.17 with the purpose of dealing with the energy ε. This is a linear relation, $d\varepsilon = (hc/2L)dn$, with a scale factor that introduces the cavity length, L. The volume element in terms of ε is:

$$dV = \pi n^2 dn = \pi \left(\frac{2L}{hc}\varepsilon\right)^2 \left(\frac{2L}{hc}d\varepsilon\right) = \pi \frac{8L^3}{h^3c^3}\varepsilon^2 d\varepsilon \tag{8.20}$$

The identity given above introduces the volume of the cavity and gives the density of states, that is, the number of energy states in the interval $[\varepsilon, \varepsilon + d\varepsilon]$. By using the second expression of Eq. 8.16, $n_i/(2L) = \nu/c$, the energy ε is $h\nu$, and the number of states in terms of the frequency is obtained.

$$dV = (8\pi L^3/c^3)\,\nu^2 d\nu \tag{8.21}$$

This expression may be compared with Eq. 8.5 of the main text referent to the number of cavity modes per frequency and per volume.

A.1.3 Energy per Volume per Frequency

It is important to determine the electromagnetic energy, U, in the cavity with the purpose of deriving the radiated power of the blackbody. In order to obtain the energy per volume, the number of available states $dV = \pi(8L^3)/(h^3c^3)\,\varepsilon^2 d\varepsilon$ should be multiplied by the average energy; this last quantity introduces the occupation probability of the photon states, Eq. 8.19.

$$\frac{U}{L^3} = \frac{1}{L^3}\int_0^\infty \left(\langle E\rangle - \frac{\varepsilon}{2}\right) dV = \frac{8\pi}{h^3c^3}\int_0^\infty \frac{\varepsilon^3}{\exp(\varepsilon/k_BT) - 1}\, d\varepsilon \tag{8.22}$$

The zero-point vibration energy $\varepsilon/2$ was subtracted because it is a constant and we require to determine energy differences. By taking the expression $\varepsilon = h\nu$ into consideration, the energy u per unit volume and unit frequency is:

$$u(\nu, T) = (8\pi\nu^2/c^3)\, h\nu \,/\, [\exp(h\nu/k_BT) - 1] \quad ; \quad \frac{U}{L^3} = \int_0^\infty u(\nu, T) d\nu \tag{8.23}$$

The energy u was proposed by Planck; however, it is presently recognized as a particular expression of statistical mechanics, which is an approach based on a family of distribution functions. Expression 8.23 uses the Bose-Einstein energy distribution function of a gas of Bose particles at maximum entropy and in thermodynamic equilibrium with the environment. The energy u is indeterminate when $\nu = 0$. On the other side, the modes with high frequencies are less likely to

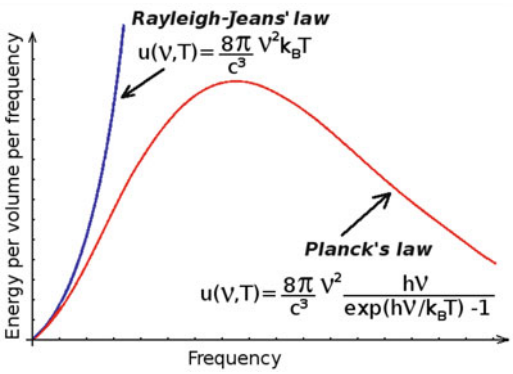


Fig. 8.5 The energy emitted by a blackbody per unit volume per unit frequency is shown in terms of the frequency. The prediction based on classical mechanics is given by the Rayleigh-Jeans law, Eq. 8.24, and the one based on quantum mechanics is given by the Planck's law, Eq. 8.23. The two predictions only match at low frequencies

occur as the energy u goes to zero when the frequency becomes large. According to classical theory, the frequency ν may take any value and grow without restrictions. The classical limit, which reproduces Eq. 8.6 and is related to the Rayleigh-Jeans law, is obtained in the limit of low frequencies, $h\nu \ll k_B T$, from Eq. 8.23. Thus, by expanding the exponential function in the expression of u in a power series and retaining the first order term, we obtain:

$$u(\nu, T) = (8\pi/c^3)\, \nu^2\, k_B T \tag{8.24}$$

The classical and quantum mechanical versions of the energy u coincide at low frequencies. The classical version of the energy grows without limits, leading to the ultraviolet catastrophe at high frequencies, Sect. 8.5. In contraposition, the quantum mechanical version of the energy shows a maximum value of a function that eventually decreases and fades away at large frequencies, Fig. 8.5.

In order to determine the energy in the blackbody box, we perform the integral of Eq. 8.23 over ε. To facilitate the integration, the dimensionless variable t is defined by $t = \varepsilon / k_B T$.

$$\begin{gathered} \int_0^\infty \frac{t^3}{\exp(t) - 1}\, dt = \pi^4/15 \quad ; \quad L^3 = V \\ \frac{U}{V} = \frac{8\pi^5}{15(hc)^3}\, (k_B T)^4 \end{gathered} \tag{8.25}$$

The energy per volume is found to be proportional to T^4, where T is the temperature of equilibrium. By introducing the Stefan-Boltzmann constant σ, Sect. 8.6, we obtain:

$$\frac{U}{V} = (4\sigma/c)\, T^4 \quad ; \quad \sigma = (2\pi^5/15)\, k_B^4 \,/\, (c^2 h^3) \tag{8.26}$$

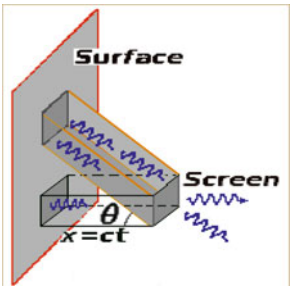


Fig. 8.6 A surface that radiates energy in the form of electromagnetic waves is in a vertical position. A parallel screen registers the radiated energy at a distance x from the surface. The detecting area of the screen is A. Some of the radiation detected by the screen makes an angle θ with respect to the horizontal direction

The energy U should not be confused with the Stefan-Boltzmann law, $J = \sigma T^4$, referent to the radiated power by the blackbody discussed in Sect. 8.6.

A.1.4 Irradiance

The irradiance or power is the energy per unit time emitted by a surface of unit area. Occasionally, the irradiance is referred to as the luminous intensity. Our interest is to compute the irradiance produced by a blackbody, which escapes through the tiny hole located in one of the cavity walls. To do this, let us consider an internal wall of the blackbody cavity. It is a surface in thermal equilibrium with the electromagnetic radiation contained in the cavity. Suppose that a screen of area A is positioned at a distance x from the wall surface with the purpose of detecting the radiation emitted by the wall. The radiation travels with speed c and makes an angle θ with respect to the horizontal direction, Fig. 8.6. The irradiance R registered on the screen is geometrically established.

$$R = u \cos\theta \,/\, (A\, t/\cos\theta) \tag{8.27}$$

The factor $t/\cos\theta$ is the time it takes the radiation to travel from the emitting surface to the screen, and $u\cos\theta/A$ is the corresponding energy per unit area registered on the detecting screen. When $\theta = 0°$, that is, when the screen directly faces the emitting surface, the traveling time is $t = x/c$, and the recorded energy per unit area is u/A. In this case, we observe the minimal time of travel and maximal energy per unit area.

The radiation detected on the screen comes from every direction. However, in thermal equilibrium, one half of the total number of waves travels toward the surface, and the other half travels away from the surface. Therefore, a factor of 1/2

should be considered in the expression of the irradiance. The total radiated power by the surface and detected on the screen is determined after integrating over all values of θ. By using the identity $\int \cos^2\theta d\theta = \theta/2 + (1/4)\sin(\theta/2)$, integrating from $-\pi/2$ to $+\pi/2$, and normalizing with respect to the integration interval, π, the result is $1/2$. The expression of the irradiance is:

$$R_{tot} = u/(4tA) = cu/(4V) \tag{8.28}$$

where the change of variables $t = x/c$ and $V = Ax$ was used. The luminous intensity I is related to R_{tot} by $I = R_{tot}/\pi$. In this regard, by using the energy u in the forms given by Eqs. 8.23 and 8.24, the intensity in quantum theory (qt) and classical theory (ct) is:

$$\begin{aligned} I_{qt}(\nu, T) &= (2\nu^2/c^2)\, h\nu\, [\exp(h\nu/k_B T) - 1]^{-1} \\ I_{ct}(\nu, T) &= (2\nu^2/c^2)\, k_B T \end{aligned} \tag{8.29}$$

The first equation is the Planck's radiation law, Eq. 8.10, and the second is the Rayleigh-Jeans radiation law, Eq. 8.7. We now assume a radiator whose luminous flux changes by direction in the form $\cos\theta$, where θ is the angle between the normal of the surface and the direction of the radiation. Such a dependence is known as the *Lambert's cosine law*. The integration of Planck's expression over all the frequencies ν, and over the angles θ and ϕ describing the hemisphere above the surface, gives the Stefan-Boltzmann law, Sect. 8.6. The following purpose is to write the Planck's equation in terms of the wavelength λ. Here, it is important to establish the relation between the spectral variables λ and ν in the form $\lambda\nu = c$ and $d\nu/d\lambda = -c/\lambda^2$. Since the energy u of Eq. 8.23 is an energy per unit volume and per unit frequency, we depart from the equation:

$$u(\lambda, T)\, d\lambda = -u(\nu, T)\, d\nu \tag{8.30}$$

The negative sign is due to the inverse relation between the spectral variables ν and λ. The final expression of the energy per unit volume and per unit wavelength is:

$$u(\lambda, T) = (8\pi hc/\lambda^5)\, \{\exp[hc/(\lambda k_B T)] - 1\}^{-1} \tag{8.31}$$

A similar transformation is obtained for the wave number k, where $k = 1/\lambda$. The next step consists on obtaining the maximum intensity from the above Planck's expression to demonstrate the Wien's displacement law, Eq. 8.8.

A.1.5 Maximum Intensity

The Planck's radiation expression gives the maximum radiation that a body can radiate in thermal equilibrium. The radiated intensity reaches a peak emission at

a finite wavelength, λ_{max} and only depends on the temperature T. Planck's law in terms of the wavelength is given in Eq. 8.31. The maximum of the energy is obtained by differentiating u with respect to λ, making the resulting expression equal to zero and solving such an equation for $\lambda = \lambda_{max}$. The process leads to the result:

$$x\,[1 - \exp(-x)]^{-1} = 5 \quad ; \quad x = hc/(\lambda k_B T) \tag{8.32}$$

The first expression is a transcendental equation whose solution is found numerically, $x = 4.965,114$. Such a value implies a constant for the product:

$$\lambda_{max} T = 2.898 \times 10^{-3}\ m\ K \tag{8.33}$$

The above expression is the Wien's displacement law, Eq. 8.8. By doing the same process for u in terms of the frequency, we obtain $3[1 - \exp(-x)] - x = 0$, with $x = h\nu/k_B T = 2.821,439$, and the maximum frequency is $\nu_{max} T^{-1} = 5.879 \times 10^{10}\ Hz\ K^{-1}$, Fig. 8.5. Thus, by supposing radiation of blackbody type, the temperature of an object, like that of a star, is determined from the wavelength or frequency of the object's electromagnetic emission. The process of establishing the source temperature in this way is known as the *color-assigned temperature*.

A.1.6 Time Interval of Temperature Cooling

It is of interest to estimate the time interval of radiative cooling of a blackbody after the energy supply is switched off. The energy of the system is supposed to be exclusively due to the temperature and given by $E = 3Nk_B T/2$, corresponding to the energy of an ideal gas where N is the number of moving particles. The dissipation rate is established by the Stefan-Boltzmann law, $dE/dt = \sigma A T^4$, Sect. 8.6. The time of heat loss is calculated by applying the chain rule, $dE/dt = (dE/dT)(dT/dt)$. This expression is equal to $\sigma A T^4$. Thus, the time of cooling depends on the way in which the body emits radiation.

$$\frac{dE}{dt} = (3Nk_B)/2\,\frac{dT}{dt} = \sigma A T^4 \tag{8.34}$$

The second identity is a differential equation of the temperature in terms of t. The solution is obtained after integration.

$$\Delta t = [Nk_B/(2\sigma A)]\,\left|T_i^{-3} - T_f^{-3}\right| \tag{8.35}$$

The fact that a real body conducts heat from its inner part to its outer surface makes the time interval of temperature cooling longer than that of a perfect blackbody. The time delay observed on the cooling of real objects may be approximated by replacing σ by $\epsilon\sigma$, where ϵ is an emission factor, $\epsilon = 1$ for a blackbody and $\epsilon < 1$ for a real object.

$$\Delta t = [Nk_B/(2\epsilon\sigma A)] \left| T_i^{-3} - T_f^{-3} \right| \tag{8.36}$$

In this regard, it is possible to establish a lower limit on the cooling time of an object from the blackbody results.

Reference

1. C. Cohen-Tannoudji, B. Diu, F. Laloe, *Quantum Mechanics*, ed. 2005, vol. 1 (Wiley, London, 1977), pp. 573.



Quantization of the Angular Momentum

9

According to classical mechanics, the angular momentum of the particles may take arbitrary values. However, the experiments have shown an opposite feature, that is, the angular momentum and the spin of the microscopic particles are quantized. The wave theory of matter accounts for the quantization of the angular momentum and spin, yet the spin remains as an enigmatic property of the elementary particles.

This chapter is structured as follows: in the first stage, the orbital angular momentum and the spin are briefly characterized from the theoretical and experimental perspectives. Particular attention is given to the Stern-Gerlach experiment on the spin of a particle. The spin value, which a particle may take, is used to classify it as a fermionic or bosonic particle and identify the collective behavior of such types of particles as well. In the second stage, we introduce new principles, like Pauli's exclusion principle and Hund's rule, which emphasize the difference between quantum mechanics and classical mechanics. A discussion on the basic relations of the spin with the magnetic moment and of the magnetic moment with both the angular momentum and the external magnetic field is presented. Finally, a brief introduction to spin functions and spin operators is provided.

9.1 Orbital Angular Momentum and Spin

A particle that rotates has angular momentum. The total angular momentum is separated onto an orbital angular momentum, $\mathbf{L}$, and an intrinsic angular momentum, $\mathbf{S}$. The intrinsic angular momentum is known as the *particle spin*. In the classical treatment, the vectors $\mathbf{L}$ and $\mathbf{S}$ are given by:

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \quad ; \quad \mathbf{S} = \bar{\mathbf{I}}\boldsymbol{\omega} \tag{9.1}$$

The variables $\mathbf{r}$ and $\mathbf{p}$ are the position and linear momentum of the particle, while $\bar{\mathbf{I}}$ and $\boldsymbol{\omega}$ are the inertia moment tensor and angular velocity of the particle. The total angular momentum constituted by the vectors $\mathbf{L}$ and $\mathbf{S}$ of an isolated system of

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particles is conserved. Such vectors are permitted to take any magnitude and any spatial direction in classical mechanics, but this is not so in quantum mechanics. In the case of an atom with N electrons, the total electronic angular momentum $\mathbf{L}$ corresponds to the vector sum of their individual contributions $\mathbf{L} = \sum_{i=1}^{N} \mathbf{L}_i$, and similarly for the total electronic spin angular momentum $\mathbf{S}$, that is, $\mathbf{S} = \sum_{i=1}^{N} \mathbf{S}_i$. The experimental observations on microscopic particles indicate that the orbital angular momentum and spin are quantized. The quantized values are denoted by ℓ and s. They take part in the eigenvalues of the operators $\hat{\mathbf{L}}^2$ and $\hat{\mathbf{S}}^2$.

$$\begin{gathered} \hbar^2 \ell(\ell+1) \qquad ; \qquad \hbar^2 s(s+1) \\ \ell = 0, 1, \cdots, (n-1) \; ; \quad s = 0, 1/2, 1, 3/2, \cdots \end{gathered} \tag{9.2}$$

The factor ℓ is the *azimuthal quantum number* of the orbital angular momentum, and s is the *principal spin quantum number*, commonly referred to as the particle spin. Like other quantum numbers, ℓ and s adopt discrete values. The allowed values of s are both half-integer and integer numbers, and those of ℓ are integer numbers with limit $n-1$, where n is the *principal quantum number*.The atoms have electronic configurations, and these are classified according to the quantum numbers. The principal quantum number takes integral values and establishes the energy state and energy level of an electron in the atom, while the other quantum numbers define the energy sub-levels. The energy level with $n = 1$ has more negative energy than the $n = 2$ energy level and so on. The *Aufbau principle* imposes a relative order by pointing out that the electronic energy levels with low energy are populated before the energy levels with high energy. For the electrons of atoms, the orbital quantum number ℓ defines the sub-shell type, leading to an orbital diagram of sub-shells with nomenclature [1]:

$$s(\ell = 0:2) \;\; ; \; p(\ell = 1:6) \;\; ; \; d(\ell = 2:10) \; ; \; f(\ell = 3:14)$$

$$g(\ell = 4:18) \; ; \; h(\ell = 5:22) \; ; \; i(\ell = 6:26) \;\; ; \qquad \cdots$$

The value of the orbital quantum number and the maximum number of electrons permitted in the atomic sub-shell are shown in parenthesis. The maximum number of orbitals in sub-shell ℓ is given by the *sub-shell multiplicity*, $2\ell + 1$. We have a single s orbital with two electrons, one with spin up and one with spin down, three p orbitals with six electrons, three with spin up and three with spin down, etc. A closed sub-shell gives zero contribution to the total electronic angular momentum and total electronic spin. The sub-shells s, p, $\cdots$, adopt different spatial shapes to accommodate the different numbers of electrons.

The quantization of $\mathbf{L}$ and $\mathbf{S}$ is also known as *space quantization* to distinguish it from the energy quantization, both observed on the microscopic particles in confinement. Space quantization should not be confused with a quantization of the spacetime framework. Space quantization and energy quantization are associated with the wave-particle nature of the microscopic particles. Any interpretation of

the intrinsic spin of the microscopic particles using classical theory contradicts the experimental results. Some of the contradictions are illustrated below when, for example, we consider a particle with spin as a rotating sphere of radius r, uniform mass m, and inertia moment $I = 2mr^2/5$.

- The radius of a tiny microscopic particle, such as that of an electron, is sufficiently small, $r \to 0$. In this case, the inertia moment ($I = 2mr^2/5$) approaches zero in quadratic form, which in turn produces a spin tending to zero. Thus, for a point particle, we essentially have $\mathbf{S} = 0$. Such a deduction contradicts the experimental results, where the electron is observed to have a non-zero spin.
- If we consider that $\mathbf{S}$ is sufficiently small but different from zero, then we require a very large angular velocity $\boldsymbol{\omega}$ to keep the product $\mathbf{S} = \bar{\mathbf{I}}\boldsymbol{\omega}$ with finite value. However, a very large angular velocity implies a mass on the sphere surface that rotates with speed greater than the speed of light, a situation which is in contradiction with relativistic mechanics.
- If we assume a relatively small angular velocity (to keep the angular speed of the sphere within bounds), then we mistakenly end up with the prediction of an electron radius that is much greater than the radius of the atom.

The classical picture of an electron as a small sphere with charge and spin is not appropriate because it leads to contradictions in quantum mechanics. The spin of the elementary particles cannot be explained using the existence of even smaller particles, since the elementary particles have no inner structure. The spin of elementary particles is a true physical attribute associated with pure quantum mechanical effects and only defined in terms of what is observed and measured: specifically, the spin is a physical property that makes the elementary particles act as unconventional magnets. In the following sections, the important features of the space quantization are distinguished after experimental observations employing magnetic fields on microscopic particles with spin. Yet no relativistic effects are considered in this frame of the wave theory of matter.

9.2 Characterizing a Particle with Spin

An electron orbiting around the nucleus resembles an electric current in a circular loop and generates a magnetic field of dipolar type. The induced *magnetic moment* is proportional to the angular momentum $\mathbf{L}$ in the form $\boldsymbol{\mu}_L = g_L\, q\mathbf{L}/(2m)$, Sect. A.1.1. The quantities q and m are the charge and mass of the particle, and g_L is the *g factor*, Sect. A.1.2; it takes the value 1 when the mass and charge of the particle are homogeneously distributed, and the particle is in orbital motion. Similarly, a charged particle with spin $\mathbf{S}$ induces a magnetic moment, in principle described by an electric current in a circular loop, Sect. A.1.5.

$$\boldsymbol{\mu}_s = g_s \, q\mathbf{S}/(2m) \tag{9.3}$$

The factor $g_s q/(2m)$ is the *gyro-magnetic ratio*, and the dimensionless number g_s is the g factor for the type of particle and that type of motion. According to the theory of quantum electrodynamics, the value of the g factor of a spinning electron is $g_s = 2.002{,}319{,}304{,}3622$. It is the result of the self-interaction of the electron using an exchange photon. The self-interaction smears the electron distribution by a small fraction of a Fermi unit in the distance, resulting on a slightly weaker interaction with the proton.

9.3 Stern-Gerlach Experiment

The Stern-Gerlach experiment is important as it gives evidence of the spin and the spin quantization of the electron. Stern and Gerlach worked with neutral silver atoms in their experiment to avoid a Lorentz force, $\mathbf{F} = -q\mathbf{v}\times\mathbf{B}$, acting on charged atoms. The silver atom is constituted by a core formed by the nucleus and 46 electrons, plus a single electron in the outermost shell. The net angular momentum and the net spin of the core are zero. The electron out of the core also has zero orbital angular momentum as it is in a s ($\ell = 0$) state; however, it has non-zero spin. In this connection, a measurement on the magnetic moment of the silver atom is in a first approximation a measurement on the magnetic moment of the electron spin.

In the Stern-Gerlach experiment, the silver atoms are ejected from a furnace with random orientations of their magnetic moments. In particular, the z component of the magnetic moment may take any value between $-|\mu_z|$ and $|\mu_z|$ before reaching the magnetic field of the experiment. The magnetic field is inhomogeneous with the purpose of creating different magnetic forces on the outermost electrons of the silver atoms. The beam of particles is collimated by a small hole on each of the frontal screens to produce a fine particle beam. The particles enter in the Stern-Gerlach apparatus traveling along the y axis. The magnetic field $\mathbf{B}$ of the experiment is axisymmetric in the z axis, Fig. 9.1. If the potential energy of a particle in the magnetic field is E_p, then the force $\mathbf{F} = -\nabla E_p$ exerted by the magnets on an atom in the z direction is, Sect. A.1.3:

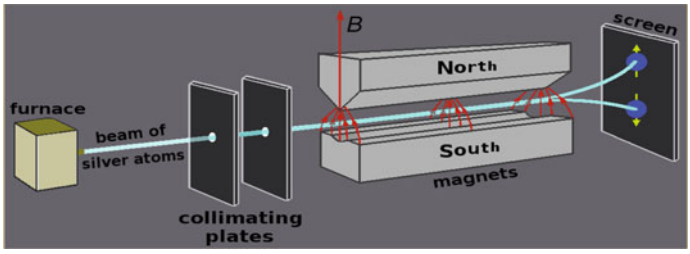


Fig. 9.1 The Stern-Gerlach experiment gives evidence of the spin and the spin quantization of the electron. The microscopic particles with spin behave as unusual tiny bar magnets in the presence of a non-uniform magnetic field

$$E_p = -\boldsymbol{\mu} \cdot \mathbf{B} \quad ; \quad \mathbf{F} = \mu_z \frac{\partial B}{\partial z} \hat{k} \tag{9.4}$$

The force on an atom depends on μ_z and the intensity B of the magnetic field. The atoms that form the beam are deflected differently due to the different atomic magnetic moments. The experimental magnetic field attempts to align the magnetic moment in the z direction, thus creating a torque on the magnetic dipole and inducing the precession of $\boldsymbol{\mu}$ in the (x, y) plane with frequency ω, Sect. A.1.4. In this respect, the x and y components of the $\boldsymbol{\mu}$ vector change with time.

$$\omega = \mu_z B/\hbar \tag{9.5}$$

The observed pattern on a screen consists of two separated blots in the z axis. In more detail, the magnetic field in the Stern-Gerlach experiment defines the spin inclination relative to the z axis with only two discrete states:

$$S_z = \pm \hbar/2$$

The electrons are half-spin particles. Under the consideration of $+$ and $-$ signs, the electron spin is pictured as a clockwise rotation or *spin up* and an anti-clockwise rotation or *spin down*, around its rotation axis. By inserting the value of S_z in the magnetic moment of Eq. 9.3, with $g = 2$, the z component of the magnetic moment associated with the electron spin is:

$$\mu_z = \pm e\hbar/(2m_e) \tag{9.6}$$

The results of the Stern-Gerlach experiment contradict the classical predictions based on the behavior of small and ordinary magnets in a magnetic field.

Problem 9.1 The mass and velocity of the particles in the Stern-Gerlach experiment are m and v. The particles travel a distance L and impact the screen at the spots $\pm z$, measured from the center of the observation screen, Fig. 9.1. Find the value of z in terms of the magnetic force, the distance L, and the kinetic energy E_k of the particles.

Solution In their journey to the screen, the particles are subjected to a magnetic force $F_z = \mu_z \, \partial B/\partial z$, Eq. 9.4, in the z direction, similar to a particle with a given mass and traveling in a gravitational field. The height is computed from the force F_z.

$$z = \frac{1}{2} (F_z/m) \, t^2 = \frac{1}{2m} \left(\mu_z \frac{\partial B}{\partial z} \right) t^2$$

(continued)

The travel time is determined by the velocity v of the particles and the distance L in the form $t = L/v$. The kinetic energy of a particle is $E_k = mv^2/2$. By using the expression $t^2 = L^2/v^2 = mL^2/(2E_k)$, the distance z is:

$$z = \frac{L^2}{4E_k}\,\mu_z\,\frac{\partial B}{\partial z}$$

The experimental observations are independent of the reference z axis defined in the previous analysis, and whichever axis is imposed in the experiment, two spin values are always detected in the respective axis. In short, the Stern-Gerlach experiment exhibits the quantization of the electron spin with specific values and spatial orientations of the spin components. Similar experiments have shown the quantization of the orbital angular momentum and spin of other particles, thus determining the expressions of the L_z and S_z components (the z projection is conventionally used here).

$$\begin{aligned} L_z = m_\ell \hbar \quad &; \quad S_z = m_s \hbar \\ m_\ell = 0, \pm 1, \cdots, \pm \ell \quad &; \quad m_s = 0, \pm 1, \cdots, \pm s \end{aligned} \tag{9.7}$$

The allowed values of the azimuthal quantum number, ℓ, are those given in expression 9.2. The m_ℓ quantity is the *magnetic quantum number*, which defines the projection of the angular momentum. On the other hand, the allowed values of the principal spin quantum number, s, are those given in expression 9.2. The m_s quantity is the *magnetic quantum number of spin*, or *spin projection quantum number*. A photon has integral spin, $s = 1$, and the only values of the spin component are $S_z = \pm\hbar$ corresponding to right circularly polarized light and left circularly polarized light; however, an electron has half-integer spin, $s = 1/2$, and the values of the spin components are $S_z = \pm\hbar/2$. The total number of states that create the same total electronic spin of a system is referred to as the *spin multiplicity*, given by $2S + 1$. An atom or molecule with one or more electrons may be in one of the $2S + 1$ possible spin states. A system is in a *singlet spin state* $(2 \times 0 + 1)$ when the total spin of the system is zero ($S = 0$), it is in a *doublet spin state* $(2 \times 1/2 + 1)$ with possible spin states $\pm 1/2$, when the total spin of the system is one half ($S = 1/2$), and so on. The spin states that integrate a multiplet are energy degenerate. Such a degeneracy in atoms may unfold into different energy sub-levels due to the electromagnetic fields of nearby atoms in a molecule. In the wave theory of matter, the spin states are formally described by spinors, Eq. 9.40. The quantum number m_s has no dependence on any other quantum number. There is not a general wave function of the spin solely due to its nature, and thus, the spin of every particle in a molecule is frequently considered in the molecular wave function. More information on the way the spin is introduced in quantum mechanics is provided in Sects. B.1 and 10.9.

9.4 Wave-Particle Duality and Spin of a Particle

Spin quantization in the Stern-Gerlach experiment is in principle considered the aftermath of the particle-wave character of the microscopic particles. According to Cresser [2], such a statement is illustrated by displaying an analogy with the Young's double-slit interference experiment. Consider a set of Stern-Gerlach (SG) devices arranged in series. The first SG apparatus has the magnetic field oriented in the z direction. The beam of particles comes out of the magnetic field with the spin components $S_z = \pm\hbar/2$. The particles with spin down, $S_z = -\hbar/2$, are blocked, and the particles with spin up, $S_z = +\hbar/2$, are allowed to travel to a second SG device, preserving the orientation of the magnetic field of the first SG apparatus. The particles that exit from the second device are the spin-up particles solely. Nothing has changed with the presence of the second apparatus. On the other hand, when the magnetic field of the second SG device is re-oriented, say in the y direction, the emerging beam of the second device shows with equal probability the spin-up and the spin-down particles in the y direction, $S_y = \pm\hbar/2$. A third SG apparatus in series with the second device, and the magnetic field oriented in the z direction, like in the case of the first apparatus, shows a beam of particles with spin components $S_z = \pm\hbar/2$. Surprisingly, the intermediate device randomized the spin orientations of the particles coming out of the first apparatus in the z direction, since particles with both spins emerge from the third device.

The fact that a Stern-Gerlach device randomizes the spin orientations indicates the loss of the particle history in the experiment and, thereby, the impossibility of measuring any pair of spin components in different directions, such as the particles with spin up in the y direction and the particles with spin up in the z direction. The impossibility to simultaneously have knowledge of the spin components in different directions is not due to the imperfection of any technique in the experiment but to the intrinsic randomness ascribed to a spin interference effect which is beyond our control. In this respect, the Stern-Gerlach spin experiment shows resemblance to Young's double-slit experiment, in spite of the different spectra. The similarities between the two experiments is illustrated after conceiving that an interference pattern occurs when, in general, a beam of light or a beam of microscopic particles is presented with the option of two or more paths to travel. Let us consider several Stern-Gerlach devices in series. In the first stage, a SG apparatus plays the role of a beam collimator producing particles with a given spin value, say $S_x = \hbar/2$. In the second stage, the beam is presented with different options in the spin of the particles, such as $S_z = \pm\hbar/2$, in resemblance to the paths presented by the two slits in Young's experiment. In the third stage, the two beams interfere. It is after the third SG device that we perform the measurements on the spin of the particles to determine the spin component, for example, $S_y = \hbar/2$, without knowing the spin components of the particles in the previous devices, and in the same way as it occurs in the Young experiment where we have no knowledge of the precise pathways and states taken by the particles in traveling toward the screen. Also, there is no formation of a diffraction pattern when we monitor any of the intermediate paths of the particles.

In conclusion, the Stern-Gerlach and Young experiments give evidence of the different properties of the microscopic world. Still, the observations share a common factor based on the general laws of nature, as such laws are in essence indifferent to the types of measuring devices; however, they determine the way in which the particles at the microscopic level manifest themselves.

9.5 Fermions and Bosons

The spin has a direction and a magnitude; therefore, it is a vector. However, it is impossible to simultaneously determine every component of the spin vector. A single spin component is quantized, and we must use its quantum number to describe it. The direction of the spin may be changed, but it is impossible to increase or decrease the spin magnitude in any experiment.

The spin is a physical observable, and we use it to classify the particles and characterize some common behaviors. The particles with half-integer spin, 1/2, 3/2, 5/2, $\cdots$, are called *fermions*, and the particles with integral spin, 0, 1, 2, $\cdots$, are called *bosons*. No elementary fermions have been discovered with spin values of $3/2, 5/2, \cdots$, neither elementary bosons with high spin values (the graviton is assumed to have a spin value of 2). The Higgs boson is an elementary particle with zero spin. The composite particles integrated by two or more microscopic particles such as the helium-4 nucleus ($s = 0$), helium-3 nucleus ($s = 1/2$), or silver atom ($s = 1/2$) of the Stern-Gerlach experiment exhibit a net spin. In this respect, a composite particle may classify as a fermion or a boson. A boson may be made of several fermions, but a fermion cannot be made of bosons. The electrons and the leptons have half-integer spin and classify as fermions. The photon ($s = 1$) and the force-carrier particles classify as bosons. The neutron and neutrino are fermionic particles without electric charge, but surprisingly, the neutron has a magnetic dipole moment because it is a composite particle formed by quarks. The fermions and bosons have different behaviors and follow different statistical rules, originating from different particle distributions. In particular, the fermions are ruled by the Fermi-Dirac statistics, and the bosons obey the Bose-Einstein statistics. Still, at high temperature, the Fermi gases and the Bose gases with a large number of particles behave classically and are ruled by the Boltzmann statistics [3].

9.6 Pauli's Exclusion Principle and Hund's Rule

The quantum numbers required to describe the electronic state of an atom are n, ℓ, m_ℓ, and m_s. They are the principal or total quantum number, n; the orbital quantum number, ℓ; magnetic quantum number, m_ℓ; and spin magnetic quantum number, m_s. Their possible values are presented in Table 9.1.

The order in which the quantum numbers n, ℓ, m_ℓ, and m_s are shown in the filling of atomic shells by electrons depends not only on the contribution of each electron to the atomic energy but also on other physical principles and rules. The

Table 9.1 Main quantum numbers of fermionic particles

Designation		Quantum numbers
Principal quantum number		$n = 1, 2, 3, \cdots$
Orbital quantum number		$\ell = 0, 1, 2, \cdots, (n-1)$
Magnetic quantum number		$m_\ell = 0, \pm 1, \pm 2, \cdots, \pm \ell$
Magnetic quantum number of spin		$m_s = \pm 1/2$
ℓ	$2\ell + 1$ (electrons)	m_ℓ
s ($\ell = 0$)	1 (2)	0
p ($\ell = 1$)	3 (6)	$-1, 0, +1$
d ($\ell = 2$)	5 (10)	$-2, -1, 0, +1, +2$
f ($\ell = 3$)	7 (14)	$-3, -2, -1, 0, +1, +2, +3$

Orbital quantum number, ℓ, subshell nomenclature, s, p, ..., multiplicity, $2\ell + 1$, and the respective m_ℓ permitted values

Pauli exclusion principle indicates that two or more identical fermions in a common system are prohibited to have the same set of quantum numbers. As a result, the fermions must occupy different physical states. In contrast, the bosons are allowed to occupy the same physical state. In addition to the Pauli exclusion principle, we have the *Hund's rule*, an important ingredient in the characterization of the electronic configurations, as well. The Hund's rule is based on a regular pattern, which has been observed in experimental works and concluded from theoretical calculations on atoms. It states that configurations with closed shells are equally populated with spin-up electrons and spin-down electrons. That is, the most convenient electron distribution in the energy levels, which are below the one being filled up, is the double occupancy of the energy levels. However, the non-closed levels of equal energy are populated by single electrons with similar spins. In this regard, the fundamental state of an atom is, according to the Hund's rule, a high-spin state; specifically, it shows the largest number of electrons with parallel spins in the partially filled shell.

In order to explain Hund's rule, let us consider a set of electrons occupying equal-energy levels with one electron in each level; they all have parallel spins. Hund's rule is explained by the reduction of the screening effects that electrons with parallel spins have on the nucleus. The reduced screening of the nucleus brings the electrons closer to the nucleus. Such a distribution increases the nucleus-electron energy, leading consequently to an energetically stable atom. Summing up, the electrons follow the Pauli exclusion principle and show an occupation order of the atomic shells under the Hund's rule and Aufbau's principle. In contraposition, the bosons follow no exclusion principle, and as the temperature approaches zero, they condense in the same energy state. The different properties between fermions and bosons is the reason that fermions satisfy the Fermi-Dirac statistics and the bosons satisfy the Bose-Einstein statistics.

Electrons and nuclei are important components in the formation of atoms and molecules. Electrons are negatively charged particles and the nuclei positively charged particles. The incomplete/complete electron population of the atomic orbitals results in unpaired/paired electrons, originating a non-zero/zero net electron

spin. This endows some materials with paramagnetic/diamagnetic characteristics. The paramagnetic materials intensify an external magnetic field, and the diamagnetic materials reduce an external magnetic field. The electrons tend to oppose the compression due to both the Coulomb repulsion and the Pauli exclusion principle [4]. In the inner core of a collapsing star, the electrons resist the gravitational confinement of the outermost shells of the star. However, at sufficiently large compression rates, an instability between the forces pointing in the inward radial direction and the forces pointing in the outward radial direction eventually builds up, resulting in the explosion of the star. On the other hand, the bosons, with no restriction on the population of the same energy states, can behave in collective form as in the case of a laser, a superfluid, or a superconductor. The applications based on the spin of the particles are different, Sect. A.1.7. There is a spectroscopy based on the magnetic properties of the electron spin with name electron spin resonance, ESR, where the spin precessions in the analyzed sample are employed as a source of information, and a spectroscopy based on the magnetic properties of the nuclei with name nuclear magnetic resonance, NMR, where the signal intensity of the sample is proportional to the nuclear magnetic moment. In addition to the imaging techniques using the particle spins, there are applications in solid-state physics (referent to materials with ferromagnetic, paramagnetic, and diamagnetic domains leading, e.g., to the scientific field of spintronics), optics (involving polarized light), etc. The fundamental investigation on the electrodynamics of the spin and the proposal of spin models, such as the Ising model with spin-up and spin-down states to explain the ferromagnetism of certain materials, are active fields of research in basic science.

A.1 Appendix: Magnetic Moment

This section presents the basic relations of the magnetic moment with other physical variables. The purpose is to clarify different aspects concerned with the Stern-Gerlach experiment of Sect. 9.3. The departure point is the existence of atoms that respond to an external magnetic field due to their magnetic moments. In this respect, we consider that our atom has a dipolar magnetic field simulating that of a small bar magnet and capable of interacting with an external magnetic field. The dipolar magnetic field is described from a classical perspective using an electric current in a circular loop. In a later stage, the quantum character of the "rotations" of the microscopic particles is introduced by establishing a link to the quantization rules of both the orbital angular momentum and the spin of the particle.

A.1.1 Electric Current in a Circular Loop

A charged point particle in motion represents an electric current, and according to the Maxwell equations, it generates a magnetic moment $\boldsymbol{\mu}$.

$$\boldsymbol{\mu} = (\mathbf{r} \times \mathbf{j}) / 2 = q\mathbf{r} \times \mathbf{v}/2 \tag{9.8}$$

The variable $\mathbf{r}$ is the instantaneous position of the particle, $\mathbf{v}$ is the velocity, and $\mathbf{j}$ is the current density. The 1/2 factor is chosen by convention. In the case of a large number of particles, the magnetic moment is expressed as an integral over the volume of the particle distribution. From the classical version of the atom, an electron with an orbit around the nucleus describes an electric current i in a circular loop and creates a magnetic moment $\boldsymbol{\mu}$. The magnetic field generated by the electric current is equivalent to that of a small bar magnet. The magnetic dipole moment of the electron in orbit is $\mu = iA$, where the area enclosed by the trajectory is $A = \pi r^2$. The current is given by $i = -e/T = -e\omega/(2\pi)$, with T and ω the period and angular frequency of the orbital motion. The magnetic dipole may be written as $\boldsymbol{\mu} = -(e\omega r^2/2)\hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the normal vector to the area A with direction defined by the right-hand rule using the electric current direction. On the other hand, the angular momentum is $\mathbf{L} = \mathbf{r} \times \mathbf{p}$. The magnitude of $\mathbf{L}$ in terms of the particle speed is $L = r(mv) = m\omega r^2$, where the velocity is $v = \omega r$. The relation between $\boldsymbol{\mu}$ and $\mathbf{L}$ is $\boldsymbol{\mu} = -e\mathbf{L}/(2m)$. Therefore, the electron magnetic moment is anti-parallel to the electron angular momentum.

$$\boldsymbol{\mu} = -e\mathbf{L}/(2m) \tag{9.9}$$

The magnetic dipole may be described as that created by either an electric current in a circular loop or a pair of magnetic poles relatively close to each other, Fig. 9.2. At far distances, the magnetic fields are the same when we use any of the above descriptions. Yet in the region close to the magnetic sources, the magnetic field produced by the electric current loop is different from the magnetic field produced by the pair of magnetic poles. We deal with the model using the electric current in a circular loop as the expressions derived from it have a general aspect and describe the magnetic moments of all the particles of an atom (the electron in orbital motion, the electron with spin, and the nucleus with spin) with similar simplicity.

The magnetic moment $\boldsymbol{\mu}$ may be additionally understood as the property of a system which establishes a relation between the torque $\boldsymbol{\tau}$ exerted on the system and

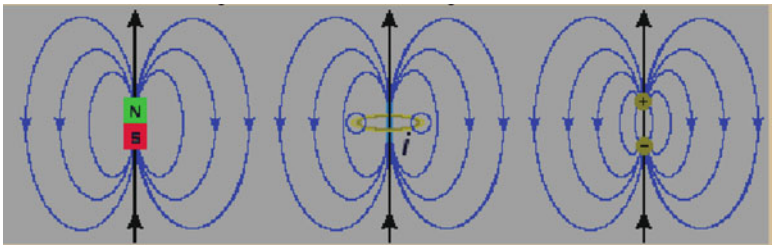


Fig. 9.2 Sketch on the left: an atom has a dipolar magnetic field, which is similar to some extent to that of a tiny magnet. Sketch on the middle: the dipolar magnetic field may be described by means of an electric current i in a circular loop. Sketch on the right: the dipolar magnetic field may be also described by two magnetic monopoles, where one is positive and the other is negative. In the far-field limit, the magnetic field is the same using any of the models that create the dipolar magnetic field. In the near field, all the magnetic fields differ among one another

the external magnetic field **B**, when the system is immersed in such an external magnetic field.

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B} \tag{9.10}$$

A.1.2 Magnetic g Factor

The fact that a magnetic moment is originated by a charged particle in angular motion brings the expression of the angular momentum **L** of a particle into consideration.

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = m\mathbf{r} \times \mathbf{v} \tag{9.11}$$

For a many-particle system such as an atom or molecule with many electrons and nucleons, the vector of total angular momentum is obtained after adding the total orbital momentum of the electrons, the total spin of the electrons, and the total spin of the nucleons. Expressions 9.8 of $\boldsymbol{\mu}$ and 9.11 of **L** also share some resemblance in those cases. The ratio of the magnetic moment to the angular momentum is known as the gyro-magnetic ratio, γ. The γ ratio is the proportionality factor between the magnetic moment $\boldsymbol{\mu}$ of the particle and the physical variable that describes the motion type of the particle, in this case **L**. If the particle has uniform mass and uniform charge distributions, the gyro-magnetic ratio is:

$$\gamma = gq\,|\mathbf{r} \times \mathbf{v}|\,/\,(2m|\mathbf{r} \times \mathbf{v}|) = gq/(2m) \tag{9.12}$$

The number g is the magnetic g factor. The value of g depends on the type of motion and the type of particle under investigation. For example, the value of g depends on either the type of momentum (angular momentum or spin) or the type of particle (electron, proton, or neutron). In the case of an electron moving in circular orbit, the charge is $q = -e$, and the magnetic moment is anti-parallel to the angular momentum.

$$\boldsymbol{\mu}_L = -e\mathbf{L}/(2m) \tag{9.13}$$

The $\boldsymbol{\mu}_L$ and **L** quantities are parallel for a positron. The magnetic moment of a molecule is determined by the molecular energy state. The orbital angular momenta of the electrons and the number of unpaired electron spins in the atoms lead to the different magnetic moments of the atoms. In turn, the magnetic properties of a macroscopic material are derived from the magnetic moments $\boldsymbol{\mu}_i$ of the individual particles that build the material. The magnetization density **M** of the material with volume V is defined by:

$$\mathbf{M} = \boldsymbol{\mu}/V = \sum_i \boldsymbol{\mu}_i/V \tag{9.14}$$

The materials where a spin resonance can be observed are many. In most instances, those materials contain spin unpaired electrons and spin unpaired nucleons, giving non-zero magnetic moments. The magnetic materials are of organic and inorganic types like the free radical molecules, polymers, liquid molecules, and matter containing transition metal atoms. By using the spin resonance techniques, it is possible to determine the inner electronic and nuclear structures, as well as the magnetic susceptibilities of different compounds.

A.1.3 Magnetic Energy and Magnetic Work

A magnetic moment in the presence of an external magnetic field has potential energy E_p. This energy is computed from the work, W, performed on the magnetic dipole by the external magnetic field in the process to align the magnetic dipole along the external magnetic field.

$$W = \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F} \cdot d\mathbf{s} \tag{9.15}$$

The infinitesimal small displacement vector is $d\mathbf{s}$. The deviation of the vector $\boldsymbol{\mu}$ from the direction of the external magnetic field $\mathbf{B}$ produces a torque on the magnetic moment, Fig. 9.3. A force couple is built when two equal forces separated by a distance act in opposite directions. In our situation, a force couple is created to align the magnetic moment, without translational motion. It is convenient to introduce the

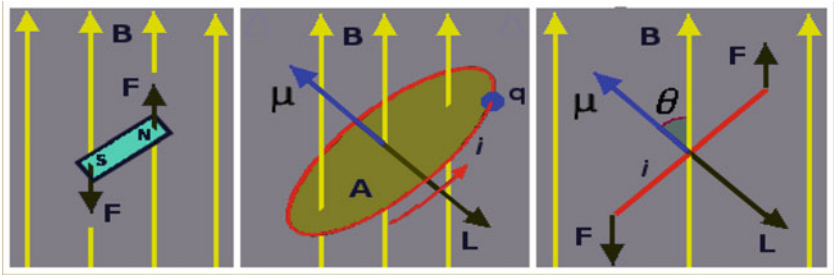


Fig. 9.3 Sketch on the left: a bar magnet with north and south poles is immersed in an external magnetic field $\mathbf{B}$. The external magnetic field exerts forces, F, at different points of the magnet producing a force couple. Sketch in the middle: a particle with charge q moves in a circular trajectory producing an electric current i. The area enclosed by the trajectory is A. The magnetic moment induced by the electric current is $\boldsymbol{\mu}$. The angular momentum vector $\mathbf{L}$ and magnetic moment $\boldsymbol{\mu}$ have opposite directions for the electron, $q = -e$. Sketch on the right: The external magnetic field produces a torque on the magnetic moment, as in the case of the bar magnet. The angle between $\boldsymbol{\mu}$ and $\mathbf{B}$ is θ

angle θ between the vectors $\boldsymbol{\mu}$ and $\mathbf{B}$ and deal with the expressions of both the arc length $d\mathbf{s} = -\hat{\mathbf{n}}_\theta\, d\theta \times \mathbf{r}$, where $\hat{\mathbf{n}}_\theta$ is the unit vector perpendicular to the plane that contains the displacement $d\mathbf{s}$, and the radius vector $\mathbf{r}$. The kernel of the integral describing W becomes $-\mathbf{F}\cdot(\hat{\mathbf{n}}_\theta\, d\theta \times \mathbf{r})$. It can be written in terms of the torque using the circular shift rules of the scalar triple product, $\hat{\mathbf{n}}_\theta\, d\theta \cdot (\mathbf{r} \times \mathbf{F}) = \boldsymbol{\tau} \cdot \hat{\mathbf{n}}_\theta\, d\theta$. Under the consideration that the torque and the angular displacement vector are parallel, the integral describing W is:

$$W = -\int_{\pi/2}^{\theta} \tau\, d\theta = -\int_{\pi/2}^{\theta} \mu B\, \sin(\theta) d\theta = \mu B \cos(\theta) = \boldsymbol{\mu} \cdot \mathbf{B} \tag{9.16}$$

The work is measured for angle changes with respect to $\theta = \pi/2$. The potential energy E_p is the work exerted against the magnetic field, which tries to align the magnetic dipole moment, thus $E_p = -W$. The lowest and the highest potential energies, $E_p = \pm\mu B$, are achieved when $\boldsymbol{\mu}$ is parallel and anti-parallel to $\mathbf{B}$.

When a sample in the Stern-Gerlach apparatus is subjected to an orthogonal electromagnetic wave of high frequency, the magnetic component of the electromagnetic wave flips the magnetic moments of the particles in the sample. The required radio-frequency energy to induce a transition of the magnetic moment is $\Delta E_p = \mu B - (-\mu B)$, being proportional to the strength of the magnetic field.

$$\Delta E_p = 2\mu B \tag{9.17}$$

By using the Planck relation $\Delta E_p = \hbar\omega$, we obtain the frequency of the electromagnetic wave that flips the magnetic moment in an external magnetic field.

$$\omega = 2\mu B/\hbar \tag{9.18}$$

The frequency ω may be tuned by changing the magnetic strength B in continuous manner. The force on the magnetic moment is obtained from the potential energy.

$$\mathbf{F} = -\nabla E_p = \nabla(\boldsymbol{\mu} \cdot \mathbf{B}) \tag{9.19}$$

If $\boldsymbol{\mu}$ is constant and the external magnetic field changes in the z direction, then the non-zero force component is $F_z = \mu_z\, \partial B/\partial z$. A magnetic moment in an external magnetic field shows a deflected trajectory according to the value of μ_z, Sect. 9.3.

It is known that diamagnetic materials create a magnetic moment to repel an external magnetic field. Such a property may be used to levitate a diamagnetic sample. To illustrate this, the force of gravity $-mg\hat{z}$ has to be counteracted by the magnetic force to reach an equilibrium. A diamagnetic material is characterized by a negative magnetic susceptibility, $\chi = -|\chi|$. The magnetization $\mathbf{M}$ of a sample is given in terms of $\boldsymbol{\mu}$, Eq. 9.14. By solving for $\boldsymbol{\mu}$, we obtain $\boldsymbol{\mu} = \mathbf{M}V$. In the case of a weak external magnetic field, the magnetization is proportional to the

external magnetic field, $\mathbf{M} \simeq -|\chi|B/\mu_0$, where $|\chi| \ll 1$ and μ_0 is the free space permeability. This leads to a magnetic moment of the sample.

$$d\boldsymbol{\mu} = -(|\chi| \,/\, \mu_0)\, V\, dB \tag{9.20}$$

The sample has no potential energy when it is on the ground, and there is no external magnetic field. We have to do work against all the forces acting on the sample to provide it with potential energy.

$$E_p = \int_0^z mg\, dz - \int_0^B d\boldsymbol{\mu} \cdot \mathbf{B} = mgz + \int_0^B (|\chi| \,/\, \mu_0)\, V\, B\, dB$$

$$E_p = mgz + (|\chi| \,/\, 2\mu_0)\, V[B(z)]^2 \tag{9.21}$$

The sample gathers energy by its position in both the gravitational field and the magnetic field. The force exerted on the sample is defined by the potential energy, $\mathbf{F} = -\nabla E_p(\mathbf{r})$. The force has rotational symmetry around the z axis, and it must be zero in the z direction to achieve the equilibrium of the sample.

$$mg = -\frac{|\chi|}{\mu_0}\, V\, B(z)\, \frac{\partial B(z)}{\partial z} \tag{9.22}$$

Such a result has been applied to hover different materials, including small living organisms such as a frog [5]. In order to levitate a human being, one requires a magnetic field slightly smaller than 45 T (in Tesla units), with 1×10^9 Watts to cool the system to operate.

A.1.4 Zeeman Effect

It is time to introduce the quantum version of the angular momentum in both the magnetic moment and the potential energy. This is done by assuming a total angular momentum with magnitude $L = \hbar\sqrt{\ell(\ell+1)}$, Eq. 9.2, and a z component of the form $L_z = m_\ell \hbar$, Eq. 9.7. By using the expression $\mu_z = -eL_z/(2m)$, Eq. 9.13, then the potential energy of Eq. 9.16 and the magnetic moment are transformed into:

$$E_p = m_\ell\, \mu_B B \qquad ; \qquad \mu_z = -m_\ell\, \mu_B \qquad ; \qquad \mu_B = e\hbar/(2m) \tag{9.23}$$

where the *Bohr magneton* μ_B, with units eV/T, is defined. The main result indicates that an external magnetic field splits the orbital energy into energy sub-levels of magnitude $m_\ell \mu_B B$. For example, when the magnetic field is applied, the energy of an electron in a p level, $\ell = 1$ and $m_\ell = 0, \pm 1$, unfolds into three equally spaced energy sub-levels.

$$E_{B=0} - \frac{e\hbar}{2m} B \quad ; \quad E_{B=0} \quad ; \quad E_{B=0} + \frac{e\hbar}{2m} B$$

The energy in the absence of the magnetic field, $E_{B=0}$, is the reference energy. The energy splitting may be increased or decreased in continuous manner since it is directly proportional to the external magnetic field and it may be tuned up. In short, the energy of an electron of an atom immersed in a magnetic field depends on the quantum numbers n and ℓ and also on the magnetic quantum number m_ℓ and the strength B of the magnetic field. The splitting of the energy spectrum with dependence on B is known as the *Zeeman effect*.

An atom or molecule is a multi-particle system with angular momentum characterized by the total orbital angular momentum, the total spin, and the corresponding eigenvalues.

$$\mathbf{L} = \sum_i \boldsymbol{L}_i \quad ; \quad \mathbf{S} = \sum_i \boldsymbol{S}_i$$

$$\hbar\sqrt{\ell(\ell+1)} \quad ; \quad \hbar\sqrt{s(s+1)}$$

Such vectors produce magnetic moments μ_L and μ_S, and precess in a magnetic field.

$$\boldsymbol{\mu}_L = -e\mathbf{L} / (2m) \quad ; \quad \boldsymbol{\mu}_S = -g\, e\mathbf{S} / (2m)$$

On the other side, the $\mathbf{J}$ vector is identified with the total angular momentum. The eigenvalues of $\mathbf{J}$ are $\hbar\sqrt{j(j+1)}$, where j satisfies $|\ell - s| \leq j \leq |\ell + s|$. The J_z component is quantized in the form $J_z = m_j\hbar$, with $m_j = 0, \pm 1, \cdots, \pm j$. The electron magnetic moment and magnetic energy are written in terms of the orbital angular momentum contribution and the spin contribution.

$$\boldsymbol{\mu}_J = \boldsymbol{\mu}_L + \boldsymbol{\mu}_S \quad ; \quad E_p = e[\mathbf{L} + 2\mathbf{S}] \cdot \mathbf{B} / (2m)$$

The coefficient 2 takes the g factor of the electron spin into consideration. By introducing the quantization of $\mathbf{J}$ and J_z, the overall magnetic moment and Zeeman splitting energy are written in terms of the quantum numbers j and m_j.

$$\mu_J = g\mu_B\sqrt{j(j+1)} \quad ; \quad E_p = gm_j\mu_B B$$

The g factor depends on J, L, and S:

$$g = 1 + [J(J+1) + S(S+1) - L(L+1)] / [2J(J+1)]$$

The g factor is usually computed using perturbation theory for the energy of an atom in a weak magnetic field, adopting the general form of a second-rank tensor. The g factor of an electron with orbital angular momentum and spin is known as the *Lande factor*. On the other hand, the selection rules determine the possible electronic transitions and frequency shifts among the new energy sub-levels of the atom or molecule in a non-zero magnetic field.

Historically, the total magnetic moment of the electron was conceived without the spin contribution. Now, it is known that the spin also contributes to the total magnetic moment of the electron. In the past, the deviations between the hypothesized results, without consideration of the spin, and the lab measurements originated the name *anomalous Zeeman effect*. Nonetheless, the anomalous Zeeman spectrum is presently realized as the Zeeman effect with the spin contribution.

There is a torque on the magnetic moment of an electron moving in orbital motion around the nucleus and in the presence of a magnetic field. The torque tends to align the magnetic moment with the external magnetic field, producing a change of the angular momentum with time, similar to that of a spinning top or gyroscope in a gravitational field. The frequency of precession is known as the *Larmor frequency*. It is computed by considering the angle θ between the angular momentum $\mathbf{L}$ and the z axis, this last axis taken as the direction of the external magnetic field. The component of $\mathbf{L}$ in the plane of precession is $L \sin\theta$, Fig. 9.4. In the time interval $[t, t + \Delta t]$ there is a directional change ΔL of the angular momentum. The corresponding angular change in the plane of precession is $\Delta\phi$. The torque is the change of angular momentum with time and has a perpendicular direction to the angular momentum.

$$\tau = \lim_{\Delta t \to 0} \Delta L \,/\, \Delta t = \lim_{\Delta t \to 0} \Delta\phi \, L \sin\theta \,/\, \Delta t \tag{9.24}$$

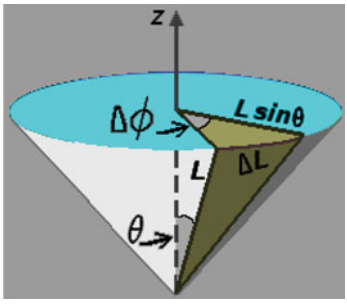


Fig. 9.4 An external magnetic field exerts a torque $\boldsymbol{\tau}$ on the magnetic moment associated with the orbital motion of an electron in an atom. The magnetic moment is anti-parallel to the orbital angular momentum $\mathbf{L}$ of the electron. Vector $\mathbf{L}$ makes an angle θ with the vertical axis, which corresponds to the direction of the magnetic field. In a time interval $[t, t + \Delta t]$, the vector $\mathbf{L}$ describes an arc segment ΔL. The torque is computed from the expression $d\mathbf{L}/dt = \lim_{\Delta t \to 0} \Delta L/\Delta t$

The torque involves the magnetic moment, $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B} = \mu B \sin(\theta)\, \hat{\mathbf{n}}$, Eq. 9.10, where $\hat{\mathbf{n}}$ is the unit vector perpendicular to the plane that contains the vectors $\boldsymbol{\mu}$ and $\mathbf{B}$. The magnitude of $\boldsymbol{\mu}$ is $eL/(2m)$, Eq. 9.13. By comparing the magnitude of $\boldsymbol{\tau}$ with that of Eq. 9.24, and solving for $d\phi/dt$, we obtain:

$$\omega_L = d\phi \,/\, dt = \lim_{\Delta t \to 0} \Delta\phi \,/\, \Delta t = eB \,/\, (2m) \tag{9.25}$$

The frequency ω_L is the Larmor frequency of the magnetic moment in precession.

A.1.5 Electron Spin

The Stern-Gerlach experiment gives evidence of the spin of the electron and the quantization of the spin by using silver atoms, where the outermost electron of every silver atom is in a S state characterized by the quantum numbers $\ell = 0$ and $m_\ell = 0$. A couple of deflections in the atomic trajectories are observed, and two projection states, $m_s = \pm 1/2$, are hypothesized for the electron spin. The existence of the electron spin makes possible to have states doubly occupied, that is, we have fermionic states characterized by the set of quantum numbers (n, l, m_ℓ), which may be occupied by one spin-up electron and one spin-down electron, in line with the Pauli exclusion principle.

The electron spin is identified as a kind of intrinsic angular momentum of a tiny charged ball, but with particular properties of quantum nature, which make impossible to provide a clear classical description. For instance, the magnetic moment of the electron involves the factor g, which is equal to 1.0 for the orbital motion of the electron and 2.002, 319, 304, 92 for the spin of the electron.[1] According to our old suppositions, the electron is extremely small and demands an exceedingly high angular velocity to compensate for the observed spin magnetic moment. However, such a picture is in contradiction with the principles of relativistic mechanics, as explained in Sect. 9.1.[2] In spite of that, the intrinsic angular momentum of the electron is modeled using an electric current in a circular loop, taking the quantization rules of the spin into consideration to account for the values of the magnetic moment, Fig. 9.5. This approach provides a "reasonable" quantitative description of the electron spin, and for such a reason, it is also applied to the nuclear spins with modifications of the g-factor magnitudes and the magnetons.

We have the factors g_s, g_p, and g_n for the spin of the electron, proton, and neutron, respectively (the g factor of the electron in circular orbit is $g_L = 1$).

[1] The value of the g factor is predicted by quantum electrodynamics. It has been verified in the Lamb shift experiment with high accuracy. Feynman [6].

[2] In principle, any incompatibility vanishes in the relativistic zitterwegegun model.

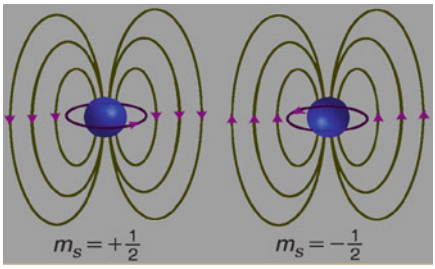


Fig. 9.5 Two spin projections of the electron are observed in the Stern-Gerlach experiment. Such states are hypothesized to involve an intrinsic angular momentum of the electron, referred to as the electron spin. The electron spin produces a dipolar magnetic moment. We use a classical approach, in conjunction with the quantization rules of the spin, in an effort to explain the dipolar magnetic moment

$$g_s = 2.002,319,304,92 \simeq 2 \quad ; \quad g_p = 5.585,694,713$$
$$g_n = -3.826,083,7 \tag{9.26}$$

The magneton factors are μ_B and μ_n for the electron spin and nuclear spin.

$$\mu_B = (e\hbar)\,/\,(2m_e) = 5.788{,}382 \times 10^{-5}\, eV/T$$
$$\mu_n = (e\hbar)\,/\,(2m_p) = 3.152{,}451 \times 10^{-8}\, eV/T \tag{9.27}$$

The magnetons are the standard units to gauge the magnetic moments of the microscopic particles. The proton mass and proton charge are used as the quantities of reference in the magnetons of the nucleons. There are differences in the values of the magnetons, $\mu_B \gg \mu_n$, due to differences in the masses of the electron, m_e, and proton, m_p. The magnetic moments μ_L and μ_s of the orbital motion and the spin of the electron, Eqs. 9.13 and 9.3, and the magnetic moment μ_p of the proton spin are, in that order:

$$\boldsymbol{\mu}_L = -g_L \left(\frac{e}{2m_e}\right)\mathbf{L} \quad ; \quad \boldsymbol{\mu}_s = -g_s \left(\frac{e}{2m_e}\right)\mathbf{S} \quad ; \quad \boldsymbol{\mu}_p = g_p \left(\frac{e}{2m_p}\right)\mathbf{I}$$
$$\mu_{L_z} = -g_L \left(\frac{e}{2m_e}\right)\hbar m_\ell \quad ; \quad \mu_{S_z} = -g_s \left(\frac{e}{2m_e}\right)\hbar m_s \quad ; \quad \mu_{I_z} = g_p \left(\frac{e}{2m_p}\right)\hbar m_I \tag{9.28}$$

The proton spin is denoted with the symbol $\mathbf{I}$, and the corresponding z component is $\hbar m_I$, where m_I takes one of the $2I + 1$ possible values of the nuclear spin. The magnetic moment of the proton spin is smaller than the magnetic moment of the electron spin. On the other hand, the magnetic moment of the proton spin is $2.792{,}847{,}350\,\mu_n$, and that of the neutron spin is $-1.913{,}041{,}8\,\mu_n$. The directions of the electron spin and its magnetic moment are anti-parallel, and those of the

proton are parallel to each other. The proton spin precesses clockwise, and the electron spin precesses counter-clockwise. The Zeeman splitting of the nuclear energy levels is computed similarly to the splitting of the electronic energy levels but using the corresponding nuclear variables.

A.1.6 Paschen-Back Effect

In real samples, particularly those containing transition metals, there are electric fields and dipolar magnetic fields of the electrons and nuclei. The Zeeman energy splitting depends not only on the external magnetic field imposed in the lab but also on the magnetic moments of the nearby particles, which are capable of modifying the dynamics of a particle due to the proximity. For example, consider an electron in orbital motion around the nucleus. In the reference frame of the electron, such an electron is subjected to a magnetic field produced by a charged nucleus, which is in relative motion with respect to the electron. The magnetic field interacts with the spin magnetic moment of the electron, generating a spin-orbit (LS) coupling with dependence on the angular momentum of the electron. In general, the interactions among the magnetic moments of different particles lead to a cross-energy term $E_{en} = am_I m_s$, where a is known as the *hyperfine coupling constant*. The cross-energy term is determined from the g factors of the electron and nucleus and the distances and orientations between the interacting magnetic moments. That interaction has analogy with the interaction of two or more electric dipole moments.[3] The Zeeman splitting of the energy level of an unpaired electron in the field of the nuclei may be approximated by $E = g_s m_s \mu_B (B + \sum_i a_i m_{I_i})$, where the summation is a correction to the imposed magnetic field B. Therefore, in this approach, an effective magnetic field is considered in the energy expression.

The splitting of energy levels has been discussed in the Zeeman regime, that is, where the external magnetic field maintains the LS coupling between the angular momentum of the orbital motion and the angular momentum of the spin motion. In the Zeeman regime, the splitting of the energy levels is small with respect to the energy difference among unperturbed levels due to a relatively weak external magnetic field. However, when the external magnetic field is strong, it modifies the LS coupling to generate a strong individual coupling of L with the external magnetic field and similarly for the spin S. As a result, we have the independent precession of L and S and the intertwining of the energy sub-levels of the particles. This response receives the name *Paschen-Back effect*. In such a domain, the energy splitting depends not only on the atom type but also on the energy-level type. Moreover, it may occur that for a given strength of the magnetic field, some energy levels split according to the Zeeman effect, but others split in anomalous form according to the Paschen-Back effect, all in the same atom. With an approximate

[3] For the interaction of two magnetic dipoles of fermionic particles, refer to: Cohen-Tannoudji et al. [7]. For the interaction of two linear charge distributions, refer to: Maitland et al. [8].

external magnetic field of $3\,T$, the Zeeman effect is observed in Na atoms, but a Paschen-Back effect in Li atoms. The Paschen-Back effect is used in astronomy to determine the magnetic fields of astronomical objects by measuring the radiation emitted by the atoms subjected to strong magnetic fields.

A.1.7 Applications of the Spin Resonance Technique

In the experiments with electron spin resonance (ESR), electron paramagnetic resonance (EPR), and nuclear magnetic resonance (NMR), the magnetic field is tuned, and the adequate frequency to induce the flip of the spin states of electrons and protons is chosen. The magnetic field applied in NMR spectroscopy may reach up to $22\,T$ (for reference, the geomagnetic field at our planet surface is between 25 and 65 μ-Tesla). The spin relaxation emits an electromagnetic signal after the device is switched off. The emitted signal depends on the external magnetic field. Thus, under a magnetic field gradient which coils around the sample, the different frequencies coming from the separate spatial areas of the sample are identified and analyzed, resulting in well-resolved images of the sample. The spin resonance techniques are specially effective on materials with radicals, where electrons with unpaired spins are abundant. Such spectroscopic techniques are non-destructive. By following a similar derivation of Eq. 9.17, and using the expressions of μ_{I_z} and g_p, Eqs. 9.28 and 9.26, the energy to flip the proton spin in a $1\,T$ magnetic field is:

$$\Delta E_p = 2\mu_{I_z} B = g_p \mu_n B = 1.7608 \times 10^{-7}\, eV \tag{9.29}$$

This energy is small with respect to the average room thermal energy of $0.026\,eV$ at $300\,K$. Therefore, the thermal energy randomizes the populations of the proton spin states. In the case of an electron, the required angular frequency of the electromagnetic wave to produce a spin transition is derived from the expression $\omega_e = \Delta E_p/\hbar = 2\mu_{S_z} B/\hbar$, Eq. 9.18. In the case of a proton, the frequency for the spin transition is $\omega_p = 2\mu_{I_z} B/\hbar$, Eq. 9.29. By assuming an external magnetic field of $1\,T$, the respective frequencies, $\nu = \omega/(2\pi)$, are $\nu_e = 27.995\ GHz$ and $\nu_p = 42.574\ MHz$.

On the other hand, the frequency to induce the spin transition of the neutron is similar to that of the proton. Yet it is surprising that the neutron has a large and negative magnetic moment since it is a neutral particle. The magnetic moment of the neutron is indirectly determined by measuring the magnetic moment of the deuteron, which is formed by a neutron and a proton with parallel spins and with previous knowledge of the magnetic moment of the proton. The magnitude of the magnetic moment of the proton and the magnitude and sign of the magnetic moment of the neutron suggest that both the proton and the neutron have internal structure and are not elementary particles. They are made of charged *quarks*, with magnetic moments resulting in those of the proton and neutron. On the other hand, the neutrino is hypothesized to have an extremely small magnetic moment, but contrary to the neutron, the neutrino is an elementary particle with zero mass. The neutrino's

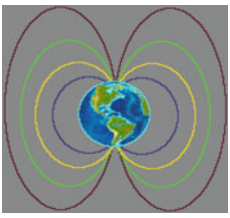


Fig. 9.6 The Earth has a large magnetic field of dipolar type, created by the geodynamo of the core particles. The far-field limit is similar to that produced by the electron spin; nevertheless, the magnetic field strengths differ by many orders of magnitude

magnetic moment is believed to be associated with quantum fluctuations, which make the neutrino show up as a mixture of two interacting particles, W^+ and e^-, with the coupling of a magnetic field, like in the case of the electron self-energy with involvement of a photon, Sect. 9.2. The muon (a lepton 207 times heavier than the electron but with equal negative charge) is a particle that shows an anomalous magnetic moment. The experimental measurements and the theoretical predictions differ by a factor of 0.1%, a relatively large number. By refining the research on the muon magnetic moment, the experiments may point to either the existence of new types of particles or the necessity to propose a more exotic theory beyond the Standard Model and based, for example, on the dark photon (a possible component of dark matter in the universe). Such conclusions are expected to be derived in the frame of quantum electrodynamics. Finally, at the other end of the spectrum, many macroscopic objects are characterized by their magnetic moments. The planet Earth is an example with an enormous magnetic moment of $\sim 8 \times 10^{22}\ Coulomb\ m^2/s$; contrariwise, the Moon has no dipolar magnetic field, Fig. 9.6.

B.1 Appendix: Spin Functions and Operators

The spin is understood as a pure quantum-mechanical property with no classical analogue. Therefore, it is important to discuss the notation and algebra that the spin demands to be formally described. The information may be complemented with the work of Cohen-Tannoudji et al. [9]. In the first stage, a linear Hermitian operator designated with the letter $\hat{\mathbf{S}}$ is assigned to the spin. The operator $\hat{\mathbf{S}}$ classifies as an angular momentum operator with similar quantization rules to the orbital angular-momentum operator, $\hat{\mathbf{L}}$. The relevant spin operators are $\hat{\mathbf{S}}^2$ and the components $\hat{S}_i$, where $i = x, y, z$. The relation between these operators is given by $\hat{\mathbf{S}}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2$. They satisfy commutation relations with cyclic permutations.

$$\begin{gathered}[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z \quad ; \quad [\hat{S}_y, \hat{S}_z] = i\hbar \hat{S}_x \quad ; \quad [\hat{S}_z, \hat{S}_x] = i\hbar \hat{S}_y \\ [\hat{\mathbf{S}}^2, \hat{S}_i] = 0 \quad ; \quad i = x, y, z\end{gathered} \tag{9.30}$$

The spin operators act on functions that belong to the space formed by the *spin states* of the particle. Since the spin operator $\hat{\mathbf{S}}^2$ and $\hat{S}_z$ commute, they have common eigenfunctions which we distinguish with the symbols α and β. For example, by using the eigenfunction α, we have:

$$\begin{aligned} \hat{S}_z\alpha(s,m_s) &= m_s\hbar\,\alpha(s,m_s) \quad ; \quad m_s = -s, -s+1, \cdots, s-1, s \\ \hat{\mathbf{S}}^2\alpha(s,m_s) &= s(s+1)\hbar^2\alpha(s,m_s) \quad ; \quad s = 0, 1/2, 1, 3/2, \cdots \end{aligned} \tag{9.31}$$

In this particular case, the functions α and β are not eigenfunctions of the spin operators $\hat{S}_x$ and $\hat{S}_y$ because these operators do not commute with $\hat{S}_z$. The arguments s and m_s of the eigenfunctions may be considered the degrees of freedom of the spin functions α and β. An elementary particle owns a unique value of s. The photon with value $s = 1$ breaks the rule of m_s stated in Eq. 9.31 as it only takes values $m_s = \pm 1$, and no value $m_s = 0$. For the electron, we only have the spin $s = 1/2$ and the following eigenvalue expressions:

$$\begin{aligned} \hat{S}_z\alpha(s,m_s) &= +\frac{1}{2}\,\hbar\alpha(s,m_s) \quad ; \quad \hat{\mathbf{S}}^2\alpha(s,m_s) = \frac{3}{4}\,\hbar^2\alpha(s,m_s) \quad ; \quad m_s = +1/2 \\ \hat{S}_z\beta(s,m_s) &= -\frac{1}{2}\,\hbar\beta(s,m_s) \quad ; \quad \hat{\mathbf{S}}^2\beta(s,m_s) = \frac{3}{4}\,\hbar^2\beta(s,m_s) \quad ; \quad m_s = -1/2 \end{aligned} \tag{9.32}$$

The nomenclature spin-up and spin-down is associated with values $m_s = 1/2$ and $m_s = -1/2$ of the electron. The functions α and β form an orthonormal basis set and generate the spin-state space of the electron with dimension 2. A general expression of a spin function is $\psi_{spin} = c_\alpha\alpha + c_\beta\beta$ where the coefficients may be complex numbers. Before discussing additional features of the spin, it is convenient to introduce a new notation to simplify the expressions. The expansion of a function ψ in the basis set w is displayed below for discrete and continuous functions.

$$\psi(\mathbf{r}) = \textstyle\sum_i c_i\, w_i(\mathbf{r}) \quad ; \quad \psi(\mathbf{r}) = \int d\alpha\; c(\alpha)\; w_\alpha(\mathbf{r}) \tag{9.33}$$

Here, the continuous variable α should not be confused with the α spin function. The expansions may be represented under the corresponding *Dirac bra-ket vector notation* (the nomenclature follows the word bra-ket).

$$\begin{aligned} |\psi\rangle &= \sum_i c_i\,|w_i\rangle \quad ; \quad |\psi\rangle = \int d\alpha\; c(\alpha)\;|w_\alpha\rangle \\ \langle\psi| &= \sum_i c_i^*\,\langle w_i| \quad ; \quad \langle\psi| = \int d\alpha\; c^*(\alpha)\;\langle w_\alpha| \end{aligned} \tag{9.34}$$

The *closure relations* displayed below involve the identity operator in terms of the basis functions.

$$\hat{I} = \sum_i |w_i\rangle \langle w_i| \quad ; \quad \hat{I} = \int d\alpha \, |w_\alpha\rangle \langle w_\alpha| \tag{9.35}$$

The closure relation is important to project a bra or ket onto the basis functions.

$$\begin{aligned} |\psi\rangle &= \sum_i |w_i\rangle \langle w_i|\psi\rangle = \sum_i c_i \, |w_i\rangle \\ |\psi\rangle &= \int d\alpha \, |w_\alpha\rangle \langle w_\alpha|\psi\rangle = \int d\alpha \, c(\alpha) \, |w_\alpha\rangle \end{aligned} \tag{9.36}$$

The space of functions that completely describe the particle state, including the spin, is constructed from the tensor product of the space formed by the spatial functions and the space formed by the spin functions. If a space-vector state is represented by the ket vector $|\mathbf{r}\rangle$, and a spin-vector state is represented by the ket vector $|\varepsilon\rangle$, corresponding to either the α or β spin function, then the vector that defines the particle state is given by the tensor product $|\mathbf{r}, \varepsilon\rangle = |\mathbf{r}\rangle \otimes |\varepsilon\rangle$. The vector $|\mathbf{r}, \varepsilon\rangle$ is a *ket vector*, and the corresponding complex conjugate vector is the *bra vector* $\langle \mathbf{r}, \varepsilon|$.

The spin operators commute with the spatial operators, and thus, they have common eigenfunctions. The eigenvalue equations that involve such operators are:

$$\begin{aligned} X|\mathbf{r}, \varepsilon\rangle = x|\mathbf{r}, \varepsilon\rangle \quad ; \quad Y|\mathbf{r}, \varepsilon\rangle = y|\mathbf{r}, \varepsilon\rangle \quad ; \quad Z|\mathbf{r}, \varepsilon\rangle = z|\mathbf{r}, \varepsilon\rangle \\ \hat{\mathbf{S}}^2|\mathbf{r}, \varepsilon\rangle = \frac{3}{4}\hbar^2|\mathbf{r}, \varepsilon\rangle \quad ; \quad \hat{S}_z|\mathbf{r}, \varepsilon\rangle = \pm\frac{\hbar}{2}|\mathbf{r}, \varepsilon\rangle \end{aligned} \tag{9.37}$$

The kets $|\mathbf{r}, \varepsilon\rangle$ form a complete basis sets. The orthonormal condition on the kets is $\langle \mathbf{r}', \varepsilon'|\mathbf{r}, \varepsilon\rangle = \delta_{\varepsilon'\varepsilon}\delta(\mathbf{r}' - \mathbf{r})$, where the symbols δ denote Dirac-delta functions. The general expression of a state function is written as a linear combination of basis functions.

$$|\psi\rangle = \sum_\varepsilon \int |\mathbf{r}, \varepsilon\rangle \langle \mathbf{r}, \varepsilon|\psi\rangle \, d^3r \tag{9.38}$$

The components of ψ in the spin states $\varepsilon = \alpha$ and $\varepsilon = \beta$ are in general indicated by $\psi_\varepsilon(\mathbf{r}) = \langle \mathbf{r}, \varepsilon|\psi\rangle$.

$$\psi_\alpha(\mathbf{r}) = \langle \mathbf{r}, \alpha|\psi\rangle \quad ; \quad \psi_\beta(\mathbf{r}) = \langle \mathbf{r}, \beta|\psi\rangle \tag{9.39}$$

Such functions define a two-dimensional *spinor*.

$$[\psi](\mathbf{r}) = \begin{pmatrix} \psi_\alpha(\mathbf{r}) \\ \psi_\beta(\mathbf{r}) \end{pmatrix} \tag{9.40}$$

When a state vector is expressed as a tensor product, $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$, the spinor becomes:

$$[\psi](\mathbf{r}) = \begin{pmatrix} \psi_\alpha(\mathbf{r}) \\ \psi_\beta(\mathbf{r}) \end{pmatrix} = \begin{pmatrix} \langle \mathbf{r}|\psi_1\rangle \ \langle \alpha|\psi_2\rangle \\ \langle \mathbf{r}|\psi_1\rangle \ \langle \beta|\psi_2\rangle \end{pmatrix} = \begin{pmatrix} \psi_1(\mathbf{r})\, c_\alpha \\ \psi_1(\mathbf{r})\, c_\beta \end{pmatrix} \tag{9.41}$$

where c_α and c_β are the expansion coefficients of the spin function. The spin operators under the spinor representation correspond to matrices with dimension 2×2. Moreover, any matrix of such a dimension may be written as a linear combination of the three Pauli spin matrices and the unit matrix. The application of an operator $\hat{A}$, with matrix representation $[\hat{A}]$, on the spinor $[\psi]$ takes the form:

$$[\psi'](\mathbf{r}) = [\hat{A}][\psi](\mathbf{r}) \tag{9.42}$$

On the other hand, the matrix element of the operator $\hat{A}$ in terms of the $[\psi]$ and $[\phi]$ spinors is given by:

$$\langle \psi|\hat{A}|\phi\rangle = \int [\psi]^\dagger(\mathbf{r})\ [A]\ [\phi](\mathbf{r})\ d^3r \tag{9.43}$$

where $[\psi]^\dagger$ is the adjoint vector of $[\psi]$.

Problem 9.2 By starting from a closure relation, like those of Eqs. 9.35, show that the scalar product between the state vectors $|\psi\rangle$ and $|\phi\rangle$ is given in terms of their spinors.

$$\langle \psi|\phi\rangle = \int [\psi]^\dagger(\mathbf{r})\ [\phi](\mathbf{r})\ d^3r$$

Obtain the normalization of the spinor $[\psi]$ when the kets are equal, $|\phi\rangle = |\psi\rangle$.

Solution The appropriate closure relation for the particle involves integration over the space variable and summation over the possible spin states.

$$\sum_\varepsilon \int d^3r\ |\mathbf{r}\varepsilon\rangle\ \langle \mathbf{r}\varepsilon| = 1$$

We introduce the closure relation in the bra vector $\langle\psi|$ without affecting such a vector because the closure relation is equivalent to the identity operator.

$$\langle \psi| = \int d^3r\ [\langle \psi|\mathbf{r}\alpha\rangle\ \langle \mathbf{r}\alpha| + \langle \psi|\mathbf{r}\beta\rangle\ \langle \mathbf{r}\beta|]$$

(continued)

The scalar product between $\langle\psi|$ and $|\phi\rangle$ is:

$$\begin{aligned}\langle\psi|\phi\rangle &= \int d^3r\ [\langle\psi|\mathbf{r}\alpha\rangle\ \langle\mathbf{r}\alpha|\phi\rangle + \langle\psi|\mathbf{r}\beta\rangle\ \langle\mathbf{r}\beta|\phi\rangle] \\ &= \int d^3r\ \left[\psi_\alpha^*(\mathbf{r})\phi_\alpha(\mathbf{r}) + \psi_\beta^*(\mathbf{r})\phi_\beta(\mathbf{r})\right] \\ &= \int d^3r\ (\psi_\alpha^*, \psi_\beta^*)\begin{pmatrix}\phi_\alpha(\mathbf{r}) \\ \phi_\beta(\mathbf{r})\end{pmatrix} \\ &= \int d^3r\ [\psi]^\dagger(\mathbf{r})\ [\phi](\mathbf{r})\end{aligned}$$

The scalar product of the state vectors $|\psi\rangle$ and $|\phi\rangle$ corresponds to the scalar product of the spinors, which proofs the first requirement. When $|\phi\rangle = |\psi\rangle$, we have the normalization of the state vector $|\psi\rangle$ from the second expression presented previously.

$$\langle\psi|\psi\rangle = \int d^3r\ \left[|\psi_\alpha(\mathbf{r})|^2 + |\psi_\beta(\mathbf{r})|^2\right] = 1$$

This satisfies the second requirement.

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Postulates of Quantum Mechanics

10

The experiments at the beginning of the 20*th* century revealed unexpected results on the behavior of the light and the particles at the atomic level. In those experiments a flux of particles showed interference, diffraction, and dispersion patterns originally attributed to waves, and, on the other hand, the light showed behavior of a matter particle by colliding with other particles in different experiments. From a classical perspective, a particle is an entity with mass, occupying a position in space, and the light is a wave with spatial extension, no mass, and no fixed position. The particles and waves represent opposite aspects, and thus, the waves are not expected to behave as particles, and the particles are not expected to behave as waves. In other words, the properties of particles and the properties of waves are irreconcilable from our classical understanding. Still, the experimental observations on the particles and the waves undeniably showed counterintuitive aspects because an entity, such as a wave or particle, may behave in some instances as a wave, and in others as a particle, thus regarding a dual character of the entity, which is in conflict with the expectations based on classical mechanics.

The diverse aspects of the particles and the waves are nowadays considered under a new philosophy, which carries the name *quantum mechanics*. The theory of quantum mechanics introduces unorthodox ideas; however, it provides the proper explanations of the microscopic phenomena. It is a theory that we do not fully understand, with unconventional interpretations. The best way to approach quantum mechanics is to create analogies with the behaviors of the particles and the waves in the classical world, but with the appropriate caution since a microscopic entity, such as a wave or a particle, may behave as a wave under given circumstances and as a particle under different circumstances. For macroscopic particles, quantum mechanics assertively reproduces the laws of classical mechanics. In the following sections, we provide the postulates that give foundations to quantum mechanics; they engulf the key results of our observations on the microscopic world and condense the fundamentals of quantum mechanics in a formal mathematical way.

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In addition to the postulates, we discuss the principles that emerge from a new and intertwined characterization of the microscopic particles and waves. We pay attention to the wave equation and wave function, the variation principle of the energy, and other rules that give frame to quantum matter.

10.1 Reformulating the Conceptual World

A new philosophy at looking nature emerged from the experiments on microscopic bodies. The new philosophy included the particle-wave duality of matter, the energy quantization, and the consideration of a new degree of freedom, the spin. It was given the name *wave mechanics*, also recognized as quantum mechanics. Quantum mechanics constitutes a general scientific philosophy that is not possible to rationalize, as it is beyond our habitual comprehension. Yet, it was to revolutionize the way to describe nature since we may deal with a particle as a wave or with a wave as a particle. The theory of quantum mechanics combines aspects which in principle were considered as opposite aspects to us into a single description, and after a crude reality that remains unclear to humans. Nevertheless, it is capable to provide the basic features of nature.

Quantum mechanics is not in contradiction with the theory of classical mechanics, but it generalizes it by introducing: (i) the particle-wave character of the microscopic entities, (ii) the quantization of the energy, and (iii) an intrinsic spin. Quantum mechanics unifies the theories of particles and radiation (this last one embodies electricity, magnetism, and optics) to provide a unique conception of nature. By using quantum mechanics, there are no more differences between what is theorized and observed in the experiments, since the wave-particle duality, energy quantization, and spin constitute the foundations of the modern theory. Wave mechanics has been under test in many different forms, and it has been successfully confirmed in all of them. In this regard, quantum mechanics is a general and fundamental theory. Still, some limitations persist. According to relativistic mechanics, the relativistic effects are important when the speeds of the particles are close to the speed of light. Relativistic mechanics also considers matter as a form of energy, and vice versa. However, it has not been possible to properly include relativistic mechanics in quantum mechanics, and therefore, there is not a general theory (with general consensus) unifying both fields up to this time.

In the following sections, the main postulates of quantum mechanics are presented. They establish the physical and mathematical foundations of the theory. The comments after each postulate provide the details on the formulation of the postulate. The consideration of the postulates in the simulations makes possible to unfold the dynamics of particles and suitably explain their behaviors at the atomic and subatomic levels.

10.2 Postulates of Quantum Mechanics

Classical mechanics is a branch of physics formulated according to our everyday experience, mostly based on observations at the macroscopic level. Nevertheless, the experiments performed at the microscopic scale at the beginning of the 20*th* century uncovered aspects of particles and waves not observed before. The experiments demonstrated the quantization of the energy, with microscopic matter absorbing and emitting radiation in discrete bundles of energy, and revealed the existence of the spin, a kind of intrinsic angular momentum of the particles. The predictions based on classical mechanics were found in contradiction with the experimental observations in the atomic domain. In this regard, the classical understandings of a particle and a wave, incompatible with each other and mutually exclusive at macroscopic scales, had to be reconciled into a new scientific theory to describe a particle and a wave as a single entity at the microscopic realm.

The postulates of quantum mechanics are provided in the sections below to establish the formal basis of quantum mechanics. We follow the common practice to state them concisely, with details provided after the formulation of each postulate.[1]

10.2.1 First Postulate

The particles show a wave character in the microscopic world leading to probabilistic predictions. In this regard, the quantum state of a particle system is expected to be described by a *wave function*. The first postulate establishes a relation between the wave function and the quantum-mechanical state of the system by associating the wave function to a vector which belongs to the space of state vectors of the system. When the state vector $|\psi(t)\rangle$ is considered, then we can deal with a representation of it by employing a wave function $\psi(t)$ of the system.

Statement 10.1 There exists a function $\psi(t)$ with name *wave function*, which describes the quantum-mechanical state of the system at the instant of time t. The wave function is a state vector $|\psi(t)\rangle$ which belongs to the space of state vectors associated with the system.

The wave function depends on the coordinates and spins of the particles and is a function of the time, t.

$$\psi(\mathbf{r}_1, \sigma_1, \mathbf{r}_2, \sigma_2, \cdots, \mathbf{r}_n, \sigma_n; t) \tag{10.1}$$

[1] More information can be found in: Cohen-Tannoudji et al. [1], Chapt. III.

The variables $\mathbf{r}_i$ and σ_i represent the spatial coordinates and spin of the ith particle.[2] The wave function is a *state function*. It is mathematically well behaved, this is, it is univaluated, continuous, with continuous derivatives, and quadratically integrable at any instant of time and satisfies the superposition principle.

$$\begin{aligned} &\int\!\!\int \psi^*(\{\mathbf{r}_i, \sigma_i; t\})\, \psi(\{\mathbf{r}_i, \sigma_i; t\})\, dv = 1 \\ &dv = d^3r_1\, d\sigma_1\, d^3r_2\, d\sigma_2 \cdots d^3r_n\, d\sigma_n \end{aligned} \tag{10.2}$$

The expression ψ^* is the complex conjugate of ψ (the spin variables σ_i are discrete, and summations should be additionally used in Eq. 10.2 for the spins; nevertheless, the integrals over the different degrees of freedom should be understood to also include the summations, for simplicity). The product $\psi^*\psi\, dv$ has a probabilistic interpretation. If we consider a single particle, the probability of finding it in the space region limited by the vectors (x, y, z) and $(x + dx, y + dy, z + dz)$ at the instant of time t is given by $\psi^*(x, y, z; t)\, \psi(x, y, z; t)\, dxdydz$. The quadratically integrable requirement ensures a finite numerical value of the spatial probability function at any instant of time. The set of all wave functions $\{\psi_1, \psi_2, \cdots\}$, referent to the different states of the system, forms a complex vector space which satisfies the algebra rules of a Hilbert space. In contraposition to a particle which is described in deterministic form in classical mechanics, the description of a particle by a wave function, which may now be written in terms of the basis functions $\{\psi_1, \psi_2, \cdots\}$, acquires a probabilistic nature in quantum mechanics.

There are cases in nature where it is impossible to apply the first postulate due to the extreme physical conditions, for example, in the very first moments of the Universe creation, when a strong and short-lived singularity was responsible for the birth of the Universe. The spacetime singularities, also produced by black holes, impede the use of quantum mechanics due to the lack of a smooth wave function, free of discontinuities in the derivatives. Also, it is neither possible to describe phenomena smaller than the Planck scale in quantum mechanics, as the Planck constant h imposes the smallest quantization scale. Fortunately, there is a myriad of other physical cases where quantum mechanics may be applied.

10.2.2 Second Postulate

We perform measurements in the laboratory to extract information of the particle system. In this respect, we require a mathematical tool in quantum mechanics to act upon the wave function to reproduce the results measured in the lab.

[2] The electronic spin is introduced in ad hoc manner in the wave function of the electrons of a molecule, Sect. 10.9, like in the Hartree-Fock-based methods or the Density Functional Theory, and after considering Pauli's exclusion principle, Sect. 10.8. The incorporation of the electronic spin leads to unique solutions of the quantum-mechanical eigenvalue equations. Unfortunately, the origins of the spin remain as a conundrum.

Statement 10.2 There is an associated operator, $\hat{A}$, to every physical observable, $\mathcal{A}$, which acts upon the vectors that belong to the state-vector space of the system. It must be Hermitian to render real eigenvalues.

There are two important aspects of the quantum operators that should be considered. The first aspect includes the way to build the operators, and the second one involves the mathematical requirements on the operators. The purpose of such aspects is to guarantee both, the correct mathematical behavior of the operators and the proper access to the physical information.

The first aspect demands operators that represent the physical observables. To do this, there are rules to associate an operator $\hat{A}$ of quantum mechanics to every physical observable $\mathcal{A}$. The rules consist of replacing the classical variables $\mathbf{r}$ and $\mathbf{p}$ that describe the physical observable $\mathcal{A}$, by the respective operators $\hat{\mathbf{R}}$ and $\hat{\mathbf{P}}$.

$$\mathcal{A}(\mathbf{r}, \mathbf{p}; t) \quad \Rightarrow \quad \hat{A}(\hat{\mathbf{R}}, \hat{\mathbf{P}}; t) \tag{10.3}$$

In most situations, we demand a symmetric expression in terms of the operators $\hat{\mathbf{R}}$ and $\hat{\mathbf{P}}$. The symmetrization is a common requirement in the proposal of operators corresponding to those physical observables where the product of the fundamental variables $\mathbf{r}$ and $\mathbf{p}$ is present. However, care should be exercised, as the dynamical variable $\mathbf{p}$ is understood as a conjugate momentum and should not be confused with other types of momenta. For example, the conjugate momentum of a particle in a vector potential $\mathbf{C}$ is $\mathbf{p} = m\mathbf{v} + q\mathbf{C}$, while the mechanical momentum is $m\mathbf{v}$. The operator $\hat{A}$ should additionally ensure a faithful representation of the physical observable $\mathcal{A}$ at the quantum-mechanical level by being mathematically well behaved. The operators are expected to be linear, a property that is linked to the superposition principle. They should avoid the breaking of the canonical commutation relations of quantum mechanics among the Cartesian components of $\hat{\mathbf{R}}$ and $\hat{\mathbf{P}}$.

$$[R_i, R_j] = [P_i, P_j] = 0 \quad ; \quad [R_i, P_j] = i\hbar\, \delta_{ij} \tag{10.4}$$

The quantum-mechanical operators are Hermitian or self-adjoint to: (i) render real eigenvalues, in agreement with the results obtained from the experimental measurements, and (ii) generate vectors that form a basis set in the corresponding Hilbert space. The Hermitian property renders the possibility to apply the Hermitian operator, $\hat{A}$, in any of the two functions ψ_1 or ψ_2.

$$\int \psi_1^* \hat{A} \psi_2 dv = \int \psi_2 (\hat{A}\psi_1)^* dv \tag{10.5}$$

The generation of basis vectors with Hermitian operators is important to describe any state of the system as a linear combination of the basis vectors in the domains that the physical observable $\mathcal{A}$ demands. There are more required properties on the operators. Still, in concise manner, when the operators are correctly proposed and they are in line with all the other quantum mechanical postulates, they form a key part of the mathematical tool box of quantum mechanics to obtain the information of the physical observables of the system.

10.2.3 Third Postulate

In the laboratory, the appropriate measuring device is applied to the system under investigation to obtain the required information. By following an analogy, the third postulate involves the appropriate operator $\hat{A}$ to extract the information of the system.

Statement 10.3 The value of a measurement of the observable $\mathcal{A}$ corresponds to an eigenvalue of the operator $\hat{A}$ associated with the observable $\mathcal{A}$.

The laboratory measurements of the physical quantity $\mathcal{A}$ are definite and correspond to the eigenvalues of $\hat{A}$. The operator $\hat{A}$ is Hermitian to guarantee that the set of eigenvalues of operator $\hat{A}$ are real numbers. A discrete spectrum $\{a_1, a_2, a_3, \cdots\}$ deduced from the operator $\hat{A}$ indicates quantization of the results measured on the system referent to the physical observable $\mathcal{A}$.[3]

10.2.4 Fourth Postulate

When the spectrum of the operator $\hat{A}$ is discrete, the eigenvectors ψ_i and the eigenvalues a_i carry a discrete subindex, i, with ψ_i and a_i satisfying the eigenvalue equation of $\hat{A}$.

$$\hat{A}\psi_i = a_i\psi_i \tag{10.6}$$

The eigenvectors are orthonormal with respect to each other and expand the Hilbert space of the system under investigation. The theory of quantum mechanics is structured in such a way that the quantization of the energy, the angular momentum, and the spin is obtained after solving the corresponding eigenvalue equations, giving solutions with discrete values. There are also operators such as $\hat{\mathbf{R}}$ and $\hat{\mathbf{P}}$, representing

[3] The discussion provided after every postulate not only complements the postulate but also provides clarity.

the physical observables **r** and **p**, that unfold a continuous spectrum. Since the eigenvectors ψ_i generate the Hilbert space, an arbitrary function ψ of such a space is described by a linear combination of eigenfunctions, which all together form a *complete orthonormal basis set*.

$$\psi = \sum_i c_i \psi_i \tag{10.7}$$

The coefficients c_i indicate the weights of the respective state functions ψ_i in the production of ψ. The weights c_i may also have time dependence. In this context, the coefficients c_i with large magnitudes infuse a large character of the corresponding state functions ψ_i to the wave function ψ. The projection of ψ onto ψ_i gives the factor c_i due to the orthogonality of the eigenvectors.

$$c_i = \langle \psi_i | \psi \rangle = \iint \psi_i^* \, \psi \, dv \tag{10.8}$$

The integration is over all the degrees of freedom, and the Dirac bra-ket notation is introduced, Sect. B.1 of Chap. 9.

Problem 10.1 Show that the set of functions $\{\psi_i\}$ constitutes a complete basis set. Every ψ_i is normalized and is solution of Eq. 10.6.

Solution In order to demonstrate that $\{\psi_i\}$ is a complete basis set, we show that it is a set of orthonormal functions. The functions ψ_j and ψ_k are solutions of expression 10.6. In this case, the equation of ψ_j is multiplied by ψ_k^* and the complex conjugate equation of ψ_k by ψ_j:

$$\psi_k^* \hat{A} \psi_j - \psi_k^* \hat{A}^* \psi_j = (a_j - a_k) \psi_k^* \psi_j \quad \Rightarrow \quad 0 = (a_j - a_k) \int_V \psi_k^* \psi_j \, dv$$

The left-hand side is zero upon integration because $\hat{A}^* = \hat{A}$. Since the left-hand side is 0, and $a_j \neq a_k$, then the integral is zero. In this respect, ψ_j and ψ_k are orthonormal, and the set of functions $\{\psi_i\}$ constitutes a complete basis set.

$$\int_V \psi_k^* \psi_j \, dv = \delta_{jk}$$

The spectral decomposition may also include degenerate functions, or continuous functions, or their combination. In the case of *degenerate functions*, several eigenvectors $\psi_n^{(i)}$ reproduce the same eigenvalue.

$$\hat{A} \psi_n^{(i)} = a_n \psi_n^{(i)} \quad ; \quad i = 1, 2, \cdots, g_n \tag{10.9}$$

The degeneracy degree is g_n. The description of ψ in terms of the eigenvectors also includes all the degenerate states.

$$\psi = \sum_n \sum_i^{g_n} c_n^{(i)} \psi_n^{(i)} \quad ; \quad c_n^{(i)} = \langle \psi_n^{(i)} | \psi \rangle \tag{10.10}$$

The eigenvectors $\psi_n^{(i)}$ constitute an orthonormal set of vectors and reproduce the vector space associated with the eigenvalue a_n. In the continuous case, the expansion of ψ in terms of the eigenvectors ψ_α, which form a continuous basis set, demands the replacement of the summation of Eq. 10.7 by an integral over the continuous variable α.

$$\psi = \int c(\alpha)\, \psi_\alpha \, d\alpha \tag{10.11}$$

Based on the mathematical formalism that has been introduced with the eigenvalue equations, we are now in the position to make contact with the probabilistic predictions of quantum mechanics. To do this, we consider the probability of finding the system in a certain state, say ψ_n, with eigenvalue a_n of the operator $\hat{A}$, in the following postulate.

Statement 10.4 If the system is described by the normalized state vector $|\psi\rangle$, and the corresponding function ψ_n is a normalized eigenvector of the operator $\hat{A}$ (associated with the physical observable $\mathcal{A}$) with eigenvalue a_n, then the probability $\mathcal{P}(a_n)$ of obtaining the value a_n in the discrete case, when the physical observable $\mathcal{A}$ is measured, is:

$$\mathcal{P}(a_n) = |\langle \psi_n | \psi \rangle|^2 \tag{10.12}$$

In the degenerate case with degeneracy degree g_n, the probability of obtaining the value a_n involves all the eigenfunctions $\psi_n^{(i)}$ with eigenvalue a_n.

$$\mathcal{P}(a_n) = \sum_{i=1}^{g_n} |\langle \psi_n^{(i)} | \psi \rangle|^2 \tag{10.13}$$

In the continuous case, the probability of obtaining the value α involves the eigenfunction ψ_α, whose eigenvalue is α in the extension $d\alpha$.

$$d\mathcal{P}(\alpha) = |\langle \psi_\alpha | \psi \rangle|^2 \, d\alpha \tag{10.14}$$

ψ is normalized to deal with probabilities equal or lower than 1.

10.2.5 Fifth Postulate

The following postulate deals with the state vector that corresponds to the system after carrying a measurement of a physical observable $\mathcal{A}$. In order to establish the postulate, it is convenient to introduce the projection of a ket vector $|\psi\rangle$ onto the subspace associated with the eigenvalue a_i of the system. We start by considering the function ψ_i as a linear combination of the degenerate functions $\{\psi_i^{(j)}\}$, with expansion factors $c_i^{(j)}$.

$$\psi_i = \sum_j^{g_n} c_i^{(j)} \psi_i^{(j)} \qquad ; \qquad c_i^{(j)} = \langle \psi_i^{(j)} | \psi \rangle \tag{10.15}$$

The *projection operator* P_i onto the subspace associated with the eigenvalue a_i is defined in terms of the degenerate functions using the bra-ket notation as:

$$P_i = \sum_j^{g_n} |\psi_i^{(j)}\rangle \, \langle \psi_i^{(j)}| \tag{10.16}$$

By applying the projector to $|\psi\rangle$, the individual ket $|\psi_i\rangle$ is recovered.

$$P_i|\psi\rangle = \sum_j^{g_n} |\psi_i^{(j)}\rangle \, \langle \psi_i^{(j)}|\psi\rangle = \sum_j^{g_n} c_i^{(j)} \, |\psi_i^{(j)}\rangle = |\psi_i\rangle \tag{10.17}$$

The ket vector $|\psi_i\rangle$ is the projection of $|\psi\rangle$ onto the subspace associated with the eigenvalue a_i. Next, suppose that the system is described by the wave function $\psi(t_0)$ at the instant t_0. It evolves with time and is described by $\psi(t)$. Eventually, the measurement of the physical observable $\mathcal{A}$ is performed on the system. The result of the measurement is a_i, an eigenvalue of operator $\hat{A}$, and the system is in state ψ_i immediately after the measurement. Such a state also evolves with time after the measurement.

Statement 10.5 When the system is described by the state function ψ, and the value obtained by measuring the observable $\mathcal{A}$ on the system is a_i, then the state function that describes the system immediately after the measurement is given by the projection of $|\psi\rangle$ onto the subspace associated with the eigenvalue a_i, namely, $P_i|\psi\rangle \,/\, \langle\psi|P_i|\psi\rangle^{1/2}$, where $\langle\psi|P_i|\psi\rangle^{1/2}$ is the normalization constant.

10.2.6 Sixth Postulate

The microscopic particles show wave character, and thus, we are required to propose an equation that governs such a behavior. A dynamic equation is proposed by assuming a wave function ψ with dependence on both the spatial variables $\{\mathbf{r}_i\}$ and the time t. The change of ψ with time replaces the trajectory concept of a classical particle. By following a relative analogy to the expression that describes the propagation of electromagnetic waves in vacuum, an equation for the particles is postulated.

Statement 10.6 The time evolution of the state function ψ is ruled by the wave equation:

$$i\hbar \frac{\partial \psi(\{\mathbf{r}_i\}, t)}{\partial t} = \hat{\mathcal{H}}(t)\psi(\{\mathbf{r}_i\}, t) \tag{10.18}$$

where $\hat{\mathcal{H}}$ is the operator corresponding to the Hamiltonian of the system.

The wave equation is a central expression in quantum mechanics since it rules the behavior of ψ in space and time, avoiding in that aspect any indeterminacy of ψ. It receives the name *time-dependent wave equation* or *Schrödinger equation* in honor of Erwin Schrödinger, who proposed it. Section A.1 derives the wave equation 10.18 for matter waves. The Hamiltonian operator introduces the characteristics of the system in the Schrödinger equation, like the number of nuclei and electrons, their masses and charges, the position variables, the kinetic energy operator, $\hat{T} = -\hbar^2\nabla^2/(2m)$, and the interaction types among the particles through the potential energy operator, $\hat{V}$. For a single particle with mass m, the time-dependent wave equation is:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{\mathcal{H}}\psi \quad ; \quad \hat{\mathcal{H}} = -\frac{\hbar^2}{2m}\nabla^2 + \hat{V} \tag{10.19}$$

The Schrödinger equation is linear and homogeneous and accounts for the spatial and temporal changes of ψ. When the wave equation is solved, the state function not only acquires information of the system from the Hamiltonian but also dictates the temporal evolution of ψ, which should be compatible with the term $\hat{\mathcal{H}}\psi$. This process is similar to that of classical mechanics, where the solutions of the Hamiltonian equations of motion acquire the information of the system from the Hamiltonian function. Yet, the wave equation is difficult to solve due to the complex interactions in the microscopic world. The exact solutions are found in the cases of model systems and the simplest atomic and molecular systems with a few

degrees of freedom. In the cases of more complex or realistic systems, usually with a large number of particles, we require to introduce approximations, and the solutions are frequently achieved in numeric form. The current methodologies that solve in approximate form the wave equation are classified according to both the simplifications in their formulations and their accuracy levels.

The most accurate approaches are in principle applicable to all kinds of particle systems, but practical limitations in the speed and storage of the computational devices make those approaches only applicable to small systems. By introducing further approximations, less elaborate schemes are created. These are used to investigate large complexes; however, some accuracy is lost when an accurate methodology is simplified. There is a trade-off in such a simplification process between the system size and the computational efficiency. In this regard, based on a balance between accuracy and computational efficiency, one may decide the approach to employ for a system with a given size. When the wave equation is solved, the state function ψ is determined, and the molecular system may be characterized by using ψ according to the quantum-mechanical procedures.

The purpose of this work is to describe the mainstream quantum-mechanical equations at different levels of theory, which have practical applications in molecular systems. Such methods are widely used to study the dynamics of atoms, molecules, clusters, and extended matter like the crystals, fluids, and gases, in the present days. We provide their mathematical foundations, paying special attention to the formulation of the suppositions, because these not only define the accuracy level and limitations of the approaches but also point out pathways for future improvements.

10.3 Time Reversibility in Quantum Mechanics

The goal of this section is to demonstrate the time reversibility in quantum mechanics. We follow similar steps to those given by Tolman [2]. Let us deal with an isolated conservative system of particles. We assume two versions of the same system: (i) the non-primed system, where the time evolution is in the forward direction, and (ii) the primed system, where the time evolution is in the backward direction. We are required to demonstrate that, by reversing the signs of the momentum and time, we obtain the same predictions from the equations of motion, probability densities, and expectation values of the physical observables, in the two systems.

The probability of finding the particle's position r in the interval dr at the instant of time t is $P(r, t)\, dr$, while the probability of finding the particle's momentum p in the interval dp at the instant of time t is $P(p, t)\, dp$. The expectation value of a physical variable $\mathcal{A}$ is represented by $\langle \hat{A}(r, p, t) \rangle$, where the operator $\hat{A}$ depends on the conjugate variables r and p. In principle, by reversing the momenta and times in the terms $P(r, t)$, $P(p, t)$, and $\langle \hat{A}(r, p, t) \rangle$ of the non-primed system, we should obtain the respective primed terms, $P'(r, t)$, $P'(p, t)$, and $\langle \hat{A}'(r, p, t) \rangle$.

$$P'(r,t) = P(r,-t) \quad ; \quad P'(p,t) = P(-p,-t)$$
$$\langle \hat{A}'(r,p,t)\rangle = \langle \hat{A}(r,-p,-t)\rangle \tag{10.20}$$

The quantum Hamiltonian $\hat{\mathcal{H}}$ is obtained from the classical Hamiltonian $\mathcal{H}$ of the system by replacing the conjugate momentum p by the momentum operator $-i\hbar\partial/\partial r$. The classical Hamiltonian has quadratic dependence on p; therefore, the quantum Hamiltonian is real and remains without changes when the complex conjugate of $\hat{\mathcal{H}}$ is considered. On the other hand, the wave function ψ depends on the conjugate coordinate r and time t and satisfies the wave equation 10.18, shown below for easy access.

$$i\hbar \frac{\partial \psi(r,t)}{\partial t} = \hat{\mathcal{H}}(t)\psi(r,t) \tag{10.21}$$

The complex conjugate of the wave equation is obtained by replacing ψ with ψ^* and changing signs of the imaginary number i.

$$-i\hbar \frac{\partial \psi^*(r,t)}{\partial t} = \hat{\mathcal{H}}(t)\psi^*(r,t) \tag{10.22}$$

The functions ψ and ψ^* deal with the time in the forward direction. We work with the primed system and its corresponding wave functions $\psi'(r,t)$ and $\psi'^*(r,t)$. These functions satisfy the following wave equations.

$$i\hbar \frac{\partial \psi'(r,t)}{\partial t} = \hat{\mathcal{H}}(t)\psi'(r,t) \quad ; \quad -i\hbar \frac{\partial \psi'^*(r,t)}{\partial t} = \hat{\mathcal{H}}(t)\psi'^*(r,t) \tag{10.23}$$

A sign change in the times establishes the relation between primed and non-primed functions.

$$\psi'(r,t) = \psi^*(r,-t) \quad ; \quad \psi'^*(r,t) = \psi(r,-t) \tag{10.24}$$

The insertion of these functions in the wave equations 10.23 results in modified expressions for the primed functions.

$$i\hbar \frac{\partial \psi^*(r,-t)}{\partial t} = \hat{\mathcal{H}}(t)\psi^*(r,-t) \quad ; \quad -i\hbar \frac{\partial \psi(r,-t)}{\partial t} = \hat{\mathcal{H}}(t)\psi(r,-t)$$
$$-i\hbar \frac{\partial \psi^*(r,-t)}{\partial(-t)} = \hat{\mathcal{H}}(t)\psi^*(r,-t) \quad ; \quad i\hbar \frac{\partial \psi(r,-t)}{\partial(-t)} = \hat{\mathcal{H}}(t)\psi(r,-t)$$

The first expression may be compared with Eq. 10.22, and the second expression with Eq. 10.21. For the time-independent Hamiltonian, the set of equations become the same when we replace t by $-t$ and vice versa. In reference to the probability $P(r,t) = \psi^*(r,t)\psi(r,t)$, it is possible to replace t by $-t$, to obtain $P(r,-t) =$

$\psi^*(r,-t)\psi(r,-t)$. By using Eqs. 10.24, one of the Eqs. 10.20 is confirmed.

$$P'(r,t) = \psi'(r,t)\psi'^*(r,t) = \psi^*(r,-t)\psi(r,-t) = P(r,-t)$$

The spatial probability $P'(r,t)$ is equivalent to the spatial probability $P(r,-t)$, which has a reversed time with respect to the probability $P(r,t)$. A similar situation is observed when dealing with the momentum functions.

$$\phi'(p,t) = \phi^*(-p,-t) \quad ; \quad \phi'^*(p,t) = \phi(-p,-t)$$

In this case, we have an equivalence between momentum probabilities.

$$P'(p,t) = \phi'(p,t)\phi'^*(p,t) = \phi^*(-p,-t)\phi(-p,-t) = P(-p,-t)$$

The momentum probability $P(-p,-t)$ has arguments with reversed signs with respect to those of the probability $P(p,t)$. This result confirms another identity of Eqs. 10.20.

Problem 10.2 The function $\psi(r,t)$ is the spatial representation of the wave function, while $\phi(p,t)$ is the momentum representation of the wave function. These functions are related by means of Fourier transforms.

$$\begin{aligned}\psi(r,t) &= h^{-n/2}\int \phi(p,t)\ \exp[i/\hbar\ (p_1r_1+\cdots+p_nr_n)]\,d\mathbf{p}\\ \phi(p,t) &= h^{-n/2}\int \psi(r,t)\ \exp[-i/\hbar\ (p_1r_1+\cdots+p_nr_n)]\,d\mathbf{r}\end{aligned} \tag{10.25}$$

Show that $\phi'(p,t) = \phi^*(-p,-t)$ and $\phi'^*(p,t) = \phi(-p,-t)$.

Solution Function $\phi'(p,t)$ is related to function $\psi'(r,t)$ by means of a Fourier transform.

$$\phi'(p,t) = h^{-n/2}\int \psi'(r,t)\ \exp[-i/\hbar\ (p_1r_1+\cdots+p_nr_n)]\,d\mathbf{r}$$

We use Eqs. 10.24 to substitute $\psi'(r,t)$ by $\psi^*(r,-t)$. In this connection, we have:

$$\phi'(p,t) = h^{-n/2}\int \psi^*(r,-t)\ \exp[-i/\hbar\ (p_1r_1+\cdots+p_nr_n)]\,d\mathbf{r}$$

On the other hand, we also know the expression of $\phi^*(-p,-t)$.

(continued)

$$\phi^*(-p,-t) = h^{-n/2} \int \psi^*(r,-t)\ \exp[i/\hbar\ (-p_1 r_1 - \cdots - p_n r_n)]\, d\mathbf{r}$$

The conclusion to this part is $\phi'(p,t) = \phi^*(-p,-t)$. Similar steps are taken to show $\phi'^*(p,t) = \phi(-p,-t)$.

$$\begin{aligned}\phi'^*(p,t) &= h^{-n/2} \int \psi'^*(r,t)\ \exp[i/\hbar\ (p_1 r_1 + \cdots + p_n r_n]\, d\mathbf{r} \\ &= h^{-n/2} \int \psi(r,-t)\ \exp[i/\hbar\ (p_1 r_1 + \cdots + p_n r_n]\, d\mathbf{r}\end{aligned}$$

On the other hand, we have:

$$\phi(-p,-t) = h^{-n/2} \int \psi(r,-t)\ \exp[-i/\hbar\ (-p_1 r_1 - \cdots - p_n r_n)]\, d\mathbf{r}$$

Thus, we conclude $\phi'^*(p,t) = \phi(-p,-t)$.

The last expression to demonstrate involves the equivalence between expectation values of physical observables $\mathcal{A}$ in the time forward and time backward directions.

$$\langle \hat{A}'(r,p,t)\rangle = \langle \psi'(r,t)|\ \hat{A}(r,p,t)\ |\psi'(r,t)\rangle$$

We employ Eqs. 10.24, $\psi'(r,t) = \psi^*(r,-t)$ and $\psi'^*(r,t) = \psi(r,-t)$ in the expression of $\langle \hat{A}'\rangle$.

$$\langle \hat{A}'(r,p,t)\rangle = \langle \psi(r,-t)|\ \hat{A}(r,-i\hbar\frac{\partial}{\partial r},t)\ |\psi^*(r,-t)\rangle$$

We may apply the complex conjugate property to the product $(\hat{A}'\psi)$.

$$\langle \hat{A}'(r,p,t)\rangle = \langle \psi(r,-t)|\ \left[\hat{A}(r,-i\hbar\frac{\partial}{\partial(-r)},t)\ \psi(r,-t)\right]^*\rangle$$

By applying the Hermitian property, Eq. 10.5, of the operators, we have:

$$\begin{aligned}\langle \hat{A}'(r,p,t)\rangle &= \langle \psi(r,-t)|\ \hat{A}(r,-i\hbar\frac{\partial}{\partial(-r)},t)\ |\psi(r,-t)\rangle \\ &= \langle \hat{A}(r,-p,-t)\rangle\end{aligned}$$

In this respect, we have the required equivalence between expectation values in the forward and backward time directions in quantum mechanics.

10.4 Stationary States

We are interested in obtaining a solution of the time-dependent wave equation from the eigenvectors of the Hamiltonian, $\hat{\mathcal{H}}$, which is supposed to be time independent. We follow the work of Cohen et al. [3]. The eigenvectors ψ_{ij} and eigenvalue E_i of $\hat{\mathcal{H}}$ are time independent. The additional index j of the state function ψ_{ij} represents in generic form the other numbers that are required to quantum mechanically define the system. The eigenvectors and eigenvalues satisfy the stationary expression:

$$\hat{\mathcal{H}}\psi_{ij} = E_i\,\psi_{ij} \tag{10.26}$$

The equation is known as the *time-independent wave equation*. The set $\{\psi_{ij}\}$ of solutions forms a complete orthonormal basis set, and any vector of the Hilbert space associated with the system can be expressed as a linear combination of these eigenvectors. Let us consider the vector $\psi(t_0)$ which describes the system at the instant of time t_0, omitting the writing of spatial variables for simplicity, but writing $\psi(t_0)$ in terms of the eigenvectors.

$$\psi(t_0) = \sum_{i,j} c'_{ij}\psi_{ij} \tag{10.27}$$

At a later instant of time t, we similarly have $\psi(t) = \sum_{i,j} c''_{ij}\psi_{ij}$. The coefficients c'_{ij} of $\psi(t_0)$ are different from the coefficients c''_{ij} of $\psi(t)$. Instead of using primes, it is convenient to introduce time-dependent coefficients. Equation 10.27 is written in the form $\psi(t_0) = \sum_{i,j} c_{ij}(t_0)\,\psi_{ij}$, while the wave function at the instant of time t is $\psi(t) = \sum_{i,j} c_{ij}(t)\,\psi_{ij}$. In order to establish the time evolution of ψ from t_0 to t, we now look for a relation between the factors $c_{ij}(t_0)$ and $c_{ij}(t)$, because the functions ψ_{ij} of the expressions of $\psi(t_0)$ and $\psi(t)$ are the same over time. To do this, we recall that ψ satisfies the time-dependent wave equation. Hence, the expression $\psi(t) = \sum_{i,j} c_{ij}(t)\,\psi_{ij}$ is introduced into the time-dependent wave equation 10.18, reaching a differential equation for the coefficients, which is similar to that of $\psi(t)$.

$$i\hbar\,\partial c_{ij}(t)\,/\,\partial t = E_i\,c_{ij}(t) \tag{10.28}$$

By considering the initial condition at the instant of time t_0, the solution of the above expression is $c_{kj}(t) = c_{kj}(t_0)\exp[-i\,E_k(t-t_0)]/\hbar$. The wave function at any instant of time t is written in terms of the parameters computed at the initial instant of time t_0.

$$\psi(t) = \sum_{kj} c_{kj}(t_0)\,e^{-i\,E_k(t-t_0)/\hbar}\,\psi_{kj} \tag{10.29}$$

This expression gives the time evolution of ψ from t_0 to t. The functions $\psi(t_0)$ and $\psi(t)$ are different due to the participation of different phase factors shown in the

exponential. On the other hand, the time evolution of a physical variable, $\mathcal{A}(t)$, is expressed in terms of the operator $\hat{A}$ and the state vectors ψ_{kj}.

$$\langle\psi(t)|\hat{A}|\psi(t)\rangle = \sum_{kj}\sum_{k'j'} e^{-i(E_k - E_{k'})(t-t_0)/\hbar}\; c^*_{k'j'}(t_0)c_{kj}(t_0)\,\langle\psi_{k'j'}|\hat{A}|\psi_{kj}\rangle \tag{10.30}$$

Let us suppose the special case where $\psi(t_0)$ is a linear combination of the state vectors ψ_{kj}, with a fixed value of k.

$$\psi(t_0) = \sum_j c_{kj}(t_0)\,\psi_{kj} \tag{10.31}$$

The function $\psi(t_0)$ is an eigenvector of $\hat{\mathcal{H}}$ with eigenvalue E_k. By multiplying the coefficients $c_{kj}(t_0)$ by the phase factor $\exp[-iE_k(t-t_0)/\hbar]$, the wave function at a later time is:

$$\sum_j c_{kj}(t_0)\,e^{-iE_k(t-t_0)/\hbar}\,\psi_{kj} = \psi(t) \qquad \Rightarrow \qquad \psi(t) = e^{-iE_k(t-t_0)/\hbar}\,\psi(t_0) \tag{10.32}$$

The functions $\psi(t_0)$ and $\psi(t)$ are physically equivalent because they differ by a single phase factor. The energy of the system at the time instants t_0 and t is E_k, indicating the energy conservation. In this regard, the wave function ψ describes a *stationary state*.

The following situation considers another facet of a stationary state function. It involves the case of a wave function expressed as the product of two functions; the first function depends on the spatial variables and the second one on the time.

$$\psi(\mathbf{r},t) = \phi(\mathbf{r})\,\chi(t) \tag{10.33}$$

In order to determine ψ, we need to know ϕ and χ. The equation that each function satisfies is proposed by inserting ψ in the time-dependent wave equation and expressing $\hat{\mathcal{H}}$ as the addition of a kinetic energy operator, $-\hbar^2\nabla^2/(2m)$, and a potential energy term, $V(\mathbf{r})$.

$$\frac{i\hbar}{\chi(t)}\,\frac{\partial\chi(t)}{\partial t} = -\frac{\hbar^2}{2m}\,\frac{\nabla^2\phi(\mathbf{r})}{\phi(\mathbf{r})} + V(\mathbf{r}) \tag{10.34}$$

The expression on the left depends on t and is independent of the expression on the right, which depends on $\mathbf{r}$. In this situation, the two expressions are equal to a constant, which we write as $\hbar\omega$, where ω is a constant angular frequency. The solution of the equation holding the time is $\chi = A\exp[-i(\hbar\omega)t/\hbar]$, and the function ϕ satisfies the time-independent equation:

$$-\frac{\hbar^2}{2m}\nabla^2\phi + V(\mathbf{r})\phi = E_\omega\phi \quad ; \quad E_\omega = \hbar\omega \tag{10.35}$$

This expression is the eigenvalue equation of the Hamiltonian, $\hat{\mathcal{H}}\phi = E_\omega\phi$, with quantized energy. The function ψ is written in terms of the amplitude A, function ϕ, and energy E_ω.

$$\psi = A\,\phi(\mathbf{r})\,e^{-i\,E_\omega t/\hbar} \tag{10.36}$$

Next, we are interested in computing the particle density, $\rho(r)$, from ψ with the purpose of showing that it is a physical variable with stationary character for a stationary state. To do this, we consider the general definition of the particle density operator of a system with n particles:

$$\hat{\rho}(\mathbf{r}) = \sum_{i=1}^{n}\delta(\mathbf{r}-\mathbf{r}_i) \tag{10.37}$$

The most probable value of $\hat{\rho}$ gives the particle density.

$$\langle\psi(\mathbf{r}_1\sigma_1, \mathbf{r}_2, \sigma_2, \cdots, \mathbf{r}_n, \sigma_n; t)|\ \hat{\rho}(r)\ |\psi(\mathbf{r}_1\sigma_1, \mathbf{r}_2, \sigma_2, \cdots, \mathbf{r}_n, \sigma_n; t)\rangle \tag{10.38}$$

By inserting the operator $\hat{\rho}$ in this expression and displaying the integrals explicitly using the notation of Eq. 10.2, we have the spatial distribution of the indistinguishable particles.

$$\begin{aligned}\rho(\mathbf{r};t) &= \sum_{s_1,\cdots,s_n}\int\sum_{i=1}^{n}\delta(\mathbf{r}-\mathbf{r}_i)\ \psi^*(\{\mathbf{r}_i\sigma_i\};t)\ \psi(\{\mathbf{r}_i\sigma_i\};t)\,dv; \qquad i=1,\cdots,n\\ &= n\sum_{s_2,\cdots,s_n}\int|\psi(\{\mathbf{r}_i\sigma_i\};t)|^2 dv \quad ; \quad i=2,\cdots,n\end{aligned} \tag{10.39}$$

If we consider ψ as given by Eq. 10.36, the particle density is $\psi^*\psi = |A\,\phi(\mathbf{r})|^2$. In this respect, ρ is time indifferent, and ψ and ρ describe a stationary state.

10.5 Superposition Principle of Quantum States

It is important to discuss the superposition principle since it is a key aspect in quantum mechanics. According to the superposition principle, the state vectors $\{\psi_i\}$ may be superposed to create a new wave function, $\sum_i c_i\psi_i = \psi$, where the coefficients c_i are in general complex numbers. Since the functions ψ_i are solutions of the equation of motion 10.18, their linear combination is also a solution of the wave equation 10.18 because such an equation is a linear differential equation.

Contrariwise, any wave function of the Hilbert space associated with the system, say ψ', may be reproduced by the superposition of state vectors, $\psi' = \sum_i d_i \psi_i$.

The wave function ψ, which results from a linear combination of state vectors ψ_i, is a state with intermediate or collective character. In this connection, it is possible to conceive that the particle whose state is ψ may be partially found in every configuration state. The superposition of state vectors introduces a factor of randomness, which is inherent in quantum mechanics.[4]

In the measurement process, the superposition of states collapses into the most likely state vector, say ψ_k, which owns a well-defined eigenvalue, a_k. For the given physical conditions of the system in the experiment, the most likely state is determined from the relative weights that the different states ψ_i have in the composition of ψ. It is in the context of the superposition principle of quantum states that we are able to describe the diffraction and interference phenomena of matter waves and fields, like the diffraction of electrons and the entanglement of quantum computing qbits, with no analog in classical mechanics. Yet, all the random factors and the probabilistic nature of quantum mechanics fade away at macroscopic scales (the derivation is given in a later chapter).

10.6 Bohr's Correspondence Principle

The correspondence principle indicates in general terms that a new and more general theory in science replaces a less general theory when: (i) the general theory is reduced to the less general theory under specific limiting conditions, and (ii) the general theory grants the correct results in the domains where the less general theory is unsuccessful. The wave theory of matter accounts for the behavior of the microscopic particles, and additionally, it reproduces the theory of classical mechanics by describing the behavior of the macroscopic objects. There are several situations that illustrate the general character of the correspondence principle. For example, statistical mechanics leads to thermodynamics by dealing with statistical averages of systems with a large number of particles. Wave optics leads to geometrical optics. The theory of general relativity leads to the Newtonian theory of gravity when the gravitational fields are weak. On the other side, the theory of special relativity leads to Newtonian theory when the velocities of the bodies are small with respect to the speed of light.

When a particle system increases in size, it is transformed from a microscopic system of particles to a macroscopic object. This is one of the important conditions to describe the objects of the macroscopic world from a microscopic theory. The condition to reach a macroscopic system is known as the *classical limit*. In this

[4] It is important to recall the uncertainty principle, which points out the impossibility to know with high accuracy and in simultaneous form the momentum and position of a particle. The position and momentum operators do not commute, and it is not possible to find common eigenfunctions of both operators.

limit, the Planck constant $\hbar$ is sufficiently small, and the potentials acting on the objects show slow changes over the objects' characteristic wavelengths. The wave theory of matter reproduces Newtonian theory in the classical limit. However, the wave theory of matter has statistical nature and provides probabilistic results, while classical mechanics gives non-probabilistic values. Also, the wave theory of matter provides quantized results, whereas classical mechanics gives continuous values of the physical observables. According to Bohr, the correspondence between wave mechanics and Newtonian theory is achieved in the limit of large quantum numbers, specifically, in the limit of large energies and large orbits. Such a correspondence is confirmed when we compare the energy and the most probable position of a quantum oscillator with the equivalent quantities of a classical oscillator. For instance, an energy value of the classical oscillator may be identified with an energy described by a large quantum number of the quantum oscillator. The energy levels of the quantum oscillator are additionally found to be extremely close among each other in the classical limit (the gap energy is small, namely, of order $\hbar\omega$), in such a way that we essentially deal with continuous values of the energy of the quantum oscillator, with little possibility of distinguishing the individual quantum energy values in that limit. In this regard, the quantized energies are blended into a continuous energy spectrum.

Problem 10.3 The energy of an atomic oscillator is $E_{at} = (n + 1/2)\hbar\omega$. The quantum number n takes the values $0, 1, 2, \cdots$, and the angular frequency is ω. On the other hand, a classical harmonic oscillator has an energy that follows Hooke's law, $E_{class} = kA^2/2$, where A is the amplitude at maximum extension and k is the force constant. Estimate the value of the quantum number n for the values $A = 0.1\,m$, mass= $0.3\ Kg$, and $\omega = 1\,rad/s$ of a macroscopic oscillator, and estimate the gap energy between contiguous energy levels.

Solution The strategy to find the quantum number of a classical oscillator is to identify the atomic and classical energies, $E = E_{at} = E_{class}$, and solve the resulting equation for the quantum number, n. The oscillation frequency is related to k by $\omega = \sqrt{k/m}$, where m is the mass of the oscillating particle. The classical energy is $E_{class} = m\omega^2 A^2/2$. After taking the values of A, m, and ω, the quantum number is found to be sufficiently large, $n \sim 1 \times 10^{30}$. The gap energy of contiguous energy levels is $E_{n+1} - E_n = \hbar\omega = 1.054 \times 10^{-34}\ J$, which is extremely small from a macroscopic perspective.

In attention to the postulates of quantum mechanics and the correspondence principle, it is possible to establish a link between the variables of quantum mechanics and their counterparts in classical mechanics. Such a correspondence is possible after recalling the probabilistic character of the wave theory of matter,

where we use probabilities to determine the values of the physical observables. According to the wave theory of matter, the expectation value $\langle\psi|\hat{A}|\psi\rangle$ of a physical variable $\mathcal{A}$ must correspond, in the limit of large quantum numbers (abbreviated in the form $n \to \infty$), to the Newtonian value of that physical observable measured on the macroscopic object.

$$\langle\psi|\hat{A}|\psi\rangle\Big|_{n\to\infty} = \mathcal{A} \tag{10.40}$$

The correspondence between both results is confirmed, once more, for a classical oscillator and a quantum oscillator with large quantum numbers. For example, when the most probable locations of both oscillators are computed, the value obtained from Newtonian mechanics and the highest probable value obtained from quantum theory are found in good agreement [4].

In conclusion, the classical limit of the wave theory of matter takes us to the theory of classical mechanics. This fact favors the use of classical mechanics over quantum mechanics in the domains where the two theories overlap, since Newtonian mechanics is computationally less demanding than quantum mechanics. By dealing with classical mechanics, we just require the positions and velocities of every particle to describe any other physical quantity, like the linear momentum, angular momentum, and energy of the particle system, even those variables of macroscopic thermodynamics. The evolution of the position and the evolution of the velocity of a given particle are defined by the respective Newtonian equations of motion, which are first-order differential equations with time dependence, Sect. 6.15.

10.7 Selection Rules

The atomic excitations are phenomena which cannot be explained in classical mechanics. However, from the quantum-mechanical perspective, the excitation of an atom to a high energy level is due to either the absorption of electromagnetic radiation or the collision with another particle. The transit from a low energy level to a high energy level depends on the initial and final atomic states, and the energy of the electromagnetic radiation as well. When the energy of the electromagnetic radiation fits the energy difference between the initial and final states, the interaction of the atom and the electromagnetic field is resonant, and the transit to a high energy state occurs. Eventually, the atom decays to a low energy level (due to the interaction of the atom with the vacuum fluctuations) with the radiation of a photon by the atom. In this decay, the emitted radiation once more matches the energy gap of energy levels. In this context, quantum mechanics considers the possibility of an atom radiating a photon, among other processes.[5]

[5] Hecht [5], Sect. 3.4.4. In some sense, the impact of a photon, which may be visualized here as a wave with specific characteristics, on an electron, which may be visualized also as another wave, is capable to produce new waves.

The electronic transitions in quantum mechanics are common and important. In an electronic transition to a lower energy state, there is a sudden distribution change of the electron cloud. This process emulates a radiating antenna: for instance, if the electron changes from the $2s$ state down to the $1s$ state, a spherical change of the electron cloud is observed, and the atom radiates a photon. In contrast, an atom in a stationary state, Eq. 10.36, shows no radiation because the average electron position, $\langle\psi|\mathbf{r}|\psi\rangle$, is time independent.

$$\langle\psi|\mathbf{r}|\psi\rangle = \int \left[\phi(\mathbf{r})e^{-iEt/\hbar}\right]^* \mathbf{r}\,\phi(\mathbf{r})e^{-iEt/\hbar}\,d^3r = \int \phi_1^*(\mathbf{r})\,\mathbf{r}\,\phi_2(\mathbf{r})\,d^3r \tag{10.41}$$

Interestingly, there is a time interval where the electron simultaneously coexists in the two energy levels of a transition. The state function in this case is given as a linear combination of the two states, $\psi = c_1\psi_1 + c_2\psi_2$. The average position with such a function is:

$$\begin{aligned}\langle\psi|\mathbf{r}|\psi\rangle = c_1^2 \int \phi_1^*\,\mathbf{r}\,\phi_1\,d^3r + c_2^2 \int \phi_2^*\,\mathbf{r}\,\phi_2\,d^3r + \\ + c_1^*c_2 \int \phi_1^* e^{iE_1t/\hbar}\,\mathbf{r}\,\phi_2 e^{-iE_2t/\hbar}\,d^3r \; + c_1c_2^* \int \phi_1 e^{-iE_1t/\hbar}\,\mathbf{r}\,\phi_2^* e^{iE_2t/\hbar}\,d^3r\end{aligned} \tag{10.42}$$

The average electron position is time dependent. By using the identity $\cos\theta = [\exp(i\theta) + \exp(-i\theta)]/2$, the average position of the electron is:

$$\langle\psi|\mathbf{r}|\psi\rangle = c_1^2 \int \phi_1^*\,\mathbf{r}\,\phi_1\,d^3r + c_2^2 \int \phi_2^*\,\mathbf{r}\,\phi_2\,d^3r + 2c_1^*c_2\cos(2\pi\nu t)\int \phi_1^*\,\mathbf{r}\,\phi_2\,d^3r \tag{10.43}$$

The vibrational frequency of the electron is $\nu = (E_2 - E_1)/h$, which corresponds to the frequency of the emitted electromagnetic radiation [6]. A short time polarization occurs in the time interval of electron density changes. If ψ_2 represents the state function of the system in state 2, ψ_1^* is the complex conjugate of the state function of the system in state 1, and $\hat{\boldsymbol{\mu}}$ is the dipole moment operator, then the value of the transition dipole moment is computed as [7]:

$$\langle\psi_1|\hat{\boldsymbol{\mu}}|\psi_2\rangle = \int \psi_1^*\,\hat{\boldsymbol{\mu}}\,\psi_2\,dv \tag{10.44}$$

We now deal with the wave functions associated with the different degrees of freedom of a molecule [8]. In many instances the system is in the *energy ground state*, and there is a subtle correlation between the dynamics of the electrons and the vibrational and rotational motions of the nuclei. For a heavy atom where the mass of the electron is substantially smaller than the mass of the nucleus, it is possible to (parametrically) separate the dynamics of the electrons from the nuclear motion without a major loss of accuracy. The state function is approximated by the product of the electronic state function, ψ_e, the nuclear vibrational state function, ψ_v, and the nuclear rotational state function, ψ_r.

$$\psi(\mathbf{r}_i, \mathbf{R}_j) \simeq \psi_e(\mathbf{r}_i, \mathbf{R}_j)\ \psi_{vr}(\mathbf{R}_j) \quad ; \quad \psi_{vr}(\mathbf{R}_j) \simeq \psi_v(\mathbf{R}_j)\ \psi_r(\mathbf{R}_j) \tag{10.45}$$

The approximation on the right assumes a vibrational motion uncoupled from the other motions. According to such separations, the transitions between the respective energy levels are classified as electronic, vibrational, and rotational transitions. There is an electronic transition when an electron changes energy levels, a vibrational transition when an oscillator jumps to a different energy level, and a rotational transition when a rotor changes energy levels. The coupled transitions are observed when an electronic transition is coupled to a vibrational mode, or a vibrational transition is coupled to a rotational mode, leading to spectra with superimposed fine structure.

In most instances, the electronic transitions occur rapidly with the nuclei being practically standstill. Yet, for a given electronic state, the nuclei may be excited to a high molecular vibrational state. The off-diagonal transition matrix element $\langle\psi_1|\hat{\boldsymbol{\mu}}_e + \hat{\boldsymbol{\mu}}_n|\psi_2\rangle$, where $\hat{\boldsymbol{\mu}}_e$ and $\hat{\boldsymbol{\mu}}_n$ are the electronic and nuclear dipole operators, is unfolded in terms of the electronic and nuclear-vibration wave functions.

$$\begin{aligned}\langle\psi_1|\hat{\boldsymbol{\mu}}|\psi_2\rangle &= \int \psi^*_{e,1}\, \hat{\boldsymbol{\mu}}_e\, \psi_{e,2}\, d^3r \left(\int \psi^*_{v,1}\, \psi_{v,2}\, d^3R\right) \\ &\quad + \left(\int \psi^*_{e,1}\, \psi_{e,2}\, d^3r\right) \int \psi^*_{v,1}\, \hat{\boldsymbol{\mu}}_n\, \psi_{v,2}\, d^3R \\ \langle\psi_1|\hat{\boldsymbol{\mu}}|\psi_2\rangle &\simeq \alpha \int \psi^*_{e,1}\, \hat{\boldsymbol{\mu}}_e\, \psi_{e,2}\, d^3r \quad ; \quad \alpha = \int \psi^*_{v,1}\, \psi_{v,2}\, d^3R\end{aligned} \tag{10.46}$$

The rotational motion was dispensed, and the orthogonality of the electronic state functions was used to simplify the above expression. The integral over the vibrational wave functions shown in the first line is denoted with α. The square of α is known as the *Franck-Condon factor*. The electronic transition induces a vibrational transition when the vibrational state functions of the Franck-Condon factor exhibit a significant overlap. On the other hand, when the electronic motion is considered to be practically uncoupled of the nuclear motion (based, e.g., on the large difference in the motion time scales of the electrons and nuclei), then the Born-Oppenheimer method of quantum mechanics may be applied.

Problem 10.4 A system of atoms has constant energy. The coordinates of the electrons and nuclei are x_i and X_α, with masses m and M_α, and momenta p_i and P_α, respectively. The system satisfies the time-independent wave equation $\hat{\mathcal{H}}\psi = E\psi$, with:

$$\hat{\mathcal{H}} = \hat{T} + \hat{V} = -\sum_i \frac{\hbar^2}{2m}\, \hat{p}_i^2 - \sum_\alpha \frac{\hbar}{2M_\alpha}\, \hat{P}_\alpha^2 + \hat{V} \tag{10.47}$$

(continued)

The potential energy of the electrons is $\hat{V}_e$, of the nuclei is $\hat{V}_n$, and $\hat{V} = \hat{V}_e + \hat{V}_n$. Find the equations and conditions that ψ_e and ψ_n should satisfy to write the function ψ of the system of atoms in a first approximation as the product $\psi_e(\{x_i\}; \{X_\alpha\})\ \psi_n(\{X_\alpha\})$.

Solution The momentum operator in quantum mechanics is $\hat{p}_x = -(i/\hbar)\,\partial/\partial x$.

$$\left[-\sum_i \frac{\hbar^2}{2m}\,\hat{p}_i^2 - \sum_\alpha \frac{\hbar^2}{2M_\alpha}\,\hat{P}_\alpha^2 + \hat{V}\right]\psi = -\sum_i \frac{\hbar^2}{2m}\,\frac{\partial^2\psi}{\partial x_i^2} + \hat{V}_e\psi$$

$$-\sum_\alpha \frac{\hbar^2}{2M_\alpha}\,\frac{\partial_\alpha^2\psi}{\partial X_\alpha^2} + \hat{V}_n\psi = E\psi$$

When the function $\psi = \psi_e\psi_n$ is introduced through ψ, and we identify $\partial^2\psi/\partial X_\alpha^2 = \psi_n\,\partial^2\psi_e/\partial X_\alpha^2 + 2\,(\partial\psi_n/\partial X_\alpha)\,(\partial\psi_e/\partial X_\alpha) + \psi_e\,\partial^2\psi_n/\partial X_\alpha^2$, we obtain:

$$\psi_n\left[-\sum_i \frac{\hbar^2}{2m}\,\hat{p}_i^2 + \hat{V}_e\right]\psi_e + \psi_e\left[-\sum_\alpha \frac{\hbar^2}{2M_\alpha}\,\hat{P}_\alpha^2 + \hat{V}_n\right]\psi_n$$

$$-\sum_\alpha \frac{\hbar^2}{2M_\alpha}\left[\psi_n\frac{\partial^2\psi_e}{\partial X_\alpha^2} + 2\frac{\partial\psi_n}{\partial X_\alpha}\,\frac{\partial\psi_e}{\partial X_\alpha}\right] = E\psi_e\psi_n$$

A time-independent expression for ψ_e, and another for ψ_n, is reached when ψ_e changes slowly with respect to X_α, as the crossed terms $\partial\psi_e/\partial X_\alpha$ and $\partial^2\psi_e/\partial X_\alpha^2$ may be in this case neglected.

$$\left[-\sum_i \frac{\hbar^2}{2m}\,\hat{p}_i^2 + \hat{V}_e\right]\psi_e = E_e\psi_e\ ;\ \left[-\sum_\alpha \frac{\hbar^2}{2M_\alpha}\,\hat{P}_\alpha^2 + \hat{V}_n\right]\psi_n = (E - E_e)\psi_n \tag{10.48}$$

There exist *selection rules* that determine whether a transition between energy levels is possible or not. More precisely, the selection rules define the transition probabilities among the different atomic and molecular states. The selection rules on the dipole moment indicate the allowed and forbidden energy transitions based on the results obtained from Eq. 10.44. The transitions involving high-order multipoles also satisfy the selection rules, but they are less probable to occur from an energetic point of view. The selection rules are frequently established from symmetry considerations of the state functions using group theory and examining

the conservation of the total angular momentum because, in the absorption and emission processes by atoms and molecules, a photon carries a unit of quantized angular momentum, in spite of its zero mass. Thereby, the selection rules take the change of the total angular momentum and the preservation of the energy into consideration. Finally, the selection rules are expressed in simple forms using *good quantum numbers*, this is, the quantum numbers that show no change due to a negligible coupling among the different degrees of freedom like, for instance, when the electronic angular momentum is weakly coupled to the nuclear spin of the molecule.

10.8 Pauli's Principle in the Electronic Wave Function

There are additional rules on the wave function to those imposed by the quantum-mechanical postulates, especially for the fermionic particles. The electrons belong to this category, and thus, we are interested in the constraints that the electronic wave function should satisfy. *Pauli's principle* is one of such constraints.

Statement 10.7 Pauli's principle or exclusion principle: two or more fermions of a system of particles cannot occupy the same quantum state defined by the quantum numbers n, l, m_l, and m_s.

Pauli's principle is also known as the *exclusion principle*. The implications of the exclusion principle in the wave function are understood by considering, for example, a system of two indistinguishable particles, 1 and 2, and the states a and b. Consider the wave function $\psi(a, b)$, where the particles 1 and 2 occupy the a and b states, respectively, and the wave function $\psi(b, a)$, where the particles 1 and 2 occupy the b and a states, respectively. Both of them satisfy the identity $|\psi(a, b)|^2 = |\psi(b, a)|^2$, because $|\psi|^2$ is a physical observable independent of the labeling of the particles. In this respect, $|\psi|^2$ is unique and represents the probability density of the indistinguishable particles to be found at a given space region. Yet, the result $|\psi|^2$ may be derived from two possible symmetries of the wave function:

$$\begin{aligned} \psi(a, b) &= \psi(b, a) \; ; \quad \text{symmetric wave function} \\ \psi(a, b) &= -\psi(b, a) \quad ; \quad \text{antisymmetric wave function} \end{aligned} \tag{10.49}$$

When the wave function ψ is expressed in terms of the individual state functions ψ_a and ψ_b, two equally probable versions of ψ are obtained.

$$\psi_I = \psi_a(1)\psi_b(2) \quad ; \quad \psi_{II} = \psi_a(2)\psi_b(1) \tag{10.50}$$

The way to build a symmetric wave function (sym) and an antisymmetric wave function (ant), satisfying Eqs. 10.49, is to linearly combine ψ_I and ψ_{II}.

$$\psi_{sym} = \psi_a(1)\psi_b(2) + \psi_a(2)\psi_b(1) \quad ; \quad \psi_{ant} = \psi_a(1)\psi_b(2) - \psi_a(2)\psi_b(1) \tag{10.51}$$

In order to determine which of the above wave functions satisfies the exclusion principle and, thereby, is the appropriate wave function to describe the system of indistinguishable fermions, let us consider that particle 1 occupies the same quantum state as particle 2. In this case, the wave function ψ_{ant} gives zero, leading to a zero probability of finding the particles with the same quantum numbers, in line with Pauli's principle. Thus, the wave function ψ_{ant} is the appropriate function to describe the pair of particles. The general conclusion is that the fermionic particles, which cannot exist in the same quantum state at the same instant of time, must be described by an *antisymmetric wave function*. According to ψ_{ant} of Eq. 10.49, the antisymmetry principle requires that an exchange of particles yields a change of signs of the wave functions. On the other hand, with respect to the bosonic particles, they are not required to satisfy Pauli's exclusion principle, and they are permitted to occupy the same quantum state.

10.9 Wave Function of the Electrons in a Molecule

If a system of n identical fermions is described by the state function ψ, then Pauli's exclusion principle ensures that the exchange of any two particles, with variables $\mathbf{x}_1$, $\mathbf{x}_2$, results in the same wave function by reversing the sign.

$$\psi(\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n; t) = (-1)^{2s}\, \psi(\mathbf{x}_2, \mathbf{x}_1, \cdots, \mathbf{x}_n; t) \tag{10.52}$$

The nomenclature $\mathbf{x}_i = (\mathbf{r}_i, \sigma_i)$ is used, where $\mathbf{r}_i$ and σ_i are the position and spin of the ith particle. We have the identity $(-1)^{2s} = -1$ for fermions ($s = 1/2$) and $(-1)^{2s} = 1$ for bosons ($s = 1$), as these last particles may occupy the same energy state.

One of the simplest representations of an antisymmetric wave function describing the electrons, fermions, of a molecule is a determinant. The elements that build the determinant are the *spin orbitals* $\phi(\mathbf{r}, \sigma)$. The spin orbitals are mathematical functions employed to describe the individual electrons and define their spatial extensions and spin states. The spin orbitals are orthogonal to each other to avoid linear dependencies and are individually normalized to keep the number of electrons described by the wave function constant. The spin orbitals form a complete basis set and guarantee the proper expansion of the Hilbert vector space associated with the system. In other words, such a feature ensures the description of the state of the system with a linear combination of spin orbitals. Still, the spin orbitals should not be understood as physical observables because they are descriptive mathematical functions with not counterpart in the real world.

If the function $\phi_i(\mathbf{x}_j)$ represents the ith spin orbital occupied by the jth electron, then the wave function of the molecular system with n electrons may adopt the form of a determinant [9].

$$\psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n) = \begin{vmatrix} \phi_1(\mathbf{x}_1) & \ldots & \phi_1(\mathbf{x}_n) \\ \phi_2(\mathbf{x}_1) & \vdots & \phi_2(\mathbf{x}_n) \\ \vdots & \phi_i(\mathbf{x}_j) & \vdots \\ \phi_n(\mathbf{x}_1) & \ldots & \phi_n(\mathbf{x}_n) \end{vmatrix} \tag{10.53}$$

The above expression is a *Slater determinant*. In a Slater determinant, all the electrons are distributed in all the orbitals; there is not a one-to-one relation between electrons and orbitals. For instance, electron 1 is located in orbitals ϕ_1, ϕ_2, $\cdots$, ϕ_n, and similarly for the other electrons, refer to the columns of the determinant. In this respect, the electrons are spatially delocalized under a Slater-determinant representation. On the other side, every electron is given equal treatment. For example, the orbital ϕ_1 provides "equal treatment" to all the electrons, because it allocates electrons $1, 2, \cdots, n$, and similarly for the other orbitals, refer to the rows of the determinant. Thereby, in spite of labeling the electrons, the orbitals do not discriminate them, and the electrons, free to occupy any spin orbital, are treated as indistinguishable fermions under a determinantal expression of the wave function. In addition, an exchange of electrons is equivalent to an exchange of columns, and, in turn, the exchange of columns demands the change of the determinant sign. The antisymmetry principle is implicitly considered in the Slater determinant following the quantum-mechanical requirements on the electrons. The Slater determinant is frequently abbreviated using the bra-ket notation, only making reference to the diagonal elements ($\phi_i = \phi_{ii} = \phi_i(\mathbf{x}_i)$).

$$\psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n) = |\phi_1(\mathbf{x}_1) \cdots \phi_n(\mathbf{x}_n)\rangle = \begin{vmatrix} \phi_1(\mathbf{x}_1) & \ldots & \phi_1(\mathbf{x}_n) \\ \vdots & \phi_i(\mathbf{x}_i) & \vdots \\ \phi_n(\mathbf{x}_1) & \ldots & \phi_n(\mathbf{x}_n) \end{vmatrix} \tag{10.54}$$

Problem 10.5 By assuming the orthonormality of the spin orbitals, find the normalization constant of the wave function 10.54.

Solution The Slater determinant contains n orthonormal spin orbitals and n electrons. In order to determine the normalization coefficient of the determinant, we consider that the spin orbitals may be populated by any electron of the system. We have n possibilities to populate the first spin orbital, $n - 1$ possibilities to populate the second spin orbital, etc. In this connection, the normalized wave function is:

(continued)

$$\psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n) = (n!)^{-1/2} \, |\phi_1(\mathbf{x}_1) \cdots \phi_n(\mathbf{x}_n)\rangle \tag{10.55}$$

The integral $\int \psi \psi^* \, dv_1 \cdots dv_n$ gives unity.

An improved representation of the electron wave function is achieved when ψ is written as a linear combination of Slater determinants, with more spin orbitals than electrons, where the spin orbitals are fractionally occupied by the electrons. The improvement of a multi-determinant wave function over a single determinant is associated with the greater spatial relaxation of the electrons due to the availability of more spin orbitals.

In order to illustrate the construction of a wave function of a 2-fermion system, let us consider that the set of spin orbitals $\{\phi_i\}$ form a basis set. The function $\psi(\mathbf{x})$ is written in terms of such spin orbitals with the coefficients c_i as $\psi(\mathbf{x}) = \sum_i c_i \phi_i(\mathbf{x})$. If $\psi(\mathbf{x}_1, \mathbf{x}_2)$ is a function of two variables, then the variable $\mathbf{x}_2$ may be fixed, this is $\mathbf{x}_2 =$ constant, to write the expansion in the form $\psi(\mathbf{x}_1, \mathbf{x}_2) = \sum_i c_i(\mathbf{x}_2)\phi_i(\mathbf{x}_1)$ for that particular value of $\mathbf{x}_2$. The same process may be repeated for every value of $\mathbf{x}_2$. However, in these cases, c_i is understood as a function of $\mathbf{x}_2$, and c_i is expanded in terms of the same basis set: $c_i(\mathbf{x}_2) = \sum_j d_{ij}\phi_j(\mathbf{x}_2)$. By inserting this expression in $\psi(\mathbf{x}_1, \mathbf{x}_2)$, we obtain:

$$\psi(\mathbf{x}_1, \mathbf{x}_2) = \sum_{i,j} d_{ij}\phi_i(\mathbf{x}_1)\phi_j(\mathbf{x}_2)$$

The antisymmetric condition should be imposed in $\psi(\mathbf{x}_1, \mathbf{x}_2)$ to define the appropriate d_{ij} factors. The function ψ is displayed in terms of diagonal and off-diagonal elements.

$$\psi(\mathbf{x}_1, \mathbf{x}_2) = \sum_{i=1} d_{ii}\phi_i(\mathbf{x}_1)\phi_i(\mathbf{x}_2) + \sum_{i<j}[d_{ij}\phi_i(\mathbf{x}_1)\phi_j(\mathbf{x}_2) + d_{ji}\phi_i(\mathbf{x}_2)\phi_j(\mathbf{x}_1)]$$

According to the antisymmetric condition, when $\mathbf{x}_1 = \mathbf{x}_2$, we should have $\psi(\mathbf{x}_1, \mathbf{x}_1) = 0$. This condition demands for every pair ij of indexes the following identity $(d_{ij} + d_{ji})\phi_i(\mathbf{x}_1)\phi_j(\mathbf{x}_1) = 0$, which is satisfied when $d_{ji} = -d_{ij}$. Also, for $i = j$, we deduce $d_{ii} = 0$. We may illustrate the process of building the correct ψ for the case of indexes $i, j = 1, 2, 3$

$$\begin{aligned}\psi(\mathbf{x}_1, \mathbf{x}_2) = {} & d_{12}\phi_1(\mathbf{x}_1)\phi_2(\mathbf{x}_2) + d_{13}\phi_1(\mathbf{x}_1)\phi_3(\mathbf{x}_2) + \\ & - d_{12}\phi_2(\mathbf{x}_1)\phi_1(\mathbf{x}_2) + d_{23}\phi_2(\mathbf{x}_1)\phi_3(\mathbf{x}_2) + \\ & - d_{13}\phi_3(\mathbf{x}_1)\phi_1(\mathbf{x}_2) - d_{23}\phi_3(\mathbf{x}_1)\phi_2(\mathbf{x}_2)\end{aligned}$$

In general, the resulting function ψ with pairwise terms is abbreviated by means of determinants.

$$\psi(\mathbf{x}_1, \mathbf{x}_2) = \sum_{i<j} d_{ij} \begin{vmatrix} \phi_i(\mathbf{x}_1) & \phi_i(\mathbf{x}_2) \\ \phi_j(\mathbf{x}_1) & \phi_j(\mathbf{x}_2) \end{vmatrix} \tag{10.56}$$

ψ is an antisymmetric wave function in terms of two variables and expressed as a linear combination of two-dimensional determinants built from the set of spin orbitals $\{\phi_i\}$. The construction of a wave function of n variables leads to determinants of $n \times n$ dimensions. Thus, the wave functions associated with the ground state and the excited states of a n-fermion system may be written in terms of all the unique determinants formed from the one-variable spin orbitals $\phi_i(\mathbf{x})$, $i = 1, \cdots, m$. The number of unique determinants is given by the *binomial coefficient* C_n^m, corresponding to the number of ways in which n objects are taken from a set constituted by m objects and $n < m$. Section B.1 provides a method to expand a Slater determinant because it is useful to evaluate matrix elements of the operators.

Problem 10.6 Write the wave function $\psi(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ of a 3-fermion system in terms of a linear combination of 3×3 determinants following the approach of the main text.

Solution The factor d_{ij} of Eq. 10.56 is a function of $\mathbf{x}$, and it is equally expanded in terms of the complete basis set $\{\phi_k\}$: $d_{ij}(\mathbf{x}) = \sum_k \alpha_{ijk}\phi_k(\mathbf{x})$, where α_{ijk} are the expansion coefficients of d_{ij}.

$$\begin{aligned} &\sum_{i<j} d_{ij}(\mathbf{x}_3)\,[\phi_i(\mathbf{x}_1)\phi_j(\mathbf{x}_2) - \phi_i(\mathbf{x}_2)\phi_j(\mathbf{x}_1)] \\ &\quad = \sum_{i<j,k} \alpha_{ijk}\phi_k(\mathbf{x}_3)\,[\phi_i(\mathbf{x}_1)\phi_j(\mathbf{x}_2) - \phi_i(\mathbf{x}_2)\phi_j(\mathbf{x}_1)] \end{aligned}$$

The permutation of the variables $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ should be considered in the expression of ψ to favor no single variable.

$$\begin{aligned} &\psi(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \\ &= \sum_{i<j,k} \alpha_{ijk}[\phi_i(\mathbf{x}_1)\phi_j(\mathbf{x}_2)\phi_k(\mathbf{x}_3) - \phi_i(\mathbf{x}_2)\phi_j(\mathbf{x}_1)\phi_k(\mathbf{x}_3)] + \quad \leftarrow \quad aei - ibd \\ &+ \sum_{i<j,k} \alpha_{kij}[\phi_i(\mathbf{x}_2)\phi_j(\mathbf{x}_3)\phi_k(\mathbf{x}_1) - \phi_i(\mathbf{x}_3)\phi_j(\mathbf{x}_2)\phi_k(\mathbf{x}_1)] + \quad \leftarrow \quad gbf - ceg \\ &+ \sum_{i<j,k} \alpha_{jki}[\phi_i(\mathbf{x}_3)\phi_j(\mathbf{x}_1)\phi_k(\mathbf{x}_2) - \phi_i(\mathbf{x}_1)\phi_j(\mathbf{x}_3)\phi_k(\mathbf{x}_2)] \quad \leftarrow \quad dhc - fha \end{aligned}$$

(continued)

In order to satisfy Pauli's exclusion principle, we separately do $\mathbf{x}_1 = \mathbf{x}_2 = \mathbf{x}$, $\mathbf{x}_1 = \mathbf{x}_3 = \mathbf{x}$, and $\mathbf{x}_2 = \mathbf{x}_3 = \mathbf{x}$, reaching the conclusion $\alpha_{ijk} = \alpha_{kij} = \alpha_{jki}$. The above elements of ψ are identified with the elements of a determinant.

$$aei + dhc + gbf - ceg - fha - ibd = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

Thereby, ψ is written as a linear combination of 3×3 determinants.

$$\psi(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = \sum_{i<j<k} \alpha_{ijk} \begin{vmatrix} \phi_i(\mathbf{x}_1) & \phi_i(\mathbf{x}_2) & \phi_i(\mathbf{x}_3) \\ \phi_j(\mathbf{x}_1) & \phi_j(\mathbf{x}_2) & \phi_j(\mathbf{x}_3) \\ \phi_k(\mathbf{x}_1) & \phi_k(\mathbf{x}_2) & \phi_k(\mathbf{x}_3) \end{vmatrix}$$

The possibility of dealing with identical rows or identical columns in every determinant is avoided with the restriction $i < j < k$ on the subindexes.

10.10 Variational Principle of the Energy

In this section we demonstrate that E_t, the energy associated with an arbitrary trial wave function ψ_t, is an upper bound on the ground-state energy E_0 of the system. If the trial wave function is solution of the Hamiltonian $\hat{\mathcal{H}}$, then ψ_t satisfies the time-independent equation.

$$\langle \psi_t | \hat{\mathcal{H}} | \psi_t \rangle = E_t \langle \psi_t | \psi_t \rangle \tag{10.57}$$

The goal is to compare the energy E_t with the ground-state energy E_0. Firstly, we take an eigenvalue of $\hat{\mathcal{H}}$, say E_k, as a reference energy. The eigenfunction ψ_k has eigenvalue E_k.

$$\hat{\mathcal{H}} \psi_k = E_k \psi_k \tag{10.58}$$

The eigenfunctions of $\hat{\mathcal{H}}$ satisfy the orthonormality condition $\langle \psi_k | \psi_l \rangle = \delta_{kl}$. We compute the energy difference between E_t and E_k to work with the deviation originated by the use of the trial wave function ψ_t.

$$E_t - E_k = [\langle \psi_t | \hat{\mathcal{H}} | \psi_t \rangle - \langle \psi_t | \psi_t \rangle E_k] \,/\, \langle \psi_t | \psi_t \rangle = [\langle \psi_t | \hat{\mathcal{H}} - E_k | \psi_t \rangle] \,/\, \langle \psi_t | \psi_t \rangle$$

When the trial function ψ_t deviates little from the eigenfunction ψ_k, the relation between ψ_t and ψ_k is written in linear form $\psi_t = \psi_k + \delta\psi$.

$$\begin{aligned} E_t - E_k &= \langle \psi_k + \delta\psi | \hat{\mathcal{H}} - E_k | \psi_k + \delta\psi \rangle \, / \, \langle \psi_t | \psi_t \rangle \\ &= [\langle \psi_k | \hat{\mathcal{H}} - E_k | \psi_k \rangle + \langle \psi_k | \hat{\mathcal{H}} - E_k | \delta\psi \rangle + \langle \delta\psi | \hat{\mathcal{H}} - E_k | \psi_k \rangle \\ &\quad + \langle \delta\psi | \hat{\mathcal{H}} - E_k | \delta\psi \rangle] \, / \, \langle \psi_t | \psi_t \rangle \end{aligned} \tag{10.59}$$

The first term in the numerator is zero because ψ_k is an eigenfunction of $\hat{\mathcal{H}}$, Eq. 10.58, and $\hat{\mathcal{H}}$ is Hermitian. The second and third terms also vanish. The last term involves the second-order contribution, which is non-zero.

$$E_t - E_k = \langle \delta\psi | \hat{\mathcal{H}} - E_k | \delta\psi \rangle \, / \, \langle \psi_t | \psi_t \rangle$$

The energy difference in the use of the trial wave function ψ_t and the wave function ψ_k depends on the term $\delta\psi$. If we wish to minimize the energy difference $E_t - E_k$, which we denote with δE, then we are required to impose the condition:

$$\delta E = 0$$

The way to obtain $\delta E = 0$ is by tuning the trial wave function, because δE depends on the difference $\delta\psi = \psi_t - \psi_k$ and, in turn, the trial wave function ψ_t depends on parameters, which can be in principle adjusted to produce $\delta E = 0$. The condition $\delta E = 0$ unfolds a *variational principle* as it imposes the condition to find the best trial wave function that minimizes the energy.

It is our next purpose to demonstrate that the trial energy E_t is an upper bound on the ground-state energy E_0. To do this, it is convenient to expand ψ_t in terms of the eigenfunctions ψ_k of the Hamiltonian.

$$\psi_t = \textstyle\sum_k c_k \psi_k$$

The energy corresponding to this function is obtained from the expectation value of the Hamiltonian $\hat{\mathcal{H}}$.

$$\langle \psi_t | \hat{\mathcal{H}} | \psi_t \rangle = \sum_{k,m} c_k^* c_m \langle \psi_k | \hat{\mathcal{H}} | \psi_m \rangle = \sum_{k,m} c_k^* c_m E_m \langle \psi_k | \psi_m \rangle = \sum_k c_k^2 E_k$$

If we replace $\hat{\mathcal{H}}$ by the identity operator $\hat{I}$ and perform the same operations, we immediately obtain the orthonormality condition in terms of the c_k coefficients.

$$\langle \psi_t | \psi_t \rangle = \sum_k c_k^2 = 1 \tag{10.60}$$

We still require to show that E_t is equal to or greater than the ground-state energy E_0. Since E_0 represents the lowest value among all possible eigenvalues E_k, then:

$$E_t = \sum_k c_k^2 E_k \geq E_0 \sum_k c_k^2 \geq E_0$$

The trial energy is an upper bound on the ground-state energy.

$$E_t \geq E_0 \tag{10.61}$$

The variational principle favors the trial wave function that gives the lowest energy value among many other wave functions with higher energy values of the particle system, because the trial wave function that exhibits the lowest energy is the wave function that better approximates the ground state of the system. The electronic wave function expressed as a single Slater determinant of the spin orbitals, in conjunction with the variational principle of the energy, gives foundation to the Hartree-Fock method. Other Hartree-Fock based methods deal with many Slater determinants and give foundation to the multi-configuration methods. Yet, they all use the variational principle to find the lowest energy of the electronic system [10].

A.1 Appendix: Proposing the Wave Equation for Matter Waves

A particle in the quantum-mechanical world has dual nature and may behave as a wave or a particle according to the circumstances. In this regard, it is necessary to establish an equation that accurately describes both possible aspects of the particle in the quantum realm. Specifically, we need to find an equation that governs the behavior not only of particles but also of *matter waves*. To accomplish this, we start with the one-dimensional equation that rules the behavior of waves in the classical world.

$$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y(x,t)}{\partial t^2} \tag{10.62}$$

This expression rules the behavior of the wave described by y, traveling with speed v along the x direction, and with dependence on time t. The changes of y in the positions are related to changes of y with times. One of the simplest solutions of the wave equation is:

$$y(x,t) = A\cos(kx - wt) \quad ; \quad k = 2\pi/\lambda \quad ; \quad \omega = 2\pi f \tag{10.63}$$

The terms A, λ and f are the amplitude, wavelength, and frequency of the wave, respectively. By inserting the trigonometric solution in Eq. 10.62, we obtain the relation $\omega = kv$ or $v = \lambda f$ between the parameters of the classical wave. There is consistency between the classical wave equation and both the traveling wave and the wave parameters, which all together establish the possibility to characterize an

entity as a wave. Still, since a particle has dual character, the equation of matter waves is expected to also include mechanical parameters to characterize the particle as a mechanical object.

The behavior of the mechanical particle is defined by the particle's energy E, which is integrated by the kinetic energy $E_k = p^2/(2m)$ and the potential energy $V(x,t)$, where the particle is immersed. In this respect, the equation governing the particle behavior should contain information of the linear momentum p, the potential energy V, and the energy E. However, before formulating the wave equation of matter waves, it is important to recall the de Broglie relations linked to the linear momentum and energy:

$$\begin{aligned} p = h/\lambda \quad &; \quad E = hf \\ p = \hbar k \quad &; \quad E = \hbar\omega \end{aligned} \tag{10.64}$$

The identities $\hbar = h/(2\pi)$ and $k = 2\pi/\lambda$ were used. The particle's energy is written in terms of the wave number k and angular frequency ω.

$$\begin{aligned} E &= p^2/(2m) + V(x,t) \\ &= \hbar^2 k^2/(2m) + V(x,t) = \hbar\omega \end{aligned} \tag{10.65}$$

In the following we consider a plane wave as given by a complex exponential function. Such a function considers the complex number i and is relatively different to the trigonometric wave 10.63.

$$y_p(x,t) = \exp[i(kx - \omega t)] \tag{10.66}$$

The quantum wave equation is expected to contain some ingredients of the classical wave expression, Eq. 10.62, like the differentials of the matter-wave function with respect to the position and time because they provide the correlated changes of the wave in space and time. Thereby, we differentiate the plane wave with respect to the position and time.

$$\begin{gathered} \frac{\partial}{\partial x} y_p(x,t) = ik \exp[i(kx - \omega t)] \quad ; \quad \frac{\partial^2}{\partial x^2} y_p(x,t) = -k^2 \exp[i(kx - \omega t)] \\ \frac{\partial}{\partial t} y_p(x,t) = -i\omega \exp[i(kx - \omega t)] \end{gathered} \tag{10.67}$$

In order to reproduce the energy 10.65 in terms of the above derivatives, we link $p = \hbar k$ and $E = \hbar\omega$, Eqs. 10.64, to the position and time derivatives.

$$\frac{-\hbar^2}{2m} \frac{\partial^2 y_p}{\partial x^2} = (\hbar k)^2 \, y_p \,/\, (2m) \quad ; \quad i\hbar \frac{\partial}{\partial t} y_p(x,t) = \hbar\omega \, y_p \tag{10.68}$$

We now have the possibility of proposing the equation for matter waves. The energy 10.65 is $E = \hbar^2 k^2/(2m) + V(x,t) = \hbar\omega$. It can be written in terms of the position and time derivatives.

$$i\hbar \frac{\partial}{\partial t} y_p(x,t) = \frac{-\hbar^2}{2m} \frac{\partial^2 y_p(x,t)}{\partial x^2} + V(x,t) y_p(x,t) \tag{10.69}$$

This is the non-relativistic $1D$ wave equation of matter waves. It is different to the classical equation 10.62 as a single time derivative is considered in Eq. 10.69 while a double time derivative is considered in Eq. 10.62, thus breaking the symmetry between spatial and time derivatives, and resembling a kind of diffusion expression. Equation 10.69 also includes the imaginary number, i. It is common to represent the wave function with the Greek letter ψ instead of y_p. The corresponding expression in $3D$ space for many particles is:

$$i\hbar \tfrac{\partial}{\partial t} \psi(\{\mathbf{r}_i\},t) = \textstyle\sum_i \frac{-\hbar^2}{2m} \nabla_i^2 \psi(\{\mathbf{r}_i\},t) + V(\{\mathbf{r}_i\},t)\, \psi(\{\mathbf{r}_i\},t) \tag{10.70}$$

The equation of matter waves is an equation of mathematical operators acting on ψ. The operators are engulfed by the Hamiltonian operator, $\hat{\mathcal{H}}$.

$$i\hbar \frac{\partial}{\partial t} \psi(\{\mathbf{r}_i\},t) = \hat{\mathcal{H}} \psi(\{\mathbf{r}_i\},t) \quad ; \quad \hat{\mathcal{H}} = \sum_i \frac{-\hbar^2}{2m} \nabla_i^2 + V(\{\mathbf{r}_i\},t) \tag{10.71}$$

The equation of matter waves satisfies the principle of superposition, since the linear combination of solutions ψ_i of the matter-wave equation is also a solution of that equation.

$$\psi(\{\mathbf{r}_k\},t) = \sum_{i=1}^{n} c_i \psi_i(\{\mathbf{r}_k\},t) \tag{10.72}$$

The superposition principle is required to have mathematical well-behaved solutions, capable to unfold the Hilbert space associated with the system, in addition to make possible the dispersion and interference phenomena of waves in the physical world.

B.1 Appendix: Expansion of a Determinantal Wave Function

From a mathematical point of view, a determinant is an abbreviated description of a sum of products of the determinantal elements without repetition. The addition of determinants also represents a summation of products, where all the elements of the determinants are considered under equal circumstances. The expansion of a Slater determinant allows to write the electronic wave function in an alternative form and demonstrate its properties. We use the *Laplace expansion approach* to

expand the Slater determinant [11]. According to this procedure, a set of smaller determinants are produced after systematically removing rows and columns of the original determinant, which is assumed to have size $n \times n$. Let us consider an element of the determinant, say a_{ij}, where the i and j indexes represent a row and a column of the determinant. A set of smaller determinants with dimension $n-1$ are produced after deleting selected rows and selected columns of the original determinant. In the example provided below, a determinant of size 3×3 is expanded in terms of three determinants of size 2×2. The expansion coefficients are the elements (2,4,6) that integrate the second column of the original determinant.

$$\begin{vmatrix} 1 & 2 & 7 \\ 3 & 4 & 8 \\ 5 & 6 & 11 \end{vmatrix} = -2 \begin{vmatrix} 3 & 8 \\ 5 & 11 \end{vmatrix} + 4 \begin{vmatrix} 1 & 7 \\ 5 & 11 \end{vmatrix} - 6 \begin{vmatrix} 1 & 7 \\ 3 & 8 \end{vmatrix} = -4$$

The signs of the expansion coefficients are determined from a checkerboard pattern whose elements are given in the general form $(-1)^{i+j}$, where i is the row and j the column of the element a_{ij}. The checkerboard pattern of a determinant of dimension 3 alternates signs.

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix} = \begin{pmatrix} (-1)^{1+1} & (-1)^{1+2} & (-1)^{1+3} \\ (-1)^{2+1} & (-1)^{2+2} & (-1)^{2+3} \\ (-1)^{3+1} & (-1)^{3+2} & (-1)^{3+3} \end{pmatrix} \tag{10.73}$$

The Laplace expansion may be performed using the elements of a row instead of the elements of a column. Such an alternative is illustrated using our example with the same checkerboard pattern of signs and taking the elements (1,2,7) that integrate the first row of the original determinant as the expansion coefficients.

$$\begin{vmatrix} 1 & 2 & 7 \\ 3 & 4 & 8 \\ 5 & 6 & 11 \end{vmatrix} = 1 \begin{vmatrix} 4 & 8 \\ 6 & 11 \end{vmatrix} - 2 \begin{vmatrix} 3 & 8 \\ 5 & 11 \end{vmatrix} + 7 \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} = -4$$

When a column of the determinant is exchanged with a neighbor column, the sign of the determinant should be changed according to the previous checkerboard. For instance, after exchanging the first and second columns, we have the antisymmetric property of the determinant.

$$\begin{vmatrix} 1 & 2 & 7 \\ 3 & 4 & 8 \\ 5 & 6 & 11 \end{vmatrix} = - \begin{vmatrix} 2 & 1 & 7 \\ 4 & 3 & 8 \\ 6 & 5 & 11 \end{vmatrix}$$

If two columns of the determinant are repeated, for example, the column (1,3,5) is replaced by the column (2,4,6) in the numeric example, we in principle have two

fermions populating the same spin orbital, and the determinant is zero, satisfying the exclusion principle.

$$\begin{vmatrix} 2 & 2 & 7 \\ 4 & 4 & 8 \\ 6 & 6 & 11 \end{vmatrix} = 2 \begin{vmatrix} 4 & 8 \\ 6 & 11 \end{vmatrix} - 2 \begin{vmatrix} 4 & 8 \\ 6 & 11 \end{vmatrix} + 7 \begin{vmatrix} 4 & 4 \\ 6 & 6 \end{vmatrix} = 0$$

The small determinants resulting from the previous expansions are reduced determinants, known as *minors*. The Laplace expansion has general character and is applied to determinants of any dimension. The general expansion in terms of the ith row of a determinant A has the form:

$$\det A = (-1)^{i+1} a_{i1} \det A_{i1} + \cdots + (-1)^{i+n} a_{in} \det A_{in} \tag{10.74}$$

And the expansion in terms of the jth column of a determinant A is:

$$\det A = (-1)^{1+j} a_{1j} \det A_{1j} + \cdots + (-1)^{n+j} a_{nj} \det A_{nj} \tag{10.75}$$

The indexes of the coefficients $a_{i1}, \cdots, a_{in}$ in Eq. 10.74 and the indexes of the coefficients $a_{1j}, \cdots, a_{nj}$ in Eq. 10.75 indicate the rows and columns that should be omitted from the original determinant A to build the minors of dimension $(n-1)$. Thus, the identities 10.74 and 10.75 show an expansion of the original determinant in terms of determinants with reduced dimension. The Laplace expansion may be again applied to the minors to achieve further reduction of the determinants. The recursive expansion of the minors leads to a final expression where only sums of products are shown (with no presence of a determinant).

The computational cost of the Laplace expansion is $n!$, where n is the dimension of the original determinant. The Laplace expansion is not an efficient technique; nevertheless, the efficiency may be improved by carrying expansions in terms of rows or columns that contain zeros (since a zero multiplied by its respective minor gives a null contribution and simplifies the expansion process). The Laplace expansion is used to obtain the matrix elements of operators representing the physical observables:

$$\mathcal{A} = \langle \phi_1(\mathbf{x}_1) \cdots \phi_n(\mathbf{x}_n) | \, \hat{A} \, | \phi_1(\mathbf{x}_1) \cdots \phi_n(\mathbf{x}_n) \rangle \tag{10.76}$$

where $\hat{A}$ is the operator associated with the physical observable $\mathcal{A}$ and where we have used the bra-ket notation, Eq. 10.54, to describe the determinantal wave function of the particle system. In conclusion, the Laplace expansion shows that a Slater determinant can be represented as a summation of smaller determinants, and such a summation satisfies Pauli's exclusion principle.

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Part III

First-Principles Molecular Dynamics



Dynamics of Electrons and Nuclei

11

Quantum mechanics is a formulation of general nature created to describe the microscopic world. It carries the attribute of quantum to distinguish it from classical mechanics, which in principle is understood as a sub-branch of quantum mechanics. At the microscopic scale, the particles have probabilistic manifestation, and it is not possible to assign well-defined positions and momenta. On the other side, at the macroscopic level, the particles have well-defined positions and momenta and exhibit trajectories with time. The energy at the atomic scale is quantized, but has continuous character at the macroscopic level. The quantum particles display features like spin, non-local entanglement and can do tunneling through high energy barriers, among other exotic features, but these features fade away at macroscopic dimensions. The principles of quantum mechanics establish the behavior of the particles and should be applied at the microscopic domain.[1]

There is a universal time-dependent wave equation in quantum mechanics that should be solved to determine the wave function in terms of the time and, in turn, obtain the time evolution of the system of particles. Yet, the time-dependent wave equation is a complex equation that defies exact solution, especially for real systems. In this respect, we need to introduce approximations to achieve the solution. The purpose of this chapter is to present a relatively general and flexible method to simplify the time-dependent wave equation of a system of atoms. We depart from the fact that most complications in the process of solving the time-dependent wave equation are attributed to the complexity of the Hamiltonian. Thus, we analyze the nature of the molecular Hamiltonian and proceed to uncouple the dynamics of the electrons from that of the nuclei by considering physical properties such as the individual mobility of the particles. In this enterprise, we reach a time-dependent wave equation for the electrons, and another for the nuclei. The equations maintain a correlation between the two types of particles in an average manner, in principle

[1] The words "macroscopic" and "microscopic" have relative meanings in our discussion, but they shall be defined in the proper context later.

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facilitating the finding of the solution, which is now defined as the product of the nuclear wave function and the electronic wave function.

11.1 The Electronic and Nuclear Dynamics Are Coupled

The behavior of the electrons is correlated with that of the nuclei due to the crossed interactions between the two sets of particles. The crossed interactions lead to the formation of atoms and molecules; however, they complicate the quantum mechanical description of the particle system. In the next sections, we follow the work of Tully and Marx on the treatment of the electrons and nuclei [1]. The approximations to simplify the quantum mechanical description of the molecular system permit a relative separation of the electron dynamics from the nuclear dynamics, resulting in a time-dependent wave equation for the electrons and another for the nuclei. The equations maintain an average correlation between the electrons and nuclei, and their solutions determine the electronic wave function, ψ_{elec}, and the nuclear wave function, ψ_{nuc}. Thus, the mathematical complexity in solving a single wave equation for the two types of particles forming the molecular system is reduced to two simpler equations. The wave function Ψ of the system is the product $\psi_{nuc}\psi_{elec}$, describing both types of particles at equal level of theory. Depending on the molecular system and the conditions imposed, the approach presented in this chapter may oversimplify the correlations between the electrons and nuclei. Thereby, for strongly correlated systems, the wave equation that portrays the behavior of the fully coupled system of particles must be solved. Currently, this topic represents a challenge to understand the behavior of particles and is beyond the scope of this work [2].

11.2 Molecular Hamiltonian

The time-dependent wave equation is the fundamental expression that rules the time evolution of molecular systems. It is stated as a differential equation on the wave function Ψ over time.

$$i\hbar \frac{\partial \Psi\left(\{\mathbf{r}_i\}_{i=1}^{n}, \{\mathbf{R}_\alpha\}_{\alpha=1}^{N}; t\right)}{\partial t} = \hat{\mathcal{H}}\Psi\left(\{\mathbf{r}_i\}_{i=1}^{n}, \{\mathbf{R}_\alpha\}_{\alpha=1}^{N}; t\right) \tag{11.1}$$

The $\mathbf{r}_i$ and $\mathbf{R}_\alpha$ symbols represent the electron and nuclear coordinates, and t is the time. In principle, there is no need to differentiate between nuclear and electronic coordinates, because they all have equal hierarchy; nevertheless, we anticipate such a nomenclature for the next sections. The spin variables of electrons and nuclei may be equally considered in the nomenclature of the coordinates (in this case, e.g., the variable $\mathbf{r}_i$ also includes the spin of the ith particle). The Hamiltonian $\hat{\mathcal{H}}$ specifies the number and types of particles participating in the system and introduces the kinds of interactions among the particles. For the case of a molecule constituted by electrons and nuclei subjected to no external forces, the non-relativistic Hamiltonian

contains the nuclear and electronic kinetic energies, and the electron-electron, nucleus-electron, and nucleus-nucleus Coulombic interactions. By assuming n electrons with mass m_e and charge e, and N nuclei with mass M_α and charge $Z_\alpha e$, the Hamiltonian terms in the previous specific order are:

$$\begin{aligned}\hat{\mathcal{H}} = &-\sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \nabla_\alpha^2 - \sum_{i=1}^{n} \frac{\hbar^2}{2m_e} \nabla_i^2 + \sum_{i=1}^{n} \sum_{j>i} \frac{e^2}{r_{ij}} \\ &- \sum_{\alpha=1}^{N} \sum_{i=1}^{n} \frac{Z_\alpha e^2}{r_{i\alpha}} + \sum_{\alpha=1}^{N} \sum_{\beta>\alpha} \frac{Z_\alpha Z_\beta}{R_{\alpha\beta}} e^2\end{aligned} \tag{11.2}$$

The electron-electron, electron-nucleus, and nucleus-nucleus interparticle distances are:

$$r_{ij} = |\mathbf{r}_i - \mathbf{r}_j| \quad ; \quad r_{i\alpha} = |\mathbf{r}_i - \mathbf{R}_\alpha| \quad ; \quad R_{\alpha\beta} = |\mathbf{R}_\alpha - \mathbf{R}_\beta| \tag{11.3}$$

The time-dependent wave equation 11.1 applies at any instant of time and depends on the Hamiltonian because it contains the interactions that rule the particle dynamics.

Problem 11.1 (i) Build the Hamiltonian of formaldehyde ($O = CH_2$) and display the different sets of terms. (ii) Determine the number of Hamiltonian terms of a protein with N nuclei and n electrons, when N is large. (iii) A computer is capable to deal with m^2 particle interactions in a period of time ΔT. According to *Moore's law*, every 18 months the computer power is duplicated. Determine the number of particles that the next generation computer may handle in the same period of time.

Solution

(i) The masses and electric charges of oxygen, carbon, and hydrogen carry the subindexes O, C, and H. The Hamiltonian consists of many sets of terms:

$$\begin{aligned}\hat{\mathcal{H}} = &-\frac{\hbar^2}{2M_O} \nabla_O^2 - \frac{\hbar^2}{2M_C} \nabla_C^2 - \frac{\hbar^2}{2M_{H1}} \nabla_{H1}^2 - \frac{\hbar^2}{2M_{H2}} \nabla_{H2}^2 - \sum_{i=1}^{16} \frac{\hbar^2}{2m_e} \nabla_i^2 \\ &+ \sum_{i=1}^{16} \sum_{j>i}^{16} \frac{e^2}{r_{ij}} - \sum_{i=1}^{16} \frac{Z_O e^2}{|\mathbf{r}_i - \mathbf{R}_O|} - \sum_{i=1}^{16} \frac{Z_C e^2}{|\mathbf{r}_i - \mathbf{R}_C|} - \sum_{i=1}^{16} \frac{Z_{H1} e^2}{|\mathbf{r}_i - \mathbf{R}_{H1}|}\end{aligned}$$

(continued)

$$
\begin{aligned}
&- \sum_{i=1}^{16} \frac{Z_{H2} e^2}{|\mathbf{r}_i - \mathbf{R}_{H2}|} \\
&+ \frac{Z_O Z_C e^2}{R_{OC}} + \frac{Z_O Z_{H1} e^2}{R_{OH1}} + \frac{Z_O Z_{H2} e^2}{R_{OH2}} + \frac{Z_C Z_{H1} e^2}{R_{CH1}} \\
&+ \frac{Z_C Z_{H2} e^2}{R_{CH2}} + \frac{Z_{H1} Z_{H2} e^2}{R_{H1H2}}
\end{aligned}
$$

(ii) The protein has N nuclei and n electrons. The respective Hamiltonian, Eq. 11.2, has $N + n + n(n-1)/2 + N(N-1)/2 + nN = N(N+1)/2 + n(n+1)/2 + nN$ terms. When N is large, n is also a big number, and the number of terms in the Hamiltonian scales quadratically with the total number of particles, $\sim (n+N)^2/2$. This shows the complexity to obtain Ψ, a function of the $3(n+N)$ spatial variables plus time.

(iii) We use the symbol cpu_1 for the present computer and cpu_2 for the next generation computer. From Moore's law, the power of cpu_2 is $2\,cpu_1$. The number of interactions calculated in ΔT with the new computer is $2m^2 = (\sqrt{2}m)^2$. If we take $n = \sqrt{2}m$, then the number of particles that the new computer may deal with is $n = 1.414\,m$. We require one further generation, cpu_3, to double the initial number of particles ($1.414^2 = 2$). The efficiency can be improved with several computers working in parallel ($cpu_{2A}, cpu_{2B}, \cdots, cpu_{2N}$) or employing novel computing hardware ($gpu_A, gpu_B, \cdots, gpu_N$). The scaling on the number of particles with computer power is more complex than our discussion.

11.3 Approximating the Total Wave Function

In this section we introduce a wave function that simplifies the time-dependent wave equation of the molecule.

$$
i\hbar \, \frac{\partial \Psi(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\}; t)}{\partial t} = \hat{\mathcal{H}} \, \Psi \left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\}; t\right) \tag{11.4}
$$

The correlations between electrons and nuclei are due to the term $-\sum_\alpha^N \sum_i^n Z_\alpha e^2 / r_{i\alpha}$, which is present in the Hamiltonian as it accounts for the electron-nucleus Coulomb interactions. However, the crossed Coulombic interactions complicate the finding of the solution of the wave equation. It is convenient to separate the kinetic energy terms from the potential energy terms in the Hamiltonian.

$$\begin{gathered}\hat{\mathcal{H}} = -\sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \nabla_\alpha^2 - \sum_{i=1}^{n} \frac{\hbar^2}{2m_e} \nabla_i^2 + \hat{V}\left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\}\right) \\ \hat{V}\left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\}\right) = \hat{V}_{ee} + \hat{V}_{Ze} + \hat{V}_{ZZ} \\ \hat{V}_{ee} = \sum_{i=1}^{n} \sum_{j>i} \frac{e^2}{r_{ij}} \quad ; \quad \hat{V}_{Ze} = -\sum_{\alpha=1}^{N} \sum_{i=1}^{n} \frac{Z_\alpha e^2}{r_{i\alpha}} \\ \hat{V}_{ZZ} = \sum_{\alpha=1}^{N} \sum_{\beta>\alpha} \frac{Z_\alpha Z_\beta}{R_{\alpha\beta}} e^2\end{gathered} \tag{11.5}$$

It is important to recall the form in which a wave function is evolved in quantum mechanics with time, $\Psi(t) = \hat{\mathcal{U}}\Psi(t_0)$. The time evolution operator $\hat{\mathcal{U}}$ takes one of the forms:

$$\hat{\mathcal{U}}(t, t_0) = \begin{cases} \hat{\mathcal{T}} \exp\left[-\frac{i}{\hbar}\int_{t_0}^{t} dt' \hat{\mathcal{H}}(t')\right] \hat{\mathcal{H}} \text{ does not commute at different times} \\ \exp\left[-\frac{i}{\hbar}\int_{t_0}^{t} dt' \hat{\mathcal{H}}(t')\right] \hat{\mathcal{H}} \text{ commutes at different times} \\ \exp\left[-\frac{i}{\hbar}(t - t_0)\,\hat{\mathcal{H}}\right] \hat{\mathcal{H}} \text{ is time independent} \end{cases} \tag{11.6}$$

$\hat{\mathcal{T}}$ is the time-ordering operator, and the Hamiltonian may or may not commute with itself at different instants of time. The function $\Psi(t) = \hat{\mathcal{U}}\Psi(t_0)$ represents a solution of the time-dependent wave equation when the time evolution operator $\hat{\mathcal{U}}$ is known. Nevertheless, it is of little help in our case due to the mathematical complications in obtaining $\hat{\mathcal{U}}$.

Problem 11.2 Suppose that a Hamiltonian, $\hat{\mathcal{H}}$, is constant during short time intervals, $[t_0, t_1]$, $[t_1, t_2]$, $\cdots$, $[t_{n-1}, t_n]$, and commutes at different times, $[\hat{\mathcal{H}}(t_i), \hat{\mathcal{H}}(t_j)] = 0$. Show that $\hat{\mathcal{U}}(t, t_0)\psi(t_0)$ satisfies the wave equation.

Solution The expression of $\hat{\mathcal{U}}$ is given in terms of an integral of the Hamiltonian over time. The integration from t_0 to $t = t_n$ is computed by intervals.

$$\begin{aligned}\int_{t_0}^{t} dt' \hat{\mathcal{H}}(t') &= \int_{t_0}^{t_1} dt' \hat{\mathcal{H}}(t') + \int_{t_1}^{t_2} dt' \hat{\mathcal{H}}(t') + \cdots + \int_{t_{n-1}}^{t_n} dt' \hat{\mathcal{H}}(t') \\ &= \hat{\mathcal{H}}(t_1)\Delta t_1 + \hat{\mathcal{H}}(t_2)\Delta t_2 + \cdots + \hat{\mathcal{H}}(t_n)\Delta t_n = \sum_{i=1}^{n} \hat{\mathcal{H}}(t_i)\Delta t_i\end{aligned}$$

Where it was used $t = t_n$. The time evolution operator $\hat{\mathcal{U}}$ adopts a new form based on the constant nature of the Hamiltonian during short time intervals.

$$\hat{\mathcal{U}}(t, t_0) = \exp\left[-\frac{i}{\hbar}\int_{t_0}^{t} dt' \hat{\mathcal{H}}(t')\right] = \exp\left[-\frac{i}{\hbar}\sum_{i=1}^{n} \hat{\mathcal{H}}(t_i)\Delta t_i\right]$$

(continued)

We require to show that $i\hbar\,\partial[\hat{\mathcal{U}}(t,t_0)\psi(t_0)]/\partial t$ is equal to $\hat{\mathcal{H}}\hat{\mathcal{U}}(t,t_0)\psi(t_0)$. We start by inserting $\hat{\mathcal{U}}(t,t_0)$ in $i\hbar\,\partial[\hat{\mathcal{U}}(t,t_0)\psi(t_0)]/\partial t$ and recalling that $\exp(x)=\sum_k x^k/k!$.

$$i\hbar\frac{\partial}{\partial t}\hat{\mathcal{U}}(t,t_0)\psi(t_0)=i\hbar\sum_{k=0}^{\infty}\frac{1}{k!}\left[-\frac{i}{\hbar}\sum_{i=1}^{n}\hat{\mathcal{H}}(t_i)\Delta t_i\right]^k\left(-\frac{i}{\hbar}\right)\hat{\mathcal{H}}(t)\psi(t_0)$$

The Hamiltonian evaluated at different times commutes with itself, and the previous power series is again identified with an exponential function, then:

$$i\hbar\frac{\partial}{\partial t}\hat{\mathcal{U}}(t,t_0)\psi(t_0)=\hat{\mathcal{H}}(t)\,\exp\left[-\frac{i}{\hbar}\sum_{i=1}^{n}\hat{\mathcal{H}}(t_i)\Delta t_i\right]\psi(t_0)=\hat{\mathcal{H}}\hat{\mathcal{U}}(t,t_0)\psi(t_0)$$

Thereby, the expression $\hat{\mathcal{U}}(t,t_0)\psi(t_0)$ satisfies the wave equation.

A molecule consists of atoms, where an atom is understood in our description as a particle formed by electrons and a nucleus, and such a nucleus is supposed to have no internal structure. The total wave function is approximated by the product of an electronic wave function, ψ_{elec}, and a nuclear wave function, ψ_{nuc}. This ansatz on the wave function also considers an exponential function in terms of the energy, E', and the time, t.

$$\begin{aligned}\Psi\left(\{\mathbf{r}_i\},\{\mathbf{R}_\alpha\};t\right)&=A(t)\,\psi_{elec}(\{\mathbf{r}_i\};\{\mathbf{R}_\alpha\},t)\,\psi_{nuc}\left(\{\mathbf{R}_\alpha\};\{\mathbf{r}_i\},t\right)\\ A(t)&=\exp\left[\tfrac{i}{\hbar}\int_{t_0}^{t}E'\left(t'\right)dt'\right]\end{aligned}\tag{11.7}$$

The electronic wave function ψ_{elec} depends on the nuclear coordinates in parametric form, and the nuclear wave function ψ_{nuc} depends on the electron coordinates in parametric form.[2] The electronic and nuclear functions also include the time. The amplitude A introduces an exponential in terms of E' and t, instead of the Hamiltonian operator $\hat{\mathcal{H}}$ which is present in $\hat{\mathcal{U}}$, Eq. 11.6. Thus, the complications

[2] The concepts of variable and parameter apparently play similar roles, but there is a subtle difference between them in mathematics. A variable is an entity which can take different numerical values after an equation is defined. On the other hand, the parameters take constant values in an equation, yet they are not constants with perpetual values, like the number π. A parameter defines a family or class of equations, or equivalently, it is an entity that reduces a general equation to a specific one. The usefulness of a parameter in an equation is related to the connection of variables, allowing the exchange of information.

related to the time evolution operator acting on the total wave function are avoided with the incorporation of an exponential factor, which only considers average effects through the introduction of E'.

$$E'(t) = \int \psi^*_{nuc}\psi^*_{elec}\hat{\mathcal{H}}_{elec}\,\psi_{elec}\,\psi_{nuc}\,d\mathbf{r}\,d\mathbf{R} \tag{11.8}$$

The symbols $d\mathbf{r}$ and $d\mathbf{R}$ abbreviate the products of the infinitesimally small volumes $d^3r_1 \cdots\ d^3r_n$ and $d^3R_1 \cdots d^3R_N$, for the electronic, $\mathbf{r}_i$, and nuclear variables, $\mathbf{R}_\alpha$. The expression of the energy $E'(t)$ is originated from the electronic Hamiltonian operator.

$$\hat{\mathcal{H}}_{elec} = -\sum_{i=1}^{n}\frac{\hbar^2}{2m_e}\nabla_i^2 + \sum_{i=1}^{n}\sum_{j>i}\frac{e^2}{r_{ij}} - \sum_{\alpha=1}^{N}\sum_{i=1}^{n}\frac{Z_\alpha e^2}{r_{\alpha i}} + \sum_{\alpha=1}^{N}\sum_{\beta>\alpha}\frac{Z_\alpha Z_\beta}{R_{\alpha\beta}}\,e^2 \tag{11.9}$$

The nucleus-nucleus repulsion term is included in the electronic Hamiltonian operator to maintain the electrostatic stability, because without the nuclear repulsions in the electronic Hamiltonian, the nuclei may coalesce. The electronic and nuclear wave functions are mathematically well-behaved, as they are quadratically integrable and normalized at every instant of time.

$$\int \psi^*_{elec}\psi_{elec}\,d\mathbf{r} = 1 \quad ; \quad \int \psi^*_{nuc}\psi_{nuc}\,d\mathbf{R} = 1 \tag{11.10}$$

After inserting the total wave function $\Psi = A\,\psi_{elec}\psi_{nuc}$ into the expression $i\hbar\,\partial\Psi/\partial t = \hat{\mathcal{H}}\Psi$, we reach an equation which displays the electronic and nuclear functions.

$$i\hbar\left(\psi_{elec}\psi_{nuc}\frac{\partial A}{\partial t} + A\psi_{nuc}\frac{\partial\psi_{elec}}{\partial t} + A\psi_{elec}\frac{\partial\psi_{nuc}}{\partial t}\right) = -A\,\psi_{elec}\sum_{\alpha=1}^{N}\frac{\hbar^2}{2M_\alpha}\nabla_\alpha^2\psi_{nuc} + A\,\hat{\mathcal{H}}_{elec}\,\psi_{elec}\psi_{nuc}$$

Multiplication from the left by the complex conjugate functions $\psi^*_{elec}\psi^*_{nuc}$, integrating over all electronic and nuclear coordinates, making use of the normalization constraints, leads to:

$$i\hbar\left(\frac{\partial A}{\partial t} + A\int \psi^*_{elec}\frac{\partial\psi_{elec}}{\partial t}\,d\mathbf{r} + A\int \psi^*_{nuc}\frac{\partial\psi_{nuc}}{\partial t}\,d\mathbf{R}\right)$$

$$= -A \sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \int \psi^*_{nuc} \nabla^2_\alpha \psi_{nuc} \, d\mathbf{R}$$

$$- A \sum_{i=1}^{n} \frac{\hbar^2}{2m_e} \int \psi^*_{elec} \nabla^2_i \psi_{elec} \, d\mathbf{r} + A \int \psi^*_{elec} \psi^*_{nuc} \hat{V} \, \psi_{elec} \psi_{nuc} \, d\mathbf{r} \, d\mathbf{R}$$

where the terms building the electronic Hamiltonian are displayed. It is important to focus attention on the energy contribution from the potential $\hat{V}$, in particular on $\hat{V}_{Ze}$, Eq. 11.5, as it is the term that introduces the coupling between electrons and nuclei (it is also the term responsible for coupling the fast and slow particles in the Born-Oppenheimer approximation). Due to the exponential form, the time derivative of A is proportional to itself.

$$\partial A / \partial t = (i/\hbar) A E' \tag{11.11}$$

The insertion of this result and the expression of E' in terms of the electronic and nuclear functions in the wave equation, let us avoid the use of the amplitude A.

$$i\hbar \left(\int \psi^*_{elec} \frac{\partial \psi_{elec}}{\partial t} \, d\mathbf{r} + \int \psi^*_{nuc} \frac{\partial \psi_{nuc}}{\partial t} \, d\mathbf{R} \right)$$

$$= \int \psi^*_{nuc} \psi^*_{elec} \hat{\mathcal{H}}_{elec} \, \psi_{elec} \psi_{nuc} \, d\mathbf{r} \, d\mathbf{R} - \sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha}$$

$$* \int \psi^*_{nuc} \nabla^2_\alpha \psi_{nuc} \, d\mathbf{R} - \sum_{i=1}^{n} \frac{\hbar^2}{2m_e} \int \psi^*_{elec} \nabla^2_i \psi_{elec} \, d\mathbf{r}$$

$$+ \int \psi^*_{elec} \psi^*_{nuc} \hat{V} \psi_{nuc} \psi_{elec} \, d\mathbf{r} \, d\mathbf{R} \tag{11.12}$$

This equation is equivalent to the total wave equation because it has been only rewritten in terms of ψ_{elec} and ψ_{nuc}. Still, it is difficult to solve due to the terms mixing the electronic and nuclear degrees of freedom.

11.4 Time-Dependent Self-Consistent Field Equations

The next step consists on deriving an equation for the electrons and another for the nuclei. To do this, it is important to introduce the constraints that guarantee the solutions of the equations. In addition to the normalization constraints, Eqs. 11.10, which involve spatial variables, the new constraints should demand well-behaved functions with time. In particular, we should ensure that integrations of products like $\psi^*_{elec} \partial \psi_{elec} / \partial t$ and $\psi^*_{nuc} \partial \psi_{nuc} / \partial t$ are not undetermined.

$$i\hbar \int \psi^*_{elec} \frac{\partial \psi_{elec}}{\partial t} d\mathbf{r} = E' \quad ; \quad i\hbar \int \psi^*_{nuc} \frac{\partial \psi_{nuc}}{\partial t} d\mathbf{R} = E \tag{11.13}$$

The quantities E' and E have dimensions of energy. It is necessary to take the physical attributes of the particles into consideration to achieve the separation of equations. In most common processes, the nuclei are the heavy-weight and slow-moving particles, and the electrons are the light-weight and fast-moving particles. The electrons and nuclei exhibit different physical properties and time scales in their motions, giving basis to the proposal of a ruling equation for the electrons and another for the nuclei.[3]

In order to establish the time-dependent wave equations that determine the electronic and nuclear wave functions, we consider Eq. 11.12 and compare the terms of the left side with the terms of the right side. The resulting equations are:

$$\begin{aligned} i\hbar \frac{\partial \psi_{elec}}{\partial t} &= -\sum_{i=1}^{n} \frac{\hbar^2}{2m_e} \nabla_i^2 \psi_{elec} + \left[\int \psi^*_{nuc} \hat{V} \psi_{nuc} \, d\mathbf{R} \right] \psi_{elec} \\ i\hbar \frac{\partial \psi_{nuc}}{\partial t} &= -\sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \nabla_\alpha^2 \psi_{nuc} + \left[\int \psi^*_{elec} \hat{\mathcal{H}}_{elec} \psi_{elec} \, d\mathbf{r} \right] \psi_{nuc} \end{aligned} \tag{11.14}$$

The first expression is the wave equation that regulates the dynamics of the electrons, and the second one is the wave equation that regulates the dynamics of the nuclei. The electrons and nuclei have equal hierarchy as the electronic and nuclear wave functions are shown in symmetric forms in both equations and are treated at the same level of quantum theory. The presence of ψ_{nuc} in the electronic wave equation, and the presence of ψ_{elec} in the nuclear wave equation, is the manner in which such equations maintain the correlation between the different sets of particles. According to such expressions, the electrons do not interact with any nucleus in particular, and the nuclei do not interact with any electron in particular. However, the electrons are understood to be immersed in an *effective field* or *average potential* created by the nuclei and, in turn, the nuclei are immersed in an effective field created by the electrons.

After introducing the concept of an effective field, the time-dependent wave equations are simplified by using the effective Hamiltonians.

$$\begin{aligned} \hat{\mathcal{H}}^{eff}_{elec}(\{\mathbf{r}_i\}; t) &= -\sum_{i=1}^{n} \frac{\hbar^2}{2m_e} \nabla_i^2 + \int \psi^*_{nuc} \hat{V}(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\}) \psi_{nuc} \, d\mathbf{R} \\ \hat{\mathcal{H}}^{eff}_{nuc}(\{\mathbf{R}_\alpha\}; t) &= -\sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \nabla_\alpha^2 + \int \psi^*_{elec} \hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha\}) \psi_{elec} \, d\mathbf{r} \end{aligned} \tag{11.15}$$

[3] The separation of variables discussed in the text is not a common separation of the independent degrees of freedom, because the position variables of the electrons and nuclei are equally important, all of them have equal priority, and satisfy the same conditions. The separation of variables proposed in the text is based on the phenomenological features of the particles and are similar to those employed in the Born-Oppenheimer approach, with the difference that time is not present in the Born-Oppenheimer approach.

Both Hamiltonians change with time by means of the wave functions. The time-dependent wave equations of the electrons and nuclei are simplified by taking the notation of the average potentials into consideration.

$$i\hbar \frac{\partial \psi_{elec}}{\partial t} = \hat{\mathcal{H}}^{eff}_{elec}(\{\mathbf{r}_i\}; t)\, \psi_{elec} \quad ; \quad i\hbar \frac{\partial \psi_{nuc}}{\partial t} = \hat{\mathcal{H}}^{eff}_{nuc}(\{\mathbf{R}_\alpha\}; t)\, \psi_{nuc} \tag{11.16}$$

These equations are difficult to solve analytically due to the particle correlations and, thereby, have to be solved numerically in a self-consistent way. For example, an initial nuclear wave function is proposed with the purpose of building the average potential acting on the electrons. Thus, the electronic wave equation may be numerically solved to obtain a first approximation of the electronic wave function. Next, the electronic wave function is inserted into the nuclear wave equation for building the average potential acting on the nuclei. Now, the nuclear wave equation may be numerically solved to obtain an approximate nuclear wave function. The required feedback demands a self-consistent iteration process to improve the wave functions at every step. The self-consistent process is terminated when the electronic and nuclear wave functions and their energies are accurate and show no significant changes from one cycle to the next one of the iterative process. Due to computational efficiency, one or another wave function may be favored to start the process. Still, the final wave functions are independent of the initial function that was used to start the self-consistent process. The expressions 11.16 are recognized as the *time-dependent self-consistent field equations*. The equations satisfy the normalization conditions 11.13 imposed on the products $\psi^*_{elec}\partial\psi_{elec}/\partial t$ and $\psi^*_{nuc}\partial\psi_{nuc}/\partial t$, which lead to the energies E' and E, respectively.

Problem 11.3 (i) Show that the plane wave $\psi(x,t) = A\exp[i(kx - wt)]$ reaches a finite value when the kernel $\psi^*\,\partial\psi/\partial t$ is integrated. (ii) Demonstrate that the wave equations of electrons and nuclei, Eqs. 11.14, are consistent with the constraints 11.13 imposed on the functions ψ_{elec} and ψ_{nuc}, resulting in the energies E' and E, respectively.

Solution

(i) The required integral is $\int_a^b(\psi^*\partial\psi/\partial t)dx = AA^*\int_a^b e^{-i(kx-wt)}(-iw t)\, e^{i(kx-wt)}dx$. When $a = -\infty$ and $b = \infty$, the integral is undetermined; however, if the integration limits are finite ($a = -\ell$ and $b = \ell$), the integral achieves a finite value. Thus, the function ψ acquires physical meaning for finite integration limits.

(continued)

(ii) If the first expression of Eqs. 11.14 is multiplied from the left by ψ^*_{elec} and integrated, then we obtain:

$$i\hbar \int \psi^*_{elec} \frac{\partial \psi_{elec}}{\partial t}\, d\mathbf{r} = -\int d\mathbf{r} \sum_{i=1}^{n} \frac{\hbar^2}{2m_e}\, \psi^*_{elec} \nabla_i^2 \psi_{elec} + \int \psi^*_{elec} \psi^*_{nuc}\, \hat{V}\, \psi_{nuc} \psi_{elec}\, d\mathbf{r}\, d\mathbf{R}$$

If we use the orthonormality condition 11.10 of the nuclear wave function to introduce both the product $\psi^*_{nuc}\psi_{nuc}$ on the first term of the right and the electronic Hamiltonian 11.9, then the energy E' of Eqs. 11.13 is obtained.

$$i\hbar \int \psi^*_{elec} \frac{\partial \psi_{elec}}{\partial t}\, d\mathbf{r} = \int \psi^*_{elec} \psi^*_{nuc} \hat{\mathcal{H}}_{elec}\, \psi_{elec} \psi_{nuc}\, d\mathbf{r}\, d\mathbf{R} = E'(t)$$

The nuclear equation is symmetric to the electronic equation, and therefore, a similar process on the nuclei leads to the energy E.

$$i\hbar \int \psi^*_{nuc} \frac{\partial \psi_{nuc}}{\partial t}\, d\mathbf{R} = \int \psi^*_{nuc} \psi^*_{elec}\, \hat{\mathcal{H}}\, \psi_{elec} \psi_{nuc}\, d\mathbf{r}\, d\mathbf{R} = E$$

Interestingly, we started from a single wave equation in the overall simplification process and ended up with two correlated wave equations, apparently doubling the difficulty in obtaining the final solution of the total wave equation of the molecular system. Nevertheless, the pair of equations are in principle easier to solve than the initial one due to the average potentials in which the electrons and nuclei are immersed. Finally, the wave functions may be propagated with time by building the time evolution operators $\hat{\mathcal{U}}_{nuc}$ and $\hat{\mathcal{U}}_{elec}$ from the effective Hamiltonians. Yet, we avoid such operators in later sections by further simplifying the wave equations and determining the time evolution of the electrons and nuclei.

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Classical Limit of the Nuclear Motion

12

In going from the microscopic world to the macroscopic one, it is possible to have a smooth transition from quantum mechanics to classical mechanics in different ways. According to the Bohr's correspondence principle, there is a connection between quantum theory and classical theory. Such a principle demands large quantum numbers to describe the particles as classical objects. The corresponding principle has been given different interpretations, yet it is difficult to have practical use of it due to the requirement of establishing the degree of quantization of the particles [1]. There is an equivalent way to determine a connection between quantum mechanics and classical mechanics. It is referred to as the classical limit. After perceiving no major obstacles in the transition from quantum mechanics to classical mechanics, the classical realm is reached from quantum mechanics by introducing the observations of the macroscopic phenomena as limiting mathematical conditions in the quantum-mechanical equations.

The quantum nature of the nuclei is important in processes such as the nuclear dispersion, the nuclear spin transitions, the tunnel effect to traverse high energy barriers, and in other phenomena. These all processes can be solely studied from the quantum-mechanical perspective. However, as we have observed in previous sections, the application of quantum mechanics is difficult in many instances. For example, we face problems to apply the quantum-mechanical theory to the nuclei of polyatomic molecules due to the many degrees of freedom required to describe these molecules. The vast demand of efforts and computational resources implied in an approach that considers electrons and nuclei at the same level of quantum-mechanical theory is only justified when a precise and detailed picture of the particle system is crucial. On the other side, the approximation of the nuclei as classical particles may be used to simulate the dynamics of particles in experiments involving, for example, low-energy collisions, temperature effects, chemical reactions, and more, where the quantum-mechanical nature of the nuclei is less relevant. Once the nuclei are described classically, their quantum character is lost in the description.

The aim of the present chapter is to provide a classical description of the nuclei while maintaining the quantum character of the electrons. In the following

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sections, we reformulate the wave equation of the nuclei and migrate the quantum observables to classical variables to reach a description in terms of classical theory. The discussion exhibits a hierarchy of approximations, starting from a quantum mechanical treatment of the nuclei and ending up on a classical one, because there are phenomena where the nuclei act as quantum particles, or the quantum nature of the nuclei plays a second role, favoring the nuclei to act as classical objects. In these last cases, a description of the nuclear particles based on analytical mechanics is possible, rendering results that are close to those obtained with quantum mechanics but with a lower computational cost.

The chapter follows the main ideas of Tully [2], and is structured as follows. We depart from the wave equation of the electrons and the wave equation of the nuclei of the previous chapter. Such equations deal with the electrons and nuclei at the same quantum-mechanical level of theory, maintaining a good degree of correlation among the particles. However, with the purpose of further reducing the complexity of the expressions, we turn the attention to the nuclear wave function. By assuming a polar form for that function, we are able to simplify the nuclear potential and make contact with classical mechanics, where the nuclear particles become classical bodies with motions governed by Newtonian mechanics. This gives formal birth to *mixed quantum-classical dynamics*. Later, by parameterizing the atomic interactions, a full classical description of the atoms is achieved, thus establishing *classical molecular dynamics*. By dealing with the classical particles, we are able to establish an equation of continuity and discuss the diffusion and flux of the classical particles by deriving the first and second Fick's laws, and linking the results to the chemical potential and the diffusion constant. On the other side, the electronic wave function is expanded employing a complete basis set of functions, with implications on the equations of the nuclei, since all the particles maintain a degree of correlation. Finally, a section is dedicated to the Bohm potential, which is one of the important intermediate results in the transition from quantum mechanics to classical mechanics. Based on the Bohm potential, a discussion is oriented to an alternative interpretation of quantum mechanics.

12.1 Polar Form of the Nuclear Wave Equation

In order to simplify the description of the nuclei in polyatomic molecules, we depart from the time-dependent wave equation of the nuclei and the time-dependent wave equation of the electrons, Eqs. 11.15 and 11.16. In the time-dependent self-consistent field method, the nuclei and electrons are described at the same level of theory by their respective wave equations.

$$i\hbar \frac{\partial \psi_{nuc}}{\partial t} = \hat{\mathcal{H}}^{eff}_{nuc}(\{\mathbf{R}_\alpha\}; t)\, \psi_{nuc} \quad ; \quad i\hbar \frac{\partial \psi_{elec}}{\partial t} = \hat{\mathcal{H}}^{eff}_{elec}(\{\mathbf{r}_i\}; t)\, \psi_{elec} \tag{12.1}$$

$$\hat{\mathcal{H}}^{eff}_{nuc}(\{\mathbf{R}_\alpha\};t) = -\sum_{\alpha=1}^{N}\frac{\hbar^2}{2M_\alpha}\nabla^2_\alpha + \int \psi^*_{elec}(\{\mathbf{r}_i\};t) \times \hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\},\{\mathbf{R}_\alpha\})\,\psi_{elec}(\{\mathbf{r}_i\};t)\,d\mathbf{r}$$

$$\hat{\mathcal{H}}^{eff}_{elec}(\{\mathbf{r}_i\};t) = -\sum_{i=1}^{n}\frac{\hbar^2}{2m_e}\nabla^2_i + \int \psi^*_{nuc}(\{\mathbf{R}_\alpha\};t) \times \hat{V}(\{\mathbf{r}_i\},\{\mathbf{R}_\alpha\})\,\psi_{nuc}(\{\mathbf{R}_\alpha\};t)\,d\mathbf{R} \tag{12.2}$$

The wave functions of nuclei and electrons are ψ_{nuc} and ψ_{elec}. The effective Hamiltonians involve the kinetic and potential energies of the corresponding particles. The potential $\hat{V}$ and Hamiltonian $\hat{\mathcal{H}}_{elec}$ are shown in the kernels of the integrals and are given by Eqs. 11.5 and 11.9. The averaged potentials where the nuclei and electrons are immersed may be written in bra-ket notation $\langle\psi_{elec}|\hat{\mathcal{H}}_{elec}|\psi_{elec}\rangle$ and $\langle\psi_{nuc}|\hat{V}|\psi_{nuc}\rangle$, respectively.

$$\langle\psi_{elec}|\hat{\mathcal{H}}_{elec}|\psi_{elec}\rangle = \int \psi^*_{elec}(\{\mathbf{r}_i\};t)\,\hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\},\{\mathbf{R}_\alpha\})\,\psi_{elec}(\{\mathbf{r}_i\};t)\,d\mathbf{r}$$

$$\langle\psi_{nuc}|\hat{V}|\psi_{nuc}\rangle = \int \psi^*_{nuc}(\{\mathbf{R}_\alpha\};t)\,\hat{V}(\{\mathbf{r}_i\},\{\mathbf{R}_\alpha\})\,\psi_{nuc}(\{\mathbf{R}_\alpha\};t)\,d\mathbf{R}$$

In the simplified description of the nuclei, the quantum character of the electrons is retained because it is of primary importance. Following similar steps in the transition from wave optics to geometrical optics, this is, from waves to rays, a function of polar type for the nuclei is proposed.

$$\psi_{nuc}\left(\{\mathbf{R}_\alpha\};t\right) = D\left(\{\mathbf{R}_\alpha\};t\right)\,\exp\left[iS\left(\{\mathbf{R}_\alpha\};t\right)/\hbar\right] \tag{12.3}$$

This equation is recognized as the action wave, Eq. 6.64. The factor D represents the amplitude and the term S is to be identified. Both functions are real and depend on the nuclear coordinates $\mathbf{R}_\alpha$ and time t. The introduction of this expression in the nuclear wave equation produces an identity involving both functions:

$$i\hbar\frac{\partial}{\partial t}D + \sum_{\alpha=1}^{N}\frac{\hbar^2}{2M_\alpha}\left[2\,\nabla_\alpha(iS/\hbar)\cdot\nabla_\alpha D + D\,\nabla^2_\alpha(iS/\hbar)\right]$$

$$= D\frac{\partial}{\partial t}S - \sum_{\alpha=1}^{N}\frac{\hbar^2}{2M_\alpha}\left[-\frac{D}{\hbar^2}\,|\nabla_\alpha S|^2 + \nabla^2_\alpha D\right] + D\,V_{elec} \tag{12.4}$$

where it was convenient to introduce the function V_{elec} to simplify the above expression.

$$V_{elec} = \int \psi^*_{elec}(\{\mathbf{r}_i\};t)\,\hat{\mathcal{H}}_{elec}\,\psi_{elec}(\{\mathbf{r}_i\};t)\,d\mathbf{r} \tag{12.5}$$

When the separation of the real and imaginary parts of Eq. 12.4 is performed, two coupled expressions which contain the functions D and S are obtained.

$$\frac{\partial D}{\partial t} + \sum_{\alpha=1}^{N} \frac{1}{M_\alpha} \nabla_\alpha D \cdot \nabla_\alpha S + \sum_{\alpha=1}^{N} \frac{1}{2M_\alpha} D \nabla_\alpha^2 S = 0$$

$$\frac{\partial S}{\partial t} + \sum_{\alpha=1}^{N} \frac{1}{2M_\alpha} |\nabla_\alpha S|^2 + \int \psi_{elec}^* \hat{\mathcal{H}}_{elec} \psi_{elec} \, d\mathbf{r} - \sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \frac{\nabla_\alpha^2 D}{D} = 0 \tag{12.6}$$

Problem 12.1 Deduce the expression 12.4 of the main text.

Solution The time-dependent wave equation of the nuclei is:

$$i\hbar \frac{\partial \psi_{nuc}}{\partial t} = \hat{\mathcal{H}}_{nuc}^{eff} \psi_{nuc}$$

When the polar form of ψ_{nuc} and the form of $\hat{\mathcal{H}}_{nuc}^{eff}$ are inserted in this equation, an expression in terms of the functions D and S is obtained.

$$i\hbar \frac{\partial}{\partial t} \left[D \exp(iS/\hbar) \right] = \left[-\sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \nabla_\alpha^2 + \int \psi_{elec}^* \hat{\mathcal{H}}_{elec} \psi_{elec} \, d\mathbf{r} \right] \times D \exp(iS/\hbar)$$

The divergence of a continuously differentiable vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ is defined as $\nabla \cdot \mathbf{F} = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$, while the Laplacian ∇^2 is the divergence of the gradient or $\nabla^2 = \nabla \cdot \nabla$. In this connection, the time and spatial differentiations give:

$$i\hbar \, e^{(iS/\hbar)} \left\{ \frac{\partial}{\partial t} D + D \frac{\partial}{\partial t} iS/\hbar \right\} = -\sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} e^{(iS/\hbar)} \left[\nabla_\alpha (iS/\hbar) \cdot \nabla_\alpha D \right.$$

$$+ \nabla_\alpha^2 D + \nabla_\alpha D \cdot \nabla_\alpha (iS/\hbar) + D \nabla_\alpha (iS/\hbar) \cdot \nabla_\alpha (iS/\hbar)$$

$$\left. + D \nabla_\alpha^2 (iS/\hbar) \right] + D e^{(iS/\hbar)} V_{elec}$$

where Eq. 12.5 was used. The exponential function represents a common factor and is dismissed.

$$i\hbar \left\{ \frac{\partial}{\partial t} D + D \frac{\partial}{\partial t} iS/\hbar \right\} = -\sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \left[\nabla_\alpha (iS/\hbar) \cdot \nabla_\alpha D + \nabla_\alpha^2 D \right.$$

$$\left. + \nabla_\alpha D \cdot \nabla_\alpha (iS/\hbar) + D \nabla_\alpha (iS/\hbar) \cdot \nabla_\alpha (iS/\hbar) + D \nabla_\alpha^2 (iS/\hbar) \right] + D V_{elec}$$

(continued)

The real and imaginary terms are separated from each other, thus reproducing Eq. 12.4. The resulting expression is only possible when each member of Eq. 12.4 equals zero. Thus, an equation for D and another for S is obtained, leading to Eqs. 12.6.

12.2 Continuity and Hamilton-Jacobi Equations

In the following lines, the first identity of Eq. 12.6 is shown to be equivalent to an equation of continuity, with no source and sink terms, thus establishing the conservation of the probability, and the second identity of Eq. 12.6 is shown to lead, under specific conditions, to a Hamilton-Jacobi classical equation. In order to unfold the first demonstration, let us multiply the nuclear wave equation and its complex conjugate by ψ^*_{nuc} and $-\psi_{nuc}$, respectively, adding the two resulting expressions to have a single one.

$$i\hbar \frac{\partial(\psi^*_{nuc}\psi_{nuc})}{\partial t} = \psi^*_{nuc}\,\hat{\mathcal{H}}^{eff}_{nuc}\,\psi_{nuc} - \psi_{nuc}\,\hat{\mathcal{H}}^{eff}_{nuc}\,\psi^*_{nuc}$$

The equation is simplified when the nuclear Hamiltonian consists of a kinetic energy operator and a potential energy operator. The latter operator contributes nothing to the equation because, contrary to the kinetic energy operator that involves differentiation over the wave function, the potential energy acts as a multiplicative factor affecting no wave function, thereby cancelling the difference.

$$i\hbar \frac{\partial(\psi^*_{nuc}\psi_{nuc})}{\partial t} = -\sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha}\left[\psi^*_{nuc}\nabla^2_\alpha\psi_{nuc} - \psi_{nuc}\nabla^2_\alpha\psi^*_{nuc}\right]$$

With the purpose of simplifying the equation, we define the *probability density* ρ and *probability flux* $\mathbf{J}$, in terms of ψ_{nuc} and $\nabla_\alpha\psi_{nuc}$.

$$\begin{aligned} \rho\,(\mathbf{R};t) &= \psi^*_{nuc}\psi_{nuc} \\ \mathbf{J}\,(\mathbf{R};t) &= \sum_{\alpha=1}^{N} \frac{\hbar}{2iM_\alpha}\left[\psi^*_{nuc}\nabla_\alpha\psi_{nuc} - \psi_{nuc}\nabla_\alpha\psi^*_{nuc}\right] \end{aligned} \tag{12.7}$$

The functions ρ and $\mathbf{J}$ depend on the variable $\mathbf{R}$ as these functions represent probability densities. The direction of the $\mathbf{J}$ vector is defined by the difference of the gradients on the wave functions. $\mathbf{J}$ is responsible for producing probability currents, in analogous manner to a current of electrons in a metal [3]. The equation that establishes the conservation of the probability is now stated in terms of the functions ρ and $\mathbf{J}$.

$$\frac{\partial \rho(\mathbf{R};t)}{\partial t} + \nabla \cdot \mathbf{J}(\mathbf{R};t) = 0 \tag{12.8}$$

Such an expression is commonly referred to as the *equation of continuity*. On the other hand, the insertion of $\psi_{nuc} = D\,\exp(iS/\hbar)$, Eq. 12.3, in the definitions of ρ and $\mathbf{J}$ makes these variables to show dependencies on D and S.

$$\rho(\mathbf{R};t) = D^2 \quad ; \quad \mathbf{J}(\mathbf{R};t) = \sum_{\alpha=1}^{N} \frac{1}{M_\alpha} D^2 \nabla_\alpha S$$

When they are introduced in the continuity equation 12.8, the differential equation in the amplitude D, Eq. 12.6, is obtained, thus demonstrating the equivalence between equations that apparently look different.

$$\begin{gathered} \frac{\partial D}{\partial t} + \sum_{\alpha=1}^{N} \frac{1}{M_\alpha} \nabla_\alpha D \cdot \nabla_\alpha S + \sum_{\alpha=1}^{N} \frac{1}{2M_\alpha} D \nabla_\alpha^2 S = 0 \\ \Updownarrow \\ \frac{\partial \rho(\mathbf{R};t)}{\partial t} + \nabla \cdot \mathbf{J}(\mathbf{R};t) = 0 \end{gathered} \tag{12.9}$$

It is time to work with the differential equation in the function S, Eq. 12.6, and show that under limiting conditions it leads to the second Newton's law. To do this, we consider the term containing Planck's constant, $h = 2\pi\hbar$. It is a reference parameter that determines the size of quanta. When the term of an expression carrying h is substantially smaller than the other terms of the expression carrying no h, we conclude that we are in the classical realm. Such a situation is frequently abbreviated in the form $h \to 0$ (it is a symbolic form because h is a constant and it never goes to zero and is similar to the transition from relativistic to non-relativistic mechanics, when we write $c \to \infty$). The term that carries h in Eq. 12.6 is the *Bohm quantum potential*, defined by:

$$V_{bohm} = -\sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \nabla_\alpha^2 D \,/\, D \tag{12.10}$$

This is an important quantity that deserves special attention, Sect. A.1, because the quantum potential is capable to take us to an interpretation of quantum mechanics different to the traditional one. By assuming that V_{bohm} is sufficiently small with respect to the other terms present in Eq. 12.6, we step into a classical description framework.

$$\frac{\partial S(\{\mathbf{R}_\alpha\};t)}{\partial t} + \sum_{\alpha=1}^{N} \frac{1}{2M_\alpha} |\nabla_\alpha S(\{\mathbf{R}_\alpha\};t)|^2 + \int \psi_{elec}^{*}\, \hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\},\{\mathbf{R}_\alpha\})\, \psi_{elec}\, d\mathbf{r} = 0 \tag{12.11}$$

This result is isomorphic to the Hamilton-Jacobi equation obtained in classical mechanics.

$$\frac{\partial S(\{\mathbf{R}_\alpha\};t)}{\partial t} + \mathcal{H}\left(\{\mathbf{R}_\alpha\}, \{\mathbf{P}_\alpha\}\right) = 0 \tag{12.12}$$

The expression 12.12 is the first-order differential equation of the *classical action function*, S, Eq. 6.49, which in turn is defined in terms of the Lagrangian, L [4].

$$S = \int L\left(\{\mathbf{R}_\alpha\}, \{\mathbf{P}_\alpha\}; t\right) dt \tag{12.13}$$

The last two terms of 12.11 constitute the classical Hamiltonian, $\mathcal{H}$, defined in terms of the kinetic and potential energy contributions.[1]

$$\begin{aligned} &\sum_{\alpha=1}^{N} \frac{1}{2M_\alpha} \left(\nabla_\alpha S\right)^2 + \int \psi^*_{elec}\, \hat{\mathcal{H}}_{elec}\, \psi_{elec}\, d\mathbf{r} \\ &\xrightarrow[\hbar \to 0]{} \quad \mathcal{H} = T\left(\{\mathbf{P}_\alpha\}\right) + V_{elec}\left(\{\mathbf{R}_\alpha\}; t\right) \end{aligned} \tag{12.14}$$

The projection of the terms on the top onto the terms on the bottom is done under physical basis: the integral represents the potential V_{elec}, Eq. 12.5, created by the electron cloud where the nuclei are immersed, and the term involving the summation represents the kinetic energy T of the nuclei. The kinetic energy is a quadratic function of the momentum $\mathbf{P}$ and is identified in terms of the action function. Thus, the classical terms are defined by:

$$\begin{aligned} V_{elec}\left(\{\mathbf{R}_\alpha\}; t\right) &= \int \psi^*_{elec}(\{\mathbf{r}_i\}; t)\, \hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\})\, \psi_{elec}(\{\mathbf{r}_i\}; t)\, d\mathbf{r} \\ \mathbf{P}_\alpha &= \nabla_\alpha S \end{aligned} \tag{12.15}$$

We now follow the ideas of the Hamilton-Jacobi theory to link the momentum to the potential energy. By applying ∇_α to each term in $\partial S/\partial t + \mathcal{H} = 0$, we obtain:

$$\frac{d\mathbf{P}_\alpha}{dt} = -\nabla_\alpha V_{elec}\left(\{\mathbf{R}_\alpha\}; t\right) \tag{12.16}$$

The change rate of the momentum is due to a gradient of the potential V_{elec} created by the electrons and the other nuclei. Thus, the particle evolves according to the second Newton's law with time. Concluding, it has been demonstrated the relation between the following equations:

$$\begin{aligned} &\frac{\partial S}{\partial t} + \sum_{\alpha=1}^{N} \frac{1}{2M_\alpha} |\nabla_\alpha S|^2 + \int \psi^*_{elec}\, \hat{\mathcal{H}}_{elec}\, \psi_{elec}\, d\mathbf{r} - \sum_{\alpha=1}^{N} \frac{\hbar^2}{2M_\alpha} \frac{\nabla^2_\alpha D}{D} = 0 \\ &\Downarrow \qquad \hbar \to 0 \\ &\frac{d\mathbf{P}_\alpha}{dt} = -\nabla_\alpha V_{elec}\left(\{\mathbf{R}_\alpha\}; t\right) \end{aligned} \tag{12.17}$$

[1] We discriminate terms by following a similar procedure as that of the WKB method, where an expansion in terms of $\hbar$ is performed.

Due to the fact that the particles have wave-particle duality, there is not a concept of force at the quantum-mechanical level. Still, the force is a consequence of quantum theory by reaching after special considerations the classical limit. In such a limit, the Newtonian force is associated with the action of a field on a particle that shows a sufficiently localized mass in space. In the second Newton's law, the force is defined implicitly as that entity that produces the change of inertia of a particle with mass, giving no explanation on the origin of the force. Therefore, in the Newtonian framework, the force is manifested by its action on a particle with sufficiently localized mass.

Problem 12.2 A particle with mass m and electric charge e is subjected to a potential V and a magnetic field, which is described by the vector potential $\mathbf{A}$. The vector potential satisfies the equation $\nabla \cdot \mathbf{A} = 0$. The wave equation of the particle is:

$$-\frac{\hbar^2}{2m}\left(\nabla - \frac{ie}{\hbar c}\mathbf{A}\right)^2 \psi + V\psi = i\hbar\frac{\partial \psi}{\partial t}$$

A solution is proposed by expanding S as a power series.

$$S = S_0 + \left(\frac{\hbar}{i}\right) S_1 + \left(\frac{\hbar}{i}\right)^2 S_2 + \cdots$$

Show that $\psi = \exp(iS/\hbar)$ is an exact solution of the wave equation when S satisfies the Hamilton-Jacobi equation.

Solution The wave function $\psi = \exp(iS/\hbar)$ is supposed to be an exact solution, and it is inserted into the wave equation. The result, using $\nabla \cdot \mathbf{A} = 0$, is an equation in the function S.

$$\frac{1}{2m}\left[(\nabla S)^2 + (\frac{e}{c})^2 \mathbf{A}\cdot\mathbf{A} - \frac{e}{c}\mathbf{A}\cdot\nabla S\right] - \frac{i\hbar}{2m}\nabla^2 S + V + \frac{\partial S}{\partial t} = 0$$

By inserting the power series of S and using the Laplace equation, the above expression leads to a system of equations, which are defined after comparing the coefficients with the same power of $\hbar$ in both sides of the equation. The expression of interest is the zero-order term S_0, as it is a Hamilton-Jacobi equation in the function S_0.

$$\frac{\partial S_0}{\partial t} + \frac{1}{2m}\left[\nabla S_0 - \frac{e}{c}\mathbf{A}\right]^2 + V = 0$$

(continued)

Interestingly, there are forms of defining the potential V with dependence on the coordinates and time, in such a way that a solution S of the Hamilton-Jacobi equation gives an exact solution, $\exp(iS/\hbar)$, of the corresponding wave equation [5]. Similarly, there are forms of defining the refractive index, in such a way that a solution S of the eikonal equation gives an exact solution, $\exp(ik_0 S)$, of the corresponding Helmholtz equation [6].

It is convenient to summarize the main equations on the dynamics of the nuclei and electrons.

$$\begin{aligned} \frac{d\mathbf{P}_\alpha}{dt} &= -\nabla_\alpha \int \psi^*_{elec}(\{\mathbf{r}_i\}; t)\, \hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\})\, \psi_{elec}(\{\mathbf{r}_i\}; t)\, d\mathbf{r} \\ i\hbar \frac{\partial \psi_{elec}}{\partial t} &= -\sum_{i=1}^{n} \frac{\hbar^2}{2m_e} \nabla_i^2\, \psi_{elec} + \left[\int \psi^*_{nuc}(\{\mathbf{R}_\alpha\}; t)\, \hat{V}\, \psi_{nuc}(\{\mathbf{R}_\alpha\}; t)\, d\mathbf{R}\right] \psi_{elec} \end{aligned} \tag{12.18}$$

Care should be exercised in not confusing the potential V_{elec}, Eq. 12.15, with the potential $\hat{V} = \hat{V}_{ee} + \hat{V}_{Ze} + \hat{V}_{ZZ}$, Eq. 11.5, which contains operators without being averaged. The equation in the first line describes from a classical point of view the behavior of the nuclei immersed in an effective potential created by the electrons. Instead, the equation in the second line describes the behavior of the electrons immersed in an effective potential created by the nuclei. In this last expression, the wave function of the nuclei is used, in spite of the fact that the nuclei are described classically in the first expression. In this regard, Eq. 12.18 is considered a transitional expression in going from a quantum-mechanical description toward a classical-mechanical description of the nuclei. In order to be consistent with a description of the nuclei as classical particles, we must avoid the use of the nuclear wave function in the final equations. This is done later.

12.3 Conditions to Describe the Nuclear Particles Classically

In the previous section, the classical limit was achieved by considering $\hbar \to 0$. Let us now be more specific on determining the physical conditions to reach the classical limit. When the objects are sufficiently massive or the particles increase in size, their quantum numbers are big, and the probabilistic outputs of quantum mechanics give a single and definite outcome. In this limit we are in the domain where the results obtained from quantum mechanics correspond to those of the classical world. Since the procedure to reach classical mechanics from quantum mechanics follows similar steps to the procedure where geometrical optics is reached from wave optics, it is convenient to briefly document this last procedure [7].

When the Hamiltonian of the system is time indifferent, *Hamilton's principal function*, S, of classical mechanics is written in terms of both *Hamilton's characteristic function*, W, which depends on the conjugate variables $\{q_i, p_i\}$, and a term which is linearly proportional to the time, Eq. 6.53.

$$S(\{q_i, p_i; t\}) = W(\{q_i, p_i\}) - \alpha t$$

In principle, it is a consequence of considering the wave function of quantum mechanics in a stationary state, Eq. 10.36:

$$\psi_{nuc}(\{\mathbf{R}_\alpha\}, t) = \psi_{nuc}(\{\mathbf{R}_\alpha\}) \exp(-i\,Et/\hbar)$$

where the coordinates and time are separated into a pair of functions. The substitution of S in the Hamilton-Jacobi equation defines the value of α, this is, $\alpha = \mathcal{H} = E$, where E is the integration constant associated with the energy. The separation of the conjugate variables from the time in the expression of S makes the Hamilton-Jacobi approach useful: if the value of the Hamilton's characteristic function at $t = 0$ is W_0, then the corresponding value of S is $S(0) = W_0$. Suppose that we work with the value $S(0)$. At a later instant of time, say t_i, the value of W is $W_{t_i} = W_0 + Et_i$, and thus, the value of S is preserved, $S(t_i) = W_{t_i} - \alpha t_i = [W_0 + Et_i] - \alpha t_i = W_0 = S(0)$. We have a surface S with fixed value that moves from the surface W_0 at the instant of time $t = 0$ to the surface W_{t_i} at the instant of time t_i. It is visualized as a wave front propagating in phase space with a velocity computed from the properties of the particles under consideration. The velocity can be shown to point perpendicularly to the direction of the wave front. The surface with constant value of S satisfies a wave equation. A procedure like this characterizes the transition from wave optics to geometrical (ray) optics establishing, among other things, the eikonal equation of geometrical optics, with ray trajectories pointing in the perpendicular direction to the wave fronts, and giving foundations to Fermat's principle and the Huygens propagation of wave fronts.

Now, we start from Eq. 12.4, and assume the separation of positions and time in the action function, $S = \sigma(\{\mathbf{R}_\alpha\}) - Et$, where σ is a function of the positions solely. It is also supposed that the amplitude D of Eq. 12.4 is a slowly varying function of the positions and time. In this last case, the spatial and time derivatives of D are negligible, and Eq. 12.4 is simplified.

$$\frac{\partial}{\partial t} S + \sum_{\alpha=1}^{N} \frac{1}{2M_\alpha} |\nabla_\alpha S|^2 - \sum_{\alpha=1}^{N} \frac{i\hbar}{2M_\alpha} \nabla_\alpha^2 S + V_{elec} = 0$$

It is time to introduce the expression of the action function in the above equation.

$$\sum_{\alpha=1}^{N} \frac{1}{2M_\alpha} |\nabla_\alpha \sigma|^2 + V_{elec} - \sum_{\alpha=1}^{N} \frac{i\hbar}{2M_\alpha} \nabla_\alpha^2 \sigma = E \tag{12.19}$$

If we use the power series expansion $\sigma = \sigma_0 + (\hbar/i)\sigma_1 + (\hbar/i)^2\sigma_2 + \cdots$, which is a consequence of writing the wave function of a particle system in the stationary state, $\psi(\{\mathbf{R}_\alpha\}, t) = \exp(i\sigma\{\mathbf{R}_\alpha\}/\hbar)$, problem 12.2, then we recover the classical equation of the energy. This process is equivalent to neglecting the term of Eq. 12.19 that contains $\hbar$, because it is sufficiently small with respect to the other terms containing no $\hbar$.

$$|\nabla_\alpha \sigma_0|^2 \gg \hbar|\nabla_\alpha^2 \sigma_0| = \hbar|\nabla_\alpha \cdot \nabla_\alpha \sigma_0| \quad \Rightarrow \quad \sum_{\alpha=1}^{N} \frac{1}{2M_\alpha} |\nabla_\alpha \sigma_0|^2 + V_{elec} = E$$

By identifying $\mathbf{P}_\alpha$ with $\nabla_\alpha S = \nabla_\alpha \sigma_0$, Eq. 12.15, the above expressions are written in terms of the momentum.

$$\mathbf{P}_\alpha^2 \gg \hbar\,|\nabla_\alpha \cdot \mathbf{P}_\alpha| \quad \Rightarrow \quad \textstyle\sum_{\alpha=1}^{N} \mathbf{P}_\alpha^2/(2M_\alpha) + V_{elec} = E \tag{12.20}$$

The first expression is equivalent to $1 \gg \hbar|\nabla_\alpha \cdot \mathbf{P}_\alpha|/\mathbf{P}_\alpha^2$. The second expression is consistent with the Hamiltonian of Eq. 12.14. When the ∇ operator is applied to the expression of the energy E, Eq. 12.20, we deduce:

$$\hbar M_\alpha\, |\partial V_{elec}/\partial X_\alpha| \ll P_\alpha^3 \tag{12.21}$$

Therefore, the classical limit of the quantum motion of a particle is achieved for large values of the particle momentum with respect to the gradient of the potential V_{elec}. An alternative form of establishing the classical behavior is by considering an action, say S_0, where $\hbar \ll S_0$ (as discussed in Sect. 12.2, where $h \to 0$ was employed, Eq. 12.17).

Let us now deal with a single particle in one dimension and use de Broglie's relation of the momentum with the wavelength, $P = 2\pi\hbar/\lambda$, to transform the inequality 12.21 written in terms of the particle momentum to an expression in terms of the wavelength λ. By differentiating the de Broglie's relation with respect to the particle position, we obtain:

$$\partial P_\alpha/\partial X_\alpha = -(2\pi\hbar/\lambda^2)\,\partial\lambda/\partial X_\alpha$$

It is deduced that $|\partial P_\alpha/\partial X_\alpha|/P_\alpha^2 = 1/(2\pi\hbar)(\partial\lambda/\partial X_\alpha)$. Since $1 \gg \hbar|\nabla_\alpha \cdot \mathbf{P}_\alpha|/\mathbf{P}_\alpha^2$, the quantity $\partial\lambda/\partial X_\alpha$ is found to be very small.

$$(2\pi)^{-1}\,\partial\lambda/\partial X_\alpha \ll 1$$

If $\partial\lambda/\partial X_\alpha$ is approximated by λ/L, with L a characteristic dimension of the system, then an alternative condition to have a classical description is:

$$\lambda \ll L \tag{12.22}$$

The degree of quantum character of the system is equally determined from the *de Broglie thermal wavelength*, λ_T.

$$\lambda_T = h/(2\pi m k_B T)^{1/2} \tag{12.23}$$

The factors h and k_B are the Planck and Boltzmann constants, and T is the temperature. By following similar steps in the transition from quantum optics to ray optics, when λ_T is smaller than the characteristic spatial variations of the interaction potential between particles, the condition of classicality on the particle system is observed [8].

Problem 12.3 (i) Suppose a classical harmonic oscillator with mass m, spring stiffness k, and angular vibrational frequency ω. The energy E_c is given by:

$$E_c = kx^2/2 \quad ; \quad \omega^2 = k/m$$

On the other hand, the energy of a quantum oscillator is $E_q = (n+1/2)\hbar\omega$, where n is the quantum number ($n = 0, 1, \cdots$), $\hbar = h/(2\pi)$, and h is the Planck constant. Estimate the quantum number of a grain of pollen with mass $2.5 \times 10^{-10}\, Kg$, vibrating with frequency $\omega = 1\, rad/s$, and amplitude $1\, mm$. (ii) Obtain the wavelength of a proton with thermal energy $3k_BT/2$ when the temperature is $300\ K$.

Solution

(i) The quantum number is solved in terms of the energy E_q, giving $n = E_q/(\hbar\omega) - 1/2$. The classical energy of the particle is $E_c = kx^2/2$. By using the expression $k = m\omega^2$, the energy becomes $E_c = m\omega^2 A^2/2$. If the energy value is associated with the quantum energy, the corresponding quantum number is $n = m\omega A^2/(2\hbar) - 1/2$. The insertion of the numeric values gives $n \simeq 1 \times 10^{+20}$. Thus, the pollen grain may be treated as a classical particle due to the big quantum number. (ii) According to de Broglie's relation, the wavelength of the particle is $\lambda = h/p$, where p is the particle's momentum. The kinetic energy is the thermal energy.

$$E = p^2/(2m) = 3k_BT/2$$

The momentum is written in terms of the wavelength, to receive $\lambda = \sqrt{h^2/(3mk_BT)}$. The higher the momentum, or temperature, the smaller λ.

(continued)

By inserting values (from the *National Institute of Standards and Technology* webpage) of h, k_B, and the proton mass m, the wavelength is determined.

$$\lambda \sim 1.45\text{Å}$$

Depending on the characteristic spatial variations of the interaction potential between particles, the proton may be considered a classical or quantum particle at 300 K.

12.4 Simplification of the Nuclear Potential

The next goal is to avoid the presence of the nuclear wave function in the nuclear potential which acts on the electrons, Eq. 12.18, to fully reach the classical limit of the nuclei.

$$\int \psi^*_{nuc}(\{\mathbf{R}_\alpha\}; t)\, \hat{V}\, \psi_{nuc}(\{\mathbf{R}_\alpha\}; t)\, d\mathbf{R} = \int \psi^*_{nuc}[\hat{V}_{ee} + \hat{V}_{Ze} + \hat{V}_{ZZ}]\psi_{nuc}\, d\mathbf{R}$$

The small spatial extension of the nucleus with respect to that of the electron, with a difference of 4 to 5 orders of magnitude, shows the nucleus as a point particle. In this regard, the spatial profile of a nucleus may be considered sufficiently localized, and the particle density may be represented by a product of Dirac-delta functions.

$$|\psi_{nuc}\left(\{\mathbf{R}_\alpha(t)\}; t\right)|^2 = \Pi_\alpha\, \delta\left(\mathbf{R}_\alpha - \mathbf{R}_\alpha(t)\right) \tag{12.24}$$

It is with this particle density that the complexity of the nuclear potential is reduced. We start by evaluating each term of the potential. Firstly, we deal with the nucleus-nucleus $\hat{V}_{ZZ}$ term.

$$\int \psi^*_{nuc}\left[\sum_{\alpha<\beta} \frac{Z_\alpha Z_\beta\, e^2}{|\mathbf{R}_\alpha - \mathbf{R}_\beta|}\right]\psi_{nuc}\, d\mathbf{R} = \sum_{\alpha<\beta} Z_\alpha Z_\beta \int \left[\frac{\Pi_\gamma \delta(\mathbf{R}_\gamma - \mathbf{R}_\gamma(t))\, e^2}{|\mathbf{R}_\alpha - \mathbf{R}_\beta|}\right] d\mathbf{R}$$

It is convenient to take the properties of the delta function into consideration for the integral evaluation.

$$\int F(\mathbf{R})\, \delta\left(\mathbf{R} - \mathbf{R}(t)\right)\, d\mathbf{R} = F(\mathbf{R}(t)) \tag{12.25}$$

The integration gives the same function F of the kernel, but evaluated on the variable $\mathbf{R}_\alpha$ which shows time dependence. In the case of the nucleus-nucleus function, we obtain a similar result. The integrations over the variables different to $\mathbf{R}_\alpha$ and $\mathbf{R}_\beta$ give, by orthonormality conditions, factors that produce unity, so we are left with the integrals that only involve the $\mathbf{R}_\alpha$ and $\mathbf{R}_\beta$ variables.

$$\sum_{\alpha<\beta} Z_\alpha Z_\beta \, e^2 \int \left[\frac{\delta \left(\mathbf{R}_\alpha - \mathbf{R}_\alpha(t) \right) \, \delta \left(\mathbf{R}_\beta - \mathbf{R}_\beta(t) \right)}{|\mathbf{R}_\alpha - \mathbf{R}_\beta|} \right] d^3 R_\alpha d^3 R_\beta$$

By integrating over the β coordinate, and later over the α coordinate, we obtain an expression where the nuclear coordinates show time dependence, inherited from the particle density.

$$\sum_{\alpha<\beta} Z_\alpha Z_\beta \, e^2 \int \left[\frac{\delta \left(\mathbf{R}_\alpha - \mathbf{R}_\alpha(t) \right)}{|\mathbf{R}_\alpha - \mathbf{R}_\beta(t)|} \right] d^3 R_\alpha = \sum_{\alpha<\beta} \frac{Z_\alpha Z_\beta \, e^2}{|\mathbf{R}_\alpha(t) - \mathbf{R}_\beta(t)|}$$

A similar integration is done for the $\hat{V}_{Ze}$ contribution, which considers the electron-nucleus interaction. The result equally exhibits time dependence through the nuclear coordinates.

$$\int \psi^*_{nuc} \left(\{\mathbf{R}_\alpha; t\} \right) \hat{V}_{Ze} \left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\} \right) \psi_{nuc} \left(\{\mathbf{R}_\alpha; t\} \right) \, d\mathbf{R} = V_{Ze} \left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha(t)\} \right) \tag{12.26}$$

Problem 12.4 Demonstrate that the electron-nucleus interaction is simplified by:

$$\int \psi^*_{nuc} \left(\{\mathbf{R}_\alpha; t\} \right) \hat{V}_{Ze} \left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\} \right) \psi_{nuc} \left(\{\mathbf{R}_\alpha; t\} \right) \, d\mathbf{R} = V_{Ze} \left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha(t)\} \right)$$

when the particle density corresponds to a product of delta functions.

Solution The demonstration follows similar steps to those given for the nucleus-nucleus interaction. Let us include the forms of the potential and the density in the equation.

$$\int \psi^*_{nuc} \hat{V}_{Ze} \left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\} \right) \psi_{nuc} \, d\mathbf{R} = - \sum_{\alpha=1}^{N} \sum_{i=1}^{n} Z_\alpha e \int \frac{\Pi_\gamma \delta(\mathbf{R}_\gamma - \mathbf{R}_\gamma(t))}{|\mathbf{r}_i - \mathbf{R}_\alpha|} \, d\mathbf{R}$$

For simplicity, we consider the β term of the summation over the nuclei and unfold the product of delta functions.

(continued)

$$\sum_{i=1}^{n} \int \frac{\prod_\gamma \delta(\mathbf{R}_\gamma - \mathbf{R}_\gamma(t))}{|\mathbf{r}_i - \mathbf{R}_\beta|}\, d\mathbf{R} = \sum_{i=1}^{n} \int \delta(\mathbf{R}_1 - \mathbf{R}_1(t))\, d\mathbf{R}_1 \cdots \int \delta(\mathbf{R}_{\beta-1} - \mathbf{R}_{\beta-1}(t))\, d\mathbf{R}_{\beta-1} *$$

$$* Z_\beta e \int \frac{\delta(\mathbf{R}_\beta - \mathbf{R}_\beta(t))}{|\mathbf{r}_i - \mathbf{R}_\beta|}\, d\mathbf{R}_\beta \int \delta(\mathbf{R}_{\beta+1} - \mathbf{R}_{\beta+1}(t))\, d\mathbf{R}_{\beta+1} \cdots \int \delta(\mathbf{R}_N - \mathbf{R}_N(t))\, d\mathbf{R}_N$$

All the integrals give unity, except the one containing the electron-nucleus potential which, by using Eq. 12.25, gives the potential at the instant of time t.

$$\sum_{i=1}^{n} Z_\beta e \int \frac{\delta(\mathbf{R}_\beta - \mathbf{R}_\beta(t))}{|\mathbf{r}_i - \mathbf{R}_\beta|}\, d\mathbf{R} = \sum_{i=1}^{n} \frac{Z_\beta e}{|\mathbf{r}_i - \mathbf{R}_\beta(t)|}$$

When we reinstall the sum over the nuclei, we recover the expression of the potential $\hat{V}_{Ze}$, but with a time dependence through the nuclear coordinates.

$$-\sum_{\beta=1}^{N} \sum_{i=1}^{n} \frac{Z_\beta e}{|\mathbf{r}_i - \mathbf{R}_\beta(t)|} \equiv \hat{V}_{Ze}\left(\{\mathbf{r}_i\}, \{\mathbf{R}_\beta(t)\}\right)$$

With the consideration of classical point-like particles, the final potential acting on the electrons is independent of the nuclear wave function.

$$\int \psi^*_{nuc}(\{\mathbf{R}_\alpha\}; t)\, [\hat{V}_{ee} + \hat{V}_{Ze} + \hat{V}_{ZZ}]\, \psi_{nuc}(\{\mathbf{R}_\alpha\}; t)\, d\mathbf{R} = \sum_{i=1}^{n} \sum_{j>i} \frac{e^2}{r_{ij}}$$

$$- \sum_{\alpha=1}^{N} \sum_{i=1}^{n} \frac{Z_\alpha e^2}{|\mathbf{r}_i - \mathbf{R}_\alpha(t)|} + \sum_{\alpha=1}^{N} \sum_{\alpha<\beta} \frac{Z_\alpha Z_\beta\, e^2}{|\mathbf{R}_\alpha(t) - \mathbf{R}_\beta(t)|}$$

By paying attention to the coordinates $\mathbf{R}_\alpha$ and their time dependence, the above result is summarized using bra-ket notation in a single line.

$$\langle \psi_{nuc}(\{\mathbf{R}_\alpha\}; t)\, |\hat{V}\, (\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha\})|\, \psi_{nuc}(\{\mathbf{R}_\alpha\}; t)\rangle \Big|_{\hbar \to 0} = \hat{V}\, (\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha(t)\}) \tag{12.27}$$

The new set of equations ruling the electron and nuclear dynamics changes to:

$$\begin{aligned} \frac{d\mathbf{P}_\alpha}{dt} &= -\nabla_\alpha \int \psi^*_{elec}\, (\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}, t)\ \hat{\mathcal{H}}_{elec}\, \psi_{elec}\, (\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}, t)\ d\mathbf{r} \\ i\hbar\, \frac{\partial \psi_{elec}}{\partial t} &= \left[-\sum_{i=1}^{n} \frac{\hbar^2}{2m_e} \nabla_i^2 + \hat{V}\, (\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}) \right] \psi_{elec}(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}, t) \end{aligned} \tag{12.28}$$

In the description of the nuclei as classical particles, the electronic wave function ψ_{elec} has acquired dependence on the instantaneous nuclear positions $\mathbf{R}_\alpha(t)$, because the potential $\hat{V}$ in the last equation has dependence on such coordinates. This is the reason to write the electronic wave function with arguments in the form $\psi_{elec}\left(\{\mathbf{r}_i\};\{\mathbf{R}_\alpha(t)\},t\right)$. The electronic Hamiltonian is defined from the above expression by:

$$\hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\};\{\mathbf{R}_\alpha(t)\})\Big|_{\hbar\to 0} = -\sum_{i=1}^{n}\frac{\hbar^2}{2m_e}\nabla_i^2 + \hat{V}\left(\{\mathbf{r}_i\};\{\mathbf{R}_\alpha(t)\}\right) \tag{12.29}$$

This Hamiltonian takes the same functional form of the electronic Hamiltonian, Eq. 11.9, but with nuclear coordinates showing time dependence. The behavior of the electrons and nuclei persists correlated because, after the previous approximations, the Newtonian nuclei remain in an average electronic potential, while the electrons are immersed in an average nuclear potential. The averaged correlation between particles permits the redistribution of the energy between the electrons and nuclei. Still, the correlation is only approximately accounted for because, in working with average potentials, only the averaged collective behavior (not the individual behavior of each particle) is considered in the interaction between the different sets of particles. The average correlation demands that the equations ruling the dynamics of both types of particles are self-consistently solved.

$$\begin{aligned}
\frac{d\mathbf{P}_\alpha}{dt} &= -\nabla_\alpha V_{elec}\left(\{\mathbf{R}_\alpha(t)\}\right)\\
V_{elec}\left(\{\mathbf{R}_\alpha\};t\right) &= \int \psi^*_{elec}(\{\mathbf{r}_i\};t)\,\hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\},\{\mathbf{R}_\alpha\})\,\psi_{elec}(\{\mathbf{r}_i\};t)\,d\mathbf{r}\\
\hat{\mathcal{H}}_{elec} &= -\sum_{i=1}^{n}\frac{\hbar^2}{2m_e}\nabla_i^2 + \sum_{i=1}^{n}\sum_{j>i}\frac{e^2}{r_{ij}} - \sum_{\alpha=1}^{N}\sum_{i=1}^{n}\frac{Z_\alpha e^2}{r_{\alpha i}}\\
&\quad + \sum_{\alpha=1}^{N}\sum_{\beta>\alpha}\frac{Z_\alpha Z_\beta}{R_{\alpha\beta}}e^2\\
i\hbar\,\frac{\partial\psi_{elec}(\{\mathbf{r}_i\};\{\mathbf{R}_\alpha(t)\},t)}{\partial t} &= \hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\};\{\mathbf{R}_\alpha(t)\})\,\psi_{elec}(\{\mathbf{r}_i\};\{\mathbf{R}_\alpha(t)\},t)
\end{aligned} \tag{12.30}$$

In the approximation of point-like particles, we may also write the results with bra-ket notation:

$$\frac{d\mathbf{P}_\alpha}{dt} = -\nabla_\alpha\langle\psi_{elec}|\hat{\mathcal{H}}_{elec}|\psi_{elec}\rangle \quad ; \quad i\hbar\,\frac{\partial\psi_{elec}}{\partial t} = \hat{\mathcal{H}}_{elec}\,\psi_{elec} \tag{12.31}$$

We may use the fact that the electrons are the tiny and fast particles, and the nuclei the heavy and slow particles to invoke the stationary state of the electrons. This is, the electrons are supposed to rapidly reach the stationary state with nuclei moving slowly. Under such a situation, the time-independent wave equation replaces the time-dependent wave equation of the electrons, Sect. 10.4.

$$\frac{d\mathbf{P}_\alpha}{dt} = -\nabla_\alpha\langle\psi_{elec}|\hat{\mathcal{H}}_{elec}|\psi_{elec}\rangle \quad ; \quad \hat{\mathcal{H}}_{elec}\,\psi_{elec} = E_{elec}\,\psi_{elec} \tag{12.32}$$

This is one of the most common sets of equations to perform mixed quantum-classical molecular dynamics. It also receives the name *quantum molecular dynamics* because, whenever the nuclei meet the conditions to satisfy the classical limit, a quantum approach on the nuclei would in principle render the same results to those obtained with Newtonian mechanics on such a nuclei. In this regard, the theory of classical mechanics can be now employed to describe the nuclei of the microscopic world.

12.5 Parameterizing the Potential Function

Eq. 12.32 of the nuclei is further simplified by working with $\langle\psi_{elec}|\hat{\mathcal{H}}_{elec}|\psi_{elec}\rangle$, which is the electronic energy E_{elec} of the molecule and works simultaneously as the potential energy for the nuclei. This energy depends in parametric form on the instantaneous atomic positions, $\{R_\alpha(t)\}$. When the energy E_{elec} is computed for different molecular conformations, $\{R_\alpha(t_1)\}$, $\{R_\alpha(t_2)\}$, $\{R_\alpha(t_3)\}$..., we are in fact scanning the potential energy surface where the nuclei are immersed. In this context, the energy E_{elec} corresponds to the potential energy function V_{elec} for the nuclei, with dependence on the nuclear coordinates.[2]

$$V_{elec}(\{\mathbf{R}_\alpha(t)\}) = E_{elec}\Big|_{\{\mathbf{R}_\alpha\}} \tag{12.33}$$

The potential energy is frequently computed on the fly by solving the quantum mechanical electronic wave equation:

$$E_{elec} = \langle\psi_{elec}|\hat{\mathcal{H}}_{elec}|\psi_{elec}\rangle$$

E_{elec} is obtained at the current conformation $\{\mathbf{R}_\alpha(t)\}$ of the nuclei in the molecular dynamics simulation. The forces over the nuclei are obtained by differentiation of the potential energy function V_{elec}, and evaluating the forces at the current nuclear positions:

$$\mathbf{F}_\alpha = -\nabla_\alpha V_{elec}\{\mathbf{R}_\alpha(t)\}$$

Once the set of forces $\{\mathbf{F}_\alpha\}$ are obtained, these are applied to the nuclei to reach their new positions $\{\mathbf{R}_\alpha(t)\}$. The process is repeated after this step is completed. The major time-consuming aspect of the simulation involves the solution of the wave equation. An alternative approach to improve the time-consuming process of the simulation consists on using a parameterized version of the potential energy

[2] In the equations of the text, the nuclear coordinates play the role of parametric variables. Note that the parametric expression of a function, say $y = y(x)$, introduces a third variable, t, referred to as the *parametric variable*, in terms of which the independent and dependent variables are expressed, reaching their values in the form $y = g(t)$, $x = f(t)$.

V_{elec}, with the purpose of avoiding the process of continuously solving the equation $E_{elec} = \langle \psi_{elec} | \hat{\mathcal{H}}_{elec} | \psi_{elec} \rangle$ at every time step. Such a procedure has important limitations (to be pointed out later), but, on the other hand, it allows to investigate structural properties of molecules in solvation and under conditions of temperature and pressure, which is a difficult task using the wave function approach, especially for large molecules, like the proteins.

In order to illustrate the parameterized approach, consider, for example, a molecule with three atoms: a, b, and c, like water. Their positions are $\mathbf{R}_a$, $\mathbf{R}_b$, $\mathbf{R}_c$, and the equilibrium distance is $\mathbf{R}_0$ for the bonds ab and ac. The $3N - 6 = 3$ corresponding degrees of freedom, dof's, may be represented by the pair distances $\mathbf{R}_{ab}$ and $\mathbf{R}_{ac}$, plus the angle θ_{abc} formed by the vectors $\mathbf{R}_{ab}$ and $\mathbf{R}_{ac}$ with atom a located at the vertex. The angle of equilibrium is θ_0. After computing a set of energies E_{elec} for the elongation and shrinking of the ab bond, with the other dof's fixed at their equilibrium values, $\mathbf{R}_{ac} = \mathbf{R}_0$ and $\theta_{abc} = \theta_0$, the potential energy of this dof may be written as:

$$V_{ab} = k_{ab}(R_{ab} - R_0)^2/2 \quad ; \quad R_{ab} = |\mathbf{R}_{ab}| \tag{12.34}$$

A similar process for the other dof's gives $V_{ac} = k_{ac}(R_{ac} - R_0)^2/2$ and $V_{abc} = k_{abc}(\theta_{abc} - \theta_0)^2/2$. The force constants, equilibrium distance, and equilibrium angle, k_{ab}, k_{ac}, k_{abc}, $\mathbf{R}_0$, θ_{abc}, are parameters determined after fitting the potential energies V_{ab}, V_{ac} and V_{abc} to the molecular energies E_{elec}, simultaneously reproducing the structure of the reference molecule. The force constants of other molecules, different from the water molecule, are in general different. By using Eqs. 12.32–12.34 and considering the total potential energy function as the addition of individual terms, $V_{elec} = V_{ab} + V_{ac} + V_{abc}$, we obtain:

$$\frac{d\mathbf{P}_\alpha}{dt} = -\nabla_\alpha V_{elec} = -\nabla_\alpha [V_{ab}(R_{ab}, R_0) + V_{ac}(R_{ac}, R_0) + V_{abc}(\theta_{abc}, \theta_0)] \tag{12.35}$$

The model potential is acceptable in the domains of the parameterized dof's solely. When the potential function is used to extrapolate the conformations of the molecule to other unexplored conformations, there is a risk of obtaining nonphysical behavior and wrong energies. For instance, when the atom c is separated from the atoms ab that integrate the example molecule, we are left with an isolated atom c and an isolated dimer ab. Yet, the parameterized potential retains the molecular bond of c to the dimer ab due to the parabolic shape of the harmonic potential function employed in this case. Such a model potential is incapable to deal with the breaking of the $c - ab$ bond.

For molecules with many atoms, the V_{elec} may be represented by a general n-body form [9].

$$V_{elec}(\{\mathbf{R}_\alpha(t)\}) = \Big[\sum E_i^{(1)} + \sum E_{ij}^{(2)} + \sum E_{ijk}^{(3)} + \sum E_{ijkl}^{(4)}$$

$$+\sum E_{ij}^{coul}+\sum E_{ij}^{vdW}\Big]_{\{\mathbf{R}_\alpha\}} \tag{12.36}$$

The expression shows the first additive contributions of a general n-body potential. The first term, $\sum E_i^{(1)}$, gives the addition of atomic energies. When the number of atoms does not change or the atoms do not mute onto different atoms in the particle system, the first term is an energy constant and may be neglected. The following contribution, $\sum E_{ij}^{(2)}$, introduces the pair interactions of neighbor atoms of the same molecule. The potentials employed to describe the pair interactions are frequently of harmonic type in the separation distance, R_{ij}. The next term, $\sum E_{ijk}^{(3)}$, represents the potential associated with trimers, comprised by three neighbor atoms of the same molecule. The potentials used to describe the trimers are frequently of harmonic type in the trimer angle, θ_{jik}. The term, $\sum E_{ijkl}^{(4)}$, is the contribution of tetramers, formed by four neighbor atoms of the same molecule. The potentials employed for tetramers frequently consist of a series of cosine functions in the tetramer torsion angle, θ_{ijkl}. The first four contributions of Eq. 12.36 describe the internal dof's of the molecule, because they involve bonded atoms in the separation distance, the trimer angle, and the tetramer torsional angle. The first four potential terms apply to small or large molecules and, due to their nature, constitute all together the molecular internal energy.

Finally, the last two energy terms of the model potential, Eq. 12.36, represent the non-bonded interactions like the Coulomb, $coul$, and van der Waals, vdW, interactions. They consider the interactions among the i and j atoms, which are distant apart from each other, usually belonging to different molecular groups. The non-bonded atomic interactions introduce the *external forces* or environmental forces acting on the molecule. For example, given a specific water molecule, the non-bonded atomic interactions introduce the forces acting on that specific water molecule by the other water molecules. In short, a model potential like that proposed in Eq. 12.36 represents a parameterization of the energy landscape of the nuclei. The model potential accounts for both the molecular dof's and the atomic interactions by defining groups of atoms and proposing the internal and external potential functions of the atomic groups. The force constants, the equilibrium distances, and the equilibrium angles represent all together a way of introducing the information referent to the structure and rigidity of the molecule in the force functions previous to the simulations. The method displayed here is known as *molecular mechanics*. It constitutes a branch of classical molecular dynamics, while the proposal of model potentials takes us to the area of forcefields, to be unfolded later. In particular, when the potentials of harmonic type are employed, and the force constants have relatively large values or the bond distances and bond angles are fixed in the simulations, Sect. 15.7, we may transit from molecular mechanics to a *ball-and-stick approximation*.

12.6 Total Energy of the Molecular System

The level of theory employed in the description of the electrons is different to that used in the description of the nuclei, because the electrons are treated as quantum particles and the nuclei as classical particles. The addition of the kinetic energy plus the potential energy gives the energy of the particle system.

$$E = \sum_\alpha \frac{\mathbf{P}_\alpha^2}{2M_\alpha} + \int \psi_{elec}^* \left(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}, t\right) \hat{\mathcal{H}}_{elec} \, \psi_{elec} \left(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}, t\right) d\mathbf{r} \tag{12.37}$$

When the energy is preserved over time, the time derivative of the energy is zero.

$$\frac{dE}{dt} = \sum_\alpha \frac{\mathbf{P}_\alpha}{M_\alpha} \cdot \frac{d\mathbf{P}_\alpha}{dt} + \int \left[\frac{d\psi_{elec}^*}{dt} \hat{\mathcal{H}}_{elec} \, \psi_{elec} + \psi_{elec}^* \hat{\mathcal{H}}_{elec} \frac{d\psi_{elec}}{dt} + \psi_{elec}^* \frac{d\hat{\mathcal{H}}_{elec}}{dt} \psi_{elec} \right] d\mathbf{r} = 0$$

If the following three conditions are met: (i) ψ_{elec} satisfies the wave equation, (ii) ψ_{elec}^* satisfies the opposite-sign wave equation, and (iii) the Hamiltonian is a Hermitian operator, then the first two terms in square brackets eliminate each other. By recalling that $\hat{\mathcal{H}}_{elec}$ depends on t by means of the nuclear coordinates, the time derivative $d\hat{\mathcal{H}}_{elec}/dt$ is evaluated by applying the chain rule.

$$\sum_\alpha \frac{\mathbf{P}_\alpha}{M_\alpha} \cdot \frac{d\mathbf{P}_\alpha}{dt} = -\int \psi_{elec}^* \sum_\alpha \nabla_\alpha \hat{\mathcal{H}}_{elec} \cdot \frac{d\mathbf{R}_\alpha}{dt} \psi_{elec} \, d\mathbf{r}$$

The factor $d\mathbf{R}_\alpha/dt$ is identified with $\mathbf{P}_\alpha/M_\alpha$, and the force on each nucleus is deduced.

$$\frac{d\mathbf{P}_\alpha}{dt} = -\int \psi_{elec}^* \nabla_\alpha \hat{\mathcal{H}}_{elec} \, \psi_{elec} \, d\mathbf{r} \tag{12.38}$$

This expression reproduces the first expression of Eqs. 12.28. The electronic Hamiltonian is in turn divided into kinetic energy and potential energy terms, Eq. 12.30. The electron kinetic energy operator may be considered as a function with the purpose to carry the derivation; nevertheless, it contains no nuclear coordinates and is neglected. On the other side, the potential energy terms of the electronic Hamiltonian are, Eq. 12.27.

$$\hat{V}\left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha(t)\}\right) = -\sum_{i=1}^{n} \frac{\hbar^2}{2m_e} \nabla_i^2 + \sum_{i=1}^{n} \sum_{j>i} \frac{e^2}{r_{ij}} - \sum_{\alpha=1}^{N} \sum_{i=1}^{n} \frac{Z_\alpha e^2}{r_{i\alpha}} + \sum_{\alpha=1}^{N} \sum_{\beta>\alpha} \frac{Z_\alpha Z_\beta}{R_{\alpha\beta}} e^2$$

By defining a force like in classical theory, this is, as the negative of the gradient acting upon the potential function, it is possible to compute the force over the αth nucleus.

$$\mathbf{F}_\alpha = -\nabla_\alpha \hat{V}\left(\{\mathbf{r}_i\}, \{\mathbf{R}_\alpha(t)\}\right) \tag{12.39}$$

This expression is the *general Hellmann-Feynman theorem*. It states that the force acting upon the αth nucleus is due to the electrostatic potentials and the electronic wave function [10]. Care should be exercised when ψ_{elec} is not the exact wave function but an approximate one, because the previous expression may produce spurious forces. By paying attention to the specific contributions of $\hat{V}$ with dependence on the nuclear coordinates, we have:

$$\mathbf{F}_\alpha = \int \psi^*_{elec} \left[-\sum_{i=1}^{n} \nabla_\alpha V_{\alpha i} - \sum_{\beta=1}^{N} \nabla_\alpha V_{\alpha\beta} \right] \psi_{elec}\, d\mathbf{r}$$

$$V_{\alpha i} = -Z_\alpha e^2 / r_{i\alpha} \quad ; \quad V_{\alpha\beta} = Z_\alpha Z_\beta e^2 / R_{\alpha\beta} \quad ; \quad \alpha \neq \beta$$

From the fact that the nuclear and electronic coordinates in the $V_{\beta i}$ potential are shown in symmetric form, it is possible to use the equivalence $\nabla_\alpha V_{\beta i} = -\delta_{\alpha\beta} \nabla_i V_{\beta i}$, with $\delta_{\alpha\beta} = 0$ for $\alpha \neq \beta$ and $\delta_{\alpha\beta} = 1$ for $\alpha = \beta$, in the expression of the force. In particular, $\nabla_\alpha V_{\alpha i} = -\nabla_i V_{\alpha i}$. The force shall be further simplified by carrying integrations over selected electronic coordinates.

$$\mathbf{F}_\alpha = \int \sum_{i=1}^{n} \nabla_i V_{\alpha i} \left[\int \psi^*_{elec} \psi_{elec}\, d^3 r_1 \cdots d^3 r_{i-1}\, d^3 r_{i+1} \cdots d^3 r_n \right] d^3 r_i + \\ - \sum_{\beta=1}^{N} \nabla_\alpha V_{\alpha\beta}$$

Problem 12.5 Demonstrate the following identities in the potentials $V_{\beta i}$ and $V_{\beta\gamma}$.

$$\nabla_\alpha V_{\beta i} = -\delta_{\alpha\beta} \nabla_i V_{\beta i} \quad ; \quad \nabla_\alpha \sum_{\substack{\beta,\gamma \\ \gamma > \beta}} V_{\beta\gamma} = \sum_{\substack{\beta \\ \beta \neq \gamma}} \nabla_\alpha V_{\alpha\beta}$$

Solution The constants of the potentials are omitted for simplicity in the derivations. Firstly, we recall the definition of the attractive potential $V_{\alpha i}$ and perform the differentiation with respect to the α coordinate.

$$\nabla_\alpha V_{\alpha i} = (\partial_{X_\alpha}, \partial_{Y_\alpha}, \partial_{Z_\alpha}) \left[-1/|\mathbf{r}_i - \mathbf{R}_\alpha| \right] = -(\mathbf{r}_i - \mathbf{R}_\alpha)/|\mathbf{r}_i - \mathbf{R}_\alpha|^3$$

By similarly operating over the coordinates of particle i, we have:

(continued)

$$\nabla_i V_{\alpha i} = (\mathbf{r}_i - \mathbf{R}_\alpha)/|\mathbf{r}_i - \mathbf{R}_\alpha|^3$$

The result is $\nabla_\alpha V_{\alpha i} = -\nabla_i V_{\alpha i}$. If the β coordinate is considered, a delta function is required, and the conclusion is $\nabla_\alpha V_{\beta i} = -\delta_{\alpha\beta}\nabla_i V_{\beta i}$. In order to proof the second identity, we exemplify a particular case to rationalize the general case. By showing the components of the summation, we have:

$$\nabla_\alpha \sum\nolimits_{\substack{\beta,\gamma \\ \gamma>\beta}} V_{\beta\gamma} = \nabla_\alpha \left(V_{12} + V_{13} + V_{14} + \cdots + V_{1N} + \right.$$
$$\left. + V_{23} + V_{24} + \cdots + V_{2N} + V_{34} + \cdots + V_{(N-1)N}\right)$$

Suppose $\alpha = 3$. The non-zero terms are those containing subindex $\alpha = 3$.

$$\nabla_{\alpha=3} \sum_{\substack{\beta,\gamma \\ \gamma>\beta}} V_{\beta\gamma} = \nabla_{\alpha=3}(V_{13} + V_{23} + V_{34} + V_{35} + \cdots + V_{3N})$$

$$= \sum_{\substack{\beta=1 \\ \alpha\neq\beta}}^{N} \nabla_{\alpha=3} V_{\alpha=3\,\beta}$$

It was considered $V_{\alpha\beta} = V_{\beta\alpha}$ in the last identity. The generalization of the result is done by replacing $\alpha = 3$ with $\alpha = \kappa$, namely, $\nabla_\kappa \sum_{\substack{\beta,\gamma \\ \gamma>\beta}} V_{\beta\gamma} = \sum_{\beta=1}^{N} \nabla_\kappa V_{\kappa\beta}$, and $\beta \neq \kappa$.

The term in square brackets of the previous expression is the *electron density*, ρ_{elec}, a function of a single variable.

$$\rho_{elec}(\mathbf{r}_i) = n \int \psi^*_{elec}\psi_{elec}\, d^3r_1 \cdots d^3r_{i-1}\, d^3r_{i+1} \cdots d^3r_n$$

$$\mathbf{F}_\alpha = \tfrac{1}{n}\left[\int \nabla_1 V_{\alpha 1}\, \rho_{elec}(\mathbf{r}_1) d^3r_1 + \int \nabla_2 V_{\alpha 2}\, \rho_{elec}(\mathbf{r}_2) d^3r_2 + \cdots\right] - \textstyle\sum_{\beta=1}^{N} \nabla_\alpha V_{\alpha\beta}$$

Variable $\mathbf{r}_i$, with $i = 1, 2, \cdots, n$, is a dummy variable due to the indistinguishability of the electrons, and instead, we may use $\mathbf{r}$ in every term. On the other hand, the factor $\nabla_i V_{\alpha i}$ is understood as the electric field $\mathbf{E}_\alpha(\mathbf{r}_i)$ produced by the electron distribution, which acts on the α nucleus. The *Hellmann-Feynman force* is defined by:

$$\mathbf{F}_\alpha = \tfrac{1}{n}\left[\int \mathbf{E}_\alpha(\mathbf{r}_1)\, \rho_{elec}(\mathbf{r}_1) d^3r_1 + \int \mathbf{E}_\alpha(\mathbf{r}_2)\, \rho_{elec}(\mathbf{r}_2) d^3r_2 + \cdots\right] - \textstyle\sum_{\beta=1}^{N} \nabla_\alpha V_{\alpha\beta}$$

$$\mathbf{E}_\alpha(\mathbf{r}_i) = \nabla_i V_{\alpha i} \tag{12.40}$$

The force on a nucleus is the classical electrostatic force exerted by the other point-like nuclei and the electron charge density. We additionally observe that force changes over the nuclei are related to electron-distribution changes and vice versa. The n individual terms of the above expression are all the same, and they are simplified to a single one with factor n. The final expression of the force is referred to as the *electrostatic theorem*, because the force on a nucleus is due to both a continuous electron distribution and a discrete nuclear distribution. In concise manner, we write:

$$\mathbf{F}_\alpha = \int \mathbf{E}_\alpha(\mathbf{r})\, \rho_{elec}(\mathbf{r}) d^3r - \sum_{\beta=1}^{N} \nabla_\alpha V_{\alpha\beta} \tag{12.41}$$

The computation of the force in the Hellmann-Feynman approach is described by functions with dependence on 3 spatial degrees of freedom instead of the $3n$ spatial degrees of freedom associated with n electrons.

12.7 Establishing the Accuracy of Atomic Forces

After considering several simplifications in the equations of the nuclei, it is important to establish the accuracy of the atomic forces, Eq. 12.38, with the purpose to reach reliable values of the physical observables. To do this, we are required to estimate the deviations of the theoretical forces from the experimental forces. There are different experiments whose results may be used for assessing the accuracy of the atomic forces of the simulations. The experiments using the atomic force microscope or optical tweezers may directly provide the atomic forces; nevertheless, those experiments are not abundant. On the other side, there is a relatively large number of experiments that provide the vibrational spectra of the individual molecules. The accuracy of the atomic forces may be indirectly determined from the vibrational spectra [11].

Suppose a molecule with N atoms moving with no constraints. The number of degrees of freedom is $3N - 6$. In the calculation of the vibrational spectrum of the molecule, the *Hessian matrix*, h_{ij}, is proposed.

$$h_{ij} = \begin{pmatrix} \frac{\partial^2 V}{\partial x_1^2} & \frac{\partial^2 V}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 V}{\partial x_1 \partial x_N} \\ \frac{\partial^2 V}{\partial x_2 x_1} & \frac{\partial^2 V}{\partial x_2^2} & \cdots & \frac{\partial^2 V}{\partial x_2 \partial x_N} \\ \vdots & & \vdots & \\ \frac{\partial^2 V}{\partial x_N x_1} & \frac{\partial^2 V}{\partial x_N x_2} & \cdots & \frac{\partial^2 V}{\partial x_N^2} \end{pmatrix} \tag{12.42}$$

The Hessian matrix depends on the second derivatives of the potential with respect to the atomic positions, x_i. It gives information of the multidimensional topography of the potential energy surface associated with the molecule. The

Hessian matrix is diagonalized to obtain the eigenvectors, known as the normal vibrational modes, and the eigenvalues, known as the vibrational frequencies. The strength of each normal vibration mode gives the corresponding vibrational intensity. The $3N-6$ normal vibration modes, and equal number of vibrational frequencies and intensities, constitute all together the *molecular vibrational spectrum*. Such a spectrum may be determined from experiments in the lab. By estimating the deviations between the theoretical and experimental spectral data, it is possible to indirectly establish the reliability of the atomic forces employed in the simulations. The atomic forces represent the first derivatives of the potential with respect to the atomic positions, and the Hessian matrix introduces the second derivatives of the potential with respect to the atomic positions. Therefore, the proper prediction of the molecular vibrational spectrum indicates acceptable second derivatives, indirectly pointing out bona fide first derivatives.[3]

The experiments with atomic force microscopes or optical tweezers provide accurate values of the atomic forces. The number of such experiments is not as large as those giving the vibrational spectra of molecules. Still, the atomic forces from those experiments or alternative experiments should be used whenever available for an additional check of the atomic forces [14].

12.8 Diffusion from the Continuity Equation

The goal of this section is to deduce an equation for the statistical behavior of a set of classical particles that diffuse in a fluid. The diffusion of particles is important in processes that involve the doping of materials, the homogenization of substances in liquids, the sedimentation of particles, etc. The discussion departs from the equation of continuity 12.8, which is a conservation law on the number of particles since it ensures that the particles penetrating an imaginary volume in the fluid are precisely the ones that come out of that volume. Thus, any density change of the diffusing particles in a volume of the fluid is only attributed to the inflow and outflow of particles. The set of particles have an initial concentration in a spatial region of the fluid, and, after some time, the particles spread evenly with random trajectories over a larger volume. The equation of continuity 12.8 is reproduced below for ease reference.

$$\frac{\partial \rho(\mathbf{R};t)}{\partial t} + \nabla \cdot \mathbf{J}\,(\mathbf{R};t) = 0 \quad ; \quad \mathbf{J}\,(\mathbf{R};t) = \sum_{\alpha=1}^{N} \frac{1}{M_\alpha}\, D^2\, \nabla_\alpha S \tag{12.43}$$

The function ρ is the density of diffusing particles in terms of $\mathbf{R}$ at the instant of time t. The function $\mathbf{J}$ is the *particle flux* passing through a unit area per unit time. $\mathbf{J}$ is also referred to as the *particle current*. The expression $\mathbf{P}_\alpha = \nabla_\alpha S$ was given in

[3] Santamaria et al. [12]. Also, for an example of a vibrational spectrum where the position of a single atom of a molecule is differentiated from the others, refer to: Mollinedo-Rosado and Santamaria [13].

Eq. 12.15; thereby, the flux is proportional to the velocities $\mathbf{V}_\alpha$ of the particles. To be consistent with the delta profile of the particle density of Eq. 12.24, and using the identity $\rho = D^2$ of Sect. 12.2, a delta profile is also assigned to the particle current.

$$\mathbf{J} = \sum_{\alpha=1}^{N} \mathbf{V}_\alpha \, \delta(\mathbf{R} - \mathbf{R}_\alpha(t)) \quad ; \quad \mathbf{V}_\alpha = d\mathbf{R}_\alpha/dt \tag{12.44}$$

Initially, the high particle concentration tends to disperse, and after some time, the particles achieve a uniform concentration in a homogeneous fluid. For slowly varying distributions of the particles, it is possible to provide an expression for the change of the particle concentration. A subtle change of concentrations in the z axis may be described as a linear relation between the concentration ρ and the position z, this is $\rho = \alpha z$, where the constant of proportionality is α. In mathematical terms, α is a slope, and in the general case, it is a gradient, $\alpha = \nabla\rho(\mathbf{R})$. This is, α represents the rate of concentration change with respect to the position. The particle flux per unit area is related to the diffusion constant, d, by $J = -\alpha d$. In this regard, we have an equation relating the particle flux to both the rate of concentration change and the particle diffusion constant.

$$\mathbf{J} = -d\,\nabla\rho(\mathbf{R}) \tag{12.45}$$

This equation relates the particle current $\mathbf{J}$ to the gradient of ρ using the diffusion constant d. It is known as *first Fick's law*. When the gradient of the density is zero, the average density is constant (there is a uniform concentration of the diffusing particles), and no particle currents or fluxes $\mathbf{J}$ are perceptible [15].

12.9 Diffusion Equation and Particle Flux

The goal of this section is to create consistency among the equations that involve parameters referent to the diffusion of the particles. It is convenient to start by introducing the expression of $\mathbf{J}$ given by the first Fick's law, Eq. 12.45, into the continuity equation, Eq. 12.43, and under the assumption of a diffusion coefficient d with no dependence on the density of the diffusing particles.

$$\partial\rho(\mathbf{R};t)/\partial t = d\,\nabla^2\rho(\mathbf{R};t) \tag{12.46}$$

This is a partial differential equation which involves key ingredients of the diffusion process. It receives the name of *diffusion equation* or *second Fick's law*, as it describes the spatial dispersion of the particles in terms of the distribution function, ρ. The interactions of the particles are not explicitly considered in the diffusion equation. Also, the particle dispersion is time irreversible, problem 15.3. The diffusion factor has units of square length over time. The solution of the equation is unique when the conditions on the spatial distribution at the instant of time t_0 are imposed. The diffusion equation may be mutated onto an equation for the heat diffusion: if the temperature plays the role of the particle concentration, and the

thermal conductivity replaces the diffusion factor, then the new expression becomes the *heat equation*. It is equally possible to deduce the wave equation from the continuity equation [16]. The diffusion equation has wide applications in different fields of science.

When the system of particles achieves a uniform density, ρ = constant, there are no spatial and temporal changes of the concentration. The first Fick's law is recovered from the second Fick's law due to the fact that after using Eqs. 12.43 and 12.46, we reproduce the first Fick's law, $\mathbf{J} = -d\nabla\rho$. More precisely, we have $\partial\rho\,(\mathbf{R};t)\,/\partial t + \nabla\cdot\mathbf{J}\,(\mathbf{R};t) = 0$, and on the other hand, $\partial\rho(\mathbf{R};t)/\partial t = d\,\nabla^2\rho(\mathbf{R};t)$. By comparing terms, we obtain $d\nabla^2\rho = -\nabla\cdot\mathbf{J}$. Thus, the first Fick's law represents a first integration of the second Fick's law. A solution of the diffusion equation is $\rho(R;t) = \exp[i(kR-\omega t)]$ with frequency:

$$\omega = -i\,d\,(2\pi/\lambda)^2 \quad ; \quad k = 2\pi/\lambda$$

The λ factor describes the periodic variation of the density. It is possible to express a general solution of Eq. 12.46 as a linear combination of plane waves. More general solutions of the diffusion equation than these described here are found elsewhere.[4]

Problem 12.6 The one-dimension diffusion equation is $\partial F/\partial t = d\,(\partial^2 F/\partial x^2)$. Find the solution using the method of Fourier transforms.

Solution We have to integrate the diffusion equation to obtain the solution F. The Fourier transform is applied on both sides of the diffusion equation.

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{i\omega x}\,\frac{\partial F}{\partial t}\,dx = \frac{d}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{i\omega x}\,\frac{\partial^2 F}{\partial x^2}\,dx$$

We define the transformed function in the form $f(\omega,t) = \int_{-\infty}^{\infty} e^{i\omega x}\,F(x,t)\,dx/\sqrt{2\pi}$ to have:

$$\frac{\partial f(\omega,t)}{\partial t} = -d\omega^2 f(\omega,t)$$

where the property of Fourier transforms on derivatives was used to replace the second-order derivative in the variable x with the multiplicative factor ω^2. The equation is now an ordinary differential equation of first order in the

(continued)

[4] For instance, refer to: Tveito et al. [17], Chapt. 7 and 8, Caretto [18] notes on engineering analysis, and Ursell [19].

variable t. The integration gives $f(\omega, t) = C \exp(-d\omega^2 t)$, with the factor $C = f(\omega, 0)$ imposing the initial conditions. We apply the inverse Fourier transform to recover F.

$$F(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\omega x}\, C(\omega)\, e^{-d\omega^2 t}\, d\omega$$

The expression is simplified when the coefficient C is constant. The goal is to write an expression in the variable ω by squaring terms and proceed to integrate.

$$F(x,t) = \frac{C}{\sqrt{2\pi}}\, e^{-x^2/(4dt)} \int_{-\infty}^{\infty} \exp\{-\left[\sqrt{dt}\,\omega + ix/(2\sqrt{dt})\right]^2\}\, d\omega$$

The integration is done using the result $\int_{-\infty}^{\infty} d\omega\, \exp[-(\alpha/2)\,(\omega - iy/\alpha)^2] = \sqrt{2\pi/\alpha}$.

$$F(x,t) = \frac{C}{\sqrt{2dt}} \exp\left(-\frac{x^2}{4dt}\right)$$

The diffusion equation in more dimensions is solved similarly, with a result that corresponds to the product of Gaussian functions in the different dimensions.

The following step is to establish an equation of $\mathbf{J}$ when there is an external force acting on the diffusing particles, thus inducing a drift current. This step shall allow to find the diffusion constant in terms of the temperature and medium viscosity. We follow the strategy of Chaikin and Lubensky [20]. Suppose a particle concentration ρ. An external force generated from the external potential V_{ext} induces a drift of the diffusing particles, pushing them into a given direction. The particles are also subjected to the random viscous forces of the fluid. Such forces are supposed to be negatively proportional to the particle's velocity.

$$\mathbf{f} = \mathbf{f}_{ext} - \xi \mathbf{V} \quad ; \quad \mathbf{f}_{ext} = -\nabla \mathbf{V}_{ext}$$

The constant ξ is the viscosity of the fluid, and the inverse of ξ is the *particle mobility*. The velocity $\mathbf{V}$ should not be confused with an energy potential. The total force in the steady state is $\mathbf{f} = 0$. By solving for the particle velocity, we have $\mathbf{V} = -\nabla \mathbf{V}_{ext}/\xi$. Therefore, the external force has effect on the particle current 12.45. Such an effect is accounted for by the term $-\rho\, \nabla V_{ext}/\xi$.

$$\mathbf{J} = -d\,\nabla\rho(\mathbf{R};t) - \rho(\mathbf{R};t)\,\nabla V_{ext}/\xi \tag{12.47}$$

There is equilibrium when the particle current is zero.

$$\nabla \ln \rho(\mathbf{R};t) = -\nabla V_{ext}\,/\,(d\xi)$$

A solution of this equation involving the temperature of the fluid is proposed as an exponential function, $\rho = \exp(-V_{ext}/k_BT)$. It is a *Maxwell-Boltzmann expression*. By using the exponential in the last equation, we obtain a diffusion coefficient which linearly depends on the equilibrium temperature T and is inversely proportional to the viscosity ξ.

$$d = k_BT/\xi \tag{12.48}$$

This expression is the *diffusion constant* initially proposed by Einstein. Equation 12.48 is one of the important *fluctuation-dissipation relations* of stochastic dynamics.

The final purpose is to connect the particle current $\mathbf{J}$ to the chemical potential μ to further characterize the diffusion process. The expression of the chemical potential, μ, is related to the change rate of the densities.

$$\mu = k_BT\ln(\rho/\rho_0) \quad ; \quad \rho_0^{-1} = \left(\frac{h^2}{2\pi M k_BT}\right)^{3/2} = \lambda_T^3$$

The term λ_T is the de Broglie thermal wavelength, Eq. 12.23. By considering the particle current 12.47, and the expression of the diffusion constant d, the new equation of $\mathbf{J}$ is:

$$\mathbf{J} = -(\rho/\xi)\,\nabla(k_BT\ln\rho + V_{ext}) = -(k_BT/\xi)\,\nabla\rho - (\rho/\xi)\,\nabla V_{ext} \tag{12.49}$$

By inserting the expression of the chemical potential in the first identity of $\mathbf{J}$ presented above, the particle current becomes the gradient of the chemical potential difference.

$$\mathbf{J} = -(\rho/\xi)\,\nabla[\mu - \mu_{ext}] \quad ; \quad \mu_{ext} = -V_{ext} \tag{12.50}$$

By considering the free energy in the form $F = Nk_BT[\ln(\rho\lambda_T^3) - 1]$, the particle current is also written in terms of the free energy changes, δF, with respect to the change in the particle number, δN, while maintaining the temperature and confining volume constants.

$$\mathbf{J} = -(\rho/\xi)\,\nabla\left(\frac{\delta F}{\delta N}\right) \tag{12.51}$$

Eqs. 12.50 and 12.51 are expressions of the particle current $\mathbf{J}$ in terms of the chemical potential μ and the free energy F, respectively. On the other side, the mobility of a particle depends not only on the fluid viscosity factor ξ but also on the particle's geometry. According to the Stokes dragging-force law, a spherical particle with radius a has an effective viscosity factor $6\pi\xi a$ [21]. By using the expression of the particle flux 12.49 with $V_{ext} = 0$, we have:

$$\mathbf{J} = -k_B T \, \nabla\rho \, / \, (6\pi\xi a) \tag{12.52}$$

The first Fick's law, Eq. 12.45, indicates that $\mathbf{J} = -d\nabla\rho$; thereby, the diffusion factor for the spherical body is:

$$d = k_B T \, / \, (6\pi\xi a) \tag{12.53}$$

The diffusion factor can be derived from the velocity autocorrelation function and shown to be related to the mean square displacement, $\langle \mathbf{r}^2(t) \rangle$.

12.10 Expansion of the Electronic Wave Equation

It is time to return our attention to the electronic wave function, since the nuclear wave function does not fit into the classical limit. In the following, the electronic wave function is expanded using a linear combination of complete basis set functions. Under this approximation, the solution of the electronic wave equation, Eq. 12.31, is relegated to find the optimal coefficients that satisfy such an equation. We start by writing the electronic wave function as a linear combination of the basis functions $\{\psi^k_{elec}\}$, maintaining the dependence on the electronic coordinates and the parametric dependence on the nuclear coordinates.

$$\psi_{elec}(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}, t) = \textstyle\sum_k c_k(t) \, \psi^k_{elec}(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}) \tag{12.54}$$

The amplitudes or coefficients c_k exhibit time dependence solely. The strategy consists in separating the quantities that depend on the time from those that depend on the spatial variables. The wave function ψ_{elec} satisfies the wave equation of the electrons, Eq. 12.31.

$$i\hbar \, \frac{\partial \psi_{elec}(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}, t)}{\partial t} = \hat{\mathcal{H}}_{elec}(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}) \, \psi_{elec}(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}, t)$$

By introducing the expansion of the wave function and using the chain rule on the time derivatives, the wave equation is presented in terms of the basis functions and their amplitudes.

$$i\hbar \sum_k \left[\psi^k_{elec} \, \frac{\partial c_k(t)}{\partial t} + c_k(t) \sum_\alpha \nabla_\alpha \psi^k_{elec} \cdot \frac{\partial \mathbf{R}_\alpha}{\partial t} \right] = \sum_k c_k(t) \, \hat{\mathcal{H}}_{elec} \, \psi^k_{elec} \tag{12.55}$$

We multiply from the left by $(\psi^j_{elec})^*$ and integrate over the electronic coordinates.

$$i\hbar \sum_k \int d\mathbf{r} \left[(\psi^j_{elec})^* \psi^k_{elec} \, \frac{\partial c_k(t)}{\partial t} + c_k(t) \sum_\alpha (\psi^j_{elec})^* \, \nabla_\alpha \psi^k_{elec} \cdot \frac{\partial \mathbf{R}_\alpha}{\partial t} \right]$$
$$= \sum_k \int d\mathbf{r} \, c_k(t) \, (\psi^j_{elec})^* \, \hat{\mathcal{H}}_{elec} \, \psi^k_{elec}$$

Further simplification is achieved when the basis functions are the *adiabatic functions* of the Hamiltonian, $\hat{\mathcal{H}}_{elec} \psi^j_{elec} = E_j \psi^j_{elec}$, and after applying the orthonormality property of the basis functions, $\int (\psi^j_{elec})^* \psi^k_{elec} \, d\mathbf{r} = \delta_{jk}$.[5]

$$i\hbar \, \frac{\partial c_j(t)}{\partial t} + i\hbar \sum_{k,\alpha} c_k(t) \left[\int d\mathbf{r} \, (\psi^j_{elec})^* \, \nabla_\alpha \psi^k_{elec} \right] \cdot \frac{\partial \mathbf{R}_\alpha}{\partial t} = c_j(t) \, E_j$$

The integral in parenthesis is a coupling term between the basis functions. It is designated by the function $\mathbf{d}^\alpha_{jk}$, which depends parametrically on the nuclear coordinates.

$$\mathbf{d}^\alpha_{jk}(\{\mathbf{R}_\alpha(t)\}) = \int (\psi^j_{elec})^* \, \nabla_\alpha \psi^k_{elec} d\mathbf{r} \tag{12.56}$$

By using this coupling term, the final equation for the c_j amplitude is established.

$$i\hbar \, \frac{\partial c_j(t)}{\partial t} = c_j(t) E_j - i\hbar \sum_{k,\alpha} c_k(t) \, \mathbf{d}^\alpha_{jk} \cdot \frac{\partial \mathbf{R}_\alpha}{\partial t} \tag{12.57}$$

There are as many coupled differential equations of the type 12.57 as there are functions ψ^k_{elec} in the expansion of the electronic wave function ψ_{elec}, Eq. 12.54. The above equation is not compelled, in principle, to satisfy the time reversibility property of dynamic systems, especially in circumstances where the particle system moves in a multi-branched energy surface because, in this case, the energy surfaces may have sufficiently similar energies and wave functions. Such similarities introduce a probabilistic factor in the equations of motion due to the presence of the coupling terms, Eq. 12.56, leading to particle trajectories in the time-forward direction which are not necessarily the same as those in the time-backward direction. Other types of expansions of the electronic wave function and total wave function are found elsewhere [22].

[5] In the adiabatic approximation, the time evolution of the potential is sufficiently slow in such a way that the wave function of the electrons practically changes instantaneously with respect to the slowly changing potential of the nuclei.

Problem 12.7 Consider the following expansion of the electronic wave function:

$$\psi_{elec}(\{\mathbf{r}_i, \mathbf{R}_\alpha, t\}) = \sum c_j(t)\, \psi_j(\{\mathbf{r}_i, \mathbf{R}_\alpha\})\, \exp[(-i/\hbar) \int^t \mathcal{H}_{jj}\, dt']$$

The basis functions ψ_i are orthonormal, $\int d\mathbf{r}\, \psi_i^* \psi_j = \delta_{ij}$, and may be adiabatic or diabatic functions. The matrix elements of the Hamiltonian are $\mathcal{H}_{ij} = \int \psi_i^* \hat{\mathcal{H}} \psi_j\, d\mathbf{r}$. The expansion is different to that of Eq. 12.54. Obtain the time evolution of the amplitudes $c_j(t)$.

Solution In order to obtain the time evolution of the coefficients c_j, we introduce the electronic wave function in the time-dependent wave equation.

$$i\hbar \frac{\partial}{\partial t} \sum_j c_j\, \psi_j\, \exp[-i \int^t \mathcal{H}_{jj}/\hbar\, dt'] = \sum_j c_j\, \hat{\mathcal{H}} \psi_j\, \exp[-i \int^t \mathcal{H}_{jj}/\hbar\, dt']$$

The partial time derivative is applied to every term of the left-hand side, and the result is multiplied by the complex-conjugate term $\psi_i^*\, \exp[i \int^t \mathcal{H}_{ii}/\hbar\, dt']$ and integrated over the whole space.

$$i\hbar \dot{c}_i + \sum_{j,\alpha} c_j\, \dot{\mathbf{R}}_\alpha \cdot \int \psi_i^* \nabla_\alpha \psi_j\, d\mathbf{r}\, \exp[-i \int^t (\mathcal{H}_{jj} - \mathcal{H}_{ii})/\hbar\, dt'] + c_i(-\frac{i}{\hbar})\mathcal{H}_{ii}$$

$$= \sum_j c_j \mathcal{H}_{ij}\, \exp[-i \int^t (\mathcal{H}_{jj} - \mathcal{H}_{ii})/\hbar\, dt']$$

The chain rule was applied in the way it was done to obtain Eq. 12.55. By solving for $\dot{c}_i$ and considering that: (i) the diagonal term which includes $\mathcal{H}_{ii}$ is cancelled with the respective diagonal term considered in the summation, and (ii) $\int \psi_i^* \nabla_\alpha \psi_i\, d\mathbf{r} = \nabla_\alpha \int (\psi_i)^2 d\mathbf{r} = 0$, the final result is obtained.

$$\dot{c}_i(t) = -\sum_{j \neq i} c_j(t) \left[\sum_\alpha \dot{\mathbf{R}}_\alpha \cdot \mathbf{d}_{ij}^\alpha + \frac{i}{\hbar} \mathcal{H}_{ij} \right] \exp[-i \int^t (\mathcal{H}_{jj} - \mathcal{H}_{ii})/\hbar\, dt']$$

This expression corresponds to an equation of the surface hopping approach and is important to work with transitions between energy surfaces. The *Landau-Zener transition* between two energy states is an approximation to the above expression.

12.11 Expansion of the Newtonian Equation of the Nuclei

The following step is to write the Newtonian equation of the nuclei in terms of the expansion amplitudes of the electronic wave function following the discussion of Tully [23]. We start from the expression of the total energy, Eq. 12.37, and recall the electronic wave function in terms of the basis functions, Eq. 12.54.

$$E = \sum_{\alpha} \frac{\mathbf{P}_{\alpha}^2}{2M_{\alpha}} + \int \left[\sum_i c_i^*(t) \, (\psi_{elec}^i)^* \right] \hat{\mathcal{H}}_{elec} \left[\sum_j c_j(t) \, \psi_{elec}^j \right] d\mathbf{r}$$

The conservation of the energy demands a time derivative of the total energy equal to zero.

$$\frac{\partial E}{\partial t} = \sum_{\alpha} \frac{\partial \mathbf{R}_{\alpha}}{\partial t} \cdot \frac{\partial \mathbf{P}_{\alpha}}{\partial t} + \frac{\partial}{\partial t} \left[\sum_{i,j} c_i^* c_j \int (\psi_{elec}^i)^* \, \hat{\mathcal{H}}_{elec} \, \psi_{elec}^j \, d\mathbf{r} \right] = 0$$

The adiabatic functions are eigenfunctions of the electronic Hamiltonian, and it is possible to replace the term $\hat{\mathcal{H}}_{elec} \, \psi_{elec}^j$ by $E_j \psi_{elec}^j$. The orthonormal property of the adiabatic eigenfunctions is used to reduce the summation with two indexes to a summation with one index.

$$\sum_{\alpha} \frac{\partial \mathbf{R}_{\alpha}}{\partial t} \cdot \frac{\partial \mathbf{P}_{\alpha}}{\partial t} = - \sum_j \left[c_j^* c_j \, \frac{\partial E_j}{\partial t} + E_j \, \frac{\partial}{\partial t} \left(c_j^* c_j \right) \right] \tag{12.58}$$

The electronic energy was differentiated because it depends on the nuclear coordinates, which show time dependence. On the other hand, by multiplying Eq. 12.57 by c_j^*, the complex-conjugate equation by c_j and subtracting one equation from the other, we have:

$$\frac{\partial}{\partial t}(c_j^* c_j) = - \sum_{k,\alpha} \left[c_j^* c_k \, \mathbf{d}_{jk}^{\alpha} + c_k^* c_j \, (\mathbf{d}_{jk}^{\alpha})^* \right] \cdot \frac{\partial \mathbf{R}_{\alpha}}{\partial t}$$

The insertion of $d(c_j^* c_j)/dt$ in the last member of the expression that imposes the conservation of the energy, Eq. 12.58, leads directly to:

$$\sum_{\alpha} \frac{\partial \mathbf{R}_{\alpha}}{\partial t} \cdot \frac{\partial \mathbf{P}_{\alpha}}{\partial t} = - \sum_j \left[c_j^* c_j \, \frac{\partial E_j}{\partial t} - E_j \, \sum_{k,\alpha} \left(c_j^* c_k \, \mathbf{d}_{jk}^{\alpha} + c_k^* c_j \, (\mathbf{d}_{jk}^{\alpha})^* \right) \cdot \frac{\partial \mathbf{R}_{\alpha}}{\partial t} \right]$$

By using the chain rule, the coefficient $\partial E_j / \partial t$ is replaced by $\sum_{\alpha} \nabla_{\alpha} E_j \cdot \partial \mathbf{R}_{\alpha} / \partial t$. The substitution leads to an expression for each element α of the summation.

$$\frac{\partial \mathbf{P}_{\alpha}}{\partial t} = - \sum_j c_j^* c_j \, \nabla_{\alpha} E_j + \sum_{j,k} E_j \left[c_j^* c_k \, \mathbf{d}_{jk}^{\alpha} + c_k^* c_j \, (\mathbf{d}_{jk}^{\alpha})^* \right]$$

By redefining indexes, the equation is transformed.

$$\frac{\partial \mathbf{P}_\alpha}{\partial t} = -\sum_j c_j^* c_j \, \nabla_\alpha E_j + \sum_{j,k} (E_j - E_k) \, c_j^* c_k \, \mathbf{d}_{jk}^\alpha \tag{12.59}$$

Problem 12.8 Deduce the force presented in Eq. 12.59.

Solution We depart from the expression previous to Eq. 12.59.

$$\frac{\partial \mathbf{P}_\alpha}{\partial t} = -\sum_j c_j^* c_j \, \nabla_\alpha E_j + \sum_{j,k} E_j \left[c_j^* c_k \, \mathbf{d}_{jk}^\alpha + c_k^* c_j \, (\mathbf{d}_{jk}^\alpha)^* \right]$$

By recalling the definition of the coupling term $\mathbf{d}_{jk}^\alpha$, Eq. 12.56, and using the identity between $(\mathbf{d}_{jk}^\alpha)^*$ and $\mathbf{d}_{kj}^\alpha$, we have:

$$\nabla_\alpha \int (\psi_{elec}^k)^* \, \psi_{elec}^j \, d\mathbf{r} = \int \psi_{elec}^j \, \nabla_\alpha (\psi_{elec}^k)^* \, d\mathbf{r} + \int (\psi_{elec}^k)^* \, \nabla_\alpha \psi_{elec}^j \, d\mathbf{r} = 0$$

$$\int \psi_{elec}^j \, \nabla_\alpha (\psi_{elec}^k)^* \, d\mathbf{r} = -\int (\psi_{elec}^k)^* \, \nabla_\alpha \psi_{elec}^j \, d\mathbf{r}$$

The last identity indicates $(\mathbf{d}_{jk}^\alpha)^* = -\mathbf{d}_{kj}^\alpha$. If we use this expression and exchange the j and k indexes, then Eq. 12.59 is recovered.

$$\frac{\partial \mathbf{P}_\alpha}{\partial t} = -\sum_j c_j^* c_j \, \nabla_\alpha E_j + \sum_{j,k} \left(E_j \, c_j^* c_k - E_k \, c_j^* c_k \right) \mathbf{d}_{jk}^\alpha$$

When the equations that rule the time evolution of the electrons and nuclei in terms of the basis set coefficients are written together, the coupling between such equations is shown.

$$\begin{aligned} \psi_{elec}(\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}, t) &= \sum_j c_j(t) \, \psi_{elec}^j (\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha(t)\}) \\ i\hbar \, \frac{\partial c_j(t)}{\partial t} &= c_j(t) \, E_j - i\hbar \sum_{k,\alpha} c_k(t) \, \mathbf{d}_{jk}^\alpha \cdot \frac{\partial \mathbf{R}_\alpha}{\partial t} \\ M_\alpha \, \frac{\partial^2 \mathbf{R}_\alpha(t)}{\partial t^2} &= -\sum_j |c_j(t)|^2 \, \nabla_\alpha E_j + \sum_{j,k} c_j^*(t) \, c_k(t) \, [E_j - E_k] \, \mathbf{d}_{jk}^\alpha \end{aligned} \tag{12.60}$$

The first term on the right-hand side of the last equation represents a classical force term, as the force over particle α is deduced from the gradient of a potential energy function, and the second term, with no classical analog, contributes to the force on particle α by taking the coupling of the different potential energy surfaces into consideration. The equation of the total energy, Eq. 12.37, is equally written in terms of the basis set functions $\{\psi_{elec}^k\}$, and the respective probability weights.

$$E = \sum_\alpha \frac{\mathbf{P}_\alpha^2}{2M_\alpha} + \sum_{i,j} c_i^*(t)\, c_j(t) \int [\psi_{elec}^i (\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha\})]^* \hat{\mathcal{H}}_{elec}\, \psi_{elec}^j (\{\mathbf{r}_i\}; \{\mathbf{R}_\alpha\})\, d\mathbf{r}$$

$$E = \sum_\alpha \frac{\mathbf{P}_\alpha^2}{2M_\alpha} + \sum_j |c_j(t)|^2 E_j \tag{12.61}$$

In principle, the equations of motion 12.60 together with the total energy 12.61 may be solved numerically. The results are independent of the basis set $\{\psi_{elec}^k\}$ when the basis set is complete. The solutions of the equations depend on the capability to solve the electronic equation $\hat{\mathcal{H}}_{elec}\psi_{elec}^k = E_k\psi_{elec}^k$ for the many values that the subindex k may take. Unfortunately, this is a main problem of technical character as it is difficult to solve the equation when one works with real systems. However, there are exceptional cases where the solution may be achieved as, for instance, in model atomic systems. If it happens that in the expansion of the electronic function, the term corresponding to the ground state is the dominant one, this is $\psi_{elec} \approx c_0\psi_{elec}^0$, then the equations deduced above in terms of the amplitudes $c_j(t)$ become more amenable to deal with.

A.1 Appendix: Bohm's Quantum Potential

The purpose of this section is to discuss the implications of the Bohm quantum potential, Eq. 12.10, in the interpretation of quantum mechanics.

There are different interpretations of quantum mechanics in physics. The most common one is based on a probabilistic description of the particles, recognized as the *Copenhagen interpretation*, proposed by Niels Bohr and Werner Heisenberg in 1925–1927. However, there are other equally interesting interpretations like that proposed by Bohm in 1952 [24]. In this approximation a particle is driven by a quantum field, ψ, which satisfies the wave equation. In Bohm's approach of quantum theory, the particle is acted not only by the classical potentials but also by the quantum potential, V_{bohm} of Eq. 12.10, in turn determined by the quantum field ψ.

$$V_{bohm} = -\sum_{\alpha=1}^N \frac{\hbar^2}{2M_\alpha} \frac{\nabla_\alpha^2 D}{D} \quad ; \quad D = |\psi|^2 \tag{12.62}$$

In Bohm's approach, a particle such as the electron is supposed to follow a continuous and causally determined trajectory. The particle is not regarded as an object with wave-particle dual character. Yet, the wave-like features are derived from the quantum field. More precisely, the quantum field is manifested with the particle's behavior, and the particle manifests the observed reality. The quantum field and the particle are never separated from each other.[6]

[6] In order to clarify the roles of the particle and the quantum field, let us suppose a radio-controlled boat. The boat plays the role of the particle. It sails on a lake and follows instructions of a radio-frequency, or an electromagnetic wave. The radio-frequency wave plays the role of the quantum

Some important properties distinguish the quantum potential. According to our derivation, the quantum potential has no presence in the Hamiltonian, as it is the outcome of using a polar function in the description of the particle. The quantum potential suffers no change by introducing a scaling factor, because the function ψ is present in the numerator and denominator of the quantum potential, producing the cancellation of the scaling factor. Therefore, contrary to classical waves, like the ocean waves which have dependence on the field intensity, the quantum potential is independent of the field strength, but it depends on the properties of ψ.

Since the quantum potential shows no dependence on the quantum field intensity, the distant environmental effects may have repercussions on the properties of the particle. In order to illustrate this last point, let us consider the Bohm potential 12.62 as part of the force exerted on the particle [25].

$$\frac{d\mathbf{P}_\alpha}{dt} = -[\nabla_\alpha V_{elec} + \nabla_\alpha V_{bohm}] \tag{12.63}$$

The first term is the classical force and the second one is the *quantum force*. In the absence of the quantum force, the classical behavior of the particle satisfies the Newtonian equation $d\mathbf{P}_\alpha/dt = -\nabla_\alpha V_{elec}$ [26]. On the other hand, in the absence of the classical force, the behavior of the particle satisfies the equation $d\mathbf{P}_\alpha/dt = -\nabla_\alpha V_{bohm}$. In a region of empty space, it is possible to have a particle traveling in such a region in a nonlinear path, because it may respond to quite distant effects, giving evidence of non-local interactions between particles, such as the *action-at-a-distance*, also called *spooky action*. In general, the non-local quantum connection is broken with small disturbances, such as those corresponding to small temperature gradients. Special conditions are required to maintain the control of a fragile quantum connection. The non-locality of Bohm's quantum mechanics is consistent with relativistic quantum field theories by avoiding the transmission of signals faster than the speed of light [27].

In the superconducting state, the quantum potential is responsible for the strong correlations that keep the atoms in an organized collective state. Nevertheless, when the temperature is increased, the particles are scattered, breaking their organized behavior, and the multidimensional character of the quantum field is partitioned. A bifurcation of the information occurs in this case, which results in the particles taking different information channels and trajectories. The quantum field is also capable to give sensible explanations of the quantum transitions, as well as explaining the fusion and fission of subatomic particles.

In the double-slit experiment, the wave travels through both slits and interferes with itself, always driving the particle toward the screen, but the particle only goes through a single slit and is deflected according to the initial conditions imposed in the experiment. The result of dealing with many particles is an ensemble of trajectories forming a diffraction pattern. When the quantum field is defined, for

field. The instructions given to the boat are independent of the wave intensity, but depend on the wave frequency.

example by solving the wave equation, the quantum field carries information of both slits, and thus, the experimental results concerning the particle are determined. According to Bohm's theory, there is a cause, with the observer playing no relevant role in such a situation. The interpretation of quantum mechanics based on the existence of a quantum potential indicates that, in a measurement process, the measuring device and the object being measured constitute an indivisible system. The quantum field is understood as the entity that carries information, while the particle is the object that carries the main portion of the energy.

The quantum potential integrates the information of all the particles by considering the non-local connection among them. In Bohmian mechanics, the state of the system is specified by the set $\{\psi, Q\}$, where ψ is the wave function of the many particle system, and Q is the set of particle positions in $3D$ space. No classical limit is required in Bohm's interpretation of quantum mechanics. However, when the classical potential is larger than the quantum potential, the classical behavior of the particles dominates the dynamics. According to the traditional interpretation of quantum mechanics, the classical limit is achieved in the limit of large quantum numbers, problem 12.3, and when the potential changes little with respect to the wavelength of the particles, Eq. 12.22. On the other hand, the quantum potential is relatively large in the case of short wavelengths, indicating that a system under special conditions may still retain a dominant quantum character at high quantum numbers, thus contradicting the Bohr's correspondence principle. In other words, the classical limit is not always achieved at the macroscopic scale due to the possibility to have a significant contribution of the quantum potential.

Bohm's interpretation of quantum mechanics is a formulation that deals with the reality at every scale level, avoiding difficulties at the overlap domain of classical mechanics and quantum mechanics, which have been observed in other interpretations. Still, the physical origins of the quantum potential are unknown. The quantum field and the particle together constitute a complex structure (recall that measurable distances in physics can reach a resolution of $10^{-18}\,m$, and the Planck distance is $10^{-35}\,m$, giving opportunity to discover new structures at smaller dimensions). The main goal of Bohm's interpretation is to provide an intuitive basis of quantum mechanics after following some similarities with classical mechanics. The new philosophy on quantum mechanics is only possible when we are open to consider new ways of thinking on the properties of matter and embrace ideas and concepts of different nature to those given by previous interpretations of quantum mechanics.

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Part IV

Classical Molecular Dynamics



Classical Molecular Dynamics

13

The complexity of quantum theory prevents its application to systems with a large number of particles. The most important problems to reach a quantum-mechanical description of molecules are concerned with the solution of difficult differential equations of motion, the optimization of a great number of degrees of freedom to determine the proper structures, the diagonalization of big matrices to establish the energies, the integration of complex functions to determine the average potentials, and other algebraic and computational complications. Fortunately, in many instances the atoms behave in the classical sense, and it is possible to use classical mechanics in such instances. The possibility to deal with classical atoms circumvents many situations where the application of quantum mechanics is difficult. In fact, under such a perspective, a new branch of molecular dynamics is created. It carries the name classical molecular dynamics. In this context, classical molecular dynamics is shown as an alternative to quantum molecular dynamics to describe many-body systems at the atomic level. Classical molecular dynamics was derived from quantum molecular dynamics under limiting conditions in the previous sections; thereby, we apply Newtonian mechanics infused with important concepts, methods, and algorithms of Lagrangian and Hamiltonian mechanics to unfold the classical molecular dynamics theory.

One of the central points in classical molecular dynamics is the approximation of the forces. The static and dynamic correlations of all the particles complicate the formulation of the atomic forces. The particle correlations, not fully understood to present, constitute a general problem known as the *many-body problem*. The theory of classical molecular dynamics employs model interatomic potentials to mimic the interactions of quantum character and derive the forces from such potentials. Thus, classical molecular dynamics (CMD) reaches simplicity by implementing model interaction potentials and classical forces.

The CMD approach is capable to deal with a large number of particles because it has been designed to be computationally efficient. In consequence, CMD allows the evaluation of different statistical and thermodynamic observables. By using CMD we are able to compute the time evolution of physical properties of many-particle

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systems and, for example, design new compounds like proteins, paying attention to their folding state to fit pharmacological purposes. The CMD approach is nowadays an integrated part of most software codes. Still, the application of less accurate methods than quantum mechanics should be always examined in detail and wise judgment in advance.

In the previous chapters, we discussed the different methods of classical mechanics and formulated their respective equations of motion. The forces were assumed to be known there with the purpose of focusing attention on the derivation of the motion equations. The interest in this chapter is to provide, in a first stage, a general classification of the model potential functions. To do this, we separate atoms of biological nature from metallic atoms for the building of the respective potential functions. In a second stage, the model potential functions are discussed for the two-body, three-body, and four-body interactions. These all functions are applied to bonded atoms. The potential functions of non-bonded atoms are also introduced to simulate the electrostatic and van der Waals distant interactions. Other types of potential functions for different purposes of the atomic phenomena are presented. The parametrization of the potential functions is discussed as well.[1]

13.1 Model Interaction Potentials

Classical molecular dynamics is an approach integrated by *model interaction potentials*, considered to be functional energy expressions. They emulate the interactions among atoms, which are used to investigate the behavior of molecules. The functional expressions of the energy represent one of the fundamental aspects to reduce the computer time and the computer memory in the simulations. The CMD approach is capable to reproduce the collective features of the atoms, reaching consistency with the laboratory observations on the physical phenomena of molecules. The model interaction potentials are formulated after solving the electronic wave equation of a system with a number of atoms.

$$\hat{\mathcal{H}}\,\psi(\mathbf{r}_1,\cdots,\mathbf{r}_N) = E\Big|_{\mathbf{r}_1,\cdots,\mathbf{r}_N}\psi(\mathbf{r}_1,\cdots,\mathbf{r}_N) \tag{13.1}$$

The electronic energy eigenvalue, E, depends in parametric form on the atomic coordinates $\mathbf{r}_i$, $i = 1, ..., N$. Thus, by departing from a quantum energy, it is possible to propose a model interaction potential, V [3].

$$E\Big|_{\mathbf{r}_1,\cdots,\mathbf{r}_N} = V(\mathbf{r}_1,\cdots,\mathbf{r}_N) \quad ; \quad \text{model interaction potential} \tag{13.2}$$

[1] We follow the works: Maitland et al. [1], Chapt. 1 and Rappe and Casewith [2].

The model interaction potentials, $MIPs$, allow to carry simulations of molecules in solution, large clusters, extended materials like polymers, ceramics, solids, fluids, etc. There are no formal rules in the classical modeling of the quantum interactions; nonetheless, the quantum interactions suggest some functional expressions of the $MIPs$. It is common to start working with simple $MIPs$, whose parameters are tuned to reproduce different experimental facts of the compounds under investigation. The $MIPs$ are designed according to the degrees of freedom which are characteristic of the molecules. The degrees of freedom frequently correspond to the motion of pairs, triplets, and quadruplets of bonded atoms. The simplest model interaction potentials that describe the interactions of atoms like H, C, N, O, P, S, etc., of biological compounds, are collected in a scheme that carries the name *forcefield approach*. This is, a forcefield represents a collection of $MIPs$ for atoms of biological nature. There are also $MIPs$ of non-biological compounds, which are mainly used in materials science. The set of $MIPs$ that simulate the behavior of atoms like Cu, Ag, Au, Pd, Pt, etc., in materials are called *many-body potentials*. The discussion and methods given below apply equally well to both biological and non-biological compounds. The incorporation of the $MIPs$ in the Newtonian equations of motion generates the forces that unfold the dynamics of the atoms. Thus, the motion equation of the ith atom employs the model interaction potential V.

$$m\,\frac{d^2\mathbf{r}_i}{dt^2} = -\nabla_i V(\mathbf{r}_1, \cdots, \mathbf{r}_N) \tag{13.3}$$

The potential function has dependence on all the atom coordinates and the parameters for the interaction of the ith atom with the other atoms of the system. Classical molecular dynamics deals with the proposal and evaluations of the potential V. The potential function is frequently approximated in piecewise form.

$$V(\mathbf{r}_1, \cdots, \mathbf{r}_N) = \sum_i V_1(\mathbf{r}_i) + \sum_{i<j} V_2(\mathbf{r}_i, \mathbf{r}_j) + \sum_{i<j<k} V_3(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) + \cdots \tag{13.4}$$

The first term contains the individual atomic energies, the second represents the two-body energies, the third contribution gives the three-body energies, and so on. The summation indexes run over the atoms of the particular interaction groups. The external potentials, like a confining potential and a gravity potential, may be equally included in the expression of V. The model interaction potentials V_1, V_2, V_3, etc., are originated by the attraction and repulsion of the atoms and are frequently proposed after employing first-principles calculations. This process implies knowing the solution of the wave equation of small systems. However, the proposal of $MIPs$ may be also achieved by observing the behavior of atoms in the molecules of the lab and establishing the parameter values of $MIPs$ from the experimental measurements.

One of the main drawbacks of using model interaction potentials is the averaging of electron effects. More precisely, the parameters of the $MIPs$ include the electronic effects in average form, restricting the use of classical molecular dynamics to

the domains of the parametrized molecular conformations. In this regard, the CMD simulations overlook the breaking of bonds, or other effects where a substantial redistribution of the electron cloud occurs with the molecular structural changes. This is the case of the electronic transitions, the tunnel effect, electric currents, the interaction of radiation with matter, chemical reactions, etc. In the following, the MIPs are discussed within the forcefield approach.

13.2 Forcefields

A forcefield is a collection of model interaction potentials for a given set of molecules of biological nature, in an effort to simplify the atomic interactions for computational purposes, but without missing the key ingredients and general features of the atomic interactions. The success of a forcefield, FF, is to a great extent based on the faithfulness of the MIPs on reproducing the atomic interactions, with significant reduction of the computational times and memory. Each MIP of a FF is associated with a degree of freedom characteristic of the molecule; however, the number of degrees of freedom described by a FF does not necessarily correspond to the number of degrees of freedom of the molecular phase space. A FF is also constituted by parameters, which are specific to both the model interaction potentials and the molecular system. The parameters are determined by reproducing particular properties of the real system. The distinct FF differ in the types and numbers of the MIPs. Thereby, a FF has its own particular composition, with its own parameters.

The model interaction potentials are classified in terms of bonded, V_b, and non-bonded, V_{nb}, interactions, or intramolecular and intermolecular interactions, respectively. The terms V_b and V_{nb} contain the parameters determined from experimental and theoretical sources of information.

$$V = V_b + V_{nb} \tag{13.5}$$

The bonded interactions consider linked atoms and are modeled using sets of two, three, and four atoms. The bonded interactions deal with bond elongation and bond compression, changes of bond angles, and torsion effects of groups of atoms with up to four atoms. On the other side, the non-bonded interactions consider long-range forces due to charged atoms. The non-bonded interactions include the electrostatic interactions and the van der Waals forces. The long-range forces are the main cause of molecules to change conformations to reach structural stability in a given environment. The MIPs and the motion equations coil together to make classical molecular dynamics a concerted approach in the investigation of the structural and energetic changes of molecules, with linkage to molecular reactivity. These all aspects are important, for example, in elucidating the structure-reactivity relationship of proteins.

The next step is to discuss the individual interaction potential functions. They are given in terms of bond distances, bond angles, atom charges, and more. It is

important to remark that there are no formal rules in the proposal of the $MIPs$. For such a reason, it is common to find a variety of expressions describing the same type of interaction in the different software codes implementing classical molecular dynamics. Thereby, classical molecular dynamics opens the opportunity for creativity.

13.3 Atom Types

The forcefield approach introduces the concept of *atom type*, where every atom of the molecule is classified not only according to the periodic table but also with respect to the close surrounding environment. For example, a carbon atom bonded to an oxygen atom, $C-O$, classifies differently in reference to a carbon atom bonded to a nitrogen atom, $C-N$, due to the neighbors O and N of C, respectively. The classification process considers not only the different elements that an atom may link to but also the types of bonds that it may establish with other atoms. For instance, there are different types of bonds, single, double, and triple bonds, among carbon atoms, and from a sharp perspective, the atoms are additionally classified according to their chemical environment, Table 13.1. The concept of atom type in the FF approach enlarges the classification provided by the periodic table of the elements. It is in such a fashion that a FF approach accounts for the behavior of atoms in molecules. In this regard, CMD is sufficiently efficient and relatively accurate. A main advantage of classical molecular dynamics over the quantum-mechanical calculations is the no necessity of re-computing the electron cloud distribution at every time step of the simulation to retain good accuracy.

In most situations, the atom types are assigned by the user based on both the classification and the nomenclature rules of the computational program in use. The atom types which are automatically assigned by some programs are an error-prone process. The assignation process of atom types is one of the initial steps in the construction of the molecular model; it is based on the correct chemical identification of the molecular fragments and represents a critical point to obtain acceptable results.

Table 13.1 Atom types of C according to the bond type with other atoms[a]

Atom type	Assignment
C	Carbonyl C, peptide backbone
CA	Aromatic C
$CT1$	Aliphatic sp^3 C for CH
$CT2$	Aliphatic sp^3 C for CH_2
$CT3$	Aliphatic sp^3 C for CH_3
CS	thiolate C

[a]The atom-type nomenclature changes from one software code to another

13.4 United Atom

The concept of atom type may be generalized to introduce the *united atom approach*. In this approximation, the terminal carbons and the terminal nitrogens, together with their respective hydrogens, are redefined as either a single atom, united atom, or super atom. It is usually applied to atomic groups like CH_3, NH_2, CH_2, and carbonyl and hydroxyl groups, treating each group as a single atom, with its own model potential function for the interaction with other backbone atoms. The purpose of the united atom is to save computer time on the atom groups that many times play second roles in the simulations. The united atom approximation is a coarse-grained approach and, in principle, is simpler to implement than the all-atom forcefield approach.

13.5 Bond Elongation and Compression

It is time to discuss the individual terms of the forcefield, Eq. 13.4. The first term, V_1, involves the independent atomic energies. It constitutes a constant energy when the numbers and types of atoms have no change in the simulation and may be neglected since the important energies are the relative energies. The second term, V_2, represents the total energy of the two-body interactions.

$$\begin{gathered} r_{ij} = |\mathbf{r}_i - \mathbf{r}_j| \\ V_2(\mathbf{r}_1, \cdots, \mathbf{r}_N) = \sum_i \sum_{j>i} V_2(r_{ij}) \quad ; \quad \mathbf{F}_i = -\sum_j \frac{\partial V_2(r_{ij})}{\partial r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \end{gathered} \tag{13.6}$$

The term $V_2(r_{ij})$ is a pairwise potential which applies to bonded atoms. To avoid confusions, it shall be denoted by V_b. The force $\mathbf{F}_i$ is derived from the pair potential. It is the force exerted by all the other atoms on atom i. According to 3rd Newton's law, the force $\mathbf{F}_{ij}$ over the ith atom due to the jth atom is the same as the force $\mathbf{F}_{ji}$ exerted over the jth atom by the ith atom, but with opposite sign, $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$. The use of 3rd Newton's law reduces the number of computations on the atomic forces.

The pair potential of a forcefield is based on the natural elongation and compression (vibration) of a molecular bond. Such a potential is usually of harmonic type. Thus, we have the *harmonic potential*, Fig. 13.1.

$$V_b = k(\mathbf{r}_{ij} - \mathbf{r}_{ij}^0)^2/2 \tag{13.7}$$

The term V_b represents the bond-stretching interaction between the ith and jth atoms. The elongation and compression are measured with respect to the equilibrium distance $\mathbf{r}_{ij}^0$. The spring stiffness or force constant is k; it depends on the types of atoms taking part in the interaction (the indexes ij in the notation of the force constant k are dropped for simplicity). The constant computations of the electron cloud re-arrangements of a bonded atomic dimer in a quantum-

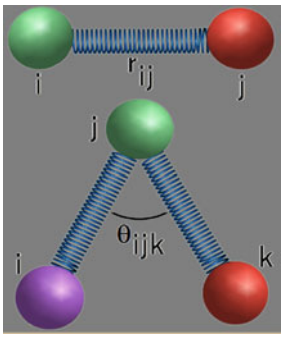


Fig. 13.1 The figures of the top and bottom illustrate the bond-distance and bond-angle vibrations. Such degrees of freedom are frequently simulated with the harmonic potential functions $V_b = k(\mathbf{r}_{ij} - \mathbf{r}_{ij}^0)^2/2$ and $V_a = k'(\theta_{ijk} - \theta_{ijk}^0)^2/2$. The parameters $\mathbf{r}_{ij}^0$ and θ_{ijk}^0 are the reference bond-distance and the reference bond-angle, and k and k' are the spring constants

mechanical calculation are greatly simplified in the classical approach with the harmonic potential, which may be corrected by introducing anharmonic effects.

$$V_b = k\left[(\mathbf{r}_{ij} - \mathbf{r}_{ij}^0)^2 - \alpha(\mathbf{r}_{ij} - \mathbf{r}_{ij}^0)^3\right]/2 \tag{13.8}$$

The constant α is the weight of the anharmonic term gauged with respect to the harmonic one. The addition of the anharmonic term makes the potential more steep at small distances and less steep at large distances, breaking the symmetry imposed by the harmonic term at large distances and, in principle, producing a more realistic behavior. The parameters k and $\mathbf{r}_{ij}^0$ of potential 13.8 are not necessarily the same as those of Eq. 13.7. The potential 13.8 may be further improved with a quartic term, $\beta(\mathbf{r}_{ij} - \mathbf{r}_{ij}^0)^4$, due to the fact that a cubic anharmonic term may overvalue the negative contribution of the potential. In most instances of molecular behavior, no extra contributions are required beyond the quartic term in the expression of V_b.

There are alternative proposals to the harmonic potential. An important proposal describes covalent bonds with pronounced elongations, like the *Morse potential*.

$$V_b = V_0\left(1 - \exp[-\alpha(r_{ij} - r_{ij}^0)]\right)^2 - V_0 \tag{13.9}$$

The parameters in the above expression are the reference energy, V_0, the reference distance, r_{ij}^0, and the exponential factor, α. Another potential for covalent bonds is the *Lennard-Jones potential* with parameters ϵ, σ.

$$V_b = 4\epsilon\left[(\sigma/r_{ij})^{12} - (\sigma/r_{ij})^6\right] \tag{13.10}$$

The *Buckingham potential*, with parameters V_0, ρ, and C_6, is another expression that may be used to describe covalent bonds.

$$V_b = V_0 \exp[-r_{ij}/\rho] - C_6/{r_{ij}}^6 \tag{13.11}$$

Once the interaction potentials are defined, the force exerted on the ith atom by the jth atom is obtained by differentiating V_b with respect to the position of atom i, and employing the chain rule.

$$\mathbf{f}_i = -\nabla_i V_b = -\frac{\partial V_b}{\partial r_{ij}} \left(\frac{\partial r_{ij}}{\partial x_i}, \frac{\partial r_{ij}}{\partial y_i}, \frac{\partial r_{ij}}{\partial z_i} \right) = -\frac{\partial V_b}{\partial r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \tag{13.12}$$

When the potential cannot be described analytically, it is possible to deal with *numeric potentials*, whose values are tabulated. There are more expressions to describe the covalent bonds in the literature; nevertheless, the ones presented previously are well-known and have been tested in many occasions, rendering acceptable results on the vibration of atom pairs.

13.6 Combination Rules

In many instances, the model interaction potentials are applied to systems integrated by different types of atoms. In these cases, we are required to deal with parameters for the interaction of atoms of different species. One of the straight ways to obtain such parameters without much additional work is to use the *combination rules*. According to those rules, the interaction parameters of atoms of type a and the interaction parameters of atoms of type b may be combined to generate the parameters of the interaction of unlike atoms, ab. There are no formal proposals in the combination of the parameters, because they are intertwined with their MIPs. Nevertheless, there are some empirical strategies. If the atomic species a and b constitute the system, and the interaction potentials of the like species are of Lennard-Jones type, Eq. 13.10, then the crossed parameters ab may be approximated in any of the following ways.

$$\begin{aligned}
\epsilon_{ab} &= (\epsilon_a \epsilon_b)^{1/2} \; ; \; \sigma_{ab} = (\sigma_a + \sigma_b)/2 \\
\epsilon_{ab} &= 2\,\frac{\epsilon_a \epsilon_b}{\epsilon_a + \epsilon_b} \; ; \; \sigma_{ab} = (\sigma_a + \sigma_b)/2 \\
\epsilon_{ab} &= (\epsilon_a \epsilon_b)^{1/2} \; ; \; \sigma_{ab} = (\sigma_a \sigma_b)^{1/2} \\
r_{ij} &= (r_i r_j)^{1/2} \; ; \; r_{ij} = (r_i + r_j)/2
\end{aligned} \tag{13.13}$$

More combination strategies are found elsewhere [4].

13.7 Bond Angle Vibration

The model bond functions, V_b, and the forces derived from them are not enough to describe the many degrees of freedom of a polyatomic molecule in most instances. We require to introduce further terms like the V_3 and V_4, which take part in Eq 13.4 to guarantee the molecular stability.[2] The purpose of this section is to discuss the term V_3 of Eq. 13.4.

When atom i is linked to atom j and also to atom k, the bond angle between the bonds ij and ik is formed. The bond angle vibrates describing a kind of scissoring motion, Fig. 13.1. The respective potential function is similar to the bond-stretching potential; however, the bond angle takes part in this case.

$$V_a = k(\theta_{ijk} - \theta^0_{ijk})^2/2 \quad ; \quad \theta_{ijk} = \arccos\left(\frac{\mathbf{r}_{ij}\cdot\mathbf{r}_{ik}}{r_{ij}\, r_{ik}}\right) \tag{13.14}$$

The above function is known as the *bond-angle harmonic potential*. The reference angle formed by vectors $\mathbf{r}_{ij}$ and $\mathbf{r}_{jk}$ of the atoms in equilibrium is θ^0_{ijk}. It is possible to introduce correction terms similar to those of the bond elongation-compression potential by considering higher-order terms up to the quartic contribution.

$$V_a = k(\theta_{ijk} - \theta^0_{ijk})^2/2 + k'(\theta_{ijk} - \theta^0_{ijk})^3/3 + k''(\theta_{ijk} - \theta^0_{ijk})^4/4 \tag{13.15}$$

We may also insert terms with different character, such as those that mix a bond-vibration term with a bond-angle vibration term in Eq. 13.14.

$$V_{mix} = V_0\ (\theta_{ijk} - \theta^0_{ijk})\ (r_{ij} - r^0_{ij})\ (r_{ik} - r^0_{ik}) \tag{13.16}$$

The expression 13.16 is a crossed-terms equation which couples a bond vibration to an angle vibration. The constants of the potential are the reference distances r^0_{ij}, r^0_{ik}, reference angle θ^0_{ijk}, force constants k, k', k'', and factor V_0. An alternative form of the potential V_a is the *harmonic cosine expression*.

$$V_a = k[\cos(\theta_{ijk}) - \cos(\theta^0_{ijk})]^2/2 \tag{13.17}$$

There are many more expressions for the bond-angle potential, frequently formulated in the general form $V_a = V_0(\theta_{ijk})\, F(r_{ij})\, F(r_{ik})\, F(r_{jk})$. The proposal of bond-angle potentials in numeric form is equally possible.

[2] For the number of terms to achieve the stability of a cluster, refer to: Kaplan et al. [5] and Kaplan et al. [6].

13.8 Plane Bending

There are groups of atoms located in planes like in the aromatic rings and other conjugated systems in many molecules. The dynamics of the molecule may induce the bending vibration of the planar atoms. It is possible to propose a potential function that describes the vibration of sets of four atoms, where three atoms of them form a plane, and other three atoms of them form a different plane, Fig. 13.2. The vibration of a plane with respect to the other plane can be described by a *dihedral-angle harmonic potential*.

$$V_p = k(\theta_{ijkl} - \theta^0_{ijkl})^2/2 \quad ; \quad \theta_{ijkl} = \arccos\left[\frac{(\mathbf{r}_{ij}\times\mathbf{r}_{jk})\cdot(\mathbf{r}_{jk}\times\mathbf{r}_{kl})}{|\mathbf{r}_{ij}\times\mathbf{r}_{jk}|\,|\mathbf{r}_{jk}\times\mathbf{r}_{kl}|}\right] \tag{13.18}$$

The variable θ_{ijkl} is the instantaneous angle between the two planes, and θ^0_{ijkl} the angle of equilibrium. When $\theta^0_{ijkl} = 0$, the potential imposes the planarity of the four atoms. This type of potential is frequently applied to the planar peptide bonds of proteins. In all the previous model potential functions, the deviations from the reference values produce a $3D$ conformational change and, consequently, an energy change. In this regard, the forcefield approach always requires an initial reference conformation of the molecule and the corresponding energy of reference. Other potentials describing the bending of planes are the *cosine potential* and *harmonic cosine potential*.

$$V_p = V_0[1 + \cos(\alpha\theta_{ijkl} - \beta)] \quad ; \quad V_p = k[\cos(\theta_{ijkl}) - \cos(\theta^0_{ijkl})]^2/2 \tag{13.19}$$

We may also deal with different short series of cosine functions to describe the torsional effects.

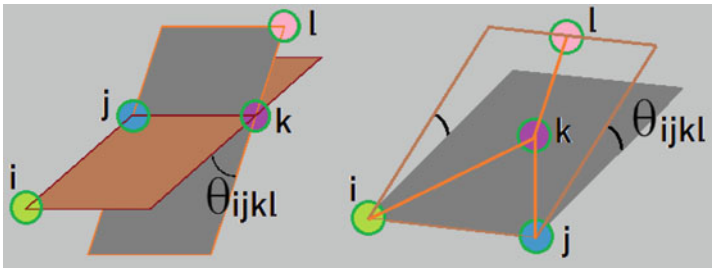


Fig. 13.2 Two different molecules with four bonded atoms are shown. In the left figure, the atoms ijk form a plane, and the atoms jkl form a different plane. In the right figure, the atoms ijk form a plane, and the atoms ijl form a different plane. In each case, the angle between planes is θ_{ijkl}. The plane-bending degree of freedom demands a model interaction potential in a forcefield. Such a potential may adopt any of the forms 13.18–13.20

$$V_p = (k/2)[1 + \cos(\theta_{ijkl})] + (k'/2)[1 + \cos(2\theta_{ijkl})] + (k''/2)[1 + \cos(3\theta_{ijkl})]$$
$$V_p = (k/2)[a_0 + a_1 \cos(\theta_{ijkl}) + a_2 \cos^2(\theta_{ijkl}) + a_3 \cos^3(\theta_{ijkl}) + a_4 \cos^4(\theta_{ijkl}) + a_5 \cos^5(\theta_{ijkl})] \tag{13.20}$$

The four-body terms are computationally more expensive than the three-body and two-body terms. The expressions of the forces that have to be derived from complicated model potential functions may be obtained using a *symbolic-algebra software* or an *artificial-intelligence software*, to directly translate onto a programming language.

13.9 Angle Inversion

We frequently have an atom surrounded by three other atoms in symmetric form, like a nitrogen atom in ammonia, NH_3. The central atom is in many occasions able to invert positions, producing a chiral change. The inversion of positions is described by an *inversion potential*, which may adopt simple forms like those of harmonic type.

$$V_{in} = k(\theta_{ijkl} - \theta^0_{ijkl})^2/2 \quad ; \quad V_{in} = k[\cos(\theta_{ijkl}) - \cos(\theta^0_{ijkl})]^2/2 \tag{13.21}$$

The inversion process of the positions may be equally described by a planar potential.

$$V_{in} = \tfrac{k}{2}[1 - \cos(n\theta_{ijkl} - \theta^0_{ijkl})] \tag{13.22}$$

If we consider that atoms j, k, l surround atom i, then the angle θ_{ijkl} corresponds to the angle described by the vector $\mathbf{r}_{ij}$ and the ikl plane.

Problem 13.1 Determine the angle θ_{ijkl} of Eqs. 13.21 and 13.22 under the supposition that atoms jkl surround atom i, in the way it happens with the hydrogens of nitrogen in the ammonia molecule.

Solution The strategy to determine θ_{ijkl} consists of determining two coplanar vectors in the plane ikl. To do this, we define the basis vectors:

$$\hat{\mathbf{u}}_{kl} = \frac{\hat{\mathbf{r}}_{ik} + \hat{\mathbf{r}}_{il}}{|\hat{\mathbf{r}}_{ik} + \hat{\mathbf{r}}_{il}|} \quad ; \quad \hat{\mathbf{v}}_{kl} = \frac{\hat{\mathbf{r}}_{ik} - \hat{\mathbf{r}}_{il}}{|\hat{\mathbf{r}}_{ik} - \hat{\mathbf{r}}_{il}|}$$

(continued)

The unit vectors $\hat{\mathbf{u}}_{kl}$ and $\hat{\mathbf{v}}_{kl}$ are described as the sum and the difference of unit vectors $\hat{\mathbf{r}}_{ik}$ and $\hat{\mathbf{r}}_{il}$. The basis vectors lie in the ikl plane. From the projection of vector $\mathbf{r}_{ij}$ onto the basis vector $(\hat{\mathbf{u}}_{kl}, \hat{\mathbf{v}}_{kl})$, we define the new vector $\mathbf{s}_{kl}$ on the ikl plane.

$$\mathbf{s}_{kl} = (\mathbf{r}_{ij} \cdot \hat{\mathbf{u}}_{kl})\hat{\mathbf{u}}_{kl} + (\mathbf{r}_{ij} \cdot \hat{\mathbf{v}}_{kl})\hat{\mathbf{v}}_{kl}$$

The dot product between the coplanar vector $\mathbf{s}_{kl}$ and the non-planar vector $\mathbf{r}_{ij}$ gives the angle between the vector $\mathbf{r}_{ij}$ and the ikl plane.

$$\theta_{ijkl} = \arccos\left(\frac{\mathbf{r}_{ij} \cdot \mathbf{s}_{kl}}{r_{ij}\, s_{kl}}\right)$$

13.10 Torsional Motion

In a molecule the rotation of atomic groups around a central bond is commonly observed. For example, for the group of atoms $X - CH_3$, we have the central bond $X - C$. The hydrogens of the CH_3 group face low energy barriers in the rotation around bond $X - C$. When a rotation of the hydrogen bonds around the central bond $X - C$ occurs, the group of atoms achieves a conformation which is indistinguishable from the initial one. In this regard, the potential function of a rotating atomic group may be described by a *periodic potential*.

$$V_t = (k/2)[1 + \cos\phi] + (k'/2)[1 + \cos(2\phi)] + (k''/2)[1 + \cos(3\phi)] \qquad (13.23)$$

The parameters of the potential are the constants k, k', and k''. The angle ϕ is a dihedral angle. The second term has a greater periodicity than the first one; in turn, the third term has a greater periodicity than the second one. The terms introduce the number and the height of peaks and subpeaks between major peaks, according to the sp^i-bond hybridization, Fig. 13.3. The potential only applies to groups of atoms linked to a central single bond, since the group of atoms show no rotation around double and triple bonds.

The model potential functions of the different degrees of freedom presented so far produce forces and energies of different magnitude, because the potentials describe interactions of different nature, contributing in different amounts to the molecular energy. The energy contributions increase from the soft to the hard degrees of freedom. It takes more energy to distort a bond distance than a bond angle; in turn, it is energetically more costly to distort a bond angle than a torsion angle with small energy barriers. Such observations have been confirmed by establishing the infrared and Raman vibrational spectra of the DNA nucleic acid bases and their Watson-Crick pair complexes, where different types of individual and collective motions have been assessed and classified with different frequencies [7].

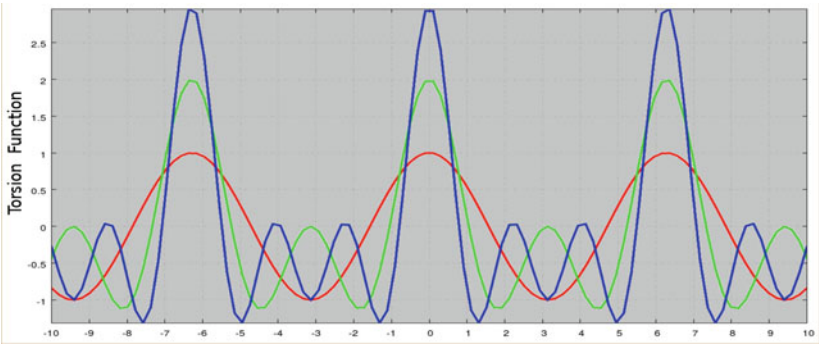


Fig. 13.3 The plots in red, green, and blue colors represent the functions $\cos(\phi)$, $[1+\cos(\phi)]+[1+\cos(2\phi)]$ and $[1+\cos(\phi)]+[1+\cos(2\phi)]+[1+\cos(3\phi)]$ in terms of the angle ϕ, respectively. The last two functions introduce one and two subpeaks between the main peaks of $\cos(\phi)$. In general, for n-fold rotational barriers, there are n additional functions for the cosine function

13.11 Electrostatic Interaction

The intramolecular and short-range interactions have been modeled using potential energy functions in the previous sections. It is time to deal with the intermolecular and long-range interactions. We start by considering that the atoms of a molecule are partially charged as they invest part of their electron charge in forming bonds (it is common to speak of *polar bonds*). The charged atoms interact with other relatively far apart charged atoms of the molecule and also with the solvent atoms according to the electrostatic *Coulomb potential*.

$$V_{el} = \frac{1}{4\pi\varepsilon}\frac{q_i q_j}{r_{ij}} \quad ; \quad r_{ij} = |\mathbf{r}_{ij}| = |\mathbf{r}_i - \mathbf{r}_j| \tag{13.24}$$

The parameters of the potential are the partial charges q_i and q_j of the ith and jth atoms. They are maintained constant in the course of a simulation. The interatomic separation distance is r_{ij}. The constant ε is the *relative permittivity* of the medium. It is a term representing the screening effects of the medium in the electrostatic interactions of atoms. In manyinstances, ε is modified to account in average manner for the interactions with distant atoms.

$$\begin{aligned} V_{el} &= [4\pi g(r_{ij})]^{-1} q_i q_j / r_{ij} \quad ; \quad g(r) = \alpha\varepsilon r \\ \mathbf{f}_i &= (2\pi\alpha\varepsilon)^{-1} q_i q_j \, \mathbf{r}_{ij} / r_{ij}^4 \end{aligned} \tag{13.25}$$

In this expressions, the dielectric g has linear dependence on the distance, as α is a constant. The force $\mathbf{f}_i$ exerted on the ith atom derived from the potential is also shown. The expressions reduce the computational times because the electrostatic energy is made to converge faster by considering the average effect of distant charged atoms.

The *monopole interactions* 13.24 represent one of the computational expensive parts of a simulation due to the pairwise nature and slow decay of the electrostatic interaction. Such effects, together with a poor computational efficiency of the mainframes, constitute some of the main reasons to create novel methods, and even new hardware, for the rapid convergence of the electrostatic energy, especially for large systems such as a molecule immersed in a liquid. The novel methods are classified within the *implicit-solvent approach* or *explicit-solvent approach*. The implicit-solvent approaches are, for instance, the continuum dielectric and static dipolar field approximations, where the presence of the liquid molecules is avoided, and only their effects on the compound under investigation are considered. The explicit-solvent approaches involve the Ewald summation, and the approximations of truncated potentials, where the explicit presence of the liquid molecules is considered, and the interactions are simplified. The different approaches may be combined to lead, for example, to the reaction field scheme together with the fast multipole approximation. Some of these relevant methods are presented later.

13.12 van der Waals Forces

The van der Waals forces are important when the interacting molecules are not too close but also not too distant apart from each other. The van der Waals forces are the result of the mutual distortions of the electron clouds of the interacting molecules at intermediate distances, because in every molecule the electrons are in constant motion and are susceptible to a spatial re-arrangement for the electric fields of the other molecules. In any particle system, the molecular interactions introduce a correlation of the fluctuating electron clouds, resulting in the formation of instantaneous multipoles (dipole, quadrupole, and more). The van der Waals interaction potential is due to the induced multipoles of the molecules interacting at intermediate distances. The interaction potential is of quantum nature as the zero-point vibration energy of a molecule provides a contribution to the dispersion energy (the calculation of the dispersion energy of a pair of model interacting molecules is illustrated in Sect. A.1). This fact cannot be explained using a classical description of the electron cloud [8].

The van der Waals interactions are *non-bonded forces* and lead to the *dispersion energy*, usually of attractive character at semi-large distances. The van der Waals potential also contains a repulsive contribution at short distances. It arises from the partial overlap of the electron clouds, which have quantum character, and with origins on the Pauli exclusion principle. The short-range and semi-long-range contributions of the van der Waals forces are, in a few words, due to the electron correlations which are not considered by the classical electrostatic forces. In this connection, the electrostatic force due to point charges or electric monopoles classifies as a first-order energy because it involves no deformation of electron clouds; however, the dispersion energy classifies as a second-order energy because it involves a small deformation of electron clouds. The proper behavior of the

interactions at the intermediate range of distances is a complicated problem to solve since, at short distances, the antisymmetric character of the wave function plays an important role and, at far distances, the asymmetric part of the wave function is the dominant contribution. Thus, at the intermediate region, where there is an overlap of the intramolecular and intermolecular forces, the mixed antisymmetric-asymmetric nature of the wave function complicates the description of the molecular interactions. A rough simplification, with the purpose of modeling the forces in the intermediate region, consists in averaging the short-distance and large-distance interactions.

The van der Waals forces are weaker than the hydrogen bonds, but they increase with the delocalization of the electron clouds, particularly for atoms with a big number of electrons or molecules with significant polarization. The van der Waals forces are anisotropic, because the polarizability depends on the orientation of the interacting molecules. This is the reason to deal with the polarizability as a second rank tensor, where the electric field component in one direction and the dipole moment component in a different direction are, in fact, related [9].

The inflection point in going from the attractive to the repulsive part of the van der Waals interaction is an equilibrium point, say r_0, in the van der Waals potential. The equilibrium point defines the sum of the *van der Waals radii* of the interacting atoms. A common representation of the van der Waals interactions between two atoms is given by the Lennard-Jones potential [10].

$$V_W = \alpha / r_{ij}^{12} - \beta / r_{ij}^{6} \tag{13.26}$$

The parameters α and β depend on the atom types and are determined from energy values at intermediate atomic distances. The term r_{ij} is the atomic separation. The repulsive part of the potential is deduced from empirical observations and corresponds to the first term of Eq. 13.26. The attractive part of the potential is designed according to the attraction of the dipoles and corresponds to the second term of Eq. 13.26. The repulsive part grows faster than the attractive part at short distances. At large distances the attractive part dominates over the repulsive contribution. The van der Waals potential given above is spherically symmetric and pairwise additive. The interactions exclusively based on the Lennard-Jones potential lead to compact structures, like the close-packed hexagonal and face-centered cubic structures. There are equivalent forms of writing the Lennard-Jones expression in terms of different pairs of parameters, making it more appealing from the computational point of view, since one of the terms is calculated as the square of the other. If we make the transformation $\alpha = 4E_0\sigma^{12}$ and $\beta = 4E_0\sigma^{6}$, then the Lennard-Jones expression is:

$$V_W = 4E_0\left[\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right] = E_0\left[\left(\frac{r_0}{r_{ij}}\right)^{12} - 2\left(\frac{r_0}{r_{ij}}\right)^{6}\right] \quad ; \quad r_0 = 2^{1/6}\sigma \tag{13.27}$$

The reference energy E_0 is the interaction strength at the equilibrium distance r_0 and corresponds to the depth of the potential well: when $r_{ij} = r_0$, the potential is $V_W = -E_0$. The quantity σ is the coordinate where the potential is null: when $r_{ij} = \sigma$, the potential is $V_W = 0$ and, for such a reason, σ is interpreted as the *hard-core van der Waals radius*.

Problem 13.2 Determine the relation $r_0 = 2^{1/6}\sigma$ between the equilibrium distance r_0 and the hard-core radius σ of the van der Waals potential energy function, Eq. 13.27.

Solution At the equilibrium distance, r_0, the force $-\partial V_W/\partial r$ over the particles is zero. Thus, by differentiating V_W with respect to r, we have:

$$\frac{\partial V_W}{\partial r} = 4E_0\left[-12\,\frac{\sigma^{12}}{r^{13}} + 6\,\frac{\sigma^6}{r^7}\right]_{r=r_0} = 0 \quad \Rightarrow \quad 12\,\sigma^{12}/r_0^{13} = 6\,\sigma^6/r_0^7$$

From the last expression, we obtain $r_0 = 2^{1/6}\sigma$.

The smoothness of the repulsive part, first term of Eq. 13.27, may be also modulated using a decaying exponential function, which is in turn based on the short-range energy behavior of an anti-bonding molecular orbital, for example, of the H_2 molecule.

$$V_W = \gamma\left[\exp(-r_0/r_{ij}) - \left(r_0/r_{ij}\right)^6\right] \tag{13.28}$$

The above equation is the Buckingham expression of the van der Waals potential. The van der Waals forces strongly depend on the environment due to the re-arrangements of the electron cloud with the nearby molecules, in consequence provoking the induction and extinction of local dipole moments. The dipoles create a screening effect, resulting in the reduction of the magnitude of the electrostatic interactions. Thus, it is common to incorporate additional terms in the forcefield to take the electronic polarization into consideration and improve the reliability of the simulations.

The van der Waals and electrostatic contributions are non-bonded potentials of pairwise nature, and their calculations represent one of the highest computational costs in the simulations. If we consider that an atom has non-bonded interactions with all the other atoms of the system, then for a total number of N atoms, there is a number of $N(N-1)/2$ interactions. For a large number N of atoms, the calculations scale quadratically, N^2. For such a reason, the non-bonded interactions are in many times calculated just for the nearest neighbors. This last process is implemented by introducing a cutoff distance, with the creation of a *list of atom neighbors*.

The purpose of the list is to keep track of the atom neighbors and compute the atomic interactions according to such a list. The list assigned to each atom should be updated on a regular basis because the atoms of the list may diffuse, getting out of the neighbor limits. Scale factors of the Coulombic and van der Waals interactions may be also included in the simulations to implicitly consider distant atoms. The scale factors are estimated by averaging the interactions with distant atoms that belong to the background. The introduction of scale factors is expected to reduce the number of non-bonded calculations in the simulations.

In many forcefields, the non-bonded interactions between atoms of a covalent pair are discarded, since such interactions are already introduced by potential functions of the type V_2, which describe the pair interactions, and similarly the functions V_3 and V_4.

13.13 Interaction Potential Functions of Water

The dynamics of molecules many times take place in a fluid. The fluid may be water, benzene, chloroform, acetone, ethanol, etc. The interaction of the fluid with the molecule may be considered explicitly or implicitly. In the explicit form, the atoms of the fluid are considered, and a model potential energy function is assigned to simulate the interactions among atoms, following the rules of the forcefield approach.

In the case of water, which is the most common fluid, the common energy potential functions are the $TIPnP$ models, with $n = 3, ..., 6$. The atoms of the water molecule are bonded by distance and angle potentials of the types discussed in Sects. 13.5 and 13.7. The $TIPnP$ models also introduce n partial charges on the water atoms and in centers that may be massless but symmetrically located in the neighborhood of the water atoms. The fictitious charged centers are due to that part of the electron cloud that has no participation in bond formation. The water molecule has C_{2v} symmetry or isosceles-triangle geometry, and the fictitious charged centers are strategically located, depending on the $TIPnP$ model, around the water atoms. In the $TIP3P$ potential energy function, there are no fictitious centers, and all the charges are situated on the atoms. In the $TIP4P$ function, three of the charges are on the atoms, and the extra charge is located on a point of the line bisecting the angle HOH, in proximity to the oxygen atom. In the $TIP5P$ function, the two charged fictitious centers are positioned in such a way that the water atoms plus the two fictitious centers resemble the geometry of the CH_4 compound with T_d or tetrahedral symmetry and the oxygen atom positioned at the center of the geometry, like the carbon of CH_4. The $TIP6P$ function includes the renormalized charged centers of the $TIP4P$ and $TIP5P$ models together. In short, the purpose of the extra charged sites is to delocalize the electronic charge of the atoms in an attempt to reproduce the quantum-mechanical charge distribution in a kind of coarse-grained approach.

In addition to the $TIPnP$ water models, other representations like the SPC (simple point charge) three-center water model are available. The parameters of all

these models are proposed after reproducing the dipole moments of the individual water molecule, and the collective properties of bulk water, such as the density, radial distribution function, relative permittivity, vaporization heat, static dielectric constant, diffusion constant, fluid shear viscosity, and the free energy of hydration, at a given pressure and temperature. The parametrization process is particular of the model interaction potentials, which may be also designed for specific purposes, for example, to investigate the different water phases.

The interactions between the solvated molecule and water atoms are of non-bonded type, due to the electrostatic and van der Waals forces.

$$V_{wat} = \gamma \frac{q_i q_j}{r_{ij}} + \frac{\alpha}{r_{ij}^{12}} - \frac{\beta}{r_{ij}^{6}} \tag{13.29}$$

The Coulomb interactions also consider the offset distances introduced by the fictitious charged centers. When the structure of the water is of second importance in the simulation, the pair of bonds OH and the angle HOH may be maintained rigid using holonomic constraints in the equations of motion. The rigid approximation of the water molecules allows for computational efficiency, but reduces the accuracy of the simulations. Further improvements referent to the computational cost are obtained by: (i) considering a cheap $TIPnP$ water model where $n = 3$ or 4, and (ii) including van der Waals interactions between the oxygen atoms solely, neglecting the van der Waals interactions among the water hydrogens. A coarse-grained approach where one, two, or more water molecules are described by just one or two charged centers, using a kind of united atom strategy, may be also applicable. A radical strategy consists on avoiding the whole water molecules and introducing their effects implicitly on the solvated molecule with the assumption of a polarizable model. However, the approaches introducing the individual water molecules are accurate and more realistic, since the solvated molecule has the opportunity to interact with the neighbor atoms of the solvent. Such approaches come with a high computational cost, as the charged centers of the molecules forming the fluid scale the number of interactions. A similar discussion applies to solvents different to water by introducing the proper atom types, interaction potential functions, and their parameters.

13.14 Atom Polarization

When an atom is immersed in an electric field (it may be an external field or the field produced by the surrounding molecules), the electron cloud is deformed. The positive nucleus or ion is displaced toward the electric field $\mathbf{E}$, while the electrons move away from the source of the electric field. The net effect of the field on the atom is the charge redistribution, producing an internal polarization that in turn builds an electric field to counteract the external field, Fig. 13.4. There are atoms with different degrees of polarization; this property defines the *induced dipole*, μ [11].

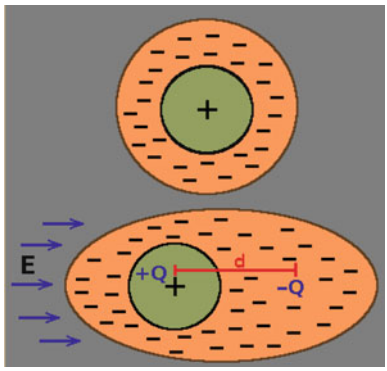


Fig. 13.4 In the top figure, an electron cloud (area depicted with many negative signs) surrounds the nucleus (area depicted with a positive sign). In the bottom figure, the atom is immersed in an electric field **E**, and as a result, the electron cloud is distorted. A dipole is created due to a shift d of the negative charge with respect to the positive charge

$$\mu = \alpha E + \beta\, E^3/3! + \cdots \tag{13.30}$$

The expression is acceptable for small distortions of the electron cloud, when the field strength is small. We usually deal with the first term, where the factor α is the *electric polarizability*. The energy of the dipole in the electric field is:

$$V = -\int_0^E \mu \cdot d\mathbf{E} = -\alpha E^2/2 \tag{13.31}$$

In a molecule, the atoms are polarized by the electric field of the other atoms. The polarizability of an atom in a forcefield is emulated by considering the atom as an entity with internal structure. The atom polarizability is approximated using fictitious charged centers. They are, in fact, *Drude particles* playing the roles of a central core and an external shell. The core and shell mimic the ion and the electron cloud, respectively. The core and shell split the atom charges and are supposed to be harmonically linked to each other.

$$V_{pol} = k r_{ij}^2/2 \tag{13.32}$$

There is no other interaction between the core and shell units as, otherwise, an additional interaction potential function has to be designed. There are neither rotational energy and individual kinetic energy contributions from the core and shell to the total energy, because the core and shell are considered to be a single unit. An instantaneous equilibrium is produced when the mechanical force kd equals the electric force qE, where d is the maximum displacement of the electron cloud. In this case, $kd = qE$. On the other hand, if we consider that the dipole is written in the form $\mu = qd$, together with Eq. 13.30, then we can establish identities between the mechanical and electrical variables $kd = qE = q\mu/\alpha = q^2 d/\alpha$. In this regard, the polarizability is written in terms of the basic constants of the atom; more precisely, the polarizability is inversely proportional to the restoring force constant k and proportional to the square of the charge.

$$\alpha = q^2/k \tag{13.33}$$

The core and shell may additionally divide the atom mass. The masses linked by a harmonic force lead to an internal vibration, with a frequency inside the atom that is substantially larger than the vibrational frequency of the whole atomic unit observed in the molecule. The internal vibration may get correlated with the motion of the other atoms, but fortunately, the correlation is frequently small. The position of the core is usually this of the atom, and the position of the shell is determined after minimizing the molecular energy with respect to the shell position at each time step of the simulation to have a self-consistent dynamic process with the other atoms. The higher-order polarization terms of the atom may be equally simulated by dealing with spherical shells around the core, which may perform breathing motions. In this case, the interaction of the polarized atom with the other atoms of the molecule depends on the shell radii.

For a couple of molecules interacting through their dipoles, the *Stockmayer potential* may be used as the model interaction potential.

$$\begin{aligned} V(r) &= 4\varepsilon\left[\left(\tfrac{\sigma}{r}\right)^{12} - \left(\tfrac{\sigma}{r}\right)^{6}\right] - \tfrac{\mu_1\mu_2}{4\pi\varepsilon_0 r^3}\,\eta(\theta_1,\theta_2,\phi) \\ \eta(\theta_1,\theta_2,\phi) &= 2\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2\cos\phi \end{aligned} \tag{13.34}$$

The dipole moments of the molecules are μ_1 and μ_2. The characteristic length and energy are σ and ε. The quantities θ_1 and θ_2 are the angles of the first dipole axis and the second dipole axis with respect to the line linking the two dipole centers, and ϕ is the azimuth angle between the two dipole axes.

In summary, the spring constants, partial electric charges, and positions of the Drude particles, these positions determined at every time step of the simulation, define the way in which the electrically polarized atoms respond to the local and external electrostatic fields. The polarizability model demands additional computational resources. The Drude particles are not expected to correctly respond to intense electrostatic fields as, in those cases, the electron cloud suffers severe distortions.

13.15 External Fields and Potentials

There are situations when we require to investigate the behavior of a molecule in the presence of an external field. The forces exerted on the ith atom by an electric field, magnetic field, and gravitational field are, in that order:

$$\mathbf{F_i} = \mathbf{f_i} + q_i\mathbf{E} \quad ; \quad \mathbf{F_i} = \mathbf{f_i} + q_i(\mathbf{v}_i \times \mathbf{B}) \quad ; \quad \mathbf{F_i} = \mathbf{f_i} + m_i\mathbf{g} \tag{13.35}$$

where the other atoms of the molecule are supposed to exert a net force $\mathbf{f}_i$ on the ith atom. By introducing a sinusoidal function, a pulsating electrostatic field is simulated.

$$\mathbf{F_i} = \mathbf{f_i} + q_i \mathbf{E} \sin(2\pi\omega t) \tag{13.36}$$

There is always the possibility to use other types of external potentials, for example, to simulate the external forces that help in the folding of a protein [12]. In this case, the external potential is of mechanical type. The effects of laser optical tweezers and those of an atomic force microscope to fold or unfold a protein can be equally simulated using mechanical potentials [13]. The possibility of creating external potentials with different purposes, for example, to stress a polymer, accelerate its relaxation, induce a particular reaction [14], confine the molecular system, examine certain regions of energy landscapes, dissociate a substrate from the receptor, follow a trajectory along a specific pathway, induce the transport of ions through membrane channels, and more, opens a new branch recognized as *steered molecular dynamics*.

Problem 13.3 Design a potential that simulates a spherical confinement.

Solution A model potential that may simulate a spherical confinement should in principle depend on the radius R_0 of the spherical confinement. In this respect, it is possible to propose an exponential function in terms of r, which is the particle's radial position.

$$V = V_0 \exp(\alpha r)/\exp(\alpha R_0)$$

The term V_0 is the potential strength. The factor $\exp(\alpha R_0)$ is a normalization term. When $r = R_0$, the potential V achieves the maximum value, $V = V_0$. The factor α determines the potential steepness. When α is large, the potential increases rapidly and strongly repels the particles close to the spherical surface, this is, the repulsive potential plays the role of a rigid wall. On the other hand, when α is small, the potential decreases slowly and softly repels the particles close to the spherical wall. By using several potentials of exponential type, the confinement of particles in a rectangular box or in a toroidal circuit can be equally designed.

13.16 Parameterization of Forcefields

A forcefield is a set of empirical functions with the required parameters to simulate the atomic interactions of a family of molecular compounds. The parameters are defined according to the atom types of the molecule and correspond to the constants present in the bonded and non-bonded interaction potential functions. In the bonded contributions, the parameters are, for example, the lengths and angles of reference

and the force constants. In the non-bonded contributions, the parameters are the effective atomic charges of the electrostatic interactions and the reference radii and energy strengths of the van der Waals interactions at the equilibrium distance. The parameters may be initially proposed from accurate first-principles calculations on small reference molecules and refined by matching the experimental measurements on bulk systems, particularly to reproduce the X-ray structures, the spectroscopic vibrational patterns, the thermodynamic energies, phonon energies, specific heats, elastic constants from compression measurements, etc. [15]. The parameters may be equally constrained to reproduce a database of well-known properties of a family of compounds. An optimal fitting process demands a global search of the parametric space to determine the best parameters.

A model interaction potential improperly parameterized in a FF may result in the inaccurate behavior of the molecule in the simulation. Thus, it is important to work with properly parametrized forcefields and pay attention to the following topics. The set of parameters are specific to that part of the phase space and molecular conformations where the parameters were obtained for. The parameters are not transferable to different molecules or applicable to (very) different conformations of the same molecule. They are neither transferable to different forcefields, although different forcefields correctly parameterized should predict similar conformations and energy changes of the same molecule. For a given forcefield, some parameters may be reused when specific molecular fragments of a large molecule are identical to the molecular fragments of another molecule that has been already parameterized. This process enlarges the list of molecules that a FF may be applied to and keeps the FF self-consistent. Any model interaction potential with its set of parameters should be extensively tested before it can be applied.

Most software packages implementing classical molecular dynamics include their own data banks, with the set of parameters for the particular molecular structures they deal with. In such data banks, there is a file containing the parameters of the MIPs. The file is frequently referred to as the *parameter file*; it comes together with the *topology file*, where the atom types and their connectivity in the molecule are indicated. In this regard, the proposal of new structures with their atom types and parameters requires the modification of such files in the software codes.

Problem 13.4 Consider a pair of interacting atoms AB, with masses $m_A = 1.2$ and $m_B = 1.4$ amu. After carrying first-principles calculations, as given in Eq. 13.1, the energies of the molecule AB at different distances are registered.

Distance(Å)	Energy(au)	Distance(Å)	Energy(au)
$R_1 = 1.20$	$E_1 = 0.062$	$R_4 = 2.00$	$E_4 = 0.023$
$R_2 = 1.40$	$E_2 = 0.022$	$R_5 = 2.40$	$E_5 = 0.123$
$R_3 = 1.60$	$E_3 = 0.002$	$R_6 = 2.80$	$E_6 = 0.303$

(continued)

Determine the values of the constants k and R_0 of a model harmonic potential from the numerical values, and predict the vibrational frequency of the molecular bond. Discuss the validity of the model potential when the molecule is immersed in water.

Solution A model harmonic potential has the form $V(R) = k(R - R_0)^2/2$, the force constant is k, the equilibrium distance is R_0, and R is the instantaneous separation between atoms A and B. By plotting the values of the table, we obtain a parabolic curve, which is reproduced by the expression $E(R) = k(R - R_0)^2/2$. The constants that best fit the set of values are $k = 0.50\ au/\text{Å}^2$ and $R_0 = 1.70$ Å. According to Eq. 13.2, we write $V(R) = E$, and the model harmonic potential that describes the oscillations of the AB bond is $V(R) = 0.50\,(R - 1.70)^2/2\,au$ for $R \epsilon [1.20, 2.80]$ Å. It is now possible to predict interparticle distances with their corresponding energies and the vibrational frequency ω of the pair AB in vacuum.

$$\omega = \sqrt{k/\mu} = 0.880\ [au/(\text{Å}^2\, amu)]^{1/2}$$
$$1/\mu = 1/m_A + 1/m_B = 1.548\ amu^{-1}$$

The oscillations of AB depend on the molecular environment, and the above model potential V is not acceptable for the case of AB in water, this is, the results are only valid in the spatial domain [1.20, 2.80] Å in vacuum, where the parameters were determined. We require new data of the molecule AB in water to carry the appropriate parametrization of the MIP.

13.17 Model Potentials of Non-biological Systems

In addition to the forcefields for biological systems, there are model interaction potentials especially designed for non-biological systems. Such potentials are referred to as many-body potentials. The metals and the alloys are described by many-body potentials. A many-body potential is a function which involves N-body interactions under the premise that the interactions between atoms depend upon the arrangement of the neighboring atoms. The many-body potentials introduce the effects of the environment in the molecular bonds, and therefore, they differ from a two-body potential, where the interaction depends on the distance of the two concerned particles solely. A many-body potential is approximated by correlating two-body terms with higher-order effects. The formulation of a many-body potential relies on the *embedded-atom approach*, which departs from pair-like terms, taking the general form:

$$V = \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i}^{N} V_{ij}(r_{ij}) + \sum_{i=1}^{N} F(\rho_i) \quad ; \quad \rho_i = \sum_{j\neq i}^{N} \rho_{ij}(r_{ij}) \tag{13.37}$$

The first term, V_{ij}, gives the two-body contributions. In the embedded-atom scheme, every atom pair is considered immersed in a bulk of atoms described by the density ρ_i. Thus, the second term of the potential introduces a functional F of the density ρ_i. It usually assumes the form $F(\rho_i) = \alpha\rho_i$, where α is a constant. The functional may be split into different shells, like the s and d shells. The density ρ_i is of local nature and depends on the atomic coordination through the functions ρ_{ij}, second expression of Eq. 13.37, which are due to atom pairs. It may be further corrected for non-local effects by considering three-body or four-body terms, and it may be also averaged over large distance interactions to achieve higher accuracy.

When there are different species of atoms in the system as, for example, in the case of alloys, and the required parameters to compute the interactions of unlike atoms are unknown, the combination rules may be applied to create the two-body potential functions and the two-body densities of the mixed-species system.

$$V_{ij}^{AB} = V_{ij}^{BA} = [V_{ij}^{AA} + V_{ij}^{BB}]/2$$
$$\rho_{ij}^{AB} = \rho_{ij}^{BA} = \left[\rho_{ij}^{AA}\ \rho_{ij}^{BB}\right]^{1/2}$$

The functions V_{ij}^{AB} and ρ_{ij}^{AB} define the interaction between the A and B atoms of Eq. 13.37. However, it is recommended to carry the appropriate parametrization process to avoid the use of the combination rules on the parameters of unlike atoms. In the following lines, some examples of metallic potentials inspired by Eq. 13.37 are presented.

13.18 Sutton-Chen Potential Function

The *Sutton-Chen function* is a many-body potential designed for metallic atoms, such as Ni, Cu, Rh, Pd, Ag, Ir, Pt, Au, Pb, Al, etc. It is used for investigating clusters and crystals. It has the form [16]:

$$V = \epsilon\left\{\frac{1}{2}\sum_{i\neq j}^{N}(a/r_{ij})^n - c\sum_{i=1}^{N}\left[\sum_{j\neq i}^{N}(a/r_{ij})^m\right]^{1/2}\right\} \tag{13.38}$$

The first term is a repulsive one and the second term is an attractive one. The factor c is a positive dimensionless parameter, and the n and m exponents are positive integers with the restriction $m < n$. The parameters ϵ and a determine the energy intensity and a characteristic length of the system. The parameters are fitted to reproduce the cohesive energies per atom, and the crystal lattice parameters of different face-centered-cubic metals, with acceptable results in the prediction of the bulk moduli and elastic constants of the materials. By restricting the values

of the parameters (m, n), it is possible to use the scaling properties of the many-body Sutton-Chen potential to deal with other materials. For example, the potential function of Cu with numbers $(m, n) = (6, 9)$ may be transferred to Ni with the same (m, n) numbers, and re-scaling the values of the energy intensity and the characteristic length, and similarly for Pt and Au.

13.19 Gupta Potential Function

The *Gupta interaction potential* is a many-body function designed to describe the interaction of transition metal atoms of a cluster or crystal [17].

$$V = A \sum_{i \neq j}^{N} \exp[-p\,(r_{ij} - r_0)/r_0] - B \sum_{i=1}^{N} \left\{ \sum_{j \neq i}^{N} \exp[-2q\,(r_{ij} - r_0)/r_0] \right\}^{1/2} \tag{13.39}$$

The form of such a potential is based on Friedel's tight-binding model of s electrons. The first term is a short-range pair potential of repulsive character, and the second one is an attractive term where the function in square brackets is an embedding or glue potential. The parameters p and q are dimensionless numbers; they are assumed to change little along a transition metal series. The other parameters of the potential are the factors A and B with units of energy, and r_0 with units of length. They all are fitted on the basis of the bulk and surface properties. Interestingly, the Morse and Lennard-Jones potentials may be applied to a similar particle system; however, they are not appropriate to describe the behavior of a metal crystal since, for the topmost surface layers, they predict an increasing interlayer separation. It is a result in contraposition to experiments, especially for the (1,0,0), (1,1,0), and (1,1,1) surfaces of a face-centered-cubic transition metal. The Gupta potential provides the correct picture by giving a contraction in the separation of the metal topmost surface layers.

Problem 13.5 Obtain the force on the ith particle according to the Sutton-Chen interaction potential, on the one hand, and the Gupta interaction potential, on the other.

Solution The force on the ith particle is obtained from the expression $\mathbf{F}_i = -\nabla_i V(\{\mathbf{r}_k\})$. In the case of the Sutton-Chen potential, the function V corresponds to Eq. 13.38. By considering the identity $\partial r_{ij}/\partial x_i = x_{ij}/r_{ij}$, the expression of the Sutton-Chen force is:

(continued)

$$\mathbf{F}_i = \frac{\epsilon}{2} \sum_{j \neq i}^{N} \left(n\, a^n / r_{ij}^{n+2} - cf\, ma^m / r_{ij}^{m+2} \right) \mathbf{r}_{ij} \quad ; \quad f = \left[\sum_{j \neq i}^{N} (a / r_{ij})^m \right]^{-1/2}$$

In the case of the Gupta potential, the function V corresponds to Eq. 13.39. The expression of the Gupta force is:

$$\mathbf{F}_i = \sum_{j \neq i}^{N} \left\{ pA \exp[-p\,(r_{ij} - r_0)/r_0] - qBf * \right.$$
$$\left. * \exp[-2q\,(r_{ij} - r_0)/r_0] \right\} \mathbf{r}_{ij} / (r_0\, r_{ij})$$
$$f = \left\{ \sum_{j \neq i}^{N} \exp[-2q\,(r_{ij} - r_0)/r_0] \right\}^{-1/2}$$

If the two-body terms of the many-body interaction potential, Eq. 13.37, are avoided, then the *perturbation potential* is obtained.

$$V = -\epsilon \sum_{i=1}^{N} \left[\sum_{j \neq i}^{N} \frac{1}{2} \frac{a}{r_{ij}^m} \left\{ 1 + \mathrm{erf}[\alpha (r_{ij} - r_0)] \right\} \right]^{1/2} \tag{13.40}$$

The presence of the error function in the potential is clarified in the Ewald approach, Sect. 14.5

13.20 Tersoff Potential Function

The last expression to discuss is the *Tersoff potential function*. It was initially proposed for silicon, which is polymorphic due to the different types of bonding with close cohesive energies under moderate pressure effects, 10–150 GPa. The Tersoff expression goes beyond the pair potentials, which are useful for describing close-packed structures, abandoning the typical N-body potential, Eq. 13.4, since the Tersoff function is designed to describe bonded structures and also open structures, like those observed in silicon materials. The *coordination number*, this is, the number of neighbors of an atom, and the *bond order* or *bond strength* are specially treated in the Tersoff function. The bond order rapidly decreases when the coordination number increases. The close-packed structures maximize the number of bonds with a bond order which exhibits a weak dependence on the coordination number, this is usually the case of metallic systems. The Tersoff potential adopts the form [18]:

$$
\begin{aligned}
V &= \tfrac{1}{2}\sum_{i\neq j}^{N}\sum_j f_c(r_{ij})\left[a_{ij}\,f_r(r_{ij}) + b_{ij}\,f_a(r_{ij})\right] \\
f_r(r) &= A\exp(-\lambda_1 r) \quad ; \quad f_a(r) = -B\exp(-\lambda_2 r) \\
f_c(r) &= \begin{cases} 1 & ; \quad r < R - D \\ \frac{1}{2} - \frac{1}{2}\sin[\pi(r-R)/(2D)] & ; \; r - D < r < R + D \\ 0 & ; \quad R + D < r \end{cases}
\end{aligned}
\tag{13.41}
$$

The f_r and f_a functions represent repulsive and attractive pair potentials, respectively. Their forms are based to some extent on the universal type of bonding given by the Morse potential. The function f_c is a cutoff function to limit the range of the potential. It is continuous with continuous derivatives in terms of the distance r, going from 0 to 1 in a small range centered at the value R. The term D is a cutoff parameter. The pair of values $R = 2.75$ Å and $D = 0.1$ Å are frequently chosen together. The Tersoff potential includes the environment of a bond by dealing with the function b_{ij}, related to the bond order. Thus, an extra summation in the particle index k, with $k \neq i, j$, is introduced.

$$
\begin{aligned}
b_{ij} &= (1 + \beta^n \xi_{ij}^n)^{-1/(2n)} \quad ; \quad a_{ij} = (1 + \alpha^n \eta_{ij}^n)^{-1/(2n)} \\
\xi_{ij} &= \textstyle\sum_{k\neq i,j} f_c(r_{ik})\; g(\theta_{ijk})\; \exp[\lambda_3^3 (r_{ij} - r_{ik})^3] \quad ; \\
g(\theta) &= 1 + \frac{c^2}{d^2} - \frac{c^2}{d^2 + (h - \cos\theta)^2} \\
\eta_{ij} &= \textstyle\sum_{k\neq i,j} f_c(r_{ik})\; \exp[\lambda_3^3 (r_{ij} - r_{ik})^3]
\end{aligned}
\tag{13.42}
$$

The b_{ij} function has a monotonic decreasing behavior according to the coordination of atoms i and j. Such a function also introduces an exponential function in the bond difference, $r_{ij} - r_{ik}$, to account for the separation of bonds. The functions b_{ij} and a_{ij} impose a limit in the interaction range to just consider the first neighbor shell atoms, and account for the atom coordination's relatively weak dependence on the cohesive energy. The function b_{ij} introduces angular terms to stabilize, for example, the diamond structure. In general $b_{ij} \neq b_{ji}$, but the symmetry in such terms may be imposed when necessary. The term a_{ij} is included for completeness; however, its presence may be omitted or employed to improve the potential.

The secondary functions ξ_{ij} and η_{ij} establish the number of bonds different to the ij bond. The quantity θ_{ijk} is the bond angle between the bonds ij and ik. The dependence on the angle is not necessarily monotonic; this is the case of arsenic compounds. The coefficient h in the expression of $g(\theta)$ may take values out of the range $[-1, +1]$, the coefficient d emphasizes or diminishes the dependence on the angle θ, and the coefficient c establishes the strength of the angular effect.

The fitting parameters of the potential are A, B, λ_1, λ_2, R, D, shown in functions f_r and f_a of Eq. 13.41. We additionally have the coefficients α, β, n, λ_3, c, d, h, shown in the bond-order functions a_{ij} and b_{ij} of Eqs. 13.42. It is occasionally assumed that $\lambda_1 = 2\lambda_2 = 2\lambda_3$ for simplicity. The Tersoff function is a short-range potential frequently acting up to 3 Å. It has been used successfully in the

study of silicon and carbon structures, like the Buckminster fullerenes, nano-tubes, nano-rods, nano-cones, diamond, graphite, surfaces, etc. Less encouraging results are obtained in the formation of point defects and vacancies, and in the description of the liquid state. The combination rules may be used for unlike atoms that interact in the system.

The parameter values of the Sutton-Chen, Gupta, and Tersoff potential functions are found in the original works. The parameters depend on the type of system under investigation. The forces are all obtained after (tedious) differentiation of the interaction potential functions, $\mathbf{f}_i = -\nabla_i V(\{\mathbf{r}_j\})$ for the ith particle. The forces may be directly derived or obtained after using a symbolic-algebra software package or an artificial-intelligence software package. There is a large number of potentials in the literature, many of them are of general use, and others are specific to one or more elements of the periodic table. In this regard, the research continues in this field.

A.1 Appendix: Harmonic Model of the Dispersion Energy

The dispersion energy is an important contribution at intermediate distances. The goal in this section is to derive the dispersion energy by using a pair of simplified molecules separated by a distance r. Each molecule is neutral; however, the ion charge $+q$ and electron charge $-q$ form a dipole in each molecule, because each molecule is immersed in the electric field of the other molecule. The two molecules lie along the x axis. The positive charges have fixed positions, and the negative charges are allowed to vibrate harmonically in the neighborhood of their ions in the x axis with frequency $\omega_0 = \sqrt{k/m}$, where m is the mass of the vibrating particle, and k the vibrational force constant. The ions are separated by the distance r, in such a way that the distance from $-q$ to $+q$ is always much smaller than r. We are interested in finding the dispersion energy of the system from the dipole-dipole interaction given by [19]:

$$V(r) = -\frac{2\mu_a\mu_b}{4\pi\epsilon\, r^3} \quad ; \quad \mu_a = x_a q \quad ; \quad \mu_b = x_b q \tag{13.43}$$

The dipole-dipole electrostatic interaction of the molecules is attractive and decays as r^{-3}. The dispersion energy is a non-classical energy and should be obtained in the quantum mechanical framework. In this regard, we have to solve the wave equation, taking the above potential function into consideration.

Problem 13.6 Suppose a molecule with a pair of atoms. One of the atoms has a positive charge q and the other a negative charge $-q$. Their separation is d. The dipole p of the molecule is $p = qd$. The origin of the reference

(continued)

system is located at the middle point of the line joining the atoms. Obtain the electrostatic potential at the position $\mathbf{r}$, far from the atoms, with $r \gg d$.

Solution Let us consider $\mathbf{r}_1$ and $\mathbf{r}_2$, with magnitudes r_1 and r_2, as the vectors starting at atom 1 and atom 2, respectively, both finishing at point $\mathbf{r}$. The electrostatic potential at the position $\mathbf{r}$ is:

$$V = \frac{1}{4\pi\epsilon}\left(\frac{q}{r_1} - \frac{q}{r_2}\right) = \frac{q}{4\pi\epsilon}\left(\frac{r_2 - r_1}{r_1\, r_2}\right)$$

The difference $r_2 - r_1$ at large distances is approximately $d\cos\theta$, where θ is the angle between $\mathbf{r}$ and the line joining the atoms. The product $r_1 r_2$ is also approximated by r^2. By using such approximations, the electrostatic potential includes the dipole p.

$$V = \frac{1}{4\pi\epsilon}\frac{p\cos\theta}{r^2}$$

The harmonic behavior of the oscillating charges and the interaction 13.43 of the dipoles are considered in the formulation of the wave equation.

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_a^2} - \frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_b^2} + \left[\frac{k}{2}\left(x_a^2 + x_b^2\right) + \frac{2x_a x_b}{4\pi\epsilon}\frac{q^2}{r^3}\right]\psi = E\psi \tag{13.44}$$

The distance from $+q$ to $-q$ in one of the molecules is x_a and in the other molecule is x_b. The complexity of the potential is reduced by introducing the new variables x_1 and x_2 and the effective force constants k_1 and k_2.

$$\begin{aligned} x_1 &= (x_a + x_b)/\sqrt{2} \quad ; \quad x_2 = (x_a - x_b)/\sqrt{2} \\ k_1 &= k\left(1 + \frac{1}{4\pi\epsilon}\frac{2q^2}{kr^3}\right) \quad ; \quad k_2 = k\left(1 - \frac{1}{4\pi\epsilon}\frac{2q^2}{kr^3}\right) \end{aligned} \tag{13.45}$$

The potential now avoids crossed terms in the new variables.

$$\frac{k_1}{2}x_1^2 + \frac{k_2}{2}x_2^2 = \frac{k}{2}\left(x_a^2 + x_b^2\right) + \frac{2x_a x_b}{4\pi\epsilon}\frac{q^2}{r^3} \tag{13.46}$$

The wave equation is given in terms of the variables x_1 and x_2 and the effective spring constants k_1 and k_2.

$$\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_1^2} + \frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_2^2} + \left[\frac{k_1}{2}x_1^2 + \frac{k_2}{2}x_2^2\right]\psi = E\psi \tag{13.47}$$

The equation corresponds to two independent harmonic oscillators. The energy and frequencies are given in terms of the old frequency ω_0 and the old force constant k.

$$\begin{aligned} E &= \hbar\omega_1\left(n_1+\tfrac{1}{2}\right)+\hbar\omega_2\left(n_2+\tfrac{1}{2}\right) \\ \omega_i &= \sqrt{k_i/m} = \omega_0\left(1 \pm \tfrac{1}{4\pi\epsilon}\,\tfrac{2q^2}{kr^3}\right)^{1/2} \quad ; \quad i=1,2 \end{aligned} \tag{13.48}$$

When the oscillators are far from each other, $r \to \infty$, they have no interaction and the energy of each oscillator in the ground state is $E = \hbar\omega_0/2$. At intermediate distances, the oscillators interact and the total energy of the system contains the dispersion energy. We apply the binomial expansion $(1+x)^{1/2} = 1+x/2-x^2/8+x^3/16+\cdots$ to the expressions of the frequencies ω_i, where the distance r is present, to deal with the energy which is now called E_{tot}.

$$E_{tot} = \tfrac{1}{2}\hbar\omega_0\left(1+\tfrac{1}{4\pi\epsilon}\,\tfrac{2q^2}{kr^3}\right)^{1/2} + \tfrac{1}{2}\omega_0\left(1-\tfrac{1}{4\pi\epsilon}\,\tfrac{2q^2}{kr^3}\right)^{1/2}$$

$$\begin{aligned} E_{tot} = {} & \tfrac{1}{2}\hbar\omega_0\left[1+\tfrac{1}{4\pi\epsilon}\,\tfrac{q^2}{kr^3}-\tfrac{1}{2}(\tfrac{1}{4\pi\epsilon}\,\tfrac{q^2}{kr^3})^2+\cdots\right] \\ & +\tfrac{1}{2}\hbar\omega_0\left[1-\tfrac{1}{4\pi\epsilon}\,\tfrac{q^2}{kr^3}-\tfrac{1}{2}(\tfrac{1}{4\pi\epsilon}\,\tfrac{q^2}{kr^3})^2+\cdots\right] \end{aligned}$$

$$E_{tot} = \hbar\omega_0 - \tfrac{1}{2}\,\hbar\omega_0\left[\tfrac{1}{(4\pi\epsilon)}\,\tfrac{q^2}{kr^3}\right]^2 \tag{13.49}$$

The expansion is consistent with Eq. 13.43 referent to the interaction of distant dipoles. We require to subtract the energy of the infinitely separated oscillators from the above energy to obtain the dispersion energy.

$$E_{disp} = E_{tot} - E\Big|_{r\to\infty} = -\hbar\omega_0\,\tfrac{q^4}{2(4\pi\epsilon)^2k^2r^6} \tag{13.50}$$

The zero-point energy, $\hbar\omega_0$, is present in E_{disp} and exhibits the quantum character of the interaction. The dispersion energy is written in equivalent form by defining the factor C_6.

$$E_{disp} = C_6/r^6 \quad ; \quad C_6 = -\hbar\omega_0\, q^4/[2(4\pi\epsilon)^2k^2] \tag{13.51}$$

By using the expression of the polarizability 13.33, the dispersion factor C_6 is expressed in terms of the molecular polarizability.

$$C_6 = -\hbar\omega_0\,\alpha^2/[2(4\pi\epsilon)^2] \tag{13.52}$$

The dispersion energy depends on the spatial orientations of the interacting dipoles and also of the individual energy states of the molecules. In many situations, the dispersion energy is an attractive energy; however, it may also be a repulsive

energy. The dispersion energy demands, in fact, higher-order multipoles to be properly described.

$$V_{disp} = C_6/r^6 + C_8/r^8 + C_{10}/r^{10} + \cdots \tag{13.53}$$

The dispersion energy of three particles is approximated with additive terms plus a non-additive contribution using the *Axilrod-Teller expression*.

$$V_{disp} = C_6^{ij}/r_{ij}^6 + C_6^{ik}/r_{ik}^6 + C_6^{jk}/r_{jk}^6 + \frac{\alpha_{ijk}}{(r_{ij}r_{ik}r_{jk})^3}\,(1 + \cos\theta_{ij} + \cos\theta_{ik} + \cos\theta_{jk}) \tag{13.54}$$

The quantities θ_{ij}, θ_{ik} and θ_{jk} are the internal angles of the triangle described by the particles i, j, and k. The angle values depend on the triangular shape (equilateral, isosceles, scalene, including a linear conformation). The cosine functions define the sign of the non-additive contribution. In this regard, the role of the angular term is to correct the additive contributions by adding/subtracting a non-additive part.

The van der Waals interaction potential of Eq. 13.26 includes the dispersion function $-\beta/r^6$ as an attractive term. Still, such a potential additionally includes a repulsive interaction of the electron clouds at short distances. This last contribution is simulated by means of the term α/r^{12} to overcome the attractive dispersion $-\beta/r^6$ at short distances. The discussion on the proper form of the van der Waals interaction potential continues in our days [20].

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Extended Systems

14

The study of extended systems is important in science. An extended system may be a gas, fluid, or solid. A crystal, an amorphous material, and a substance with no homogeneous structure and no periodicity are considered to be $3D$ extended systems. Other physical systems such as a surface or ceramic are $2D$ extended systems. A large $1D$ structure, such as a polymer, may be considered an extended system of reduced dimensionality.

In many instances, we require to investigate the properties and the behavior of extended matter or the features of a compound immersed in an extended sample, like a protein in water, with the water extending all over the available space. Yet, we have a limit on the number of particles that can be simulated in a computer to study extended systems.[1] This limitation compels to implement approximations to investigate the behavior of extended matter. The purpose of this chapter is to discuss the methods that allow the characterization of extended systems by dealing with a relatively small number of atoms of the sample. In a first stage, we consider a finite material sample enclosed by walls. Later, the walls are avoided to work with a material with no apparent borders, but with high repetitive structure. In this last case, the Ewald summation approach for systems with periodic boundary conditions is introduced. The electrostatic interactions in the Ewald approach are divided according to short-range and long-range interactions, leading to closed formulae which are computationally accurate and efficient. In the final part of the chapter, alternative approaches are discussed to compute the electrostatic interactions from modified potentials and forces. The required mathematical tools in every case are presented.

[1] A mole of substance is considered the smallest piece of matter representative of a substance; still, it contains a huge number of particles, specifically 6.0×10^{23}, for a molecular simulation.

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14.1 Fixed and Flexible Boundaries

When many particles interact, they may scatter in all directions. In a molecular dynamics simulation, it is important to introduce boundaries to confine the particles and avoid their dispersion. The confinement of particles is of help to study a small or large fraction of the atoms of an extended system. The confinement may be imposed in different ways. The case of rigid walls is one of the straight forms to confine particles. Such a case evolves to a more realistic model when the boundaries are simulated using repulsive walls. The repulsive walls may be simulated by introducing exponential functions with positive arguments.

$$V(\{\mathbf{r}_i\}) = \sum_{i=1}^{N} \exp(\alpha r_i)/\exp(\alpha r_0) \quad ; \quad r_i = \left(x_i^2 + y_i^2 + z_i^2\right)^{1/2} \tag{14.1}$$

This expression gives the spherical confinement of the particles. The summation is over the confined particles. The parameter α determines the growing rate of the exponential potential, V; big values of α make the potential rising sharply, and small values of α make the potential increasing slowly. The exponential in the numerator is weighted with a similar exponential in the denominator. This last function represents a normalization factor because the potential acting on a particle becomes unity at r_0, which corresponds to the radial position of the wall. Thus, the increasing exponential plays the role of a repulsive potential that keeps the particles confined and away from the confining surface. There are other forms of repulsive potentials. Such barriers may additionally exhibit texture by using different or changing values of r_0; an oscillating wall is obtained by doing $r_0 = Af(t)$, where A is the position of the wall and the factor $f(t)$ is, for example, $[1 + (1/10)\sin(\theta(t))]$. When θ is written in terms of the time, $f(t)$ introduces a pulsating behavior of the wall with 10% fluctuations of the value of A. Other forms of $f(t)$ may be proposed.

The function playing the role of a repulsive potential may be replaced by particles linked to each other to form a relatively hermetic container with atomic structure. The particles at the interior of the container move and interact with the particles of the container and cannot escape from the confinement. In response to the interaction with the confined particles, the particles forming the container may be permitted to vibrate in the neighborhood of their positions, without missing the shape of the confining cage. The links between the container atoms may be of harmonic type for simplicity, Fig. 14.1.

$$V(\mathbf{r}_i) = k\left(r_{ij} - r_{ij}^0\right)/2 \tag{14.2}$$

The spring stiffness is k, and the equilibrium distance between the ith and jth particles is r_{ij}^0. When k or the masses of the boundary particles are sufficiently large, the fixed-particle boundary scheme is recovered, because the wall particles barely move. The dynamics of the particles enclosed by rigid walls is different to the dynamics of the particles enclosed by flexible walls. This last type of approach is known as the *semi-rigid boundary scheme*. In principle, a large number of atoms

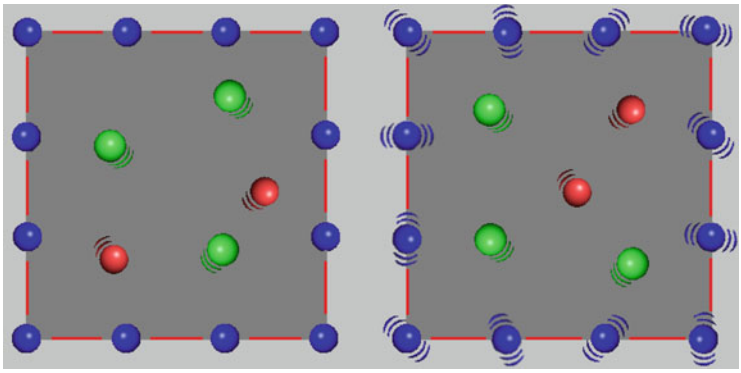


Fig. 14.1 The inset on the left shows a system of particles; they are confined by rigid walls formed by particles with fixed positions. The inset on the right shows flexible walls, formed by particles that vibrate in the neighborhood of their original positions

(that build a sample) may be imprisoned by walls to investigate the properties of extended matter.

Problem 14.1 Consider a torus formed by sweeping a disk around a circular axis. The torus lies in the xy plane with radius R and disk radius r_d. Create a potential energy function that confines N particles in a container with toroidal geometry from exponential functions.

Solution Suppose that the position of the ith particle is $\mathbf{r}_i$. It is possible to find a factor λ in such a way that $\lambda(x_i^2 + y_i^2)^{1/2}$ gives the torus radius R, this is, $\lambda = R/(x_i^2 + y_i^2)^{1/2}$. Once the factor λ for particle i is determined, the torus vector radius is $\mathbf{R} = \lambda(x_i, y_i, 0)$. Next, consider the disk around the ith particle. The position of the ith particle with respect to the central point of the disk is $\mathbf{r}_i' = \mathbf{r}_i - \mathbf{R} = (x_i - R_x, y_i - R_y, z_i - 0)$. The toroidal confinement is now built from exponential functions.

$$V = V_0 \sum_{i=1}^{N} \exp\left(\alpha r_i'\right) / \exp(\alpha r_d)$$

When r_i' reaches the disk radius, this is $r_i' = r_d$, the potential exerted on particle i is a maximum since $V_0 \exp(\alpha r_i')/\exp(\alpha r_d) = V_0$; otherwise $V_0 \exp(\alpha r_i')/\exp(\alpha r_d) < V_0$. Therefore, particle i is confined in a toroidal geometry when the potential V is applied.

14.2 Periodic Boundary Conditions

A common approximation to simulate an extended system using a minimum of computing resources is the *periodic boundary conditions* (PBC) approach. It consists in taking a representative portion of the system to build a *central unit cell*, and repeat it in every direction to fill the entire space, Fig. 14.2.

A requirement to apply the PBC approach is that the shape of the central cell covers the $3D$ space by employing translational operations on the cell. The resulting system is a perfect periodic system, representing an idealization of the real extended system. The most common cell shapes are the cubic cell, hexagonal prism, truncated octahedron, and rhombic dodecahedron, Fig. 14.3. Other shapes may not satisfy the required conditions, like a cell with spherical symmetry. The favored cell shapes are those that not only tile the $3D$ space but also involve the lowest number of particles per cell, without missing the relevant aspects of the system, because they reduce the computational costs.

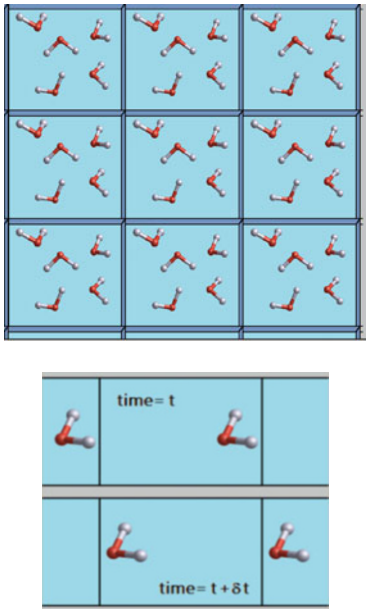


Fig. 14.2 Top inset: in the periodic boundary conditions approach, a central unit cell is indefinitely repeated in every direction, and there are no borders. The repetition of the nanoscopic piece of matter simulates the crystal. In this $2D$ periodic lattice representation, the central and image cells may be enumerated in the form [(-1,-1), (-1,0), (-1,1)], [(0,-1), (0,0), (0,1)], [(1,-1), (1,0), (1,1)], where the indexes run from the lower-left corner to the upper-right corner. Bottom inset: the particles of the boxes may travel from one box to another crossing the imaginary borders. A molecule is located in the central unit cell at the instant of time t. After an interval of time δt, the molecule moves out of such a cell, but an image molecule concurrently moves into the central unit cell to keep the particle density of the cell constant

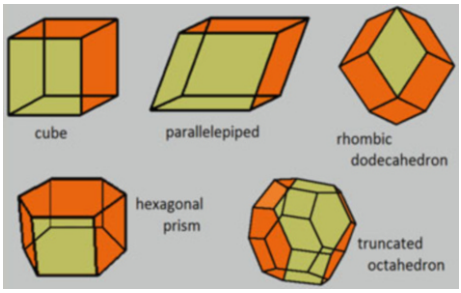


Fig. 14.3 The depicted polyhedra are some of the geometrical structures that completely tile the $3D$ space. They are commonly used in the periodic boundary conditions approach

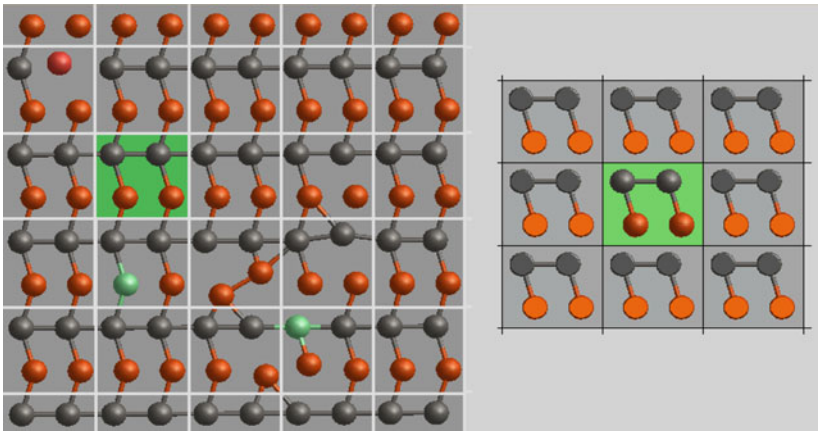


Fig. 14.4 The rendering of the left illustrates a real crystal with impurities, particle vacancies, and local irregular forms, randomly located, breaking the symmetry of the crystal. The rendering of the right shows the central unit cell with different background color to the image cells. The simulated crystal is a crystal with perfect symmetry in the PBC approach since the unit cell is repeated in every direction

The central cell contains a finite number of particles and reproduces the particle density of the real sample. The copies of the central cell are known as the *image cells* and have the same size and shape of the central cell. The image cells contain an equal number of particles and the same positions and momenta of the particles as those of the central cell. In this regard, the PBC approach simulates a perfect bulk sample with an infinite number of replicas of the central unit cell. However, it is important to remark that the simulated crystal is not necessarily the case of a real crystal because the real crystal may contain impurities, vacancies, localized irregular forms, and more, with no periodicity, Fig. 14.4.

A particle in the central cell is considered to interact with the other particles of the central cell, plus the closest particles in the neighbor image cells. The force exerted on the ith particle by the jth particle, plus the corresponding jth particles of the neighbor image cells is:

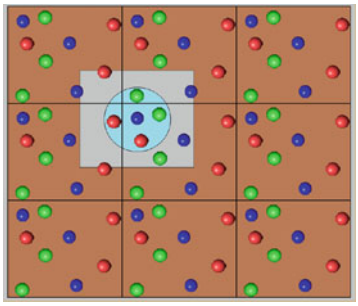


Fig. 14.5 According to the minimum image convention, the particle in blue color and centered in the gray square interacts with the other particles located in that square solely. In a different approximation, the same blue particle may only interact with the other particles located inside the blue sphere, which has a given cutoff radius. In this case, the number of particles in the blue sphere is lower than the number of particles in the gray square

$$\mathbf{f}_{ij} = -\left[\nabla_i V(r_{ij^{(0)}}) + \nabla_i V(r_{ij^{(1)}}) + \cdots + \nabla_i V(r_{ij^{(26)}})\right] \tag{14.3}$$

The nomenclature $j^{(k)}$, for $k = 0, 1, \cdots, 26$, stands for the jth particle of the central cell, $k = 0$, and the image particles of the neighbor cells, $k = 1, \cdots, 26$, in $3D$ space. In order to minimize the number of calculations, the interactions may be truncated. In this case, the *minimum image convention* is assumed, where a particle located in the central unit cell only interacts with the surrounding particles positioned inside an *imaginary cell*, iL^3, with equal dimensions to the central unit cell, and where the particle of interest, namely, the ith particle is located at the center of such an imaginary cell. In this case, the force exerted on the ith particle is due to a reduced number of particles, Fig. 14.5.

$$\mathbf{f}_{ij} = -\sum_{j\,\varepsilon\, iL^3} \nabla_i V(r_{ij}) \tag{14.4}$$

The summation runs over the particles of the imaginary cell. It may happen that a moving particle in the central cell travels from the central cell to a contiguous image cell during the dynamics. This situation changes, among other things, the particle density of the simulated material. Hence, in order to maintain the properties of the material, an equivalent particle with equal velocity must enter the central cell from an opposite face, Fig. 14.2.

In principle, by using the PBC approach, we work in the thermodynamic limit, making possible to apply the thermodynamic concepts to the extended sample. Thus, the PBC scheme is frequently used to investigate the thermodynamic aspects of extended systems. Still, there are some drawbacks of the PBC approach. One of these, previously mentioned, is referent to the repetition of a unit cell describing a perfect material. In fact, the real material is not perfect in composition, neither in the atomic spatial arrangements. In the case of simulating compounds in solution, like a DNA molecule or a protein in bulk water, the solvated compound interacts with all

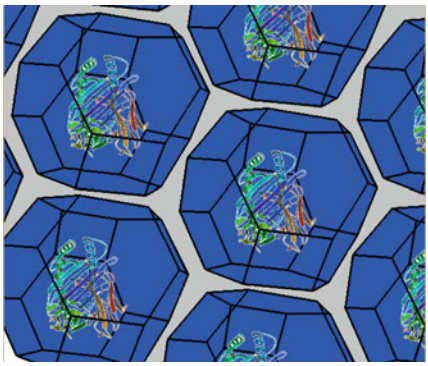


Fig. 14.6 A solvated protein encapsulated in a truncated octahedron cell is shown in the PBC approach. The water molecules are not depicted for simplicity, and the unit cells forming the crystal are separated for better visualization. In a molecular simulation of the crystal, there may be spurious forces due to interactions between the protein and its images. Such interactions are negligible when the distance between contiguous proteins is sufficiently large. This demands unit cells of large sizes

its own solvated images, especially when the central unit cell is small, introducing spurious interactions that should be avoided. A practical way to deal with such a situation is to introduce large separations between the solvated compound and its image compounds, this is, dealing with a big central unit cell to diminish the non-bonded Coulomb and van der Waals interactions, Fig. 14.6. Also, the multipolar electric moments are not preserved in the PBC simulation for the introduction of fictitious electric fluxes. More precisely, the charged atoms move in and out of the cells, producing charge-distribution changes and inducing uncommon fluctuations of the multipolar electric moments in the central cell; details are given later. It is recommended to have a zero total charge to avoid an overcharged material due to the charges of every image cell. To do this, ions or charged particles may be introduced to impose the charge neutrality of the central cell.

The long-range forces decay by their own nature slowly in the material. In the PBC approach, it is convenient to have long-range forces decaying relatively fast to avoid discontinuities due to the finite size of the central cell.

For example, the van der Waals interactions become null at relatively far distances. Yet, the van der Waals potential may be forced to decay faster by subtracting a constant term. If we require the potential to be zero at the cutoff distance $r_c = a\sigma$, where r_c is a scale factor of the hard-core van der Waals radius σ, and $1 < a$, then the truncated van der Waals potential, V_W^t, may be used under the definition:

$$V_W^t = \begin{cases} V_W(r) - V_W(r_c) \; ; \; r \leq r_c \\ 0 \qquad\qquad ; \; r_c < r \end{cases} \tag{14.5}$$

The truncated potential reaches zero at the imposed cutoff point r_c. In this respect, it is convenient to deal with a central cell whose size is larger than r_c to avoid the interaction of a solvated compound with its image counterparts. The truncated potential brings a small additional computational cost, Fig. 14.5. On the other side, measurements on the properties of traveling waves in the substance demand large box sizes. The collective fluctuations, especially near critical points, are not properly simulated when the central box contains a small number of particles due to the existence of long-range correlations. In this regard, the central cell is expected to be greater than the typical correlation lengths observed on bulk matter.

14.3 The PBC System Is an Open System

In the following lines, it is shown that there is a non-zero flux of mass, angular momentum, and energy in the central unit cell of the PBC method, and therefore, such a cell constitutes an open system. The strategy is firstly exemplified with the no-conservation of the angular momentum, $\mathbf{L}$, following the work of Kuzkin [1]. For simplicity we consider image cells with cubic shapes of length ℓ and a single particle traveling with momentum $\mathbf{p} = m\mathbf{v}$ and negligible interactions with the image particles. The particle moves in the central unit cell and eventually leaves the cell at the instant of time t. At this precise instant of time, an image particle enters the central unit cell with the same momentum from the opposite cell face. The linear momentum of the particle is preserved with time, but the angular momentum $\mathbf{L}$ is not necessarily preserved with time because $\mathbf{L}$ depends on the origin of coordinates. If δt is the time step of the particle dynamics, the change of angular momentum in this time interval is:

$$\mathbf{L}(t+\delta t) - \mathbf{L}(t) = m\mathbf{r} \times \mathbf{v} - m(\mathbf{r} - \boldsymbol{\ell}) \times \mathbf{v} \quad \Rightarrow \quad \Delta\mathbf{L} = m\boldsymbol{\ell} \times \mathbf{v} \tag{14.6}$$

The first term of the first expression corresponds to the image particle that entered in the central cell at the instant of time $t+\delta t$, and the second term corresponds to the particle in the central cell that leaves the central cell at the instant of time t, Fig. 14.7. There is a difference between $\mathbf{L}(t+\delta t)$ and $\mathbf{L}(t)$ which depends on the cell size and the particle mass and velocity. When the surface vector $\boldsymbol{\ell}$ is parallel to $\mathbf{v}$, this is, the particle moves in perpendicular form to the cell surface, the cross product is zero, and the angular momentum is preserved in time. However, the angular momentum is discontinuous when the cross product between $\boldsymbol{\ell}$ and $\mathbf{v}$ is different from zero, namely, when $\boldsymbol{\ell}$ and $\mathbf{v}$ make an angle different from zero. Therefore, when the particle stays in the central cell or it travels from one cell to another in perpendicular way to the cell surface, the angular momentum is conserved. The possibility to have a discontinuous angular momentum makes the particle system in the central unit cell of the PBC approach an open system.

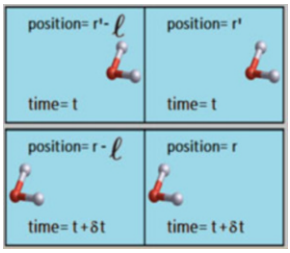


Fig. 14.7 Two instances of the dynamics of a molecule are depicted. The size of the square box is ℓ. At the instant of time t, the molecule is in the central cell with position $\mathbf{r}'$, and an image molecule is located at $\mathbf{r}' - \ell$ (upper right and upper left insets, respectively). In a later step, $t + \delta t$, the molecule leaves the central unit cell, and an image molecule enters the central unit cell. The molecule's position is $\mathbf{r}$, and the image molecule is present at $\mathbf{r} - \boldsymbol{\ell}$ (lower right and lower left insets, respectively). The angular momenta of the molecule in the central unit cell at the instants of time t and $t + \delta t$ are in general different when $\boldsymbol{\ell} \times \mathbf{v} = |\boldsymbol{\ell}|\ |\mathbf{v}|\ \sin(\theta)$ is different from zero, Eq. 14.6

The angular momentum is additionally changed by the torques exerted on the particles of the central cell due to the interaction with the image particles. The torque $\mathbf{T}$ depends on the positions and forces.

$$\mathbf{T} = \sum_{i,j} \mathbf{r}_i \times \mathbf{f}_{ij} \quad ; \quad i \ \varepsilon \ L^3(t) \quad ; \quad j \ \varepsilon \ iL^3(t) \tag{14.7}$$

The index i considers the particles of the central cell, L^3, and the index j considers the particles of the image cells, iL^3. The change of the total angular momentum in time is given by $d\mathbf{L}/dt = \mathbf{T} + \mathbf{Q}$, where the contribution $\mathbf{T}$ is given by Eq. 14.7, and the contribution $\mathbf{Q}$ represents the *flux of angular momentum* due to the incoming and outgoing particles of the central unit cell according to Eq. 14.6. Instead of writing the flux of angular momentum in terms of a differential equation, it may be written as a *balance expression* on the angular momentum.

$$\Delta\mathbf{L} = \int_{t'}^{t'+\delta t} (\mathbf{Q} + \mathbf{T})\, dt = \int_{t'}^{t'+\delta t} \left(\sum_i \mathbf{r}_i \times m\mathbf{v}_i - \sum_i' \mathbf{r}_i \times m\mathbf{v}_i + \sum_{i,j} \mathbf{r}_i \times \mathbf{f}_{ij} \right) dt \tag{14.8}$$

The first summation runs over the particles entering the central cell, and the second summation with a prime runs over the particles leaving the central cell in the time interval $[t', t' + \delta t]$. The last summation considers the interaction with the image particles. Similar types of fluxes occur for other physical variables. For example, we deal with a charge flux when the charged particles travel from one cell to another.

Problem 14.2 Consider a pair of particles $p1$ and $p2$ in the central unit cell and their images $i1$ and $i2$, respectively. The particles are allowed to move and repel each other at short distances. Demonstrate that when the particle interactions $p1-i2$ and $p2-i1$ occur, the angular momentum is discontinuous in time.

Solution Suppose that the particle $p1$ is close to the image particle $i2$, and they interact. At that instant of time, the particle $p2$ is also close to the image particle $i1$ for symmetry, and they also interact. The total torque in that instant of time is:

$$\mathbf{T} = \mathbf{r}_{p1} \times \mathbf{F}_{p1,i2} + \mathbf{r}_{p2} \times \mathbf{F}_{p2,i1}$$

In this case, $\mathbf{T}$ is not zero. Since $\mathbf{T} = d\mathbf{L}/dt$, the angular momentum is not a constant. However, when the distance between the particles $p1$ and $i2$ is large, they have no interaction, the forces $\mathbf{F}_{p1,i2}$ and $\mathbf{F}_{p2,i1}$ are zero, and the angular momentum is preserved with time as $\mathbf{T} = 0$ and $\mathbf{L} =$ constant.

The flux of angular momentum may be important for systems in the gas phase state. In liquids, such a flux produces the torques that disrupt the angular momentum of the particles in the central unit cell. In solids, the torques may break the conservation of the angular momentum. The PBC approach also induces the breaking of the equipartition principle of the energy, which applies to particles in thermodynamic equilibrium. The breaking of that principle is the consequence of imposing constraints on the position and momentum of the mass center of the system to keep it in the central unit cell of the PBC scheme. The constraints affect light-weight and heavy-weight particles in different forms, resulting in different average energies, $\langle E \rangle$, for the light and heavy particles in the dynamics. In the NVE ensemble. the average energies are related by the different particle masses [2].

$$\langle E \rangle_{\text{heavy}} = \langle E \rangle_{\text{light}} \left[1 - (m_{\text{heavy}} - m_{\text{light}})/M_{tot}\right] + \mathcal{O}\left(M_{tot}^{-2}\right)$$

Care should be exercised when thermodynamic quantities are derived in the PBC approach.

14.4 Electrostatics in the PBC Approach

The electrostatic interactions are important in the behavior of extended structures. In the PBC approach, the atoms of the central cell are repeated periodically, building an endless $3D$ grid of atoms, and the infinite number of image charges interact

with the charged atoms of the central unit cell, because the electrostatic interactions decay slowly, which complicates the computation of the total electrostatic energy. In the following sections, we introduce the *Ewald summation approach*, understood as a method that permits the computation of the electrostatic energy in the PBC scheme with the required numerical accuracy [3]. To do this, we base the discussion on the work of C. Kittel, complemented with the work of Lee and Cai [4].

The purpose of the Ewald sum approach is to compute the electrostatic energy of the PBC system under the consideration of an ordered array of charged atoms. The atomic distribution has periodicity, say d, in the interplanar distances of a crystal, with frequency $1/d$ in the *reciprocal space*. The Ewald sum approach employs Fourier transformations of the electrostatic interactions to work in the reciprocal space and obtain the total electrostatic energy. The approach is based on key aspects to simplify the electrostatic computations; the slow decay of the electrostatic potential is recognized as it is proportional to r^{-1}, and it is possible to separate the short distance interactions from the large distance interactions to individually deal with each of them. The electrostatic interactions between nearby atoms change rapidly with distance, becoming singular at $r = 0$, but they can be rapidly computed in the real space. In contrast, the electrostatic interactions between distant atoms change slowly, but they can be rapidly computed in the reciprocal space.[2] The Ewald sum approach requires the particle system in the central unit cell to be globally neutral. In order to make the summations of the electrostatic interactions convergent, Gaussian functions are introduced and located at every grid point to deal with an atomic charge distribution. The Gaussian functions are manageable under the Fourier transforms and are implemented in such a way to avoid any spurious contribution to the final energy. The Gaussian functions are essential to produce the screening effects and achieve a rapid convergence of the electrostatic summations. In this respect, the Ewald proposal is important, computationally efficient, and provides accurate results.

14.5 Ewald Sum Approach

Let us consider the presence of N charged atoms in the central unit cell. The atoms have positions $\mathbf{r}_i$ with charges q_i, $i = 1, \cdots, N$. The central cell is neutral, $\sum_{i=1}^{N} q_i = 0$; otherwise, there is no convergence in the Ewald electrostatic summations. The central cell is repeated in every direction. The *fundamental*

[2] The Yukawa approach explains the long-range nature of the Coulomb potential. In such an approach, the neutron and proton have a short-range interaction by exchanging a particle. The Yukawa approach predicted the existence of the meson. The Yukawa potential is $V_Y = -g^2 \exp(-r\mu c/\hbar)/r$, where μ is the mass of the exchanged particle and g is a scalar. The Coulomb potential is deduced from V_Y when the particle of exchange is the photon with zero mass and $g^2 = q_i q_j/(4\pi\varepsilon_0)$, this is, $V_Y = q_i q_j/(4\pi\varepsilon_0) r^{-1}$. The deduction is consistent with the uncertainty principle, as $1/R = \mu c/\hbar = \mu c^2/(\hbar c) = \Delta E/(\hbar c)$ and, on the other hand, $1/R = 1/(c\Delta t)$, which takes to $\Delta E \Delta t \sim \hbar$.

translational vectors of the lattice are $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. They are not necessarily orthogonal to each other. The numbers n_i, $i = 1, 2, 3$, are integers running without limits. The positions of the image charges are $\mathbf{r}_i + n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$, which are abbreviated by $\mathbf{r}_i + \mathbf{n}L$, with $\mathbf{n} = (n_1, n_2, n_3)$ and L the cell length. The central cell is defined with $\mathbf{n} = 0$. A charge q_i located at $\mathbf{r}_i$ of the central cell interacts with the charges q_j positioned in the central cell, with the condition $j \neq i$, plus all the periodic image charges, including its own periodic image charges. The electrostatic energy of the particle system of interest is:

$$E_{el} = \frac{1}{2}\frac{1}{4\pi\varepsilon_0}\sum_{n1=-\infty}^{\infty}\sum_{n2=-\infty}^{\infty}\sum_{n3=-\infty}^{\infty}\sum_{i=1}^{N}\sum_{j=1}^{N*}\frac{q_i q_j}{|\mathbf{r}_i - \mathbf{r}_j + \mathbf{n}L|}$$

All the charged atoms of the lattice are included in the summations. The denominator gives the distances between the different atom pairs, where one atom is located in the central cell and the other may be located in the same central cell or in an image cell. The asterisk in the last summation recalls to avoid the self-interactions in the central cell. In this respect, the diagonal terms $i = j$ are omitted when $\mathbf{n} = 0$. The asterisk may also indicate other restrictions such as the electrostatic interactions that should be avoided between bonded atoms. The factor 1/2 avoids the double counting of the interactions. The vacuum permittivity is:

$$\varepsilon_0 = 1/(\mu_0 c^2) = 8.854 \times 10^{-12}\, C^2 N^{-1} m^{-2}$$

where c is the speed of light and μ_0 the vacuum permeability. The energy E_{el} converges slowly. For a system with N atoms, the calculation of pairwise interactions to determine the energy E_{el} is computationally costly, scaling in quadratic form, $N(N-1)/2$. In the periodic boundary conditions approach, a $1D$ grid of alternating positive-negative charges separated by a distance d demands 800,000 evaluation terms to reach an electrostatic energy with accuracy up to six decimal digits. By using the Ewald summation for such a system, the same accuracy is achieved with less than a dozen evaluations. The goal is to reduce the calculation cost of the electrostatic energy of a many-particle system employing the Ewald sum scheme.

A charged atom can be described by a distribution function ρ_i centered at the point $\mathbf{r}$. In our approach, the ith charge is the distribution positioned at $\mathbf{r}_i$ with charge q_i, described by a delta function.

$$\rho_i(\mathbf{r}) = q_i\,\delta(\mathbf{r} - \mathbf{r}_i) \tag{14.9}$$

The electrostatic potential energy is now written in terms of the distribution functions.

$$E_{el} = \tfrac{1}{2}\,\tfrac{1}{4\pi\varepsilon_0}\sum_{\mathbf{n}}\sum_{i,j=1}^{N*}\iint\frac{\rho_i(\mathbf{r})\,\rho_j(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'+\mathbf{n}L|}\,d^3r\,d^3r' \tag{14.10}$$

The electrostatic energy expressed in either of the above forms may be evaluated for a system with a finite number of particles. However, for an infinite number of particles, like in the PBC case, the summations converge in relative form, but not in absolute form. The expression is said to be a *conditionally convergent sum* and it is divergent for non-neutral systems. One of the solutions to avoid complications in the evaluation of the PBC electrostatic energy is to fade the interactions away in the form proposed by Eq. 14.5 using a truncated potential. Such an approximation is efficient, but some loss of accuracy is introduced in the calculation of the electrostatic energy. Another possibility is to consider the decaying behavior of the electrostatic energy in terms of the interatomic distance, without the intervention of a truncated potential, but performing the summations for the charged atoms which are just located within a spherical threshold distance r_c. A spherical cutoff radius of approximately 8–12 Å may be chosen for every atom of interest, while the interactions beyond r_c may be neglected. This process is flexible to impose the required accuracy in the calculation of the electrostatic energy according to the radius r_c; nevertheless, some care should be exercised since the sums are conditionally convergent for big cutoff values.

14.6 Using the Poisson Equation

The lack of efficiency in the calculation of the electrostatic energy is attributed to the Coulomb expression, because it does not discriminate close atoms from distant atoms. The goal in this section is to deduce an expression that allows to differentiate close atoms from distant atoms, maintaining computational efficiency and accuracy, and making the summations of the electrostatic energy convergent.

The electrostatic potential at point $\mathbf{r}$ created by a source charge q_j positioned at point $\mathbf{r}_j$, and using the distribution 14.9, is given in the form:

$$\frac{1}{4\pi\varepsilon_0}\frac{q_j}{|\mathbf{r}-\mathbf{r}_j|} \quad \Rightarrow \quad \frac{1}{4\pi\varepsilon_0}\int \frac{\rho_j(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,d^3r'$$

The electrostatic potential $v_{(i)}$ at point $\mathbf{r}$ and created by all the charged particles, excluding the ith particle, is:

$$\begin{aligned} v_{(i)}(\mathbf{r}) &= \frac{1}{4\pi\varepsilon_0}\sum_{\mathbf{n}}\sum_{j=1}^{N*}\int \frac{\rho_j(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'+\mathbf{n}L|}\,d^3r' \\ &= \frac{1}{4\pi\varepsilon_0}\left[\sum_{\mathbf{n}}\sum_{j=1}^{N}\int \frac{\rho_j(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'+\mathbf{n}L|}\,d^3r' - \int \frac{\rho_i(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,d^3r'\right] \\ E_{el} &= \frac{1}{2}\sum_{i=1}^{N}\int \rho_i(\mathbf{r})\,v_{(i)}(\mathbf{r})\,d^3r \end{aligned} \tag{14.11}$$

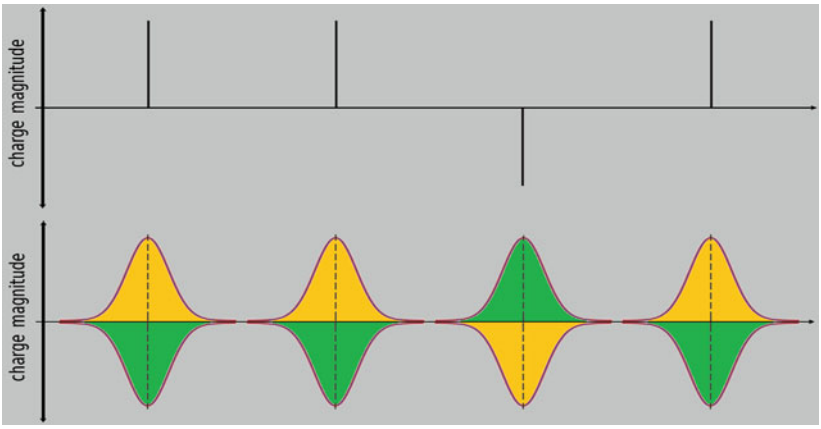


Fig. 14.8 The Ewald charge distribution functions ρ^s and ρ^l are illustrated in the $1D$ case, Eq. 14.13. The up and down vertical lines in the top inset are delta functions representing the positive and negative charges of the atoms in the $1D$ grid. The charge density ρ^s is built by both the delta functions and Gaussian functions, depicted in green color and with opposite signs to the delta functions. The charge density ρ^l is only built by Gaussian functions, in yellow color and with equal signs to the delta functions. The addition of positive and negative Gaussian functions gives zero total charge

The total energy E_{el} is written in terms of the interaction potential $v_{(i)}$. The energy excludes the self-interaction energy of the ith atom with the negative term in square brackets. The above expression of $v_{(i)}$ establishes an integral relation between the interaction potential and the charge density. Yet, there is an alternative relation of differential type between those two quantities using the *Poisson equation*.

$$\nabla^2 v_{(i)}(\mathbf{r}) = -\rho_i(\mathbf{r})/\varepsilon_0 \tag{14.12}$$

In order to discriminate close atoms from distant atoms and make the summations of the energy convergent, it is convenient to introduce distribution functions of Gaussian type centered at the grid points $\mathbf{r}_i$. They are introduced as a "smart zero," by simultaneously adding and subtracting the Gaussian functions, Fig. 14.8, in the short-range distribution function ρ_i^s and the long-range distribution function ρ_i^l. The attributes "short-range" and "long-range" shall take sense later in the discussion.

$$\rho_i^s(\mathbf{r}) = q_i\,\delta(\mathbf{r}-\mathbf{r}_i) - q_i\,g(\mathbf{r}-\mathbf{r}_i) \quad ; \quad \rho_i^l(\mathbf{r}) = q_i\,g(\mathbf{r}-\mathbf{r}_i)$$

$$g(\mathbf{r}) = \alpha\exp(-\beta r^2) \quad ; \quad \alpha = 1/(2\pi\sigma^2)^{3/2} \quad ; \quad \beta = 1/(2\sigma^2) \tag{14.13}$$

$$\rho_i(\mathbf{r}) = \rho_i^s(\mathbf{r}) + \rho_i^l(\mathbf{r})$$

The Gaussian distributions split the charge density ρ_i, but they have no contribution to the final electrostatic energy. The role of the Gaussian functions in

the charge density ρ_i^s is to screen the point charges as the signs of the Gaussian functions are opposite to the signs of the delta functions, and their role in the charge density ρ_i^l is to balance the interactions produced by the Gaussians introduced in ρ_i^s. The electrostatic potential of a Gaussian function, say $g(r) = \alpha \exp(-\beta r^2)$ with parameters α and β, Eq. 14.13, is given in terms of the *error function*.

$$v(r) = \tfrac{\alpha}{4\beta\varepsilon_0} \sqrt{\pi/\beta}\, r^{-1} \operatorname{erf}(\sqrt{\beta} r) \quad ; \quad \operatorname{erf}(t) = \tfrac{2}{\sqrt{\pi}} \int_0^t \exp(-x^2)\, dx \tag{14.14}$$

Problem 14.3 The Poisson equation is $\nabla^2 v(r) = -g(r)/\varepsilon_0$, where ε_0 is the vacuum permittivity; the constants may change after using different units. Show that the electrostatic potential, v, of a Gaussian charge density $g(r) = \alpha \exp(-\beta r^2)$, with $\alpha = 1/(2\pi\sigma^2)^{3/2}$, $\beta = 1/(2\sigma^2)$, is given in terms of the error function:

$$v(r) = (\gamma/r) \operatorname{erf}(\sqrt{\beta} r)$$

where γ depends on the parameters α, β, and ε_0. Discuss the asymptotic limit of the potential at large distances.

Solution The electrostatic potential depends on the distance r and is symmetric in the spherical coordinates θ and ϕ. Thus, the Laplacian operator is:

$$\nabla^2 v(r) = \tfrac{1}{r^2} \tfrac{\partial}{\partial r} \left(r^2 \tfrac{\partial v}{\partial r} \right) = \tfrac{1}{r} \tfrac{\partial^2 (rv)}{\partial r^2}$$

We need to integrate twice to obtain v. The first integral is:

$$\frac{1}{r} \frac{\partial^2 (rv)}{\partial r^2} = -\frac{\alpha}{\varepsilon_0} e^{-\beta r^2} \quad \Rightarrow \quad \frac{\partial (rv)}{\partial r} = -\frac{\alpha}{\varepsilon_0} \int_0^r r' e^{-\beta r'^2}\, dr'$$

$$= -\frac{\alpha}{2\beta\varepsilon_0} \int_0^{\beta r^2} e^{-t}\, dt$$

The change of variables $t = \beta r'^2$ was used in the last identity. The integral of the right involves an exponential, and the result is also an exponential function.

$$\frac{\partial (rv)}{\partial r} = \frac{\alpha}{2\beta\varepsilon_0} e^{-\beta r^2} \quad \Rightarrow \quad v(r) = \frac{\alpha}{2\beta\varepsilon_0} \frac{1}{r} \int_0^r e^{-\beta r'^2}\, dr'$$

$$= \frac{\alpha}{2\beta\varepsilon_0 \sqrt{\beta}} \frac{1}{r} \int_0^{\sqrt{\beta} r} e^{-x^2}\, dx$$

(continued)

The change of variables $x = \sqrt{\beta} r'$ was used in the last identity. The second integral is over the Gaussian function, and it corresponds to the error function 14.14.

$$v(r) = \gamma \, \mathrm{erf}(\sqrt{\beta} r)/r \quad ; \quad \gamma = \alpha/(4\beta\varepsilon_0)\sqrt{\pi/\beta}$$

The presence of the error function is due to the impossibility of directly computing the integral of the Gaussian expression in terms of basic functions. When the expressions of α and β are introduced in γ, we have $\gamma = 1/(4\pi\varepsilon_0)$. The error function approaches unity fast enough in the distance. For example, when the argument of the erf function is 3, the value of the function is over 99% close to the asymptotic limit 1, Fig. 14.9. In such a limit, the potential is that of a point charge, $v(r) = 1/(4\pi\varepsilon_0 r)$.

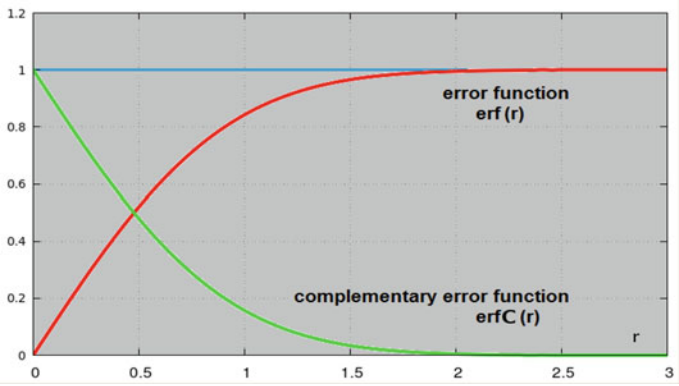


Fig. 14.9 The error and complementary error functions are depicted. The addition of such functions gives 1 (line in blue color). The potential $1/r$ may be expressed in the form $1/r = [\mathrm{erf}(r) + \mathrm{erfC}(r)]/r$. The error function tends to 1 in the limit $r \to \infty$, while the complementary function tends to 0 in that limit. On the other hand, the product $r^{-1}\mathrm{erf}(r)$ is finite when $r \to 0$ and exhibits a long tail when $r \to \infty$. The product $r^{-1}\mathrm{erfC}(r)$ is singular when $r \to 0$ and fades exponentially away with increasing values of r

14.7 Short-Range Interactions

The short-range interaction potential 14.11 in terms of the charge distribution ρ_i^s, Eq. 14.13, is:

$$v_{(i)}^s(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{\mathbf{n}}\sum_{j=1}^{N*}\int \frac{\rho_j^s(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'+\mathbf{n}L|}\,d^3r'$$

$$= \frac{1}{4\pi\varepsilon_0}\sum_{\mathbf{n}}\sum_{j=1}^{N*}\int \frac{q_j\delta(\mathbf{r}'-\mathbf{r}_j) - q_j g(\mathbf{r}'-\mathbf{r}_j)}{|\mathbf{r}-\mathbf{r}'+\mathbf{n}L|}\,d^3r'$$

The term with the delta function can be integrated. The terms are separated to reach a simplified expression.

$$v_{(i)}^s(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{\mathbf{n}}\sum_{j=1}^{N*}\frac{q_j}{|\mathbf{r}-\mathbf{r}_j+\mathbf{n}L|} - \frac{1}{4\pi\varepsilon_0}\sum_{\mathbf{n}}\sum_{j=1}^{N*}\int \frac{q_j\, g(\mathbf{r}-\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'+\mathbf{n}L|}\,d^3r'$$

According to problem 14.3, the potential derived from a Gaussian function is:

$$v(r) = 1/(4\pi\varepsilon_0)\,\mathrm{erf}\left(\sqrt{\beta}r\right)/r$$

This result is employed in the expression of the short-range potential.

$$v_{(i)}^s(\mathbf{r}) = \tfrac{1}{4\pi\varepsilon_0}\textstyle\sum_{\mathbf{n}}\sum_{j=1}^{N*} q_j \,/\, |\mathbf{r}-\mathbf{r}_j+\mathbf{n}L| +$$
$$-\textstyle\sum_{\mathbf{n}}\sum_{j=1}^{N*}\frac{q_j}{4\pi\varepsilon_0}\,\mathrm{erf}\left(\sqrt{\beta}\,|\mathbf{r}-\mathbf{r}_j+\mathbf{n}L|\right)\,/\,|\mathbf{r}-\mathbf{r}_j+\mathbf{n}L|$$

The short-range potential and energy E^s are written in terms of the *complementary error function*, erfC.

$$\begin{aligned} v_{(i)}^s(\mathbf{r}) &= \tfrac{1}{4\pi\varepsilon_0}\textstyle\sum_{\mathbf{n}}\sum_{j=1}^{N*}\frac{q_j}{|\mathbf{r}-\mathbf{r}_j+\mathbf{n}L|}\,\mathrm{erfC}\left(\sqrt{\beta}\,|\mathbf{r}-\mathbf{r}_j+\mathbf{n}L|\right) \\ E^s &= \tfrac{1}{2}\textstyle\sum_{i=1}^{N} q_i\, v_{(i)}^s(\mathbf{r}_i) \quad ; \quad \mathrm{erfC}(x) = 1-\mathrm{erf}(x) \end{aligned} \tag{14.15}$$

The initial contribution of the delta function in the charge density ρ_i^s is considered by the unit term that builds the complementary error function. The function $v_{(i)}^s$ is the electrostatic potential at position $\mathbf{r}$, avoiding the contribution of atom i indicated by the asterisk in the summation. The potential $v_{(i)}^s$ is a short-range potential due to the fact that the complementary error function of Eq. 14.15 rapidly fades away in the distance, Fig. 14.9. On the other hand, the energy E^s has equal expression to the electrostatic energy of point charges, except for the consideration of the erfC function. It enhances the convergence of E^s through the parameter $\beta = 1/(2\sigma^2)$, and for such a reason, it is computed in real space.

14.8 Electrostatic Self-Interactions

The next step is to compute the interaction of a charge with itself. This contribution is included in the long-range interactions with the purpose to avoid disruption of the closed expression of the long-range interactions. Yet, the self-interaction energy is a spurious term that should be dismissed in a later stage from the total electrostatic energy by direct subtraction to compensate for its insertion in the long-range interactions. The self-interaction potential and self-interaction energy are established by: (i) finding the solution of the Poisson equation for the charge distribution ρ_i^l of Gaussian type, problem 14.3,

$$v_i^l(\mathbf{r}_i) = \frac{1}{4\pi\varepsilon_0}\frac{q_i}{|\mathbf{r}-\mathbf{r}_i|}\,\mathrm{erf}\left[\frac{|\mathbf{r}-\mathbf{r}_i|}{\sqrt{2}\sigma}\right] \quad ; \quad E_{self} = \tfrac{1}{2}\sum_{i=1}^{N} q_i\, v_i^l(\mathbf{r}_i) \tag{14.16}$$

and (ii) taking the zero limit in the argument of the error function, $\mathrm{erf}(x \to 0) = 2x/\sqrt{\pi}$.

$$v_i^l(\mathbf{r}_i) = \frac{1}{4\pi\varepsilon_0}\frac{q_i}{\sigma}\left(\frac{2}{\pi}\right)^{1/2} \quad ; \quad E_{self} = \frac{1}{4\pi\varepsilon_0}\frac{1}{\sqrt{2\pi\sigma^2}}\sum_{i=1}^{N} q_i^2 \tag{14.17}$$

E_{self} is a constant energy, which should be subtracted from the total electrostatic energy.

14.9 Long-Range Interactions

The last energy to compute is obtained from $\rho^l(\mathbf{r}) = \sum_{\mathbf{n}}\sum_{i=1}^{N}\rho_i^l(\mathbf{r}+\mathbf{n}L)$, originated from the Gaussian functions ρ_i^l, Eq. 14.13. The electrostatic potential of a Gaussian function is proportional to erf/r, which is non-singular and contains the self-interaction. In the limit of large distances, it leads to $\mathrm{erf}(r \to \infty) = 1$, indicating that the electrostatic potential of ρ^l is a long-range potential. The slow decay of the potential in real space turns into a rapid decay in reciprocal space. Thus, it is convenient to work in the reciprocal space. The steps to follow are succinctly sketched below.

real space		**reciprocal space**
electrostatic potential (Poisson equation)	Fourier transformation $\longrightarrow$	electrostatic potential (Poisson equation)
		$\downarrow$
		solve the equation
		$\downarrow$
solution	inverse Fourier transformation $\longleftarrow$	solution

We require to perform Fourier transforms to work in the reciprocal space. The representation of a function in reciprocal space is defined in terms of exponential functions with imaginary arguments. If the function in real space is $f(\mathbf{r})$, and the function in reciprocal space is $\tilde{f}(\mathbf{k})$, then their relation is given by a Fourier transform.

$$f(\mathbf{r}) = \sum_k \tilde{f}(\mathbf{k})\, e^{i\mathbf{k}\cdot\mathbf{r}} \quad ; \quad \tilde{f}(\mathbf{k}) = \frac{1}{V}\int_V f(\mathbf{r})\, e^{-i\mathbf{k}\cdot\mathbf{r}} d^3r$$

The functions of the reciprocal space are denoted with a tilde. A periodic condition on f demands $\exp(i\mathbf{k}\cdot\mathbf{R}) = 1$, with $\mathbf{R}$ the lattice translational vector.

$$f(\mathbf{r}+\mathbf{R}) = \sum_k \tilde{f}(\mathbf{k}) \exp[i\mathbf{k}\cdot(\mathbf{r}+\mathbf{R})] = \sum_k \tilde{f}(\mathbf{k}) \exp(i\mathbf{k}\cdot\mathbf{r}) = f(\mathbf{r})$$

The periodic condition on the function f is achieved when $\mathbf{k}\cdot\mathbf{R} = 2\pi m$, with m an integral number. In this regard, for the lattice in the real space characterized by the translational vector $\mathbf{R}$, written in the form $\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$, we may generate an alternative geometrical lattice with the conjugate vector $\mathbf{k} = m_1\mathbf{a}_1^* + m_2\mathbf{a}_2^* + m_3\mathbf{a}_3^*$. The numbers $m_k, k = 1, 2, 3$, are integers similar to the numbers n_i of the real space grid. The vectors $\mathbf{a}_1^*, \mathbf{a}_2^*, \mathbf{a}_3^*$ may be found from their dot product with the primitive real space lattice vectors, $\mathbf{a}_i \cdot \mathbf{a}_j^* = 2\pi\delta_{ij}$, where δ_{ij} is the Kronecker delta function. An expression that satisfies this condition is:

$$(\mathbf{a}_1^*, \mathbf{a}_2^*, \mathbf{a}_3^*) = 2\pi\left(\frac{\mathbf{a}_2\times\mathbf{a}_3}{\mathbf{a}_1\cdot(\mathbf{a}_2\times\mathbf{a}_3)}, \frac{\mathbf{a}_3\times\mathbf{a}_1}{\mathbf{a}_2\cdot(\mathbf{a}_3\times\mathbf{a}_1)}, \frac{\mathbf{a}_1\times\mathbf{a}_2}{\mathbf{a}_3\cdot(\mathbf{a}_1\times\mathbf{a}_2)}\right)$$

The primitive vectors of the conjugate cell are $\mathbf{a}_1^*, \mathbf{a}_2^*, \mathbf{a}_3^*$, obtained from vectors in real space and vice versa. These primitive vectors have units $1/length$ and are called *reciprocal vectors*. The scalar triple product $\mathbf{a}_1\cdot(\mathbf{a}_2\times\mathbf{a}_3)$ in the first denominator is the volume of the real cell. The other denominators also represent that volume, since the scalar triple product remains the same with the circular shift of vectors.

$$V = \mathbf{a}_1\cdot(\mathbf{a}_2\times\mathbf{a}_3)$$

The periodic condition $\exp(i\mathbf{k}\cdot\mathbf{R}) = 1$ in terms of the lattice vectors in real and reciprocal space is:

$$\exp(i\mathbf{k}\cdot\mathbf{R}) = \exp\left[i\left(\sum m_i\mathbf{a}_i^*\right)\cdot\left(\sum n_j\mathbf{a}_j\right)\right] = \exp\left(2\pi i\sum\delta_{ij}m_in_j\right) = 1$$

The solutions of the differential equations in reciprocal space are of particular interest. The differentiation of a function in real space corresponds to a multiplication by the conjugate variable of the reciprocal space, thus facilitating the finding of the solution in reciprocal space, as shown in the initial sketch. The total distribution

function ρ^l and the Poisson equation in real space, together with their counterparts in reciprocal space, are:

$$\begin{aligned} \rho^l(\mathbf{r}) &= \textstyle\sum_{\mathbf{n}} \sum_{i=1}^{N} \rho_i^l(\mathbf{r}+\mathbf{n}L) \quad ; \quad \nabla^2 v^l(\mathbf{r}) = -\rho^l(\mathbf{r})/\varepsilon_0 \\ \tilde{\rho}^l(\mathbf{k}) &= \tfrac{1}{V}\textstyle\int_V \rho^l(\mathbf{r})\, \exp(-i\mathbf{k}\cdot\mathbf{r})\, d^3r \quad ; \quad k^2 \tilde{v}^l(\mathbf{k}) = \tilde{\rho}^l(\mathbf{k})/\varepsilon_0 \end{aligned} \tag{14.18}$$

The electrostatic potential $\tilde{v}^l$ in reciprocal space is determined by the charge density $\tilde{\rho}^l$ divided by k^2, Eq. 14.18. Thus, it is convenient to simplify the charge density $\tilde{\rho}^l$ to latter retake $\tilde{v}^l$ and, after applying a Fourier transform, determine v^l in real space. The periodic attributes of the exponential function allow to simplify the expression of $\tilde{\rho}^l$. Specifically, we use the translational property with the integral number n and the arbitrary real number x_0.

$$\begin{aligned} &\rho(\mathbf{r}) = \rho(\mathbf{r}+\mathbf{R}) \quad ; \quad \exp[i(x+2\pi n)] = \exp(ix) \\ &\text{if} \quad h(x) = f(x-x_0) \quad \Rightarrow \quad \tilde{h}(k) = \exp(-i2\pi x_0 k)\, \tilde{f}(k) \end{aligned} \tag{14.19}$$

R is the translational vector.

Problem 14.4 Consider a periodic charge density ρ^l given in terms of the individual densities ρ_j^l.

$$\rho^l(\mathbf{r}) = \sum_{\mathbf{n}} \sum_{j=1}^{N} \rho_j^l(\mathbf{r}+\mathbf{n}L)$$

The sums run over the particles of the central unit cell and the lattice vectors $\mathbf{n} = (n_1, n_2, n_3)$ to produce the particle grid. Obtain the density $\tilde{\rho}^l$ in reciprocal space.

Solution The Fourier transform of the charge density is given as a combination of exponential functions with continuous arguments that demand integration, and each integral is defined over a unit cell volume.

$$\tilde{\rho}^l(\mathbf{k}) = \frac{1}{V} \sum_{\mathbf{n}} \sum_{j} \int_V e^{-i\mathbf{k}\cdot\mathbf{r}} \rho_j^l(\mathbf{r}+\mathbf{n}L)\, d^3r$$

The volume of integration is V; it may be considered in the reciprocal space, or in real space, or shared in the form $V^{-1/2}$ by both spaces. The expression is simplified when the change of variables $\mathbf{t} = \mathbf{r} + \mathbf{n}L$ is done and using the identity $\exp[i\mathbf{k}\cdot\mathbf{n}L] = 1$, which gives the translational invariance requirement

(continued)

of the unit cell.

$$\tilde{\rho}^l(\mathbf{k}) = \frac{1}{V}\sum_{\mathbf{n}}\sum_j \int_{V+\mathbf{n}L} e^{-i\mathbf{k}\cdot\mathbf{t}}\,\rho_j^l(\mathbf{t})\,d^3t = \frac{1}{V}\sum_j \int_{\mathbb{R}^3} e^{-i\mathbf{k}\cdot\mathbf{t}}\,\rho_j^l(\mathbf{t})\,d^3t$$

The summation over integrals covering the image-cell volumes corresponds to an integral over the $3D$ space, $\mathbb{R}^3$.

The previous periodic and translational properties avoid the summation over $\mathbf{n}$, which is the vector referent to the image cells in the expression of the charge density, problem 14.4.

$$\begin{aligned}\tilde{\rho}^l(\mathbf{k}) &= \tfrac{1}{V}\textstyle\sum_{j=1}^N \int_{\mathbb{R}^3} e^{-i\mathbf{k}\cdot\mathbf{t}}\,\rho_j^l(\mathbf{t})\,d^3t = \tfrac{1}{V}\int_{\mathbb{R}^3}\sum_{j=1}^N q_j\, g(\mathbf{r}-\mathbf{r}_j)\, e^{-i\mathbf{k}\cdot\mathbf{r}}\,d^3r \\ &= \tfrac{1}{V}\textstyle\sum_{j=1}^N q_j \int_{\mathbb{R}^3} \alpha\exp[-\beta(\mathbf{r}-\mathbf{r}_j)^2]\ \exp[-i\mathbf{k}\cdot\mathbf{r}]\,d^3r \\ &\alpha = 1/(2\pi\sigma^2)^{3/2} \quad ; \quad \beta = 1/(2\sigma^2)\end{aligned} \tag{14.20}$$

The expression 14.20 is the Fourier transform of a Gaussian function. In order to carry the integration, we require to apply the result:

$$\int_{-\infty}^{\infty} \exp[-a^2(x-x_0)^2]\exp(-ikx)\,dx = (\pi^{1/2}/a)\exp(-ikx_0)\exp[-k^2/(4a^2)]$$

which applies to the $1D$ case and is generalized to the $3D$ case.

$$\tilde{\rho}^l(\mathbf{k}) = V^{-1}\textstyle\sum_{j=1}^N q_j\ \exp(-i\mathbf{k}\cdot\mathbf{r}_j)\ \exp(-\sigma^2k^2/2) \tag{14.21}$$

When $\mathbf{k}=\mathbf{0}$, we have the neutrality of the system $\sum_{j=1}^N q_j = 0$, and $\tilde{\rho}^l(0) = 0$. According to Eq. 14.18, by dividing the above expression by $\varepsilon_0 k^2$, the electrostatic potential in reciprocal space is determined.

$$\tilde{v}^l(\mathbf{k}) = (V\varepsilon_0)^{-1}\textstyle\sum_{j=1}^N (q_j/k^2)\, e^{-i\mathbf{k}\cdot\mathbf{r}_j}\ e^{-\sigma^2k^2/2} \tag{14.22}$$

There is no divergence in the charge density and electrostatic potential for $\mathbf{k}=\mathbf{0}$, since $\lim_{k\to 0}\tilde{\rho}^l(\mathbf{k})/k^2 =$ finite value. We are interested in the expression of the potential in real space to compute the electrostatic energy. The expression in real space is obtained by inverse-Fourier transforming the potential from the reciprocal space to the real space.

$$v^l(\mathbf{r}) = \sum_{\mathbf{k}\neq 0} \tilde{v}^l(\mathbf{k}) \exp(i\mathbf{k}\cdot\mathbf{r}) = \frac{1}{V\varepsilon_0} \sum_{\mathbf{k}\neq 0} \sum_{j=1}^{N} (q_j/k^2)\, e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}_j)}\, e^{-\sigma^2 k^2/2} \tag{14.23}$$

The energy due to the potential v^l in real space is deduced from $E^l = \sum_i q_i v^l(\mathbf{r}_i)/2$.

$$E^l = (2V\varepsilon_0)^{-1} \sum_{\mathbf{k}\neq 0} \sum_{i,j=1}^{N} (q_i q_j/k^2)\, e^{-\sigma^2 k^2/2}\, e^{i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)} \tag{14.24}$$

The summations show no asterisk as the self-interaction terms are included. Also, the running index i should not be confused with the imaginary number of the exponential function. The $3D$ array of points allows to introduce the *structure factor*, S.

$$S(\mathbf{k}) = \sum_{j=1}^{N} q_j\, \exp(i\mathbf{k}\cdot\mathbf{r}_j) \tag{14.25}$$

The expression of the energy E^l is now simplified in terms of the structure factor using the identity $|S(\mathbf{k})|^2 = S(\mathbf{k})S^*(\mathbf{k}) = S(\mathbf{k})S(-\mathbf{k})$.

$$\begin{aligned} E^l &= (2V\varepsilon_0)^{-1} \sum_{\mathbf{k}\neq 0} |S(\mathbf{k})|^2\, k^{-2} \exp(-\sigma^2 k^2/2) \\ |S(\mathbf{k})|^2 &= |\sum_{j=1}^{N} q_j \cos(\mathbf{k}\cdot\mathbf{r}_j)|^2 + |\sum_{j=1}^{N} q_j \sin(\mathbf{k}\cdot\mathbf{r}_j)|^2 \end{aligned} \tag{14.26}$$

The final expression of the long-range electrostatic energy is given in real space, in spite of the fact that the summation runs over the variable $\mathbf{k}$. The summation is symmetric in $\mathbf{k}$ and $-\mathbf{k}$, and only the positive reciprocal variable $\mathbf{k}$ requires consideration. The summation is rapidly convergent because the exponential function decays with an increasing value of $|\mathbf{k}|$. The parameter σ of the Gaussian functions is used to control the summation convergence. The efficiency of the calculations improves from N^2 to $N^{3/2}$ [5].

14.10 Ewald Electrostatic Energy

We are now in the position of gathering all the energy contributions, Eqs. 14.15, 14.17, and 14.26, and recast the electrostatic energy 14.10 of the system.

$$\begin{aligned} E^s &= \frac{1}{2} \frac{1}{4\pi\varepsilon_0} \sum_{n_1} \sum_{n_2} \sum_{n_3} \sum_{i=1}^{N} \sum_{j=1}^{N*} \frac{q_i q_j}{|\mathbf{r}_i - \mathbf{r}_j + \mathbf{n}L|} \operatorname{erfC}\left[|\mathbf{r}_i - \mathbf{r}_j + \mathbf{n}L|/(\sqrt{2}\sigma)\right] \\ E^l &= \frac{1}{2V\varepsilon_0} \sum_{\mathbf{k}\neq 0} k^{-2}\, \exp(-\sigma^2 k^2/2)\, |S(\mathbf{k})|^2 \\ E_{self} &= \frac{1}{4\pi\varepsilon_0} (\sqrt{2\pi\sigma^2})^{-1} \sum_{i=1}^{N} q_i^2 \\ E_{el}(\mathbf{r}_1, \cdots, \mathbf{r}_N) &= E^s - E_{self} + E^l \end{aligned} \tag{14.27}$$

The energy depends on the Gaussian standard deviation, σ, Eq. 14.13. The efficiency in the computation of E^s and E^l is due to the erfC function and

factor $\exp(-\sigma^2 k^2/2)$. The value of σ is a trade-off in the computational cost of the terms E^s and E^l; a small σ produces the rapid convergence of E^s and a slow convergence of E^l, and vice versa. Physically, there is a weakening of the electrostatic interactions in one of the contributions while the other is strengthened, and technically, there is a scaling of order $\mathcal{O}(N)$ in one of the contributions and a scaling of order $\mathcal{O}(N^2)$ in the other. The σ value may be decided after weighting the E^s and E^l energy contributions with respect to σ.

$$r^{-1}\,\mathrm{erfC}\left[r/(\sqrt{2}\sigma)\right] < \epsilon_r \quad ; \quad k^{-2}\,\exp\left(-\sigma^2 k^2/2\right) < \epsilon_k$$

It is common to establish the σ value that satisfies the inequality on the left by assigning a value to r larger than the cutoff distance required in the E^s term. In contrast, the inequality on the right imposes a limit on the conjugate reciprocal variable, k, to rapidly perform the summation of E^l. The tolerances ϵ_r and ϵ_k depend on the charged atoms of the grid, but are expected to be sufficiently small, taking values from 10^{-4} to 10^{-8} depending on a small, medium, or high accuracy. In many instances, the cell dipole moment may be included in the Ewald sum method. In the case of considering an exclusion list, where the electrostatic interactions of selected pairs of atoms, usually separated by a few bonds, are excluded, it is necessary to subtract the respective contributions from the total energy. This is done in analogy to the self-interaction energy of Eq. 14.17, eliminating the selected non-diagonal terms, $i \neq j$.

$$E_{excl} = -\frac{1}{2}\,\frac{1}{4\pi\varepsilon_0}\sum_i\sum_j \frac{q_i q_j}{|\mathbf{r}_i - \mathbf{r}_j|}\,\mathrm{erfC}\left[\frac{|\mathbf{r}_i - \mathbf{r}_j|}{\sqrt{2}\sigma}\right] \;;\; (i,j)\,\epsilon\,\{\text{exclusion list}\}$$

The electrostatic forces over the ith atom are computed by differentiation of the energies presented in Eqs. 14.27 and under the considerations that $d\,[\mathrm{erfC}(r)]/dx = -d\,[\mathrm{erf}(r)]/dx$, and $d\,[\mathrm{erfC}(r)]/dx = -(2/\pi)^{1/2}\,\sigma^{-1}\exp(-r^2)\,(x_i - x_j + n_x L)\,/\,r$ [5].

$$\mathbf{F} = -\nabla_i (E^s - E_{self} + E^l) \;\; ; \;\; r = |\mathbf{r}| = |\mathbf{r}_i - \mathbf{r}_j + \mathbf{n}L| \;\; ; \;\; \ell = x, y, z$$

$$F^s_\ell = \tfrac{1}{4\pi\varepsilon_0}\,q_i \sum_{j\neq i}^{N}\sum_{\mathbf{n}}\tfrac{q_j}{r^2}\left[(r_\ell/r)\,\mathrm{erfC}\left(r\,/\,[\sqrt{2}\sigma]\right) + \sqrt{2/\pi}\,(r_\ell\,/\,\sigma)\,\exp[-r^2\,/\,(2\sigma^2)]\right]$$

$$F^l_\ell = (2V\varepsilon_0)^{-1}\,q_i \sum_{\mathbf{k}\neq 0}\sum_{j\neq i}^{N} q_j (k_\ell/k^2)\,\exp[-\sigma^2 k^2/2]\,\sin[\mathbf{k}\cdot(\mathbf{r}_i - \mathbf{r}_j)]$$

The expression 14.24 of the energy E^l simplifies the computation of the force F^l_ℓ. The self-interaction energy is constant and gives no contribution to the force components. The total force over the ith atom should additionally include the other terms derived from the total energy of the system, such as the bond force, angle force, dihedral force, van der Waals force, etc. When all the forces are computed for every atom, the time evolution of the system may be unfolded.

14.11 Smooth Particle Mesh Ewald Approach

With the purpose of improving the efficiency of the Ewald sum approach, a multidimensional piecewise interpolation of the structure factors 14.25 and 14.26 may be employed. The approach applies a Cardinal B-spline interpolation to a regular grid to produce an approximation to the structure factors, Fig. 14.10. The B-splines permit analytic differentiation of the energy, leading to smooth changes of the forces with the positions of the particles. When the energy and forces of the reciprocal space are written as convolutions and evaluated with fast Fourier transforms, the calculations are reduced from $\mathcal{O}(N^2)$ down to an almost linear scaling, $\mathcal{O}(N \log N)$. The approach preserves the total energy, but not the total momentum of the particle system since the forces on the atoms are not exactly zero. The deviations are estimated to be close to the root mean square error of the forces. The formulation is of general character and receives the name of *smooth particle mesh Ewald method*. It equally applies to potentials decaying in the general form $1/r^x$, $1 \leq x$. The smooth particle mesh Ewald method is also used in the calculation of the London dispersion energy, this is, the long-range part of the van der Waals interaction when the contributions are given by pairwise interactions:

$$E = \frac{1}{2} \sum_{\mathbf{n}} \sum_{i,j=1}^{N*} C_6^{ij} / |\mathbf{r}_i - \mathbf{r}_j + \mathbf{n}L|^6 \tag{14.28}$$

The summation is absolute convergent, but computationally costly. In this regard, it is important to apply the Ewald summation approach as in the electrostatic case [6].

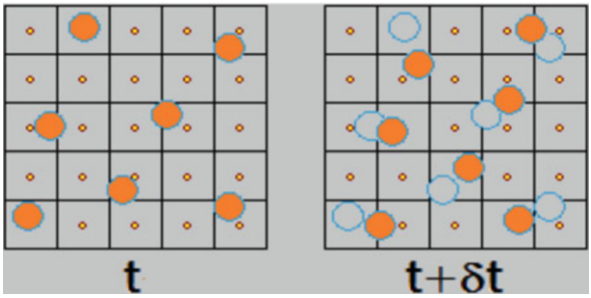


Fig. 14.10 The evolution of a system of particles from the instant of time t to $t + \delta t$ is depicted. The large filled circles indicate the locations of the particles at the instants of time t (inset on the left) and $t + \delta t$ (inset on the right). The large empty circles in the rendering of the right indicate the locations of the particles at the previous instant of time. The tiny circles represent a regular grid of points. The atom charges are projected and interpolated onto that grid at every instant of time. The electrostatic potential is evaluated from such a grid, and the computed forces from the electrostatic potential are applied to the particles

$$E_{dir} = \frac{1}{2}\sum_{\mathbf{n}}\sum_{i=1}^{N}\sum_{j=1}^{N*} C_6^{ij}\,\frac{g(\beta|\mathbf{r}_i-\mathbf{r}_j+\mathbf{n}L|)}{|\mathbf{r}_i-\mathbf{r}_j+\mathbf{n}L|^6} \quad ; \quad E_{self} = (\beta^6/12)\sum_{i=1}^{N} C_6^{ii}$$

$$E_{rec} = \pi^{3/2}\beta^3/(2V)\sum_{\mathbf{k}}^{*}\sum_{i=1}^{N}\sum_{j=1}^{N} f(\pi k/\beta)C_6^{ij}\,\exp\left[-2\pi i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)\right]$$

$$E = -E_{dir} + E_{rec} - E_{self} \tag{14.29}$$

The dispersion coefficients are approximated using the combination rules. There are several proposals for such rules; one of them is $C_6^{ij} = (C_6^{ii}\,C_6^{jj})^{1/2}$. The β parameter plays the role of σ^{-1} in Eqs. 14.27. The auxiliary functions f and g introduce the complementary error function and the Gaussian function.

$$\begin{aligned} f(r) &= \left[(1-2r^2)\,\exp(-r^2) + 2r^3\sqrt{\pi}\,\mathrm{erfC}(r)\right]/3 \\ g(r) &= \exp(-r^2)\,(1+r^2+r^4/2) \end{aligned} \tag{14.30}$$

The order of the B-splines may be chosen from the $3rd$ up to the 7th order for small, medium, and high accuracy. The accuracy is also established with the grid density. If the PBC simulations require constant pressure, then the stress tensor P should be computed. It is calculated by differentiating the electrostatic energy with respect to the atomic coordinates, keeping in mind that $d[\mathrm{erf}(r)]/dr = (2/\sqrt{\pi})\exp(-r^2)$.

$$P_{\alpha\beta}^{s}V = \frac{1}{2}\sum_{\mathbf{n}}\sum_{i,j=1}^{N*} q_i q_j\left[r^{-3}\,\mathrm{erfC}(\beta r) + 2\beta\pi^{-1/2}\,r^{-2}\,\exp(-\beta^2 r^2)\right]\mathbf{r}_\alpha\mathbf{r}_\beta$$

$$P_{\alpha\beta}^{l}V = (2\pi V)^{-1}\sum_{\mathbf{k}\neq 0} k^{-2}\exp[-(\pi k/\beta)^2]\,|S(\mathbf{k})|^2$$

$$\times\left[\delta_{\alpha\beta} - 2[1+(\pi k/\beta)^2]\,\mathbf{k}_\alpha\mathbf{k}_\beta/k^2\right]$$

$$P = P_{\alpha\beta}^{s} + P_{\alpha\beta}^{l} \quad ; \quad r = |\mathbf{r}| = |\mathbf{r}_i - \mathbf{r}_j + \mathbf{n}L| \tag{14.31}$$

The stress tensor is given in terms of fractional coordinates, or coordinates normalized with the system volume. The terms of an exclusion list may be introduced in the above identity by reproducing the expression of $P_{\alpha\beta}^{s}V$ with negative sign and avoiding the summation over the lattice numbers n_1, n_2, n_3. The smooth particle mesh Ewald approach is accurate and efficient. A cutoff imposed in the real space summation of 9 Å results in van der Waals energy deviations lower than the errors observed in the electrostatic energy. The pressure is more sensitive to cutoff distances, showing deviations of hundreds atm units. The deviations are associated with the fact that the pressure is proportional to the negative derivative of the energy with respect to the volume, making the pressure highly sensitive to energy changes. Other physical variables sensitive to a cutoff radius are the diffusion constant, d, and the dielectric constant, ϵ, approximated by:

$$d(t) = \sum_{i=1}^{N} d_i^2(t) \,/\, (6Nt) \quad ; \quad \epsilon = 1 + 3\left[4\pi\rho_N\mu^2 \,/\, (9k_BT)\right] \langle \mathbf{m}^2 \rangle \,/\, (N\mu^2)$$

$d_i(t)$ is the distance travelled by the mass center of particle i in the simulation. The diffusion constant is calculated in long-time simulations, $t \to \infty$. The diffusion constant depends on the solvent and its density. For values of the water density close to 1 g/cm^3, the diffusion constant takes values from 2.5×10^{-5} to 6.0×10^{-5} cm^2/s in a simulation of several ps. On the other hand, in the expression of the dielectric constant, $\mathbf{m}$ is the dipole moment. The dipole moment of the individual molecule is denoted by μ, the number of molecules is N, and ρ_N is the numeric density. The dielectric constant depends on the dipole moment variations, and henceforth, of the relatively long-range interactions. At ambient temperature and density of 1 g/cm^3, we may find $\epsilon = 60 - 70$, also depending on the water model applied in the simulation.

The electrostatic potential and energy of a large system of charged particles may be also calculated efficiently using the *fast multipole algorithm* of Greengard and Rokhlin [7]. Here, a cell with large size is divided in subcells, and the interactions of the charged atoms located in the same subcell and similar subcells in the neighborhood are calculated. For charged atoms in distant subcells, a multipole expansion is employed to determine the long-range electrostatic interactions. The approach is equally applicable to gravitational forces.

Finally, a common shortcoming of the methods dealing with periodic charges is the transmission of the electrostatic interaction, considered to occur instantaneously when, in fact, there is a retardation effect imposed by the speed of light. Fortunately, the retardation effect is only important at very large interparticle distances, when the electrostatic interactions are sufficiently small.[3]

14.12 Shifted Potentials and Forces

The use of the PBC approach is justified by the impossibility of dealing with a large number of particles and computing their non-bonded interactions. However, with the advent of powerful hardware, we are now able to deal with a relatively large number of particles. This fact has implications in the simulation models, as those exposed by Fennell and Gezelter [8]. For instance, if we consider that the electrostatic interactions are effectively short-ranged in condensed phases, then we face the possibility of directly calculating those interactions with minor deviations using the present mainframes. In a more precise parlance, it is possible to replace the PBC approach and use a large cutoff radius for the non-bonded interactions, with the additional consideration of a spherical charged background whose role is to

[3] The light speed is $\sim 3 \times 10^8$ m/s, and a typical time step of molecular dynamics is 1 fs. In this time interval the light has traveled a distance of $\sim 300\,\eta$m.

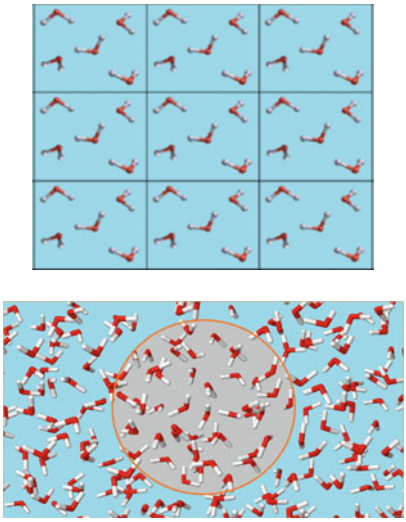


Fig. 14.11 Top inset: due to the low efficiency of old mainframes, the Ewald sum approach was implemented for the rapid calculations of the electrostatic forces of extended systems. In this approach, a crystal is studied using a relatively small number of atoms in a central unit cell and repeating it indefinitely according to the PBC scheme. Bottom inset: with the advent of more efficient mainframes, it is possible to deal with a big number of atoms and compute the non-bonded interactions within a large cutoff radius. Approximations for the residual interactions of atoms located far away may be used

neutralize the total charge of the particle system contained within the cutoff radius, Fig. 14.11. In order to unfold such a proposal, it is convenient to start from a general expression similar to that given in Eq. 14.5, written below for easy reference, with a spherical cutoff radius r_c in the shifted potential V_{sp} and force $\mathbf{F}_{sp}$.

$$V_{sp} = \begin{cases} V(r) - V(r_c) \ , \ r \leq r_c \ ; \ \mathbf{F}_{sp} = -\nabla V(r) \\ 0 \qquad\quad , \ r_c < r \ ; \ \mathbf{F}_{sp} = 0 \end{cases} \tag{14.32}$$

The force obtained from expression 14.32 has pairwise nature, but it is not a continuous function. In order to avoid discontinuities in the force, a new potential V_{sf} and new force $\mathbf{F}_{sf}$ are proposed in the region $r \leq r_c$, with a shift in the force.

$$V_{sf} = V(r) - V(r_c) - \frac{\partial V(r)}{\partial r}\bigg|_{r=r_c} (r - r_c) \quad ; \quad \mathbf{F}_{sf} = -\frac{\partial V(r)}{\partial r} + \frac{\partial V(r)}{\partial r}\bigg|_{r=r_c} \tag{14.33}$$

The discontinuity in the force of Eq. 14.32 is avoided in the force of Eq. 14.33, because the potential and the force presented in Eqs. 14.33 introduce derivative corrections, making the potential and the force going smoothly to zero at r_c, avoiding spurious energy drifts. By introducing the form of V in terms of the

distance r and the charges q_i and q_j, the electrostatic potential and the new force, paying no attention to constant units, are:

$$V_{el} = q_i q_j \left[\frac{1}{r} - \frac{1}{r_c} + \frac{1}{r_c^2}(r - r_c)\right] \quad ; \quad \mathbf{F}_{el} = q_i q_j \left(\frac{1}{r^2} - \frac{1}{r_c^2}\right) \tag{14.34}$$

Based on our experience referent to the Ewald sum approach, the Ewald expression that carries the complementary error function should be used because this term includes the charge screening in the potential. We have two versions, the first one is derived from Eqs. 14.32 and is referred to as the *damped shifted potential* (dsp) approach.

$$V_{dsp} = q_i q_j \left[\frac{\text{erfC}(\alpha r)}{r} - \frac{\text{erfC}(\alpha r_c)}{r_c}\right] \quad ; \quad \mathbf{F}_{dsp} = q_i q_j \left(\frac{\text{erfC}(\alpha r)}{r^2} + \frac{2\alpha}{\pi^{1/2}} \frac{\exp(-\alpha r^2)}{r}\right) \tag{14.35}$$

We have $V_{dsp} = 0$ when $r = r_c$. The force $\mathbf{F}_{dsp}$ is applied in the region $r \leq r_c$, zero otherwise, but it is discontinuous at $r = r_c$. When we write $\alpha = 0$ and use the fact $\text{erfC}(0) = 1$ in the above expressions, the dsp equations of the electrostatic potential and the electrostatic force are:

$$V_{dsp}(r)\Big|_{\alpha=0} = q_i q_j \left[\frac{1}{r} - \frac{1}{r_c}\right] \quad ; \quad \mathbf{F}_{dsp}(r)\Big|_{\alpha=0} = \frac{q_i q_j}{r^2} \quad ; \quad r \leq r_c$$

The second version of the electrostatic expressions is derived from Eqs. 14.33. The expressions attempt to solve the previous shortcoming on the discontinuity at $r = r_c$ of the force; they are referred to as the *damped shifted force* (dsf) approach.

$$\begin{aligned} V_{dsf} &= q_i q_j \Big[r^{-1}\,\text{erfC}(\alpha r) - r_c^{-1}\,\text{erfC}(\alpha r_c) \\ &\quad + \left(r_c^{-2}\,\text{erfC}(\alpha r_c) + \beta r_c^{-1}\,\exp(-\alpha r_c^2)\right)(r - r_c)\Big] \\ \mathbf{F}_{dsf} &= q_i q_j \Big[\left(r^{-2}\,\text{erfC}(\alpha r) + \beta r\,\exp(-\alpha r^2)\right) \\ &\quad - \left(r_c^{-2}\,\text{erfC}(\alpha r_c) + \beta r_c^{-1}\,\exp(-\alpha r_c^2)\right)\Big] \end{aligned} \tag{14.36}$$

These two expressions, with $\beta = 2\alpha/\pi^{1/2}$, gradually vanish toward the boundary $r = r_c$, thus avoiding discontinuities. When $\alpha = 0$, the dsf equations of the electrostatic potential and the electrostatic force are obtained.

$$\begin{aligned} V_{dsf}(r)\Big|_{\alpha=0} &= q_i q_j \left[\frac{1}{r} - \frac{1}{r_c} + \frac{1}{r_c^2}(r - r_c)\right] \\ \mathbf{F}_{dsf}(r)\Big|_{\alpha=0} &= q_i q_j \left[\frac{1}{r^2} - \frac{1}{r_c^2}\right] \quad ; \quad r \leq r_c \end{aligned}$$

The sets of Eqs. 14.35 and 14.36 avoid the summation in the conjugate reciprocal variable that characterizes the Ewald sum approach. Typical values of α are 0.1–0.3 Å^{-1}, usually combined with a cutoff radius in the range 9–15 Å. The expressions of the potentials and forces provided in Eqs. 14.35 and 14.36 have been tested in representative simulations of chemically relevant systems like liquid water, crystalline water, $NaCl$ crystals, $NaCl$ melts, ionic solutions, etc. The study avoids any bias by the types of systems that were tested toward favorable conditions on the application of any expression. The method of reference is the particle mesh Ewald method. The properties that have been studied using the damped shifted potentials and the damped shifted forces are the velocity correlation function $C_v(t)$ and its Fourier transform $I(\omega)$, recognized as the power spectrum of the velocity correlation function [9].

$$C_v(t) = \langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle \,/\, \langle \mathbf{v}(0) \cdot \mathbf{v}(0) \rangle \quad ; \quad I(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} C_v(t)\, e^{-i\omega t}\, dt \tag{14.37}$$

In contrast to pair distribution functions and diffusion constants, the C_v and $I(\omega)$ quantities are more sensitive to the long-range interactions and the low frequency behavior of the substance. After monitoring $C_v(t)$ in short- and intermediate-time dynamics and $I(\omega)$ in long-time dynamics with large-scale collective motions, it is found that the results of each trajectory using the damped shifted proposals with moderate values, $r_c = 9$ Å and $\alpha = 0.2$ Å^{-1}, are encouraging, closely matching the results obtained with the smooth particle mesh Ewald scheme, where a convergence parameter of 0.3119 Å^{-1} was used.

An alternative model to abbreviate the computation of long-range forces, especially the electrostatic ones in an electrically neutral system, is the *reaction field approach*. In this approximation, a cutoff radius r_c is defined around a molecule, and the interactions are calculated within the cutoff radius. The external particles to the cutoff radius are assumed to form a dielectric. The dielectric interacts with the charges enclosed by the cutoff radius and is permitted to be polarized. The reaction field potential V_{rf} is included in the expression of the interaction of the charges: [10]

$$\begin{aligned} V_{rf} &= \tfrac{1}{4\pi\varepsilon_0} \textstyle\sum_{i<j} q_i q_j \left[\tfrac{1}{r_{ij}} + (\varepsilon - 1) \,/\, (2\varepsilon + 1)\, r_{ij}^2 / r_c^3 \right] \\ \mathbf{f}_{rf}^{i} &= \tfrac{1}{4\pi\varepsilon_0} \textstyle\sum_{j \neq i} q_i q_j \left[\tfrac{1}{r_{ij}^3} - 2(\varepsilon - 1) \,/\, (2\varepsilon + 1)\, r_c^{-3} \right] \mathbf{r}_{ij} \end{aligned} \tag{14.38}$$

The last term of the potential is the contribution of the reaction field, with dielectric constant ε, to the electrostatic interactions. The second expression is the force on atom i calculated from the potential V_{rf}. Care should be exercised when an atom located at the boundary of the cutoff radius interacts with the central atoms. In this case, an amendment for the pair interactions should be introduced in the potential.

$$\frac{q_i q_j}{4\pi\varepsilon_0}\left[\frac{1}{r_c}+(\varepsilon-1)/(2\varepsilon+1)\,\frac{1}{r_c}\right]=\frac{1}{4\pi\varepsilon_0}\,\frac{q_i q_j}{r_c}\,[1+(\varepsilon-1)/(2\varepsilon+1)] \qquad (14.39)$$

The reaction field model simplifies the computations by considering the local electrostatic interactions explicitly and simplifying the distant electrostatic interactions with implicit terms.

The general conclusion is that the damped shifted approaches and the reaction field model represent alternative schemes to those employing lattice summations. They are capable to maintain both the efficiency since the computations scale linearly and are friendly parallelizable, and the accuracy of the calculations. Finally, the damped shifted expressions and reaction field model avoid the artificial stabilization of over-compacted proteins, occasionally observed in the PBC approach and associated with the uniform periodic images in the Ewald summation method.

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Part V

Time Evolution Operators



Integrating the Equations of Motion 15

There are different formulations of classical mechanics like that of Newton, Lagrange, Hamilton, Hamilton-Jacobi, and Poisson. They all have the common goal of providing in one way or another the equations of motion that rule the behavior of the classical particles. Yet, the different formulations provide no general or analytical solutions to the equations of motion, because the atomic forces are intricate and depend on the characteristics of the atoms. In this context, there are complications in the process of finding the solutions of the motion equations.

The purpose of the present chapter is to discuss the *discrete time propagators*, also known as *numerical time integrators*. By using them, we are able to evolve the positions and velocities of the particles with time, thus avoiding the commitment of analytically solving the equations of motion, which in most conditions of realistic particle systems is impossible. The discussion starts by introducing the Poisson propagator, because it represents a formal way to evolve the system of particles in time. However, since the Liouville operator is fundamental to build the basic blocks of the discrete time propagators, a connection of the Poisson brackets is established with the Liouville operator. In the simulations, the initial conditions of the particles are supposed to be known, and the forces over each particle are assumed to be computable at every instant of time from either first-principles calculations or model potential functions, with the goal of determining the unique particle trajectories, and thereby, establishing the history of the particle system.

15.1 Liouville Operator as a Time Propagator

The method based on Poisson brackets, Sect. 6.22, gives the possibility of creating a classical time propagator to investigate the time evolution of the particles. For practical reasons, it is convenient to work with the Liouville operator, $\hat{\mathcal{L}}$, as defined in terms of the Poisson brackets in Eq. 6.69. If $\mathcal{H}$ is the system Hamiltonian, and $F(\{q_i\}, \{p_i\}, t)$ represents an arbitrary function in terms of the generalized coordinates q, p, and time, then the action of $\hat{\mathcal{L}}$ on function F is defined by:

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$$\hat{\mathcal{L}}F = \{F, \mathcal{H}\} = \frac{\partial F}{\partial q}\frac{\partial \mathcal{H}}{\partial p} - \frac{\partial \mathcal{H}}{\partial q}\frac{\partial F}{\partial p}$$

Suppose that F depends on the positions and velocities solely. The time derivative of F is written in terms of the Liouville operator, Eq. 6.72.

$$\frac{dF}{dt} = \{F, \mathcal{H}\} = -\{\mathcal{H}, F\} = \hat{\mathcal{L}}F \tag{15.1}$$

The solution of the equation is of exponential type.

$$\begin{gathered}\frac{d}{dt}F(t) = \frac{d}{dt}\left[\exp(\hat{\mathcal{L}}t)\, F(t_0)\right] = \hat{\mathcal{L}}\, \exp(\hat{\mathcal{L}}t)\, F(t_0) = \hat{\mathcal{L}}F(t) \\ F(t) = \exp(\hat{\mathcal{L}}t)\, F(t_0)\end{gathered} \tag{15.2}$$

The exponential operator $\exp(\hat{\mathcal{L}}t)$ is a symbolic representation of a power series in terms of $\hat{\mathcal{L}}t$. The imaginary symbol i may be employed under convention in the definition of the Liouville operator, like $i\hat{\mathcal{L}}$ instead of $\hat{\mathcal{L}}$, and in analogy to the time propagators of quantum mechanics. The function $F(t_0)$ includes the initial conditions on the positions and momenta at the instant of time t_0. The expression $\exp(\hat{\mathcal{L}}t)$ is the *classical time propagator* giving the propagation of F from time t_0 to time t. If the function $F(t)$ represents a position or momentum, it may be advanced forward in time. To illustrate this, we write the Hamiltonian in terms of the kinetic energy and potential energy, $\mathcal{H} = \mathbf{p}^2/(2m) + V(\mathbf{r}_i)$. Under such a representation, the Liouville operator is:

$$\hat{\mathcal{L}}F = \{F, \mathcal{H}\} = \frac{\mathbf{p}}{m} \cdot \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right] F + \mathbf{f} \cdot \left[\frac{\partial}{\partial p^x}, \frac{\partial}{\partial p^y}, \frac{\partial}{\partial p^z}\right] F \tag{15.3}$$

The net force acting on the particle is $\mathbf{f}$. We deal with the Liouville operator in the form discussed by Tuckerman et al. [1] The action of $\hat{\mathcal{L}}$ is divided into $\hat{\mathcal{L}}_1$ to include the effect of the momenta on the positions, and $\hat{\mathcal{L}}_2$ to include the effect of the forces on the momenta.

$$\hat{\mathcal{L}}_1 = \frac{\mathbf{p}}{m} \cdot \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right] \quad ; \quad \hat{\mathcal{L}}_2 = \mathbf{f} \cdot \left[\frac{\partial}{\partial p^x}, \frac{\partial}{\partial p^y}, \frac{\partial}{\partial p^z}\right] \tag{15.4}$$

The Liouville operators $\hat{\mathcal{L}}_1$ and $\hat{\mathcal{L}}_2$ may adopt different expressions according to the types of coordinates. A summation is introduced to describe the system of n particles.

$$\hat{\mathcal{L}}_1 = \sum_{i=1}^{n} \frac{\mathbf{p}_i}{m_i} \cdot \nabla_i \quad ; \quad \hat{\mathcal{L}}_2 = \sum_{i=1}^{n} \mathbf{f}_i \cdot \nabla_{p_i} \tag{15.5}$$

The first operator evolves the positions and the second one the momenta.

Problem 15.1 Demonstrate that $\hat{\mathcal{L}}_1$ and $\hat{\mathcal{L}}_2$, Eqs. 15.5, are non-commuting operators.

Solution For simplicity, we work with operators expressed in a single dimension. The commutator of $\hat{\mathcal{L}}_1$ and $\hat{\mathcal{L}}_2$ is $[\hat{\mathcal{L}}_1, \hat{\mathcal{L}}_2] = \hat{\mathcal{L}}_1\hat{\mathcal{L}}_2 - \hat{\mathcal{L}}_2\hat{\mathcal{L}}_1$. By introducing the expressions of the operators in the commutator, we obtain:

$$[\hat{\mathcal{L}}_1, \hat{\mathcal{L}}_2] = \frac{p}{m}\frac{\partial}{\partial x}\left(f(x)\frac{\partial}{\partial p}\right) - f(x)\frac{\partial}{\partial p}\left(\frac{p}{m}\frac{\partial}{\partial x}\right) = \frac{p}{m}\frac{\partial f}{\partial x}\frac{\partial}{\partial p} - \frac{f}{m}\frac{\partial}{\partial x}$$

Therefore, $[\hat{\mathcal{L}}_1, \hat{\mathcal{L}}_2] \neq 0$, and $\hat{\mathcal{L}}_1$ and $\hat{\mathcal{L}}_2$ of Eqs. 15.5 do not commute.

15.2 Discretizing the Time Propagator

In principle, the Liouville operator has the capability to evolve the system from time t_0 to t in a single time step, Eq. 15.2. Nevertheless, we can use a concise series expansion to advance the system in small time increments.

The result of applying the operators $\hat{\mathcal{L}}_1$ and $\hat{\mathcal{L}}_2$ on a function F depends on the application order. In this regard, a symmetrical factorization similar to the symmetrization process of operators in quantum mechanics is used. Based on the *Trotter expansion*, the symmetrical factorization is:[1]

$$\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t] = \lim_{N\to\infty} [e^{\hat{\mathcal{L}}_2 t/2N}\, e^{\hat{\mathcal{L}}_1 t/N}\, e^{\hat{\mathcal{L}}_2 t/2N}]^N \tag{15.6}$$

We have an approximation to the Trotter expansion.

$$\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t] = [e^{\hat{\mathcal{L}}_2 t/2N}\, e^{\hat{\mathcal{L}}_1 t/N}\, e^{\hat{\mathcal{L}}_2 t/2N}]^N + \mathcal{O}(t^3/N^2) \tag{15.7}$$

The expression is accurate up to second order. It is convenient to apply the Nth root on both sides of the expression, with the simultaneous definition of the time integration step $\Delta t = t/N$.

$$\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)\Delta t] = e^{\hat{\mathcal{L}}_2 \Delta t/2}\, e^{\hat{\mathcal{L}}_1 \Delta t}\, e^{\hat{\mathcal{L}}_2 \Delta t/2} \tag{15.8}$$

[1] According to the Trotter theorem, the exponential $\exp(P + Q)$ may be decomposed in the form $\exp(P + Q) = \lim_{N\to\infty}[\exp(Q/2N)\exp(P/N)\exp(Q/2N)]^N$ when N is a large integer. Trotter [2].

The time step Δt is the time interval defined for the numerical integration of the equations of motion. Different time steps may be introduced in the particle dynamics by dealing with the exponential factor in the middle of the expression.

$$\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t] = e^{\hat{\mathcal{L}}_2 \Delta t/2} \, [e^{\hat{\mathcal{L}}_1 \delta t}]^M \, e^{\hat{\mathcal{L}}_2 \Delta t/2} \quad ; \quad \delta t = \Delta t / M \tag{15.9}$$

The time step δt is a fraction of Δt, and therefore, $\delta t < \Delta t$. When working with particles whose masses are notably different, or the forces exhibit substantially different magnitudes, then different time steps are usually required. The different time steps in the integration of the equations of motion are particularly helpful for particle systems where we observe multiple time scales [3]. The approach may be used as many times as required in a simulation, with multiple applications depending on the type of system under investigation. The time propagator is now discretized based on the Liouville operator 15.8.

$$U(\Delta t) = U_2(\Delta t/2) \, U_1(\Delta t) \, U_2(\Delta t/2) \quad ; \quad U_i(\Delta t) = e^{\hat{\mathcal{L}}_i \Delta t} \tag{15.10}$$

Problem 15.2 From the general expansion of $\exp(\hat{A}t) = 1 + \hat{A}t + \hat{A}^2 t^2/2! + \cdots$, show that the following approximations are satisfied up to first- and second-order terms, respectively.

$$\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t] \simeq \exp[\hat{\mathcal{L}}_1 t] \exp[\hat{\mathcal{L}}_2 t]$$

$$\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t] \simeq \exp[\hat{\mathcal{L}}_2 t/2] \, \exp[\hat{\mathcal{L}}_1 t] \, \exp[\hat{\mathcal{L}}_2 t/2]$$

Solution The common term in both approximations is $\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t]$. Thus, we expand the operator $\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t]$.

$$\begin{aligned}
&\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t] \\
&\quad = 1 + \left(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2\right) t + \left(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2\right)^2 t^2/2! + \left(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2\right)^3 t^3/3! + \cdots \\
&\quad = 1 + \left(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2\right) t + \left[\hat{\mathcal{L}}_1^2 + \hat{\mathcal{L}}_1 \hat{\mathcal{L}}_2 + \hat{\mathcal{L}}_2 \hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2^2\right] t^2/2 + \cdots
\end{aligned}$$

On the other hand, the product of individual expansions gives:

$$\begin{aligned}
\exp[\hat{\mathcal{L}}_1 t] \exp[\hat{\mathcal{L}}_2 t] &= \left[1 + \hat{\mathcal{L}}_1 t + \hat{\mathcal{L}}_1^2 t^2/2 + \cdots\right] \left[1 + \hat{\mathcal{L}}_2 t + \hat{\mathcal{L}}_2^2 t^2/2 + \cdots\right] \\
&= 1 + \left(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2\right) t + \left[\hat{\mathcal{L}}_1^2 + 2\hat{\mathcal{L}}_1 \hat{\mathcal{L}}_2 + \hat{\mathcal{L}}_2^2\right] t^2/2 + \cdots
\end{aligned}$$

(continued)

Thereby, the approximation $\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t] \simeq \exp(\hat{\mathcal{L}}_1 t)\exp(\hat{\mathcal{L}}_2 t)$ holds up to the first-order term, because the following terms break the symmetry between $\hat{\mathcal{L}}_1$ and $\hat{\mathcal{L}}_2$ in the expansion of $\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t]$. A similar demonstration is provided in the case of $\exp[(\hat{\mathcal{L}}_1 + \hat{\mathcal{L}}_2)t] \simeq \exp(\hat{\mathcal{L}}_2 t/2)\exp(\hat{\mathcal{L}}_1 t)\exp(\hat{\mathcal{L}}_2 t/2)$. A symbolic algebra package may facilitate the demonstration, reaching the conclusion that the approximation holds up to second-order terms in this last case.

The inverse operator is obtained by replacing $+t$ by $-t$. In this connection, the classical propagator 15.10 is demonstrated to be unitary, allowing time reversibility.

$$\begin{aligned} U(\Delta t)\,U(-\Delta t) &= \left(e^{\hat{\mathcal{L}}_2\Delta t/2}\,e^{\hat{\mathcal{L}}_1\Delta t}\,e^{\hat{\mathcal{L}}_2\Delta t/2}\right)\left(e^{-\hat{\mathcal{L}}_2\Delta t/2}\,e^{-\hat{\mathcal{L}}_1\Delta t}\,e^{-\hat{\mathcal{L}}_2\Delta t/2}\right) \\ &= e^{\hat{\mathcal{L}}_2\Delta t/2}\,[e^{\hat{\mathcal{L}}_1\Delta t}e^{-\hat{\mathcal{L}}_1\Delta t}]\,e^{-\hat{\mathcal{L}}_2\Delta t/2} = e^{\hat{\mathcal{L}}_2\Delta t/2}e^{-\hat{\mathcal{L}}_2\Delta t/2} = \mathcal{I} \end{aligned} \tag{15.11}$$

The time reversibility is a property of all the particle systems, except in the dissipative particle approximation, where a portion of the system energy is transformed into heat and vice versa, making the time evolution irreversible [4].

15.3 Evolving Positions and Momenta

In the previous section, we briefly discussed the Trotter expression and the way it enriches the time propagator expressions. The next step is to apply the classical time propagator to implement numerical time integrators. We start from the operators $\hat{\mathcal{L}}_1$ and $\hat{\mathcal{L}}_2$ of Eqs. 15.5.

$$\hat{\mathcal{L}}_1 = \sum_{i=1}^{n} \frac{\mathbf{p}_i}{m_i} \cdot \nabla_i \quad ; \quad \hat{\mathcal{L}}_2 = \sum_{i=1}^{n} \mathbf{f}_i \cdot \nabla_{p_i}$$

The operators are deduced from a Hamiltonian of type $\mathcal{H} = \sum_{i=1}^{n} \mathbf{p}_i^2/2m_i + V(\mathbf{r}_i)$. The time propagator 15.2 with inclusion of Eq. 15.8 evolves the function f from time t_0 to time t.

$$F(t) = e^{\hat{\mathcal{L}}\Delta t}\,F(t_0) = \left[e^{\hat{\mathcal{L}}_2\Delta t/2}\,e^{\hat{\mathcal{L}}_1\Delta t}\,e^{\hat{\mathcal{L}}_2\Delta t/2}\right]F(t_0) \tag{15.12}$$

The insertion of the individual operators $\hat{\mathcal{L}}_1$ and $\hat{\mathcal{L}}_2$ in the exponential functions gives:

$$F(t) = \left[\Pi_{i=1}^{n} \exp\left(\frac{\Delta t}{2}\,\mathbf{f}_i \cdot \nabla_{p_i}\right)\right]\left[\Pi_{i=1}^{n} \exp\left(\Delta t\,\frac{\mathbf{p}_i}{m_i} \cdot \nabla_i\right)\right] \left[\Pi_{i=1}^{n} \exp\left(\frac{\Delta t}{2}\,\mathbf{f}_i \cdot \nabla_{p_i}\right)\right] F(t_0) \tag{15.13}$$

where the summations present in the $\hat{\mathcal{L}}_1$ and $\hat{\mathcal{L}}_2$ operators were transformed onto products of exponential functions in the expression of $F(t)$. We now evaluate the action of the exponential operators on function F, which is considered to be either $\mathbf{r}_i$ or $\mathbf{p}_i$, to advance the position and momentum of the ith particle. In this process, the exponential function $\exp[a\,\partial/\partial g(t)]$ is approximated:

$$\exp[a\,\partial/\partial g(t)] \simeq 1 + a\,\partial/\partial g(t)$$

where a is independent of t. The first operator to be applied is the rightmost expression of Eq. 15.13.

$$\mathbf{p}_i(\Delta t/2) = \exp\left(\frac{\Delta t}{2}\,\mathbf{f}_i \cdot \nabla_{p_i}\right)\mathbf{p}_i(t_0) = \mathbf{p}_i(t_0) + \frac{\Delta t}{2}\,\mathbf{f}_i(\mathbf{r}_i(t_0))$$

The term $(\Delta t/2)\mathbf{f}_i$ was identified with a and the function g with the momentum of the particle. The operator in the middle of Eq. 15.13 acts on the positions. The computations are performed in a similar way as the previous one.

$$\mathbf{r}_i(\Delta t) = \exp\left(\Delta t\,\frac{\mathbf{p}_i}{m_i} \cdot \nabla_i\right)\mathbf{r}_i(t_0) = \mathbf{r}_i(t_0) + \Delta t\,\mathbf{p}_i(\Delta t/2)/m_i$$

The leftmost operator of Eq. 15.13 acts on the momentum. It is computed in a similar form, but shifted in time.

$$\mathbf{p}_i(\Delta t) = \exp\left(\frac{\Delta t}{2}\,\mathbf{f}_i \cdot \nabla_{p_i}\right)\mathbf{p}_i(\Delta t/2) = \mathbf{p}_i(\Delta t/2) + \frac{\Delta t}{2}\,\mathbf{f}_i(\mathbf{r}_i(\Delta t))$$

It is convenient to gather all the results with the proper time order.

$$\begin{aligned}
&\mathbf{p}_i(\Delta t/2) = \mathbf{p}_i(t_0) + \tfrac{\Delta t}{2}\,\mathbf{f}_i(\mathbf{r}_i(t_0)) \\
&\mathbf{r}_i(\Delta t) = \mathbf{r}_i(t_0) + \Delta t\,\mathbf{p}_i(\Delta t/2)/m_i \quad ; \quad \mathbf{f}_i(\mathbf{r}_i(\Delta t)) \text{ is evaluated} \\
&\mathbf{p}_i(\Delta t) = \mathbf{p}_i(\Delta t/2) + \tfrac{\Delta t}{2}\,\mathbf{f}_i(\mathbf{r}_i(\Delta t))
\end{aligned} \tag{15.14}$$

The set of Eqs. 15.14 represents a process in the evaluation of the positions and momenta in terms of the time in the simulation; given the initial conditions $\{\mathbf{r}_i(t_0), \mathbf{p}_i(t_0)\}$, the momentum should be first obtained at half time step, $\mathbf{p}_i(\Delta t/2)$, to later proceed with the computation of the position $\mathbf{r}_i(\Delta t)$ at the immediate full time step, and finish with the momentum at the full time step, $\mathbf{p}_i(\Delta t)$. The

force is evaluated at full time steps, after $\mathbf{r}_i(\Delta t)$ is determined. The strategy based on the Liouville time propagator to create time integrators is known in the literature as the *RESPA approach*, acronym of reversible reference system propagator algorithm. The *RESPA* approach is characterized by its stability, with the possibility of simultaneously dealing with different time scales, ensuring time reversibility, accelerating the integration of the equations of motion, and sufficiently versatile to include other integration strategies [5].

The Eqs. 15.14 may be further simplified. To do this, the first identity is inserted in the second and third expressions, to end up with just a pair of equations, one for the position and the other for the momentum.

$$\begin{aligned}
\mathbf{r}_i(\Delta t) &= \mathbf{r}_i(t_0) + \mathbf{p}_i(t_0)\ \Delta t/m_i + \mathbf{f}_i(\mathbf{r}_i(t_0))\ \Delta t^2/(2m_i) \\
&\quad \mathbf{f}_i(\mathbf{r}_i(\Delta t)) \text{ is evaluated} \\
\mathbf{p}_i(\Delta t) &= \mathbf{p}_i(t_0) + [\mathbf{f}_i(\mathbf{r}_i(t_0)) + \mathbf{f}_i(\mathbf{r}_i(\Delta t))]\ \Delta t/2
\end{aligned} \tag{15.15}$$

This integrator is recognized as the *velocity-Verlet time integrator*. In the next time step, the position and velocity may be advanced forward in time by a time amount Δt.

$$\begin{aligned}
\mathbf{r}_i(2\Delta t) &= \mathbf{r}_i(\Delta t) + \Delta t\ \mathbf{v}_i(\Delta t) + \tfrac{(\Delta t)^2}{2}\ \mathbf{a}_i(\mathbf{r}_i(\Delta t)) \\
&\quad \mathbf{f}_i(\mathbf{r}_i(2\Delta t)) \text{ is evaluated} \\
\mathbf{v}_i(2\Delta t) &= \mathbf{v}_i(\Delta t) + \tfrac{\Delta t}{2}\ [\mathbf{a}_i(\mathbf{r}_i(\Delta t)) + \mathbf{a}_i(\mathbf{r}_i(2\Delta t))]
\end{aligned} \tag{15.16}$$

We are equally interested in traveling in the reverse time direction employing the velocity-Verlet equations. To do this, we suppose that several time steps have been performed in the simulation. The goal is to reach point $[\mathbf{r}_i(\Delta t), \mathbf{v}_i(\Delta t), \mathbf{a}_i(\Delta t)]$ from point $[\mathbf{r}_i(2\Delta t), \mathbf{v}_i(2\Delta t), \mathbf{a}_i(2\Delta t)]$. The negative time step, $-\Delta t$, is considered in Eq. 15.16 to build the new position and velocity, denoted by $\mathbf{r}_i(t')$ and $\mathbf{v}_i(t')$.

$$\begin{aligned}
\mathbf{r}_i(t') &= \mathbf{r}_i(2\Delta t) - \Delta t\ \mathbf{v}_i(2\Delta t) + \tfrac{(\Delta t)^2}{2}\ \mathbf{a}_i(\mathbf{r}_i(2\Delta t)) \\
\mathbf{v}_i(t') &= \mathbf{v}_i(2\Delta t) - \tfrac{\Delta t}{2}\ [\mathbf{a}_i(\mathbf{r}_i(\Delta t)) + \mathbf{a}_i(\mathbf{r}_i(2\Delta t))]
\end{aligned} \tag{15.17}$$

We require to identify the variables $\mathbf{r}_i(t')$ and $\mathbf{v}_i(t')$. We insert the velocity-Verlet Eqs. 15.16 in terms of $\mathbf{r}_i(\Delta t)$ and $\mathbf{v}_i(\Delta t)$ in the above expressions.

$$\begin{aligned}
\mathbf{r}_i(t') &= \{\mathbf{r}_i(\Delta t) + \Delta t\ \mathbf{v}_i(\Delta t) + \frac{(\Delta t)^2}{2}\ \mathbf{a}_i(\Delta t)\} \\
&\quad - \{\mathbf{v}_i(\Delta t) + \frac{\Delta t}{2}[\mathbf{a}_i(\Delta t) + \mathbf{a}_i(2\Delta t)]\}\ \Delta t + \mathbf{a}_i(2\Delta t)\ \Delta t^2/2 \\
\mathbf{v}_i(t') &= \{\mathbf{v}_i(\Delta t) + \frac{\Delta t}{2}[\mathbf{a}_i(\Delta t) + \mathbf{a}_i(2\Delta t)]\} \\
&\quad - [\mathbf{a}_i(\Delta t) + \mathbf{a}_i(2\Delta t)]\Delta t/2
\end{aligned}$$

The results of such equations are $\mathbf{r}_i(t') = \mathbf{r}_i(\Delta t)$ and $\mathbf{v}_i(t') = \mathbf{v}_i(\Delta t)$. Therefore, the equations permit time reversibility because the old variables can be recovered systematically from the new ones.

$$\mathbf{r}_i(\Delta t), \mathbf{v}_i(\Delta t), \mathbf{a}_i(\Delta t) \quad \underleftarrow{-\Delta t} \quad \mathbf{r}_i(2\Delta t), \mathbf{v}_i(2\Delta t), \mathbf{a}_i(2\Delta t)$$

Problem 15.3 Demonstrate that $F(t + \Delta t)$ is the result of evolving $F(t)$ in time after using the operator $\exp(\Delta t\, \partial/\partial t)$ on $F(t)$. On the other hand, consider the diffusion expression $\partial\rho/\partial t = d\,\nabla^2\rho$, Eq. 12.46, and the Newtonian equation $f_x(t) = m\, d^2x/dt^2$. They are written in terms of finite differences:

$$[\rho(t + \delta t) - \rho(t)] / \delta t = (d/2) \left[\nabla^2\rho(t + \delta t) + \nabla^2\rho(t)\right]$$
$$f_x = m\, [x(t + \delta t) - 2x(t) + x(t - \delta t)] / \delta t^2$$

Are these equations time reversible?

Solution The exponential function is expanded in a power series and applied to F at the instant of time t.

$$\exp\left[\Delta t \frac{\partial}{\partial t}\right] F(t) = \left[1 + \frac{\Delta t}{1!}\frac{\partial}{\partial t} + \frac{\Delta t^2}{2!}\frac{\partial^2}{\partial t^2} + \frac{\Delta t^3}{3!}\frac{\partial^3}{\partial t^3} + \cdots\right] F(t)$$

$$= F(t) + F^{(1)}(t)\, \Delta t + \frac{1}{2!} F^{(2)}(t)\, \Delta t^2 + \cdots$$
$$= F(t + \Delta t)$$

Therefore, $\exp(\Delta t\, \partial/\partial t)\, F(t)$ advances the function F forward in time by an amount Δt. In reference to the second point, the time evolution proceeds from t to $t + \delta t$. In order to travel backward in time, we evolve the system from t to $t - \delta t$ in the diffusion equation and from $t + \delta t$ to $t - \delta t$ in the Newtonian expression.

$$\frac{\rho(t - \delta t) - \rho(t)}{\delta t} = \frac{d}{2}\left[\nabla^2\rho(t - \delta t) + \nabla^2\rho(t)\right] ;$$
$$f_x = m\, \frac{x(t - \delta t) - 2x(t) + x(t + \delta t)}{\delta t^2}$$

We rename variables by writing $t = t' + \delta t$ in both equations.

(continued)

$$-\frac{[\rho(t'+\delta t)-\rho(t')]}{\delta t}=\frac{d}{2}\left[\nabla^2\rho(t'+\delta t)+\nabla^2\rho(t')\right] \quad ;$$

$$f_x = m\,\frac{x(t'+2\delta t)-2x(t'+\delta t)\,+\,x(t')}{\delta t^2}$$

After doing the time-reverse process, the initial and final diffusion equations differ in signs, but the initial and final Newtonian equations are unchanged. The diffusion equation is time irreversible, and the Newtonian equation is time reversible.

15.4 Simplified Time Integrators

In the following sections, we derive some integrators of common type from the equations of the velocity-Verlet approach, maintaining the character of a series expansion. The differences among the integrators are due to the way the positions and the velocities are processed. In some of those integrators, the positions and velocities are shown in synchronous or retarded forms, namely, the dynamical variables may be defined at the same instant of time, like $\mathbf{r}(t)$ and $\mathbf{v}(t)$, or one of them may be delayed (retarded) with respect to the other, like the velocity $\mathbf{v}(t-\Delta t)$ from the position $\mathbf{r}(t)$.

In order to abbreviate the notation, we use $\mathbf{r}$ to represent the position variable of any particle; however, the notation $\mathbf{r}_\alpha$, with a Greek index α, makes reference to a particular atom. The simulation time is partitioned in small intervals dt. The time step dt in a simulation is frequently proposed as the oscillation period of the fastest particle divided by a factor of ten. For example, if the hydrogen atom of the HCl molecule vibrates with a period of $1.13\times10^{-14}\,s\approx10\,fs$, then the time step to choose in the simulation is $dt=1\,fs$. A variable evaluated at the ith instant of time means evaluation at $t_i=i\,dt$, with $i=0,1,\cdots$. The evaluation at half-integer numbers is equally possible, $t_{i+1/2}=i\,dt+dt/2$.

$$t_i = i\,dt \quad ; \quad t_{i+1/2}=i\,dt+dt/2 \quad ; \quad i=0,1,\cdots \tag{15.18}$$

The representation $\mathbf{r}(t_i)$ is replaced by $\mathbf{r}_i$, which should not be confused with the position of particle i. The same treatment is given to the other dynamical variables, like the velocity $\mathbf{v}_i$ and acceleration $\mathbf{a}_i$. We start with the leapfrog algorithm and continue with the Verlet algorithm. Their basic characteristics reside on the simplicity, computational efficiency, accuracy using sufficiently small time steps, stability, and time reversibility.

15.5 Leapfrog Algorithm

There are alternative time integrators that may be obtained from the expressions of the velocity-Verlet integrator, Eqs. 15.15. In order to derive one of those integrators, we start from Eqs. 15.15, which are written with the new notation here.

$$\mathbf{r}_{i+1} = \mathbf{r}_i + \mathbf{v}_i\, dt + \mathbf{a}_i\, dt^2/2 \quad ; \quad \mathbf{v}_{i+1} = \mathbf{v}_i + (\mathbf{a}_i + \mathbf{a}_{i+1})\, dt/2 \tag{15.19}$$

All the variables are evaluated at integral time steps. We suppose the equivalences $\mathbf{v}_{i+1/2} = \mathbf{v}_i + \mathbf{a}_i\, dt/2$ and $\mathbf{a}_{i+1/2} = (\mathbf{a}_i + \mathbf{a}_{i+1})/2$. They correspond in that order to the last two terms of $\mathbf{r}_{i+1}$ and $\mathbf{v}_{i+1}$ of Eqs. 15.19. Thus, they are inserted in such equations, but writing $j = i + 1$.

$$\mathbf{r}_j = \mathbf{r}_{j-1} + \mathbf{v}_{j-1/2}\, dt \quad ; \quad \mathbf{v}_j = \mathbf{v}_{j-1} + \mathbf{a}_{j-1/2}\, dt$$

These expressions may be interpreted as follows: the evolution of the position is governed by the velocity evaluated at a half-integer time step, while the evolution of the velocity is governed by the acceleration evaluated at a half-integer time step. Still, the equation of $\mathbf{r}_j$ is given in terms of $\mathbf{v}_{j-1/2}$, while the equation of the right gives $\mathbf{v}_j$. The equations are not linked to each other because the speeds are shifted in time. In order to make them self-consistent, we change the j index of the velocity by $j + 1/2$.

$$\mathbf{r}_j = \mathbf{r}_{j-1} + \mathbf{v}_{j-1/2}\, dt \quad ; \quad \mathbf{v}_{j+1/2} = \mathbf{v}_{j-1/2} + \mathbf{a}_j\, dt \tag{15.20}$$

The position and velocity are now coupled through velocities computed at half-integer time steps. The above equations constitute the *leapfrog time integrator*. It is based on a time leap since the position and velocity differ by a half-integer time step. Also, the acceleration is shown in the expression of the velocity, but it is not present in the expression of the position. The leapfrog integrator is accurate up to second order. In the case of accuracy or stability problems, it is always possible to consider sufficiently small time steps to avoid such problems, with the consequent computational cost of performing more steps in the simulation. In order to illustrate the propagation of variables in this model, let us assume that the initial values $(\mathbf{r}_0, \mathbf{v}_0, \mathbf{a}_0)$ are given, Fig. 15.1. An initial expression of the velocity is proposed:

$$\mathbf{v}_{1/2} = \mathbf{v}_0 + \mathbf{a}_0\, dt/2$$

There is no general consensus to offset the first velocity $\mathbf{v}_{1/2}$; nevertheless, the above proposal goes along the initial suppositions on the formulation of the leapfrog algorithm. Once the first velocity is determined, we obtain the first position $\mathbf{r}_1$ in terms of the known quantities according to the leapfrog expression 15.20.

$$\mathbf{r}_1 = \mathbf{r}_0 + \mathbf{v}_{1/2}\, dt$$

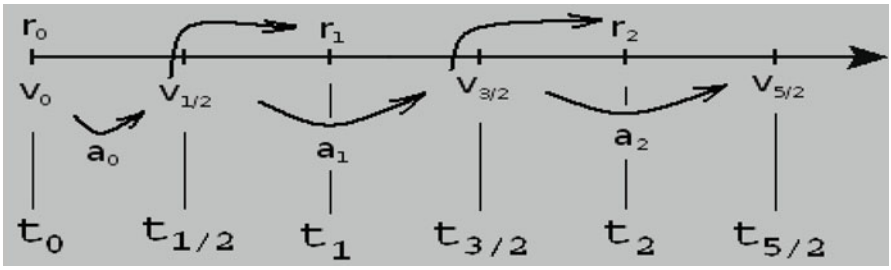


Fig. 15.1 The leapfrog algorithm is illustrated. The initial conditions are (r_0, v_0, a_0) at the instant of time t_0. Once the velocity $v_{1/2}$ is obtained, the new position r_1 is determined. At this stage, the process is repeated, computing the particle acceleration from the force

Table 15.1 Propagation of the position and velocity forward and backward in time in the leapfrog approach. The force f_i is immediately computed after the position r_i is known. The initial conditions (r_0, v_0, a_0) are supposed to be given*

	r_0, v_0, a_0	r_0, v_0, a_0	
	$\vdots$	$\vdots$	
↓	$r_i = r_{i-1} + v_{i-1/2}\,dt$	$r_i = r_{i+1} - v_{i+1/2}\,dt$	↑
	$v_{i+1/2} = v_{i-1/2} + a_i\,dt$	$v_{i+1/2} = v_{i+3/2} - a_{i+1}\,dt$	
T	$r_{i+1} = r_i + v_{i+1/2}\,dt$	$r_{i+1} = r_{i+2} - v_{i+3/2}\,dt$	T
i	$v_{i+3/2} = v_{i+1/2} + a_{i+1}\,dt$	$v_{i+3/2} = v_{i+5/2} - a_{i+2}\,dt$	i
m	$r_{i+2} = r_{i+1} + v_{i+3/2}\,dt$	$r_{i+2} = r_{i+3} - v_{i+5/2}\,dt$	m
e	$v_{i+5/2} = v_{i+3/2} + a_{i+2}\,dt$	$v_{i+5/2} = v_{i+7/2} - a_{i+3}\,dt$	e
	$r_{i+3} = r_{i+2} + v_{i+5/2}\,dt$	$\vdots$	
↓	$v_{i+7/2} = v_{i+5/2} + a_{i+3}\,dt$	$r_{i+3}, v_{i+7/2}, a_{i+3}$	↑
	$\vdots$	$\vdots$	

*The leapfrog equations in the forward and backward time directions are similar because in both situations the previous position, velocity, and acceleration are required, except for the change of signs of dt

The force is calculated at the new position $\mathbf{r}_1$, giving $\mathbf{a}_1$. This gives the opportunity to compute the next speed from the leapfrog proposal.

$$\mathbf{v}_{3/2} = \mathbf{v}_{1/2} + \mathbf{a}_1\,dt$$

At this stage the first cycle is completed in the time evolution process because we know $\mathbf{r}_1$, $\mathbf{v}_{3/2}$, $\mathbf{a}_1$. The second cycle similarly follows the leapfrog algorithm. The general time-forward process is sketched in Table 15.1.

The leapfrog algorithm also allows time reversibility. Thus, after advancing a certain number of time steps forward in time, it is possible to return to the original position and velocity by reversing the algorithm. The time reversibility is ascribed to the symmetry in the expressions of both the position and velocity in the leapfrog

approach. In order to exemplify the time reversibility, we suppose that the last known vectors are $(\mathbf{r}_{i+3},\ \mathbf{v}_{i+7/2},\ \mathbf{a}_{i+3})$, Table 15.1. According to the leapfrog equations, we firstly solve for the velocity $\mathbf{v}_{i+5/2}$ and later for the position $\mathbf{r}_{i+2}$.

$$\mathbf{v}_{i+5/2} = \mathbf{v}_{i+7/2} - \mathbf{a}_{i+3}\, dt \quad ; \quad \mathbf{r}_{i+2} = \mathbf{r}_{i+3} - \mathbf{v}_{i+5/2}\, dt$$

These operations give the new set of values $(\mathbf{r}_{i+2}, \mathbf{v}_{i+5/2}, \mathbf{a}_{i+2})$, where the acceleration $\mathbf{a}_{i+2}$ is computed at the position $\mathbf{r}_{i+2}$. This process gives the first reversed cycle because the old positions and velocities, $(\mathbf{r}_{i+2}, \mathbf{v}_{i+5/2}, \mathbf{a}_{i+2})$, are recovered from $(\mathbf{r}_{i+3}, \mathbf{v}_{i+7/2}, \mathbf{a}_{i+3})$. By repeating a number of similar steps, it is possible to reach the initial conditions from which the dynamics was originally started, Table 15.1. In short, the leapfrog expressions are invariant under time reversibility, as in going backward in time the functional forms of the equations are retained, except for the corresponding signs of the time steps.

The time reversibility is an important feature of time integrators because such a property allows to verify the conservation of physical quantities like the energy, linear and angular momenta, and more, in the reversed time direction. The lack of conservation of such variables may be attributed to many possible situations that are capable of producing a long-term drift of the particle trajectories. However, it is rarely associated with the leapfrog time integrator. A machine performs millions of mathematical operations with small rounding errors, which accumulate in the molecular simulation leading to slightly different energies of the particle system. In this case, the machine, not the numeric integrator, produces the deviations. On the other hand, the systematic errors may be also present. These may be estimated by knowing the number of reliable digits of the energy. Be also aware of the cancellation of errors which, in principle, may mask important deviations. Fortunately, this is an infrequent situation.

15.6 Verlet Algorithm

The Verlet algorithm is a time integrator which allows to unfold the dynamics to create the trajectory and history of each particle with time. In order to derive the equations of the Verlet approach, we start from the velocity-Verlet expression of the position given in Eqs. 15.15. It is written in the forward and backward time directions.

$$\begin{aligned} \mathbf{r}_{i+1} &= \mathbf{r}_i + \mathbf{v}_i\, dt + \tfrac{1}{2}\,\mathbf{a}_i\, dt^2 + \tfrac{1}{6}\,\mathbf{b}_i\, dt^3 + \mathcal{O}(dt^4) \\ \mathbf{r}_{i-1} &= \mathbf{r}_i - \mathbf{v}_i\, dt + \tfrac{1}{2}\,\mathbf{a}_i\, dt^2 - \tfrac{1}{6}\,\mathbf{b}_i\, dt^3 + \mathcal{O}(dt^4) \end{aligned} \tag{15.21}$$

The sum of the previous equations gives the position at the instant of time t_{i+1}.

$$\mathbf{r}_{i+1} = 2\mathbf{r}_i - \mathbf{r}_{i-1} + \mathbf{a}_i\, dt^2 + \mathcal{O}(dt^4) \tag{15.22}$$

The new position is given in terms of the previous positions and acceleration. In contraposition to the leapfrog position 15.20, the Verlet position has no direct dependence on the velocity, but the acceleration. The velocity is required to evaluate the kinetic energy, and thus, a proposal of the velocity is needed. The velocity is given by the difference of Eqs. 15.21.

$$\mathbf{v}_i = [\mathbf{r}_{i+1} - \mathbf{r}_{i-1}] / (2dt) - \mathbf{b}_i dt^3 / [3\,(2dt)] + \mathcal{O}(dt^5) \tag{15.23}$$

The term with the contribution $\mathbf{b}_i$ may be neglected to deal with a simplified expression of the velocity.

$$\mathbf{v}_i = (\mathbf{r}_{i+1} - \mathbf{r}_{i-1}) / (2dt) + \mathcal{O}(dt^2) \tag{15.24}$$

In order to compute the velocity at time t_i, we require to have the positions at the instants of time t_{i+1} and t_{i-1}. The computation of the velocity $\mathbf{v}_i$ is delayed by one time step from the position $\mathbf{r}_{i+1}$. Equations 15.22 and 15.24 constitute the *Verlet time integrator* of the position and velocity; they are written below for easy access.

$$\begin{aligned} \mathbf{r}_{i+1} &= 2\mathbf{r}_i - \mathbf{r}_{i-1} + \mathbf{a}_i\, dt^2 + \mathcal{O}(dt^4) \\ \mathbf{v}_i &= (\mathbf{r}_{i+1} - \mathbf{r}_{i-1}) / (2dt) + \mathcal{O}(dt^2) \end{aligned} \tag{15.25}$$

The acceleration is derived from the force exerted on the particle, which in turn is computed on the fly from the first-principles mechanical equations, involving a quantum electronic-structure method, or derived from a classical model potential.

It is of interest to estimate the accumulation of errors in the Verlet approach in a time interval τ of the simulation. We know that the position 15.25 is accurate up to third order.

$$Error\,(\mathbf{r}_{i+1}) = \mathcal{O}(dt^4) \tag{15.26}$$

The simulation is supposed to be initiated from the position $\mathbf{r}_i$, at time t_i. The error in the position at this instant of time is null, $Error\,(\mathbf{r}_i) = 0$. The error accumulated in the following step is calculated from expression 15.25, namely, $\mathbf{r}_{i+2} = 2\mathbf{r}_{i+1} - \mathbf{r}_i + \mathbf{a}_{i+1}\, dt^2 + \mathcal{O}(dt^4)$.

$$Error\,(\mathbf{r}_{i+2}) = 2\,Error\,(\mathbf{r}_{i+1}) - Error\,(\mathbf{r}_i) + \mathcal{O}(dt^4) = 3\,\mathcal{O}(dt^4)$$

No error is attributed to the force. At step t_{i+3}, the computation of the error is similarly worked out. We have $\mathbf{r}_{i+3} = 2\mathbf{r}_{i+2} - \mathbf{r}_{i+1} + \mathbf{a}_{i+2}\, dt^2 + \mathcal{O}(dt^4)$, and there is an error of $\mathbf{r}_{i+2}$, an error of $\mathbf{r}_{i+1}$, Eq. 15.26, and the carrying error $\mathcal{O}(dt^4)$.

$$\begin{aligned} Error\,(\mathbf{r}_{i+3}) &= 2\,Error\,(\mathbf{r}_{i+2}) - Error\,(\mathbf{r}_{i+1}) + \mathcal{O}(dt^4) \\ &= 2[3\,\mathcal{O}(dt^4)] - \mathcal{O}(dt^4) + \mathcal{O}(dt^4) = 6\,\mathcal{O}(dt^4) \end{aligned}$$

We follow a similar procedure at the instant of time t_{i+4}, with $\mathbf{r}_{i+4} = 2\mathbf{r}_{i+3} - \mathbf{r}_{i+2} + \mathbf{a}_{i+3}\, dt^2 + \mathcal{O}(dt^4)$.

$$Error\,(\mathbf{r}_{i+4}) = 2\,Error\,(\mathbf{r}_{i+3}) - Error\,(\mathbf{r}_{i+2}) + \mathcal{O}(dt^4)$$

$$= 2[6\,O(dt^4)] - 3\,\mathcal{O}(dt^4) + \mathcal{O}(dt^4) = 10\,\mathcal{O}(dt^4)$$

The generalization of the result to n simulation steps is:

$$Error\,(\mathbf{r}_{i+n}) = \tfrac{n(n+1)}{2}\,\mathcal{O}(dt^4) \tag{15.27}$$

By supposition, n time steps cover the time interval τ, this is, $ndt = \tau$. The error gathered in such an interval of time is obtained by inserting $n = \tau/dt$ in Eq. 15.27.

$$Error\,(\mathbf{r}_{i+n}) = \tfrac{1}{2}\left(\tfrac{\tau^2}{dt^2} + \tfrac{\tau}{dt}\right)\mathcal{O}(dt^4) = \mathcal{O}(dt^2) \tag{15.28}$$

The error in the velocity, Eq. 15.25, is of order $\mathcal{O}(dt^2)$ because the expression of the velocity is based on the difference of contiguous positions. For example, for $i + 1$ simulation steps, we have:

$$\begin{aligned} Error\,(\mathbf{v}_i) &= \frac{Error\,(\mathbf{r}_{i+1}) - Error\,(\mathbf{r}_{i-1})}{2dt} + \mathcal{O}(dt^2) \\ &= \frac{1}{2dt}\left[\frac{(i+1)(i+2)}{2} - \frac{(i-1)i}{2}\right]\mathcal{O}(dt^4) + \mathcal{O}(dt^2) \\ &= \frac{1}{4dt}\,[4i+2]\,\mathcal{O}(dt^4) + \mathcal{O}(dt^2) \end{aligned}$$

The error of the Verlet velocity remains of order $\mathcal{O}(dt^2)$. The Verlet scheme is one of the most commonly used integrators in molecular dynamics. It is numerically stable and exhibits time reversibility when small time steps are employed. Other numerical time integrators are available such as the Runge-Kutta, or predictor-corrector schemes. They all find wide applications in computer animation to produce ripples in water, cloth movements, articulated bodies, etc. The utility of any integrator is recognized when the trajectory of each particle of a many-particle system is accomplished in an extended time interval, and thus, the history of the particle system is known.

Problem 15.4 The position of the center of mass, cm, of a particle system is $\mathbf{r}_{cm}$. The velocity of the mass center obtained from the positions $\mathbf{r}_{cm}$ in the Verlet approach is:

(continued)

$$\mathbf{v}_{cm}(t) = [\mathbf{r}_{cm}(t+dt) - \mathbf{r}_{cm}(t-dt)] \,/\, (2dt)$$

Obtain $\mathbf{v}_{cm}$ from the set of particles with positions $\mathbf{r}_\alpha$ and velocities $\mathbf{v}_\alpha$.

Solution The speed of the mass center deduced from the particles, p, is:

$$\mathbf{v}_{cm}^{(p)}(t) = \sum_\alpha m_\alpha \mathbf{v}_\alpha(t)/M \quad ; \quad M = \sum_\alpha m_\alpha$$

The velocity $\mathbf{v}_\alpha$, Eq. 15.25, of the α particle is $[\mathbf{r}_\alpha(t+dt) - \mathbf{r}_\alpha(t-dt)]/2dt$.

$$\begin{aligned}\mathbf{v}_{cm}^{(p)}(t) &= \frac{\sum_\alpha m_\alpha[\mathbf{r}_\alpha(t+dt) - \mathbf{r}_\alpha(t-dt)]}{2dt\, M} \\ &= \frac{\sum_\alpha m_\alpha \mathbf{r}_\alpha(t+dt)}{2dt\, M} - \frac{\sum_\alpha m_\alpha \mathbf{r}_\alpha(t)}{2dt\, M}\end{aligned}$$

The terms on the right correspond to the cm position, $\mathbf{r}_{cm} = \sum_\alpha m_\alpha \mathbf{r}_\alpha / M$.

$$\mathbf{v}_{cm}^{(p)}(t) = [\mathbf{r}_{cm}(t+dt) - \mathbf{r}_{cm}(t-dt)]/(2dt)$$

The expression $\mathbf{v}_{cm}^{(p)}$ deduced from the set of positions $\{\mathbf{r}_\alpha\}$ is equal to the velocity $\mathbf{v}_{cm}$ directly deduced from $\mathbf{r}_{cm}$, this is, $\mathbf{v}_{cm}^{(p)} = \mathbf{v}_{cm}$.

15.7 Bond Constraints

In many instances, we are required to constrain the bond of a given pair of atoms. A constraint of such a type demands a force to keep the atoms with the required separation; however, the force also has effect on the speeds of the atoms. We follow the algorithm of Andersen, known as the *rattle model*, to determine the equations that impose bond constraints in molecular dynamics [6]. A holonomic constraint between the ith and jth atoms starts from the distance condition σ_{ij} and the time derivative $\dot{\sigma}_{ij}(t)$.

$$\begin{aligned}\sigma_{ij}(t) &= [\mathbf{r}_i(t) - \mathbf{r}_j(t)]^2 - d_{ij}^2 \\ \dot{\sigma}_{ij}(t) &= 2[\mathbf{r}_i(t) - \mathbf{r}_j(t)] \cdot [\dot{\mathbf{r}}_i(t) - \dot{\mathbf{r}}_j(t)]\end{aligned} \tag{15.29}$$

The conditions $\sigma_{ij}(t) = 0$ and $\dot{\sigma}_{ij}(t) = 0$ are used to impose the required distance d_{ij} on the ij-bond at every instant of time in the simulation. The Lagrangian equation of motion that considers the net intermolecular force $\mathbf{f}_i$ exerted by the other atoms of the molecule on the ith atom, plus the constraining force $\mathbf{g}_i$, is:

$$
\begin{aligned}
m_i\,\partial^2\mathbf{r}_i(t)\,/\,\partial t^2 &= \mathbf{f}_i(t)+\mathbf{g}_i(t) \quad ; \quad \mathbf{f}_i(t)=-\nabla_i U \\
\mathbf{g}_i(t) &= -\sum_j{}' \lambda_{ij}(t)\,\nabla_i\sigma_{ij}(t) = -\sum_j{}' \lambda_{ij}(t)\,2(\mathbf{r}_i-\mathbf{r}_j)
\end{aligned}
\tag{15.30}
$$

The force $\mathbf{f}_i$ is obtained from the interaction potential U. The constraints are introduced using the Lagrange multipliers λ_{ij}, which depend on the time. The summation considers the atoms forming constrained bonds with the ith atom. If we have $3N$ equations of the type 15.30 and a total of n constraints, then there are $3N+n$ variables of the type $(\{\mathbf{r}_i(t)\},\{\lambda_{ij}(t)\})$ to be determined. It is with the velocity-Verlet integrator approach, Eqs. 15.15, that we unfold the dynamics of the particles satisfying the previous constraints. In this case, the velocity-Verlet equations are:

$$
\begin{aligned}
\mathbf{r}_i(t+\delta t) &= \mathbf{r}_i(t)+\delta t\,\dot{\mathbf{r}}_i(t)+\frac{(\delta t)^2}{2m_i}\left[\mathbf{f}_i(t)-2\sum_j{}'\lambda_{rij}(t)\,\mathbf{r}_{ij}(t)\right] \\
\mathbf{r}_{ij}(t) &= \mathbf{r}_i(t)-\mathbf{r}_j(t) \\
\dot{\mathbf{r}}_i(t+\delta t) &= \dot{\mathbf{r}}_i(t)+\frac{\delta t}{2m_i}\left[\mathbf{f}_i(t)-2\sum_j{}'\lambda_{rij}(t)\,\mathbf{r}_{ij}(t)+\right. \\
&\qquad \left.+\,\mathbf{f}_i(t+\delta t)-2\sum_j{}'\lambda_{vij}(t+\delta t)\,\mathbf{r}_{ij}(t+\delta t)\right]
\end{aligned}
\tag{15.31}
$$

Under Lagrangian theory, Sect. 6.8, the unknown factors λ_{rij} and λ_{vij} are used to impose the first and second conditions given in Eqs. 15.29. The velocity-Verlet integrator is particularly helpful in this task. The bond constraint, first expression of Eq. 15.29, is maintained at the instants of time $t+\delta t$ and t. The last terms of the first expression of Eqs. 15.31 are written as:

$$
\delta t\left\{\dot{\mathbf{r}}_i(t)+\tfrac{\delta t}{2m_i}\left[\mathbf{f}_i(t)-2\textstyle\sum_j{}'\lambda_{rij}(t)\,\mathbf{r}_{ij}(t)\right]\right\}
\tag{15.32}
$$

It is convenient to introduce the new variable $\mathbf{q}$ and the new Lagrangian multipliers g_{ij} and k_{ij} with the purpose of simplifying Eq. 15.32.

$$
\begin{aligned}
\mathbf{q}_i &= \dot{\mathbf{r}}_i(t)+\tfrac{\delta t}{2m_i}\,\mathbf{f}_i(t)-\tfrac{1}{m_i}\textstyle\sum_j{}'\,g_{ij}\,\mathbf{r}_{ij}(t) \\
g_{ij}(t) &= \delta t\,\lambda_{rij}(t) \quad ; \quad k_{ij}(t+\delta t)=\delta t\,\lambda_{vij}(t+\delta t)
\end{aligned}
\tag{15.33}
$$

The velocity-Verlet expressions are concisely written in terms of $\mathbf{q}_i$, g_{ij}, and k_{ij}.

$$\begin{aligned} \mathbf{r}_i(t+\delta t) &= \mathbf{r}_i(t) + \delta t\, \mathbf{q}_i \\ \dot{\mathbf{r}}_i(t+\delta t) &= \mathbf{q}_i(t) + \tfrac{\delta t}{2m_i}\, \mathbf{f}_i(t+\delta t) - \tfrac{1}{m_i} \sum_j{}' k_{ij}\, \mathbf{r}_{ij}(t+\delta t) \end{aligned} \tag{15.34}$$

A bond constraint is under examination by computing $\mathbf{b}_{ij}^2$ and comparing with d_{ij}^2 in the form shown in Eq. 15.29.

$$\mathbf{b}_{ij} = [\mathbf{r}_i(t) + \delta t\, \mathbf{q}_i(t)] - [\mathbf{r}_j(t) + \delta t\, \mathbf{q}_j(t)] \tag{15.35}$$

The positions satisfy the constraint when $\mathbf{b}_{ij}^2 = d_{ij}^2$. If this last identity is not satisfied, then a correction term proportional to $\mathbf{r}_{ij}$ is introduced to reproduce the required bond distance.

$$\begin{aligned} \mathbf{b}_{ij} = \mathbf{r}_i(t) + \delta t\, \mathbf{q}_i(t) - \delta t\, \tfrac{g}{m_i}\, \mathbf{r}_{ij}(t) \\ - \left[\mathbf{r}_j(t) + \delta t\, \mathbf{q}_j(t) + \delta t\, \tfrac{g}{m_i}\, \mathbf{r}_{ij}(t)\right] \end{aligned} \tag{15.36}$$

The value of g is defined by the condition $\mathbf{b}_{ij}^2 = d_{ij}^2$. The modified positions of the particles are:

$$\begin{aligned} \mathbf{r}_i(t+\delta t) &= \mathbf{r}_i(t) + \delta t\, \mathbf{q}_i(t) - \delta t\, \tfrac{g}{m_i}\, \mathbf{r}_{ij}(t) \\ \mathbf{r}_j(t+\delta t) &= \mathbf{r}_j(t) + \delta t\, \mathbf{q}_j(t) + \delta t\, \tfrac{g}{m_j}\, \mathbf{r}_{ij}(t) \end{aligned} \tag{15.37}$$

A similar procedure is followed for the atom speeds. The velocities in the velocity-Verlet scheme satisfy the constraint when $[\mathbf{r}_i(t) - \mathbf{r}_j(t)] \cdot [\dot{\mathbf{r}}_i(t) - \dot{\mathbf{r}}_j(t)] = 0$, Eq. 15.29. When this constraint is not satisfied, a correction term proportional to the vector $\mathbf{r}_{ij}$ is again introduced to reproduce the required constraint.

$$\dot{\mathbf{r}}_i = \dot{\mathbf{r}}_i(t+\delta t) - \tfrac{k}{m_i}\, \mathbf{r}_{ij}(t+\delta t) \quad ; \quad \dot{\mathbf{r}}_j = \dot{\mathbf{r}}_j(t+\delta t) + \tfrac{k}{m_j}\, \mathbf{r}_{ij}(t+\delta t) \tag{15.38}$$

The value of k is defined by the condition $[\mathbf{r}_i - \mathbf{r}_j] \cdot [\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_j] = 0$, where the modified velocities of the particles at the instant of time $t + \delta t$ are the expressions 15.38.

Problem 15.5 Determine the values of factors g and k of Eqs. 15.37 and 15.38 from the conditions $[\mathbf{r}_i(t+\delta t) - \mathbf{r}_i(t+\delta t)]^2 = d_{ij}^2$ and $[\mathbf{r}_i - \mathbf{r}_j] \cdot [\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_j] = 0$, respectively.

Solution In order to determine g, we deal with the first condition and use the modified Eqs. 15.37 to compute d_{ij}^2.

$$d_{ij}^2 = \left[\mathbf{r}_i(t) + \delta t\, \mathbf{q}_i(t) - \delta t\, \frac{g}{m_i}\, \mathbf{r}_{ij}(t) - \left(\mathbf{r}_j(t) + \delta t\, \mathbf{q}_j(t) + \delta t\, \frac{g}{m_j}\, \mathbf{r}_{ij}(t)\right)\right]^2$$

The expression is simplified by using $\mathbf{b}_{ij}$ of Eq. 15.35.

$$d_{ij}^2 = \left[\mathbf{b}_{ij} - \delta t\, \frac{g}{\mu_{ij}}\, \mathbf{r}_{ij}(t)\right]^2 \quad ; \quad \frac{1}{\mu_{ij}} = \frac{1}{m_i} + \frac{1}{m_j}$$

The first expression is solved for g by just considering linear terms in δt:

$$g = \left[\mathbf{b}_{ij}^2 - d_{ij}^2\right] / (2\delta t)\, \mu_{ij} / \mathbf{b}_{ij} \cdot \mathbf{r}_{ij}$$

In regard to factor k, we deal with the condition $[\mathbf{r}_i - \mathbf{r}_j] \cdot [\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_j] = 0$ and use the modified velocities 15.38.

$$\mathbf{r}_{ij}(t+\delta t) \cdot \left[\dot{\mathbf{r}}_i(t+\delta t) - \dot{\mathbf{r}}_j(t+\delta t) - \frac{k}{\mu_{ij}}\, \mathbf{r}_{ij}(t+\delta t)\right] = 0$$

By solving for factor k, but identifying $\mathbf{r}_{ij}^2 = d_{ij}^2$, we have:

$$k = [\mathbf{r}_{ij} \cdot (\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_j)] / d_{ij}^2\, \mu_{ij}$$

where the previous variables are evaluated at the instant of time $t + \delta t$.

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